The Engineer Factory

How India Priced Its Engineers by the Hour, and What AI Does to That Bargain

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A note on how this book was made

This book was researched, drafted and checked with the help of an AI model made by Anthropic. The same company's products, Claude Code and Claude Cowork, appear in sources discussed in Chapter 19, which deals with AI coding tools and the Indian software industry. Readers should weigh what the book says about AI with that in mind. The products are described there as the sources describe them, and the book makes no claim about them beyond what those sources say. Much of what Chapter 19 describes happened after the AI model's training ended; those passages rest on documents read in October 2026, not on the model's memory.

The work followed a fixed method. A register of sources was compiled first, and every chapter was drafted only from research dossiers built on that register. Each chapter was then checked by a separate pass that re-opened the sources behind every quotation, number, date, name and causal claim, corrected or softened what did not hold, and logged the result. A revision pass removed repetition and contradictions across chapters. The checking logs, the source register and the data files are kept in the book's public repository.

A note on sources

No one was interviewed for this book. Every person who speaks in it does so through a published or archived record: a memoir, an oral history, a company filing, a court judgment, a parliamentary record, a government report, a scholarly study, or a newspaper or magazine report. The notes give the source and, where one exists, the page.

Some people appear only under names given to them by others. Where a reporter or a scholar used a pseudonym, the book uses the same one and says so where the person first appears. No character in the book is a composite, and no scene, thought or line of dialogue has been invented.

Gaps in the record, such as a report that could not be located or a date on which sources disagree, are recorded in the notes rather than in the text. Quotations are exact and kept short. Where a source could be read only in summary, the book paraphrases it rather than quoting. Figures published by companies, by the industry's lobby group Nasscom, or by consultancies that sell to the firms they count are labelled as theirs. Disagreements between sources are shown rather than resolved where the evidence does not settle them. Material published after mid-2025 was checked against the live source in October 2026, and the notes say so.

Ordinary workers speak less often in this book than its subject deserves. The published record of Indian software work is dominated by founders, executives, officials and scholars; the accounts of trainees, testers, call-centre agents and engineers who never became prominent are few, and most are anonymous or pseudonymous. Where such accounts exist, the book uses them and labels them. Where they do not, the gap is part of the story.

Rupee amounts are converted to US dollars at the average rate for the year in question, and Indian fiscal years, which run from April to March, are written in the form FY2023-24. Cities are given the names in use at the time, with the current name at first mention in each chapter.

Prologue. The joining date (2026)

On Thursday 6 February 2025 a 24-year-old trainee at the Infosys training campus in Mysuru, whom the reporter who told her story for *The News Minute* called Gayatri (not her real name), received an email summoning her to a meeting the next day.[1] The email concerned the assessment held at the end of her training, which she had failed a month earlier. It said: 'You are required to maintain confidentiality, hence do not discuss this, or share this calendar with anyone.'[2] 'I was devastated after receiving the email,' she told the reporter. Her seniors, she said, had explained that such an email meant she would be asked to leave.[3]

The next morning, according to her account, security staff at the campus entrance took her electronic devices and her bag, as they did with the other trainees summoned that day. She was taken into a cabin, where a member of the human resources staff asked her to sign a 'mutual separation agreement'.[4] The report says she was coerced. The agreement itself said the opposite: 'You have knowingly and voluntarily chosen to enter into this agreement and have not been forced or pressured in any way to sign it.'[5] She signed. 'I signed it unwillingly. I did not want to quit the company. They wasted two and a half years of my career. It was my dream job,' she said.[6]

What followed took a few hours. The company took back the trainees' laptops, put them on a bus to their accommodation and told them to leave the premises by 5 pm that day.[7] Gayatri, the report says, had no time to arrange travel, so she booked a hotel for the night before going home to Andhra Pradesh.[8] The separation payment Infosys offered was one month's salary, which the reporter put at Rs 25,000.[9] Another trainee, given the name Raju, described the day in six words: 'We were thrown on the road.'[10] A third, given the name Rishi, aged 25, said he had waited nearly two years to be onboarded and had been removed after four months: 'I regret joining the company, they have traumatised me.'[11]

The two and a half years she spoke of had begun with success. Gayatri graduated in 2022 and in the same year cleared Infosys's interview for the post of system engineer, the company's entry-level job for new graduates.[12] Then nothing happened. In the reporter's words she 'was left hanging for nearly two years' and was onboarded in October 2024 'only after numerous emails and a legal intervention'.[13] From July 2024 she and others in her position took part in online 'pre-training', for which they were not paid. 'Pre-training was unheard of earlier,' she said.[14] The unpaid sessions and assessments ran for four weeks in July and six weeks in August, before any of the graduates was an employee.[15]

The offer had mattered beyond her. 'My parents and teachers were very proud when I got the offer letter from Infosys,' she told the reporter. After she failed the test three times, she said, her parents asked whether she had cheated in her college exams.[16] Her own summary of what the termination did was brief: 'The termination has taken away all my self-esteem and confidence.'[17]

This book is about the system that made her offer letter worth waiting two years for. For half a century India built a machine for turning young people into software engineers: colleges that taught them, companies that hired them by the thousand and trained them on campuses like the one at Mysuru, and an export trade that paid for it all. Her story shows that machine from the bottom, at the moment when it seemed to need fewer of the people it had been built to produce.

The waiting

Gayatri was not alone, and her wait did not begin with her. In October 2022 *Business Today* reported that freshers placed by Accenture, Infosys, Wipro and HCLTech were seeing their joining dates pushed back, in some cases into the following year.[18] One graduate told the magazine that the company had postponed the date again two days before

joining: 'I had packed my bags and booked all tickets to move to the new city.'[19] By June 2024 the Nascent Information Technology Employees Senate (NITES), a union of IT workers based in Pune, claimed that more than 10,000 graduates were still waiting for joining dates from Wipro, Infosys, Tata Consultancy Services (TCS), Zensar and LTIMindtree, a figure no company confirmed.[20]

Infosys's case is the best documented, because NITES took it to the government. On 20 August 2024 the union complained to the Union Ministry of Labour and Employment about more than 2,000 campus recruits from 2022 who had not been onboarded.[21] According to NITES, Infosys issued about 2,000 offer letters on 2 and 3 September with joining dates in October.[22] Infosys told the state's labour department that the union's allegations were 'patently false' and that the graduates had already received joining dates.[23]

The recruits arrived in Mysuru that October.[24] *Republic Business* reported that most had joined under the National Apprenticeship Training Scheme, a government programme for on-the-job training, which would matter later.[25] Trainees told the reporter that the pass mark had risen to 65 per cent from 50 per cent in earlier years, and that the syllabus had grown, to 200 hours of material by one magazine's account.[26] A trainee given the name Rohan said the earlier batches had been treated leniently: 'The fact that they made it difficult for us shows that, since the beginning, they never wanted to hire us.'[27] By the report's count, which is not a company figure, of 930 who joined in one October intake, 160 passed at the first attempt and 140 at the second.[28]

Two accounts

Infosys's account deserves to be stated at its strongest. In a statement quoted in the report and by *Deccan Herald*, the company said that all freshers, after foundational training at Mysuru, are expected to clear internal assessments. 'All freshers get three attempts to clear the assessment, failing which they will not be able to continue with the organisation, as is also mentioned in their contract,' it said, adding that the process had been in place 'for over two decades' and ensured 'a high quality of talent availability for our clients'.[29] On this view the test is quality control, in a training system that, later chapters argue, the firms built because colleges did not teach what they needed. In a later round the termination email listed what the company had provided: 'additional preparation time, doubt-clearing sessions, several mock assessments and three attempts'.[30] Infosys offered those let go help with finding other work, courses through the training firms NIIT and UpGrad, and a route into its business-process arm.[31]

The trainees and the union saw a company reducing a batch it no longer wanted. *Outlook Business* reported that termination rates at the assessment had risen from under 10 per cent to between 30 and 40 per cent, and unnamed current Infosys employees told the magazine that 'an excess of resources and fewer projects' lay behind it.[32] The two accounts are not simply opposed: the process could be old, as the company said, and the bar could have been raised, as the trainees said. Alagunambi Welkin, general secretary of the Union of IT and ITES Employees, said that companies hire large numbers 'as a result of poor decision making by the management' and, when money is short, 'sack the employees instead of taking accountability'.[33]

Even the number let go is disputed: Infosys said about 300 in February, the report estimated 400, and NITES, in its complaint to the Labour Ministry, said about 700.[34] The departures did not stop. By one count in the financial press, about 35 more trainees went in March, 240 on 18 April and 195 later that month, almost 800 since February.[35]

The state did not agree with itself. On 13 February 2025 the Union Labour Ministry directed Karnataka's Labour Department to take 'urgent necessary actions to resolve the dispute'.[36] But the Karnataka department, according to an unnamed source quoted by the financial news site Moneycontrol, concluded that no layoff had occurred at all: 'They were all apprenticeship trainees, not employees.'[37] If that was right, the law on retrenchment did not apply. NITES then complained to the skills ministry that the apprenticeship law itself had been broken, and no outcome has been reported.[38] Whether the trainees were wronged in law is unsettled. What is clear is that two levels of the Indian state disagreed about whether this was a labour question at all.

The same years, from the top

From the top, the same years looked like growth. In February 2026 Nasscom (the National Association of Software and Service Companies), the industry's lobby group, reported that India's technology industry would earn $315 billion in the year to March 2026, up 6.1 per cent, with 5.95 million direct employees after 135,000 net additions.[39] Its president, Rajesh Nambiar, acknowledged that jobs were growing more slowly than revenue, 2.3 per cent against 6.1, a gap he called a non-linearity.[40]

That word, in this book's reading, names the bargain on which the whole industry was built, at the moment it began to come apart. For half a century Indian software firms sold, above all, the time of their engineers. A firm that sells time can grow only by selling more of it, so revenue rose with headcount, and the factory had to keep hiring graduates like Gayatri by the tens of thousands. Colleges, training campuses and offer letters were all arranged around that fact. A non-linearity is the opposite: revenue rising faster than the number of people needed to earn it. A gap of 2.3 per cent against 6.1 is small, but it runs against the logic of everything that came before. Artificial intelligence tools press on the bargain at its centre, because what they promise to shrink is the time it takes to write and test code, and that time is exactly what was being billed.

The largest firm shows how unsettled the picture still is. In April 2026 K. Krithivasan, chief executive of TCS, said the company had made 25,000 offers to freshers for the coming year and that clarity on demand would lead to more hiring, and by the end of September, by the company's figures as reported, about 24,000 freshers had joined in the year's first two quarters.[41]

The factory, then, had not stopped. The waits that Gayatri and thousands of others endured began in 2022, in what the report called the 2022 lull in the IT industry.[42] Infosys's annual reports to the American regulator mentioned AI in general terms before then, but did not mention AI code generation until the one for the year to March 2024, filed in June 2024.[43] So the first answer to the obvious question, whether artificial intelligence caused all this, is that the waits began about two years before AI code generation appeared in Infosys's own account of the risks to its business. That does not make the question smaller. On this evidence the bargain was already under strain when the tools arrived, and the tools work on the very hours the bargain sold. How much they have added since is weighed in Chapter 19, and it is the question that makes the history worth telling.

The question

The question this book asks is how India came to build software engineering, as an education and as a profession, and what AI does to what was built. Gayatri's offer letter was the product of a long chain of decisions, made in planning offices and boardrooms, in classrooms, on export contracts that priced Indian programmers by the hour, and in visa offices abroad. The training campus where her career ended is one of the last links. It exists because colleges and companies spent decades arguing over who should teach the engineer what the job required.

The chain starts long before there was an industry, in two research institutes where computers were scientific instruments and nobody yet thought of programming as a job. In 1955, the computer scientist V. Rajaraman records in his history of Indian computing, only a few dozen scientists and engineers in India knew about computers; by 2010 more than 2.4 million people were employed in computer-related jobs.[44] Between those two numbers lies a bargain that held for half a century: India came to train its engineers by the hundred thousand and sold their time to the world. Gayatri's offer letter was a promise that the selling would go on. Whether it does, now that machines can shorten the hours themselves, will decide what the next offer letter is worth, and whether a joining date comes with it.

Notes to Prologue. The joining date (2026)

[1] Ganeshan, '"Thrown on the Road"' (2025) (online article, unpaginated; pseudonyms as given by the reporter). The report is Balakrishna Ganeshan's for *The News Minute*, republished by *Newslaundry*. Gayatri is the reporter's pseudonym, not her real name, and everything known about her comes from this one report.

[2] Ganeshan, '"Thrown on the Road"' (2025) (online article, unpaginated; pseudonyms as given by the reporter).

[3] Ganeshan, '"Thrown on the Road"' (2025) (online article, unpaginated; pseudonyms as given by the reporter).

[4] Ganeshan, '"Thrown on the Road"' (2025) (online article, unpaginated; pseudonyms as given by the reporter).

[5] Ganeshan, '"Thrown on the Road"' (2025) (online article, unpaginated; pseudonyms as given by the reporter).

[6] Ganeshan, '"Thrown on the Road"' (2025) (online article, unpaginated; pseudonyms as given by the reporter).

[7] Ganeshan, '"Thrown on the Road"' (2025) (online article, unpaginated; pseudonyms as given by the reporter). A *Deccan Herald* headline gave the deadline as 6 pm: Press reports on the NITES complaint against Infosys (2024-2025) (Deccan Herald, 14 February 2025, online, unpaginated).

[8] Ganeshan, '"Thrown on the Road"' (2025) (online article, unpaginated; pseudonyms as given by the reporter).

[9] Ganeshan, '"Thrown on the Road"' (2025) (online article, unpaginated; pseudonyms as given by the reporter).

[10] Ganeshan, '"Thrown on the Road"' (2025) (online article, unpaginated; pseudonyms as given by the reporter).

[11] Ganeshan, '"Thrown on the Road"' (2025) (online article, unpaginated; pseudonyms as given by the reporter).

[12] Ganeshan, '"Thrown on the Road"' (2025) (online article, unpaginated; pseudonyms as given by the reporter).

[13] Ganeshan, '"Thrown on the Road"' (2025) (online article, unpaginated; pseudonyms as given by the reporter).

[14] Ganeshan, '"Thrown on the Road"' (2025) (online article, unpaginated; pseudonyms as given by the reporter).

[15] Choudhary, 'IT Professionals' Union Accuses Karnataka Govt of Bias' (2024) (online, unpaginated).

[16] Ganeshan, '"Thrown on the Road"' (2025) (online article, unpaginated; pseudonyms as given by the reporter).

[17] Ganeshan, '"Thrown on the Road"' (2025) (online article, unpaginated; pseudonyms as given by the reporter).

[18] Press reports on delayed onboarding of graduates (2022) (Business Today, 5 October 2022, online, unpaginated).

[19] Press reports on delayed onboarding of graduates (2022) (Business Today, 5 October 2022, online, unpaginated; the graduates in this reporting are anonymous).

[20] Press reports on the NITES complaint against Infosys (2024-2025) (HR Katha, 13 June 2024, online, unpaginated; union's figure); *The News Minute*, 'Infosys to Face Govt Probe over Delayed Recruitment' (2024) and Choudhary, 'IT Professionals' Union Accuses Karnataka Govt of Bias' (2024) (online, unpaginated; NITES as 'Pune-based').

[21] *The News Minute*, 'Infosys to Face Govt Probe over Delayed Recruitment' (2024) (online, unpaginated).

[22] Ganeshan, '"Thrown on the Road"' (2025) (online article, unpaginated; pseudonyms as given by the reporter).

[23] Choudhary, 'IT Professionals' Union Accuses Karnataka Govt of Bias' (2024) (online, unpaginated).

[24] Ganeshan, '"Thrown on the Road"' (2025) (online article, unpaginated; pseudonyms as given by the reporter), which gives the joining date as 7 October; a trainee quoted by *Republic Business* gave 21 October, and there may have been two batches: Rajput, 'Hired, Trained, Terminated' (2025) (online, unpaginated).

[25] Rajput, 'Hired, Trained, Terminated' (2025) (online, unpaginated).

[26] Ganeshan, '"Thrown on the Road"' (2025) (online article, unpaginated; pseudonyms as given by the reporter); Tripathi, 'Why Infosys "Forcibly" Laid Off a Batch of Recruits' (2025) (*Outlook Business*, online, unpaginated; reports the same changes, with 200 hours of material).

[27] Ganeshan, '"Thrown on the Road"' (2025) (online article, unpaginated; pseudonyms as given by the reporter).

[28] Ganeshan, '"Thrown on the Road"' (2025) (online article, unpaginated; pseudonyms as given by the reporter). The report gives the count in its own voice, apparently drawn from the trainees, for the batch that joined on 7 October.

[29] Ganeshan, '"Thrown on the Road"' (2025) (online article, unpaginated; pseudonyms as given by the reporter); Press reports on the NITES complaint against Infosys (2024-2025) (Deccan Herald, 14 February 2025, online, unpaginated).

[30] Rajput, 'Hired, Trained, Terminated' (2025) (online, unpaginated); *Business Today*, 'Infosys Layoffs: 195 More Trainees Fired' (2025) (online, unpaginated; relaying Moneycontrol).

[31] Rajput, 'Hired, Trained, Terminated' (2025) (online, unpaginated); *Business Today*, 'Infosys Layoffs: 195 More Trainees Fired' (2025) (online, unpaginated; relaying Moneycontrol).

[32] Tripathi, 'Why Infosys "Forcibly" Laid Off a Batch of Recruits' (2025) (online, unpaginated; *Outlook Business* cites Moneycontrol for the termination rates).

[33] Ganeshan, '"Thrown on the Road"' (2025) (online article, unpaginated; pseudonyms as given by the reporter).

[34] Tripathi, 'Why Infosys "Forcibly" Laid Off a Batch of Recruits' (2025) (online, unpaginated); Press reports on the NITES complaint against Infosys (2024-2025) (Deccan Herald, 14 February 2025, online, unpaginated); Ganeshan, '"Thrown on the Road"' (2025) (online article, unpaginated; pseudonyms as given by the reporter).

[35] *Business Today*, 'Infosys Layoffs: 195 More Trainees Fired' (2025) (online, unpaginated; *Business Today* relaying Moneycontrol's count). *Republic Business* put the 18 April figure at 370: Rajput, 'Hired, Trained, Terminated' (2025) (online, unpaginated).

[36] Press reports on the NITES complaint against Infosys (2024-2025) (Deccan Herald, 14 February 2025, online, unpaginated).

[37] *Business Today*, 'Infosys Layoffs: 195 More Trainees Fired' (2025) (online, unpaginated; relaying Moneycontrol). No written order of the Karnataka department has been located.

[38] Rajput, 'Hired, Trained, Terminated' (2025) (online, unpaginated).

[39] PTI, 'India FY26 Tech Revenues Seen 6.1 Pc Up' (2026) (PTI wire report of Nasscom's figures, online, unpaginated; post-cutoff material, checked in October 2026.); Rajaraman, *History of Computing in India* (2012), p. 75 (Nasscom's full name, in the chronology).

[40] PTI, 'India FY26 Tech Revenues Seen 6.1 Pc Up' (2026) (PTI wire report of Nasscom's figures, online, unpaginated; post-cutoff material, checked in October 2026.).

[41] *Business Today*, 'TCS Fresher Hiring Hits 25,000 in FY27' (2026) (online, unpaginated; post-cutoff material, checked in October 2026.). Krithivasan told the Press Trust of India: 'We have made 25,000 offers for freshers in FY27. Clarity on demand will lead to more hiring.' *Business Standard* framed the 25,000 as the year's target against a usual figure of about 40,000: Das, 'Top 5 IT Cos' Net Hiring Down for FY26' (2026) (online, unpaginated; post-cutoff material, checked in October 2026.). *Business Today* reported it as the offers made so far, with more to depend on demand, which is closer to his words: *Business Today*, 'TCS Fresher Hiring Hits 25,000 in FY27' (2026). Freshers joined: about 14,000 in the first quarter of the year and 10,000 in the second: Thapliyal, 'TCS Adds 4,258 Employees in Q2' (2026) (online, unpaginated; post-cutoff material, checked in October 2026.); Shinde, 'TCS Onboards 10,000 Freshers in Q2 FY27' (2026) (online, unpaginated; post-cutoff material, checked in October 2026.; the paper's 'June 30, 2026' headcount date is an error, see Thapliyal, 'TCS Adds 4,258 Employees in Q2' (2026)).

[42] Ganeshan, '"Thrown on the Road"' (2025) (online article, unpaginated; 'The 2022 lull'); Press reports on delayed onboarding of graduates (2022) (Business Today, 5 October 2022, online, unpaginated).

[43] Infosys, Form 20-F FY2024 (Infosys Form 20-F for fiscal 2024, Item 3.D risk factors, filed 24 June 2024); Infosys, Form 20-F FY2022 and Infosys, Form 20-F FY2023 (Forms 20-F for fiscal 2022 and 2023: AI and generative AI mentioned, AI code generation not mentioned; full texts searched 2026-10-14).

[44] Rajaraman, *History of Computing in India* (2012), p. 7.

Part I: Before software (1940s-1977)

Chapter 1. The statistician's machines (1931-1966)

In February 1959 a small team of physicists and engineers in Bombay (Mumbai since 1995) finished assembling a computer, and then had to stop work on it. The machine stood in the Tata Institute of Fundamental Research (TIFR), which had been working on it since 1955. It held 2,700 vacuum tubes and drew around 18 kilowatts, and the institute had no suitably air-conditioned room in which to run it.[1] 'Unfortunately, owing to the lack of air-conditioning facilities, the work had to be suspended till almost the end of 1959,' its designer, R. Narasimhan, wrote a few weeks after the machine finally entered service.[2] Testing began only in mid-November, with an auxiliary air-conditioning system brought in for the purpose, and the computer was commissioned for routine work in the third week of February 1960.[3]

The men who built it had learned as they went. When the project began, the team numbered about six, most with a master's degree in physics and a specialism in electronics. Narasimhan described them in 1999: 'Except for one member (namely, myself), none had been outside India, and none of the participants in the project, including myself, had any experience in either using a computer or operating one, let alone building one.'[4] Their machine would later be named TIFRAC, and Narasimhan would call it 'the first fullscale electronic digital computer designed, built, and operated in India'.[5]

A year's delay for want of an air conditioner is a small story, but it holds much of what follows. India's first computers were acquired, and in one case built, by a handful of scientists working for the state. They were limited less by ideas than by things that had to be paid for in foreign currency or manufactured at scale: machines, spare parts, cooling, memory. And the most lasting thing they produced was not hardware. It was a small group of people who now knew how computers worked.

The state had already named the value of such people, though not with computers in mind. In March 1958, while the TIFR team was still wiring its machine, the Government of India adopted a Scientific Policy Resolution. It declared that 'Science and technology can make up for deficiencies in raw materials by providing substitutes, or, indeed, by providing skills which can be exported in return for raw materials.'[6] The resolution was about science in general. Beyond it, notes V. Rajaraman, a computer scientist who later wrote a history of Indian computing, 'there was no government policy on computers'.[7] Read with hindsight, though, the sentence describes, decades early, the bargain on which India's software industry would be built: not machines, but skills, sold abroad in return for what the country lacked.

Two institutions, in two cities, took two different routes to the first computers. In Calcutta (Kolkata since 2001), the Indian Statistical Institute (ISI) bought its machines, first from Britain and then from the Soviet Union. In Bombay, TIFR decided to build one. By the mid-1960s both had arrived at the same place. ISI's attempt to build a machine of its own failed, and TIFR, having built TIFRAC, chose to buy an American computer to replace it.

Computers before machines

The Indian Statistical Institute was founded in Calcutta in 1931 under Prasanta Chandra Mahalanobis.[8] In its early decades the word 'computer' meant a person: Mahalanobis designated some of the institute's statistical staff as computers, and they calculated and tabulated its large sample surveys, beginning with those of the jute crop.[9] By late 1951 the institute had two new models of IBM tabulators, the punched-card machines then used for counting and sorting large volumes of records.[10]

What turned this into a demand for electronic computers was national planning. Mahalanobis helped set up the National Sample Survey in 1950, a continuing household survey that produced data on a scale no team of clerks

could tabulate quickly.[11] The historian Nikhil Menon argues that the decision to plan India's economy from the centre, and the 'explosion of official statistics' that came with it, provided the justification for the pursuit of computers; ISI's influence culminated in its effectively authoring the Second Five Year Plan of 1956-61, a plan with an emphasis on heavy industry in the public sector.[12] India sought computers for planning and development, not for military use.[13] A plan of that size, built from survey data, needed arithmetic that a room of clerks with tabulators could not supply quickly enough. That, more than any interest in the machines themselves, is why a statistician became India's first buyer of computers.

Mahalanobis first saw an electronic computer at Harvard in 1947.[14] By 1953 ISI's computing laboratory, opened three years earlier under the engineer S.K. Mitra, had built an analogue computer from Second World War surplus parts that could solve simultaneous linear equations in up to ten variables.[15] An analogue machine represents numbers as physical quantities such as voltages, rather than as digits, and it could not do the general-purpose arithmetic that planning required. Mahalanobis suspected that it would not be adequate.[16] He wanted a digital computer, and India could not make one.

A Cold War shopping trip

For four years Mahalanobis tried to buy an American machine. Between 1950 and 1954 he held informal talks in Washington about acquiring a UNIVAC, a machine famous in American popular culture, on which he had been fixed since the 1940s; the best American machines would cost a million dollars.[17] The talks failed. One reason, Menon found, was a secret dispatch sent from the US Embassy in Delhi to the State Department in 1953, which described Mahalanobis as a Soviet sympathiser.[18]

He turned east. In February 1954, at a lunch hosted by Prime Minister Jawaharlal Nehru, Mahalanobis pressed the Soviet ambassador, Mikhail Menshikov, for help with a computer, arguing in effect that the Soviet Union could hardly object to sharing what the Americans already knew.[19] That spring and summer he travelled through Geneva, London and Prague to Moscow, writing about computers to his colleague Pitambar Pant as he went.[20] In Moscow he, Mitra and a delegation of Indian scientists visited the Institute of Precision Mechanics and Computer Engineering, which was building a computer with the backing of Nikita Khrushchev, and Mahalanobis persuaded the Soviet authorities to supply India with a machine.[21] Some eight months after the Menshikov meeting, according to Kris Gopalakrishnan and his co-authors, the Soviet Union and the United Nations agreed to fund India's purchase of a digital computer, which they identify as the British HEC-2M.[22]

A computer without support

The first electronic digital computer to work in India was British, and it came on hard terms. Mahalanobis ordered it from the British Tabulating Machine Company (BTM), which sold keypunches, sorters and mechanical calculators in India. 'BTM agreed to sell a computer to the ISI in 1954 though without any technical support,' Rajaraman writes. 'The ISI was to install and maintain the machine using its own scientists and engineers.'[23] BTM would not install and maintain a single machine in India, he later explained, since only one was coming to the country.[24]

So ISI sent two engineers, Mohi (Monimohan) Mukherjee and Amresh Roy, to England in December 1954 to watch the machine being made. The Hollerith Electronic Computer Model 2M had been designed by A.D. Booth at Birkbeck College in London and was assembled in BTM's workshop, and the two men gathered there the knowledge they would need to install and keep it running.[25] It cost 18,500 pounds sterling, paid by the Government of India, which lent it to ISI.[26] It reached Calcutta in 1955 or early 1956.[27] Its memory, a rotating magnetic drum, held 1,024 words, and it had to be programmed in binary machine code through punched cards.[28] 'Around a dozen people learnt to program the computer which was used for some statistical calculations,' Rajaraman writes.[29] It also handled requests from the Planning Commission, the National Sample Survey and other scientific institutions, and, in his words, it 'was found to be a very difficult machine to use'.[30]

The dozen people who learned to program it make up most of India's computing profession in the mid-1950s. 'In 1955 only a few dozen scientists and engineers in India knew about computers,' Rajaraman writes.[31] There were no courses to take. What they knew came from watching machines being assembled abroad and from operating manuals.

The URAL and the Russian manuals

Mahalanobis started looking for a second machine, and the obstacle was money from abroad. It was difficult to get funds, Rajaraman notes, 'particularly foreign exchange funds'.[32] A computer had to be paid for in dollars or pounds, not rupees. The solution again came from the Soviet side. A URAL-1, a first-generation vacuum-tube machine that could perform about 12,000 operations a second, was ordered in 1958 with United Nations assistance.[33] Eight Russian engineers installed it.[34] 'All the manuals were in Russian and the ISI engineers learnt Russian to enable them to maintain the machine,' Rajaraman writes.[35] When it was unveiled in Calcutta in December 1958, the *Times of India* reported that it could calculate 600 times faster than a single man.[36]

With two working computers and a group of people who could program and maintain them, ISI was, in Rajaraman's phrase, 'the de-facto National Computer Centre' of India in 1958.[37] Its users included universities, the atomic energy establishment and defence laboratories, among them Andhra University, the Indian Institute of Science in Bangalore (Bengaluru since 2014), the Indian Institute of Technology (IIT) at Kharagpur and TIFR.[38] India had taken a British machine with no support and a Soviet one with eight engineers and Russian manuals, because its attempt to obtain the American machine it wanted had been hampered by Cold War suspicion. Non-alignment allowed it to buy from both sides, but on terms the sellers set.

Expertise was so thin that Mahalanobis went looking for it in odd places. In February 1959 Morton Nadler, a 37-year-old American engineer, flew from Prague to Calcutta to take a two-year contract at ISI.[39] Nadler, who had joined the Communist Party in 1936 and renounced his US citizenship after the American embassy seized his passport in 1950, had worked at Czechoslovakia's Institute of Mathematical Machines and, by Menon's account, was under suspicion of espionage by both the American and the Czechoslovak authorities. Disillusioned with communism, he could neither leave Czechoslovakia easily nor go home, and Mahalanobis's offer gave him a way out.[40] His lectures at ISI in the spring of 1959 became a book, *Topics in Engineering Logic*, and in 1960 he left for the French computer maker Bull.[41]

Bombay decides to build

The physicist Homi Bhabha founded TIFR in Bombay in 1945.[42] Computing work began there in 1954, and towards the end of that year TIFR decided, as Narasimhan put it, 'to design and build a fullscale, general purpose, electronic digital computer using contemporary technology'.[43] Narasimhan led the project, with Bhabha's 'acceptance and active support'.[44] He had an engineering degree from the College of Engineering, Guindy, in Madras (Chennai since 1996), and had spent 1948 to 1954 in the United States on a part-loan scholarship from the J.N. Tata Endowment.[45] With no one who had built a computer, the team worked in two stages: first a small pilot machine, to try out circuit and logic designs and to train the team on the job, and then a full-scale computer.[46]

The pilot was completed in 1956 and kept running for about a year.[47] Design of the full-scale machine began early in 1957, and its final assembly was completed in February 1959.[48] It was a stored-program machine with a memory of 1,024 forty-bit words held in ferrite cores, tiny magnetic rings that each stored one bit, and it could add two numbers in 45 microseconds.[49] The team made that memory with their own hands. P.V.S. Rao, a member of TIFR's computer group, described how they strung the matrix together themselves, threading hair-thin enamelled wire through magnetic cores less than a millimetre across, held in Plexiglass frames.[50]

They did not work entirely alone. 'The TIFRAC control logic design benefited crucially through design details kindly made available by the ILLIAC team,' Narasimhan wrote in 1999, referring to the computer group at the University of Illinois.[51] Mahalanobis could not buy an American computer in Washington; Narasimhan was given American design details, scientist to scientist.

A name, and a verdict

The machine entered routine service in February 1960, and when Nehru inaugurated TIFR's new building in 1962, he named it TIFRAC.[52] It ran until early 1964, by the end working two shifts a day.[53] Its users were TIFR's own scientists, the Atomic Energy Establishment and some universities, and the analysis of TIFR's cosmic-ray air-shower experiments, Narasimhan wrote, would not have been feasible without it.[54] Its storage was too small to attempt a compiler, the program that translates instructions written in a human-readable language into machine code.[55]

Narasimhan's judgement of his own machine was unsentimental. The pilot, he wrote in 1999, had been close to the state of the art in 1954, and the 1957 design not very far behind it. But by the time it was commissioned in 1960, TIFRAC was 'an obsolete first generation machine'.[56] The second generation of computers, built with transistors, needed production know-how that only large manufacturers had.[57] A research institute could build one machine. It could not build the next generation, and it could not build many.

What TIFRAC left behind was people. The computer section had grown from about half a dozen members to some two dozen, with a 'hard core of specialists' who could design digital equipment and a group that had learned to maintain complex electronic systems.[58] And, in Narasimhan's words, 'many of the computer users in India, handling sophisticated and advanced computational techniques, had their first introduction to programming through the use of TIFRAC.'[59]

The first to leave

Some of those people left. Between 1959 and 1962 about half a dozen of the scientists who had designed and built TIFRAC took visiting appointments at American universities. Three of them settled permanently in the United States; others came back.[60] Narasimhan himself spent the early 1960s as a visiting scientist at the Digital Computer Laboratory of the University of Illinois.[61] Three people are not a brain drain. But the pattern would recur at much larger scale. India trained engineers at public expense, often on projects that could not employ them for long, and the United States offered the next stage of their careers.

From building to buying

The engineers who went to America also helped TIFR decide what to do next. Because some of its staff now knew American machines at first hand, TIFR could set up an expert committee, chaired by Narasimhan, to compare the large commercial systems on offer. It recommended the CDC 3600, of which only one was then in operation, at its maker's plant in Minneapolis.[62] Rajaraman calls it 'a bold decision'.[63] The machine was bought for $1.5 million, with funds from the United States Agency for International Development that Bhabha persuaded the Government of India to arrange.[64] Installed in mid-1964, it ran from October that year as a National Computation Centre, maintained and operated by TIFR staff once the manufacturer's early support ended, and over 150 institutions from across India used it for more than a decade.[65] In the first year users had the machine free of charge. The only demand on them was that they write and debug their own programs, and to help them TIFR's staff programmers ran 'many short courses on programming'.[66]

So the institute that had built India's first computer bought an American one with American aid money, on the recommendation of the man who had led the building. Narasimhan saw no contradiction. Once second-generation machines required production know-how that only large manufacturers had, he judged, TIFR was 'wise' to look to the world market.[67]

The end of building, for now

ISI tried once more. In the early 1960s it set out to build a transistorised computer jointly with Jadavpur University in Calcutta. The machine, called ISIJU, was completed in 1966, but in Rajaraman's words 'It was not a successful project as the computer was unreliable and did not have appropriate software.'[68] Menon counts 15 digital computers in India by the end of 1964.[69] The state still had no policy on computers; a national committee on electronics was appointed only in 1963, under Bhabha, after the 1962 war with China.[70] In the autumn of 1965 the Planning Commission opened its own computer centre in Delhi, an IBM 1620 bought with a Ford Foundation grant of Rs 2.3 million (about $462,000), run 'open shop' for outside users.[71] The planners who had once queued for time on ISI's machines now had one of their own, made by IBM.

Two explanations

Why did these beginnings not lead to an Indian computer industry? Narasimhan's answer was the economics of manufacturing: a research group could build a first-generation machine, but only large manufacturers could build the second generation, and build it in numbers.[72] A different answer comes from Dwaipayan Banerjee of the

Massachusetts Institute of Technology, in *Computing in the Age of Decolonization* (2026). As its publisher and reviewers describe it, the book argues that India's attempt to build a self-reliant computing industry was blocked, chiefly from outside, by Cold War politics, trade imbalances and the monopolies of American corporations, and also by the preference of India's technocratic elite for profitable technical services; Angela Saini quotes its judgement that by the mid-1960s IBM's dominance of Indian computing was 'nearly absolute'.[73] Tracie Farrell reports that it contrasts IBM with Control Data, the maker of TIFR's CDC 3600, which in the book's words, as she quotes them, was 'treating Indian researchers as peers'.[74] The record of the 1950s fits parts of both answers.

Coda: from building machines to teaching people

By 1966 the experiment of the 1950s had reached its end. ISI's own machine had failed, TIFRAC had been retired, and the best computer in the country was American. What remained was the people who had learned their trade on TIFRAC and the URAL, and the users who had learned to write their own programs on the CDC 3600. The resolution of 1958 had spoken of skills that could be exported in return for what India lacked. In 1966 nothing was being exported, but the one thing India's first computers had reliably produced was exactly those skills. The country had learned that it could make computer people more easily than it could make computers.

The next step was to train more of them, in classrooms rather than workshops. Chapter 2 tells how a new campus at Kanpur, in Uttar Pradesh, became the place where India first taught computing as a subject, and where the engineer factory began.

Notes to Chapter 1. The statistician's machines (1931-1966)

[1] Narasimhan, 'On the System and Engineering Design' (1960), p. 48 (tubes and power; the machine also held 1,700 germanium diodes and about 12,500 resistors); Narasimhan, 'On the System and Engineering Design' (1960), p. 47 (work since 1955; no suitably air-conditioned room).

[2] Narasimhan, 'On the System and Engineering Design' (1960), p. 47.

[3] Narasimhan, 'On the System and Engineering Design' (1960), p. 47; Narasimhan, 'Men, Machines and Ideas' (1999), p. 448 agrees on February 1960; TIFR Archives, 'India's First Digital Computer' likewise.

[4] Narasimhan, 'Men, Machines and Ideas' (1999), p. 448.

[5] Narasimhan, 'Men, Machines and Ideas' (1999), p. 447.

[6] *Scientific Policy Resolution* (1958), Resolution No. 131/CF/57, 4 March 1958, para. 5 and list of aims (text at nstmis-dst.org/SPResolution.aspx, consulted 2026-10-13); also quoted in Rajaraman, *History of Computing in India* (2012), p. 15.

[7] Rajaraman, *History of Computing in India* (2012), p. 18.

[8] Rajaraman, *History of Computing in India* (2012), p. 15; Gopalakrishnan, Dayasindhu and Narayanan, 'How Eastern India Saw the First Computers' (2022).

[9] Gopalakrishnan, Dayasindhu and Narayanan, 'How Eastern India Saw the First Computers' (2022).

[10] Gopalakrishnan, Dayasindhu and Narayanan, 'How Eastern India Saw the First Computers' (2022), quoting the first general report of the National Sample Survey (1953).

[11] Gopalakrishnan, Dayasindhu and Narayanan, 'How Eastern India Saw the First Computers' (2022).

[12] Menon, '"Fancy Calculating Machine"' (2018), abstract; Rajaraman, *History of Computing in India* (2012), p. 13.

[13] Menon, 'Chasing the Machine' (2018); Menon, '"Fancy Calculating Machine"' (2018) (SIGCIS Mahoney Prize citation, 2019).

[14] Menon, 'Chasing the Machine' (2018).

[15] Menon, 'Chasing the Machine' (2018); Gopalakrishnan, Dayasindhu and Narayanan, 'How Eastern India Saw the First Computers' (2022) calls it 'India's first electrical analogue computing machine'.

[16] Menon, 'Chasing the Machine' (2018).

[17] Menon, 'Chasing the Machine' (2018).

[18] Menon, 'Chasing the Machine' (2018), which quotes the 1953 dispatch.

[19] Menon, 'Chasing the Machine' (2018), which quotes Mahalanobis's words; Menon's original source not identified, so paraphrased here.

[20] Menon, '"Fancy Calculating Machine"' (2018), nn. 46-69 (letters to Pant of 12 April, 23 and 27 June, and 7 and 17 July 1954, sent from Geneva, London, Prague and Moscow; Pitambar Pant Papers, Nehru Memorial Museum and Library).

[21] Menon, 'Chasing the Machine' (2018).

[22] Gopalakrishnan, Dayasindhu and Narayanan, 'How Eastern India Saw the First Computers' (2022).

[23] Rajaraman, *History of Computing in India* (2012), p. 15, drawing on the memoir of Mohi Mukherjee, one of the engineers involved (Mukherjee, 'The First Computer in India').

[24] Gopalakrishnan, Dayasindhu and Narayanan, 'How Eastern India Saw the First Computers' (2022), quoting Rajaraman in Gopalakrishnan, Dayasindhu and Narayanan, *Against All Odds* (2022).

[25] Rajaraman, *History of Computing in India* (2012), p. 15.

[26] Menon, 'Chasing the Machine' (2018).

[27] Rajaraman, *History of Computing in India* (2012), pp. 15-16, following Mukherjee's memoir, gives July 1955 and has the machine working in August 1955. Menon, working from archival sources, writes that it arrived in two crates in early 1956 (Menon, 'Chasing the Machine' (2018)).

[28] Rajaraman, *History of Computing in India* (2012), p. 16.

[29] Rajaraman, *History of Computing in India* (2012), p. 16.

[30] Menon, 'Chasing the Machine' (2018); Rajaraman, *History of Computing in India* (2012), p. 16.

[31] Rajaraman, *History of Computing in India* (2012), p. 7.

[32] Rajaraman, *History of Computing in India* (2012), p. 16.

[33] Rajaraman, *History of Computing in India* (2012), p. 16; Gopalakrishnan, Dayasindhu and Narayanan, 'How Eastern India Saw the First Computers' (2022) names the United Nations Technical Assistance Administration.

[34] Rajaraman, *History of Computing in India* (2012), p. 16 (Russian engineers); Menon, 'Chasing the Machine' (2018) and Gopalakrishnan, Dayasindhu and Narayanan, 'How Eastern India Saw the First Computers' (2022) for the number eight.

[35] Rajaraman, *History of Computing in India* (2012), p. 16.

[36] Menon, 'Chasing the Machine' (2018), quoting the *Times of India*; date and page of the original not identified.

[37] Rajaraman, *History of Computing in India* (2012), p. 16.

[38] Rajaraman, *History of Computing in India* (2012), p. 16; Menon, '"Fancy Calculating Machine"' (2018), n. 92.

[39] Menon, 'Chasing the Machine' (2018).

[40] Menon, 'Chasing the Machine' (2018), which also records his loss of a radio-engineering job in Chicago over his politics and his move to Prague in 1948. Menon describes his disillusionment with communism; Nadler's autobiography, as Menon cites it, names the Soviet suppression of the Hungarian uprising in 1956 as the final straw.

[41] Menon, '"Fancy Calculating Machine"' (2018), n. 27.

[42] Rajaraman, *History of Computing in India* (2012), p. 15.

[43] Narasimhan, 'Men, Machines and Ideas' (1999), p. 448.

[44] Narasimhan, 'Men, Machines and Ideas' (1999), p. 447.

[45] Narasimhan, 'Men, Machines and Ideas' (1999), p. 448.

[46] Narasimhan, 'Men, Machines and Ideas' (1999), p. 448.

[47] Narasimhan, 'On the System and Engineering Design' (1960), p. 47 (completed September 1956). Narasimhan's 1999 memoir gives October or November 1956 (Narasimhan, 'Men, Machines and Ideas' (1999), pp. 448-449); the date here follows the paper he wrote in 1960.

[48] Narasimhan, 'On the System and Engineering Design' (1960), p. 47.

[49] Narasimhan, 'On the System and Engineering Design' (1960), p. 47 (stored program; parallel, binary and asynchronous, with single-address instructions; ferrite-core memory of 1,024 words); Narasimhan, 'On the System and Engineering Design' (1960), pp. 47-48 (40-bit words; punched paper tape for input, printed or punched output); Narasimhan, 'On the System and Engineering Design' (1960), p. 48 (apart from two bias supplies, no circuit used a transistor); Narasimhan, 'Men, Machines and Ideas' (1999), p. 449, table (addition in 45 microseconds, multiplication in about 500); TIFR Archives, 'India's First Digital Computer' (memory later doubled to 2,048 words).

[50] Gopalakrishnan, Dayasindhu and Narayanan, 'When India Made Its First Computer' (2022) (*Against All Odds*), quoting P.V.S. Rao; on Rao's membership of the group, Rajaraman, *History of Computing in India* (2012), p. 17 and ref. 25. Rao's own account was probably first published as Rao, 'TIFRAC, India's First Computer' (2008).

[51] Narasimhan, 'Men, Machines and Ideas' (1999), p. 449.

[52] Narasimhan, 'Men, Machines and Ideas' (1999), p. 448; TIFR Archives, 'India's First Digital Computer'; Rajaraman, *History of Computing in India* (2012), p. 16. TIFR's archive and Rajaraman expand the name as 'TIFR Automatic Calculator'; Narasimhan, writing in 1999, gave it as 'TIFR Automatic Computer' (TIFR Archives, 'India's First Digital Computer'; Rajaraman, *History of Computing in India* (2012), p. 16; Narasimhan, 'Men, Machines and Ideas' (1999), p. 447).

[53] Narasimhan, 'Men, Machines and Ideas' (1999), p. 449; TIFR Archives, 'India's First Digital Computer'.

[54] Narasimhan, 'Men, Machines and Ideas' (1999), p. 449.

[55] Narasimhan, 'Men, Machines and Ideas' (1999), p. 450.

[56] Narasimhan, 'Men, Machines and Ideas' (1999), p. 449.

[57] Narasimhan, 'Men, Machines and Ideas' (1999), p. 449.

[58] Narasimhan, 'Men, Machines and Ideas' (1999), pp. 449-450.

[59] Narasimhan, 'Men, Machines and Ideas' (1999), p. 449.

[60] Narasimhan, 'Men, Machines and Ideas' (1999), p. 450.

[61] Narasimhan, 'Men, Machines and Ideas' (1999), p. 450 (1961-1964); Rajaraman, *History of Computing in India* (2012), p. 17 gives 1961 to 1963.

[62] Narasimhan, 'Men, Machines and Ideas' (1999), p. 450.

[63] Rajaraman, *History of Computing in India* (2012), p. 17.

[64] Rajaraman, *History of Computing in India* (2012), p. 17; Menon, '"Fancy Calculating Machine"' (2018), n. 124.

[65] Rajaraman, *History of Computing in India* (2012), p. 17 ('mid 1964'); Narasimhan, 'Men, Machines and Ideas' (1999), p. 450 (mid-June 1964; National Computation Centre from October 1964; maintained by TIFR staff); Menon, '"Fancy Calculating Machine"' (2018), n. 124 gives May 1964. Control Data assisted with installation and early maintenance; later maintenance was done in-house by TIFR (Rajaraman, *History of Computing in India* (2012), p. 17).

[66] Gopalakrishnan, Dayasindhu and Narayanan, 'When India Made Its First Computer' (2022).

[67] Narasimhan, 'Men, Machines and Ideas' (1999), pp. 449-450.

[68] Rajaraman, *History of Computing in India* (2012), pp. 16-17. Start: 1960 (Rajaraman, *History of Computing in India* (2012)), 1961 (Gopalakrishnan, Dayasindhu and Narayanan, 'How Eastern India Saw the First Computers' (2022)), 1962 (Menon, '"Fancy Calculating Machine"' (2018), n. 122). Completion year: Rajaraman, *History of Computing in India* (2012) and Menon, '"Fancy Calculating Machine"' (2018), n. 122 (1966); Gopalakrishnan, Dayasindhu and Narayanan, 'How Eastern India Saw the First Computers' (2022) gives 1965-66.

[69] Menon, 'Chasing the Machine' (2018).

[70] Rajaraman, *History of Computing in India* (2012), p. 3 (Bhabha Committee, 1963); p. 21 (after the 1962 border war with China). Chapter 3 tells the committee's story.

[71] Menon, 'Chasing the Machine' (2018) for Rs 23 lakh; Menon, '"Fancy Calculating Machine"' (2018), n. 129 for $462,000, the opening (*Times of India*, 24 September 1965) and operating terms (open to outside users on a no-profit, no-loss basis, one shift a day until 1968).

[72] Narasimhan, 'Men, Machines and Ideas' (1999), p. 449.

[73] Banerjee, *Computing in the Age of Decolonization* (2026) (publisher's description); Dizikes, 'The Tech Revolution That Wasn't' (2026) (MIT News, May 2026; also for Banerjee's post, 'associate professor of science, technology, and society', and for his description of the argument as 'very much an external constraint story'); Saini, 'Why Didn't India Build Its Own Silicon Valley?' (2026) (Angela Saini, *Los Angeles Review of Books*, 5 August 2026; the 'nearly absolute' phrase is quoted there from the book, no page given); Farrell, review of Banerjee, *Computing in the Age of Decolonization* (2026) (Tracie Farrell, *International Journal of Asian Studies*); Menon, 'India's Pursuit of Tech Self-Reliance' (2026) (Nikhil Menon, *Issues in Science and Technology* 42, no. 4, Summer 2026, pp. 98-100). The book's argument is given here as these accounts summarise it. Among the constraints they name are limits on the flow of American designs, a foreign-exchange crisis in 1958 that made larger memories unaffordable, and rescue loans led by the World Bank that were conditional on opening markets to Western capital. Post-cutoff material: Banerjee, *Computing in the Age of Decolonization* (2026) and Dizikes, 'The Tech Revolution That Wasn't' (2026) live-checked 2026-10-13, the reviews 2026-10-15.

[74] Farrell, review of Banerjee, *Computing in the Age of Decolonization* (2026), pp. 2-3 (the 'peers' phrase is cited there to p. 153 of the book). Reviewer's report only; post-cutoff material, checked in October 2026.

Chapter 2. Kanpur, 1963 (1946-1978)

In July 1963 a chartered DC-7 landed at a military airbase in Kanpur, a large city on the Ganges in Uttar Pradesh. In its hold was an IBM 1620, a small transistorised computer of a model common in American universities.[1] It was bound for the Indian Institute of Technology (IIT) at Kanpur, which was then three years old. At the institute, in the historian Ross Bassett's words, 'a group of young Indian men muscled the machine into place'.[2] Waiting for it was Harry Huskey of the University of California at Berkeley, who had consulted on the pioneering American computer ENIAC and had worked with Alan Turing in England.[3]

The machine had been planned before there was a building to put it in. Its delivery date had been set by the earliest moment that a room could hold it.[4] The computer room was still unfinished. 'This is monsoon season, by the way, in the middle of July,' Huskey recalled for the Computer History Museum in 2006, and the crated machine was set down in about an inch of water.[5] Because the 1620 had been paid for with American aid under a 'buy America clause', it was built for 60-cycle power, while India's supply ran at 50 cycles. It took telegrams to IBM and a converter found in Germany before, 'in about three weeks', the machine was running.[6] With two colleagues from Princeton, Forman Acton and Irving Rabinowitz, Huskey showed it to the institute's Board of Governors in late August.[7] V. Rajaraman, who joined the Kanpur faculty that year and later wrote a history of Indian computing, calls it the first computer with a FORTRAN compiler in any Indian educational institution. A compiler is software that translates a program written in a high-level language such as FORTRAN into the machine's own instructions.[8]

Once it ran, the three Americans began an informal course for twenty-five first-year students, chosen for the purpose. After their ordinary classes, the students 'would stay long into the night programming the computer'.[9]

Kanpur was not the first place in India with a computer, although Huskey came to remember it that way. 'So I think our computer center was the first such in India,' he said.[10] The Indian Statistical Institute and the Tata Institute of Fundamental Research had working machines years earlier (Chapter 1). Kanpur was the first place where a computer was put at the centre of an engineering education, where every student was required to program, and where computer science was first taught as a subject of its own. It was also, from its first graduating year, a door to the United States. This is where the factory's raw material begins: an engineer trained in English, to an American standard, on American machines, and therefore exportable. The education was built to serve India, and a large share of its best products went to America.

The standard of MIT

The engineering colleges the British founded in India had a limited purpose: to prepare Indians for subordinate posts under British rule.[11] The idea of something grander came from a British scientist. In 1944 A.V. Hill, a Nobel laureate who was advising the government of British India, noted that India had no institution like the Massachusetts Institute of Technology (MIT), and recommended one or two technical institutes 'of the highest possible standing'.[12] A committee chaired by N.R. Sarker proposed four in 1946, spread across the regions of India, to train students to the standard of MIT.[13] The first Indian Institute of Technology opened at Kharagpur, west of Calcutta (Kolkata since 2001), in 1951, with help from the United States, Britain, the Soviet Union and UNESCO.[14]

For the next four, the government gave each institute a single foreign patron. The Soviet Union supported Bombay (Mumbai since 1995), which opened in 1958; West Germany supported Madras (Chennai since 1996), opened in 1959; the United States took Kanpur, opened in 1960; and Britain supported Delhi, which began in 1963.[15] For a while the IITs were different experiments under one name. Madras built a curriculum heavy with practical training in manual skills such as blacksmithing and woodworking, and its first computer was a surplus analogue

machine that had belonged to an American university.[16] At Bombay the first director said in 1961 that he had turned down a Soviet Ural-I computer because it cost too much.[17] The common belief that the IITs were modelled on MIT is therefore half right. MIT was the standard the planners named, but four of the five early institutes had non-American patrons. Kanpur came closest to the American model, and it was the American model that won.[18]

The turn to America predated Kanpur. Before the Second World War between one and two thousand Indian students a year studied in Britain, and at most 208 in the United States. In 1944 the government of British India began paying for Indian students at American universities, and by 1949-50 there were 1,680 Indian students in America and fewer than 700 in Britain.[19] Bassett's larger argument is that the alignment was chosen, not imposed. An Indian technical elite educated in America built ties with American firms, universities and, at times, the American government, and kept them through the worst years of relations between the two states.[20] Kanpur was that elite's most important project.

A director reads a book

By 1958 India and the United States had informally agreed that America would help build the IIT at Kanpur. Delays meant that the institute opened in 1960 without American help.[21] Its first director, P.K. Kelkar, was an electrical engineer with a 1936 doctorate from the University of Liverpool. He had taught at the Victoria Jubilee Technical Institute in Bombay and had been chief planning officer for IIT Bombay, working closely with its Soviet sponsors.[22]

In a memoir written after he retired, and never published, Kelkar described the moment his thinking turned. He read the proceedings of a 1957 conference of American engineering deans, which described how their schools were reorganising teaching around 'engineering science': a common scientific foundation beneath the separate branches of engineering.[23] The volume's other theme was the computer. One contributor called it 'the embodiment of a new attitude toward the utilization of ideas', which would become, like the library, the intellectual hub of a university.[24] Another, Gordon Brown, the dean of engineering at MIT, defined the modern engineer as someone who 'builds mechanisms to program, to control or to compute with lightning-like speed'.[25]

The engineering-science model had won in America partly because engineering schools wanted military research funding. Its rhetoric, Bassett argues, was broad enough to appeal to Kelkar, an idealist who had come to engineering through physics.[26] The computer was central to his plan from the start. Even before any American programme was in place, he planned to buy an IBM machine with American funds and to have physicists at Delhi University train his faculty to use it.[27]

Kelkar also brought with him a connection that would matter later. In the late 1950s, at the Victoria Jubilee Technical Institute, he and F.C. Kohli, an MIT-trained control engineer working for Tata Electric, had started a master's course in control engineering. Brown, the MIT dean, had been one of Kohli's professors.[28]

Arrival at Kanpur Central

The first hundred undergraduates arrived in July 1960.[29] The institute had no campus of its own. Abhay Bhushan, who holds roll number 60001 and so counts as Kanpur's first student, later wrote a three-part account of those years. He remembers arriving at Kanpur Central station with a few coins in his pocket and being collected by truck with other boys from all over the country.[30] They lived in a hostel borrowed from the Government Central Textile Institute in Gwaltoli, and attended classes and laboratories at the Harcourt Butler Technological Institute.[31] Fewer than seventy lived in the hostel, two to a room; the food, from a contractor, was by his account generally poor.[32] The first classes on the new campus were held in its workshop building in 1963.[33]

Not one of the hundred who started was a woman. When students put on plays, boys took the women's parts.[34] The absence was so ordinary that it shaped the language of the people describing the place. Peter Fay, a Caltech historian who taught at Kanpur from 1964 to 1966, described the last two undergraduate years as professional training in the branch of engineering 'which the boy or the boy's father has chosen'.[35]

Entry was by a common examination taken across India.[36] Teaching was in English, and Fay reported that only a fraction of the students were thoroughly comfortable in it. The English paper in the entrance examination had a pass mark set deliberately low, so as not to exclude promising engineers, and the first-year English course had largely become a language course.[37] Bhushan's first-year teachers, he wrote, had been borrowed from the colleges of Uttar Pradesh and taught by memory and rote, even in mathematics; in the second year new teachers arrived, many with doctorates from the United States, Canada and Britain.[38]

Influence, not power

American help arrived after the students. In January 1961 MIT sent three professors to India, and their report proposed a consortium of American universities with MIT informally at its head.[39] Later that year the two governments and nine universities, among them MIT, Caltech, Princeton, Berkeley and Purdue, agreed to set up the Kanpur Indo-American Program, known as KIAP. Stanford was asked and declined.[40] Early in 1962 the US Agency for International Development (USAID) contracted with the nine through an agent, and American faculty began to arrive.[41] The programme was meant to last ten years.[42] It ended in 1972.[43]

Fay's report from Kanpur is the fullest contemporary description of what this meant on the ground. The Americans were contract employees of the aid agency. They lived in houses it rented for them, drew flour, sugar and Scotch from the commissary in New Delhi, and were paid in dollars through their home universities.[44] At their peak there were about thirty of them; by 1966 there were twenty, and falling.[45] They sat on the committees that appointed faculty, but without a vote, and were often not even consulted on promotions. Where an Indian could head a department, an American did not.[46] They also held periodic morning meetings of their own, to which their Indian colleagues were not invited. Fay's examples of what came up there include equipment 'lying in the rain at Delhi airport waiting for customs', and how to get out of Kanpur in a hurry during 'the fighting last September, when the city was blacked out and Americans in the Punjab were all ordered to Delhi', the war with Pakistan of September 1965. In his own case, he added, the problem was leaving without abandoning 'a four-foot Benares brass tray'.[47]

A first-year student from Delhi told Fay that he had chosen Kanpur over the IIT near his home because 'you Americans run this one'. Fay replied that he sometimes wished they did. 'For the fact is we have influence; we do not have power.'[48] He was frank about the hurry the arrangement imposed: 'For we have always been in a hurry. The program is to run for only ten years, and individuals never stay for more than two.'[49]

Influence still bought a great deal. Drawing on a 1972 dissertation by Kim Patrick Sebaly, Bassett estimates that the American contribution to Kanpur, in equipment and visiting faculty, was twice what the Soviet Union gave Bombay or West Germany gave Madras, and three times what Britain gave Delhi.[50] By November 1965 more than $4 million of equipment and books had been ordered from the United States. Fay judged that Kanpur was 'already the best equipped engineering and science school in India', and that this had helped it attract faculty.[51] The student thought the Americans ran Kanpur; Fay thought they had influence and not power; Bassett reads the alignment as chosen by Indian elites for their own ends. All three can be true at once: the money and the model were American, the appointments and the institution were Indian.[52]

Learning on the fly

Kelkar chose most of his permanent faculty from among Indians with American doctorates, many of them recruited while they were working or teaching in the United States.[53] By 1966 Kanpur had about seventy-five faculty at assistant professor level or above. All but four were under forty. All held doctorates or equivalent degrees: over half from the United States and Canada, a dozen from Britain or elsewhere in Europe.[54]

The people who would run the computer centre came by the same route. H.K. Kesavan, who had a doctorate from Michigan State and had been a professor at Waterloo in Canada, arrived in January 1964 to head electrical engineering, and in May became the first head of the computer centre. Rajaraman had arrived in 1963, with a master's from MIT and a doctorate from Wisconsin in analogue computation. Two of Kesavan's former colleagues at Waterloo, H.N. Mahabala and T.R. Viswanathan, followed.[55] None of them had been trained as computer scientists, because almost nobody had. Rajaraman and Mahabala obtained their computing education, in Bassett's phrase, 'on the fly, without formal training'.[56] The first teachers of computing in India were electrical engineers who taught themselves, and then taught others.

Switching it on

Among the students who took to the machine was Ravi Sethi, a first-year in the autumn of 1963. In a recollection published in 2021 in *The Spark*, a student magazine of the 1960s revived by IIT Kanpur alumni, he describes taking an evening course from Acton after the midterm examinations, and being one of six students selected to operate the 1620. He remembers being authorised to switch it on as an event in itself.[57] Programs were punched on cards, one line of code to a card. From the blinking lights on the console, the students learned to tell when a program had fallen

into a loop. Rabinowitz once challenged Sethi to write a program that would copy a deck of cards and fit on a single card, which meant six instructions or fewer. The machine needed its generators running, and then its air conditioning, before it could be turned on. Sethi spent the following summer working for Huskey on a compiler.[58]

By February 1964 the 1620 was running twenty-four hours a day, and by April the centre was open to students around the clock.[59] Undergraduates worked there as programmers. Among them were Sethi, who later became a senior manager at Bell Labs; Jayadev Misra, later head of electrical engineering at the University of Texas at Austin; and Raj Kanodia, who would help connect MIT to the ARPANET.[60]

From the spring of 1964 every student at Kanpur was trained in computing.[61] Rajaraman describes it as a compulsory course in programming and numerical methods for all engineering students, and claims it as a first in India.[62] Kelkar, speaking at Berkeley in 1968, described the effect: 'we soon became aware of the compelling power of the computer. It exercised a unique fascination over the minds of not only post-doctorals, but also freshmen'.[63]

The faculty moved quickly to make the subject a discipline. An M.Tech programme in computers followed in the mid-1960s, which IIT Kanpur's own account calls the first time computer science was offered as an academic discipline in India.[64] The institute's catalogue for 1969 listed a full range of undergraduate and graduate courses, including software engineering, information theory, hardware design and image processing.[65] Rajaraman published an introductory textbook, *Principles of Computer Programming*, in 1969, and later texts on FORTRAN and COBOL.[66]

Missionaries

The three Americans who set up the 1620 saw their job as spreading the gospel of computing across India. They lectured at research institutes and universities including the Tata Institute of Fundamental Research, IIT Madras and the Indian Statistical Institute, and gave a special session for fifteen members of the Planning Commission.[67]

Their lasting contribution was a ten-day course for people from institutions that might one day use computers. Huskey opened it to scientists, engineers and teachers from across India rather than restricting it to Kanpur's own faculty.[68] The course taught programming, numerical methods and computer logic, in a variant of FORTRAN called FORGO, with hands-on laboratory sessions.[69] Each course had about fifty or sixty participants, and it ran three or four times a year.[70] By August 1965, 600 people from fifty institutions had taken it, from other IITs, engineering colleges, national laboratories, government agencies and private firms.[71] Indian faculty kept it going after the Americans left; Rajaraman says it ran until 1975 and trained over 1,500 people.[72]

The Kanpur faculty taught the course, as Bassett puts it from Mahabala's recollections, 'with an almost religious sense of mission', in many cases driving to the railway station ten miles away themselves to collect participants, while their wives cooked the meals.[73] Kelkar's claim at Berkeley in 1968 was that 'there is hardly a computer installation where there is nobody who had been to Kanpur at least once.'[74] The claim cannot be tested, but it was not far-fetched. A government committee later counted thirty computers in the whole of India in 1965, and a course that had trained 600 people by August of that year had reached a large part of the small world of Indian computing.[75]

A dead crow on the world market

The 1620 was a good first machine, small, simple and needing little staff, but within a year Kanpur had used up its capacity.[76] It had cost roughly $320,000 before IBM's 60 per cent educational discount.[77] The institute's second computer was a much bigger purchase: an IBM 7044, which arrived in 1966 and made Kanpur home to one of the most powerful computers in India.[78]

How it was paid for matters to the rest of this book. The 7044's list price was $3 million. IBM gave Kanpur its standard educational discount, and also 'allowed it to pay in rupees', which Bassett calls 'a major concession to a country for which foreign exchange was precious'.[79] The money came from the Government of India.[80] Foreign exchange was the binding constraint on Indian computing for the next quarter-century. A machine bought in rupees escaped it.

The concession had a cost. The 7044 had been introduced in 1961 as a stopgap while IBM developed its System/360, and by 1969 IBM was phasing it out. Neither IBM India nor its American parent helped the institute obtain new software.[81] Kanpur students had been learning FORTRAN II, which the student newspaper, *Spark*, called 'almost a dead crow on the world market'.[82] The response was the first of many Indian adaptations to old hardware. In 1970 Kanpur undergraduates wrote what Bassett calls 'the first compiler written in India', which brought features

of newer versions of FORTRAN to a machine never designed for them.[83] KIAP ended in 1972 with no second phase, and with no American money for a replacement Kanpur kept its 7044 until 1979.[84]

Other IITs fared worse. At Bombay a Soviet Minsk II computer, delivered through UNESCO in 1965, stood in storage for two years while the Soviet Union and UNESCO negotiated who would pay for the technicians to install it; it began working in January 1968, running a Soviet programming language rather than FORTRAN.[85] Madras overtook Kanpur by a different route. In 1971 West Germany agreed to give IIT Madras a computer, and Mahabala, who had moved there from Kanpur, persuaded the Germans to buy an IBM machine. The IBM 370/155 inaugurated at Madras in November 1973 was the most powerful computer in India.[86]

Kanpur's enthusiasm was not the national mood: Indian unions called computers 'job eaters', and the government did not study the question seriously until the late 1960s.[87] What connected Kanpur to the wider country was IBM. A 1974 survey counted 217 computers installed in India: 65 per cent made by IBM, 13 per cent by the British firm ICL and 11 per cent by the Indian public-sector firm Electronics Corporation of India Limited (ECIL).[88] 'Simultaneity is not causality,' Bassett writes, 'but an American-oriented program of engineering education in India and an American computing firm did reinforce each other.'[89] Kanpur taught people to program IBM machines, and IBM sold the machines they programmed.

The road out

Kelkar and the Americans who designed KIAP had meant to build an Indian institute that would train engineers to solve Indian problems. The students partly defeated that aim by making Kanpur what its name already implied: 'the Indian MIT', and a step towards the American one.[90] In 1965-66 almost 7,000 Indians were studying in the United States, and at MIT in 1966 only Canada sent more students than India's eighty-two.[91] For its best undergraduates Kanpur was often only a step towards the best American graduate programmes, while its own graduate courses drew mainly students from further down India's educational hierarchy.[92]

Law made the step permanent. Before 1965 national quotas severely restricted permanent Indian immigration to the United States, and Indian engineers could study in America but were generally not allowed to stay and work. In 1965, the year of Kanpur's first graduating class, President Lyndon Johnson signed a law that abolished the quotas and made it much easier for Indians to settle.[93] Sixty-six of Kanpur's first hundred students graduated that May.[94] Bhushan went to MIT on a research assistantship.[95] An email list of the first batch showed that roughly a third were living in the United States in 2005.[96] The most thorough study of IIT emigration, made at Bombay by S.P. Sukhatme, found that slightly over 40 per cent of the institute's electrical engineering graduates of 1973 to 1977 had ended up abroad, the great majority in the United States.[97] No comparable figure exists for Kanpur.[98]

Kanpur's graduates reached the centre of American computing within a few years. On 14 June 1971 the ARPANET's Network Working Group issued Request for Comments 176, a protocol document. Its four authors were listed as A. Bhushan and R. Kanodia of MIT, Robert Metcalfe of MIT, later the inventor of Ethernet, and Jon Postel of UCLA.[99] Bhushan chaired the working group that defined the File Transfer Protocol, a standard still used to move files between computers.[100] Two of the four names on that document had learned to program on a 1620 in Kanpur. In 2008 Bassett examined the faculty of the ten American computer science departments ranked highest in 1993. Thirty-eight had taken their first degree in India, ten of them from IIT Kanpur. It was, he wrote, 'enough to fill a good-sized department'.[101]

Twenty seats

Until 1978 computing at Kanpur lived inside electrical engineering. That year the institute created a bachelor's degree in computer science, the first at any IIT.[102] Rajaraman, who led the effort, recalled that the senate had 'grudgingly allowed 20 students' into the programme, because it did not think computer science was mainstream engineering.[103]

Admission to every IIT seat was by rank in the Joint Entrance Examination. In 1978 over 100,000 students sat it for about 1,500 places.[104] The highest-ranked candidate chose any seat at any IIT, the next chose from what remained, and so on down.[105] Arjun Malhotra, later a co-founder of the computer firm HCL, described the arithmetic to the Computer History Museum in 2018. Sitting the examination in the mid-1960s, he wanted one of the 30 places in IIT Kharagpur's electronics department and told himself that with 'a rank of 30 or less, then no one can take my first choice away from me'. He came 32nd, and got in because candidates ranked above him chose other courses.[106]

In 1978 the students' choice was decisive. 'Students and their parents were very apprehensive about this new programme,' Rajaraman recalled in 2018. 'The greatest surprise came when the JEE results were declared and the B.Tech programme closed at an AIR of 40,' AIR being the all-India rank.[107] About half of India's top forty candidates had chosen a subject the senate had not considered mainstream engineering. The other IITs, Rajaraman writes, were surprised, and all of them started degrees in computer science.[108] Ten years later Kanpur's fourteen computer science seats were filled by rank 38.[109]

Where those students went is documented for the first class. Nineteen graduated in 1983. In 2007 Bassett traced seventeen through an alumni website. One had died. Of the other sixteen, fourteen lived in the United States, one in the Czech Republic and one in India.[110] Nine had doctorates and five held faculty posts in American computer science departments. Six lived in the San Francisco Bay Area, and at least three had founded companies. One, Rajeev Motwani of Stanford, was an early academic mentor to Larry Page and Sergey Brin in the work that led to Google.[111] Some Kanpur teachers, frustrated by the outflow, tried to stop it by refusing to write recommendation letters for students applying to American graduate schools. Bassett calls the attempt futile.[112]

The road to industry

Kanpur never had a computer industry of its own, unlike MIT and Stanford, each of which seeded one nearby.[113] Its influence on Indian software ran through people.

The first of them was Kohli, Kelkar's old colleague from Bombay. Kelkar persuaded him to sit on Kanpur's electrical engineering selection committee and to visit the institute regularly. Kohli got his practical introduction to computing on Kanpur's 1620, and took computing back to Tata Electric in Bombay.[114] When the Tata group set up a computing business, Kohli moved to head it.[115] In the early years of what became Tata Consultancy Services, Mahabala spent his summers there as a consultant, and Kohli often hired Kanpur's master's graduates in computing.[116]

The second was N.R. Narayana Murthy. He had studied electrical engineering at the University of Mysore, meaning to become a hydroelectric engineer, and then took a master's degree at Kanpur, where he got 'hooked' on computing.[117] In his own account the turn came after a breakfast-table talk by a visiting professor from Princeton; within months he was running simulations through the night on the 7044. 'It was a macho thing to make the six tape units rotate fast.'[118] Each of the twelve students in his graduate class, he told the Computer History Museum in 2018, had four or five job offers from companies that had installed IBM 1401s. He took a job at half the salary with Jashwant Krishnayya, an MIT-trained professor at the Indian Institute of Management in Ahmedabad who had come to Kanpur to recruit engineers with computing skills, because Rajaraman advised him that he would learn how a time-sharing system worked.[119] It is the first price for programming skill in this book: a class of twelve choosing among some fifty offers. Murthy later joined Narendra Patni, whose firm was one of the first in India to get computing work from American companies, and left it in 1981 to found Infosys with colleagues.[120] Kanpur gave Murthy the ability; men like Patni and Kohli, already connected to American business, gave that ability its commercial force.[121]

The third large firm of the period has no Kanpur connection at all. Wipro was built by Azim Premji, a Stanford-educated electrical engineer who turned his family's cooking-oil business towards computers.[122] Kanpur was one route into the industry, not the only one.

Coda: from the classroom to the state

In 1972 the American programme ended, and Kanpur's computer was already old. Kelkar had left two years earlier to direct IIT Bombay.[123] The government in Delhi had created a Department of Electronics in 1970 and was setting out to have India's computers designed and built in India by a public-sector company.[124] The state wanted machines made at home. Kanpur had already made the thing India would one day sell to the world: an engineer trained in English, to an American standard, on American machines. For now, many of the best of them left.

Notes to Chapter 2. Kanpur, 1963 (1946-1978)

[1] Bassett, 'Aligning India in the Cold War Era' (2009), p. 783 and p. 792 n. 15; Fay, 'Report from Kanpur' (1966), p. 11. Bassett (Bassett, 'Aligning India in the Cold War Era' (2009), p. 783) gives a DC-7. Harry Huskey, who was waiting for the machine, remembered the aircraft as a chartered DC-8, and recalled that it reached Delhi without a manifest, so that a customs officer sent it on to the military airfield at Kanpur (Computer History Museum, 'Oral History of Harry Huskey' (2006), p. 41: Oral History of Harry Huskey, interviewed by William Aspray, 7 February 2006). The two accounts cannot be reconciled on the aircraft.

[2] Bassett, 'Aligning India in the Cold War Era' (2009), p. 783.

[3] Bassett, 'Aligning India in the Cold War Era' (2009), p. 783.

[4] Bassett, 'Aligning India in the Cold War Era' (2009), p. 791.

[5] Computer History Museum, 'Oral History of Harry Huskey' (2006), p. 42.

[6] Computer History Museum, 'Oral History of Harry Huskey' (2006), p. 42 (60-cycle and 50-cycle power, telegrams, converter from Germany, 'in about three weeks we had the machine up and running').

[7] Bassett, 'Aligning India in the Cold War Era' (2009), p. 792.

[8] Rajaraman, *History of Computing in India* (2012), p. 17; Bassett, 'Aligning India in the Cold War Era' (2009), pp. 783, 792, and p. 794 n. 23 (definition of a compiler). Rajaraman dates the installation to August 1963. The crate arrived in July (Bassett, 'Aligning India in the Cold War Era' (2009), p. 783), and Huskey recalled the machine running 'in about three weeks' (Computer History Museum, 'Oral History of Harry Huskey' (2006), p. 42): the two dates describe two moments, the arrival and the first running.

[9] Bassett, 'Aligning India in the Cold War Era' (2009), p. 783.

[10] Computer History Museum, 'Oral History of Harry Huskey' (2006), p. 42.

[11] Bassett, 'Aligning India in the Cold War Era' (2009), p. 786.

[12] Bassett, 'Aligning India in the Cold War Era' (2009), p. 787.

[13] Bassett, 'Aligning India in the Cold War Era' (2009), p. 787; *Interim Report of the Sarker Committee* (1946).

[14] Bassett, 'Aligning India in the Cold War Era' (2009), p. 787.

[15] Bassett, 'Aligning India in the Cold War Era' (2009), p. 787.

[16] Bassett, 'Aligning India in the Cold War Era' (2009), p. 797. Bassett calls the Madras curriculum in some ways the antithesis of Kanpur's; the analogue machine had been built by Bell Labs.

[17] Bassett, 'Aligning India in the Cold War Era' (2009), p. 796.

[18] Bassett, 'Aligning India in the Cold War Era' (2009), pp. 787, 796-798.

[19] Bassett, 'Aligning India in the Cold War Era' (2009), p. 788.

[20] Bassett, 'Aligning India in the Cold War Era' (2009), p. 786.

[21] Bassett, 'Aligning India in the Cold War Era' (2009), p. 789.

[22] Bassett, 'Aligning India in the Cold War Era' (2009), p. 789.

[23] Bassett, 'Aligning India in the Cold War Era' (2009), p. 790 (Bassett calls it an epiphany).

[24] Bassett, 'Aligning India in the Cold War Era' (2009), p. 791.

[25] Bassett, 'Aligning India in the Cold War Era' (2009), p. 791.

[26] Bassett, 'Aligning India in the Cold War Era' (2009), p. 790.

[27] Bassett, 'Aligning India in the Cold War Era' (2009), p. 791.

[28] Bassett, 'Aligning India in the Cold War Era' (2009), p. 791.

[29] Fay, 'Report from Kanpur' (1966), p. 11.

[30] Bhushan, 'Vignettes of History' (2014), Part I.

[31] Bhushan, 'Vignettes of History' (2014), Part I; Bassett, 'Aligning India in the Cold War Era' (2009), p. 792 n. 15.

[32] Bhushan, 'Vignettes of History' (2014), Part II.

[33] Bhushan, 'Vignettes of History' (2014), Part III.

[34] Bhushan, 'Vignettes of History' (2014), Parts I and III. No record has been located of when the first woman was admitted to Kanpur, or to its computing courses.

[35] Fay, 'Report from Kanpur' (1966), p. 13.

[36] Fay, 'Report from Kanpur' (1966), p. 11.

[37] Fay, 'Report from Kanpur' (1966), p. 15.

[38] Bhushan, 'Vignettes of History' (2014), Part I.

[39] Bassett, 'Aligning India in the Cold War Era' (2009), p. 790.

[40] Bassett, 'Aligning India in the Cold War Era' (2009), p. 790 n. 12; Fay, 'Report from Kanpur' (1966), p. 11. The nine were MIT, Caltech, the Carnegie Institute of Technology, Case, Princeton, the University of California at Berkeley, Michigan, Ohio State and Purdue.

[41] Fay, 'Report from Kanpur' (1966), p. 11. The agent was Educational Services Incorporated.

[42] Fay, 'Report from Kanpur' (1966), p. 16.

[43] Bassett, 'Aligning India in the Cold War Era' (2009), p. 797.

[44] Fay, 'Report from Kanpur' (1966), pp. 11-12.

[45] Fay, 'Report from Kanpur' (1966), p. 12.

[46] Fay, 'Report from Kanpur' (1966), p. 13.

[47] Fay, 'Report from Kanpur' (1966), p. 13 (the article appeared in June 1966, p. 11, so 'last September' is September 1965); Rajaraman, *History of Computing in India* (2012), p. 81 (the 1965 war with Pakistan).

[48] Fay, 'Report from Kanpur' (1966), p. 13.

[49] Fay, 'Report from Kanpur' (1966), p. 16.

[50] Bassett, 'Aligning India in the Cold War Era' (2009), p. 795 and n. 25.

[51] Fay, 'Report from Kanpur' (1966), p. 16.

[52] Fay, 'Report from Kanpur' (1966), p. 13; Bassett, 'Aligning India in the Cold War Era' (2009), p. 786.

[53] Bassett, 'Aligning India in the Cold War Era' (2009), p. 792.

[54] Fay, 'Report from Kanpur' (1966), p. 14.

[55] Bassett, 'Aligning India in the Cold War Era' (2009), p. 792. Kesavan also held a master's degree from Illinois; Mahabala and Viswanathan both had doctorates from Saskatchewan.

[56] Bassett, 'Aligning India in the Cold War Era' (2009), p. 793.

[57] Sethi, 'Our Experiences on the 1620' (2021).

[58] Sethi, 'Our Experiences on the 1620' (2021).

[59] Rajaraman and Joshi, 'Professor Harry Huskey: Planting a Tree' (2021) (same issue of *The Spark*, A. Shukla, 'The IBM 1620'); Rajaraman, *History of Computing in India* (2012), p. 18; Bassett, 'Aligning India in the Cold War Era' (2009), p. 793.

[60] Bassett, 'Aligning India in the Cold War Era' (2009), p. 793 n. 18.

[61] Bassett, 'Aligning India in the Cold War Era' (2009), p. 793.

[62] Rajaraman, *History of Computing in India* (2012), p. 18.

[63] Bassett, 'Aligning India in the Cold War Era' (2009), p. 795.

[64] Bassett, 'Aligning India in the Cold War Era' (2009), p. 793 (plans for a master's concentration in computer science in place by 1965); Rajaraman, *History of Computing in India* (2012), p. 70 (Rajaraman's chronology dates the M.Tech programme in computers to 1966); IIT Kanpur, obituary of V. Rajaraman (2025), post-cutoff material, checked in October 2026. (IIT Kanpur's account of Rajaraman's career dates it to 1965 and calls it the first time computer science was offered as an academic discipline in India).

[65] Bassett, 'Aligning India in the Cold War Era' (2009), p. 793.

[66] Bassett, 'Aligning India in the Cold War Era' (2009), pp. 793-794 and n. 20.

[67] Bassett, 'Aligning India in the Cold War Era' (2009), p. 799. The others were the University of Roorkee, the University of Bombay and the Physical Research Laboratory in Ahmedabad.

[68] Rajaraman, *History of Computing in India* (2012), pp. 17-18.

[69] Rajaraman and Joshi, 'Professor Harry Huskey: Planting a Tree' (2021); Bassett, 'Aligning India in the Cold War Era' (2009), p. 799.

[70] Rajaraman and Joshi, 'Professor Harry Huskey: Planting a Tree' (2021); Rajaraman, *History of Computing in India* (2012), p. 18.

[71] Bassett, 'Aligning India in the Cold War Era' (2009), pp. 799-800.

[72] Rajaraman, *History of Computing in India* (2012), p. 18.

[73] Bassett, 'Aligning India in the Cold War Era' (2009), p. 800 and n. 41.

[74] Bassett, 'Aligning India in the Cold War Era' (2009), p. 799.

[75] Ministry of Labour and Rehabilitation, *Report of the Committee on Automation* (1972), p. 5; Bassett, 'Aligning India in the Cold War Era' (2009), p. 800.

[76] Bassett, 'Aligning India in the Cold War Era' (2009), p. 794.

[77] Bassett, 'Aligning India in the Cold War Era' (2009), p. 792 n. 15.

[78] Bassett, 'Aligning India in the Cold War Era' (2009), p. 794.

[79] Bassett, 'Aligning India in the Cold War Era' (2009), p. 794 n. 21.

[80] Rajaraman, *History of Computing in India* (2012), p. 18.

[81] Bassett, 'Aligning India in the Cold War Era' (2009), p. 794 and n. 21.

[82] Bassett, 'Aligning India in the Cold War Era' (2009), p. 794.

[83] Bassett, 'Aligning India in the Cold War Era' (2009), p. 794.

[84] Bassett, 'Aligning India in the Cold War Era' (2009), p. 797 and n. 32. KIAP ended during what Bassett describes as the worst years of relations between India and the United States. The 7044 was replaced in 1979 by a Digital Equipment Corporation DEC-10.

[85] Bassett, 'Aligning India in the Cold War Era' (2009), p. 796. The Soviet technicians arrived in July 1967 and repaired the damage done in storage; the machine was the only one of its kind in India.

[86] Bassett, 'Aligning India in the Cold War Era' (2009), pp. 797-798; Gopalakrishnan and Narayanan, *Computers in India* (1975), p. 25, as quoted in Bassett, 'Aligning India in the Cold War Era' (2009), p. 798 n. 33.

[87] Bassett, 'Aligning India in the Cold War Era' (2009), p. 799. Bassett adds that the computer, imported and labour-saving, was 'a problematic technology in a country that had a surplus of labor, prized self-sufficiency, and put little stock in international trade' (Bassett, 'Aligning India in the Cold War Era' (2009), p. 798). The government's inquiry into automation is the subject of Chapter 3.

[88] Bassett, 'Aligning India in the Cold War Era' (2009), p. 800.

[89] Bassett, 'Aligning India in the Cold War Era' (2009), p. 800.

[90] Bassett, 'Aligning India in the Cold War Era' (2009), p. 802.

[91] Bassett, 'Aligning India in the Cold War Era' (2009), p. 802.

[92] Bassett, 'Aligning India in the Cold War Era' (2009), pp. 802-803.

[93] Bassett, 'Aligning India in the Cold War Era' (2009), p. 803 and n. 49.

[94] Fay, 'Report from Kanpur' (1966), p. 11.

[95] Bhushan, 'Vignettes of History' (2014), Part III.

[96] Bassett, 'Aligning India in the Cold War Era' (2009), pp. 803-804 n. 50 (the list was obtained by Bassett).

[97] Bassett, 'Aligning India in the Cold War Era' (2009), p. 803 and n. 50, citing Sukhatme, *The Real Brain Drain* (1994), pp. 17-18.

[98] Bassett, 'Aligning India in the Cold War Era' (2009), p. 803. Bassett expects the Kanpur figure was similar, and notes that whatever an IIT's foreign patron, those who left went overwhelmingly to America (Bassett, 'Aligning India in the Cold War Era' (2009), p. 807).

[99] Bassett, 'Aligning India in the Cold War Era' (2009), p. 805.

[100] Bassett, 'Aligning India in the Cold War Era' (2009), p. 805.

[101] Bassett, 'Aligning India in the Cold War Era' (2009), pp. 805-806. The thirty-eight were eleven from IIT Madras, ten from IIT Kanpur, five from IIT Delhi and four from IIT Bombay, with the rest from other Indian colleges.

[102] Bassett, 'Aligning India in the Cold War Era' (2009), pp. 805-806; Rajaraman, *History of Computing in India* (2012), p. 34.

[103] Rajaraman, *History of Computing in India* (2012), p. 34.

[104] Rajaraman, *History of Computing in India* (2012), p. 34.

[105] Bassett, 'Aligning India in the Cold War Era' (2009), p. 806.

[106] Computer History Museum, 'Oral History of Arjun Malhotra' (2018), p. 4 (Oral History of Arjun Malhotra, recorded 15 November 2018; he left school in December 1964, p. 4).

[107] Jayshankar, 'Interview with Dr Rajaraman', *Vox Populi* (2018) (Rajaraman's interview with *Vox Populi*, an IIT Kanpur student magazine, 7 June 2018, updated 15 May 2021; consulted 2026-10-15); Rajaraman, *History of Computing in India* (2012), p. 34; Bassett, 'Aligning India in the Cold War Era' (2009), p. 806 and n. 53. Rajaraman's own history also gives rank 40 for the last student admitted (Rajaraman, *History of Computing in India* (2012), p. 34). Bassett, citing an interview with Rajaraman, gives rank 44, so that close to half of the top forty students in India had chosen the course (Bassett, 'Aligning India in the Cold War Era' (2009), p. 806 and n. 53). All three accounts rest on memory, recorded decades later. No copy of the institute's annual report for 1978-79, which would settle the figure, has been located.

[108] Rajaraman, *History of Computing in India* (2012), p. 34.

[109] Bassett, 'Aligning India in the Cold War Era' (2009), p. 806 n. 53.

[110] Bassett, 'Aligning India in the Cold War Era' (2009), p. 807.

[111] Bassett, 'Aligning India in the Cold War Era' (2009), p. 807.

[112] Bassett, 'Aligning India in the Cold War Era' (2009), p. 807.

[113] Bassett, 'Aligning India in the Cold War Era' (2009), p. 800.

[114] Bassett, 'Aligning India in the Cold War Era' (2009), pp. 800-801.

[115] Bassett, 'Aligning India in the Cold War Era' (2009), p. 800.

[116] Bassett, 'Aligning India in the Cold War Era' (2009), p. 801.

[117] Bassett, 'Aligning India in the Cold War Era' (2009), p. 801.

[118] Computer History Museum, 'Oral History of N. R. Narayana Murthy', Part 1 (2018), pp. 7-8.

[119] Computer History Museum, 'Oral History of N. R. Narayana Murthy', Part 1 (2018), pp. 8-9 (Oral History of N.R. Narayana Murthy, Part 1, recorded 13 May 2018; founder's recollection). Krishnayya was installing a Hewlett-Packard computer bought with Ford Foundation money (Bassett, 'Aligning India in the Cold War Era' (2009), p. 801).

[120] Bassett, 'Aligning India in the Cold War Era' (2009), p. 801. After a brief spell in France, Murthy worked for Krishnayya again, at a research institute modelled on the RAND Corporation, before joining Patni. The founding of Infosys is told in Chapter 6.

[121] Bassett, 'Aligning India in the Cold War Era' (2009), p. 802 (Bassett's summary of the Kanpur, Patni and Kohli connections).

[122] Bassett, 'Aligning India in the Cold War Era' (2009), p. 801 n. 44.

[123] Bassett, 'Aligning India in the Cold War Era' (2009), p. 798.

[124] Rajaraman, *History of Computing in India* (2012), p. 3.

Chapter 3. Self-reliance and its limits (1962-1977)

On 18 July 1968 the government of India called a special meeting of its Standing Labour Committee in New Delhi to discuss a single subject: automation. It was one of the tripartite bodies in which government, employers and trade unions argued out labour policy.[1] For this session the Ministry of Labour also invited workers' and employers' representatives from the Life Insurance Corporation (LIC), the State Bank of India, the railways and the oil companies, organisations that would not normally have had a seat at the table.[2] The machines that had brought them there were computers, and there were not many of them. The country had 30 computers in 1965 and 88 in 1968.[3]

The workers' side asked for something close to a ban. Their statement, as the government later printed it, argued that 'since the country had an enormous volume of unemployment and was short of technological and capital resources the general orientation of policy should be against automation'. Exceptions could be allowed where there was a real compulsion, but until guidelines were agreed there should be a freeze, and employers who had already installed automatic machinery should be asked to stop using it.[4] The employers replied that some concerns had become so large and complicated that computers were a necessity, that automation created more employment across the economy in the end, and that fears of large-scale displacement were unwarranted because automatic processes had been introduced in only a few establishments.[5] The Labour Minister of Maharashtra proposed a tripartite sub-committee, with experts, to draw up national guidelines.[6]

The meeting ended without agreement. Summing up, the Labour Minister said there was no consensus, that the government would bear the different views in mind, and that the existing procedure for screening computer imports would continue. In the meantime the government went on allowing automation 'on a selective basis'.[7]

The argument in that room did not end in 1968. It set a machine that could do part of the work of salaried office staff against the jobs of those staff, and it asked who should bear the cost and who should keep the gain. The employers said growth would absorb the people displaced. The workers said a poor country could not afford to make them redundant. Nearly six decades later the same argument returned, over software that could write software, and the office workers whose jobs were in question were the programmers whose industry this book follows. The book comes back to it in its last chapter.

Between the war with China in 1962 and the general election of 1977, the Indian state tried to do two things with computers at once. It wanted India to design and build its own, as a matter of national self-reliance. And it wanted to stop computers from taking the jobs of the country's organised clerical workers. Both aims grew out of scarcity: of foreign exchange, which India needed to buy machines abroad, and of jobs, which the computer seemed to threaten.

Repainted machines

Before 1970, V. Rajaraman, the historian of Indian computing, writes, there was no government policy on computers beyond the 1958 Scientific Policy Resolution.[8] The rule that mattered was about money. 'Foreign companies were permitted to manufacture and sell computers provided foreign exchange was not required. Computers were not considered a priority area that deserved foreign exchange outflow.'[9]

IBM found a way to work within that rule. It had had a commercial outlet in India since 1950 and began manufacturing in 1963, making punched cards at a factory in Bangalore (Bengaluru since 2014) and, at Kurla in Bombay (Mumbai since 1995), the model 029 key punch (a machine for punching data onto cards) and other unit-record equipment.[10] The key punches were exported to IBM branches in South-East Asia, and the foreign exchange they earned paid for imports of used, disassembled IBM 1401 computers.[11] Around 1965 many 1401s were being

replaced in America and Europe by IBM's new System/360, and IBM could import the old machines at a low inter-company price. In Bombay they were dismantled, their worn parts replaced, and then reassembled, painted and rented to Indian customers.[12] The government licensed IBM to recondition 68 of them over three years, from 1968 to 1970. IBM had applied to recondition System/360s instead.[13] Its British rival ICL ran the same trade from a plant in Poona (now Pune), exporting card equipment and using the proceeds to sell its systems in India.[14]

By 1970 the 1401 was the most popular computer in India, with about 80 installations. Indian Railways had more than a dozen; most Tata group companies and the big textile firms had them; the first computer at Tata Consultancy Services (TCS) was a 1401. For smaller firms IBM ran data centres in the major cities.[15] ICL, by comparison, had 48 of its 1901A machines assembled by the public-sector Bharat Electronics between 1965 and 1975.[16]

The machines were not cheap. The annual rental for a high-end 1401, with four tape drives and two disks, was about Rs 850,000, some $113,000 at the 1970 exchange rate.[17] The devaluation of the rupee in 1966 gave IBM a windfall, because its rentals were quoted in dollars.[18] IBM 'was in fact accused of profiteering', since rentals in America were much lower. Its defence was that the rental included free maintenance and the services of a systems engineer who analysed the customer's needs, designed the software and even helped with programming: IBM was selling a computing service, not renting a machine.[19] In that arrangement the engineer's time had no price of its own. It came bundled into the rent of the machine. The industry that grew up later would turn the arrangement round, and sell the time itself.

That service is where many Indians learned the trade. Rajaraman, who is not uncritical of IBM's prices, credits it with recruiting and training very good technical and sales staff. 'A whole generation of maintenance engineers and programmers were trained and nurtured by IBM. In short IBM set a bench mark for professional operations in India.'[20]

The state was already moving against the trade in repainted machines. By the early 1970s the new Department of Electronics had declared in its annual report that 'no further manufacturing programmes based on the renovation of 'As is' machines will be accepted in future as a matter of national policy'.[21] The question was what would replace them.

A plan for 500 computers a year

The 1962 border war with China persuaded the government that electronics and communications mattered for defence and for long-run industrial growth. In 1963 it appointed a committee on electronics under Homi Bhabha, the chairman of the Atomic Energy Commission.[22] The committee's report was ready by 1966, but Bhabha was killed in an air crash before formally submitting it.[23] It was presented to the government only after his death, and a committee under Vikram Sarabhai was set up in 1966 to carry out its recommendations.[24] The central ones were a Department of Electronics in the central government and an Electronics Commission with wide financial and executive powers.[25]

The committee's computer subcommittee was chaired by R. Narasimhan, head of the computer group at the Tata Institute of Fundamental Research (TIFR) in Bombay.[26] It projected India's needs for ten years ahead. One very large computer would be imported for a national facility. Three to five medium-large machines would be imported and run as Regional Computer Centres, shared like power utilities. All smaller computers would be made in India from Indian components, about 500 a year for ten years.[27] Its report argued that 'attaining self-sufficiency in systems engineering and fabrication is of fundamental importance from the point of view of the defence and security of our country'.[28]

In 1970 the plan was debated at a National Electronics Conference at TIFR, attended by people from universities, companies, government, laboratories and the armed forces.[29] Rajaraman, then on the faculty of the Indian Institute of Technology (IIT) Kanpur, argued that India lacked a home market large enough to build up in the textbook order, from components to circuits to systems. India should go backwards, design systems first and import the components, and pay for the imports with exports. 'Foreign exchange can also be earned by software export,' he said. In his history of 2012 he added: 'However, my comment was not taken seriously.'[30]

The chairman of the conference, Sarabhai, who had succeeded Bhabha, took a larger view. In his concluding remarks he called computing 'far more fundamental, of wider significance than any other field of electronics', and of 'particular relevance to our developing country where resources are scarce'. Multinational companies had a contribution to make, but not, he said, at the cost of a major national effort.[31] On that thinking the government

started a computer division at the Electronics Corporation of India Ltd (ECIL), a public-sector company set up at Hyderabad in 1967 under the Department of Atomic Energy to commercialise electronics developed at the atomic research centre at Trombay.[32]

The gatekeeper

The Department of Electronics (DoE) was established in 1970 and the Electronics Commission, by most accounts, in 1971. M.G.K. Menon, the director of TIFR, became secretary of the department and chairman of the commission.[33] Narasimhan later described the two big decisions that followed. The first was to develop and manufacture small and medium computers in India, so that within a set time most of the country's needs in that range could be met from Indian production. The second was to centralise all computer purchasing through global tenders.[34]

The DoE also became the gate through which every imported computer had to pass. In 1970 responsibility for import licences for computers was transferred to it as a 'single window clearance', and it drafted rules that judged each request by its essentiality, whether an equivalent machine was made in India and how much foreign exchange it needed.[35] The rupee was not convertible and the trade balance was permanently in deficit, so that 'obtaining foreign exchange for import of equipment required elaborate justification'.[36] The quadrupling of oil prices in 1973 made foreign exchange scarcer still. Unless the United Nations Development Programme (UNDP) or a foreign government was paying, clearance to import a computer usually took one to three years.[37] The delays hurt universities, laboratories, industry and especially companies that wanted to import computers in order to export software.[38]

Foreign policy closed other doors. During the 1971 war with Pakistan the United States, which favoured Pakistan, embargoed exports of electronics and computers to India. India's first nuclear test in 1974 brought a further embargo on advanced electronic equipment, including high-end computers and scientific software.[39]

Private Indian firms, several of them already making electronic calculators, wanted to build computers. The DoE kept delaying the policy that would set the rules for them, partly because many components and peripherals would have to be imported and partly to protect the ECIL programme it was funding.[40] In the meantime, Rajaraman writes, the DoE 'became very unpopular' as legitimate requests from universities, government science departments and industry were 'inordinately delayed'; and 'no private Indian company was allowed to manufacture computers'.[41]

A cottage industry in Hyderabad

ECIL's computer division was headed by S. Srikantan, who had designed the Trombay Digital Computer 12 (TDC-12) at the atomic research centre. It was a 12-bit real-time machine, similar in design to the PDP-8 made by the American firm Digital Equipment Corporation, and it was commissioned at Trombay in 1969.[42] ECIL had no commercial computer of its own on which to develop software, so it sent much of the work out to TIFR, IIT Kanpur and the Indian Institute of Management at Ahmedabad; with ECIL's own engineers, they produced an operating system, an assembler and BASIC and FORTRAN compilers.[43] The TDC-12 was followed by the TDC-312 in 1974, the 16-bit TDC-316 in 1975 and, in 1978, the Micro 78, built around a microprocessor.[44]

Between 1971 and 1978 ECIL sold 98 computers.[45] A TDC-316 cost Rs 3 million, about $360,000 at the 1975 rate.[46] ECIL had shown, in Rajaraman's judgement, that computers could be made in India from local components with imported chips and peripherals. 'It however worked more like a cottage industry', because its volumes were small. Most of its machines went to atomic energy establishments, government departments and universities that depended on government grants and were 'constrained to buy' from it. 'Only 4 machines were sold to the private sector.' The machines were overpriced and short of software.[47]

ECIL's most important work was hidden from view. The Air Defence Ground Environment System (ADGES), a chain of radar stations along India's borders to detect hostile aircraft, could not be bought abroad without revealing sensitive information to foreign vendors. TIFR designed the system and its software. Each installation used three ruggedised TDC-316s and cost Rs 30 million; 25 were deployed by the Indian Air Force between 1969 and 1984, and the income sustained ECIL's other development work.[48] Besides the 98 machines sold commercially, 75 ruggedised TDC-316s went into this project.[49] 'In the published literature this important contribution of ECIL is not mentioned,' Rajaraman writes.[50]

The most lasting product was people. ECIL trained 'over a thousand engineers in designing systems and developing systems software', and many of them later joined private computer makers and software companies to

lead groups there.[51] For an Indian engineer in the 1970s who wanted to build computers rather than operate them, ECIL was one of very few possible employers.

The clerks' case

The computer met its strongest opposition not in factories but in offices. The labour movement already had a template for dealing with new machines. In 1957 the 15th Indian Labour Conference had adopted a Model Agreement on rationalisation, which promised no retrenchment and no loss of earnings for existing workers, an equitable sharing of the gains among the community, the employer and the worker, and prior consultation with the unions before new equipment was installed.[52] Automation in new process industries and in manufacturing aroused little opposition. Computers in offices, the government's Committee on Automation later found, 'evoked serious misgivings among the employees'.[53]

According to that committee, the issue gained national attention because of the 'unilateral decision' of LIC's management, in the mid-1960s, to introduce a computer. The All India Insurance Employees' Association started an agitation, and the central trade union organisations took up the wider question of automation.[54] The unions' case, as the committee summarised it, was that automation was 'a painful luxury, and ... a labour saving device which has no place in a developing economy marked by labour surpluses and scarce capital resources'. They held that the loss of jobs, existing and potential, was built into any scheme of automation, and that using an expensive computer for routine office work was a misuse of a costly tool.[55] The employers answered that automation was simply an advanced stage of mechanisation, indispensable for industrialisation and better management.[56]

The dispute reached Parliament. On 24 November 1966, the day before the insurance employees' one-day token strike against the computers, the Finance Minister, Sachindra Chaudhuri, told the Lok Sabha that LIC had decided on computers in 1964 and installed the first at Bombay in November 1965. The switch would take some three years. '383 jobs would disappear but the very process of computerisation would create 225 new jobs.' The net loss of 158 was, he said, insignificant beside the 1,500 or so new jobs LIC created every year, and no one would be retrenched.[57] Four months later George Fernandes, then a member of the Lok Sabha, asked how many computers LIC had installed, at what cost, and how many employees had lost their jobs. The Minister of State for Finance, K.C. Pant, answered that one computer, made by IBM, had been installed at Bombay, and that no one had lost a job.[58] When another member asked whether the computers would harm the corporation's capacity to create jobs, and whether that was wise, the new Finance Minister, Morarji Desai, replied: 'It will be an unwise policy if we do not do this.'[59]

The tripartite meetings did not settle the matter. The Indian Labour Conference discussed automation in July 1966 and agreed that the Model Agreement should apply. The Standing Labour Committee returned to it in May 1967 and could not agree on clerical automation at all.[60] Union conventions in 1966 and 1967 demanded an immediate ban.[61] The one safeguard the government had, a requirement that any import using foreign exchange be cleared by the Labour Ministry, barely touched the problem. Most new computers were assembled in India, not imported, so they escaped screening. Between 1967 and 1971, 94 computers were introduced in India and only five proposals were referred to the Labour Ministry.[62] IBM's way around the foreign-exchange rule was also, in practice, a way around the labour rule.

Inside the computer rooms

The Committee on Automation, which grew out of the July 1968 meeting, produced the best surviving picture of where India's computers were and who worked around them. By August 1971 it counted 140 computers in the country: 10 in government departments, 39 in public-sector undertakings including the railways, 55 in private firms and 36 in research institutions, technical institutes and universities.[63] Bombay alone had 35, a quarter of the total.[64] Nearly all were new. 'As many as 124 of the 140 computers in India have been installed only since 1965,' the committee wrote.[65]

Most did office work. Payroll, billing, invoices and financial statistics dominated, partly because these jobs had already been partly mechanised on punched-card machines.[66] The railways used their computers for wagon movements and accounts; the State Bank of India reconciled some 70,000 inter-branch transactions a day; Delhi Cloth Mills and Calico Mills used linear programming to choose the cheapest blend of cotton.[67] The committee heard the charge that some firms bought computers as status symbols, 'to keep up with the 'Joneses''.[68] When it asked 40

public and private organisations why they had installed computers, most named better service, faster tabulation and better management decisions. Only five named a reduction in labour costs.[69]

The committee's case studies show how the work changed hands. Computer departments created new posts, many at higher grades: systems analysts and programmers above all. Punch-card operators and machine operators were moved into the new sections, and much of the computer staff was recruited from inside the organisation and trained. Several existing employees, the committee found, 'got the benefit of better jobs and better pay', while the pay of the remaining clerks did not rise.[70] Some clerical workers now had to work second and third shifts to keep the expensive machines busy, 'a practice associated usually with blue-collar jobs'. Union representatives told the committee that computerisation had put clerical staff under more strain, requiring greater speed and fixed schedules.[71]

The question of jobs was harder to measure than either side admitted. A works study team in Maharashtra, looking at six establishments, followed 741 employees displaced by computers: 531 were given other jobs, 93 were retrained for the computer section, 71 were absorbed elsewhere in the organisation and 46 in expansion.[72] Almost nobody was sacked. The unions' point was that this was not the whole story. In one unnamed establishment studied by the committee, the staff on branch clearance and government accounts fell from 380 in November 1966 to 304 in January 1970, while the volume of work rose; the old methods would have needed 516 people for the 1970 volume.[73] The committee concluded that although there had been no direct retrenchment, the immediate effect of the computer was a fall in the demand for clerks, because vacancies left by natural wastage were filled by surplus staff rather than new recruits.[74] It was the next generation of clerks, not the present one, who lost.

Few managements asked their workers first. Of 42 units for which the committee had information, 35 had introduced a computer without prior consultation, and only seven had consulted their unions.[75] 'No instance was brought to our notice in which a proposal to instal a computer was dropped due to disagreement with union.'[76]

The Committee on Automation

The committee was set up by the Ministry of Labour in July 1969 under R. Venkataraman, a member of the Planning Commission, and finished its work under the economist V.M. Dandekar. Its members included industrialists, an economist, a computer specialist and two members whose notes to the report speak for labour.[77] It was candid about its limits. It was 'greatly handicapped because of the inadequacy of available data', had found no comprehensive study of the effects of automation in India, and relied on questionnaires, case studies and oral evidence.[78]

Its recommendations, signed on 18 May 1972, tried to give each side something.[79] Computers in education, research, government statistics and defence should be allowed, with protection for existing staff.[80] Commercial and industrial users, public or private, and expressly including consultancy services, should be examined case by case.[81] Where possible their needs should be met from shared computer service centres, since a computer facility was 'in the nature of a public utility' and should preferably be in the public sector.[82] And 'all use of computers by commercial and industrial establishments ... should be subject to a prior agreement with the workers'.[83] A firm wanting a computer would write a Justification Report setting out the jobs to be computerised, the expected benefits and the likely effects on employment and pay. Experts chosen jointly by management and workers would assess it, and disputes would go to a National Bipartite Machinery of employers and unions.[84] The committee urged that India's needs in hardware and software be met 'as far as possible from indigenous sources', which would save foreign exchange and create jobs.[85]

Every member signed, but three explanatory notes show how far apart they remained. Satish Loomba wrote that in a capitalist economy automation served profit, so Indian workers had to oppose it in general; he agreed to the recommendations 'as a practical policy at the present juncture'.[86] G. Ramanujam warned that automation could reduce employment: 'It can rob labour of its personality, creative effort, sense of achievement and drag down work itself to boredom.' He signed because of the prior-consent safeguard and the public-sector computer centres.[87] The industrialists Naval H. Tata, Babubhai M. Chinai and B.D. Somani argued from the report's own figures that the effect on employment would be negligible, a couple of thousand people over a decade or more even if half the country's large companies installed computers, enough 'to dispel the exaggerated apprehensions entertained by some sections of the trade unions'. They too signed.[88]

How far the recommendations were applied is not clear from the record. Rajaraman, writing as a participant in the period, judges that the committee 'prescribed in 1972 strict controls' and that its report 'constrained the

introduction of computers in banks, insurance companies and many private companies'. He attributes its hostility partly to the 1401s themselves: small machines used for payroll, accounts and stores, which vocal white-collar workers saw as labour-saving devices.[89] In 1975 Parliament's Public Accounts Committee added its own caution: in a country with large-scale unemployment the use of computers for labour-saving work 'may not be desirable or may even be detrimental'.[90] The DoE had to move cautiously.[91]

Training the state's own

The state that restricted computers also trained people to use them, and much of the education of these years ran through its institutions rather than through universities. IBM had trained the first generation of programmers and maintenance engineers on its rented 1401s, ECIL its thousand systems engineers, and IIT Kanpur ran short programming courses for people from other institutions until 1975.[92]

The DoE's largest training venture was the National Centre for Software Development and Computing Techniques (NCSDCT) at TIFR, set up in the mid-1970s with money from the DoE and UNDP and directed by Narasimhan from 1975 to 1985.[93] Narasimhan saw the training of enough software engineers as a major challenge. 'Formal educational channels are wholly inadequate to cope with a demand of this magnitude,' he wrote. NCSDCT's answer was a one-year evening course in computer science, held at the Victoria Jubilee Technical Institute in central Bombay and linked by telephone lines to the PDP-10 computer at TIFR. Preference went to people already working in software, employers were persuaded to pay, entry was by a fairly rigorous test, and classes took 60 to 80 students at a time, taught largely through self-study, assignments and imported textbooks.[94]

The DoE built other institutions too. The National Informatics Centre, founded in 1975, set up a large computer centre in New Delhi for central government data processing.[95] Regional Computer Centres followed: at Poona in 1974, with an ICL mainframe offering cheap computer time to students and software exporters; at Jadavpur University in Calcutta (Kolkata since 2001) in 1977, where students in eastern India had free access to a Burroughs mainframe; and at Chandigarh.[96] In 1976 the DoE created the Computer Maintenance Corporation (CMC) to maintain imported computers and, foreseeing IBM's departure, the machines IBM would leave behind.[97] These bodies employed engineers and programmers on government terms. Of NCSDCT's successor, Rajaraman notes that a core group of its scientists 'later became leaders in the profession'.[98]

A small door marked export

One DoE decision of these years mattered more than its authors could have known. Rajaraman records that, 'recognizing the potential of software export', the department allowed computers to be imported exclusively for software export, and that the first policy paper on this was published in 1974.[99] It was the idea Rajaraman had put to the 1970 conference, turned into an import rule: a computer could be justified if it earned foreign exchange by producing software for foreign clients. TCS, the Tata group's computer services unit, used the rule to import a mainframe and earn foreign exchange from programming work done for clients abroad.[100]

What matters here is the mechanism. In a decade when almost every computer import was suspect, the one sure justification was export earnings, and the export that India could make was programming. The machine was to be paid for with the time of the people who programmed it. That is the first form of the bargain this book follows.

Four explanations

The record agrees on what happened between 1962 and 1977. It disagrees on why India did not build a computer industry. Rajaraman's verdict is that 'as a poor country with a low demand for computers it was strategically incorrect to try to design computers starting at the component level during 1970-1980 and hope to be completely "self-reliant"'.[101] Narasimhan had argued that building second-generation computers required production know-how that only large-scale manufacturers had or could command.[102] The historian Dwaipayan Banerjee's 2026 history of Indian computing, as its publisher and reviewers describe it, blames Cold War politics, trade imbalances, the monopolies of American corporations and the technocratic elite's preference for profitable technical services; one reviewer, Nikhil Menon, sums up the outcome as 'reliance, not self-reliance'.[103] The Committee on Automation's records add a fourth force. Organised clerical labour made the computer politically costly in exactly the sectors, insurance, banking and the railways, that would later be among its largest users.[104] These explanations do not

exclude one another. The state was at once an obstacle and a builder, and both roles appear in the same documents.[105]

Coda: 1977

In 1975 Indira Gandhi declared a state of emergency after a court ruled her election invalid; civil rights were suspended and opposition leaders arrested. The emergency ended in 1977, and in the general election that followed the Congress party lost power for the first time since independence. A coalition led by the Janata Party formed the government, with Morarji Desai as prime minister.[106] Ten years earlier, as Finance Minister, Desai had told the Lok Sabha of LIC's computer: 'There is going to be no retrenchment on this account.'[107]

The change reached computer policy quickly. M.G.K. Menon was replaced as secretary of the DoE, and in 1978 the government at last announced a minicomputer policy.[108] The same year IBM left India.[109]

The self-reliance programme had produced fewer than a hundred commercial computers and more than a thousand trained engineers. The labour settlement had made the office computer something a firm had to justify to its workers. And the import rules had left one opening. In the summer of 1970, when the Committee on Automation took evidence in Bombay, one of the delegations in front of it came from a small Tata unit that called itself a consultancy, and among its members was F.C. Kohli.[110] The clerks' unions had made the computer hard to justify at home. Kohli's unit would justify it by selling the work abroad.

Notes to Chapter 3. Self-reliance and its limits (1962-1977)

[1] Ministry of Labour and Rehabilitation, *Report of the Committee on Automation* (1972), pp. 3, 5.

[2] Ministry of Labour and Rehabilitation, *Report of the Committee on Automation* (1972), p. 5.

[3] Ministry of Labour and Rehabilitation, *Report of the Committee on Automation* (1972), p. 4.

[4] Ministry of Labour and Rehabilitation, *Report of the Committee on Automation* (1972), p. 5.

[5] Ministry of Labour and Rehabilitation, *Report of the Committee on Automation* (1972), pp. 5-6.

[6] Ministry of Labour and Rehabilitation, *Report of the Committee on Automation* (1972), p. 6.

[7] Ministry of Labour and Rehabilitation, *Report of the Committee on Automation* (1972), p. 6.

[8] Rajaraman, *History of Computing in India* (2012), pp. 15, 18.

[9] Rajaraman, *History of Computing in India* (2012), p. 18.

[10] Ministry of Labour and Rehabilitation, *Report of the Committee on Automation* (1972), p. 33.

[11] Ministry of Labour and Rehabilitation, *Report of the Committee on Automation* (1972), p. 33.

[12] Rajaraman, *History of Computing in India* (2012), pp. 18-19.

[13] Ministry of Labour and Rehabilitation, *Report of the Committee on Automation* (1972), p. 33.

[14] Ministry of Labour and Rehabilitation, *Report of the Committee on Automation* (1972), p. 33.

[15] Rajaraman, *History of Computing in India* (2012), p. 19.

[16] Rajaraman, *History of Computing in India* (2012), p. 19.

[17] Rajaraman, *History of Computing in India* (2012), p. 19; Penn World Table 11.0, exchange rate series (1970 annual average, Rs 7.50 per dollar). Rajaraman converts the rental to $190,000 'at the prevailing exchange rate' (Rajaraman, *History of Computing in India* (2012), p. 19), which roughly matches the rate before the rupee was devalued in 1966 (Rajaraman, *History of Computing in India* (2012), p. 70; Penn World Table 11.0, exchange rate series: 1965, Rs 4.76 per dollar).

[18] Rajaraman, *History of Computing in India* (2012), p. 70.

[19] Rajaraman, *History of Computing in India* (2012), p. 19.

[20] Rajaraman, *History of Computing in India* (2012), p. 19.

[21] Ministry of Labour and Rehabilitation, *Report of the Committee on Automation* (1972), p. 34.

[22] Rajaraman, *History of Computing in India* (2012), p. 21.

[23] Rajaraman, *History of Computing in India* (2012), p. 21.

[24] Gopalakrishnan, Dayasindhu and Narayanan, 'When India Made Its First Computer' (2022) (Kris Gopalakrishnan, N. Dayasindhu and Krishnan Narayanan; web extract, unpaginated).

[25] Rajaraman, *History of Computing in India* (2012), p. 21.

[26] Narasimhan, 'Men, Machines and Ideas' (1999), p. 450. Narasimhan's earlier work on TIFRAC is described in Chapter 1.

[27] Rajaraman, *History of Computing in India* (2012), p. 21.

[28] Rajaraman, *History of Computing in India* (2012), p. 24.

[29] Rajaraman, *History of Computing in India* (2012), p. 21.

[30] Rajaraman, *History of Computing in India* (2012), pp. 21-22; Bassett, 'Aligning India in the Cold War Era' (2009), p. 792.

[31] Rajaraman, *History of Computing in India* (2012), p. 22.

[32] Rajaraman, *History of Computing in India* (2012), p. 22.

[33] Rajaraman, *History of Computing in India* (2012), pp. 24, 70; Gopalakrishnan, Dayasindhu and Narayanan, 'When India Made Its First Computer' (2022).

[34] Narasimhan, 'Men, Machines and Ideas' (1999), p. 450.

[35] Rajaraman, *History of Computing in India* (2012), p. 23.

[36] Rajaraman, *History of Computing in India* (2012), p. 22.

[37] Rajaraman, *History of Computing in India* (2012), p. 23.

[38] Rajaraman, *History of Computing in India* (2012), p. 23.

[39] Rajaraman, *History of Computing in India* (2012), p. 23.

[40] Rajaraman, *History of Computing in India* (2012), p. 24. A DoE panel on minicomputers, chaired by P.V.S. Rao of TIFR, reported in 1973 that minicomputers could be made in India without foreign collaboration and projected demand for about 1,400 of them between 1974 and 1979 (Rajaraman, *History of Computing in India* (2012), pp. 24, 71). The policy itself, announced in 1978, is described in Chapter 5.

[41] Rajaraman, *History of Computing in India* (2012), p. 27.

[42] Rajaraman, *History of Computing in India* (2012), pp. 25, 70.

[43] Rajaraman, *History of Computing in India* (2012), p. 25.

[44] Rajaraman, *History of Computing in India* (2012), p. 25.

[45] Rajaraman, *History of Computing in India* (2012), p. 25.

[46] Rajaraman, *History of Computing in India* (2012), p. 25; Federal Reserve, G.5A Foreign Exchange Rates (1975 annual average, Rs 8.41 per dollar).

[47] Rajaraman, *History of Computing in India* (2012), pp. 25-26. ECIL's 98 sales were 21 TDC-12s, 32 TDC-312s, 35 TDC-316s and 10 Micro 78s; a TDC-12 typically cost Rs 600,000 and a TDC-312 Rs 1 million (Rajaraman, *History of Computing in India* (2012), p. 25). ECIL's share of the installed base in 1974 (11 per cent) is given in Chapter 2 (Bassett, 'Aligning India in the Cold War Era' (2009), p. 800).

[48] Rajaraman, *History of Computing in India* (2012), p. 26. The work at TIFR was led by P.V.S. Rao. ECIL machines also logged data for atomic energy, monitored satellite launches, tested gas turbines and controlled the cutting of billets in steel mills, all places where export controls made imports difficult; the DoE gave ECIL grants and loans of Rs 37.44 million (Rajaraman, *History of Computing in India* (2012), p. 26).

[49] Rajaraman, *History of Computing in India* (2012), p. 25.

[50] Rajaraman, *History of Computing in India* (2012), p. 26.

[51] Rajaraman, *History of Computing in India* (2012), p. 26. No private Indian company was then allowed to make computers (Rajaraman, *History of Computing in India* (2012), p. 27). What ECIL's engineers were paid is not recorded in the published histories.

[52] Ministry of Labour and Rehabilitation, *Report of the Committee on Automation* (1972), pp. 1-2.

[53] Ministry of Labour and Rehabilitation, *Report of the Committee on Automation* (1972), p. 2.

[54] Ministry of Labour and Rehabilitation, *Report of the Committee on Automation* (1972), p. 2; pp. 62-63. The OCR text of the committee's report garbles the year of LIC's decision; elsewhere the committee says that until 1965 computers had aroused little opposition, and that serious opposition developed 'since 1965' after a big public-sector undertaking decided to install a computer in one of its offices without consulting its workers. The Lok Sabha record dates the decision to 1964 and the first installation to November 1965 (*Lok Sabha Debates*, 24 November 1966, col. 5287). A short biographical note on Ajoy Dasgupta, a leader of the insurance employees, dates the insurance workers' 'struggles against computerisation' earlier, to 1959-61, and says that 'the computers were installed step by step and the agitation died down' (Indian Labour Archives, 'Ajoy Dasgupta', p. 2). The two dates may describe two waves of protest, one against earlier office machinery and one against the computer itself; the record does not settle which.

[55] Ministry of Labour and Rehabilitation, *Report of the Committee on Automation* (1972), pp. 2-3. Bassett records that Indian unions called computers 'job eaters' (Bassett, 'Aligning India in the Cold War Era' (2009), p. 799; see Chapter 2).

[56] Ministry of Labour and Rehabilitation, *Report of the Committee on Automation* (1972), p. 3.

[57] *Lok Sabha Debates*, 24 November 1966, cols. 5286-88 (Lok Sabha Debates, Third Series, 24 November 1966, calling-attention statement on the proposed strike by LIC employees; read in the printed record via the archive.org mirror of the Parliament Digital Library, 2026-10-15). The quotation is at cols. 5287-88; the strike, called by the All-India Insurance Employees' Association for 25 November, is at col. 5286; the comparison with 'the 1500 or so new jobs created in the L.I.C. every year' and the assurance that none of the employees whose jobs became redundant would be retrenched are at col. 5288.

[58] *Lok Sabha Debates*, 23 March 1967, cols. 675-76 (Lok Sabha Debates, Fourth Series, 23 March 1967, oral question *40, 'Computers in LIC'). Pant gave the Bombay system's purchase price as about Rs 42 lakh (Rs 4.2 million), excluding import charges and sales tax, paid for under a 'blocked rupee' arrangement; a second computer, from International Computers and Tabulators of Britain, was contracted for Calcutta. His answer to the question how many employees had lost their jobs was 'None'.

[59] *Lok Sabha Debates*, 23 March 1967, col. 678. The question was put by Bal Raj Madhok.

[60] Ministry of Labour and Rehabilitation, *Report of the Committee on Automation* (1972), pp. 3-4.

[61] Ministry of Labour and Rehabilitation, *Report of the Committee on Automation* (1972), p. 4.

[62] Ministry of Labour and Rehabilitation, *Report of the Committee on Automation* (1972), p. 4.

[63] Ministry of Labour and Rehabilitation, *Report of the Committee on Automation* (1972), pp. 24-25.

[64] Ministry of Labour and Rehabilitation, *Report of the Committee on Automation* (1972), p. 25.

[65] Ministry of Labour and Rehabilitation, *Report of the Committee on Automation* (1972), p. 37.

[66] Ministry of Labour and Rehabilitation, *Report of the Committee on Automation* (1972), p. 30.

[67] Ministry of Labour and Rehabilitation, *Report of the Committee on Automation* (1972), p. 29.

[68] Ministry of Labour and Rehabilitation, *Report of the Committee on Automation* (1972), p. 37.

[69] Ministry of Labour and Rehabilitation, *Report of the Committee on Automation* (1972), p. 40.

[70] Ministry of Labour and Rehabilitation, *Report of the Committee on Automation* (1972), pp. 52, 55.

[71] Ministry of Labour and Rehabilitation, *Report of the Committee on Automation* (1972), p. 56.

[72] Ministry of Labour and Rehabilitation, *Report of the Committee on Automation* (1972), pp. 52-53.

[73] Ministry of Labour and Rehabilitation, *Report of the Committee on Automation* (1972), pp. 53, 160.

[74] Ministry of Labour and Rehabilitation, *Report of the Committee on Automation* (1972), p. 54.

[75] Ministry of Labour and Rehabilitation, *Report of the Committee on Automation* (1972), pp. 63-64.

[76] Ministry of Labour and Rehabilitation, *Report of the Committee on Automation* (1972), p. 65.

[77] Ministry of Labour and Rehabilitation, *Report of the Committee on Automation* (1972), pp. 7, 89. Venkataraman resigned on 22 March 1971 and Dandekar, director of the Gokhale Institute of Economics and Politics in Poona, became chairman on 24 July 1971 (Ministry of Labour and Rehabilitation, *Report of the Committee on Automation* (1972), p. 7); Rajaraman therefore calls it the Dandekar Committee and notes that it was set up under Venkataraman (Rajaraman, *History of Computing in India* (2012), p. 22). The members included G. Ramanujam and Satish Loomba, the industrialists Naval H. Tata, B.D. Somani and Babubhai M. Chinai, the economist V.M. Dandekar and V.R. Rao of the Computer Centre in Delhi. Its witnesses included LIC's management, the State Bank of India, the railways, IBM, ICL, the Computer Society of India, the main union federations and three representatives of the All India Insurance Employees' Association (Ministry of Labour and Rehabilitation, *Report of the Committee on Automation* (1972), pp. 136-140).

[78] Ministry of Labour and Rehabilitation, *Report of the Committee on Automation* (1972), p. 35. Since planning and installing a computer took about two years and full use two or three more, the committee thought most installations were still in their early phases (Ministry of Labour and Rehabilitation, *Report of the Committee on Automation* (1972), pp. 36-37).

[79] Ministry of Labour and Rehabilitation, *Report of the Committee on Automation* (1972), p. 75.

[80] Ministry of Labour and Rehabilitation, *Report of the Committee on Automation* (1972), p. 67.

[81] Ministry of Labour and Rehabilitation, *Report of the Committee on Automation* (1972), pp. 67-68.

[82] Ministry of Labour and Rehabilitation, *Report of the Committee on Automation* (1972), p. 69.

[83] Ministry of Labour and Rehabilitation, *Report of the Committee on Automation* (1972), p. 70.

[84] Ministry of Labour and Rehabilitation, *Report of the Committee on Automation* (1972), pp. 70-72.

[85] Ministry of Labour and Rehabilitation, *Report of the Committee on Automation* (1972), pp. 73-74.

[86] Ministry of Labour and Rehabilitation, *Report of the Committee on Automation* (1972), pp. 79-80.

[87] Ministry of Labour and Rehabilitation, *Report of the Committee on Automation* (1972), pp. 81-82.

[88] Ministry of Labour and Rehabilitation, *Report of the Committee on Automation* (1972), pp. 83-85. Their arithmetic: if half of India's 445 companies with paid-up capital of Rs 10 million or more installed computers (129 more than the 94 already in place), and each new computer affected about 16 employees, some 2,064 people would be affected over a decade or more.

[89] Rajaraman, *History of Computing in India* (2012), pp. 22-23. No record of how far the committee's recommendations were applied in practice has been located.

[90] Rajaraman, *History of Computing in India* (2012), p. 23.

[91] Rajaraman, *History of Computing in India* (2012), p. 23.

[92] Rajaraman, *History of Computing in India* (2012), p. 19 (IBM); p. 26 (ECIL); Rajaraman, *History of Computing in India* (2012), p. 18 (Kanpur, three times a year), Rajaraman and Joshi, 'Professor Harry Huskey: Planting a Tree' (2021) (four times a year); see Chapter 2.

[93] Narasimhan, 'Men, Machines and Ideas' (1999), pp. 450-451; Rajaraman, *History of Computing in India* (2012), p. 27. The founding date differs between the accounts. Rajaraman dates the centre to 1972, with Rs 2.85 million from the DoE and $2 million from UNDP (Rajaraman, *History of Computing in India* (2012), pp. 27, 71). Narasimhan says a UNDP mission in May 1973 recommended placing it at TIFR, which had run the National Computation Centre on its CDC 3600 since 1964; the centre began working informally in late 1974 and was formally launched in 1975, when a DEC 1077 system was installed (Narasimhan, 'Men, Machines and Ideas' (1999), p. 450). At full strength it had a research staff of about 30 (Narasimhan, 'Men, Machines and Ideas' (1999), p. 451). Narasimhan's account does not give the year of the first evening course.

[94] Narasimhan, 'Men, Machines and Ideas' (1999), p. 451.

[95] Rajaraman, *History of Computing in India* (2012), p. 27 (Rs 31.72 million from the DoE and $4.4 million from UNDP).

[96] Rajaraman, *History of Computing in India* (2012), pp. 28, 71.

[97] Rajaraman, *History of Computing in India* (2012), p. 28 (at a cost of Rs 53 million); Narasimhan, 'Men, Machines and Ideas' (1999), p. 450.

[98] Rajaraman, *History of Computing in India* (2012), p. 27.

[99] Rajaraman, *History of Computing in India* (2012), p. 28.

[100] Rajaraman, *History of Computing in India* (2012), p. 28. Chapter 4 tells the story.

[101] Rajaraman, *History of Computing in India* (2012), p. 59.

[102] Narasimhan, 'Men, Machines and Ideas' (1999), p. 449.

[103] Banerjee, *Computing in the Age of Decolonization* (2026) (publisher's description); Dizikes, 'The Tech Revolution That Wasn't' (2026) (MIT News report, May 2026); Farrell, review of Banerjee, *Computing in the Age of Decolonization* (2026); Saini, 'Why Didn't India Build Its Own Silicon Valley?' (2026); Menon, 'India's Pursuit of Tech Self-Reliance' (2026), pp. 98-100 (Menon, review in *Issues in Science and Technology* 42, no. 4, Summer 2026). The summary follows the publisher's description and the reviews, not the book itself. Menon's fuller sentence is his own: 'India locked itself into cheap, skilled, backend service provision: reliance, not self-reliance.' Post-cutoff material, checked against the live pages 2026-10-13 and 2026-10-15.

[104] Ministry of Labour and Rehabilitation, *Report of the Committee on Automation* (1972), pp. 2-4, 63.

[105] Rajaraman, *History of Computing in India* (2012), pp. 27-28.

[106] Rajaraman, *History of Computing in India* (2012), p. 82.

[107] *Lok Sabha Debates*, 23 March 1967, col. 678.

[108] Rajaraman, *History of Computing in India* (2012), p. 24. The policy is described in Chapter 5.

[109] Rajaraman, *History of Computing in India* (2012), p. 24.

[110] Ministry of Labour and Rehabilitation, *Report of the Committee on Automation* (1972), pp. 136-139.

Chapter 4. Tata's experiment (1968-1978)

In January 1975 the Computer Society of India held its annual convention at Ahmedabad. Its president that year was F.C. Kohli, a power engineer who ran Tata Consultancy Services (TCS), a small computer-services company in Bombay (Mumbai since 1995) owned by the Tata group. In his presidential address, as he quoted it himself more than thirty years later, Kohli told the delegates: 'Many years ago, there was an industrial revolution. We missed it due to reasons over which we had no control.'[1] The new revolution, in information technology, was one 'which requires the capability to think clearly. This we have in abundance.' He went on: 'We have an opportunity to participate in this revolution on an equal footing; we even have the opportunity to be leaders.'[2]

It was a large claim to make in a country with a few hundred computers, most of them IBM's.[3] Getting permission to import another one could take between one and three years.[4] The company Kohli ran had begun as a bureau doing bank reconciliation, payroll and accounts for Tata companies and other Indian firms, and by its own later account its revenue passed $1 million in 1976.[5]

In the decade behind that speech, a data-processing unit set up to serve one business group found something it could sell abroad: the work of Indian programmers, done partly in Bombay on machines the company had to fight to import, and partly on the premises of American clients. A second, smaller venture sent American documents to Poona (now Pune) to be typed onto tape. Neither firm set out to create an export industry. The rules on foreign exchange, the scarcity of computers and the shape of the American market pushed both of them towards one.

Half-hour slots

The people who did the early work are known mainly through the memories of men who later ran the companies, but the memories are specific. S. Ramadorai joined TCS in 1972 and became its chief executive in 1996. In 2008 *The Economist*, drawing on him and other long-serving staff, described the computers of his early years as bulky and temperamental, in need of assembly, installation and constant repair. 'You had to work with your own hands,' a colleague recalled.[6] Jayant Pendharkar, later the company's chief marketing officer, remembered a new computer arriving in three truckloads.[7] Programmers booked half-hour slots on the Burroughs terminals, and the magazine adds that 'One hour's computer time cost more than a programmer earned in a month.'[8] Seturaman Mahalingam, later the chief financial officer, worked on an early offshore project in Feltham, outside London, and posted his instructions to the programmers in Bombay twice, a day apart, in case one postbag went astray. A telephone call from London to Bombay, the magazine adds, would be interrupted every few minutes by the operator, checking that the caller wanted to keep talking.[9]

In Cambridge, Massachusetts, in 1972, Narendra Patni and his wife, Poonam, tested a business of their own in their apartment. Dinesh Sharma, a journalist and historian of the industry, writes that they made the living room stand for the United States and the bedroom for India.[10] At the end of the decade Nandan Nilekani, as he recalled for the Computer History Museum, was interviewed with puzzles by N.R. Narayana Murthy in a small cabin at Nariman Point in Bombay and hired on the spot for Patni's software division. The firm's Data General computer was often idle at night, 'and for those of us who had grown up sending these cards, to have a terminal where you could do whatever you wanted was unbelievable luxury.' He worked all night on it, learning, and part of the job was teaching the customers who had bought the machine.[11]

The magazine's comparison of an hour of computer time with a month of a programmer's pay holds the economics of the decade in one line. The machine was dear and the programmer was cheap. The business that grew out of these years would be built on the difference.

A bureau for the group

TCS began as a computer centre for the Tata group's own companies. The likeliest reading of the record is that the centre existed, with an IBM 1401, by late 1967, and that the company built on it was constituted as Tata Consultancy Services in 1968.[12]

The founding staff came from management schools and statistics more than from engineering. Sharma names three graduates of MIT's management school who had worked as interns with the Tatas, Lalit Surajmal Kanodia, Nitin Patel and Ashok Malhotra, and a fourth founding member, Yashpal Sahni, a postgraduate of the Indian Statistical Institute.[13] The Tata group's tribute to Kohli says that he recruited the most promising engineers, chartered accountants, and science and mathematics postgraduates.[14] Whatever the industry later became, its first company was not staffed only by engineers, still less only by graduates of the Indian Institutes of Technology (IITs).

The work was the work of a service bureau: bank reconciliation, payroll and accounting.[15] A profile of Kohli in *MIT Technology Review*, based on interviews with him, adds two jobs: reconciling transactions between the branches of a nationalised bank, and producing Bombay's telephone directory.[16] These were the standard uses of a commercial computer in India at the time. The Committee on Automation, appointed by the Ministry of Labour to study what computers would do to jobs, noted that one of the important uses of the State Bank of India's computer was to reconcile inter-branch transactions, about 70,000 of them a day.[17] Most Tata companies already rented IBM 1401s, so a Tata bureau that sold computing to Tata companies was doing, inside the group, what IBM did for its clients.[18]

The control engineer

Faqir Chand Kohli came to computers through electric power. Born in Peshawar in 1924, he took a physics degree at Government College, Lahore, and in 1946, 'before the partition', as he put it, went on a scholarship to Queen's University in Canada to study electrical engineering.[19] He worked for about a year at General Electric, took a master's degree at MIT on a scholarship, and learned load dispatch, the scheduling of power stations' output across a grid, at the New England Power Company.[20] In 1951 he came back to India and joined Tata Electric in Bombay, which wanted to set up what he called 'the first load dispatch in India'.[21]

Ross Bassett, a historian of technology who interviewed Kohli, describes him as an MIT-trained control engineer, a specialist in the regulation of systems, which was one of the fields from which computing grew.[22] Kohli's ties to IIT Kanpur gave him his practical introduction to computing, and he brought digital computers into Tata Electric's system control in the late 1960s.[23] In 1973 he became the first Indian to sit on the board of directors of the IEEE, the international but American-oriented society of electrical engineers.[24]

The Tata group's chairman, J.R.D. Tata, asked him to take on the new computer company.[25] In 1969 he moved from Tata Electric to TCS as general manager, succeeding Kanodia.[26] In 1974, according to the *Technology Review* profile, the director of TCS died and Kohli succeeded him.[27] He stayed for three decades. In 2020 he corrected an interviewer who dated his retirement to 1996: he had worked until 1999, when he was seventy-five.[28]

Machines on a budget

The company's first machine was a rented IBM 1401, in place from September 1967. The first it bought, on the likeliest reading, was an ICL 1903, made by the British firm International Computers, in use from 1971.[29]

How it bought the ICL is the best story in the *Technology Review* profile, and it rests on that profile, which drew on Kohli's own account. According to the profile, the machine, never used, had belonged to a government-owned insurance company where a labour union had blocked its installation, calling it a monster that would eat jobs. TCS bought it for a fraction of its original price.[30] If the story is accurate, it connects the two halves of the Indian argument about computers. A union's veto in one public enterprise produced a cheap computer for a private firm, which would come to use machines like it for work that displaced no Indian clerks, because the clients were abroad.

New machines were harder to get than old ones. Every import needed a licence from the Department of Electronics (DoE), and the delay, V. Rajaraman, the historian of Indian computing, writes, 'particularly hurt companies which wanted to import computers for software export.'[31] Old machines had been the way round the problem. Sharma writes that IBM and ICL had imported used computers until an audit by the Comptroller and Auditor General led the government to rein them in and give the DoE control of the matter.[32]

Up, but not running

The domestic market could not keep the company going. 'TCS was up, but it wasn't running,' Kohli told the *Technology Review*. The company struggled, and took on management-consulting work to stay afloat.[33]

Near the end of his life Kohli told a simpler story. In an interview for the Computer History Museum in January 2020, when he was ninety-five, he said: 'First few years it was built only on exports. People didn't understand.'[34] Asked about strategy, he said that he had built strong links with the engineering colleges, starting with IIT Kanpur, so that people knew about the company: 'There was no question of my going for business. The people came to us, "Could you do this for us?"'[35] The better evidence points the other way. The company's own histories list domestic bureau work as its first business,[36] and Sharma found that Kohli, courting Burroughs, visited the United States at least nine times between 1972 and 1974.[37]

A second witness, from outside TCS, confirms how thin the Indian market was. Murthy, who later founded Infosys, had worked for a French company in Paris in the early 1970s. Within a couple of years of his return he started a firm to write software for Indian customers. Within seven or eight months, he told the Computer History Museum, he realised 'that there was no domestic market in India for software. There were hardly any computers in India'. He closed the firm and set out to learn whether his market would have to be global, 'a market particularly in the United States'.[38] His is a founder's retrospective, but it comes from a different firm and points the same way.

Burroughs

TCS tied its fortunes to Burroughs, a computer maker based in Detroit and, in Kohli's words, the 'number two computer company at that time'.[39] TCS became Burroughs' exclusive agent in India, in a relationship formed in 1973-74.[40]

In the account Kohli gave the *Technology Review*, Burroughs wanted a healthcare software package to sell with its new computers, and TCS had no Burroughs machine in India to write it on. Its programmers wrote the software on the ICL and built a filter, a program that converted it to run on Burroughs hardware. The job made Burroughs a large repeat client.[41] Other accounts give other first jobs: Sharma describes operating-system work for a new series of Burroughs mainframes, and Rajaraman the conversion of software written for other makers' computers.[42] The Tata group's own timeline counts the conversion of a hospital-accounting system for a Detroit client in 1973 as the first recorded case of offshore software delivery, which is the company's claim about itself.[43] Sharma lists the clients who followed as the Detroit Police, the City of Detroit, the State University of New York and several banks.[44]

Whichever account is closest, the work had a pattern that would last. Programmers in Bombay wrote code for machines they could not see, and the code was carried to the United States to run. Software was the one product a computer-starved country could make without many computers of its own, so long as it had one machine to write on and a way to move the result.

The partnership spread beyond services. In 1975 Tata and Burroughs formed a company to manufacture computers in Bombay. They also set up a separate software unit, Tata Burroughs Limited, which, with TCS, remained India's top software exporter for a long time.[45] TCS also looked well beyond Detroit. The company lists a project for an electric utility in Iran in 1971 and one for a British housing society, done for Burroughs, in 1974.[46] Rajaraman says TCS 'actively scouted for programming work' in Iran, Switzerland, South Africa and Canada as well as the United States, and that it 'was able to demonstrate that software services export from India was feasible and profitable'.[47] In 1976, according to the Tata group, Kohli received an award as a leading exporter from the Engineering Export Promotion Council.[48] The Tata group's tribute to Kohli adds that customers valued the company's quality of service at a low cost, the more so after the 1973 oil crisis.[49] That is the company's own statement of the cost advantage on which the industry would be built.

Programmers on site

Not all the work could be done in Bombay. William Aspray, an American historian of computing, writes that Burroughs asked TCS to send programmers to the United States to install systems software for Burroughs customers, and he treats that request as the birth of India's software export business.[50] A manufacturer installing machines at its customers' premises needed programmers there, and TCS supplied them. The practice came to be called, derisively, 'body shopping', and once TCS had succeeded, other Indian firms began to offer the same service.[51]

How TCS charged Burroughs or its American clients in the 1970s, by the person-month, the hour or a fixed price for each job, is not shown by any contract or rate card that has come to light.[52] The conversion and package work suggests fixed deliverables; the installation work on site suggests time-based billing. So it cannot be said that billing by the hour began here. A review of Ramadorai's own history of the company says only that the arrangement with Burroughs let TCS do part of the development work in India and part at client locations abroad.[53] The industry's first exports combined two things: code written at home and people's time spent at a client's site. Chapter 6 follows how the second came to dominate.

Data by air

The second experiment was smaller, and in one way it went further. Narendra Patni had studied at MIT and, according to Bassett, worked with the computer pioneer and entrepreneur Jay Forrester, doing consulting work and taking part in his business ventures.[54] Patni formed Data Conversion Inc., the forerunner of his later Patni Computer Systems, to convert paper records into data on paper and magnetic tape.[55] After the test in the Cambridge apartment, a team of data-entry operators was hired in Poona, two dozen by Sharma's account, twenty by another.[56] The work was keying American documents onto tape in India and sending the tapes back. It was an early form of the offshore 'process' work that would fill the back offices of the 1990s, and it began two decades before them.

Patni then moved into software, forming Patni Computer Systems with his brothers Gajendra and Ashok, both graduates of IIT Delhi, in the late 1970s. It began by reselling minicomputers made by the American firm Data General and then wrote software for Data General's customers.[57] In August 1977 Patni hired Murthy to head its software division. By the time he arrived, he recalled in 2018, the Patnis had 'also started a business to develop software for export to the United States'.[58] The division's staff by the end of the decade included Nilekani and Kris Gopalakrishnan; in 1981 Murthy and six colleagues left to found Infosys.[59]

Who did the work

The profession at this stage was defined by the task, whether systems programming, conversion or data entry, rather than by a degree. TCS's recruits included chartered accountants and science and mathematics postgraduates as well as engineers; the operators in Poona needed typing, not engineering; and the people who joined Patni's software division at the end of the decade had a year or so of experience each.[60] Most of those in India who could already program had been trained by IBM, the public-sector Electronics Corporation of India, Kanpur's short courses or the state's software training centre in Bombay.[61] TCS drew on these pools and trained its own staff on the job. Kohli said in 2009 that the company had judged early on that technology became obsolete 'in less than 24 months', so that its staff would need continuing education.[62] His view of the raw material, as he put it in 2020, was simple: 'Indians are capable of doing a lot of things. And only thing is they have to receive some training.'[63]

Three readings

Three explanations of this decade can be drawn from the record, and they are not exclusive. The first is the founders' own. Kohli saw the potential of the technology, recruited the best people and, in 1975, told the profession what India could become. The Tata group's tributes tell the story this way, and so, mostly, did Kohli.[64] It is a self-description, and Kohli himself resisted its strongest form. When the Computer History Museum's interviewer called him the father of the industry, he replied: 'Father is a misnomer.' He was, he said, the father of his three children, and that was all. Then he added: 'I did set up the computer industry in this country, and that is that. People know it.'[65] The second reading, which Rajaraman's and Aspray's accounts support, is that the export industry was a by-product of import controls and foreign-exchange rules: firms exported because exports were the price of getting machines, and because there was too little demand at home.[66] The third is Bassett's. What mattered, he argues, was the pairing of an American-style engineering education, at Kanpur and elsewhere, with men such as Kohli and Patni who had American training and American contacts.[67]

The record bears on two claims that recur in histories of the industry. The first is that TCS was India's first software company. It began as a bureau selling computing to its own group, as IBM's data centres in the big cities already did for smaller firms. As a firm that exported software services it was among the first, with Patni's venture beside it.[68] The second is that the first offshore development centre in India was opened by Texas Instruments in

Bangalore (Bengaluru since 2014) in 1985 (Chapter 8). Software written in Bombay for Burroughs in 1973-74, and data keyed in Poona from 1972, show Indian firms doing work for foreign clients in India more than a decade earlier, though without satellite links and not in a centre owned by a foreign company.[69]

Twice the value

The rule that turned scarcity into an export business took shape in the early 1970s. The DoE allowed computers to be imported exclusively for software export. Rajaraman had told a conference in 1970 that foreign exchange could also be earned by software export.[70] Its working is clearest in TCS's own purchase of a Burroughs computer in 1974. The machine cost $340,000. With Indian tariffs TCS paid more than twice that.[71] *The Economist* itemised the tariff at 101.25 per cent: import duty, auxiliary duty, countervailing duty and a levy to help pay for the war in Bangladesh.[72] The government let the machine in on one condition. In the *Technology Review*'s words, 'Kohli had to commit to exporting twice the value of the machine over a five-year period.'[73]

On the machine's price alone, that meant at least $680,000 of earnings abroad in five years, or about $136,000 a year. The size of the commitment shows against the national figures. In 1977, by a contemporary West German estimate, the whole of India's software exports came to about Rs 10 million, roughly $1.1 million, and the firms named as reaching markets abroad were TCS and the Operations Research Group of Baroda (now Vadodara).[74] In 1980, Aspray writes, twenty-one Indian firms were exporting about $4 million a year, mostly by sending people to client sites.[75] One import licence could commit a single firm to earning abroad, each year, more than a tenth of the whole country's software exports in 1977.

A firm in Bombay had few ways to earn that money. It could not sell computers, which it could barely import. It could sell the work of its staff, as code written in Bombay or as programmers sent to the client. The machine was the dear part of the business and the programmer the cheap one, and the obligation could be met only by selling the cheap part abroad. Each new computer had to be paid for with engineers' work sold abroad, and each addition to the stock of machines meant more people whose time would pay for them. This is the bargain the rest of the book follows, written down for the first time in an import licence: a computer priced in the future labour of Indian programmers.

The rule was not watertight. Aspray concludes that by 1980 most of the hardware imported into India was being used for domestic purposes, not for producing software exports.[76] But for TCS, and for the firms that copied it, the condition fixed the shape of the business.

1978

The decade closed with a break. The Tata group dates the end of TCS's partnership with Burroughs to 1978.[77] The reviewer Rashmi Bansal writes that Ramadorai's history of the company presents that year as the gravest test TCS had faced, because Tata Sons formed a formal joint venture with Burroughs and left TCS to fend for itself.[78] The same year IBM, which had installed most of India's computers and trained many of its programmers, left the country, and the government at last let private Indian companies make minicomputers (Chapter 5).[79] India had fewer than 1,000 computers.[80] In 1979 TCS sent Ramadorai to the United States to open its first overseas office. Free now to work with other computer makers, it had to find its own customers.[81] What it offered them was what the export licence had already taught it to sell: the time of its programmers.

Notes to Chapter 4. Tata's experiment (1968-1978)

[1] Kohli, *Towards Inclusive Information Technology Revolution in India* (2009), PDF p. 4.

[2] Kohli, *Towards Inclusive Information Technology Revolution in India* (2009), PDF p. 4. No text of the 1975 address has been located. The wording here is Kohli's own quotation of it in a speech of 2008-09 (Kohli, *Towards Inclusive Information Technology Revolution in India* (2009)); later writers' versions differ. Dinesh Sharma retells the speech in his obituary of Kohli, 'F.C. Kohli and the Eureka Moment of India's Software Industry', though the 'Eureka moment' of his title is TCS's breakthrough with Burroughs in 1974, not the address (Sharma, 'F.C. Kohli and the Eureka Moment' (2020)); a Tata group tribute of 2024 quotes yet another wording (Tata Group, 'The IT Revolutionary' (2024)).

[3] Bassett, 'Aligning India in the Cold War Era' (2009), p. 800.

[4] Rajaraman, *History of Computing in India* (2012), p. 23.

[5] Tata Group, TCS timeline and 'Five Decades of Transformation' (2021).

[6] *The Economist*, 'Ram Drive' (2008), para. 1 ('Ram drive', *The Economist*, 21 August 2008, course-page mirror); para. 6 (chief executive in 1996).

[7] *The Economist*, 'Ram Drive' (2008), para. 3. The paragraph says 'the computer'; the preceding paragraph concerns the Burroughs machine TCS bought in 1974.

[8] *The Economist*, 'Ram Drive' (2008), para. 5.

[9] *The Economist*, 'Ram Drive' (2008), paras 4-5.

[10] Sharma, 'When Patni Experienced the Eureka Moment' (2011) (Sharma). The living-room test rests on Sharma's account alone; a *Business Standard* profile of 2007 confirms only that the Patnis began from their apartment in 1972 (*Business Standard*, 'The Original IT Man' (2007)).

[11] Computer History Museum, 'Oral History of Nandan Nilekani' (2020), p. 7 (Nilekani, Computer History Museum oral history; title page dated 8 January 2019; CHM reference X9234.2020). He joined Patni in January 1979, and part of his job was teaching the customers who had bought the machine.

[12] Sharma, 'F.C. Kohli and the Eureka Moment' (2020); Bassett, 'Aligning India in the Cold War Era' (2009), p. 800; Tata Group, TCS timeline and 'Five Decades of Transformation' (2021); Tata Group, 'The IT Revolutionary' (2024); Kohli, *Towards Inclusive Information Technology Revolution in India* (2009), PDF p. 4; Ministry of Labour and Rehabilitation, *Report of the Committee on Automation* (1972), App. IX, p. 145. The founding is dated differently. Sharma writes that the unit was established in 1967 as the Tata Computer Centre, to handle the data processing of all the Tata companies (Sharma, 'F.C. Kohli and the Eureka Moment' (2020)). Bassett says the group 'established a computing enterprise in 1967', which by 1969 was called Tata Consultancy Services (Bassett, 'Aligning India in the Cold War Era' (2009), p. 800). The Tata group's own histories give 1968 for the company, though one dates the group's shared computer centre to 1966 (Tata Group, TCS timeline and 'Five Decades of Transformation' (2021); Tata Group, 'The IT Revolutionary' (2024)). Kohli said in 2009: 'TCS was set up in 1968 to partake in the new technology' (Kohli, *Towards Inclusive Information Technology Revolution in India* (2009)). The Committee on Automation's list of computers in India as of August 1971 has two Bombay entries for 'Tata Computer Centre Consultancy Services': an IBM 1401 installed in September 1967 and an ICL 1903 installed in 1971 (Ministry of Labour and Rehabilitation, *Report of the Committee on Automation* (1972), App. IX, p. 145). On this reading both dates are right about different things.

[13] Sharma, 'F.C. Kohli and the Eureka Moment' (2020).

[14] Tata Group, 'The IT Revolutionary' (2024).

[15] Tata Group, TCS timeline and 'Five Decades of Transformation' (2021); Tata Group, 'The IT Revolutionary' (2024).

[16] Venkatraman, 'India's IT Guy' (2013).

[17] Ministry of Labour and Rehabilitation, *Report of the Committee on Automation* (1972), p. 29.

[18] Rajaraman, *History of Computing in India* (2012), pp. 18-19. IBM's rental business is described in Chapter 3.

[19] Computer History Museum, 'Oral History of FC Kohli' (2020), pp. 2-3.

[20] Computer History Museum, 'Oral History of FC Kohli' (2020), p. 3.

[21] Computer History Museum, 'Oral History of FC Kohli' (2020), p. 3.

[22] Bassett, 'Aligning India in the Cold War Era' (2009), p. 791.

[23] Bassett, 'Aligning India in the Cold War Era' (2009), pp. 800-801 (Kanpur; see Chapter 2); Kohli, *Towards Inclusive Information Technology Revolution in India* (2009), PDF p. 3 (Kohli dates Tata Electric's digital computers for system control and load dispatching to 1968); Computer History Museum, 'Oral History of FC Kohli' (2020), p. 3 (in 2020 he gave 1969).

[24] Bassett, 'Aligning India in the Cold War Era' (2009), p. 801 n. 43.

[25] Computer History Museum, 'Oral History of FC Kohli' (2020), p. 3.

[26] Sharma, 'F.C. Kohli and the Eureka Moment' (2020); Venkatraman, 'India's IT Guy' (2013); Bassett, 'Aligning India in the Cold War Era' (2009), p. 800. All sources agree on 1969; Sharma says he succeeded Kanodia (Sharma, 'F.C. Kohli and the Eureka Moment' (2020)).

[27] Venkatraman, 'India's IT Guy' (2013). The profile does not name the director, and no other source records the 1974 succession.

[28] Bassett, 'Aligning India in the Cold War Era' (2009), p. 800 (Bassett: Kohli led TCS for decades); Kohli, *Towards Inclusive Information Technology Revolution in India* (2009), PDF p. 3 (eighteen years at Tata Electric and more than thirty at TCS, as he put it in 2009); Computer History Museum, 'Oral History of FC Kohli' (2020), p. 5. Rajaraman has S. Ramadorai succeeding Kohli as chief executive in 1996 (Rajaraman, *History of Computing in India* (2012), p. 28).

[29] Ministry of Labour and Rehabilitation, *Report of the Committee on Automation* (1972), App. IX, p. 145; Venkatraman, 'India's IT Guy' (2013); Rajaraman, *History of Computing in India* (2012), p. 19. Rajaraman says the company's first computer was an IBM 1401 (Rajaraman, *History of Computing in India* (2012), p. 19). The *Technology Review* profile, drawing on Kohli, says its first mainframe was an ICL 1903, supplemented by two leased IBM machines (Venkatraman, 'India's IT Guy' (2013)). The Committee on Automation's list settles the order: the 1401 from September 1967, the ICL from 1971 (Ministry of Labour and Rehabilitation, *Report of the Committee on Automation* (1972)). The accounts agree if the IBMs were rented and the ICL was the first machine the company bought.

[30] Venkatraman, 'India's IT Guy' (2013). The profile does not name the insurer, and no other source identifies it; the Committee on Automation's 1971 list does not say where the Tata centre's ICL came from (Ministry of Labour and Rehabilitation, *Report of the Committee on Automation* (1972), App. IX, p. 145).

[31] Rajaraman, *History of Computing in India* (2012), pp. 22-23. The licensing system is described in Chapter 3.

[32] Sharma, 'F.C. Kohli and the Eureka Moment' (2020).

[33] Venkatraman, 'India's IT Guy' (2013) (the quotation is Kohli's, as the profile reports it).

[34] Computer History Museum, 'Oral History of FC Kohli' (2020), p. 5.

[35] Computer History Museum, 'Oral History of FC Kohli' (2020), p. 4.

[36] Tata Group, TCS timeline and 'Five Decades of Transformation' (2021); Tata Group, 'The IT Revolutionary' (2024). In the same interview Kohli corrected the interviewer's dates more than once (Computer History Museum, 'Oral History of FC Kohli' (2020), pp. 2, 5).

[37] Sharma, 'F.C. Kohli and the Eureka Moment' (2020). Rests on one source.

[38] Computer History Museum, 'Oral History of N.R. Narayana Murthy, Part 2' (2020), pp. 3-4. The firm was called Softronics; Murthy closed it in less than a year. Chapter 6 tells how he went on to Patni and Infosys.

[39] Computer History Museum, 'Oral History of FC Kohli' (2020), pp. 4-5 (Kohli: built up while he was in the United States; not a legal partnership); Sharma, 'F.C. Kohli and the Eureka Moment' (2020) (Sharma: Kohli decided to work with IBM's closest rival).

[40] Aspray, 'The Role of Policy' (2010), section 'Trade, industrial regulation, and foreign investment policy'; *The Economist*, 'Ram Drive' (2008), para. 4 (Burroughs sold its machines in India through TCS). The Tata group dates the partnership to 1973 (Tata Group, TCS timeline and 'Five Decades of Transformation' (2021)); the *Technology Review* profile places the first contract in 1974 (Venkatraman, 'India's IT Guy' (2013)).

[41] Venkatraman, 'India's IT Guy' (2013).

[42] Sharma, 'F.C. Kohli and the Eureka Moment' (2020); Rajaraman, *History of Computing in India* (2012), p. 28.

[43] Tata Group, TCS timeline and 'Five Decades of Transformation' (2021). Sharma credits a group of young engineers with the filter that made ICL software work on the Burroughs platform (Sharma, 'F.C. Kohli and the Eureka Moment' (2020)); Rajaraman says TCS imported a Burroughs B6700 and that Burroughs outsourced many conversion jobs to TCS (Rajaraman, *History of Computing in India* (2012), p. 28). The versions may describe a sequence, conversion first, then application packages, then systems software; no contract has been located to settle it.

[44] Sharma, 'F.C. Kohli and the Eureka Moment' (2020).

[45] Rajaraman, *History of Computing in India* (2012), p. 28 (the manufacturing company, which failed and closed after four years); Sharma, 'F.C. Kohli and the Eureka Moment' (2020) (peripherals and Tata Burroughs Limited). Whether Rajaraman's manufacturing company and Sharma's peripherals venture were the same is not clear.

[46] Tata Group, TCS timeline and 'Five Decades of Transformation' (2021).

[47] Rajaraman, *History of Computing in India* (2012), p. 28.

[48] Tata Group, TCS timeline and 'Five Decades of Transformation' (2021).

[49] Tata Group, 'The IT Revolutionary' (2024).

[50] Aspray, 'The Role of Policy' (2010), section 'Trade, industrial regulation, and foreign investment policy' (the online journal has no page numbers).

[51] Aspray, 'The Role of Policy' (2010), same section; Rajaraman, *History of Computing in India* (2012), p. 29. Body shopping is described in Chapter 6.

[52] No contract, rate card or first-hand account showing how TCS priced its work in the 1970s has been located.

[53] Ramadorai, *The TCS Story ... and Beyond* (2011), as summarised in a published review (*Deccan Herald*, 29 October 2011).

[54] Bassett, 'Aligning India in the Cold War Era' (2009), p. 802 n. 45.

[55] Sharma, 'When Patni Experienced the Eureka Moment' (2011).

[56] Sharma, 'When Patni Experienced the Eureka Moment' (2011) (two dozen); *Business Standard*, 'The Original IT Man' (2007) (*Business Standard*, 2007: the Poona office was started by Poonam Patni with twenty people).

[57] Sharma, 'When Patni Experienced the Eureka Moment' (2011); *Business Standard*, 'The Original IT Man' (2007); Bassett, 'Aligning India in the Cold War Era' (2009), p. 801. Sharma dates the firm to 1976 (Sharma, 'When Patni Experienced the Eureka Moment' (2011)); the *Business Standard* profile gives 1978 (*Business Standard*, 'The Original IT Man' (2007)). Murthy's recollection that the export business had begun by 1977 (Computer History Museum, 'Oral History of N. R. Narayana Murthy', Part 1 (2018), pp. 14-15) fits 1976 better, though it does not show when the company was constituted. Bassett calls the firm Patni Data Systems and describes it as one of the first Indian firms to get computing work from American companies (Bassett, 'Aligning India in the Cold War Era' (2009), p. 801).

[58] Sharma, 'When Patni Experienced the Eureka Moment' (2011) (August 1977); Computer History Museum, 'Oral History of N. R. Narayana Murthy', Part 1 (2018), pp. 14-15 (Murthy, Computer History Museum oral history, 2018).

[59] Sharma, 'When Patni Experienced the Eureka Moment' (2011); Computer History Museum, 'Oral History of Nandan Nilekani' (2020), p. 7 (Nilekani, January 1979); Computer History Museum, 'Oral History of N.R. Narayana Murthy, Part 2' (2020), p. 5; Bassett, 'Aligning India in the Cold War Era' (2009), p. 801. The founding of Infosys is told in Chapter 6.

[60] Tata Group, 'The IT Revolutionary' (2024); Computer History Museum, 'Oral History of N.R. Narayana Murthy, Part 2' (2020), p. 5.

[61] Rajaraman, *History of Computing in India* (2012), pp. 18-19, 26-27. These training routes are described in Chapter 3.

[62] Kohli, *Towards Inclusive Information Technology Revolution in India* (2009), PDF p. 4.

[63] Computer History Museum, 'Oral History of FC Kohli' (2020), p. 2.

[64] Kohli, *Towards Inclusive Information Technology Revolution in India* (2009), PDF p. 4; Tata Group, 'The IT Revolutionary' (2024).

[65] Computer History Museum, 'Oral History of FC Kohli' (2020), p. 7.

[66] Rajaraman, *History of Computing in India* (2012), pp. 23, 28; Aspray, 'The Role of Policy' (2010). Aspray's finding that most imported hardware went to domestic use (see 'Twice the value' below) is a reminder that the rule was not a clean success.

[67] Bassett, 'Aligning India in the Cold War Era' (2009), pp. 801-802. Bassett notes that no IT industry grew up in Kanpur itself; its graduates found work in firms such as TCS (Bassett, 'Aligning India in the Cold War Era' (2009), pp. 800-801).

[68] Tata Group, TCS timeline and 'Five Decades of Transformation' (2021); Sharma, 'When Patni Experienced the Eureka Moment' (2011); Rajaraman, *History of Computing in India* (2012), p. 19. Patni's Data Conversion was exporting data-entry work from 1972 (Sharma, 'When Patni Experienced the Eureka Moment' (2011)).

[69] Tata Group, TCS timeline and 'Five Decades of Transformation' (2021); Venkatraman, 'India's IT Guy' (2013); Sharma, 'When Patni Experienced the Eureka Moment' (2011). The Texas Instruments claim is examined in Chapter 8.

[70] Rajaraman, *History of Computing in India* (2012), pp. 21-22, 28; Aspray, 'The Role of Policy' (2010). The date of the rule differs between the sources. Rajaraman writes that the DoE 'allowed import of computers exclusively meant for software export. The first policy paper on this was published in 1974' (Rajaraman, *History of Computing in India* (2012), p. 28). William Aspray dates a Software Export Scheme to a 1972 bill sponsored by the DoE that eased hardware imports for companies that would use them to produce software for export (Aspray, 'The Role of Policy' (2010), section 'Trade, industrial regulation, and foreign investment policy'). No primary text of either has been located; the two dates may describe an announcement and a formal policy.

[71] Venkatraman, 'India's IT Guy' (2013) (the profile rests on Kohli's own account).

[72] *The Economist*, 'Ram Drive' (2008), para. 2. The magazine states the condition in the same terms: 'The government required TCS to earn twice as much from exports as it spent on its imported machines' (para. 4). Its 1974 date dates TCS's own licence, not the general rule.

[73] Venkatraman, 'India's IT Guy' (2013). Neither account makes clear whether the duty counted towards the sum to be earned back; no copy of the import licence has been located. The figures in the next paragraph are this book's arithmetic on the profile's price.

[74] *Computerwoche*, 'ICL und Ostblockrechner stoßen in eine Lücke' (1978), paras 1 and 9 (*Computerwoche*, April 1978; paragraphs numbered in the online text; translated from the German, 'dürfte... rd. 10 Millionen indische Rupien erreicht haben'); Federal Reserve, G.5A Foreign Exchange Rates, annual average rupees per US dollar for 1977 (Federal Reserve G.5A, Rs 8.7683), at the year's average rate. Rajaraman judges India's software exports insignificant in 1978 (Rajaraman, *History of Computing in India* (2012), p. 29).

[75] Aspray, 'The Role of Policy' (2010), section 'Trade, industrial regulation, and foreign investment policy' (*The Rutherford Journal* 3, 2010). Aspray adds that TCS and Tata Infotech held 63 per cent of the market. Richard Heeks also estimated gross software exports at $4 million in 1980 (Heeks, 'The Uneven Profile of Indian Software Exports' (1998), p. 2; see Chapter 6).

[76] Aspray, 'The Role of Policy' (2010), same section.

[77] Tata Group, TCS timeline and 'Five Decades of Transformation' (2021).

[78] Bansal, 'Take A Techie Like That!', *Outlook*, para. 2 (the reviewer's summary, not Ramadorai's words).

[79] Rajaraman, *History of Computing in India* (2012), pp. 24, 29; Bassett, 'Aligning India in the Cold War Era' (2009), p. 800.

[80] Rajaraman, *History of Computing in India* (2012), p. 29.

[81] *The Economist*, 'Ram Drive' (2008), para. 6.

Part II: Bodies and wires (1977-1991)

Chapter 5. The year IBM left (1973-1984)

Some time in 1977, a group of senior executives from International Business Machines (IBM) went to see India's new minister of industry. They had already been to the top. The prime minister, Morarji Desai, had heard them out and told them to see the minister, George Fernandes.[1] The meeting, as the science journalist Dinesh C. Sharma later wrote, turned out to be decisive.[2]

'IBM was very cocky,' Fernandes told Sharma in 2006, in the only account of the meeting that survives. The executives, he recalled, told him they had refused to accept what the French president, General Charles de Gaulle, had told them about diluting their equity. His reply, as he remembered it: 'if you think the General succumbed to you, I am telling you that I am not succumbing to you. You get out.' And then, for emphasis, he added that yes, he had said it: you get out.[3]

On 1 June 1978 IBM closed its Indian operations.[4] The company that had brought most of India's computers into the country, rented them out, maintained them and trained the people who ran them was gone.

What India lost, and what it gained, was not what either side expected. A rule written to save foreign exchange pushed out the largest foreign computer firm. A licensing policy then let a handful of small Indian companies build computers from imported chips. Because those companies were not allowed to import software, they wrote their own, and the engineers who did that work became, by the account of the period's leading participant-historian, the skilled programmers India later sold to the world. The hardware itself did not last.

The renting business

IBM had sold its machines in India since 1950.[5] From 1965 it built a business on key punches made in Bombay (Mumbai since 1995) and exported, whose foreign-exchange earnings paid for used IBM 1401 computers, which IBM refurbished, repainted and rented to Indian customers with maintenance and help with software included.[6] The rents drew charges of profiteering, but V. Rajaraman, the historian of Indian computing, also credits IBM with training a whole generation of India's maintenance engineers and programmers (Chapter 3).[7] The model gave IBM about two-thirds of the 217 computers counted in India in 1974, and by Sharma's later estimate about 80 per cent of the market.[8]

The same model made IBM a target. Sharma calls it IBM's 'As is' policy, and judges that it 'kept the technology in Indian market one or two generations behind the latest'.[9] He dates the government's pressure to 1967, when the Electronics Committee under the physicist Vikram Sarabhai wanted IBM to stop importing old machines and leasing them to government departments at inflated rents.[10]

Forty per cent

The legal hinge was a law about money. In 1973 India amended the Foreign Exchange Regulation Act (FERA). Its purpose, in Rajaraman's words, was 'to conserve foreign exchange', and under it foreign companies 'except those considered essential were to dilute equity to 40% and take an Indian partner'.[11] Any company with more than 40 per cent foreign equity had to obtain fresh approval from the Reserve Bank of India to continue operating.[12] FERA was one more of the devices, like import licences and export obligations, by which the Indian state rationed its dollars.

Two years later Parliament added a charge of its own. After the Comptroller and Auditor General raised questions about IBM's dealings with the Railways, the Public Accounts Committee of Parliament, chaired by Hirendra Nath Mukherjee, opened 'a full-scale enquiry into the functioning of IBM' in November 1975.[13] The committee found, as

Sharma reports it, that IBM had 'defrauded the country of enormous revenues by resorting to various unfair practices like transfer pricing under the garb of inter-company billing system, misuse of import entitlements'.[14] Transfer pricing here meant setting the price at which IBM's own affiliates sold machines to its Indian arm, which decided how much foreign exchange left the country. 'IBM was found to dump used and obsolete mainframe computers in India at inflated prices,' Sharma writes of the same enquiry.[15]

IBM bargained. Rajaraman says it first refused to dilute, 'as it was not its global policy'. It then proposed a manufacturing unit, wholly owned and wholly for export, whose earnings would pay for computers imported for sale in India, and offered to invest 26 per cent of its profit in a research centre in India.[16] Sharma describes the offer in more detail. IBM would split its Indian business in two. Card manufacture and the data centres would go into one company with 60 per cent Indian equity, as FERA required; the export factory would be the other, fully owned. 'But it was not willing to dilute its equity in its marketing and maintenance business', which Sharma calls its core business.[17]

IBM, Rajaraman writes, was 'very confident that it would be considered "essential"'. It had over 150 installations in India, at the Railways, the Planning Commission, textile and petrochemical companies, and many research laboratories and educational institutions. 'IBM's data centres were widely used and its service was excellent'.[18] The Department of Electronics (DoE) still hoped for a negotiated solution. Then, in March 1977, the government changed.[19]

IBM's British rival made the other choice. ICL diluted its equity, formed International Computer India Manufacturing Ltd (ICIM) with an Indian partner, and in 1977 began making a computer series called the ICL 2900 in India.[20] It shows that a foreign computer firm could stay on FERA's terms, and so that leaving was at least partly IBM's choice. It does not show whether anyone in Delhi wanted IBM to stay.

The minister

The 1977 election ended thirty years of continuous Congress rule. A coalition led by Morarji Desai took office.[21] George Fernandes was made minister of communications and moved very soon afterwards to industry, replacing Brij Lal Verma.[22]

He brought a history with him. 'As a trade Union leader in Bombay,' Sharma writes, 'Fernandes had led anti-computer agitation of workers in the 1960s'.[23] The man who now held IBM's file had been on the other side of the union campaigns against office computers. During the election campaign he had said that India did not need 'foreign technology for making brassieres and tooth paste'.[24] He was also pressing Coca-Cola to share its secret formula if it wanted to stay in India.[25]

In his 2006 interview Fernandes gave two reasons for his stand. One was IBM's attitude in the meeting. The other was that, in his view, IBM's computers 'were taking away jobs of Indian workers'.[26] The second reason ties the exit to the long Indian argument about automation and work.

Sharma's own verdict is careful. The audit, the parliamentary enquiry, the FERA rules and pressure from the Department of Electronics had already made it hard for IBM to stay, and Fernandes's arrival was the last nail in the coffin. 'The file on IBM and Coca Cola came to him as FERA cases, and he took the decision based on recommendations of the FERA committee. It was well within the purview of policy and law, but helped Fernandes score a political brownie point'.[27] What the minister did, in Sharma's view, was to shut the window in which IBM and Coca-Cola might have renegotiated.[28] By 1978 Coca-Cola, IBM, Mobil and Kodak had quit India or applied to do so.[29]

Who left whom

Whether IBM was thrown out or chose to go has been argued ever since. Rajaraman's account is that it chose to leave rather than give up control: 'After prolonged negotiations the government was unwilling to relax FERA for IBM. Consequently IBM decided to withdraw from India in 1978'.[30] Sharma's is that FERA only eased an exit whose real cause was a power struggle between a multinational and the government.[31] A report written at the time points the same way. In April 1978 the West German trade weekly *Computerwoche* wrote that specialists thought the Electronics Commission might have brought on the confrontation deliberately, accusing IBM of making excessive profits from outdated machines and of largely blocking the local development of new systems. IBM replied that it

had proposed replacing its 1400 series in 1970, and again in 1972, and that the government had taken up neither plan.[32]

Shiv Nadar, a founder of one of the Indian firms that grew in IBM's absence, gave a third account to Knowledge at Wharton in 2006. When the interviewer began to say that Fernandes had evicted IBM in 1977, Nadar cut in: 'IBM wasn't evicted. The real story was different.' In his account the government had ruled that if a product was made in India, its source code had to reside in the country. Equity dilution, he said, did not hurt companies much, but the source-code requirement did. 'That was the main reason why IBM decided to leave India, as did Coca-Cola'.[33] No other source mentions such a rule for computers; Coca-Cola's secret formula was certainly an issue, and Nadar may have run the IBM and Coca-Cola cases together.[34]

The fourth account is expulsion. The historian Dwaipayan Banerjee's 2026 history of Indian computing is described by its publisher as treating IBM's departure as a decisive expulsion, one act in a project of technological independence, and a reviewer summarises it as saying that IBM was 'eventually expelled in 1977 by the union-backed Janata Party'.[35] Fernandes's own words support that reading at the level of political intent.[36] The accounts agree on most of the facts. The rule made IBM's way of doing business in India impossible unless it gave up control of its core business; IBM would not; the minister closed the door loudly. 'Expelled' and 'chose to leave' describe the same event from opposite sides.

June 1978

The unwinding was quick. In April 1978 *Computerwoche* reported that IBM, which under FERA could have kept only 40 per cent of its Indian subsidiary, against the company's worldwide policy, would end all its activities in India by the end of the year.[37] The refurbishing work in Bombay stopped. The computers IBM had rented out 'were sold to the customers at their book value which was almost zero'.[38] A group of former IBM employees formed a company called International Data Machines (IDM) and took over the card factory and the data centres.[39]

Maintenance went to a state company that had been waiting for it. The Department of Electronics had set up the Computer Maintenance Corporation (CMC) in 1976, Rajaraman writes, 'to maintain imported computers foreseeing such a contingency'.[40] It had cost Rs 53 million to establish,[41] about $5.9 million at the 1976 average exchange rate.[42] IBM sold CMC the spare parts for its 1401s and other machines at a nominal price, and CMC took over the maintenance of IBM's installations across the country.[43] Indian users, *Computerwoche* reported, were soon paying much more for data processing, and CMC was said to be using its new monopoly to charge fees even higher than IBM's.[44]

What happened to IBM's people is disputed. Rajaraman's account is smooth: 'Most employees of IBM joined CMC. There was thus no disruption of operations of computers'.[45] The trade magazine Dataquest's chronology says that T.B. Finn, who became IBM India's general manager in 1976, laid off a third of the workforce within four months, and concludes that IBM had already made up its mind to leave.[46] A 2018 report in the Deccan Chronicle on a reunion of former IBM India staff in Mumbai, forty years after the exit, said there had been around 150 employees at the end, and that many had been left with almost nowhere to go. Some went back to their home towns to look for work, some found jobs, and some started businesses.[47] One of them, K.R. Naik, laid off by IBM, later worked for ORG Systems, one of the Indian computer makers that grew in IBM's place, and helped organise the reunion.[48]

When IBM left there were fewer than 1,000 computers in India, and software exports were, in Rajaraman's word, insignificant.[49]

The Sondhi policy

The new government reviewed the electronics industry. A committee chaired by Mantosh Sondhi, with Biswajit Nag as its secretary, found 'long delays in importing computers due to the cumbersome procedures of the DoE'. It proposed simpler procedures, and it proposed that private companies be allowed to make computers.[50] In 1978 Nag became secretary of the Department of Electronics.[51]

The minicomputer policy that followed opened a door but kept a hand on it. It favoured what it called systems engineering companies, which need not make their own processors or peripherals. Small entrepreneurs were preferred for dedicated machines for data entry, accounting, invoicing and typesetting. Foreign collaboration was in general not allowed, though foreign equity of up to 40 per cent was permitted in special cases. Every manufacturer needed a licence. Production was capped at Rs 20 million in value,[52] about $2.4 million at the 1978 average rate,[53] and

there was a limit on the number of machines. Only the public-sector Electronics Corporation of India Limited (ECIL) would be allowed to make 32-bit computers, the larger machines of the time.[54]

One clause carried the labour settlement of the previous decade into the new one. The policy 'would continue to be restrictive in allowing use of computers in areas where labour displacement was likely to be involved'.[55] The state that let private firms build computers still limited where those computers might replace clerks.

Rajaraman's judgement on the policy is double. It 'did not completely free the industry from the "licence-permit raj"', but it let private companies into a field that had been reserved for the state.[56] Many licences went to traders who never made anything. The firms that did manufacture, by his list, were DCM Data Products in Delhi, ORG Systems in Baroda (now Vadodara), Hindustan Computers Limited in Delhi, PSI Data Systems in Bangalore (Bengaluru since 2014), Wipro in Bangalore, Patni Computers in Pune, IDM in Bombay and Zenith Computers in Bombay.[57] Two of them left founders' and institutional accounts detailed enough to show what the work was like.

An attic in Delhi

Shiv Nadar and his colleagues were engineers in the calculator division of Delhi Cloth Mills (DCM), a textile company. The calculators had moved from digital circuits to chips, and the more advanced chips went into programmable calculators, which Nadar later called the forerunners of personal computers. A few of the engineers thought they could build a computer.[58]

The state, in Nadar's telling, was not only an obstacle. The government's Electronics Commission 'was interested in making entrepreneurs out of executives and engineers. They asked us to write a report' saying they could build an eight-bit machine, and they did. 'Youthful exuberance and age were on our side'.[59] They launched Hindustan Computers Limited (HCL) in August 1976, two years before IBM left. Getting permission to make computers was the hard part. They could not expect a licence themselves, so they found a partner that had one, the government of Uttar Pradesh, and formed a joint venture. 'Interestingly, our toughest challenge was not designing or making computers but getting the government's approval to produce them'.[60]

The money came from the founders, who had first set up a company to sell calculators from their own savings and used its earnings for HCL. In 1975 six of them, among them Nadar and Arjun Malhotra, pooled Rs 175,000 in a calculator-trading company, Microcomp, most of which went on buying an office in Bombay. Once they had the licence they incorporated Hindustan Computers Limited with equity of Rs 2 million (20 lakh), about $223,000 at the 1976 average rate. A quarter came from the Uttar Pradesh government, and the 49 per cent meant for the public was taken up by Microcomp.[61]

The people were young. 'I was 31, and I was the oldest,' Nadar said. When candidates came for interviews, the founders told them first to consider the size of the company they were joining. 'We offered a monthly salary of Rs. 2,000,' he said, a sum 'more than what Citibank was paying at that time. Otherwise, how do you get someone to work in an attic with a bedroom, a table and a chair?'[62] Rs 2,000 a month was about $223 at the 1976 average rate.[63] It is the first salary this book can document for a job in an Indian computer company, and it is worth holding in mind. The export trade that followed was built on the gap between a rupee salary like this one and the dollar price the same skill could fetch abroad. Each hire added 5 per cent to the company's overheads, Nadar said, so they judged each one closely. They took five students from the Indian Institute of Management in Calcutta (Kolkata since 2001) and told them that if they did things right they should become entrepreneurs themselves. 'Did any of them become entrepreneurs? No one did'.[64]

HCL's engineers wrote software as well as building hardware. Before they could build their computer they had to create the system software that made the parts run together.[65] In the 1980s they wrote a relational database, which they called Genesis, and Nadar insisted they wrote it 'from scratch'.[66] HCL's first computer, the HCL 8/C of 1977, was meant to replace IBM 1401s, Malhotra recalled for the Computer History Museum. Because 1401 users were used to punched cards, HCL sold separate data-entry machines that recorded onto floppy disks. To ride out power cuts, the computer was wired to a truck battery that held its state through a cut, and a maintenance engineer had to call every two weeks to top up the battery's distilled water.[67]

Nadar's view of IBM's departure is the opposite of the nationalist story. 'If IBM had stayed on, we would have grown faster. See, IBM was fundamentally creating the market for computers in India.' After IBM left, HCL had to create the market by itself, and 'At that time no one saw the computer as a necessity. It was a convenience'.[68]

Vegetable oil and the 8086

Wipro's road into computers ran through a university. In the late 1970s Wipro Products made vegetable and refined oils.[69] It decided to move into computers, and in July 1979 it signed a consultancy agreement with the Indian Institute of Science (IISc) in Bangalore, through the institute's consultancy centre, worth about Rs 1.4 million (14 lakh),[70] about $172,000 at the 1979 average rate.[71] On 5 October 1979 the institute's registrar wrote to Ashok Narasimhan, Wipro's general manager for finance and planning, that the institute had 'undertaken a project for the design and development of microprocessor-based minicomputer systems' for Wipro, with the know-how to be handed over 'in another fifteen months'.[72]

Sridhar Mitta, who joined Wipro as research and development manager in 1980, explained in 2019 why a company like his could now enter the business at all. The world, he said, was moving 'From the expensive vertically integrated systems where computer vendors had to develop components' themselves towards 'more open and cheaper, horizontally integrated systems', in which a maker could buy sub-systems and peripherals from different vendors and put them together.[73] A microprocessor bought from Intel could be the heart of a machine designed in Bangalore.

Buying even the tools was hard. N.J. Rao, who ran the institute's Digital Systems Lab, recalled that in 1979 Intel released a development system for its new 8086 processor. 'It would have been really difficult for a company to import the Intel Microprocessor Development System,' he said. 'So IISc procured it'.[74] The university served as a way round the import licence. The Wipro team worked out of Rao's lab. Azim Premji, Wipro's chairman, said the institute set up a committee 'to advise us on what computer we should make'.[75]

The Wipro Series-86 came out in 1981. It used the Intel 8086 processor and Intel's Multibus, a standard set of connections for plugging circuit boards into the machine, and its operating system was licensed from Sentinel Computer Corporation, an American firm. A team at the institute's Centre for Electronics Design and Technology, led by Kishore Babu, designed the case.[76] By Mitta's account the first year's revenue was Rs 20 million (2 crore),[77] about $2.3 million at the 1981 average rate.[78] The new company moved to its first office at 45 Dickenson Road in Bangalore.[79] Forty years later Premji told the institute's convocation that 'it would be quite accurate to say that the Wipro IT business was born in IISc',[80] a founder's tribute on a ceremonial day.

The Wipro story also names some of the first employees, which founders' stories rarely do. Y. Narahari, then an engineering undergraduate at the institute and later the chair of one of its engineering divisions, remembered that Victor Jayakaran, from the Digital Systems Lab, was one of Wipro's first hires. 'Out of the first few employees of Wipro, several were trained in this lab,' he said. 'IISc students were given an opportunity to work for a start-up like Wipro'.[81] Many of the institute's graduates joined Wipro in its first two years, among them Vallab Kulkarni, Srikant Seshadri, Anand Talwai and Ravi Bail.[82] Narahari himself was offered a job at Wipro early in 1982. He chose instead to stay for a master's degree in the institute's School of Automation.[83]

Programmers made by scarcity

Rajaraman's description of what these firms did is the core of this chapter. They 'designed their systems from scratch using imported microprocessors, support chips, peripherals, and local components'. They 'designed and fabricated printed circuit boards, mounted and soldered all the components and tested the systems'.[84]

Then came the constraint that turned hardware makers into software writers. 'As software could not be imported they had to develop operating systems, device drivers, compilers and application programs. UNIX was the operating system of choice as it was easily available'.[85] Unix, developed by Ken Thompson and Dennis Ritchie at AT&T in 1969, was itself written in C, a high-level programming language, and because American regulators did not allow AT&T to market it, versions circulated almost freely. The one developed at the University of California at Berkeley was popular, and Indian minicomputer makers modified and adopted it.[86] A committee of the Reserve Bank of India chaired by Rangarajan, which recommended that every bank computerise its back offices, made Unix the standard operating system for bank computers.[87] The banks' demand, Rajaraman writes, was large 'and could not be ignored'.[88]

His conclusion follows. 'Thus during 1978-1984 a large number of systems programmers in India became experts in UNIX development and usage. They also adopted C widely as UNIX was written in C. This expertise came in handy when India entered the software services export market in the late 80s'.[89] Systems programmers write the software

closest to the machine: operating systems, compilers that translate programs into machine instructions, and the drivers that let a computer talk to its disks and printers.

The only reference Rajaraman gives for this passage is a personal communication in 2012 from Vinay Deshpande, an engineer of the period. It is a participant's judgement, not a count.[90] A second participant, independent of that source, makes the same link. Malhotra said that HCL wrote its own operating systems and compilers 'because we technically weren't allowed to import software unless it was in source code form', and that this was 'why UNIX became so popular in India over time'.[91] He states the rule more precisely than Rajaraman: software could come in, but only as source code. The claim is plausible and partly evidenced.

The new firms also drew people from the state, among them engineers trained at ECIL.[92] After 1978, Rajaraman writes, 'ECIL found it difficult to compete with these companies and lost many of their designers to the private sector companies'.[93] In his account the years from 1978 to 1984 saw 'considerable local design and development'. Firms 'designed mother boards and device drivers, modified the UNIX OS and wrote compilers from scratch', and competition drove prices down.[94] The profession was being built from three directions at once: IBM's former staff moving to CMC, IDM and the Indian makers; ECIL's engineers moving to private firms; and new graduates from IISc, the IITs and the management institutes taking jobs in attics and converted offices. The universities began to respond in the year IBM left, when IIT Kanpur started India's first undergraduate degree in computer science; a national panel proposed a new Master of Computer Applications degree in 1980, and the private training company NIIT was founded in 1981 (Chapter 7).[95] ## Screwdrivers

The 1978 policy did not end the permit system. Infosys, founded in 1981 to sell programming abroad rather than build computers, found the import rules as hard as HCL had found the manufacturing rules: by N.R. Narayana Murthy's later account, a licence to import a single computer took about three years.[96] In 1981 the Electronics Commission appointed another committee under Rajaraman, which 'recommended liberalization of import of computers for genuine exporters'.[97] The trade was still tiny.[98] Nor had the argument about computers and jobs gone away. Seven years after Fernandes told IBM that computers took Indian jobs, the bank unions, who read the Rangarajan committee's push to computerise as a warning sign, observed 1984 as an 'anti-computerization year' (Chapter 8).[99]

The period ended with a new prime minister and a new policy. Indira Gandhi was assassinated in 1984 and Rajiv Gandhi succeeded her.[100] He had organised the computerisation of the 1982 Asian Games in Delhi, using locally made DCM terminals, and had heard there, in Rajaraman's account, the private makers' complaints about the rules that hampered them.[101] That year's computer policy, announced on 19 November,[102] let any manufacturer make 32-bit machines and removed the limits on numbers.[103] It also allowed assembled boards, with their microprocessors and interface electronics already mounted, to be imported, and cut the duty on them.[104]

Rajaraman saw what that meant for the engineers who had been soldering their own boards. 'It changed manufacturing to assembly and system integration.' Imported boards were cheaper and more reliable than boards made in India from local parts and imported chips. 'Thus local manufacturing of boards slowly disappeared'.[105] Within two years the growth of computers went up by 100 per cent and their cost fell by half, and the old guard called the new kind of production 'screw driver' manufacturing.[106] A friend of Rajaraman's described his own job as a consultant to Indian PC makers: finding firms in Taiwan, Singapore and Hong Kong to supply motherboards, disk drives, monitors, power supplies, cables and cases, and timing the deliveries to local 'manufacturers' who assembled them, 'put their labels on them' and printed a manual.[107] The ideal of self-reliance, the cornerstone of the Bhabha Committee's report, was in Rajaraman's words 'truly and completely buried'.[108]

He found one failure larger than the rest. 'The only major failing was not investing enough resources to develop a local integrated circuit manufacturing industry.' Indian engineers could design large integrated circuits for Texas Instruments and other foreign firms, 'but had no foundry to fabricate them in India'.[109] A committee on semiconductors had reported in 1973, and nothing tangible was done until 1978; even then, he writes, the investment was too small.[110]

What India kept was the skill of making computers work. The engineers who had written operating systems and compilers because software could not be imported, and the graduates trained to design business systems, were about to find a market abroad. The same 1984 policy that ended board-making recognised software as an industry, cut the customs duty on computers imported for software export, which had been over 100 per cent, and made one more change that would shape the next decade. 'Sending engineers abroad to develop and maintain software for clients at their sites and profits earned thereby was recognized as "software export"'.[111] By FY1990-91 software exports,

insignificant when IBM left, had reached $131.2 million on the estimate of Richard Heeks of the University of Manchester.[112] Rajaraman dates the start of that trade to these years and gives it the name the industry used, 'body shopping'.[113] India had tried to make its own computers and was now assembling other people's. What it would sell instead was an engineer's time, and from 1984 an Indian engineer's hours at a client's desk abroad counted as an Indian export.

Notes to Chapter 5. The year IBM left (1973-1984)

[1] Sharma, 'George Fernandes and the Infamous IBM Exit' (2019).

[2] Sharma, 'George Fernandes and the Infamous IBM Exit' (2019).

[3] Sharma, 'George Fernandes and the Infamous IBM Exit' (2019). Fernandes's recollection, given to Sharma in an interview in 2006, nearly thirty years after the event, is the only account of what was said. Whether the de Gaulle exchange happened as he told it cannot be checked: it is a politician's recollection of an episode that brought him credit. Sharma's text adds '(to dilute their equity)' inside the quotation; that part is paraphrased here.

[4] *Dataquest*, '1976-1985: The Big Blue Impact' (2006); Sharma, 'Rise, Fall and Rise of IBM in India' (2011). The date of 1 June 1978 is from Dataquest's later chronology (*Dataquest*, '1976-1985: The Big Blue Impact' (2006)); a contemporary report (*Computerwoche*, 'ICL und Ostblockrechner stoßen in eine Lücke' (1978), para. 1) confirms that IBM was winding up its Indian activities in 1978.

[5] Ministry of Labour and Rehabilitation, *Report of the Committee on Automation* (1972), p. 33 (para. 2.54).

[6] Rajaraman, *History of Computing in India* (2012), pp. 18-19.

[7] Rajaraman, *History of Computing in India* (2012), p. 19.

[8] Bassett, 'Aligning India in the Cold War Era' (2009), p. 800 (the 1974 survey; see Chapter 2); Sharma, 'Rise, Fall and Rise of IBM in India' (2011) (Sharma, 2011, on the 80 per cent share).

[9] Sharma, 'George Fernandes and the Infamous IBM Exit' (2019).

[10] Sharma, 'Rise, Fall and Rise of IBM in India' (2011).

[11] Rajaraman, *History of Computing in India* (2012), p. 29; Sharma, 'George Fernandes and the Infamous IBM Exit' (2019).

[12] Sharma, 'George Fernandes and the Infamous IBM Exit' (2019).

[13] Sharma, 'George Fernandes and the Infamous IBM Exit' (2019).

[14] Sharma, 'George Fernandes and the Infamous IBM Exit' (2019). No copy of the committee's report has been located; its finding is quoted as Sharma gives it.

[15] Sharma, 'George Fernandes and the Infamous IBM Exit' (2019). IBM's own reply at the time, as reported in April 1978, was that it had offered to replace its 1400 series in 1970 and again in 1972 and that the government had taken up neither plan (*Computerwoche*, 'ICL und Ostblockrechner stoßen in eine Lücke' (1978), para. 2).

[16] Rajaraman, *History of Computing in India* (2012), p. 29.

[17] Sharma, 'George Fernandes and the Infamous IBM Exit' (2019).

[18] Rajaraman, *History of Computing in India* (2012), p. 29.

[19] Sharma, 'George Fernandes and the Infamous IBM Exit' (2019).

[20] Rajaraman, *History of Computing in India* (2012), p. 30.

[21] Rajaraman, *History of Computing in India* (2012), p. 30.

[22] Sharma, 'George Fernandes and the Infamous IBM Exit' (2019).

[23] Sharma, 'George Fernandes and the Infamous IBM Exit' (2019).

[24] Sharma, 'George Fernandes and the Infamous IBM Exit' (2019).

[25] Sharma, 'George Fernandes and the Infamous IBM Exit' (2019).

[26] Sharma, 'George Fernandes and the Infamous IBM Exit' (2019).

[27] Sharma, 'George Fernandes and the Infamous IBM Exit' (2019).

[28] Sharma, 'George Fernandes and the Infamous IBM Exit' (2019).

[29] Panchal, 'Economic Milestone: Exit of the MNCs (1977)' (2014).

[30] Rajaraman, *History of Computing in India* (2012), p. 29; *Dataquest*, '1976-1985: The Big Blue Impact' (2006) (Dataquest's 2006 retrospective says much the same: that IBM left because it could not agree to the FERA rules).

[31] Sharma, 'Rise, Fall and Rise of IBM in India' (2011).

[32] *Computerwoche*, 'ICL und Ostblockrechner stoßen in eine Lücke' (1978), para. 2 (paragraphs numbered in the online text; translated from the German).

[33] Aron, 'HCL's Shiv Nadar' (2006).

[34] Sharma, 'George Fernandes and the Infamous IBM Exit' (2019) (Coca-Cola's formula). Nadar's source-code rule rests on his 2006 interview alone (Aron, 'HCL's Shiv Nadar' (2006)).

[35] Banerjee, *Computing in the Age of Decolonization* (2026) (publisher's description, not the book); Dizikes, 'The Tech Revolution That Wasn't' (2026). Post-cutoff material, checked in October 2026. Farrell, review of Banerjee, *Computing in the Age of Decolonization* (2026), p. 3 (Tracie Farrell's review; the reviewer's summary, not Banerjee's words; post-cutoff material, checked against the live source in October 2026). The review gives 1977, the year of the Fernandes meeting; the contemporary record places the closure itself in 1978 (*Computerwoche*, 'ICL und Ostblockrechner stoßen in eine Lücke' (1978); *Dataquest*, '1976-1985: The Big Blue Impact' (2006)).

[36] Sharma, 'George Fernandes and the Infamous IBM Exit' (2019).

[37] *Computerwoche*, 'ICL und Ostblockrechner stoßen in eine Lücke' (1978), para. 1 (translated from the German).

[38] Rajaraman, *History of Computing in India* (2012), p. 29.

[39] Rajaraman, *History of Computing in India* (2012), pp. 29-30.

[40] Rajaraman, *History of Computing in India* (2012), p. 30.

[41] Rajaraman, *History of Computing in India* (2012), p. 28.

[42] Federal Reserve, G.5A Foreign Exchange Rates, annual average rupees per US dollar for 1976 (Federal Reserve G.5A), at the year's average rate.

[43] Rajaraman, *History of Computing in India* (2012), p. 30 (spare parts); *Dataquest*, '1976-1985: The Big Blue Impact' (2006) (Dataquest: CMC took over 'maintenance of 800 IBM installations'); Rajaraman, *History of Computing in India* (2012), p. 29 ('over 150' installations); *Computerwoche*, 'ICL und Ostblockrechner stoßen in eine Lücke' (1978), para. 5 ('der 150 IBM-1400', translated from the German). Dataquest's 800 is far above Rajaraman's 'over 150'. The contemporary *Computerwoche* figure of 150 IBM 1400-series machines that would need replacing suggests that Rajaraman's figure counts those computers. Dataquest's larger number may count every piece of IBM equipment, including older punched-card machines, but it does not explain its count.

[44] *Computerwoche*, 'ICL und Ostblockrechner stoßen in eine Lücke' (1978), paras 1 and 3 (translated from the German).

[45] Rajaraman, *History of Computing in India* (2012), p. 30.

[46] *Dataquest*, '1976-1985: The Big Blue Impact' (2006).

[47] *Deccan Chronicle*, 'Indian Ex-IBMers Reunite' (2018). The Deccan Chronicle's background paragraphs closely follow Sharma's 2011 article (Sharma, 'Rise, Fall and Rise of IBM in India' (2011)), so it is not a fully independent witness. The employee numbers in these accounts cannot be reconciled.

[48] *Deccan Chronicle*, 'Indian Ex-IBMers Reunite' (2018) (Kamlaksha Rama Naik; in the paper's paraphrase, the former staff still shared a strong bond and affection for IBM); Rajaraman, *History of Computing in India* (2012), p. 40 (ORG Systems among the makers under the post-1978 policy).

[49] Rajaraman, *History of Computing in India* (2012), p. 29.

[50] Rajaraman, *History of Computing in India* (2012), p. 30.

[51] Rajaraman, *History of Computing in India* (2012), p. 30.

[52] Rajaraman, *History of Computing in India* (2012), p. 31.

[53] Federal Reserve, G.5A Foreign Exchange Rates, annual average rupees per US dollar for 1978 (Federal Reserve G.5A), at the year's average rate.

[54] Rajaraman, *History of Computing in India* (2012), p. 31.

[55] Rajaraman, *History of Computing in India* (2012), p. 31.

[56] Rajaraman, *History of Computing in India* (2012), pp. 39-40.

[57] Rajaraman, *History of Computing in India* (2012), p. 40.

[58] Aron, 'HCL's Shiv Nadar' (2006).

[59] Aron, 'HCL's Shiv Nadar' (2006).

[60] Aron, 'HCL's Shiv Nadar' (2006).

[61] Aron, 'HCL's Shiv Nadar' (2006) (the founders' own savings; HCL's capital of Rs 2 million, a quarter from the Uttar Pradesh government); Federal Reserve, G.5A Foreign Exchange Rates, annual average rupees per US dollar for 1976 (Federal Reserve G.5A), at the year's average rate; Computer History Museum, 'Oral History of Arjun Malhotra' (2018), pp. 9-10, 13 (Malhotra, recorded 15 November 2018; a founder's recollection: the Rs 175,000, the Bombay office, Microcomp named after the programmable calculator the founders had designed at DCM, and the 49 per cent). Dataquest's chronology (*Dataquest*, '1976-1985: The Big Blue Impact' (2006)) says Nadar started Microcomp with six friends and Rs 175,000 and that it was renamed Hindustan Computers. Malhotra's account shows instead that Microcomp was the founders' first firm and HCL's largest shareholder.

[62] Aron, 'HCL's Shiv Nadar' (2006).

[63] Federal Reserve, G.5A Foreign Exchange Rates, annual average rupees per US dollar for 1976 (Federal Reserve G.5A), at the year's average rate.

[64] Aron, 'HCL's Shiv Nadar' (2006).

[65] Aron, 'HCL's Shiv Nadar' (2006).

[66] Aron, 'HCL's Shiv Nadar' (2006).

[67] Computer History Museum, 'Oral History of Arjun Malhotra' (2018), pp. 13-14 (Malhotra, Computer History Museum oral history, 2018).

[68] Aron, 'HCL's Shiv Nadar' (2006).

[69] IISc, 'IISc's Role in Helping Wipro Launch Its Minicomputers' (2019).

[70] IISc, 'IISc's Role in Helping Wipro Launch Its Minicomputers' (2019).

[71] Federal Reserve, G.5A Foreign Exchange Rates, annual average rupees per US dollar for 1979 (Federal Reserve G.5A), at the year's average rate.

[72] IISc, 'IISc's Role in Helping Wipro Launch Its Minicomputers' (2019).

[73] IISc, 'IISc's Role in Helping Wipro Launch Its Minicomputers' (2019).

[74] IISc, 'IISc's Role in Helping Wipro Launch Its Minicomputers' (2019).

[75] IISc, 'IISc's Role in Helping Wipro Launch Its Minicomputers' (2019).

[76] IISc, 'IISc's Role in Helping Wipro Launch Its Minicomputers' (2019).

[77] IISc, 'IISc's Role in Helping Wipro Launch Its Minicomputers' (2019).

[78] Federal Reserve, G.5A Foreign Exchange Rates, annual average rupees per US dollar for 1981 (Federal Reserve G.5A), at the year's average rate.

[79] IISc, 'IISc's Role in Helping Wipro Launch Its Minicomputers' (2019).

[80] IISc, 'IISc's Role in Helping Wipro Launch Its Minicomputers' (2019).

[81] IISc, 'IISc's Role in Helping Wipro Launch Its Minicomputers' (2019).

[82] IISc, 'IISc's Role in Helping Wipro Launch Its Minicomputers' (2019).

[83] IISc, 'IISc's Role in Helping Wipro Launch Its Minicomputers' (2019).

[84] Rajaraman, *History of Computing in India* (2012), p. 40.

[85] Rajaraman, *History of Computing in India* (2012), p. 40.

[86] Rajaraman, *History of Computing in India* (2012), p. 40.

[87] Rajaraman, *History of Computing in India* (2012), pp. 36, 40.

[88] Rajaraman, *History of Computing in India* (2012), p. 40.

[89] Rajaraman, *History of Computing in India* (2012), p. 40.

[90] Rajaraman, *History of Computing in India* (2012), pp. 40, 66 (ref. 73). No count has been located of how many Unix or C programmers the hardware firms trained, and no study has been found that follows individual engineers from these firms into the export trade. HCL's Genesis database (Aron, 'HCL's Shiv Nadar' (2006)) is a founder's example, not a census.

[91] Computer History Museum, 'Oral History of Arjun Malhotra' (2018), p. 16.

[92] Rajaraman, *History of Computing in India* (2012), p. 26.

[93] Rajaraman, *History of Computing in India* (2012), p. 40.

[94] Rajaraman, *History of Computing in India* (2012), p. 40.

[95] Rajaraman, *History of Computing in India* (2012), p. 34 (Kanpur's degree, 1978, and the other IITs following); pp. 34-35 (the 1980 Panel on Computer Manpower Development under Rajaraman; the MCA), 42 (NIIT).

[96] Computer History Museum, 'Oral History of N.R. Narayana Murthy, Part 2' (2020), p. 5 (the transcript prints this answer under the label 'Brock:', but its content, 'we founded Infosys in 1981', shows it is Murthy's). Chapter 9 gives his fuller account of the licence.

[97] Rajaraman, *History of Computing in India* (2012), p. 35.

[98] Heeks, 'The Uneven Profile of Indian Software Exports' (1998), p. 2 (Table 1). Richard Heeks estimated India's gross software export earnings at $4 million in 1980 and $25 million in 1984 (gross foreign-exchange earnings, calendar years; net earnings estimated at about 40% of gross).

[99] Rajaraman, *History of Computing in India* (2012), p. 36.

[100] Rajaraman, *History of Computing in India* (2012), pp. 30-31.

[101] Rajaraman, *History of Computing in India* (2012), p. 31. Locally made DCM computers served as terminals at the venues, connected to a Hewlett-Packard server, and engineers of the National Informatics Centre wrote the whole software system in six months; the Games brought N. Seshagiri, the centre's director general, close to Rajiv Gandhi. Chapter 8 tells the story.

[102] *Dataquest*, '1976-1985: The Big Blue Impact' (2006) (Dataquest's chronology gives the date).

[103] Rajaraman, *History of Computing in India* (2012), p. 31.

[104] Rajaraman, *History of Computing in India* (2012), p. 32.

[105] Rajaraman, *History of Computing in India* (2012), p. 32.

[106] Rajaraman, *History of Computing in India* (2012), p. 40.

[107] Rajaraman, *History of Computing in India* (2012), p. 41.

[108] Rajaraman, *History of Computing in India* (2012), p. 41. The Bhabha Committee's report is discussed in Chapter 3.

[109] Rajaraman, *History of Computing in India* (2012), p. 42.

[110] Rajaraman, *History of Computing in India* (2012), p. 28.

[111] Rajaraman, *History of Computing in India* (2012), p. 32.

[112] Heeks, 'The Uneven Profile of Indian Software Exports' (1998), p. 2 (gross export earnings, FY1990-91). Other series give different figures; Chapter 9 and the data appendix compare them.

[113] Rajaraman, *History of Computing in India* (2012), p. 29.

Chapter 6. Body shopping (1978-1991)

In 1990 a computer programmer from Bombay (Mumbai since 1995) went to California on a B1/B2 visa. He had been hired by the Indian computer company Hindustan Computers Limited (HCL) and sent to its offices in Palo Alto. In 1995 the American magazine *Multinational Monitor* told his story under a name it made up, 'Rajesh Ahuja (not his real name)', and that pseudonym is the only name the record gives him.[1] The magazine described the visa he travelled on as one 'intended for foreign consultants to work in the United States for relatively short stints'.[2]

The work was not at HCL. Ahuja 'worked anywhere HCL sent him, at whatever company he was subcontracted to', and he was 'required to work overtime without extra pay'.[3] Most of his time in America was spent at Sun Microsystems in Mountain View. 'I shared a two-bedroom apartment and a car with three other programmers, and got $40 a day for expenses,' he told the magazine. His salary, about $800 a month, was paid into his bank account in India, and he called it 'all savings'.[4] HCL had rented an apartment complex for its programmers, and one person handled 'all the paperwork, transportation and living arrangement for 160 of us'.[5]

He knew the terms were hard and said he did not like them. He accepted them for a reason he put plainly: 'All my life I have dreamed about coming to the United States and being able to go to New York and Disneyland, and listen to bands like the Rolling Stones.' He would have two years of it, and his flight was paid.[6]

The magazine recorded one more figure, and it is the reason this programmer opens the chapter. 'For Ahuja's services, Sun paid HCL $50 an hour, about half of which remained with HCL.'[7] When the assignment ended he went home for what he meant to be a short visit. He then learned, by the magazine's account, that HCL 'had not bothered to renew his visa', and he could not go back. 'That's typical of them,' he told the magazine.[8]

The arithmetic that follows is this book's, not the magazine's. At a working month of 160 hours, $50 an hour comes to about $8,000 billed to Sun each month. Ahuja's salary and a thirty-day expense allowance together came to about $2,000, with the apartment, the shared car and the flights on top. The magazine's own statement is narrower: about half the hourly fee stayed with HCL.

Ahuja's year in California stood for a whole trade. Through the 1980s, most of what India sold abroad as 'software' was the time of programmers like him, working on clients' computers in clients' offices. The firms that sent them priced that time by the hour or the month, and the founders of one small company spent most of the decade doing the same work.

A name the trade gave itself

The trade's own name for the business was not flattering. V. Rajaraman, the historian of Indian computing, defines body shopping as 'sending software engineers to client sites abroad to develop software', and the historian William Aspray notes that the service came, derisively, to be called by that name.[9] Richard Heeks, a British researcher who did fieldwork in Indian software firms from the late 1980s, was more exact. In the Indian context, he wrote, 'the combination of onsite and programming work has come to be known as "body shopping"'.[10] The work was done at the client's site (onsite), and it was programming rather than analysis or design. Both halves mattered. Later scholars, among them Carol Upadhya of the National Institute of Advanced Studies in Bangalore (Bengaluru since 2014), used the word both for the industry's whole early model and, for the 2000s, for agencies that placed engineers abroad and took a cut of their pay. This book uses it in Heeks's wider sense.[11]

The model was older than the decade. Dinesh Sharma, a journalist and historian of the industry, writes that Tata Consultancy Services (TCS) and Patni Computer Systems had been sending staff to customer sites in the 1970s, and that this 'became a dominant way of doing software business in the 1980s'.[12] By 1980, by Aspray's count, twenty-one

Indian firms exported about $4 million of software a year, and the industry was mainly a body-shopping one.[13] Heeks's series gives the same $4.0 million for 1980.[14] When the Americans first noticed the trade is disputed, and no source names a first contract, but TCS had programmers in the United States by 1974.[15]

The early firms clustered in Bombay, around Nariman Point and later the Santa Cruz Electronics Export Processing Zone (SEEPZ), until at least the mid-1980s.[16] An office in Bombay, though, was often only where the contracts were signed. Heeks's 1988 survey found that 65 per cent of export contracts were carried out wholly at the client's site. Counting the work itself, just under three-quarters of export development took place overseas and a quarter in India. That was true even of firms in the export zones, which had been set up as bases for work done in India.[17]

Why ship people

Why would a country export programmers rather than programs? The sources give the same answer from different angles. Sharma's is the simplest. Indian firms did not have the hardware, the software tools or the basic infrastructure for serious export work, so 'the best way to leverage it was go to the customer site and work there using their computers'.[18]

Heeks turned the point into an explanation of the whole industry. Because the work was done onsite, he wrote, it needed no purchase of hardware or software by the Indian company and no technology infrastructure in India. Firms had chosen 'a path of almost "inputless" exports which requires only a contact overseas, a little finance, and the names of some local programmers who can be hired if a contract is forthcoming'. Such a path could be followed 'relatively independent of government policy measures, particularly those on imports, because no imports are needed'.[19]

That last clause links body shopping to the import rules. India rationed foreign exchange, and so it rationed imported machines. N.R. Narayana Murthy, a founder of Infosys, told the Computer History Museum that before 1991 a licence to import a computer took years of visits to Delhi (Chapter 9 gives his account).[20] He also recalled banks that did not understand what software was and would not lend against it, no venture capital, a wait of years for a telephone line, and the need for the Reserve Bank of India's permission even for a day's business trip abroad.[21] Before 1984, Rajaraman adds, customs duty on computers imported for software export 'used to be over 100%'.[22]

Computers stayed scarce. India had fewer than 1,000 of them in 1978 and 80,000 in 1990.[23] Against that background, a programmer on a plane was the cheapest export India had. The interviewer at the Computer History Museum, David Brock, put it to Murthy that despite these obstacles Infosys had got going quickly 'because you could send your colleagues to go work in the U.S. at customer sites'. Murthy answered, 'Yeah.'[24]

Policy was the push. Heeks's interviews found a pull as well. Indian managers liked onsite programming because it 'produces quick revenue for low investment', which suited a business culture wary of risk in uncertain times.[25] Sharma calls it simply 'a high margin business'.[26] The workers pulled too. 'Many Indian staff also want to work overseas and, if denied the opportunity by one company, will simply join another which will send them abroad.'[27] Ahuja's dream of New York and the Rolling Stones was not unusual. It was one of the things the business ran on.

The minicomputer route

The firms that did this work had come to it by several roads. One ran through the American computer makers that replaced IBM in India after 1978. These minicomputer firms found the Indian market small and slow, and to keep their operations viable they began using local engineers to write software for the machines they sold locally. They found, in Sharma's words, 'that not only were Indian engineers cheaper but the quality of software they wrote was very good', and they began using them for customers back home.[28]

Narendra Patni, whose company sold Data General minicomputers in India, described how his firm's software business began. 'There was a lot of business exchange, and we started doing software work for them including operating systems. That's how our software stream started.'[29] The engineers who did such work are rarely heard in the record. One who is, through a note he sent Sharma in 2006, is K.R. Trilokekar, who joined ORG Systems in 1978. ORG, a division of the Ambalal Sarabhai group, distributed Sperry's mainframes in India. 'We developed P1024 protocol, which Sperry used in its mainframes worldwide,' Trilokekar recalled. 'We also developed several systems software products for them.'[30] His is a reminder that the trade was not only coding to other people's designs. Some Indian engineers wrote the systems software that sat at the heart of American machines.

The largest road ran through Tata. TCS had learned its trade in the 1970s on work for the American computer maker Burroughs, and the two firms' joint software company, Tata Burroughs, was renamed Tata Unisys in 1986.[31] Sharma reports that these two Tata companies 'accounted for nearly 70 per cent of exports through body shopping'.[32] The World Bank's table of top exporters, from data gathered by the National Association of Software and Service Companies (Nasscom), the industry's lobby group, and published in 1994, shows how far ahead they stayed. TCS's exports were Rs 1,754.8 million and Tata Unisys's Rs 560.0 million. The next firm, Digital Equipment (India), had Rs 317.8 million.[33]

Below these few firms, the industry was tiny. Over three-quarters of Indian software companies had fewer than twenty-five professionals, and about 95 per cent had software revenue under $100,000. Some, the World Bank noted, were in the business of renting out programmers abroad and operated with 'a one-room office, telephone and fax'.[34] So many start-ups took this road for a simple reason. A new firm founded by someone who had left a big company could trade on that person's contacts. It could not so easily overcome the barriers of technology, skills and finance, and so it began with onsite programming, where there were few economies of scale. Selling anything else took patience: a few firms had spent two or three years on export marketing before winning a first order.[35]

Seven founders and an apparel package

Infosys was one of those start-ups. After a first company of his own, Softronics, had failed for want of computers in India, Murthy joined Patni Computer Systems in Bombay as general manager of software and visited the United States many times.[36] Patni, he recalled in an earlier museum interview in 2018, sold Data General computers in India, wrote application software for Indian customers and exported software to the United States, and he built a group of 'about 100 youngsters' there.[37] Among his colleagues were Nandan Nilekani and Kris Gopalakrishnan, young engineers with a year or a year and a half of experience. When he left to try again, he told them he was 'convinced that there's going to be a great future for a company that develops bespoke software for a large market like the U.S.'[38]

The company was incorporated in July 1981 as Infosys Consultants Private Limited.[39] Its 1999 filing with the US Securities and Exchange Commission says it was founded by seven people: Murthy and six software professionals.[40] They had, Murthy said in 2018, 'lots of hopes, lots of dreams, some confidence, but no money', and what carried them was 'a deal with our client who gave us a reasonably good advance'.[41] The founding is usually remembered for a small sum: Sudha Murty, Murthy's wife, has said she gave him Rs 10,000 of her savings of Rs 10,250 to start it, keeping Rs 250 back, and that she saw it as a risk because his previous venture had failed.[42] At the 1981 average exchange rate that was about $1,150.[43] The story is told often. What the company did next is told less.

Murthy described the first customer to the Computer History Museum. It was an American firm, Data Basics, which sold a software package to apparel and shoe companies, among them Jockey and Reebok. Most of its customers ran the package on small 16-bit minicomputers, the Data General NOVA or the DEC PDP-11. Infosys proposed to move it to a 32-bit Data General MV/8000 and later to an IBM 4381 mainframe, so that Data Basics could sell to bigger companies. 'It was a godsend for us,' Murthy said.[44] The job was a port, the rewriting of an existing program to run on different machines: exactly the kind of conversion work that the World Bank would later list as typical of Indian contracts.[45]

The clearest record of where the founders spent the 1980s is the management section of that 1999 filing. It reads like the biography of a body shop. 'From 1981 to 1987, Mr. Nilekani was in the United States managing the marketing and development efforts of Infosys.' K. Dinesh, from 1981 to 1990, 'managed projects for Infosys in the United States'. S.D. Shibulal, from 1981 to 1991, 'worked for Infosys in the United States on projects in the retail and manufacturing industries'. From 1987 to 1994 Gopalakrishnan managed 'all projects at the U.S.-based KSA/Infosys', a joint venture with the American consultancy Kurt Salmon Associates.[46] Three of the seven founders, in other words, spent most or all of the decade working in America, and a fourth ran the projects of an American-based joint venture for the last part of it.

Nilekani's own recollection matches the filing. Murthy, he said, opened an office in his flat in Poona (now Pune), and 'we all went abroad, so he was the only guy here. We were all sitting in Florida or somewhere doing some programming work for a company which was a client'. 'Essentially from 1981 to 1987,' he said, 'I was in the U.S. working on various projects for Infosys for clients in the U.S.' His explanation was the one Sharma and Heeks gave: 'in the early days when the technology was not there yet, we had to actually go and do this work physically.'[47]

The same filing describes Infosys as one of the pioneers of a different model. 'The Indian offshore development model was initiated in the mid-1980's', it says, as a way of dividing a project between a facility in India and the client's site.[48] That is the company's own claim, made to investors. The founders' careers show that for most of the decade the Infosys model, like everyone else's, was onsite. By the early 1990s Infosys was twelfth in the World Bank's table of exporters, with exports of Rs 120.6 million.[49] In 1994, Murthy recalled, the whole company, perhaps 200 people, could still meet in its canteen.[50]

Selling hours

How was this work priced? The question matters to the argument of this book, which is that the Indian industry was built to sell engineers' time rather than finished products. The best sources for the 1980s converge on an answer, though none of them is a contract.

The first is a World Bank study by Nagy Hanna, published in 1994, which drew on Nasscom data and on surveys of foreign buyers made in 1991 and 1992. It described Indian export growth as coming mainly from contract programmers doing low-level work, '"body-shopping" or "manpower contracts" in which the overseas buyer typically buys "hours" and not a total project or product'. It went on: 'Typically, programmers are contracted out to work for a client at an agreed hourly or monthly rate for six to eighteen months.' The jobs were data conversion and coding, such as moving systems from batch processing to online processing, or converting code to run on Unix.[51] The contracts were 'short in term, low in risk, low in value added, and low in investment', and they made up '70-80 percent of the industry's export revenues'.[52]

The second is Heeks's survey work, and it measures something different from his onsite figure: not where contracts were done but how they were billed. 'During the period 1988-1998,' he wrote, 'at least 65% of export contracts were solely for programming work billed on a "time and materials" basis, with programming figuring strongly in the remaining one-third of contracts.'[53] Time and materials is the contract term for billing a client for staff time at agreed rates, plus expenses. The rate could be set per hour, per day or per person per month. The opposite is a fixed-price contract, where the supplier promises a result for a set sum and carries the risk if it takes longer.

The third is Ahuja's case, a single documented price: $50 an hour, billed by HCL to Sun in 1990.[54]

A fourth, from inside HCL, is a recollection. Arjun Malhotra, one of its founders, told the Computer History Museum in 2018 that he was sent to HCL's new office in Sunnyvale, taken in 1989, to wind down the company's attempt to sell its own computers in America. He laid off the hardware team, but the American software companies whose compilers and databases his engineers had been adapting for HCL's machines asked for the work to continue and offered to pay for it. That gave him 'a nice positive cashflow, because they were paying me $45+ an hour in those days. I'm going back to 1989.' He thought he had 'about 10, 15, maybe 20 people' there, and he dated HCL's services business from it: 'That is how HCL Technologies started.'[55] His figure is close to the $50 an hour that Sun paid HCL for Ahuja.

Together these sources support the claim that Indian software exports were priced by time and not by product, at least from 1988 and very likely through the decade. Two limits apply. For the years before 1988 the evidence is the World Bank's general description, written in 1994. And the time was sold by the hour or by the month; only Ahuja's case documents a true hourly rate, with Malhotra's recollection beside it.[56]

Why did Indian firms not sell products instead? Heeks's answer, written in 1998, was that the barriers to packaged software were higher than those to services. Indian firms were far from the foreign markets they wanted to serve and found it hard to follow changing needs. Software multinationals could spend 40-50 per cent of their annual revenue on selling and marketing packages, and experience suggested that 'only about 1-5% of software packages succeed'. Neither government policy nor managers' attitudes favoured such high-risk, long-term bets.[57] Some firms did custom work cheaply in the hope of selling the result as a package, but such 'packages' often became platforms for more custom work.[58]

The pricing had consequences that Heeks spelled out. A firm that sold time could grow only by selling more of it: in custom services, 'earnings are directly proportional to the number of people working in the industry'.[59] This is the bargain the rest of this book follows. Revenue rose with headcount, so a firm that wanted to grow had to keep hiring, and every hour of work it saved was an hour it could not bill. The time it sold was also the cheapest part of the job. Citing a 1991 study by Bhatnagar and Jain, Heeks wrote that 'Coding is the lowest paid part of the software

development process, earning only 10-15% of total project revenue'.[60] Foreign buyers knew what they were paying for. A survey of 150 American and European hardware and software firms, cited by the World Bank, ranked Indian software services as having 'the lowest labor cost after China'. Citing a separate 1991 study of European attitudes, the report also noted 'a low perception of Indian software skills in OECD markets', particularly in project management, and observed that programmers who returned from onsite work had gained little or no experience of managing large projects.[61] The pattern lasted. In the nine months to December 1998, Infosys earned 63.7 per cent of its revenue from time-and-materials contracts and 36.3 per cent from fixed-price ones.[62]

The policy catches up

The government did not invent body shopping, but in the 1980s it began to recognise it. The shift was slow. Looking back in 1994, Murli Menon, editor of the Indian magazine *C&C*, wrote that 'In 1984, the government still believed in self-reliance, indigenisation and protection of the domestic [computer] industry'.[63] Yet the computer policy of that year counted the work of engineers sent to clients' sites abroad as software export (Chapter 5). The state now counted the body shop as an exporter.[64]

With machines easier to obtain, some firms began working from India on clients' systems, though high telecommunications costs held this back.[65] Two foreign arrivals showed what the alternative could look like: Citibank's software unit at SEEPZ, inaugurated in 1985 as Citicorp Overseas Software Limited (COSL), which by the year to March 1990 was India's third-largest software exporter and even sold a banking product, priced as a thing rather than as hours, and the centre that Texas Instruments set up in Bangalore with its own satellite link to Dallas, which Heeks called 'the first proper offshore software facility' (Chapter 8 tells TI's story).[66]

The World Bank saw better data links as the way out of body shopping and the brain drain.[67] In the 1980s, though, TI and COSL were exceptions. An estimate by Rakesh Basant and Uma Rani, cited by Upadhya and Vasavi, shows how rare they remained: in 1990-91, '90 per cent of Indian software and service export revenues came from onsite work and only 5 per cent from offshore'.[68] Heeks traced the persistence of onsite work into the 1990s to clients' distrust of unknown Indian suppliers, their focus on cost, projects too small to be worth sending offshore, the need for face-to-face contact and the shortage of experienced managers in India.[69] Even in the year to March 1998, more than half of software services export earnings still came from onsite work.[70] Pratap Chatterjee, the *Multinational Monitor* reporter, wrote in 1995 that until about two years earlier most Indian software exported to the United States had been produced on American soil.[71]

Who went

The people who went were, overwhelmingly, young men. Heeks found that 'Only some 10% of Indian software developers are female'. Women faced particular obstacles, because the work often meant unsocial hours and, above all, going overseas. One manager described onsite programming to him as 'the young, single man's game'. Yet the women Heeks interviewed felt that software offered 'a more relaxed and less discriminatory atmosphere than many other Indian employment options'.[72]

The papers they travelled on are thinly documented for the decade. Heeks wrote that only when American rules tightened in the 1990s were Indian workers channelled into temporary work visas, 'rather than student or training or even tourist visas as in the past'.[73] Ahuja's B1/B2 visa fits that description.[74] The work visa that later defined the trade, the H-1B, belongs to Chapter 13.

The firms invested in the people they sent, and they tried to keep them. Sharma describes the routine: firms 'had to hire the right people, train them for a brief while, send them to American clients to execute projects and then they would come back'.[75] Not all came back. The World Bank observed that 'once the personnel are abroad, the temptations to remain abroad are strong', and that the cost of endlessly training replacements had to be set against low profit margins.[76]

Some firms used the courts. Chatterjee described a programmer whom TCS had brought to San Francisco. When she tried to move to another company, TCS 'sued to enforce her contract, which called for her to pay more than $30,000 to the company if she quit'. The firm's lawyer, the magazine reported, said TCS had sued sixteen former employees successfully and had won her case too.[77]

Ahuja's $800 a month in 1990 is the earliest known figure for the pay of an Indian programmer sent abroad.[78] In the 1990s the US Department of Labor found Indian firms paying their programmers in America '$15,000 a year plus

housing and travel', against '$70,000 plus benefits' for an average American counterpart.[79] Chatterjee wrote in 1995 that a good software engineer in California could earn $100,000 a year while the best in India earned $10,000.[80] Infosys's 1999 filing, citing the consultancy Software Productivity Research, put the average Indian software wage at about 15 per cent of the American rate.[81] The gap was the business. It explains both why clients bought Indian hours and why Indian programmers wanted to be the ones sent to sell them.

Many stayed abroad for good. The World Bank estimated that migration of graduates from 'premier institutions like the IITs' was 'as high as 50 percent', and that nearly 60 per cent of computer science graduates of IIT Bombay and IIT Madras left the country for study or jobs.[82] It placed part of the blame on the trade itself: 'The orientation of the sector toward renting out programmers for the export market has led to the ubiquitous problem of "brain drain"'.[83]

The ledger

By the numbers the decade was a success. Heeks's series, compiled from his own research and the trade magazine *Dataquest*, shows exports rising from $4.0 million in 1980 to $131.2 million in the year to March 1991.[84] The World Bank calculated that exports grew by 35 per cent a year between 1987 and 1991, against 21 per cent for the industry as a whole.[85] The number of firms with sales above $1 million rose from fifteen in 1990 to forty-four in 1992.[86] The exporters had also organised. Nilekani was a co-founder of Nasscom, and Murthy was its president from 1992 to 1994.[87]

The headline numbers overstated what India kept. They were gross foreign-exchange earnings. Net of the foreign exchange spent on Indian workers' travel and living allowances abroad, on marketing, on profits sent home by multinationals and on imported hardware and software, earnings were estimated at about 40 per cent of the gross.[88] The estimate, made in the late 1990s, covers more than travel, but it qualifies the case for the trade. An industry valued as a source of foreign exchange spent much of what it earned abroad, and a good part of that went on the costs of sending people there.

How to judge the decade is a matter of interpretation, and the sources allow three readings. Heeks headed his section on the trade 'Body shopping prison'. In that reading body shopping was low-value labour export that built few higher skills, lost the best people to emigration and left India's industry stuck at the bottom of the work.[89] Heeks also offered a second reading in the same paper: 'Some have therefore been able to use onsite "body shopping" services as a stepping stone to better things.'[90] He set out a possible path from exporting labour, to exporting services done in India, to exporting products, and found support for the first step among individual firms though little in the industry as a whole.[91] The third reading is about policy. Body shopping was a rational answer to import controls and scarce foreign exchange, and it outlived them because clients and managers liked it.[92] Heeks's own summary was that much of India's export growth may have come from the industry's 'dogged following of the low-risk, low-barrier first path'.[93]

A warning in 1994

The World Bank's study named a risk in the model that this book returns to at its end. Among the drawbacks of body shopping it listed this: 'Demand for these skills is likely to be eroded by tools that automate many of these tasks.' Cheaper countries, it added, were entering the low end of the trade.[94] 'Globally,' the report went on, 'a long-term decline in demand for routine or low-level programmers has been forecasted.'[95] Heeks, four years later, also named 'automation from new technology' as one of the things that could hurt this kind of export.[96] He noticed a trap in it. Clients would let work go offshore only when it was tightly specified, and tightly specified work was exactly what automated software tools could do.[97]

These warnings did not come true in the decades that followed. The tools of the 1990s did not erode the demand for Indian programmers. Exports kept growing fast, reaching $1,650 million in the year to March 1998.[98] The epilogue of this book returns to them, and to the caution they suggest about any forecast that machines will soon do the work.

Making programmers

Body shopping changed what a software education in India was for. Heeks found that '85% of workers in Indian software exports are programmers', although the average software project needed less than half its labour from programmers.[99] The industry hired, and the colleges supplied, people for the lowest stages of the work. The trade also drew them away from everyone else. The World Bank estimated that software export needs could 'cannibalize' other sectors 'by taking almost one-half of all new computing course graduates from government-initiated courses'.[100]

The 1980s had shown that India could sell its programmers' time abroad, and that the time sold best when there was plenty of it. The next chapter asks who made those programmers: the new degrees, the private institutes that sold computer courses to anyone who could pay, and the private engineering colleges whose fees became a political fight.

Notes to Chapter 6. Body shopping (1978-1991)

[1] Chatterjee, 'Invasion of the "Body Shoppers"' (1995); Rajaraman, *History of Computing in India* (2012), p. 40 (HCL's full name).

[2] Chatterjee, 'Invasion of the "Body Shoppers"' (1995).

[3] Chatterjee, 'Invasion of the "Body Shoppers"' (1995).

[4] Chatterjee, 'Invasion of the "Body Shoppers"' (1995).

[5] Chatterjee, 'Invasion of the "Body Shoppers"' (1995).

[6] Chatterjee, 'Invasion of the "Body Shoppers"' (1995).

[7] Chatterjee, 'Invasion of the "Body Shoppers"' (1995).

[8] Chatterjee, 'Invasion of the "Body Shoppers"' (1995).

[9] Rajaraman, *History of Computing in India* (2012), p. 29 (Rajaraman's definition); Aspray, 'The Role of Policy' (2010) (Aspray, paraphrased).

[10] Heeks, 'The Uneven Profile of Indian Software Exports' (1998), p. 7.

[11] Upadhya, 'A New Transnational Capitalist Class?' (2004), p. 5141 (Upadhya 2004: the software industry as a whole 'was established mainly on the basis of labour contract services, a system known as "body shopping"'); Upadhya and Vasavi, *Work, Culture, and Sociality in the Indian IT Industry* (2006), p. 21 (Upadhya and Vasavi 2006, who use the term for the lowest end of the trade in the 2000s: agencies in India that recruited engineers and placed them with clients abroad, 'taking a large cut of their salaries as their fee', paying a basic allowance while an engineer was between assignments ('on the bench') and the contracted rate only for time actually worked). The 1980s sources use the word in Heeks's wider sense, for onsite programming of any kind.

[12] Sharma, 'The Long Revolution' (YaleGlobal excerpt, c. 2009).

[13] Aspray, 'The Role of Policy' (2010).

[14] Heeks, 'The Uneven Profile of Indian Software Exports' (1998), p. 2.

[15] Pratap Chatterjee of *Multinational Monitor* wrote in 1995 that 'The first "body shoppers" arrived in Silicon Valley in 1987', citing no source (Chatterjee, 'Invasion of the "Body Shoppers"' (1995)). Sharma puts TCS and Patni staff at customer sites in the 1970s (Sharma, 'The Long Revolution' (YaleGlobal excerpt, c. 2009)), and Aspray dates TCS's first programmers in the United States, sent to install systems software for Burroughs customers (Chapter 4), to 1974 (Aspray, 'The Role of Policy' (2010)). Chatterjee's date and the earlier ones can both be true only if he meant Silicon Valley alone, which he does not say. Nandan Nilekani, who joined Patni in January 1979 and left in July 1981, recalled that Patni 'did a lot of work for clients in the U.S.', and that he spent almost a year at Data General, living in Cambridge, Massachusetts, and taking the bus to its offices in Westborough to work on the company's own software (Computer History Museum, 'Oral History of Nandan Nilekani' (2020), pp. 7-8; title page dated 8 January 2019; CHM reference X9234.2020). Upadhya and Vasavi, citing Balaji Parthasarathy, call bodyshopping 'the primary form of outsourcing in the 1980s' (Upadhya and Vasavi, *Work, Culture, and Sociality in the Indian IT Industry* (2006), p. 10, n. 12, citing Parthasarathy 2005, p. 208). None of these sources names a first contract.

[16] Heeks, 'The Uneven Profile of Indian Software Exports' (1998), p. 12. In Heeks's account the industry began around Nariman Point in the late 1970s, spread north to SEEPZ, and kept Bombay as its centre until at least the mid-1980s. Chapter 10 follows the later move to Bangalore.

[17] Heeks, 'The Uneven Profile of Indian Software Exports' (1998), p. 6.

[18] Sharma, 'The Long Revolution' (YaleGlobal excerpt, c. 2009).

[19] Heeks, 'The Uneven Profile of Indian Software Exports' (1998), p. 14.

[20] Computer History Museum, 'Oral History of N.R. Narayana Murthy, Part 2' (2020), p. 9.

[21] Computer History Museum, 'Oral History of N.R. Narayana Murthy, Part 2' (2020), pp. 5, 9 (Murthy, who remembered the wait for a telephone as about seven years, and the Reserve Bank rule on travel); Computer History Museum, 'Oral History of Nandan Nilekani' (2020), p. 8 (Nilekani remembered a two-year wait); Rajaraman, *History of Computing in India* (2012), p. 33 (Rajaraman gives over two years for 1982).

[22] Rajaraman, *History of Computing in India* (2012), p. 32.

[23] Rajaraman, *History of Computing in India* (2012), p. 29.

[24] Computer History Museum, 'Oral History of N.R. Narayana Murthy, Part 2' (2020), p. 5.

[25] Heeks, 'The Uneven Profile of Indian Software Exports' (1998), p. 8.

[26] Sharma, 'The Long Revolution' (YaleGlobal excerpt, c. 2009).

[27] Heeks, 'The Uneven Profile of Indian Software Exports' (1998), p. 8.

[28] Sharma, 'The Long Revolution' (YaleGlobal excerpt, c. 2009).

[29] Sharma, 'The Long Revolution' (YaleGlobal excerpt, c. 2009), n. 60.

[30] Sharma, 'The Long Revolution' (YaleGlobal excerpt, c. 2009), n. 61.

[31] Sharma, 'The Long Revolution' (YaleGlobal excerpt, c. 2009).

[32] Sharma, 'The Long Revolution' (YaleGlobal excerpt, c. 2009).

[33] Hanna, *Exploiting Information Technology for Development* (1994), p. 32. The table does not state its year; it is probably FY1991-92, so the rupee figures are not converted.

[34] Hanna, *Exploiting Information Technology for Development* (1994), p. 31.

[35] Heeks, 'The Uneven Profile of Indian Software Exports' (1998), p. 11.

[36] Computer History Museum, 'Oral History of N.R. Narayana Murthy, Part 2' (2020), pp. 4-5.

[37] Computer History Museum, 'Oral History of N. R. Narayana Murthy', Part 1 (2018), p. 14.

[38] Computer History Museum, 'Oral History of N.R. Narayana Murthy, Part 2' (2020), p. 5.

[39] Infosys, Form F-1 (1999), p. 7.

[40] Infosys, Form F-1 (1999), pp. 40, 54.

[41] Computer History Museum, 'Oral History of N. R. Narayana Murthy', Part 1 (2018), p. 15.

[42] *Business Standard*, 'Took Risk When I Gave Narayana Rs 10,000' (2024).

[43] Federal Reserve, G.5A Foreign Exchange Rates.

[44] Computer History Museum, 'Oral History of N.R. Narayana Murthy, Part 2' (2020), p. 6.

[45] Hanna, *Exploiting Information Technology for Development* (1994), p. 31.

[46] Infosys, Form F-1 (1999), p. 55.

[47] Computer History Museum, 'Oral History of Nandan Nilekani' (2020), pp. 9-10.

[48] Infosys, Form F-1 (1999), p. 45.

[49] Hanna, *Exploiting Information Technology for Development* (1994), p. 32.

[50] Computer History Museum, 'Oral History of N.R. Narayana Murthy, Part 2' (2020), p. 7.

[51] Hanna, *Exploiting Information Technology for Development* (1994), p. 31.

[52] Hanna, *Exploiting Information Technology for Development* (1994), p. 31.

[53] Heeks, 'The Uneven Profile of Indian Software Exports' (1998), p. 7.

[54] Chatterjee, 'Invasion of the "Body Shoppers"' (1995).

[55] Computer History Museum, 'Oral History of Arjun Malhotra' (2018), pp. 20-21, 23.

[56] Hanna, *Exploiting Information Technology for Development* (1994), p. 31; Heeks, 'The Uneven Profile of Indian Software Exports' (1998), p. 7; Chatterjee, 'Invasion of the "Body Shoppers"' (1995). One scholar describes the trade differently. AnnaLee Saxenian wrote in 2001 that in body-shopping contracts 'the amount of software code is specified in advance and revenue is earned per line of code' (Saxenian, 'Bangalore: The Silicon Valley of Asia?' (2001), p. 10). She cited no source for that sentence, and it disagrees with the direct evidence of the World Bank and Heeks cited above.

[57] Heeks, 'The Uneven Profile of Indian Software Exports' (1998), p. 3.

[58] Heeks, 'The Uneven Profile of Indian Software Exports' (1998), pp. 3-4.

[59] Heeks, 'The Uneven Profile of Indian Software Exports' (1998), pp. 14-15.

[60] Heeks, 'The Uneven Profile of Indian Software Exports' (1998), p. 15.

[61] Hanna, *Exploiting Information Technology for Development* (1994), p. 33.

[62] Infosys, Form F-1 (1999), p. 30.

[63] Chatterjee, 'Invasion of the "Body Shoppers"' (1995).

[64] Rajaraman, *History of Computing in India* (2012), p. 32.

[65] Sharma, 'The Long Revolution' (YaleGlobal excerpt, c. 2009).

[66] Sharma, 'The Long Revolution' (YaleGlobal excerpt, c. 2009) (Citibank set up the SEEPZ unit in 1984 with capital spending of Rs 2 crore, Rs 20 million or about $1.8 million at the 1984 average rate, Federal Reserve, G.5A Foreign Exchange Rates; a senior vice-president, Paul Glaser, hoped it could write much of the $250 million of software the bank bought each year; COSL was inaugurated on 28 October 1985 and by the year to March 1990 was the third-largest software exporter, behind TCS and Tata Unisys; its MicroBanker branch-banking software was priced at $120,000 for ten terminals and $200,000 for twenty); Heeks, 'The Uneven Profile of Indian Software Exports' (1998), p. 13 (TI). Heeks also credits TI with strengthening Bangalore's reputation for software (Heeks, 'The Uneven Profile of Indian Software Exports' (1998), pp. 12-13).

[67] Hanna, *Exploiting Information Technology for Development* (1994), p. 28.

[68] Upadhya and Vasavi, *Work, Culture, and Sociality in the Indian IT Industry* (2006), p. 19, citing Basant and Rani 2004, p. 5318.

[69] Heeks, 'The Uneven Profile of Indian Software Exports' (1998), pp. 7-8. Heeks adds the managers' own taste for quick returns, and puts the threshold for offshore work at about $100,000 a project.

[70] Heeks, 'The Uneven Profile of Indian Software Exports' (1998), p. 6.

[71] Chatterjee, 'Invasion of the "Body Shoppers"' (1995).

[72] Heeks, 'The Uneven Profile of Indian Software Exports' (1998), p. 7. Heeks cites a 1985 comparative study of men and women in software from SNDT Women's University in Bombay (Jayanthi and Madhavan 1985); no copy of it has been located. No count of women working onsite in the 1980s has been located.

[73] Heeks, 'The Uneven Profile of Indian Software Exports' (1998), p. 9.

[74] Chatterjee, 'Invasion of the "Body Shoppers"' (1995).

[75] Sharma, 'The Long Revolution' (YaleGlobal excerpt, c. 2009).

[76] Hanna, *Exploiting Information Technology for Development* (1994), p. 31.

[77] Chatterjee, 'Invasion of the "Body Shoppers"' (1995). The report names the programmer, a private individual; she is not named here. Her own words are not recorded. The article, published in 1995, does not date the case, which may fall in the early 1990s rather than the 1980s. The claims about the lawsuits are the firm's lawyer's, as reported.

[78] Chatterjee, 'Invasion of the "Body Shoppers"' (1995). No figure for an Indian programmer's salary in the 1980s has been located.

[79] Chatterjee, 'Invasion of the "Body Shoppers"' (1995).

[80] Chatterjee, 'Invasion of the "Body Shoppers"' (1995).

[81] Infosys, Form F-1 (1999), p. 42.

[82] Hanna, *Exploiting Information Technology for Development* (1994), pp. 74, 98.

[83] Hanna, *Exploiting Information Technology for Development* (1994), p. 98.

[84] Heeks, 'The Uneven Profile of Indian Software Exports' (1998), p. 2. The same series gives $25.3 million for 1984 and $54.1 million for 1987. Other series for these years differ; Chapter 9 and the data appendix compare them.

[85] Hanna, *Exploiting Information Technology for Development* (1994), p. 28.

[86] Hanna, *Exploiting Information Technology for Development* (1994), p. 31.

[87] Infosys, Form F-1 (1999), pp. 54-55.
[88] Heeks, 'The Uneven Profile of Indian Software Exports' (1998), p. 2.
[89] Heeks, 'The Uneven Profile of Indian Software Exports' (1998), p. 6.
[90] Heeks, 'The Uneven Profile of Indian Software Exports' (1998), p. 14.
[91] Heeks, 'The Uneven Profile of Indian Software Exports' (1998), pp. 15-16.
[92] Heeks, 'The Uneven Profile of Indian Software Exports' (1998), pp. 8, 14.
[93] Heeks, 'The Uneven Profile of Indian Software Exports' (1998), p. 15.
[94] Hanna, *Exploiting Information Technology for Development* (1994), p. 33.
[95] Hanna, *Exploiting Information Technology for Development* (1994), p. 33.
[96] Heeks, 'The Uneven Profile of Indian Software Exports' (1998), p. 14.
[97] Heeks, 'The Uneven Profile of Indian Software Exports' (1998), p. 7.
[98] Heeks, 'The Uneven Profile of Indian Software Exports' (1998), p. 2.
[99] Heeks, 'The Uneven Profile of Indian Software Exports' (1998), p. 8.
[100] Hanna, *Exploiting Information Technology for Development* (1994), p. 98.

Chapter 7. The training economy (1980-1995)

In 1971 Vinod Kumar Bansal, a newly qualified mechanical engineer from Banaras Hindu University, went to work for JK Synthetics in Kota, an industrial town in Rajasthan. He was already helping neighbourhood students prepare for the entrance tests for engineering courses. In the early 1980s, according to a 2023 book by Pradip K. Saha about the education company Byju's, he began teaching eight students sitting around his dining table, and in 1983, when his company shut down, he started teaching full-time.[1] Some of his students got into the Indian Institutes of Technology (IITs), and, as Saha tells it, word spread that '"Bansal sir" had a magic touch'. Parents in Kota sought his help, and then parents from outside Rajasthan. Bansal Classes, as a named institution, dates from 1991.[2] By 2023, on Saha's figures, Kota had more than 150 coaching institutes and about 200,000 students, and more than 1.1 million candidates competed for 17,385 IIT places.[3]

Bansal's dining table stood at the narrow top of a much larger market. In the fifteen years after 1980 the public universities could not grow fast enough to make the people the software trade needed, and three other suppliers grew up beside them, each with a hand from the state. A government panel invented a new degree. Four state governments let private trusts open engineering colleges that charged their students the cost of their education and, the courts later found, sometimes more. And private companies began to sell programming in four-week and twelve-week courses, and from 1987, through franchises, in smaller towns.

This book reads that market as a bet on hours. The export trade sold engineers' time abroad by the hour. A family that paid for a coaching class, a capitation seat or a course in COBOL was buying a claim on hours its child might one day bill. The training companies sold the hours of teaching that made the billable hours. And the courts, in deciding what a college might charge, decided who could afford to place the bet.

A degree for commerce graduates

The state saw the shortage coming. In 1980, V. Rajaraman, the historian of Indian computing, later wrote, the Electronics Commission 'foresaw the problem of scarcity of human resources for the emerging computer industry'.[4] Rajaraman, then the Kanpur professor who had built India's first academic computer science programme, was a member of the Commission, and he chaired the Panel on Computer Manpower Development that it set up.[5]

The panel made two proposals. The first was to expand undergraduate computer science at the IITs and other engineering colleges. The second, which Rajaraman called novel, was a new postgraduate course, the Master of Computer Applications (MCA).[6] It was aimed at the large number of Indian students who took three-year B.Sc degrees in physics, chemistry and mathematics, or B.Com degrees in commerce and economics. These degrees, in Rajaraman's words, 'do not prepare a student for a profession which requires specific skills'.[7] The panel felt that 'India needed a large number of persons to develop business/management information systems for use in organizations both in India and abroad'.[8] The phrase 'and abroad' is worth noticing. In 1980 the state's manpower planners already had foreign employers in view.

The curriculum combined three subjects: computer science 'with emphasis in software engineering and systems analysis', mathematics for operations research and statistics, and management science, so that graduates could talk to managers in their own language. It included 'a semester of actual system design in an organization', a placement in a working office.[9] Where a B.Tech in computer science trained people to design hardware and software, the MCA trained them to design software for business. The government decided at first to fund ten institutions in full to run it.[10] Rajaraman published his curriculum proposal in *CSI Communications*, the journal of the Computer Society of India, in October 1981.[11]

Three decades later Rajaraman judged his own creation generously. 'The availability of well trained MCAs was one of the contributing factors for the rapid growth of software services industry in India,' he wrote.[12] That is the designer assessing the design. The first firm count of MCAs in his history comes from much later: about 7,000 graduated in 1998.[13]

The other half of the panel's plan ran into the narrowness of the elite system. In 1978, the year IIT Kanpur opened the first undergraduate computer science course at any IIT, more than 100,000 students sat the entrance examination for 1,500 IIT places in all.[14] The other IITs soon started computer science degrees of their own, but IIT places stayed scarce, and the World Bank later estimated that as many as half of their graduates emigrated.[15] Rajendra Pawar, a co-founder of the training company NIIT, recalled the position in 1981 this way: 'there were a handful of people coming out of the IITs, but they were not the ones writing programs.'[16]

'A bold decision'

The decisive change came from the states. In the Sixth Five-Year Plan (1980-85), according to a 2010 paper from the National University of Educational Planning and Administration, quoted by the journalist Thomas Manuel, the central and state governments were 'finding it difficult to expand technical education', and 'a few state governments', especially Karnataka, Maharashtra, Tamil Nadu and Andhra Pradesh, 'took a bold decision to permit private registered societies and trusts to establish and run technical institutions on a self-financing basis'.[17] Self-financing meant that the students paid.

The official count of colleges shows how fast the decision worked. In the academic year 1980-81 India had 149 engineering and technology colleges. Five years later, in 1985-86, it had 242, an increase of 62 per cent. Growth then slowed: there were 289 colleges in 1990-91 and 337 in 1994-95.[18] The great expansion of engineering colleges came after 1996 (Chapter 12). The surge of the early 1980s was the first, smaller rehearsal for it.[19] These are counts of institutions, not of students.[20]

New colleges needed teachers, and the people who could teach computing were the people the industry wanted most. In 1985 the Department of Electronics set up a committee under S. Sampath, which 'suggested training teachers in Computer Science for the colleges which were being newly established'.[21] The difficulty did not go away. A World Bank study of 1994 listed 'the scarcity of teachers and trainers, particularly in public education institutes' among the causes of the shortage of software staff.[22] It also blamed the trade's habit of 'renting out programmers' for shortages of teaching staff as well as for the brain drain.[23]

Four weeks of BASIC

Outside the degree system, a different kind of supplier appeared. NIIT was founded in 1981 by Pawar and Vijay Thadani.[24] In a 2022 interview with the magazine *PCQuest*, Pawar recalled the state of computer education when they began: 'At that time, it was non-existent. But the manufacturing industry was already anticipating a problem of talent shortage.' NIIT, he said, 'was born out of that threat'.[25]

The timetable he remembered is precise. 'The first two courses that NIIT announced in 1981 were one, an entry level BASIC. It was a 4-week programme, 5 days a week and 2 hours a day.' The second was a longer programme: 'Entry level COBOL, for 12 weeks.'[26] BASIC was a simple teaching language. COBOL was the language of business data processing, and Pawar called it 'state of the art' at the time. NIIT also taught FORTRAN, the scientific language, to engineers and scientists.[27] Forty hours of BASIC was not an education in computer science. It was an introduction to a trade.

The curriculum grew with its customers. NIIT ran a course for company executives called 'How to buy a computer', and added systems analysis, design and databases. It made an arrangement with the company of Edward Yourdon, whom Pawar called 'the father of software engineering', to teach Yourdon's structured methodology, and Pawar recalled that 'TCS was a very big participant'.[28] Tata Consultancy Services (TCS), a far older firm, was buying training from a company that had existed for only a few years. But the core of the business stayed where it began: 'quite clearly the largest need was for people who could write programs.'[29]

NIIT's growth came from franchising. Rajaraman dates the start of its franchise model, 'to spread their courses to smaller towns in India', to 1987.[30] Pawar described it as an invention: 'We found like-minded entrepreneurs in cities and motivated them to go back to their towns and set up an NIIT centre.' He recalled that 'at its peak (there were

thousands)' every centre had the same look and feel, and that the *Far Eastern Economic Review* 'talked of it as the McDonald's of education'.[31]

The franchisees were small businesspeople, and a 2002 profile in the trade magazine *Dataquest* named some of them. Pratap Agarwal, who had worked as a software engineer in the United States, opened a single NIIT centre in Chandigarh in 1988 and said that by 2002 he ran more than sixty. Others opened centres in Dehradun in 1990 and Faridabad in 1992; the Faridabad franchisee, Captain Virendar Singh, had worked in human resources at TCS and Wipro. Ludhiana followed in 1993 and Amritsar in 1994.[32] The same profile reported that in 1988 the magazine *Asiaweek* had called NIIT 'the biggest computer trainer this side of the Suez'.[33]

NIIT was not alone. Aptech, founded by Atul Nishar in 1986 according to a 2014 profile in *Outlook Business*, was the other company that, on that profile's account, 'defined the space' with NIIT. Looking back, that profile described the 1980s and 1990s as years of queues of students waiting to sign up for multi-year computer courses.[34] Rajaraman called Aptech and NIIT 'pioneers' of 'non-formal training in software development and computer use', which 'also provided the much needed human resources'.[35]

None of these companies has published its enrolments or fees for the 1980s. The best independent picture comes from the World Bank's 1994 study by Nagy Hanna, which reviewed the trade in the early 1990s, drawing on industry surveys. It found that 'an estimated 1,000 proprietary organizations turn out about 80,000 trainees annually from courses ranging from two weeks to two years.' Fees ran from Rs 800 for two months to Rs 8,000 for a one-year course, roughly $25 to $250 at the 1993 exchange rate.[36] The verdict on quality was harsh. 'Industry specialists estimate that 98 percent of the training organizations are below standard,' Hanna wrote, and 'Few of the graduates find employment'. Only ten to fourteen of the larger organisations offered good programmes. Between them they trained about 40,000 people a year, some 30,000 of them in basic two-year programmes meant to qualify them for entry-level programming jobs.[37]

The study also gave the private trade's share. The proprietary organisations, it said, 'now produce almost 85 percent of the programmers and 40 percent of the systems analysts', and the large ones had been doubling their capacity and output over the previous few years.[38] If that estimate is right, by the early 1990s the typical Indian programmer had not been trained by a university at all. He, or less often she, had been trained by a company that sold courses, often in a franchised centre in a provincial town.

The state tried, late, to bring some order to this market. The Department of Electronics created a scheme called DOE Accreditation of Computer Courses (DOEACC), under which accredited private institutes taught standard courses, from basic programming, the 'O level', to a master's-level qualification, the 'C level'. Hanna called the scheme 'recently-developed' in 1994 and saw in it 'flexibility for recycling graduates from various fields into information technology'.[39] Recycling graduates from other fields was exactly what NIIT, Aptech and the MCA had been doing for a decade.

What the employers found

The people who came out of this system went into a job market that was short of them and suspicious of them at the same time. Hanna's study found that in India 'the demand for trained software personnel far outstrips the supply, and the shortage is a major constraint to the industry's expansion'. The shortage placed a heavy training burden on firms and caused high staff turnover.[40] The same study warned that manpower forecasts made by the Department of Electronics and others were unreliable for lack of basic data.[41]

Yet employers trusted few of the public institutions. Only 275 industrial training institutes, polytechnics, colleges and universities were qualified and supported to run computer training. Most of them were short of teachers, equipment and books, and had weak links with industry. In 1991 the number of MCA graduates rose by 50 per cent, yet only 5 per cent were hired by software exporters. One large software house considered students from only thirty universities.[42] The MCA was producing graduates, but few of its graduates were going to the exporters.

The large firms trained their own. The World Bank reproduced, from a background paper written by the National Association of Software and Service Companies (Nasscom), the industry's lobby group, a short account of in-house training at TCS, which had just completed twenty-five years. TCS said that it recruited able graduates from leading institutions and turned them into systems professionals through a demanding initial training programme. Its corporate training centre in Bombay (Mumbai since 1995) had run 6,000 programme days in the previous year, about 120,000 person-days of training, and the company said it spent 7 to 8 per cent of its revenue on training.[43]

Hanna judged that large firms did 'a satisfactory job' of in-house training, while small and medium firms did very little, for fear that trained staff would leave for better-paid jobs.[44]

The World Bank's remedy for the shortage was more of what the 1980s had already produced. It proposed 'recycling graduates in other disciplines into software engineering', inviting multinationals and non-resident Indians into computer education, and building 'centers of excellence in IT education on the lines of IITs'.[45]

A seat with a price

Mohini Jain, a student from Meerut, was told by the management of Sri Siddhartha Medical College in Tumkur, in Karnataka, that she could join its medical degree course in the session beginning in February or March 1991. The management asked her to deposit Rs 60,000 as the tuition fee for the first year and to furnish a bank guarantee for the fees of the years that followed.[46] That was about $2,600 at the year's exchange rate.[47] The Supreme Court later recorded what happened next in a single sentence: 'The petitioner's father informed the management that it was beyond his means to pay the exorbitant annual fee of Rs.60,000 and as a consequence she was denied admission to the medical college.'[48] She also alleged that the college had asked for a further 'capitation fee of rupees four and a half lakhs', that is Rs 450,000. The management 'vehemently denied the same'.[49]

Mohini Jain wanted to be a doctor, not a programmer, but her case belongs in a book about software because of what the judges had to describe in deciding it. By 1991 a seat in a professional college in southern and western India, in engineering as in medicine, had a price, sometimes several prices, and the states had written them into notifications under the same laws. Within a year a larger bench of the Supreme Court heard private engineering and medical colleges from four states say that the ruling in her favour would force them to shut.[50]

Regulating a surge

The states that opened the door to private colleges also tried to police what came through it. In Karnataka an ordinance of 1983 against capitation fees was replaced by the Karnataka Educational Institutions (Prohibition of Capitation Fee) Act, 1984, which took effect from 11 July 1983.[51] A capitation fee was a payment for admission, over and above tuition, and the laws defined it broadly. Maharashtra's Act of 1987, which received the governor's assent on 13 April 1988, defined it as 'any amount, by whatever name called', in excess of the fees the government had prescribed or approved.[52]

Maharashtra's legislators explained themselves in the preamble. Capitation, they wrote, 'is on the increase in the State', and 'this undesirable practice has been contributing to large scale commercialisation of education which is not conducive to the maintenance of educational standards'. They called it 'this evil practice'.[53] Tamil Nadu passed its own Act in 1992, and Andhra Pradesh already had one, from 1983.[54] Each of the four states that had licensed the private colleges thus also had a law against what the colleges were doing.

National policy said the same thing in two minds. The National Policy on Education of 1986, as the Union government quoted it to the Supreme Court, declared that 'the commercialisation of technical and professional education will be curbed', and in the next sentence promised that 'An alternative system will be devised to involve private and voluntary effort in this section of education'.[55] Maharashtra cited the first half of this in its preamble.[56]

In 1987 Parliament gave the All India Council for Technical Education (AICTE) statutory powers.[57] Its functions read like a list of the problems of the 1980s. The Council was to 'make forecast of the needed growth and development in technical education', the manpower planning of the 1980 panel now written into law. It was to 'fix norms and guidelines for charging tuition and other fees' and to 'take all necessary steps to prevent commercialisation of technical education'.[58] Its definition of technical education did not name computer applications or software, and a technical institution was 'an institution, not being a university'.[59] On the text of the Act, then, the franchised courses of NIIT and Aptech fell outside the regulator's reach unless the government notified otherwise. In the early 1990s, Hanna found, the government was 'considering regulating the proprietary training organizations and setting quality standards, but nothing has been done up to now.'[60]

The regulator also came after the surge it was meant to discipline: the number of engineering colleges had already risen from 149 to 242 before the Act was passed.[61] By 1992 the government's Programme of Action admitted an urgent need to strengthen the AICTE. It asked the Council to work towards a 'realistic fee structure' based on the actual cost of education, to encourage private participation in technical education, and to add a 'penal clause' to its own Act to curb malpractices.[62] The Union government told the Supreme Court that the AICTE was approving new

engineering colleges on the condition that they collect no capitation fee, and that under its guidelines an unaided college could recover about 70 per cent of the cost of a student's education in fees, against 20 per cent in a government institution.[63]

Three prices for one seat

The fees on paper were set by the states. In Maharashtra the government prescribed a uniform fee of Rs 6,500 a year in private unaided engineering colleges, raised to Rs 8,500 in 1991 (about $375 at that year's rate). In 1992 it raised the fee only for students from outside the state, to Rs 17,000. The government filled 90 per cent of the seats in private engineering colleges with its own nominees and left 10 per cent to the managements.[64] In Andhra Pradesh the permitted fee in private engineering colleges for 1991-92 was Rs 10,000 a year.[65] These were the official fees. Capitation payments, by their nature, were rarely recorded.

The clearest record of how a seat was priced comes from medicine, in the notification at the centre of Mohini Jain's case. On 5 June 1989 Karnataka fixed the fees that private medical colleges could charge. Students admitted to 'Government seats' paid Rs 2,000 a year. Other Karnataka students could be charged up to Rs 25,000. 'Indian students from outside Karnataka' could be charged up to Rs 60,000.[66] The same course, in the same college, had three prices, and the highest was thirty times the lowest.

The colleges defended the scale. Their counsel told the Court that the students who paid more belonged to a different class: those on government seats were 'meritorious' and the rest were not. The association of private medical colleges argued that they received no aid from the central or state governments, that a five-year medical course cost them about Rs 5 lakh (Rs 500,000) per student, and that because 40 per cent of the seats paid only Rs 2,000, the cost had to fall on the rest. The fees, on this argument, were not profit but recovery of cost.[67]

The bench of two judges, Kuldip Singh and R.M. Sahai, rejected this entirely on 30 July 1992. In Justice Kuldip Singh's words: 'Capitation fee is nothing but a price for selling education. The concept of "teaching shops" is contrary to the constitutional scheme and is wholly abhorrent to the Indian culture and heritage.'[68] 'Education in India has never been a commodity for sale,' he added.[69] The judgment named the social effect plainly. Capitation brought 'a clear class bias': 'A poor student with better merit cannot get admission because he has no money whereas the rich can purchase the admission.'[70] The Court held that the right to education flowed directly from the right to life under Article 21 of the Constitution. The Rs 60,000 charged to outsiders was 'not tuition fee but in fact a capitation fee', and a private college run with government recognition should be prohibited from charging more than the government's Rs 2,000 from any student.[71]

The case concerned a medical college, but its reasoning applied to every recognised private professional college, and the engineering colleges understood that at once.

Hyderabad, 1992

What the private engineering colleges did when the law left them room is recorded in the judgment that followed. Andhra Pradesh had 'a number of private engineering colleges', the Supreme Court noted. Until the academic year 1992-93 all their seats were filled by the convenor of the state's common entrance test. The managements had no say in admissions and were allowed only to charge a fee 'relatively higher than the fees charged in the Government Engineering Colleges'.[72] On 15 April 1992 the state inserted a new section, 3-A, into its Act. The colleges took it to mean that they could admit half their students 'according to their choice, irrespective of merit, so long as they have qualified in the entrance test'.[73] The Court's comment was dry: 'It is obvious that such a stand meant collection of capitation fee as much as they could.'[74]

'There was an uproar among the student and teaching community against such admissions,' the judgment continues. The government told the colleges on 26 July 1992 not to admit anyone until rules had been made. By then the colleges said they had already filled their half. The convenor, for his part, had allotted students to only half of each college's seats, 'thereby sending an explicit signal that the colleges were free to fill up the rest on their own'.[75] The colleges then charged the same Rs 10,000 fee to both halves. 'Thus they have reaped a double advantage,' the Court observed.[76]

The students admitted in this way were the rank and file of the training economy, and the judgment records them only as a class. Their counsel, Shanti Bhushan, argued that 'they were innocent parties and had obtained admission in a bona fide belief that their admissions were being made properly'. By the time the Supreme Court

ruled they had been studying for months.[77] The Court let them stay. It ordered the colleges to charge 'the 50% "free students"', for the rest of their course, only the fee of the university engineering colleges, which it called the 'government fee', and to pay the balance already collected that year into the government's account within six weeks or lose their recognition and affiliation.[78]

The free seat and the payment seat

The larger question reached a bench of five judges in *Unni Krishnan, J.P. v. State of Andhra Pradesh*, decided on 4 February 1993. The petitioners, Justice B.P. Jeevan Reddy wrote, were 'running medical/engineering colleges in the States of Andhra Pradesh, Karnataka, Maharashtra and Tamil Nadu', and they said that if *Mohini Jain* was followed 'they will have to close down; no other option is left to them'.[79]

Both sides put their case in strong terms. The colleges argued that the law of supply and demand should be given free play, and that 'only when there was profit motive that persons with means would come forward to open more and more schools and colleges'. At the least, they said, a citizen had the right to set up 'a self-financing educational institution, which institution might also be described as one providing cost-based education'.[80] Against them stood, among others, the Indian Medical Council and the AICTE itself. Their argument, as the headnote summarised it, was that 'a capitation fee did not cease to be a capitation fee just because it was called as cost-based education or by calling the Institution concerned as a self-financing Institution'.[81] The Union government took a middle line. Private institutions could not be forced to charge only government fees, it argued, and 'It would be unrealistic and unwise to discourage private initiative' in providing education.[82]

The Court partly overruled *Mohini Jain*. It agreed that a right to education flowed from the right to life, but held that its content was limited by what the state could afford. Citizens could not 'demand that the State provide adequate number of medical colleges, engineering colleges and other educational institutions to satisfy all their educational needs'.[83] The right to free education ran to the age of fourteen. Above that, private colleges without state aid could charge more than government colleges, within a ceiling, but 'commercialisation of education' was still not allowed.[84]

To give this effect the Court wrote a scheme, to apply from 1993-94. It covered every private professional college, defined to include 'colleges of engineering and colleges and institutions imparting technical education including electronics, computer sciences'.[85] At least half of the seats in each college were to be 'free seats', filled by the government or university on merit at the government fee. The rest were 'payment seats', filled by those prepared to pay a fixed higher fee, but still in order of merit. 'There shall be no quota reserved for the management or for any family, caste or community which may have established such college.'[86] All admissions would run through one state notification for each kind of college.[87] The payment half, in effect, was to pay for the free half.

The scheme was a compromise between the market reading and the commercialisation reading of the 1980s. It accepted that private colleges would charge their students much more than the state did, and it tried to keep merit as the rule for deciding who could pay. It governed admissions to private professional colleges for the rest of the period this chapter covers, and for some years after. On 31 October 2002 eleven judges of the Supreme Court, in *T.M.A. Pai Foundation v. State of Karnataka*, held that the scheme 'was not correct' and overruled it, leaving unaided colleges to fix their own fees, though still without capitation fees or profiteering (Chapter 12).[88]

Who paid

The question of who could enter the new profession has a partial answer in the legal record, and the answer is those whose families could pay. Karnataka's notification charged an outsider thirty times what a government-seat student paid for the same medical course.[89] The Andhra colleges filled half their engineering seats at their own discretion, and the Court found it 'not possible to investigate or verify for what consideration those admissions were made'.[90] A family that wanted an engineering seat in the south or west, and could not win one on merit in a government college, could buy one. Mohini Jain's father, in the one sentence the record gives him, said that he could not.[91]

The training companies offered a cheaper door. Their courses cost from a few hundred to a few thousand rupees, and in the World Bank's phrase they recycled graduates from other fields into computing.[92] They were also, on Hanna's evidence, a gamble: most institutes were below standard and few of their students found jobs.[93] The franchise spread the gamble to towns like Chandigarh, Dehradun and Ludhiana, where local entrepreneurs opened centres under the NIIT name.[94]

By the mid-1990s the training economy had made software a career that could be bought, at every price from a two-month course at a franchised centre to a payment seat in a private engineering college. Each purchase was a bet that the hours a graduate would one day sell would repay the hours bought to make the graduate. *Unni Krishnan* had settled who could place the larger bet, and on what terms. Whether it paid would be settled by the trade.

Notes to Chapter 7. The training economy (1980-1995)

[1] Saha, 'How Byju's Exploited the Indian "Rat Race" Mentality' (2023); live-checked 2026-10-13.

[2] Saha, 'How Byju's Exploited the Indian "Rat Race" Mentality' (2023); live-checked 2026-10-13. Saha's account is the only one of Bansal's beginnings used here.

[3] Saha, 'How Byju's Exploited the Indian "Rat Race" Mentality' (2023); live-checked 2026-10-13.

[4] Rajaraman, *History of Computing in India* (2012), p. 34.

[5] Rajaraman, *History of Computing in India* (2012), p. 34.

[6] Rajaraman, *History of Computing in India* (2012), p. 34.

[7] Rajaraman, *History of Computing in India* (2012), p. 34.

[8] Rajaraman, *History of Computing in India* (2012), p. 34.

[9] Rajaraman, *History of Computing in India* (2012), p. 34.

[10] Rajaraman, *History of Computing in India* (2012), p. 35.

[11] Rajaraman, *History of Computing in India* (2012), pp. 35, 66.

[12] Rajaraman, *History of Computing in India* (2012), p. 35.

[13] Rajaraman, *History of Computing in India* (2012), p. 56. No copy of the 1980 panel's report has been located, and no series of MCA admissions for the 1980s.

[14] Rajaraman, *History of Computing in India* (2012), p. 34.

[15] Rajaraman, *History of Computing in India* (2012), p. 34; Hanna, *Exploiting Information Technology for Development* (1994), p. 74.

[16] Rajguru, 'Taking Computer Education to the Masses' (2022), p. 48.

[17] Manuel, 'India's Engineering Education Story' (2015); live-checked 2026-10-13.

[18] MoSPI, 'Number of Recognised Educational Institutions', Table 1.17.

[19] MoSPI, 'Number of Recognised Educational Institutions', Table 1.17; Manuel, 'India's Engineering Education Story' (2015); live-checked 2026-10-13. The national series is not broken down by state, so it cannot show how much of the surge came from the four states the NUEPA paper names.

[20] Ministry of Human Resource Development, *National Policy on Education 1986: Programme of Action 1992*, p. 77; MoSPI, 'Number of Recognised Educational Institutions', Table 1.17. The government's own figure for seats is uncertain. The Programme of Action that accompanied the education policy, published in 1992, said there were 'over 200' recognised technical institutions at degree level with an annual admission capacity of 40,000, yet the official series already counted 289 engineering colleges in 1990-91 and 300 in 1992-93. The Programme's numbers match the mid-1980s count better and probably repeat older data. No reliable series of engineering admissions by year and state for this period has been located.

[21] Rajaraman, *History of Computing in India* (2012), p. 36.

[22] Hanna, *Exploiting Information Technology for Development* (1994), p. 50.

[23] Hanna, *Exploiting Information Technology for Development* (1994), p. 98.

[24] Rajaraman, *History of Computing in India* (2012), p. 42; Rajguru, 'Taking Computer Education to the Masses' (2022), p. 48; NIIT University, 'Founders' (Thadani as co-founder).

[25] Rajguru, 'Taking Computer Education to the Masses' (2022), p. 48. Pawar was speaking some forty years after the founding.

[26] Rajguru, 'Taking Computer Education to the Masses' (2022), p. 48.

[27] Rajguru, 'Taking Computer Education to the Masses' (2022), p. 48.

[28] Rajguru, 'Taking Computer Education to the Masses' (2022), p. 48.

[29] Rajguru, 'Taking Computer Education to the Masses' (2022), p. 48.

[30] Rajaraman, *History of Computing in India* (2012), p. 42.

[31] Rajguru, 'Taking Computer Education to the Masses' (2022), pp. 48-49. No copy of the *Far Eastern Economic Review* article has been located. *Dataquest* dates the comparison to November 2000 and gives it as 'McDonald's of Software Business' (Narayan, 'The Men Behind NIIT' (2002); live-checked 2026-10-13).

[32] Narayan, 'The Men Behind NIIT' (2002); live-checked 2026-10-13.

[33] Narayan, 'The Men Behind NIIT' (2002); live-checked 2026-10-13.

[34] Mahalingam, 'Why Aptech Is a Good Bet' (2014); live-checked 2026-10-13.

[35] Rajaraman, *History of Computing in India* (2012), p. 42.

[36] Hanna, *Exploiting Information Technology for Development* (1994), p. 51; Reserve Bank of India, *Handbook of Statistics on Indian Economy*, 1993 calendar-year average rate (Rs 31.45 per US dollar); Hanna does not state the survey year.

[37] Hanna, *Exploiting Information Technology for Development* (1994), p. 51.

[38] Hanna, *Exploiting Information Technology for Development* (1994), p. 51.

[39] Hanna, *Exploiting Information Technology for Development* (1994), p. 71.

[40] Hanna, *Exploiting Information Technology for Development* (1994), p. 50.

[41] Hanna, *Exploiting Information Technology for Development* (1994), p. 50.

[42] Hanna, *Exploiting Information Technology for Development* (1994), p. 51.

[43] Hanna, *Exploiting Information Technology for Development* (1994), p. 72.

[44] Hanna, *Exploiting Information Technology for Development* (1994), pp. 72, 52.

[45] Hanna, *Exploiting Information Technology for Development* (1994), p. 73.

[46] *Mohini Jain v. State of Karnataka* (1992), p. 664.

[47] Reserve Bank of India, *Handbook of Statistics on Indian Economy*, 1991 calendar-year average rate (Rs 22.69 per US dollar).

[48] *Mohini Jain v. State of Karnataka* (1992), p. 664.

[49] *Mohini Jain v. State of Karnataka* (1992), p. 664.

[50] *Unni Krishnan, J.P. v. State of Andhra Pradesh* (1993), pp. 623-24.

[51] *Mohini Jain v. State of Karnataka* (1992), p. 663.

[52] *Maharashtra Educational Institutions (Prohibition of Capitation Fee) Act, 1987*.

[53] *Maharashtra Educational Institutions (Prohibition of Capitation Fee) Act, 1987*.

[54] *Unni Krishnan, J.P. v. State of Andhra Pradesh* (1993), pp. 686, 633.

[55] *Unni Krishnan, J.P. v. State of Andhra Pradesh* (1993), p. 673.

[56] *Maharashtra Educational Institutions (Prohibition of Capitation Fee) Act, 1987*.

[57] Ministry of Human Resource Development, *National Policy on Education 1986: Programme of Action 1992*, p. 77.

[58] *All India Council for Technical Education Act, 1987*, s. 10.

[59] *All India Council for Technical Education Act, 1987*, ss. 2(g), 2(h). Section 2(g) defines technical education as 'programmes of education, research and training in engineering, technology, architecture, town planning, management, pharmacy and applied art and crafts', with any other field the central government adds by notification. How such courses were treated in practice is not recorded; the reading of the statute is supported by Hanna (Hanna, *Exploiting Information Technology for Development* (1994), p. 51).

[60] Hanna, *Exploiting Information Technology for Development* (1994), p. 51.

[61] MoSPI, 'Number of Recognised Educational Institutions', Table 1.17; *All India Council for Technical Education Act, 1987*.

[62] Ministry of Human Resource Development, *National Policy on Education 1986: Programme of Action 1992*, pp. 77-78.

[63] *Unni Krishnan, J.P. v. State of Andhra Pradesh* (1993), p. 673.

[64] *Unni Krishnan, J.P. v. State of Andhra Pradesh* (1993), p. 639; Reserve Bank of India, *Handbook of Statistics on Indian Economy*, 1991 calendar-year average rate (Rs 22.69 per US dollar).

[65] *Unni Krishnan, J.P. v. State of Andhra Pradesh* (1993), p. 691. No contemporary figure for capitation payments in engineering has been located.

[66] *Mohini Jain v. State of Karnataka* (1992), pp. 663-64.

[67] *Mohini Jain v. State of Karnataka* (1992), pp. 664-65.

[68] *Mohini Jain v. State of Karnataka* (1992), p. 670.

[69] *Mohini Jain v. State of Karnataka* (1992), p. 672.

[70] *Mohini Jain v. State of Karnataka* (1992), p. 673.

[71] *Mohini Jain v. State of Karnataka* (1992), pp. 669, 680-81.

[72] *Unni Krishnan, J.P. v. State of Andhra Pradesh* (1993), p. 636.

[73] *Unni Krishnan, J.P. v. State of Andhra Pradesh* (1993), p. 636.

[74] *Unni Krishnan, J.P. v. State of Andhra Pradesh* (1993), p. 636.

[75] *Unni Krishnan, J.P. v. State of Andhra Pradesh* (1993), pp. 636-37.

[76] *Unni Krishnan, J.P. v. State of Andhra Pradesh* (1993), p. 691.

[77] *Unni Krishnan, J.P. v. State of Andhra Pradesh* (1993), p. 691.

[78] *Unni Krishnan, J.P. v. State of Andhra Pradesh* (1993), p. 692. The judgment does not spell out which half it meant by 'free students'.

[79] *Unni Krishnan, J.P. v. State of Andhra Pradesh* (1993), pp. 623-24.

[80] *Unni Krishnan, J.P. v. State of Andhra Pradesh* (1993), headnote, p. 597.

[81] *Unni Krishnan, J.P. v. State of Andhra Pradesh* (1993), headnote, p. 599.

[82] *Unni Krishnan, J.P. v. State of Andhra Pradesh* (1993), p. 673.

[83] *Unni Krishnan, J.P. v. State of Andhra Pradesh* (1993), pp. 654-55.

[84] *Unni Krishnan, J.P. v. State of Andhra Pradesh* (1993), headnote, p. 594.

[85] *Unni Krishnan, J.P. v. State of Andhra Pradesh* (1993), p. 682.

[86] *Unni Krishnan, J.P. v. State of Andhra Pradesh* (1993), pp. 683-84.

[87] *Unni Krishnan, J.P. v. State of Andhra Pradesh* (1993), pp. 684-85.

[88] *T.M.A. Pai Foundation v. State of Karnataka* (2002), paras. 45, 56-57.

[89] *Mohini Jain v. State of Karnataka* (1992), pp. 663-64.

[90] *Unni Krishnan, J.P. v. State of Andhra Pradesh* (1993), p. 692.

[91] *Mohini Jain v. State of Karnataka* (1992), p. 664.

[92] Hanna, *Exploiting Information Technology for Development* (1994), pp. 51, 71.

[93] Hanna, *Exploiting Information Technology for Development* (1994), p. 51.

[94] Narayan, 'The Men Behind NIIT' (2002); live-checked 2026-10-13.

Chapter 8. Satellite and permit (1984-1991)

On a day in 1985 that he called memorable when he wrote about it in 2013, the journalist Janardhan Roye found his Ambassador car stuck behind a bullock cart outside his office on Miller Road in Bangalore (Bengaluru since 2014). The cart was heavily laden, and it was struggling up the steep slope at the gate of an office block called Sona Towers. Traffic piled up and horns blared until security staff came to the aid of the cart's driver.[1] The people in the neighbouring offices, he recalled, were less interested in the traffic than in the wooden boxes being unloaded.[2]

The boxes were addressed to the Bangalore office of Texas Instruments (TI), the semiconductor company from Dallas. They held the equipment for a two-way satellite link between India and the United States, for data transmission and software development.[3] Among the onlookers, Roye recalled, was Sam Pitroda, the engineer who led the government's telecommunications research centre, C-DOT, and who was often to be seen at its offices on the floor above Roye's. When the contents of the boxes became known, Roye remembered him shaking his head and smiling: 'This can happen only in India.'[4] The building's owner, the Sona Group, tells the same story of a satellite dish arriving by bullock cart.[5]

The dish was small by later standards. Roye gives its capacity as 64 kilobits a second, and he described the antenna as 'like a whale surfacing'.[6] What made it remarkable was not its speed but its permission. Until then, telephone networks and overseas links had been a government monopoly, run by a department whose staff were civil servants. TI's dish was a private line out of the country, approved by the state.[7]

The years from the computer policy of November 1984 to the first Software Technology Parks of 1990 and 1991 were the years of Rajiv Gandhi's government and its immediate successors, and they are often remembered as the moment India's information technology industry began. The record supports a narrower claim. The permit regime was loosened in pieces, and much of what changed was paperwork: duties, licences and the definitions of what counted as an export. The changes that moved software work into India, rather than sending Indian programmers abroad to do it, were mostly about telecommunications, and they came late and slowly. A satellite dish on a bullock cart is a fair emblem of both halves.

The permit, loosened

Rajiv Gandhi had learned what computers could do at a sporting event. When the Asian Games were held in Delhi in 1982, he, the son of the prime minister, Indira Gandhi, and in the account of V. Rajaraman, the historian of Indian computing, 'an amateur electronics buff', was given overall charge of organising them. He decided that computers should draw up the schedules, keep the event records, announce the results and do the clerical work. Locally made DCM computers served as terminals at the venues, connected to a Hewlett-Packard server, and software engineers of the government's National Informatics Centre wrote the whole system in six months. The computerisation of the Games was, in Rajaraman's words, 'a resounding success', and it brought the centre's director general, N. Seshagiri, close to Rajiv Gandhi.[8] Organising the Games also brought him into contact with the private computer makers, who told him of their difficulties with 'myriad rules and regulations'.[9] After Indira Gandhi's assassination in 1984 he became prime minister, with Seshagiri as an informal adviser.[10] Rajaraman describes him as a man of the post-independence generation who 'was open minded regarding private enterprise'.[11] The new computer policy followed within weeks. Dataquest's chronology dates it to 19 November 1984.[12]

Its effect on hardware was to turn Indian computer making into assembly. For exporters it did three things. It recognised software as an 'industry', which opened the way to bank loans and duty exemptions. It cut the customs

duty on computers imported to write software for export, which, Rajaraman writes, 'Earlier it used to be over 100%'. And it counted engineers sent to work at clients' sites abroad as software exporters.[13]

The policy also carried a promise. According to a World Bank study of 1994 by Nagy Hanna, the government announced its Computer Policy Plan in November 1984 and 'expected software exports to reach US$258 million (or 0.5 percent of the world market) by 1990'.[14] It was a target built, the Bank thought, on 'limited but alluring data' about the world market.[15]

Two years later, in December 1986, came the Policy on Computer Software Export, Software Development and Training; in the Bank's account, 'A major reform in computer software' followed its adoption.[16] Its terms show how far the old logic of foreign exchange still governed the trade. A firm could import a computer duty-free to write software for export, but it had to 'earn 250% of the cost of the imported computer within 4 years'. Where no foreign exchange had been spent, for example 'a non resident Indian bringing a computer with his earnings', the obligation fell to 150 per cent.[17] The rule descends from the bargain Tata Consultancy Services (TCS) had struck over its Burroughs machine in the 1970s: the state would allow imports if they paid for themselves in dollars.

Two clauses pointed somewhere new. 'Software developed in India could be exported using communication systems such as satellite and cable.' And the state-owned Computer Maintenance Corporation's (CMC's) network of IBM mainframes, called Indonet, was opened to exporters at a concessional tariff.[18] Foreign capital was eased too. From 1986 foreign firms could own all of an export-oriented software project, though they were still held to 40 per cent of firms selling software in India.[19] That 40 per cent was the figure that had driven IBM out in 1978. It was now waived for anyone who exported.

The historian Dinesh Sharma sums up the effect of the two policies cautiously. Imports of hardware and software were liberalised, 'though the actual import of hardware still took time'. Bringing software under the Open General Licence, the list of goods that could be imported without a separate permit, meant that 'New software packages and tools started becoming available'.[20] The World Bank was less kind about the system that administered all this. In its account, policy had 'fundamentally focused on ways to make export figures rise', while 'troublesome delays exist for import of computers and spare parts'. 'On the positive side,' it concluded, 'it has pushed firms into exporting. On the negative side, this pushing has been short-term oriented'.[21]

The dish

TI's link required a negotiation inside the government before it required a bullock cart. Rajaraman, who had served on the government's Electronics Commission, records it briefly. 'The DoE negotiated with the Department of Telecommunication (DoT) in 1985 to allow Texas Instruments (TI) to set up a dedicated satellite communication link to its centre at Dallas, USA.' Over that link, he adds, TI's Bangalore centre sent the software tools it developed to Dallas.[22] The Department of Electronics (DoE), headed by scientists and engineers since 1970, promoted computing; the Department of Telecommunications ran the telephones.[23] Rajaraman judged the result 'the first step in breaking the monopoly of the DoT in communications', with far-reaching consequences for offshore development centres and, later, the Software Technology Parks.[24]

The World Bank added a detail. TI 'obtained a private Intelsat Business Service earth station so that its programmers can communicate with other TI operations using satellite links'. The special measures and approvals TI won, the Bank wrote, 'helped make India more attractive' to other foreign firms, among them Hewlett-Packard, Digital Equipment Corporation (DEC), British Telecom, Philips and Westinghouse.[25] Richard Heeks of the University of Manchester, who studied the industry through fieldwork in India, reported that 'TI was known to have cut through many red tape barriers, particularly on imports', and that its arrival was 'constantly reported in the home and international media'.[26]

The date is disputed by a year. TI's own milestone list puts the start of its Indian operations in 1985, the year of the negotiation and of Roye's bullock cart; Heeks dates TI's arrival to 1986. The likeliest reading is that TI arrived in 1985 and its satellite link was working by 1986.[27]

TI's claim to have been first is narrower than the one often made for it. In 1985, its press release says, TI became 'the first multinational to establish a software design center in India and the first to install and use a satellite dish for exporting software'.[28] The World Bank's account of the private earth station supports the second half. The first half needs care. Citibank had set up a software unit in the Santa Cruz Electronics Export Processing Zone (SEEPZ) in Bombay (Mumbai since 1995) in 1984, and it was inaugurated as Citicorp Overseas Software Limited on 28 October

1985; in its early years it wrote software for the bank's own use in America and other regions.[29] What TI had first was the private line.

What TI did with it was not the work most Indian exporters did. Its Bangalore engineers wrote software tools, in Rajaraman's account.[30] From 1989, according to the company, the centre designed mixed-signal integrated circuits, chips that combine analogue and digital functions, and built libraries for computer-aided chip design; by 2005 its Bangalore engineers had filed close to 300 patent applications.[31] Rajaraman notes the irony that Indian engineers 'could design large scale integrated circuits, including combined analog and digital systems for TI and other overseas companies but had no foundry to fabricate them in India'.[32] TI showed that a multinational could put design work of a high order in India, which was a different thing from Indian firms doing it.

Heeks thought TI's arrival helped settle the industry in Bangalore rather than Bombay, a shift whose causes Chapter 10 examines.[33] Roye quotes C.S. Murali of the Indian Institute of Science on the dish: 'It turbocharged the IT revolution'.[34] Vijay Chandru, a scientist whom Roye also quotes, argues that Bangalore had been a technical city well before the multinationals came.[35]

The wires that were missing

The dish mattered because the ordinary wires were so poor. 'In 1982 there was a waiting time of over 2 years to get a land line telephone connection,' Rajaraman writes. Telephones were run by a government department whose staff were civil servants and whose budget was part of the national budget, which in his account meant low investment and poor service.[36] Under Rajiv Gandhi two reforms followed. The telephones of Bombay and Delhi were handed to a public company, Mahanagar Telephone Nigam Ltd (MTNL), with 'operational autonomy'. A second company, Videsh Sanchar Nigam Ltd (VSNL), was set up in 1986 to run overseas communication.[37]

The more visible change came from C-DOT. Formed in 1984 under Pitroda, it was given the task of designing small digital telephone exchanges that could survive rural India's poor power supply and its temperatures of 45 degrees in summer and below freezing in winter. C-DOT 'recruited a highly motivated group of engineers', and they produced a rugged exchange 'in a record time of 3 years'.[38] Alongside it spread the public call office, a telephone booth with a microprocessor that printed the cost of each call so that an attendant could collect it. Because a booth cost little to set up, many small entrepreneurs opened them in towns, villages and on highways. 'Every nook and corner of India was connected within 5 years,' Rajaraman writes. 'The power of microcomputer was visible to all'.[39]

None of this gave a software exporter a cheap line to a client in New Jersey. The overseas gateways stayed with VSNL, and the history published by Software Technology Parks of India, the agency that later took over the parks, records that data links carried a substantial cost.[40] In 1994 the World Bank still listed poor communications links among the industry's weaknesses.[41] It argued that faster, more open data services 'would open major opportunities for off-shore software services activities in India, thereby reducing dependence on body-shopping (on-site delivery of services) and the consequent brain-drain'.[42]

The numbers show how little the policies of 1984 and 1986 had moved the work. Heeks found that in 1988 about 65 per cent of export contracts were carried out wholly at the client's site abroad.[43] Sharma records that after 1986 some firms began 'remote software development', working from India on clients' machines, 'But still there were impediments such as high telecom costs'.[44] What India sold, in other words, was still mostly engineers' time, billed where the client was. The dish and, later, the parks would change where those hours were worked. They did not change what was being sold.

Fifty counters

Inside India, the most important computer of the decade sold railway tickets. The project matters to this book for what it did to clerks.

The argument over computers and jobs was still live in 1984. When the Reserve Bank of India's Rangarajan committee recommended that every bank do its back-office work by computer, the growth of computing in the banks was, in Rajaraman's words, 'a red flag to the trade unions', and they observed 1984 as an 'anti-computerization year'.[45] Dataquest's chronology adds that in 1984 the Bharatiya Mazdoor Sangh declared an 'Anti Computer Year' in protest against the bank proposal, with 1 March observed as Anti-Computer Day.[46] The unions were also bargaining. A biographical note on Suresh Dhopeshwarkar, a bank union leader, held by the Indian Labour Archives, records that in 1983 his union 'reached the first ever agreement on computerisation to meet the new

challenges'.[47] The state itself had been cautious. The minicomputer policy of 1978 had said it would 'continue to be restrictive in allowing use of computers in areas where labour displacement was likely to be involved'.[48]

The railways faced a problem of scale that no one could solve by hand. In 1984, Rajaraman writes, Indian Railways handled 'over 5 million passengers travelling in over 600 long distance trains with around 50,000 reservation requests'. Passengers stood in long queues. 'Clerks kept numerous ledgers, one for each train', and juggled between them as passengers changed their minds.[49] CMC proposed a computerised system. The railways accepted. The project began in 1984, and 'by 1986 a reservation system was developed and implemented at the New Delhi booking office. It had 50 counters and customers could go to any counter to get a reservation done for any train'.[50] The official audit of the system, by the Comptroller and Auditor General, gives an exact start for a smaller first step: the pilot, called IMPRESS, 'was launched on 15 November 1985' at New Delhi. The government dates the introduction of the reservation system proper to 1986.[51]

To the public, Rajaraman writes, it was 'almost a miracle'. Instead of waiting hours and running from counter to counter, a passenger could get a confirmed reservation within 30 minutes, book connecting and return journeys, and cancel for an immediate refund.[52] For the clerks the change came in the evening's accounts. 'The counter clerks were also happy,' Rajaraman writes, 'as they could reconcile the collected cash against issued tickets with the help of the computer within 30 minutes after closing time whereas in the manual system it used to take at least two hours.'[53] And they kept their jobs. 'There was also no reduction of staff; in fact there was an increase as more counters were operated with more clerks working simultaneously.'[54]

The mechanism is worth stating because the book returns to it. Demand for reservations had been held down by the queues. When the computer let any counter sell a ticket on any train, the railways opened more counters and needed more clerks to staff them. The machine raised each clerk's output, and the extra output was absorbed by demand that had been waiting. Whether that pattern holds for a new technology depends on how much demand is waiting, and that question runs through the last part of this book.

The system ran on a cluster of DEC VAX minicomputers, with terminals at the counters, and was written in FORTRAN. 'The entire software effort was by Indian software engineers', Rajaraman writes, with no foreign consultants involved.[55] He also claims it was the first such transaction system in the world to run on a VAX cluster.[56] Its effect on opinion he does not doubt. 'There was an attitudinal change among both the general public and the white collar workers about computerization', and some bank unions accepted the computerisation they had opposed.[57] He groups the reservation counters with the public call offices and the first electronic voting machines, used in a by-election in Kerala in 1982. Together, he writes, they 'changed the perception of policy makers and the general public about computers and their relevance to India'.[58]

The Cray that did not come

The decade's best-known story about computers and national pride concerns a machine India did not get. It is usually told as a simple sequence: the United States refused India a Cray supercomputer, so India built its own. The record is more tangled.

In 1984 the Ministry of Human Resource Development gave the Indian Institute of Science (IISc) in Bangalore Rs 500 million, about $42 million at the rate Rajaraman uses of Rs 12 to the dollar, to set up a Supercomputer Education and Research Centre.[59] A national committee visited two companies in the United States and one in Japan and chose a Cray Y-MP. Cray signed a contract to install it at the institute by 1989, and 'Cray was confident of getting a licence to export the computer'. A team led by a US State Department official visited the institute and agreed conditions for its use.[60] Rajaraman, who chaired the centre from 1982 to 1994, 'participated in the entire exercise'. He records the outcome flatly: 'The US government did not give an export licence to Cray before the date specified in the contract'.[61] IISc cancelled the order and bought other machines instead, among them a Control Data Cyber 992 mainframe and, in 1992, an Indian-made parallel computer.[62] 'The denial of Cray was a blessing in disguise', Rajaraman concluded, 'as the money saved was used to buy state-of-the-art powerful parallel computers'.[63]

A second sale, for weather forecasting, was argued over in public in the United States. After the *Christian Science Monitor* reported the planned sale in December 1986, it printed on 8 January 1987 a reply from Dean Churchill, a meteorologist at the University of Washington who studied the monsoon. He thought that 'acquisition of a Cray or Cyber supercomputer might not solve the problem' of forecasting it, because of the lack of observations over the Indian Ocean and the gaps in scientific understanding.[64] That sale appears to have gone ahead in a smaller form: a

Cray X-MP/14 was reportedly imported in October 1988 for the National Centre for Medium Range Weather Forecasting.[65] If so, India was both denied one Cray and sold another.

In 1986 the Science Advisory Committee to the Prime Minister, chaired by the chemist C.N.R. Rao, set up a committee under Rajaraman on designing high-performance computers. It was formed, he writes, 'as a consequence of the difficulties encountered by India in buying supercomputers from the USA and Japan', and it proposed 'a mission mode project to build parallel computers with a speed of a giga-flop or above'.[66] A parallel computer divides a problem among many processors working at once; a gigaflop is a billion arithmetic operations on fractional numbers a second. In 1988 the DoE set up the Centre for Development of Advanced Computing (C-DAC) at Pune with Rs 300 million, about $22 million at that year's average rate, to build such machines. 'The project was successfully completed in 1991.'[67]

C-DAC was not alone. Between 1985 and 1992 the National Aeronautical Laboratory in Bangalore built FLOSOLVER for fluid dynamics, the Defence Research and Development Organisation built PACE in Hyderabad, and the Bhabha Atomic Research Centre built ANUPAM in Bombay.[68] The popular version, in which a single refusal leads straight to a single Indian supercomputer, owes much to later journalism; a 2017 web feature on C-DAC and PARAM, for example, says that the effort began when 'the US stopped the export of a Cray supercomputer'.[69] The defensible version is that American export controls, among other difficulties, pushed the government to fund parallel computing, and that several laboratories were already working on it.

What the episode produced, besides machines, was people. All these projects, Rajaraman writes, 'generated a large pool of engineers adept in designing and programming parallel computers'. Around ten IISc students earned doctorates in the field.[70] C-DAC also built a chip, called GIST, that let personal computers display most Indian scripts.[71]

Laboratories as schools

The state projects of these years taught more people than they sold products. Most were paid for partly by the United Nations Development Programme (UNDP), and the pattern began with chip design. In 1984 the DoE used UNDP money, and a much larger sum of its own, to set up centres for computer-aided design (CAD) at two of the Indian Institutes of Technology (IITs), at Kanpur and Bombay, at IISc and at Jadavpur University. 'Many scientists went abroad to get trained in the design and use of CAD tools', Rajaraman records, and came back to build tools of their own and teach short courses for industry. They had little choice: 'During the mid 80s the US Government had banned the export of CAD tools in electronics to even educational institutions in India.'[72] The same model ran the Knowledge Based Computer Systems programme from 1985 at five institutions, which worked on parallel computers, medical expert systems and the recognition of Indian scripts and speech.[73]

The longest-lived of these programmes was a network. In 1986 the DoE obtained four years of UNDP funding for Education and Research in Computer Networks (ERNET). It linked the five IITs, IISc, the National Centre for Software Technology (NCST) in Bombay and the DoE 'for exchange of email and other documents', and it was meant to train people in networking.[74] 'ERNET also brought the Internet to India under the leadership of S.Ramani of NCST, Mumbai,' Rajaraman writes, and after the project ended it became the internet provider for India's educational and research institutions.[75]

In all of them the money bought machines and sent scientists abroad, and the scientists came back, mostly to public laboratories and universities, and trained others. Rajaraman's summary is that by 1990 'a good foundation had been laid' in computer-aided design, parallel computing, networking and software engineering.[76] The published accounts do not say where the people they trained went next: into public research, to TI and the export firms, or abroad. The wider supply of engineers grew in the same years: recognised engineering and technology colleges nearly doubled in number between 1980-81 and 1990-91, by the statistics ministry's count.[77]

The lobby and the department

The industry organised itself in these years. The Manufacturers' Association for Information Technology (MAIT) was founded in 1982, and the National Association of Software and Service Companies (Nasscom), the software firms' lobby, in 1988, with 38 members.[78] Rajaraman thought Nasscom 'fortunate in employing Dewang Mehta as its President in 1991. Mehta was a superb lobbyist for software companies'.[79]

The department the lobby dealt with changed character at the same time. From 1970 to 1988, Rajaraman writes, the DoE had been headed by technocrats and treated as a scientific department, like Atomic Energy and Space. In 1989 'A difference of opinion between two technocrats in the DoE' led to its being taken over by 'a generalist bureaucrat' of the Indian Administrative Service (IAS), and it ceased to count as a scientific department.[80] Yet he credits the generalist who followed. N. Vittal, an IAS officer who took over the DoE in 1990, was in his account 'dynamic and more business savvy' than his predecessors, and 'instrumental in setting up STPs and obtaining a host of fiscal concessions to the software industry by persuading the bureaucrats of the Ministry of Finance'.[81]

The parks

The Software Technology Parks turned TI's privilege into a public service. Their stated purpose, in the World Bank's words, was to give small and medium firms 'an opportunity to enter the international software export market'.[82] Rajaraman describes what they offered: 'buildings, work stations and continuous uninterrupted power supply', and satellite links through which firms could 'develop software on the computers of their overseas customers from access terminals located in their respective premises'.[83] The World Bank adds a 'single window' for government clearance of imported equipment.[84] 'As the investment required to set up a company in STP was low,' Rajaraman writes, 'this initiative allowed many small entrepreneurs to enter the software services export business.'[85]

Rajaraman says 'The first STP was established in Bangalore in 1990'.[86] By the World Bank's dates Pune came first, in August 1990, two months ahead of Bangalore, followed by Bhubaneswar in December 1990, Hyderabad in February 1991, and Thiruvananthapuram, Gandhinagar and Noida in July 1991.[87] Software Technology Parks of India itself was registered on 5 June 1991.[88]

When the World Bank looked at the parks, about 160 companies had been approved to join, with estimated exports of $7.28 million by 1994, and it judged that 'it may be too early to judge' whether they would live up to expectations.[89] Rajaraman, writing with hindsight, put the parks beside the breaking of VSNL's monopoly and cheaper satellite links as the changes that 'allowed many entrepreneurs to enter the software export business'.[90]

Half the target

By the end of the decade the export trade had grown fast from a small base. The World Bank's table of the industry shows software exports rising from $24 million in FY1985-86 to $128 million in FY1990-91, the year to March 1991.[91] These, like the other series of the period, count gross earnings, as Heeks warned, before the foreign exchange the trade spent abroad.[92] Even gross, the trade in FY1990-91 was about half of the $258 million that the government had expected by 1990 when it announced its policy in 1984.[93]

That gap is the best test of the claim that Rajiv Gandhi started India's information technology revolution. The strongest case for it is real. The policies of 1984 and 1986 came under his government, Rajaraman ties them to what he learned at the Asian Games, and the telephone reforms taken 'Under the leadership of Rajiv Gandhi', which created MTNL and VSNL, came in the same years as C-DOT's exchanges and the call offices that showed Indians what a microprocessor could do.[94] Roye's account of Bangalore in 2013 repeats the popular version, in which the business climate 'dramatically changed' once Rajiv Gandhi became prime minister.[95]

The case against is also documented. The export trade had begun in the 1970s, and the private computer industry that made his reforms possible had been opened by the policy of 1978, after IBM left. Heeks's series shows no sharp break at 1984. Exports grew by 99 per cent in 1982, 35 per cent in 1983 and 39 per cent in 1984, and by 9 per cent in 1985, 40 per cent in 1986 and 39 per cent in 1987.[96] The trade the policies counted as exports was still mostly programmers working abroad, and the share done in India rose only slowly. Sharma places 'The real breakthrough in software exports' after 1991, 'with the advent of the Software Technology Park scheme'.[97] The World Bank's reading is harsher still: the policies pushed firms to export, but in a 'short-term oriented' way, 'without any specific focus or direction on building sustainable competitive advantage and capabilities'.[98]

A third reading, which the evidence here supports best, puts the weight on infrastructure. The decisive changes were the ones that let work stay in India: TI's private link, negotiated in 1985, VSNL in 1986, the parks in 1990 and 1991. They came in pieces, across several governments, pushed by a technocratic department and then by a generalist one.[99] Rajiv Gandhi's government began that process. It did not finish it.

The permit, the old system of licences and obligations, was loosened between 1984 and 1986 but not abolished; a computer imported duty-free still had to earn back two and a half times its cost in dollars. The satellite, which could

make an Indian office part of a foreign company's network, was in 1990 still a privilege granted case by case. By the time the first parks were registered, Rajiv Gandhi was out of office. His party lost the election of 1989, and the coalition that replaced it fell in 1990.[100] He was assassinated during the campaign for the election of 1991.[101] The government that took office after that election inherited a country, in Rajaraman's words, in 'a near default economic situation'.[102]

Notes to Chapter 8. Satellite and permit (1984-1991)

[1] Roye, 'Revolution on a Bullock Cart' (2013).

[2] Roye, 'Revolution on a Bullock Cart' (2013).

[3] Roye, 'Revolution on a Bullock Cart' (2013).

[4] Roye, 'Revolution on a Bullock Cart' (2013); Rajaraman, *History of Computing in India* (2012), p. 33. Roye wrote in 2013, nearly three decades after the event.

[5] Sona Group, 'Satellite Dish for India's First IT Company Arrived on a Bullock Cart' (company self-description).

[6] Roye, 'Revolution on a Bullock Cart' (2013).

[7] Rajaraman, *History of Computing in India* (2012), pp. 33, 36.

[8] Rajaraman, *History of Computing in India* (2012), p. 31.

[9] Rajaraman, *History of Computing in India* (2012), p. 31.

[10] Rajaraman, *History of Computing in India* (2012), p. 31.

[11] Rajaraman, *History of Computing in India* (2012), pp. 29, 37.

[12] *Dataquest*, '1976-1985: The Big Blue Impact' (2006).

[13] Rajaraman, *History of Computing in India* (2012), pp. 31-32, 40 (any firm could now build 32-bit machines, the limits on numbers were removed, and populated circuit boards could be imported; the manufacture of superminicomputers and mainframes stayed reserved for the Electronics Corporation of India, and imported computers had to be maintained by CMC or by their users, not by their makers); Rajaraman, *History of Computing in India* (2012), p. 32 (software as an industry, the duty cut and onsite work counted as exports). Chapter 5 tells the policy's effect on the manufacturers.

[14] Hanna, *Exploiting Information Technology for Development* (1994), p. 96.

[15] Hanna, *Exploiting Information Technology for Development* (1994), p. 96.

[16] Hanna, *Exploiting Information Technology for Development* (1994), p. 106.

[17] Rajaraman, *History of Computing in India* (2012), p. 32.

[18] Rajaraman, *History of Computing in India* (2012), pp. 32-33.

[19] Hanna, *Exploiting Information Technology for Development* (1994), p. 98.

[20] Sharma, 'The Long Revolution' (YaleGlobal excerpt, c. 2009).

[21] Hanna, *Exploiting Information Technology for Development* (1994), p. 96.

[22] Rajaraman, *History of Computing in India* (2012), pp. 34-36.

[23] Rajaraman, *History of Computing in India* (2012), pp. 33, 42.

[24] Rajaraman, *History of Computing in India* (2012), p. 36.

[25] Hanna, *Exploiting Information Technology for Development* (1994), p. 98.

[26] Heeks, 'The Uneven Profile of Indian Software Exports' (1998), p. 13.

[27] Texas Instruments, 'TI Celebrates 20 Years in India' (2005) (TI's milestone list, published for its twentieth anniversary in 2005: began operating in India in 1985); Rajaraman, *History of Computing in India* (2012), p. 36 (negotiation of the link, 1985); Roye, 'Revolution on a Bullock Cart' (2013) (the dish, 1985); Heeks, 'The Uneven Profile of Indian Software Exports' (1998), p. 13 (TI's arrival and the start of its operations, 1986). A primary record of the earth station's licence would settle the date.

[28] Texas Instruments, 'TI Celebrates 20 Years in India' (2005).

[29] Sharma, 'The Long Revolution' (YaleGlobal excerpt, c. 2009). Chapter 6 gives COSL's later standing among the exporters. Indian firms had also done some work for foreign clients from India since the 1970s (Chapter 4).

[30] Rajaraman, *History of Computing in India* (2012), p. 36.

[31] Texas Instruments, 'TI Celebrates 20 Years in India' (2005).

[32] Rajaraman, *History of Computing in India* (2012), p. 42.

[33] Heeks, 'The Uneven Profile of Indian Software Exports' (1998), pp. 12-13.

[34] Roye, 'Revolution on a Bullock Cart' (2013).

[35] Roye, 'Revolution on a Bullock Cart' (2013).

[36] Rajaraman, *History of Computing in India* (2012), p. 33.

[37] Rajaraman, *History of Computing in India* (2012), p. 33.

[38] Rajaraman, *History of Computing in India* (2012), p. 33.

[39] Rajaraman, *History of Computing in India* (2012), p. 33.

[40] Software Technology Parks of India, 'Background' (agency self-description).

[41] Hanna, *Exploiting Information Technology for Development* (1994), p. 96.

[42] Hanna, *Exploiting Information Technology for Development* (1994), p. 28.

[43] Heeks, 'The Uneven Profile of Indian Software Exports' (1998), p. 6. Chapter 6 gives his fuller figures.

[44] Sharma, 'The Long Revolution' (YaleGlobal excerpt, c. 2009).

[45] Rajaraman, *History of Computing in India* (2012), p. 36.

[46] *Dataquest*, '1976-1985: The Big Blue Impact' (2006).

[47] Indian Labour Archives, 'Suresh Dhopeshwarkar'. The union's name is garbled in the scanned note, and the terms of the agreement are not given.

[48] Rajaraman, *History of Computing in India* (2012), pp. 31, 39.

[49] Rajaraman, *History of Computing in India* (2012), p. 38.

[50] Rajaraman, *History of Computing in India* (2012), p. 38.

[51] Comptroller and Auditor General, Report No. 11 of 2007 (Railways), para 1.3, p. 4 (the full sentence: 'The pilot project of PRS was launched on 15 November 1985, over Northern Railway with the installation of the Integrated Multiple Train Passenger Reservation System (IMPRESS)', developed by the railways with CMC at New Delhi). Rajaraman, *History of Computing in India* (2012), p. 38 (the project began in 1984; the New Delhi system was working by 1986). The two fit: the pilot ran at New Delhi from late 1985, and the system was in service there by 1986. *Dataquest*, 'Happy Journey!' (2009), and PIB backgrounder (2026): *Dataquest* (2009) says the pilot covered 'a few popular trains' and that the whole New Delhi reservation system was computerised in May 1987; the Press Information Bureau's backgrounder of 6 August 2026 says the reservation system was 'Introduced in 1986' and gives no day or month (post-cutoff material, checked against the live source on 15 October 2026). A Lok Sabha answer of 8 May 1985 on CMC's turnkey system for Northern Railway (Lok Sabha, written answer, 8 May 1985) bears on the contract; no copy of its text has been located.

[52] Rajaraman, *History of Computing in India* (2012), p. 38.

[53] Rajaraman, *History of Computing in India* (2012), p. 38.

[54] Rajaraman, *History of Computing in India* (2012), p. 38.

[55] Rajaraman, *History of Computing in India* (2012), p. 38.

[56] Rajaraman, *History of Computing in India* (2012), p. 38. No independent confirmation of the claim has been found.

[57] Rajaraman, *History of Computing in India* (2012), p. 38.

[58] Rajaraman, *History of Computing in India* (2012), p. 39.

[59] Rajaraman, *History of Computing in India* (2012), p. 36.

[60] Rajaraman, *History of Computing in India* (2012), p. 36.

[61] Rajaraman, *History of Computing in India* (2012), p. 37.

[62] Rajaraman, *History of Computing in India* (2012), p. 37.

[63] Rajaraman, *History of Computing in India* (2012), p. 37.

[64] Churchill, 'Monsoon Hunters' (1987).

[65] Rajya Sabha written answer, 12 November 1987, and Lok Sabha written answer, 12 May 1993, a parliamentary answer of 1993, is reported to record the import of a Cray X-MP/14 in October 1988 for the centre. No copy of the answer has been located; the model and the date rest on that report.

[66] Rajaraman, *History of Computing in India* (2012), p. 37.

[67] Rajaraman, *History of Computing in India* (2012), p. 37; Reserve Bank of India, *Handbook of Statistics on Indian Economy*.

[68] Rajaraman, *History of Computing in India* (2012), pp. 37-38.

[69] *The Better India*, 'Who Made India's First Supercomputer?' (2017) (Sanchari Pal, 13 January 2017). Low reliability; cited only as an example of the popular version.

[70] Rajaraman, *History of Computing in India* (2012), p. 38.

[71] Rajaraman, *History of Computing in India* (2012), p. 37.

[72] Rajaraman, *History of Computing in India* (2012), p. 35 (funding: $1.5 million from the UNDP and Rs 340 million from the DoE).

[73] Rajaraman, *History of Computing in India* (2012), p. 36 (funding: $5.2 million from the UNDP and Rs 140 million from the DoE; the five institutions were IISc, IIT Madras, the Indian Statistical Institute, the Tata Institute of Fundamental Research and NCST; 'A large number of scientists' went abroad).

[74] Rajaraman, *History of Computing in India* (2012), pp. 38-39 (funding: $6 million over four years from the UNDP, with Rs 41 million from the government).

[75] Rajaraman, *History of Computing in India* (2012), p. 39.

[76] Rajaraman, *History of Computing in India* (2012), p. 42.

[77] MoSPI, 'Number of Recognised Educational Institutions', Table 1.17, p. 19. Chapter 7 tells the growth of the colleges.

[78] Rajaraman, *History of Computing in India* (2012), p. 42.

[79] Rajaraman, *History of Computing in India* (2012), p. 42.

[80] Rajaraman, *History of Computing in India* (2012), p. 42.

[81] Rajaraman, *History of Computing in India* (2012), p. 42.

[82] Hanna, *Exploiting Information Technology for Development* (1994), p. 96.

[83] Rajaraman, *History of Computing in India* (2012), p. 39.

[84] Hanna, *Exploiting Information Technology for Development* (1994), p. 106.

[85] Rajaraman, *History of Computing in India* (2012), p. 39.

[86] Rajaraman, *History of Computing in India* (2012), p. 39.

[87] Hanna, *Exploiting Information Technology for Development* (1994), p. 97. Rajaraman (Rajaraman, *History of Computing in India* (2012), p. 39) names Bangalore as the first park; by the Bank's dates Pune preceded it by two months.

[88] Software Technology Parks of India, 'Background' (agency self-description). The agency says three parks, at Pune, Bangalore and Bhubaneswar, were set up as autonomous societies in 1990.

[89] Hanna, *Exploiting Information Technology for Development* (1994), p. 97.

[90] Rajaraman, *History of Computing in India* (2012), p. 41.

[91] Hanna, *Exploiting Information Technology for Development* (1994), p. 100 (the table also gives $52 million for FY1987-88). Other series give slightly different figures; the data appendix compares them.

[92] Heeks, 'The Uneven Profile of Indian Software Exports' (1998), p. 2. Chapter 6 gives Heeks's estimate of net earnings.

[93] Hanna, *Exploiting Information Technology for Development* (1994), p. 96.

[94] Rajaraman, *History of Computing in India* (2012), pp. 29, 31, 33.

[95] Roye, 'Revolution on a Bullock Cart' (2013).

[96] Heeks, 'The Uneven Profile of Indian Software Exports' (1998), p. 2.

[97] Sharma, 'The Long Revolution' (YaleGlobal excerpt, c. 2009).

[98] Hanna, *Exploiting Information Technology for Development* (1994), p. 96.

[99] Rajaraman, *History of Computing in India* (2012), pp. 33, 36, 39, 42; Hanna, *Exploiting Information Technology for Development* (1994), p. 28.

[100] Rajaraman, *History of Computing in India* (2012), p. 30.

[101] Rajaraman, *History of Computing in India* (2012), p. 43.

[102] Rajaraman, *History of Computing in India* (2012), p. 43.

Part III: The export machine (1991-2008)

Chapter 9. 1991 (1990-1997)

On 24 July 1991 Manmohan Singh, a professional economist who had been finance minister for about a month, rose in the Lok Sabha to present the budget for the year to March 1992 (FY1991-92). He began with a dedication to Rajiv Gandhi, the former prime minister, who had been assassinated during the election campaign that spring.[1] Then he described what the new government had inherited. 'The new Government, which assumed office barely a month ago, inherited an economy in deep crisis. The balance of payments situation is precarious.'[2] India had borrowed from the International Monetary Fund in July 1990 and again in January 1991, and its foreign-exchange reserves had still fallen sharply. 'We have been at the edge of a precipice since December 1990 and more so since April 1991,' he said. 'We have not experienced anything similar in the history of independent India.'[3]

On his second page he gave the number that made the crisis concrete. 'The current level of foreign exchange reserves, in the range of Rs.2500 crores, would suffice to finance imports for a mere fortnight.'[4] Rs 2,500 crore was Rs 25 billion, about $1.1 billion at that year's average exchange rate.[5]

On page 20 of a speech that ran to thirty-one pages, the minister came to software. It got one paragraph. 'Our software industry has made considerable progress in recent years. However, there is still a vast unexploited potential for growth. It is time we make all-out efforts to capture the overseas software market.' To that end he proposed to extend to software exports the income-tax concession under section 80HHC of the Income-tax Act, which gave a deduction for profits from exports. 'With this concession, the exports of this industry should register rapid growth.'[6] Eleven pages later he closed by quoting Victor Hugo, that 'no power on earth can stop an idea whose time has come', and named the idea as India's emergence as a major economic power. 'Let the whole world hear it loud and clear. India is now wide awake. We shall prevail. We shall overcome.'[7]

The prediction came true. Software exports, by one count, rose from $128 million in the year to March 1991 to $1,759 million in the year to March 1998, an average of about 45 per cent a year.[8] That growth, and the date at which it began, has made 1991 the most famous year in the story of Indian software. Founders date their industry's take-off from it, and a common view holds that the industry is a product of the liberalisation that began that summer. On the documents, 1991 changed money: the exchange rate, the right to spend foreign currency, and the right to raise capital. It did much less to change the work itself, which was still mostly sold by the hour or the month, and much of it still done at the client's site abroad. It also changed, more quietly, who was recruited into the work and how.

A fortnight of imports

Singh traced the crisis to deficits at home and abroad. The central government's fiscal deficit was more than 8 per cent of gross domestic product in 1990-91. External debt had reached about 23 per cent of GDP, and debt service took 'about 21 per cent of current account receipts in 1990-91'. These strains, he said, 'were stretched to a breaking point on account of the Gulf crisis last year'.[9] V. Rajaraman, the historian of Indian computing, puts it more bluntly. India 'was about to default on foreign loans', and it was bailed out by the IMF on conditions that included devaluing the rupee and opening the country to foreign investors.[10]

The response came in stages. The rupee was devalued on 1 and 3 July 1991. The budget speech described those 'exchange rate adjustments' and a liberalised system of import licences as 'the beginning of a transition from a regime of quantitative restrictions to a price based mechanism'.[11] Annual averages show the size of the change: about Rs 17.5 to the dollar in 1990 and about Rs 25.9 in 1992.[12]

On foreign capital the minister turned a long tradition around. 'After four decades of planning for industrialisation, we have now reached a stage of development where we should welcome, rather than fear, foreign

investment. Our entrepreneurs are second to none. Our industry has come of age.' Foreign equity of up to 51 per cent in high-priority industries would get prompt approval.[13] He also promised full statutory powers to the Securities and Exchange Board of India (SEBI), transferring to it powers then held by the Controller of Capital Issues and the government, so that it could regulate the stock exchanges.[14]

The software paragraph sat inside this argument about foreign exchange. Two paragraphs after it, Singh extended the same logic to professionals generally. Their 'scientific, technical and professional skills, knowledge and experience' had, he said, 'an increasing capacity to earn foreign exchange for the country'.[15] In July 1991 software entered national economic policy as a way to earn dollars. It was not described as an industrial capability to be built for its own sake. That framing was continuous with the export obligations of the 1970s and 1980s, when firms were allowed to import computers only if they promised to export.

What came before July

The reforms did not create the infrastructure on which the next decade's offshore work would run. The Software Technology Parks (STPs), government-run sites that offered exporters shared satellite earth stations and a single window for official permissions, had opened at Pune, Bangalore (Bengaluru since 2014) and Bhubaneswar in 1990, and the body that ran them, Software Technology Parks of India, says it was registered as an autonomous society on 5 June 1991, seven weeks before the budget.[16] The economists Gaurav Khanna and Nicolas Morales date the setting up of satellite links in the parks to 1992, which they say began 'negating the need for some kinds of on-site work'.[17] The parks were planned before the reforms and began to matter after them.

The government's export targets had a longer history of disappointment. Nagy Hanna, in a World Bank study published in 1994, recorded that it had set targets of $400 million by 1992 and $800 million by 1995.[18] On Rajaraman's figures exports were $225 million in 1992-93 and $734 million in 1995-96; the $800 million mark was passed only in 1996-97, when exports reached $1,100 million.[19] The targets were missed, and then the trade overtook them.

Fifty visits to Delhi

The best-known account of what 1991 did for software comes from N.R. Narayana Murthy, who founded Infosys in 1981 with six colleagues.[20] In an oral history recorded for the Computer History Museum in Bangalore in January 2020, he said 'there is absolutely no doubt that for an industry to take off and demonstrate rapid growth you need the blessings of the government'. He dated his industry's growth from 1992 and credited the reforms of 1991 with three changes.[21]

The first was the end of licensing for computer imports. Before it, he said, 'it would take us fifty visits to Delhi and about three years of elapsed time before we got a license to import a computer of about 300,000 US dollars'.[22] The second was convertibility on the current account, the freedom to buy foreign currency for trade and travel. 'Before that, if I wanted to travel abroad even for a day, I had to apply to the Reserve Bank of India (the Central Bank), wait for about 15 days, get the permission and then travel.' He told of a fellow entrepreneur who had permission for two days in Paris and one in Frankfurt. A prospect moved, so he spent one day in Paris and two in Frankfurt, and reported this honestly on his return. 'He promptly got a show cause notice from the Reserve Bank!'[23]

The third change concerned the stock market. Until then an official in Delhi, the Controller of Capital Issues, decided whether a company could offer shares to the public and at what price. In Murthy's words, 'That government officer had no idea what the capital markets were'. The controller priced issues on past performance, so most were at par, the shares' face value. 'Therefore, most entrepreneurs were uncomfortable about going public because they gave away a lot of their equity and raised very little money.' When the office was abolished and companies could set their own prices with their bankers, Murthy said, 'that made life so much easier for us and the industry took off'.[24]

Rajaraman, drawing on Murthy, adds that convertibility let firms 'send software engineers to clients' sites at short notice'.[25] He also credits the devaluation, which, he writes, 'enabled Indian software companies to be competitive in selling their services', and a run of smaller concessions that followed it, from ten years' freedom from tax on export earnings to private satellite links to overseas customers; multinationals could now own their Indian companies outright.[26]

Murthy is an interested witness, a founder whose company went public under the new regime. But the three changes he names have something in common. None of them changed what Indian firms sold. They changed what it

cost a firm to move people and money across borders and to raise capital at home. For a business that sold the time of programmers, many of whom worked at clients' sites abroad, the right to send an engineer to America at short notice, without a fortnight's wait for the Reserve Bank, was close to the heart of the business.

Only 64 brilliant young engineers

Four months after the budget, on Tuesday 26 November 1991, Infosys took half a page in the Bombay (Mumbai since 1995) edition of the *Times of India*. Above a photograph of a crowd on a city street, with crosshairs on the face of one young man, the headline read: 'Only 64 brilliant young engineers are destined to conquer the world of software. Find out if you are one of them this Saturday.'[27]

The advertisement shows in unusual detail how the profession was assembled in these years. It asked for 'young male/female engineers in any discipline (or science graduates plus MCAs)', the last being holders of the Master of Computer Applications degree. The firm wanted people who had been 'among the top 10 rank holders in their discipline' with 'an aggregate score of 65%. Nothing less will do.' Then came the line that matters most to the argument of this book: 'Ideally we expect you to have up to one year of work experience in any field of engineering. A computer software background is not essential.'[28]

Infosys was buying general ability and proposing to make programmers of it. The advertisement described the conversion. Training, it said, would be 'on live and simulated projects using our in-house facilities', which it listed as two IBM AS/400 minicomputers, a VAX 3400 and several Unix machines running Oracle. Recruits would share the experience of two holders of doctorates from the Indian Institute of Management who were dedicated to training. 'Training at Infosys, as some of our past recruits have discovered, is like doing your post graduation and being paid for it.'[29] A World Bank study three years later would recommend exactly this kind of conversion. Infosys was already doing it, and selling the training as a reason to join.

The pay was stated plainly. 'You will join us as a Software Engineer Trainee and will be initially based at Bangalore. The gross annual pay package is Rs.45,000 plus.' After twelve months' probation the trainee would be confirmed as a software engineer, and 'The best can expect to earn a gross annual compensation of Rs.57,000 on completion of the one year training period.'[30] Rs 45,000 was about $2,000 at the 1991 average exchange rate, and Rs 57,000 about $2,500.[31] A separate box offered a higher starting salary to engineers with one or two years in the software industry; the figure is illegible in the surviving copy.[32]

The rest of the page sold a life. 'The job is transferable, so you can look forward to jet setting not only in India, but all over the world.' The recruit would join a team of 400 people drawn from all over India. The firm said it had offices in Bangalore, Delhi, Bombay, Atlanta, Boston and Paris, a joint venture with the American consultancy Kurt Salmon Associates, and clients that included General Electric, Reebok and Polo Ralph Lauren. Forty engineers in Bangalore, it said, were working on a large project for Reebok UK. Under a heading promising fast promotion without a change of employer, it claimed that people who had joined as software engineers had risen to be vice presidents within six years.[33] Of Bangalore it said: 'The garden city, as it is called, has a quaint ambience of its own.'[34]

The selection was compressed into a weekend. Candidates were to come to the Leela Kempinski hotel in Bombay at 9 a.m. on Saturday 30 November, with their bio-data and certificates. After an initial screening on qualifications came 'a round of brain teasers, IQ and aptitude tests'; a box of tips warned that the test had sections on arithmetical reasoning and analytical skills. Those who did well would be interviewed on the Sunday, and the successful were guaranteed an appointment letter on the Monday. The heading over that promise read: 'Your girlfriend is sure going to miss you next Monday'.[35] Despite 'male/female' in the call, the page seems to have imagined its reader as a young man. Senior professionals were invited to see Nandan Nilekani or K. Dinesh, two of the founders, at the same hotel.[36]

How many came is not documented. A 2017 article on a business website, which reproduced the advertisement, says Murthy recalled about 1,500 people turning up and a crush at the hotel, but it gives no source for the recollection, and it misreads the Rs 57,000 as Rs 52,000.[37]

Seventy thousand applications

By the end of the decade the Saturday test at a Bombay hotel had become a national sorting machine. Infosys's registration statement for its American share listing, filed in February 1999, described it. The company focused its recruiting 'on the top 20% of students from engineering departments of Indian schools' and chose them through tests and interviews, selecting 'on the basis of their ability to learn as well as their academic achievement'. In the

year to March 1998 it received about 70,870 applications, tested about 21,350, interviewed about 5,800, made about 1,790 offers and saw about 1,300 accepted. A faculty of 33, twenty of them with master's degrees or doctorates, ran fourteen-week training courses for new recruits.[38]

The numbers say something about the education that fed the industry. Fewer than one applicant in fifty became an Infosys employee. The test, not the degree, did the sorting, and the firm's own course, not the college, taught the work. The same filing gave the national picture as an argument for investors: according to the National Association of Software and Service Companies (Nasscom), the industry's lobby group, the number of software professionals had grown from about 56,000 in the year to March 1990 to about 200,000 in the year to March 1998, and India had more than 1,800 engineering colleges and technical institutes training about 68,000 graduates a year in information technology.[39] Other counts, Rajaraman's among them, differ by source and definition.[40] The firm itself grew from about 480 IT professionals in the year to March 1994 to about 2,200 in the year to March 1998.[41]

The selling point to clients was price. Rajaraman writes that 'The salary of software engineers in India was about 20% that of an equivalent western engineer.'[42] The Infosys filing, citing the consultancy Software Productivity Research, put the average Indian software wage at 'approximately 15% of the average U.S. rate'.[43] A sharper comparison, which Khanna and Morales quote from a 1992 study, is that in 1989 'an Indian computer professional earned about $5486, whereas Microsoft was offering $40,000 plus relocation benefits and a green card'.[44] The trainee's Rs 45,000 in 1991 was well below even the 1989 figure, though the two are not the same measure.

Going public

Infosys had meant to sell shares to the public in 1991. Murthy said in 2020 that he had wanted to list within ten years of founding, as American companies did. The assassination of Rajiv Gandhi in May 1991 and the arrival of the reforms led the firm to wait. In 1992 the stock market was shaken by what he called 'a huge stock exchange scam', the securities scam associated with Harshad Mehta, and on 6 December 1992 a mosque, the Babri Masjid, was demolished. Each time, Murthy recalled, the firm was advised to wait.[45]

'So, we went public on 24th of February 1993. It was a very tough time. We barely managed it. Nobody had any idea what a software company was.' Infosys was, he said, the second software company to go public, after Mastek in December 1992. The offer ran from 24 to 27 February, but the shares were listed only in June. 'It opened pretty well at about 50 percent premium.'[46] The company's 1999 filing says that in 1993 it raised 'approximately $4.4 million in gross aggregate proceeds' from its initial public offering, and it names the Bangalore Stock Exchange.[47]

The listing mattered to Murthy for reasons beyond the money. He said that from the start the company would give its accounts to any journalist, and that going public was meant 'to convince the decision makers of our customer organizations in the U.S. that we were a transparent company'. From 1995 it voluntarily prepared accounts under American accounting rules.[48] A firm that sold its engineers' time to American corporations was also selling trust, and a public listing was a way to show it.

The hotel rooms

In May 1995 Infosys's largest customer came to renegotiate. Murthy's account of the episode, in the same oral history, shows in detail how the trade was priced. General Electric (GE) had chosen four Indian firms, Infosys among them, after a presentation to Rajiv Gandhi's government in the 1980s. Infosys began working for GE in 1990. GE renegotiated its contracts every five years, and by 1995, Murthy said, it 'formed about 25 percent of our revenue'.[49]

'They sequestered the vendors in different rooms at a five-star hotel in the city. They would go from room to room and would negotiate to get better and better terms.' The talks ran over a Thursday and a Friday. 'All other competitors of ours were ready to sign on the dotted line.' Infosys would not cut its price. Murthy said he told GE that 'if we reduced our prices to them we would have to reduce prices to our other customers too', and that lower prices would leave less money for pay, tools, technology and productivity.[50] Infosys withdrew and handed the work over to competitors, and, he recalled, 'my colleagues were unhappy with me'.[51]

The company's filing tells the same story without naming the client. In 1995 Infosys 'chose to reduce significantly the services provided to its then-largest client rather than to accept the price reductions and increased Company resources sought by such client'.[52] That client's share of revenue fell from 21.5 per cent in the year to March 1996 to 6.7 per cent in the year to March 1997. Total revenue still rose by 48.8 per cent, to $39.6 million, in the year to March 1997.[53]

The episode shows who held the power in a trade sold by time. The buyer set the terms in a hotel, room by room, and played its suppliers against one another. A supplier could refuse only if it had other buyers, and Infosys, on its own account, did. Murthy's further claim, that Infosys long charged more than its Indian rivals, is taken up in Chapter 11.

What was being priced was mostly time. In the year to March 1996, 76 per cent of Infosys's revenue came from time-and-materials contracts, under which the client pays for hours or months worked, and 24 per cent from fixed-price contracts. In the year to March 1997 the split was 63 per cent and 37 per cent.[54] In its 1999 filing the company said it was 'seeking to increase the proportion of projects that are undertaken on a fixed-price, fixed-time frame rather than a time-and-materials basis'. Its reason was that such projects offered 'the Company the potential benefit of enhanced margins for projects that are performed efficiently'.[55] Under time-and-materials, a firm that finished faster billed less. Under a fixed price, it kept the saving. The filing added that 'The Company's policy is to decline or discontinue projects that do not offer the potential to meet the Company's profit margin targets.'[56] Selling time also meant that revenue could grow only as fast as the number of engineers billed, which is why the recruiting machine had to keep running. That is the bargain the rest of this book follows. The question of who keeps the gains when work is done faster, the client or the supplier, returns at the end of the book, when the tools doing the work faster are machines.

Onsite, offshore and the visa

The reforms did not bring the work home quickly. On Nasscom data reported by Khanna and Morales, 'The fraction of on-site work went from 90% in 1988 to 66% in 1995 and 56% in 2000', measured as a share of software revenue.[57] Richard Heeks, a researcher at the University of Manchester who studied the trade through the 1990s, found that 'by 1997/98, it was still true that more than half of all software services export earnings came from onsite work'.[58] Other series, which count exports or development effort rather than revenue, differ by a few points but agree on the direction and the pace.[59]

Two forces pushed it towards India. One was the satellite links. The other was American visa policy, whose shift of 1993 Chapter 13 describes. The H-1B visa, created in 1990 for workers in 'specialty occupations', required the employer to pay the greater of the wage paid to its other employees in the job or the prevailing wage for the occupation. Its cap was set at 65,000 a year. 'In the early years, the cap was never reached', but by the mid-1990s it 'started binding'.[60] Heeks dates the main change to 'the arrival of the Clinton administration, partly in response to the desire of local labour unions to halt or reduce the use of foreign workers in the US'. Indian workers 'were channelled into taking temporary work visas (rather than student or training or even tourist visas as in the past)', and the changes 'pulled a number of Indian companies out of their complacency about onsite work'.[61]

The leading firms built a method around the shift. Rajaraman calls it the Global Delivery Model and, citing Murthy, says Indian software companies invented it. In his description, the part of a project that needed face-to-face work with the customer, typically 20 to 30 per cent of the time, was done at the client's premises. Coding and testing, 70 to 80 per cent of the time, was done in India, in secure areas set aside for a single client and joined to it by a dedicated satellite link.[62] The time difference became an asset: India's working day fell in the American night, so bugs reported at the end of an American shift could be fixed by morning.[63] Murthy said in 2020 that he 'formalized' the model in the late 1980s and early 1990s, aiming at 'only about 20 percent of our effort at customer site'.[64]

Infosys's 1999 filing described the mechanics as the firm practised them: a small team at the client's site and a much larger one in India, with the client's systems linked by satellite to as many as four Indian centres. 'By pursuing this model, the Company completes approximately 80% of its project work at its offshore software development facilities in India.'[65] At any time about 15 per cent of its professionals were working at client sites abroad. In December 1998, 231 of its people in the United States held H-1B visas and 111 held L-1 visas, which were open only to employees who had worked for the company for at least a year.[66]

That 80 per cent was the leading firm's account of its own work at the end of the decade. The industry as a whole still earned more than half its export revenue onsite.[67] The model described where the best-organised firms were going, not where the trade already was.

To persuade distant clients that people they never saw would do the work properly, the leading firms also collected certificates of process, and by 1999 Infosys described itself as 'Certified under ISO 9001, TickIT and at Level

4 of the Capability Maturity Model', a five-level grading of software development; Chapter 11 follows how such certificates were sold.[68]

Not only the reforms

Two series of export figures cover the decade, one compiled by Subhash Bhatnagar and one by Heeks from his own research and the trade magazine *Dataquest*. They differ by up to about a tenth in some years, but both show growth of between about a quarter and two-thirds a year, faster from 1993 than in 1990-93; the data appendix sets them side by side.[69]

Not all of the growth came from policy. Rajaraman lists the decade's 'fortuitous circumstances' alongside policy: the millennium bug, the conversion of European banking software to the euro, and 'a technically savvy influential Indian diaspora in the USA'.[70] The millennium bug, or Y2K, already brought Infosys 23.3 per cent of its revenue in the year to March 1998.[71]

On the diaspora he writes that 'The "brain drain" of the 60s and the 70s became "opportunity gain" in the 90s'. Indians in senior posts at American companies gave Indian firms 'small contracts at the beginning which grew as the companies delivered good quality software applications'.[72] He offers no data for the claim.

Foreign firms came too, beginning with a Tata joint venture with IBM, which had left India in 1978, that Rajaraman dates to 1992.[73] On his dating, American Express, Oracle and GE Capital International Services followed between 1994 and 1997, and in Bangalore engineers at Texas Instruments designed a signal-processing chip, Ankur, announced in 1997 and 'entirely designed in India'.[74] Chapter 15 follows these foreign-owned centres.

The boom was concentrated. Heeks estimated that by 1998 there were 400 to 500 active software export firms, most of them small firms with a single contract. The top five firms earned more than 40 per cent of all exports and the top twenty about 70 per cent.[75] It was concentrated in place as well. Of 558 software companies whose headquarters *Dataquest* recorded in 1998, 152 were in Bangalore, 122 in Bombay, 93 in Madras and 86 in Delhi.[76] By the close of the period, Rajaraman writes, 'the voice of NASSCOM was strong and it got the ear of the government'.[77]

What 1991 changed

There are three ways to read the years after the crisis, and the evidence supports parts of each.

The first is the founders' reading. Murthy credits the reforms with the industry's take-off. The export series gives him only partial support: growth was already fast in the late 1980s, slowed in the first years of the decade, and quickened again from 1993.[78] The reforms that mattered most to software were the ones Murthy named, about money and movement: devaluation, convertibility, the end of licensing for imports, and the freedom to price a share issue.

The second is a reading of continuity. The leading firms, the business of sending programmers to clients' sites, the duty-free import regime for exporters and the Software Technology Parks all predated July 1991. The budget's own contribution to software was a tax concession, and it justified that concession as a way to earn foreign exchange.[79] The onsite series and the Infosys contract split show that where the work was done, and how it was sold, changed only slowly. Heeks's verdict in 1998 was that much of the growth 'may have come from its industry's dogged following of the low-risk, low-barrier first path', the export of labour.[80]

The third reading looks to America. Khanna and Morales argue, using an economic model, that the prospect of American jobs through the H-1B visa drew Indian students into engineering, and that those who could not get visas because of the cap stayed home and staffed the Indian industry.[81] On this reading, American demand and American immigration law shaped the Indian profession at least as much as Indian policy did. It is an argument from a model, calibrated on data.

What none of the readings disputes is the change in who did the work. In 1991 a firm of 400 people advertised for sixty-four engineers of any discipline, promised to make them programmers, and paid them about $2,000 a year. By 1998 the same firm was testing more than twenty thousand applicants a year and hiring fewer than one applicant in fifty. Its recruits were trained in a company faculty, worked to documented processes, and spent most of their time in India while their clients slept. The advertisement of 1991 had told them where they would start. 'Say cheers to Bangalore', it said.[82] The next chapter follows them there.

Notes to Chapter 9. 1991 (1990-1997)

[1] Singh, *Budget 1991-92* speech (1991), p. 1; Rajaraman, *History of Computing in India* (2012), p. 43.

[2] Singh, *Budget 1991-92* speech (1991), p. 1.

[3] Singh, *Budget 1991-92* speech (1991), p. 1.

[4] Singh, *Budget 1991-92* speech (1991), p. 2.

[5] Reserve Bank of India, *Handbook of Statistics on Indian Economy*, calendar-year average rate for 1991 (Rs 22.69 per US dollar); conversion by the author.

[6] Singh, *Budget 1991-92* speech (1991), p. 20.

[7] Singh, *Budget 1991-92* speech (1991), p. 31.

[8] Rajaraman, *History of Computing in India* (2012), pp. 44, 48.

[9] Singh, *Budget 1991-92* speech (1991), p. 2.

[10] Rajaraman, *History of Computing in India* (2012), p. 43.

[11] Singh, *Budget 1991-92* speech (1991), p. 4.

[12] Reserve Bank of India, *Handbook of Statistics on Indian Economy*, calendar-year average rates: Rs 17.50 per US dollar (1990), Rs 25.92 (1992).

[13] Singh, *Budget 1991-92* speech (1991), p. 4.

[14] Singh, *Budget 1991-92* speech (1991), p. 6.

[15] Singh, *Budget 1991-92* speech (1991), p. 20.

[16] Software Technology Parks of India, 'Background' (agency self-description); Hanna, *Exploiting Information Technology for Development* (1994), p. 97.

[17] Khanna and Morales, 'The IT Boom and Other Unintended Consequences' (2016), p. 5.

[18] Hanna, *Exploiting Information Technology for Development* (1994), p. 96.

[19] Rajaraman, *History of Computing in India* (2012), p. 48.

[20] Rajaraman, *History of Computing in India* (2012), p. 44.

[21] Computer History Museum, 'Oral History of N.R. Narayana Murthy, Part 2' (2020), p. 9.

[22] Computer History Museum, 'Oral History of N.R. Narayana Murthy, Part 2' (2020), p. 9.

[23] Computer History Museum, 'Oral History of N.R. Narayana Murthy, Part 2' (2020), p. 9.

[24] Computer History Museum, 'Oral History of N.R. Narayana Murthy, Part 2' (2020), pp. 9-10.

[25] Rajaraman, *History of Computing in India* (2012), p. 44.

[26] Rajaraman, *History of Computing in India* (2012), p. 44.

[27] Infosys Consultants, recruitment advertisement, *Times of India*, 26 November 1991, p. 17.

[28] Infosys Consultants, recruitment advertisement, *Times of India*, 26 November 1991, p. 17.

[29] Infosys Consultants, recruitment advertisement, *Times of India*, 26 November 1991, p. 17.

[30] Infosys Consultants, recruitment advertisement, *Times of India*, 26 November 1991, p. 17.

[31] Infosys Consultants, recruitment advertisement, *Times of India*, 26 November 1991, p. 17; Reserve Bank of India, *Handbook of Statistics on Indian Economy*, converted at the 1991 calendar-year average rate (Rs 22.69 per US dollar).

[32] Infosys Consultants, recruitment advertisement, *Times of India*, 26 November 1991, p. 17.

[33] Infosys Consultants, recruitment advertisement, *Times of India*, 26 November 1991, p. 17.

[34] Infosys Consultants, recruitment advertisement, *Times of India*, 26 November 1991, p. 17.

[35] Infosys Consultants, recruitment advertisement, *Times of India*, 26 November 1991, p. 17.

[36] Infosys Consultants, recruitment advertisement, *Times of India*, 26 November 1991, p. 17.

[37] *OfficeChai*, 'This Infosys Job Advertisement from 1991' (2017).

[38] Infosys, Form F-1 (1999), p. 50.

[39] Infosys, Form F-1 (1999), p. 42.

[40] Rajaraman, *History of Computing in India* (2012), p. 43 (160,000 software engineers by about 1997); Khanna and Morales, 'The IT Boom and Other Unintended Consequences' (2016), p. 6 (Nasscom's figures as Khanna and Morales report them: about 160,000 software professionals in 1996 and about 340,000 in 2000).

[41] Infosys, Form F-1 (1999), p. 40.

[42] Rajaraman, *History of Computing in India* (2012), p. 45.

[43] Infosys, Form F-1 (1999), p. 42.

[44] Khanna and Morales, 'The IT Boom and Other Unintended Consequences' (2016), p. 5 n. 5, quoting Subramanian (1992); the 1992 study is cited here through Khanna and Morales.

[45] Computer History Museum, 'Oral History of N.R. Narayana Murthy, Part 2' (2020), pp. 17-18.

[46] Computer History Museum, 'Oral History of N.R. Narayana Murthy, Part 2' (2020), p. 18.

[47] Infosys, Form F-1 (1999), pp. 36, 40.

[48] Computer History Museum, 'Oral History of N.R. Narayana Murthy, Part 2' (2020), p. 18.

[49] Computer History Museum, 'Oral History of N.R. Narayana Murthy, Part 2' (2020), p. 19.

[50] Computer History Museum, 'Oral History of N.R. Narayana Murthy, Part 2' (2020), pp. 19-20.

[51] Computer History Museum, 'Oral History of N.R. Narayana Murthy, Part 2' (2020), p. 20.

[52] Infosys, Form F-1 (1999), p. 14.

[53] Infosys, Form F-1 (1999), p. 33. Murthy's 'about 25 percent' (Computer History Museum, 'Oral History of N.R. Narayana Murthy, Part 2' (2020), p. 19) and the filing's 21.5 per cent, which is for the year to March 1996, probably refer to different periods.

[54] Infosys, Form F-1 (1999), p. 33.

[55] Infosys, Form F-1 (1999), p. 44.

[56] Infosys, Form F-1 (1999), p. 44.

[57] Khanna and Morales, 'The IT Boom and Other Unintended Consequences' (2016), p. 5.

[58] Heeks, 'The Uneven Profile of Indian Software Exports' (1998), p. 6.

[59] AnnaLee Saxenian, citing a study by Balaji Parthasarathy, gives onsite production as 90 per cent of exports in 1990 and 58 per cent in 1998 (Saxenian, 'Bangalore: The Silicon Valley of Asia?' (2001), p. 10), and 61 per cent for 1995 in an earlier report (Saxenian, *Silicon Valley's New Immigrant Entrepreneurs* (1999), p. 66 n. 24). Heeks measured development effort rather than revenue and found the same pattern for 1988 (Heeks, 'The Uneven Profile of Indian Software Exports' (1998), p. 6). The series start in different years and measure revenue, exports or effort, which is why they differ by a few points.

[60] Khanna and Morales, 'The IT Boom and Other Unintended Consequences' (2016), pp. 7-8.

[61] Heeks, 'The Uneven Profile of Indian Software Exports' (1998), p. 9.

[62] Rajaraman, *History of Computing in India* (2012), pp. 44-45.

[63] Rajaraman, *History of Computing in India* (2012), p. 45.

[64] Computer History Museum, 'Oral History of N.R. Narayana Murthy, Part 2' (2020), pp. 16-17.

[65] Infosys, Form F-1 (1999), p. 45.

[66] Infosys, Form F-1 (1999), p. 50.

[67] Heeks, 'The Uneven Profile of Indian Software Exports' (1998), p. 6; Khanna and Morales, 'The IT Boom and Other Unintended Consequences' (2016), p. 5.

[68] Rajaraman, *History of Computing in India* (2012), p. 45 (ISO 9001 and the five levels of the Software Engineering Institute's Capability Maturity Model); Infosys, Form F-1 (1999), p. 46.

[69] Rajaraman, *History of Computing in India* (2012), p. 48 (Bhatnagar's series as reproduced by Rajaraman: $128 million in 1990-91, $164 million in 1991-92, $330 million in 1993-94, $734 million in 1995-96, $1,759 million in 1997-98); Heeks, 'The Uneven Profile of Indian Software Exports' (1998), p. 2 (Heeks, from his own research and Dataquest: $131.2 million, $173.9 million, $480.9 million in 1994-95, $1,650 million in 1997-98). Heeks's series shows growth of 29 to 51 per cent a year in the late 1980s and 24 to 33 per cent in 1990-93.

[70] Rajaraman, *History of Computing in India* (2012), p. 43.

[71] Infosys, Form F-1 (1999), pp. 15, 33.

[72] Rajaraman, *History of Computing in India* (2012), pp. 46-47.

[73] Rajaraman, *History of Computing in India* (2012), p. 47; Khanna and Morales, 'The IT Boom and Other Unintended Consequences' (2016), p. 5 n. 3.

[74] Rajaraman, *History of Computing in India* (2012), pp. 47-48.

[75] Heeks, 'The Uneven Profile of Indian Software Exports' (1998), p. 11.

[76] Heeks, 'The Uneven Profile of Indian Software Exports' (1998), p. 12.

[77] Rajaraman, *History of Computing in India* (2012), p. 47.

[78] Computer History Museum, 'Oral History of N.R. Narayana Murthy, Part 2' (2020), p. 9; Heeks, 'The Uneven Profile of Indian Software Exports' (1998), p. 2; Rajaraman, *History of Computing in India* (2012), p. 48.

[79] Singh, *Budget 1991-92* speech (1991), p. 20; Hanna, *Exploiting Information Technology for Development* (1994), p. 97.

[80] Heeks, 'The Uneven Profile of Indian Software Exports' (1998), p. 15.

[81] Khanna and Morales, 'The IT Boom and Other Unintended Consequences' (2016), abstract.

[82] Infosys Consultants, recruitment advertisement, *Times of India*, 26 November 1991, p. 17.

Chapter 10. Bangalore (1947-2010)

In the early 1970s the journalist Janardhan Roye used to see a young engineer named Vinay Deshpande working long hours in a basement. The basement belonged to the Classic Building on Richmond Road in Bangalore (Bengaluru since 2014), and the firm that occupied it was called, in Roye's spelling, Processors Systems India. One evening Roye went down to see him. Deshpande was holding a soldering iron over a circuit board. He looked up, smiled, and explained that he was putting together 'hi-tech firmware, bound for the US'.[1]

Roye wrote the scene down forty years later, in 2013, in an article about how Bangalore had become a technology city. He dated it to about twelve years before the day in 1985 when Texas Instruments' satellite equipment arrived at Sona Towers by bullock cart. He thought Deshpande typical of the engineers of the 1970s, who worked quietly on their own or alongside the government's industrial, defence and science laboratories.[2] His point is one the rest of the evidence supports. Bangalore was building electronics for American customers before any American software company had an office there.

In November 1991 Infosys advertised for sixty-four young engineers to start their careers in Bangalore.[3] There are three answers to why the software trade gathered in that city. The first is older than the trade. Bangalore was a city the Indian state had filled with engineers, laboratories and colleges in the forty years after independence. The second is a set of decisions by firms, some of them about engineers and some of them about landlords. The third is a policy that made the city's software parks a kind of export enclave, with their own power and their own links to the world, and that required each firm in them to earn back its Indian wage bill one and a half times in exports every year.[4] That rule tied what a firm had to sell to what its engineers cost. What the enclave did for the rest of Bangalore is the hardest question.

The state's city

N.R. Narayana Murthy, the founder of Infosys, gave his own account of the city's origins in an oral history for the Computer History Museum, recorded in Bangalore in January 2020. 'Right from 1947 when we got independence,' he said, 'the government of India decided to locate its defense oriented high-tech companies in Bangalore, which was away from the border with Pakistan and with China.'[5] He added a reason of geography: the city lay about 250 miles from the Arabian Sea and about 200 from the Bay of Bengal, so that a plane coming to bomb it would be shot down first. That was why, in his telling, Jawaharlal Nehru and his cabinet made Bangalore a defence-oriented, high-technology place.[6] That is Murthy's explanation; no cabinet record of it has surfaced.

The institutions themselves are not in doubt. Murthy named the Electronics and Radar Development Establishment (LRDE), a defence laboratory; Bharat Electronics Limited (BEL); and Hindustan Aeronautics Limited (HAL). His interviewer added the Indian Space Research Organisation (ISRO), and Murthy agreed that space research came 'in the late '70s'.[7] A social scientist studying the city's poor, Supriya RoyChowdhury, describes the same city from another angle. In the decades after independence Bangalore was known as 'a garden city of retirees' bunglows [sic], and for a significant public sector presence', with central and state government enterprises set up from the early to mid-1950s and 'premier scientific research institutions' founded by the central government. 'The city's development in the first four decades was closely aligned with the development of the state sector,' she writes.[8]

The city's growth followed the same timetable. A research group at the Indian Institute of Science (IISc) reports that Bangalore's population grew by 76 per cent between 1971 and 1981, a rise it attributes to the public-sector industries and defence establishments of the 1970s. Between 1991 and 2001, the decade of the software boom, growth

was 38 per cent, and the urban agglomeration's population at the 2001 census was 5.7 million.[9] Software did not make Bangalore big. It arrived in a city that the public sector had already made large and full of engineers.

The private technology community grew inside that public one. Vijay Chandru, a former professor of computer science at IISc who later founded a life-sciences company, told Roye: 'Bangalore was a tech city well before the arrival of the MNCs (multi national corporations)'.[10] He called it 'the original Beantown', after Boston, and said that from the 1960s 'a host of engineers were engaged in creating hardware and software solutions' for avionics, defence, telecommunications and weather forecasting, mostly in the public sector.[11]

The education half of the story sits in the same institutions. When Wipro, then a maker of vegetable oils, decided to make computers in 1979, its research team worked inside the Indian Institute of Science's Digital Systems Lab, on equipment the institute could import more easily than a company could, and several of its first employees had trained there.[12] The computer business of a company that later sold programmers to the world began in a university laboratory.

A flat in Pune

The software trade did not start in Bangalore. Richard Heeks of the University of Manchester, who studied the industry from the late 1980s, wrote in 1998 that it began in Bombay (Mumbai since 1995), 'in the business district around Nariman Point in the late 1970s'. When prices and commuting times rose, part of the industry moved north to the Santa Cruz Electronics Export Processing Zone (SEEPZ), and Bombay remained the centre of the trade 'until at least the mid-1980s'.[13] Then its firms began to run short of people, partly because programmers were leaving for jobs abroad, and the city's high prices and poor quality of life told against it. Firms, foreign investors and would-be founders began to compare Bombay with other places.[14]

Murthy's account of one such comparison turns on housing law. Infosys was founded in Pune in 1981. 'We pretty soon realized that Maharashtra was not the best place to have a startup,' he said, because of the state's tenancy laws.[15] A tenant who had lived in an apartment for seven years could not be made to leave and could claim it as his own. 'With the result, most landlords in Maharashtra refused to give an apartment on rent because of this fear. Unless you were a big company, unless you gave some big bank guarantee, etcetera, etcetera. But no startup could do that.'[16] The founders wanted a state where the law was 'fair between the landlord and the tenant', and Karnataka was one. So the choice narrowed to Madras (Chennai since 1996), in Tamil Nadu, or Bangalore.[17]

They chose Bangalore, Murthy said, 'because of the availability of engineering talent'. Part of the reason was the defence public sector. The other part was education. In the 1970s and 1980s, he said, 'when there were very few engineering colleges in the country, Karnataka had a disproportionately higher percentage of engineering talent and engineering students in the country.'[18] A later study put the regional supply in numbers. Writing in 2001 for a Stanford University conference on India's economy, the scholar of Silicon Valley AnnaLee Saxenian reported a finding that 60 per cent of India's computer science graduates came from three southern states, Andhra Pradesh, Karnataka and Tamil Nadu, and that the industry was concentrated there 'largely because of their concentration of engineering manpower'.[19] The private engineering colleges described in Chapter 7 were part of that supply.

Beer, climate and perception

A common answer to the question of why Bangalore won is that companies liked the weather. The evidence makes this partly true but incomplete. Heeks gave the most systematic answer, using a model of how software firms choose locations. He found three factors at work in the mid-1980s and a fourth that took over later.[20]

The first was labour. Bangalore was seen to have 'abundant labour that could be drawn from its long-standing research laboratories, educational institutes and public sector electronics firms', among them IISc and the Indian Institute of Management.[21] The second was quality of life. Bangalore had a reputation 'for a good climate and social life', and IT workers were quickly drawn into the city's social life and its 'pub culture'. Heeks recorded a semi-serious club of senior industry managers, the Beer-drinkers Association of Information Technology, whose members met to discuss strategy on condition that each drank at least six mugs of beer in the evening. Bombay, by contrast, had 'a long and troubled history of both communal and labour relations strife'.[22] The third was infrastructure. 'You could get across town in a matter of minutes in a car,' where Bombay's peak-hour traffic could take hours, and Bangalore promised plentiful power and water.[23]

Heeks added a caution from his interviews: 'the perception of these factors as much as their reality has coloured locational decisions'.[24] Some insiders gave the weather more weight. C.S. Murali of IISc, quoted by Roye in 2013, listed policy, skilled workers and the public-sector giants and then, 'above all', the city's 'inviting salubrious climate and cosmopolitan character'. Murthy, asked what had made Bangalore a hub, began with 'a pretty salubrious climate' and put engineering talent second.[25] Both named skilled people too. As the main reason for Bangalore's rise, the climate story is mostly false; as one reason among several, it is true.

The infrastructure advantage was the least durable of Heeks's factors. By 1998 a rival city was advertising Bangalore's traffic as its weakness. By 2005 the concentration of technology production had brought fast-rising wages and worsening congestion, and Bangalore's engineering salaries were among the highest in India, yet firms kept clustering there rather than moving to cheaper places.[26]

Twenty-five rules

Heeks treated the arrival of Texas Instruments (TI) as 'a key event in Bangalore's history'. Once TI 'set up operations in 1986, it was the first proper offshore software facility, using satellite-based telecommunication links, and this was constantly reported in the home and international media'. The company was also 'known to have cut through many red tape barriers, particularly on imports'. From then on, he wrote, the Bangalore 'snowball effect' began.[27]

Saxenian put a number on the red tape. Citing an interview given in 1996 by N. Seshagiri, a former additional secretary in the Department of Electronics, she wrote: 'When Texas Instruments set up the first earth station in Bangalore in 1986, for example, the process involved removing or breaking 25 different government rules.'[28] Without such links, she added, Indian firms 'had no alternative to providing contract programming on-site', at the customer's premises and usually in the United States.[29]

Other multinationals followed TI, and they explained themselves in terms of skill and risk. One American employer in Bangalore, in an interview of 1996 cited by the researcher Balaji Parthasarathy and quoted by Saxenian, said: 'we came here because of the skills. We expanded because of the skills.' The risks of distance, of the satellite link, the telephones and the flights, 'were offset by the costs'.[30]

Some of the work was hard engineering. TI's Bangalore centre began by designing tools for chip design. Its engineers went on to design a digital signal processing chip, announced in 1997 under the code name Ankur, Sanskrit for seedling, which V. Rajaraman, the historian of Indian computing, says was designed entirely in India and widely used in disk drives and instruments.[31] It became a standard.[32] Oracle opened a software development centre in Bangalore in 1994.[33]

By 1998 the snowball could be counted. A Dataquest survey of 558 Indian software companies, reproduced by Heeks, found 152 with headquarters in Bangalore, against 122 in Bombay, 93 in Madras, 86 in Delhi, 34 in Hyderabad and Secunderabad, 27 in Calcutta (Kolkata since 2001) and 22 in Pune.[34] Heeks warned that the table understated Bombay, which was still home to several of the largest firms. But since the mid-1980s, when Bombay had led on numbers, Bangalore had grown at its expense and had come to be called India's 'Silicon Plateau'.[35]

The parks and the bargain

The state that had built Bangalore's laboratories now built its export infrastructure too. Bangalore's Software Technology Park (STP) opened in October 1990.[36] An STP was in effect an export processing zone for software. Firms in it received five years' exemption from tax, guaranteed access to high-speed satellite links and reliable electricity, ready office space, duty-free imports of equipment and permission for full foreign ownership, 'in exchange for a sizable export obligation'.[37]

The obligation puts the book's bargain into the language of regulation. A park firm had to earn, net, 150 per cent of the value of its imported hardware within four years, and 'a net amount equal to 150% of their wage bill on an annual basis'.[38] Every year, then, a firm's Indian payroll had to come back one and a half times over in foreign exchange. The rule measured exports against the cost of the people who produced them. On its arithmetic, every engineer a park firm hired raised the exports it owed, and a firm that sold time could earn more chiefly by billing more people's time. That is the bargain the rest of this book follows: India sold engineers' time, and revenue grew with headcount.

From January 1995 park firms could sell a quarter of their output in India, and from 1999 half.[39] By 1998 the scheme had spread to 25 parks at various stages of planning, besides the government's own, and some were private

ventures. The Information Technology Park in Bangalore was a partnership of the Karnataka government, Tata Industries and a consortium of Singapore firms.[40]

The parks moved work from America to India, slowly. The share of Indian software exports earned at clients' sites abroad fell through the 1990s but was still more than half in 1998.[41] By 1995, 435 companies were registered under the scheme, accounting for more than 16 per cent of exports.[42]

Karnataka's numbers grew fastest. According to the state's development report of 2007, cited by RoyChowdhury, the number of software export companies in Karnataka rose from 13 to 1,154 between 1991 and 2003, and the state's software exports from Rs 5.6 crore (Rs 56 million, about $2.5 million) to Rs 12,350 crore (Rs 123.5 billion, about $2.65 billion).[43] In the second half of the 1990s Karnataka was, she writes, the only state to grow faster than 8 per cent a year.[44]

What the work paid

The economics behind the cluster were simple, and Saxenian set them out without ornament. 'India's greatest asset is a large, educated, English-speaking workforce that is willing to work at relatively low wages.' In 1994, she wrote, 'wages for software programmers and systems analysts in India were less than one-tenth of those for their US counterparts, and lower even than other developing countries like Mexico'.[45]

Low wages meant low revenue per worker. On the figures Saxenian used, the industry employed about 180,000 people in 1998, and annual revenue per employee was $15,000 to $20,000, against about $100,000 per employee in Israel and Ireland.[46] Revenue per head is the bargain's own measure: a firm that sold time at a low price could grow only by adding heads. Wages rose fast from their low base: the National Association of Software and Service Companies (Nasscom), the industry's lobby group, estimated that software pay rose 21 per cent a year in the late 1990s.[47] That eroded the cost advantage that had brought the work. Ashank Desai, chairman of Mastek, a leading Indian software company, said in 1999 that Indian firms were finding it 'difficult for them to compete only on one differential', which was cost.[48] The largest firms, among them Wipro and Infosys, were seeking bigger contracts, often on a turnkey basis, in which the firm delivers a finished system. 'Rather than competing on the basis of hourly productivity,' Saxenian wrote, 'they aim to accumulate intellectual property.'[49]

For the programmers, the city was often a waiting room for America. Those who worked abroad for Indian firms were 'paid their salaries in rupees and provided with minimal allowances for housing and expenses', and some called the business 'resume selling'.[50] 'Many thus aspire to work for foreign companies,' she wrote, 'not only for the relatively high wages but also for the opportunity be transferred overseas.'[51] By 1999 Indians were receiving the largest share of the H-1B temporary work visas that the United States issued.[52]

Behind them came a large training system. By Nasscom's count, Indian colleges and polytechnics trained more than 67,000 computer science professionals a year, and another 200,000 people enrolled each year in the private software training institutes that had multiplied in the 1990s.[53] Bangalore drew on that system and on the colleges of the three southern states that supplied most of the country's computing graduates.

The gap between those who got the jobs and everyone else stayed wide. Writing in 2013, Roye reported that the average software professional in Bangalore, outside the call-centre trade, earned Rs 50,000 a month, which he put at about $1,000, more than eight times the state's monthly income per head of Rs 6,000.[54] The figure is a journalist's unsourced estimate, but it points the same way as the evidence of a decade earlier: fast-rising pay for a skilled minority in a poor state.

Cyberabad

In November 1998 the chief minister of the neighbouring state of Andhra Pradesh, the 48-year-old Nara Chandrababu Naidu, inaugurated a building called CyberTowers near Madhapur, on the edge of Hyderabad. It was to be the centrepiece of a planned Hi-Tech City. 'Until now, the world has heard of the twin cities of Hyderabad-Secunderabad,' he said. 'From today, the world will know of the triplet cities of Hyderabad-Secunderabad-Cyberabad.'[55]

Business Today's cover story the next month, written by an economist based in Hyderabad, set the challenge in context. Hyderabad had once been called the nation's second capital, and public-sector firms such as Bharat Heavy Electricals and the Electronics Corporation of India had set up there in the 1970s. But 'the late 1980s saw Bangalore overtaking Hyderabad' as South India's most dynamic and corporate-friendly city.[56] Naidu's answer was to court the

largest names in software. 'When Naidu got Microsoft's CEO, Bill Gates, to locate its R&D centre at CyberTowers, he made sure no other company could ignore the city.'[57] Oracle, BaaN, Tata IBM, Satyam, Wipro and others were already there.[58]

Hyderabad's pitch was aimed at Bangalore's weak point. J.A. Chowdhury, a director of the Software Technology Parks of India and, in the magazine's words, a key player in Naidu's Project Cyberabad, put it bluntly: 'Hyderabad is preparing for the future. Other cities will be left behind when traffic-congestion and urban chaos overtakes them.'[59] He described the city as a 'knowledge hub'. It had the first Indian Institute of Information Technology, research institutes, defence laboratories and six universities, so that the supply of computing skills matched the demand.[60] 'Infotech professionals would like to live close to work so that they can work early morning or late into the night,' Chowdhury said, and the magazine added that housing could always be found within ten minutes of the workplace.[61] The magazine, making Naidu's case, also cited a Nasscom survey of IT professionals that scored Hyderabad 3.40 out of 4 for the availability of qualified people and 3.20 for quality of life, but only average for climate, commuting and its educational facilities.[62]

The same article recorded doubts. Drinking water was short. Large parts of the city had 'grown haphazardly'. V.K. Bawa, one of its best-regarded urban planners, had drawn up a plan in the 1970s that no government had implemented, and he complained: 'The model of Hyderabad's development is regressive; it does not conform to modern concepts of urban growth.'[63] The whole project depended on one politician. If Naidu won the next state election, due in November 1999, the direction was clear. 'If he loses, the imponderables are many.'[64]

Saxenian saw in Naidu something larger than one city's ambition. While the central government had begun liberalisation, she wrote, 'the state governments have pioneered some of the most far-reaching policy innovations in the IT sector', and Naidu had made 'e-governance', the use of computers to deliver public services, his signature.[65] His campaign set off competition from his neighbours. Karnataka's new government announced plans to upgrade its 'overburdened roads', and Tamil Nadu and Kerala drew up IT policies of their own.[66] By 2001 software employment was concentrated in Karnataka (Bangalore), Andhra Pradesh (Hyderabad), Tamil Nadu (Chennai), Maharashtra (Bombay) 'and areas surrounding Delhi'.[67]

The enclave

The sharpest contemporary warning about the cluster came from Saxenian herself, in a paper that asked whether Bangalore was 'the Silicon Valley of Asia'. The comparison, she answered, was 'not only misleading but also distracts attention from the deeper challenges and opportunities that IT affords the Indian economy'.[68] Most IT development already took place in parks 'insulated from the day-to-day challenges of doing business in India by dedicated communications links, private power sources, and liberal rules for investment and taxation'.[69] The insulation answered real shortages. In 1997 India had 18.6 telephone lines per 1,000 people, against 55.7 in China, and a new connection took more than twelve months to arrive.[70]

Its cost was disconnection. 'Professionals in IT enclaves could become better connected, both economically and socially, to distant regions than to the rest of the Indian economy.'[71] The policy of the day, she concluded, 'risks accelerating the growth of IT as a small, modern enclave in a poor and backward economy'.[72]

The road not taken

Saxenian did not think the enclave was the only possible Bangalore. India, she wrote, 'already has the design capabilities' to build products of its own. 'However India lacks the environment needed to support experimentation with the application of these skills to new markets.'[73] Her example was the Simputer, a very low-cost mobile computer priced at under Rs 9,000, about $200 by her conversion, built by a Bangalore team from IISc and Encore Software.[74] Encore was the company of the same Vinay Deshpande whom Roye had seen with a soldering iron on Richmond Road.[75] Her prescription was a level field, in which products and services sold at home would 'enjoy the same tax benefits as those currently enjoyed by software export units'.[76] A regime built to earn foreign exchange had made it easier to sell software abroad than at home.

Three readings

The evidence supports three readings of what happened in Bangalore, and they are not exclusive. The first is a story of agglomeration. A city filled with public-sector engineers and laboratories attracted a few early movers, TI, Wipro and Infosys among them, and the cluster fed itself as employees left to found firms nearby.[77] On this reading Bangalore's software industry was the delayed return on forty years of state investment in defence, science and engineering education.

The second reading is Saxenian's, that Bangalore depended on cheap skill and an export regime, and that its best engineers were drawn towards America.[78] TI's chip showed that the capability was real. Revenue per worker showed how little of it was being used. What filled the parks at the end of the 1990s was routine work done at great volume, as hundreds of American and European companies outsourced 'coding, maintenance and Y2K solutions' to Indian firms.[79]

The third reading is about the city. The industry created a well-paid professional economy with private power, private links and few ties to the city's poor.[80]

K.S. Garden

What did the boom do for the people of Bangalore who did not write software? The most careful answer comes from RoyChowdhury's study of six slums in the inner city, on J.C. Road close to Lalbagh, which she calls the central commercial hub of the city. She surveyed 300 households and held focus groups.[81] She begins with the transformation. The city had 'become the icon of India's information technology revolution'. Real estate prices soared. The large campuses of the public-sector firms became 'shadows of a past': much of their land was put up for sale, units closed, and their permanent workforces were 'downsized and contractualized'.[82]

The new economy grew, she writes, 'through the expansion of capital intensive rather than labour intensive industries, as well as skill-based industries, which offered limited scope of employment, and only to technically skilled personnel'.[83] The poor found work in construction and in the export garment trade, at low wages, and in new service jobs: driving the cars and vans of call centres and outsourcing firms, guarding and cleaning malls and supermarkets, waiting at tables. Whether in factories or services, that work remained 'in the casual/contract mode with no security of service, insurance or pension'.[84] 'In Bangalore, nearly one million poor live in slums,' she reports, and about a third of slum dwellers lived below the poverty line, on less than Rs 2,500 a month, which she converts as $55.[85]

The IT economy reached one of her slums, K.S. Garden, in a particular way. It had the lowest poverty ratio of the six. Families in its better-paid occupations, from skilled trades to permanent jobs in public bodies, could send their children to private English-medium schools or to computer training.[86] 'About 50 youths in the slum had managed to get some kind of computer training. Of these, some 25-30 women were working as receptionist in small offices, or showrooms, at cash counters in small super markets.'[87] They had left the low-paid work of their parents. But the gain was fragile. In the focus groups, 'even those with some computer training frequently remained unemployed or only partially employed', and where someone had found a foothold, 'a connection through family or network had worked in most cases'. RoyChowdhury's conclusion is plain: 'there had not been any systemic or policy based effort to incorporate the urban underclass into the city's growing economy'.[88] In the same slum, coolie labour and domestic service still accounted for 52.3 per cent of households surveyed.[89]

Roye's article of 2013 names two individuals, both young women from the villages and towns around the city. Anusaya, from Immadihalli village, worked at an ice-cream kiosk in a shopping mall in Whitefield, one of the new technology districts, for Rs 6,000 a month, about $100 by Roye's conversion. 'I work here in the day, go to evening college,' she told him. 'The money I earn meets some of my family expenses and funds college fees.'[90] Ramya Gowda, from Malur, worked at a doughnut counter. 'I got this job because I learnt English and have finished school,' she said. 'This is a part time job. I'd like to study more, especially computers, and work in a big company like Yahoo!'[91] Neither was a software engineer. Both were working at the edge of the economy that software had built, and both described education, English first and computers next, as the way in.

Roye recorded what the corridors had replaced. The campuses of Whitefield, Electronic City and the Outer Ring Road stood largely on what had been farmland or swamp; many farmers on the city's edge had sold their land; the city had fewer trees, lakes and open spaces, and was by his account the third most expensive in India.[92] He ended the

article in a bar on Lavelle Road, where a young software worker from Thanisandra told him: 'Seriously uncle, the city has arrived!'[93]

Notes to Chapter 10. Bangalore (1947-2010)

[1] Roye, 'Revolution on a Bullock Cart' (2013).

[2] Roye, 'Revolution on a Bullock Cart' (2013). The scene rests on Roye's own memory, written about forty years after the event. The bullock-cart arrival at Sona Towers opens Chapter 8.

[3] Infosys Consultants, recruitment advertisement, *Times of India*, 26 November 1991, p. 17. The advertisement, published on 26 November 1991, is described in Chapter 9.

[4] Saxenian, 'Bangalore: The Silicon Valley of Asia?' (2001), p. 7 n. 10. The rule is discussed below, in 'The parks and the bargain'.

[5] Computer History Museum, 'Oral History of N.R. Narayana Murthy, Part 2' (2020), p. 10.

[6] Computer History Museum, 'Oral History of N.R. Narayana Murthy, Part 2' (2020), p. 10.

[7] Computer History Museum, 'Oral History of N.R. Narayana Murthy, Part 2' (2020), pp. 10-11.

[8] RoyChowdhury, 'Livelihood and Income' (c. 2011), p. 1.

[9] Energy and Wetlands Research Group, IISc, 'Demography and Economy' (live-checked 2026-10-13). These are the research group's figures as published on its website, not the census tables themselves.

[10] Roye, 'Revolution on a Bullock Cart' (2013).

[11] Roye, 'Revolution on a Bullock Cart' (2013).

[12] IISc, 'IISc's Role in Helping Wipro Launch Its Minicomputers' (2019).

[13] Heeks, 'The Uneven Profile of Indian Software Exports' (1998), p. 12.

[14] Heeks, 'The Uneven Profile of Indian Software Exports' (1998), p. 12.

[15] Computer History Museum, 'Oral History of N.R. Narayana Murthy, Part 2' (2020), p. 10.

[16] Computer History Museum, 'Oral History of N.R. Narayana Murthy, Part 2' (2020), p. 10.

[17] Computer History Museum, 'Oral History of N.R. Narayana Murthy, Part 2' (2020), p. 10.

[18] Computer History Museum, 'Oral History of N.R. Narayana Murthy, Part 2' (2020), pp. 10-11. Murthy gives no year for the move to Bangalore, and no primary record of the date has been located.

[19] Saxenian, 'Bangalore: The Silicon Valley of Asia?' (2001), p. 14 n. 27.

[20] Heeks, 'The Uneven Profile of Indian Software Exports' (1998), pp. 12-13.

[21] Heeks, 'The Uneven Profile of Indian Software Exports' (1998), p. 13.

[22] Heeks, 'The Uneven Profile of Indian Software Exports' (1998), p. 13.

[23] Heeks, 'The Uneven Profile of Indian Software Exports' (1998), p. 13.

[24] Heeks, 'The Uneven Profile of Indian Software Exports' (1998), p. 13.

[25] Roye, 'Revolution on a Bullock Cart' (2013); Computer History Museum, 'Oral History of N.R. Narayana Murthy, Part 2' (2020), p. 24. Murali's list begins with a liberalised economy, friendly state policies, a skilled English-speaking workforce and the public-sector giants. Murthy's earlier account of why Infosys itself moved to Bangalore does not mention the weather (Computer History Museum, 'Oral History of N.R. Narayana Murthy, Part 2' (2020), pp. 10-11). Heeks's interviews and Murthy's account of the move both put engineers first; the climate was a real and perceived amenity that made the city easier to sell to engineers and managers.

[26] Saxenian, 'From Brain Drain to Brain Circulation' (2005), p. 7.

[27] Heeks, 'The Uneven Profile of Indian Software Exports' (1998), p. 13. The sources date the link to 1985 or 1986; Chapter 8 tells how it was negotiated and weighs the dates.

[28] Saxenian, 'Bangalore: The Silicon Valley of Asia?' (2001), p. 6.

[29] Saxenian, 'Bangalore: The Silicon Valley of Asia?' (2001), p. 6.

[30] Saxenian, 'Bangalore: The Silicon Valley of Asia?' (2001), p. 11. The employer is anonymous and speaks through two layers of citation, so the words are evidence of what managers said rather than of what the work was. A second passage on the same page, quoted in Chapter 15, has the chief executive of an offshore development centre claiming parity with his company's headquarters in Silicon Valley; Saxenian's footnotes trace both passages to the same interview of July 1996, so they may be one man.

[31] Rajaraman, *History of Computing in India* (2012), p. 48.

[32] Saxenian, 'Bangalore: The Silicon Valley of Asia?' (2001), p. 11 n. 21.

[33] Rajaraman, *History of Computing in India* (2012), p. 48.

[34] Heeks, 'The Uneven Profile of Indian Software Exports' (1998), p. 12.

[35] Heeks, 'The Uneven Profile of Indian Software Exports' (1998), p. 12.

[36] Hanna, *Exploiting Information Technology for Development* (1994), p. 97. The origins of the scheme are told in Chapter 8.

[37] Saxenian, 'Bangalore: The Silicon Valley of Asia?' (2001), pp. 6-7.

[38] Saxenian, 'Bangalore: The Silicon Valley of Asia?' (2001), p. 7 n. 10.

[39] Saxenian, 'Bangalore: The Silicon Valley of Asia?' (2001), p. 7 n. 10.

[40] Saxenian, 'Bangalore: The Silicon Valley of Asia?' (2001), p. 7.

[41] Saxenian, 'Bangalore: The Silicon Valley of Asia?' (2001), p. 10. Chapter 9 sets out and compares the series for the onsite share.

[42] Saxenian, 'Bangalore: The Silicon Valley of Asia?' (2001), p. 10 n. 19.

[43] RoyChowdhury, 'Livelihood and Income' (c. 2011), p. 9 (citing the Karnataka Development Report 2007); dollar figures converted at the calendar-year average rates for 1991 (Rs 22.69) and 2003 (Rs 46.58) from Reserve Bank of India, *Handbook of Statistics on Indian Economy*; the source does not say whether its years are calendar or fiscal.

[44] RoyChowdhury, 'Livelihood and Income' (c. 2011), p. 9.

[45] Saxenian, 'Bangalore: The Silicon Valley of Asia?' (2001), p. 9.

[46] Saxenian, 'Bangalore: The Silicon Valley of Asia?' (2001), p. 12. Other counts of the workforce differ; Chapter 9 compares them.

[47] Saxenian, 'Bangalore: The Silicon Valley of Asia?' (2001), p. 13.

[48] Saxenian, 'Bangalore: The Silicon Valley of Asia?' (2001), p. 13.

[49] Saxenian, 'Bangalore: The Silicon Valley of Asia?' (2001), p. 12.

[50] Saxenian, 'Bangalore: The Silicon Valley of Asia?' (2001), p. 10.

[51] Saxenian, 'Bangalore: The Silicon Valley of Asia?' (2001), p. 12.

[52] Saxenian, 'Bangalore: The Silicon Valley of Asia?' (2001), p. 12. Chapter 13 takes up the visa.

[53] Saxenian, 'Bangalore: The Silicon Valley of Asia?' (2001), p. 10.

[54] Roye, 'Revolution on a Bullock Cart' (2013). Roye gives no source for the figure.

[55] *Business Today*, 'A Cyber View' (1998).

[56] *Business Today*, 'A Cyber View' (1998).

[57] *Business Today*, 'A Cyber View' (1998).

[58] *Business Today*, 'A Cyber View' (1998). The Microsoft decision is given as the magazine reported it; Microsoft's own announcement has not been located.

[59] *Business Today*, 'A Cyber View' (1998).

[60] *Business Today*, 'A Cyber View' (1998).

[61] *Business Today*, 'A Cyber View' (1998).

[62] *Business Today*, 'A Cyber View' (1998).

[63] *Business Today*, 'A Cyber View' (1998).

[64] *Business Today*, 'A Cyber View' (1998).

[65] Saxenian, 'Bangalore: The Silicon Valley of Asia?' (2001), p. 13.

[66] Saxenian, 'Bangalore: The Silicon Valley of Asia?' (2001), p. 14.

[67] Saxenian, 'Bangalore: The Silicon Valley of Asia?' (2001), p. 20.

[68] Saxenian, 'Bangalore: The Silicon Valley of Asia?' (2001), p. 2.

[69] Saxenian, 'Bangalore: The Silicon Valley of Asia?' (2001), pp. 20-21.

[70] Saxenian, 'Bangalore: The Silicon Valley of Asia?' (2001), p. 16.

[71] Saxenian, 'Bangalore: The Silicon Valley of Asia?' (2001), p. 20.

[72] Saxenian, 'Bangalore: The Silicon Valley of Asia?' (2001), p. 20.

[73] Saxenian, 'Bangalore: The Silicon Valley of Asia?' (2001), p. 24.

[74] Saxenian, 'Bangalore: The Silicon Valley of Asia?' (2001), pp. 23-24.

[75] Rajaraman, *History of Computing in India* (2012), p. 55; Roye, 'Revolution on a Bullock Cart' (2013).

[76] Saxenian, 'Bangalore: The Silicon Valley of Asia?' (2001), p. 23.

[77] Heeks, 'The Uneven Profile of Indian Software Exports' (1998), p. 13; Computer History Museum, 'Oral History of N.R. Narayana Murthy, Part 2' (2020), pp. 10-11.

[78] Saxenian, 'Bangalore: The Silicon Valley of Asia?' (2001), pp. 2, 12, 19.

[79] Saxenian, 'Bangalore: The Silicon Valley of Asia?' (2001), pp. 10-11.

[80] Saxenian, 'Bangalore: The Silicon Valley of Asia?' (2001), p. 20; RoyChowdhury, 'Livelihood and Income' (c. 2011), pp. 8, 26. Two urban histories bear most directly on this reading: Janaki Nair's history of twentieth-century Bangalore (Nair, *The Promise of the Metropolis* (2005)) and James Heitzman's study of the city's planning in the IT era (Heitzman, *Network City* (2004)). Neither is drawn on here, and no ethnography of Bangalore's software workers is used in this chapter, so the professionals' own account of their city is absent from it.

[81] RoyChowdhury, 'Livelihood and Income' (c. 2011), pp. 3, 17. The paper, published by Oxford University's Contemporary South Asian Studies Programme, is undated but was released around 2011.

[82] RoyChowdhury, 'Livelihood and Income' (c. 2011), p. 1.

[83] RoyChowdhury, 'Livelihood and Income' (c. 2011), p. 8.

[84] RoyChowdhury, 'Livelihood and Income' (c. 2011), pp. 8-9.

[85] RoyChowdhury, 'Livelihood and Income' (c. 2011), p. 12.

[86] RoyChowdhury, 'Livelihood and Income' (c. 2011), pp. 24-25.

[87] RoyChowdhury, 'Livelihood and Income' (c. 2011), p. 25.

[88] RoyChowdhury, 'Livelihood and Income' (c. 2011), p. 26.

[89] RoyChowdhury, 'Livelihood and Income' (c. 2011), p. 26.

[90] Roye, 'Revolution on a Bullock Cart' (2013).

[91] Roye, 'Revolution on a Bullock Cart' (2013).

[92] Roye, 'Revolution on a Bullock Cart' (2013).

[93] Roye, 'Revolution on a Bullock Cart' (2013).

Chapter 11. Y2K and Nasdaq (1995-2002)

At twenty past six on a Tuesday morning in late 2000, a correspondent for *Business Today*, R. Sukumar, stood on a street in BTM Layout, a growing neighbourhood in south Bangalore (Bengaluru since 2014), opposite a restaurant called Udipi Garden. He was waiting for a company bus. It was one of the buses that carried close to 4,000 people a day to the campus of Infosys Technologies on Hosur Road, and he had been told that many of its staff lived nearby. Staff from outside the south, he learned later, often clubbed together in threes or fours to sub-let a flat or house in neighbourhoods like this one.[1] On the bus, almost everyone was young and almost nobody talked. People read newspapers, one young woman read what looked like a technical book, and a few at the front fell asleep. The company discouraged its junior staff from riding motorcycles to work, because Hosur Road was so dangerous.[2] The crowd was more mixed than he had expected: many from the south of India, as he had assumed, but also faces from the west, the north and the north-east.[3]

The bus reached the campus at a quarter past seven. Sukumar had visited in the summer of 1995, when it had one office block, one canteen and one small gymnasium. Now it covered 50 acres and had five office blocks, three canteens, a branch of the old Bangalore eating house Mavalli Tiffin Rooms, a coffee bar, a pizza outlet, a swimming pool, a sauna, a medical centre and a small golf green.[4] It also had a three-storey training centre, built for Rs 23.6 crore (about $5 million), where every batch of new engineers went through a three-month course. The whole campus, the magazine reported, would eventually cost Rs 320 crore, about $71 million. Its chief financial officer, T.V. Mohandas Pai, showed off a building whose curved walls, he said, had taken more than 4,000 equations to get right.[5] The cars that drove in a little before eight were mostly small Marutis and Santros. Everyone paid market prices for food in the canteens.[6]

The magazine had named Infosys the best employer in India, and its article was written in that spirit.[7] But its numbers were specific. The share price had risen from Rs 63.75 in early April 1996 to Rs 7,160 on 15 December 2000. Of the company's 7,925 employees at the end of September 2000, 7,853 had been offered stock options, creating, the magazine reported, '1,773 rupee millionaires and 213 dollar millionaires'.[8]

Five and a half years earlier this firm had walked out of a negotiation with its largest customer and lost a quarter of its business. In between, the leading Indian services firms turned a way of organising work into something they could certify, sell and list on an American stock exchange. A programming error in old American software, the millennium bug, brought them inside the largest companies in the United States. Then the market that had made their employees rich turned and fell.

A quarter of the revenue

In May 1995, in the telling of N.R. Narayana Murthy, the company's co-founder, Infosys alone among the Indian suppliers of General Electric (GE) refused its demand for lower prices and handed its GE work to competitors (Chapter 9).[9]

Two lessons can be read in the company's later filings. The first was about dependence. When Infosys registered its shares with the US Securities and Exchange Commission (SEC) in February 1999, its largest client provided 10.5 per cent of revenue in the year to March 1998, and it had about 130 clients.[10] The same document listed among the causes of uneven results 'the size, timing and profitability of significant projects'.[11] By the year to March 2002 the largest client provided 6.1 per cent of revenue and the five largest together 24.1 per cent. In both filings the company described its mix of work as chosen 'in an effort to avoid service or client concentration'.[12] The second lesson, in Murthy's own account, was about price. He said in 2020 that 'at least until 2007, 2008, probably 2009, Infosys always

got 20-25 percent higher price compared to our competitors in India'. He credited a sales method the company called Business Value Leverage, which meant talking to clients about what the work would be worth to them rather than about technology.[13] The claim is the founder's alone; no independent price data for the period have been located.

What the filings do show is how fast the firm grew from its small base. Revenue rose from $9.5 million in the year to March 1994 to $68.3 million in the year to March 1998.[14] Over the same years the number of its IT professionals rose from about 480 to about 2,200, and its development centres in India from two to nine.[15] Revenue grew at a compound rate of 63.6 per cent a year.[16]

Part of that growth came from the currency. In the year to March 1998, 90.0 per cent of Infosys's revenue was earned in US dollars, and the filing said that 'a significant portion' of its costs, including pay, would continue to be in rupees. Between December 1994 and December 1998 the rupee lost about 39.5 per cent of its value against the dollar.[17] A firm that earned dollars and paid rupees gained from every fall in the rupee unless it raised salaries to match. Process, Y2K and good management all contributed to the margins of these years. So did the exchange rate.

Selling the process

To sell work that would be done ten thousand miles from the client, the firms had to show that it would be done properly. They sought two certificates. The international quality standard ISO 9001 was awarded after auditors inspected a firm's documentation, testing and project methods, and the Capability Maturity Model (CMM) of the Software Engineering Institute (SEI) at Carnegie Mellon University graded the 'maturity' of a firm's development process on five levels.[18] In the account of V. Rajaraman, the historian of Indian computing, ISO 9001 certification was 'essential for bidding for software projects in Europe whereas the CMM maturity level is valued more in the USA'.[19]

Motorola's Global Software Group, set up in India in 1991, says that its Indian centre reached Level 5 in 1993 and was 'the first commercial organization in the world' to do so, a claim Rajaraman repeats.[20] A *Business Today* report of January 2003 says only that the Indian unit was the first company in India to be certified, in 1994, at an unstated level, so the world first remains Motorola's own claim.[21]

By the end of the decade Indian firms held a large share of the highest ratings, though the sources do not agree on how large. The researcher AnnaLee Saxenian, writing in 2001, said that in December 1999, 137 Indian companies had ISO 9000 or CMM Level 2 certification, 'and ten companies were certified at Level 5 (the highest level, at which only six U.S. companies are certified.)'[22] Rajaraman gives a different figure for the same year: 'In 1999, 6 out of 12 CMM level 5 certified companies in the world were in India.'[23] Either way, a handful of Indian firms held a large share of the top grade, and the industry said so loudly. 'Quality certification', Saxenian wrote, 'serves as an important marketing device'.[24] The share kept growing. By January 2003, *Business Today* reported, 54 of the 78 organisations assessed at Level 5 worldwide were Indian companies or the Indian operations of multinationals, against 19 American companies.[25] The magazine also recorded what the grade was for. At a Nasscom-Gartner summit in Mumbai in September 2002, one of the chief executives present, whom it did not name, explained how it helped: 'When you tell customers that you are a Level 5 company,' he said, 'they shut up and listen.'[26]

Infosys described its own standing in the February 1999 filing: it had 'been certified at Level 4, a level achieved by only 2% of the more than 1,000 software companies tested under the Capability Maturity Model'.[27] Both the 2 per cent and the word 'certified' are the company's. The SEI's process was an appraisal, and the industry described the result as a certificate.

Rajaraman also explained why process mattered to the firms themselves. Both certificates, he wrote, 'require emphasis on well documented processes which every software engineer in the organization is expected to follow'. And 'The process orientation allowed companies which were recruiting large numbers of employees each year to quickly train them to become productive members of a development team. Staff attrition also could be managed better due to the disciplined software development process.'[28] Rajaraman presents this as a benefit, and it was one. But it also describes what process did to the engineer. A firm that hired thousands of graduates a year, and lost many of them to rivals or to American visas, needed work broken into documented steps that any trained graduate could follow and any other could take over. The certificate reassured the client. The process underneath it made individual engineers easier to replace. Rajaraman does not draw the conclusion, but his sentence points to it: documented, standardised work is also the kind machines learn most easily. Chapter 19 sets that reading against what the firms' own filings later said about tools for coding, testing and documentation.

The pyramid

The work itself was organised as a pyramid. Infosys's 1999 filing described a typical project: a 'small team of two to five IT professionals' visited the client to set the scope, then its managers went back to India 'to supervise a much larger team of 10 to 50 IT professionals', with the client's systems 'linked via satellite' to the Indian centres.[29] A few experienced people faced the client. Many junior people did the work in India. The filing said the company did about 80 per cent of its project work in India.[30]

The firm described its own model as an industry practice. 'The Indian offshore development model was initiated in the mid-1980's,' the filing said (Chapter 6 tests that dating against the founders' own careers), and it listed what attracted clients: a large pool of English-speaking professionals, 'relatively low labor costs', international quality standards, the 'capability to work on specific projects on a 24-hour basis by exploiting time zone differences', and the ability to run phases of a project in parallel.[31] By 2002 the language had changed. The annual report for the year to March 2002 said that 'Our clients leverage our proprietary Global Delivery Model to achieve higher quality, rapid time-to-market and cost-effective solutions.'[32] In three years an industry practice had become a company brand.

For the industry as a whole, the move offshore was real but slower than the brand suggests. On each of the period's series, more than half of the industry's export work was still done at clients' sites abroad, which is why the story of the H-1B visa (Chapter 13) runs alongside this one.[33]

The pyramid was efficient at turning cheap hours into exports, but by international standards it produced little revenue per head (Chapter 10 gives Saxenian's comparison with Israel and Ireland). Saxenian saw the largest firms, such as Wipro and Infosys, trying to escape that trap: winning larger contracts 'often on a turnkey basis', charging higher rates, and turning what they learned on one project into software components they could reuse for the next.[34] The same pages record the other way out. Indian programmers, aware of what their skills earned abroad, were leaving, many of them on American visas.[35]

The pyramid had a consequence that a senior Infosys manager described in the *Business Today* article, published in January 2001. Sales had more than doubled in the first half of the year to March 2001, and the magazine noted that growth widened the middle and top of the organisation. 'We are growing so fast that many people with two to three years experience are getting into managerial roles,' said Hema Ravichandar, the senior vice-president for human resources.[36] Fast promotion for early recruits depended on a base that grew faster still. That was the book's bargain in miniature: revenue grew with the number of engineers billed, so the base of the pyramid had to keep widening. Not everyone wanted to climb. 'I don't have to become a project manager if I don't want to,' said Sushil, who worked in the company's mobile-commerce practice and is identified in the article only by his first name. 'I can focus on the technology and become a technical consultant to projects.'[37]

The millennium bug

The work that filled the base of the pyramid in the late 1990s was Y2K. Rajaraman explains the problem simply. Programs written in the 1970s and 1980s, 'mostly in COBOL', stored the year as two digits to save memory, on the assumption that the first two were 19. When the year became 2000 they would give wrong results, and every such program had to be found and corrected. 'This bug fix was labour intensive. There were not enough software engineers in the USA and Europe to fix the bug and it had to be outsourced. This was a godsend opportunity to many fledgling software services companies in India.'[38]

At Infosys, the 1999 filing shows how quickly the work arrived. Y2K projects made up about 7.5 per cent of revenue in the year to March 1997 and 23.3 per cent in the year to March 1998.[39] In the nine months to December 1998 the share was 21.4 per cent.[40] The often repeated figure of 23 per cent 'in 1997' is the same number, since the company's fiscal year 1998 ran from April 1997 to March 1998.[41] Y2K tripled as a share of the business in a single year and had already begun to shrink as a share by the end of 1998.

The company was frank with American investors about what kind of work this was. 'The high demand for these time-sensitive projects results in pricing and margins that are favorable to the Company,' the filing said. It expected demand to 'begin to diminish rapidly after fiscal 1999', and it could not promise to replace the work with other work at the same margins.[42] Then came a sentence about people. 'There is an additional risk that the Company may be unable to retrain and redeploy IT professionals who are currently assigned to Year 2000 conversion projects involving legacy computer systems after such projects are completed.'[43] In a document filed under American

securities law, the leading Indian firm said that a wave of demand had trained a cohort of its engineers for work that would soon end, and that it might not find other work for them.

For the industry, the figures come from the National Association of Software and Service Companies (Nasscom), the industry's lobby group, as reported by the trade newsletter *Computergram* in June 1999. Indian software exports in FY1998-99 were $2.65 billion, up 51.4 per cent, and Y2K work accounted for $560 million of that, or 21.1 per cent. Nasscom said that 203 of the Fortune 1000 companies used Indian software services, and that 25 firms earned 63 per cent of all exports.[44] Two weeks earlier the same newsletter had reported a larger Nasscom estimate: that Indian firms had earned $2.3 billion from the bug, might reach $3 billion by the end of 1999, and would earn only 'a few hundred million dollars' from it the following year.[45] The cumulative figure is hard to reconcile with $560 million in the peak year unless it spans several years or other kinds of work; the share of a single year's exports, about a fifth, is the safer number. The same report described the trap that the end of the work would set. Large companies around the world, it said, were holding back from buying new computer systems until the millennium had passed, and were no longer spending much on the bug either, because most of it had been dealt with. Indian firms that had relied on Y2K feared that their profits would fall.[46]

Nasscom already had an answer to the end of the work. Its president, Dewang Mehta, spoke of 'three Es', electronic commerce, enterprise resource planning (ERP, the integrated software that large companies were then installing to run their accounts, stock and staff) and euro conversion, and of an export target of $50 billion by FY2007-08.[47] The euro was a second conversion job. The European Union announced in 1995 that the single currency would be used from 1 January 1999, and banks and other financial institutions had to change their programs before then. In Rajaraman's words, this 'was a labour intensive job and provided software conversion contracts to numerous Indian software companies'.[48]

The millennium arrived without a crisis. In early 2000 the news magazine *Outlook* asked in a headline whether the whole thing had been 'The Rs 1,800 Cr Hoax?'. It put the world's spending on the problem at $600 billion, about $150 billion of it in the United States. It reported Nasscom's estimate that Y2K preparations had cost India Rs 1,800 crore across government and private industry, of which Rs 600 crore was for software and services.[49] The magazine converted the total to $3.6 billion. At the 1999 average exchange rate it was about $420 million.[50] The article also recorded what the night had been like for the firms. Rajeev K. Arora, resident manager of Tata Consultancy Services (TCS), a company that by the magazine's account had written more than 650 million lines of code for Y2K correction, said TCS had bought overseas tickets and arranged visas so that it could fly its engineers to clients if anything went wrong, and had stocked diesel for its generators.[51] Mehta's own verdict, as *Outlook* quoted it, was that 'Y2K was a godsend and might prove to be the turning point for the industry.'[52]

Was it the turning point? The export figures say not. In the series Rajaraman reproduces, Indian software exports had grown by between about a quarter and two-thirds a year in every year from 1990-91 to 1997-98, most of that before Y2K work became large.[53] At its peak Y2K was about a fifth of the industry's exports and about a fifth of the leading firm's revenue, and that firm expected it to fade quickly. What Y2K did, on the best evidence, was open doors. Rajaraman writes that many firms 'earned the goodwill of clients by not only fixing the Y2K bug but also improving the software (when appropriate) at no extra cost', and that 'This opportunity gave many software services companies entry to the Fortune 500 companies and enabled them to get software development contracts later.'[54] N. Chandrasekaran, who had joined TCS in 1987 and became its chief executive in 2009, gave a similar account to the Computer History Museum in 2020, two decades after the event. The Year 2000 work, he recalled, 'gave us scale, because everyone wanted a solution', and clients who saw it done well began to ask what else the firm could do for them.[55] He also gives part of the credit for the quiet rollover to the programmers 'working diligently in India between 1993 and 1999'.[56] That is his judgement, and the *Outlook* article shows that it was not universally shared at the time.[57] The common belief that Y2K made the Indian industry is half right. It accelerated growth that had already begun, and it put Indian engineers inside the systems of large American companies, some of which kept them on for other work.

Two hundred thousand applications

The engineers on the bus in 2000 had been chosen from a very large crowd: in the year to March 2002 Infosys received about 193,640 applications and about 1,550 people joined, about one in 125, against about one in 55 four years earlier.[58] These are the company's figures, and it used them to advertise itself, but they also show a supply of

graduates who wanted these jobs growing far faster than even the fastest-growing firm could take them. Infosys still selected by its own tests and taught its own methods.[59]

The firm explained its appeal in its own terms. In 1999 it told American investors that, because of its stock option plan, it had 'become one of the most sought after employers for Indian engineering graduates', with attrition 'significantly below the industry average'. It added that 'India is home to the world's second largest population of English-speaking technical talent'.[60]

The education that mattered most happened after the degree, in the three-month course that every batch of new engineers took in the training centre the reporter had been shown.[61] The firm selected from a vast pool, taught its recruits its own methods, and placed them into a documented process.

The people inside it were mostly engineers. The magazine counted 4,616 employees on the Bangalore campus in late 2000, and said that apart from the 840 people in the company's staff functions, such as human resources, finance, marketing and administration, everyone was an engineer.[62] The magazine noticed the strain as well as the perks. 'Every time things get a little too much for me,' said an employee who, it reported, 'not infrequently puts in 16 hours a day at work', 'I go out for a smoke, or a walk, or play a few holes of golf.' Muthamma Acharya, 'a twenty-something marketing executive', described the pace of hiring: 'You come in one morning and find the breakfast line 400-people longer'.[63] Suparna Shankar, an assistant manager in marketing for the Asia-Pacific region, summed up the place for the magazine: 'You can create a role for yourself.'[64] These were the voices a magazine chose for a story about the best employer in India.

Nasdaq

Infosys had listed in India in 1993, in an offering Murthy remembered as 'a very tough time'.[65] Six years later it went to New York. The final prospectus, dated 10 March 1999 and filed the next day, offered 1.8 million American Depositary Shares (ADSs), certificates traded in the United States that each represented half an Indian share, at $34.00 each. The offering raised $61.2 million before an underwriting discount of $1.68 a share. The shares were to be quoted on the Nasdaq National Market under the symbol INFY.[66] The American price was set a little below the equivalent price in Mumbai, and until then, the prospectus noted, there had been no public market for the company's shares in the United States.[67]

Murthy gave a specific reason for the listing. Infosys had prepared its accounts under American accounting rules voluntarily from 1995, so that clients could read them. 'Later on, we said, "Look. We have to create an employee stock option plan in the U.S. currency." Therefore, we wanted ADRs [American Depository Receipts], and that is how we got listed on NASDAQ in 1999.'[68] In his account, the listing was as much about paying staff as about raising money.

Others followed. In June 1999 Mehta of Nasscom said he expected up to ten Indian firms to list in the United States over the next twelve to eighteen months.[69] The next was not a software services firm but an internet company. Satyam Infoway, the internet subsidiary of Satyam Computer Services, priced its ADSs on 18 October 1999 at what the filing gives as $4.50 each, and they began trading on Nasdaq the next day.[70] Satyam Computer Services owned about 52.5 per cent of it in 2002.[71] A rival Indian content provider, Rediff.com, completed its own American offering in June 2000.[72]

Rupee millionaires

The listing turned paper into wealth for employees because Infosys had given them paper. 'In 1994, the Company established the ESOP, one of the first employee stock option plans in India,' the 1999 filing said. A trust set up for the purpose held warrants for shares, and employees received rights to buy shares at Rs 100 each, $2.35 at the exchange rate of February 1999, which the filing called 'substantially below the market price'. The rights vested over five years.[73] The filing explains the plan at length because under American accounting rules the discount was a cost, and under Indian rules it was not.[74] Stock options were so unusual in India that the government's information technology task force of 1998, among its 108 recommendations, proposed allowing 'sweat equity' and employee stock option plans, which Rajaraman describes as 'alien to other industries in India'. That required a change in company law.[75]

A rupee millionaire, in the magazine's count, held at least Rs 10 lakh (Rs 1 million), about $22,000 at the 2000 exchange rate.[76] That was a large sum for an engineer in his or her twenties, but not a fortune. A dollar millionaire was another matter. Infosys's sales had grown from Rs 88.97 crore in the year to March 1996 to Rs 884.35 crore in the

year to March 2000.[77] The new wealth reached the city. 'Indeed, a couple of enterprising Bangalore-based real-estate developers make a living primarily from selling apartments to people who work for Infosys,' the magazine wrote.[78] In 2020 Murthy put the scale of the programme in his own words: Infosys had distributed 19 per cent of its equity to employees through stock options, worth about $9 billion at that time's prices.[79]

The money was very unevenly spread. It went to those who had joined early, at the leading firm, and stayed long enough to vest. A graduate who joined a smaller firm, or who joined Infosys in 2001, was in a different position.

The bust

The bust looked different depending on where one stood. For investors in Indian internet shares it was a collapse. Satyam Infoway's annual report for the year to March 2002 said that since the offering its ADSs had traded between 'a high of $113 per ADS' and a low of $0.87.[80] The price table in the same report shows a low of $0.72 in the quarter to September 2001.[81] On 5 November 2001 a securities class action was filed in New York on behalf of everyone who had bought its ADSs between 20 October 1999 and 6 December 2000. It named the company, some of its executives and some of the underwriters of its offering, and alleged that the underwriters had failed to disclose commissions and arrangements with investors to whom they allocated shares.[82] In October 2001 Satyam Computer Services announced that it intended to sell its shares in the company.[83] The parent's own later history belongs to Chapter 16.

For the largest services firm the bust was a slower line, not a fall. From the year to March 1998 to the year to March 2002, Infosys's revenue grew at a compound 68.1 per cent a year and its IT staff rose from about 2,200 to about 9,400.[84] But its outlook for the following year was cautious. 'The economic slowdown in the U.S. has impacted the growth prospects of companies that are in the insurance, banking and financial services and telecom segments,' it said, and companies 'may cut their IT spending or postpone decisions regarding new IT expenditures'.[85] The industry's exports kept rising. The Reserve Bank of India recorded software services receipts of $6.3 billion in the year to March 2001 and $7.6 billion in the year to March 2002.[86]

For new graduates, the bust arrived on campus. In January 2002 *The Hindu Business Line* reported that Anna University had placed 'only around 320 students with various companies, down from 1,200 around the same time last year'. The year before had been a record: 65 per cent of the 70 companies that visited the university had been IT firms. This time about 35 companies came. Wipro and Satyam, which had taken 49 and 23 students the year before, did not come at all. TCS recruited 54 students against 177, Infosys 46 against 126, and Cognizant Technology Solutions 69 against 105.[87] Dr N. Baluswamy, who directed the university's Centre for University-Industry Collaboration, which ran placements, said that some of the previous year's graduates had seen their offers postponed indefinitely, and that the centre had negotiated with the companies. 'A few companies have delayed the date of joining for last year students by over a year,' he said. Some IT companies, 'except big players such as TCS, Infosys and CTS', had cut the pay promised on campus 'by 50-60 per cent'.[88]

At the Indian Institute of Technology (IIT) in Madras (Chennai since 1996), a little over 400 of the 719 students registered with the placement centre had been recruited by 51 companies, 30 of them in IT, with ten companies still to come, against about 750 placed the year before. About 30 per cent of students stayed away from campus interviews that year, the paper reported, because of 'shocking stories of lay-offs, delayed appointments and salary cuts' among the previous year's recruits. Most engineering students now wanted 'to consider only engineering companies, rather than shift to IT', and some said they wanted stable jobs.[89] Companies from outside IT were coming back. Maruti Udyog returned after about four years and the Gas Authority of India after fifteen, and NTPC was expected for the first time. At a management institute in Tiruchi, the highest offer came from Citibank, at Rs 6.5 lakh a year, about $13,600.[90]

The pattern matters for the rest of this book. In the placement director's account, the largest firms did not cut the pay they had promised; other firms cut salaries, and some put off joining dates by more than a year. The colleges went on producing graduates while the firms at the end of the pipeline cut their campus intake, the two largest at Anna University by more than half. Later chapters show the same pattern returning.

Hours and outcomes

One thread runs through the filings of these years and connects them to the end of this book. Most of Infosys's revenue in the mid-1990s had come from time-and-materials contracts, under which the client pays for the hours or months worked, and by 1999 the company said it wanted more fixed-price work.[91] Y2K added a lesson of its own. It

paid well because it was urgent: the company said that 'high demand for these time-sensitive projects' brought favourable prices and margins.[92] The risk ran the other way too: the 1999 filing counted 'the accuracy of estimates of resources and time required to complete ongoing projects, particularly projects performed under fixed-price, fixed-time frame contracts' among the causes of uneven results.[93] Under a fixed price, a project that overran cost the firm and not the client.

By 2002 the company reported progress. 'As compare [sic] to the previous year, we have increased the proportion of projects that are undertaken on a fixed-price, fixed-timeframe, rather than a time-and-material basis, which reduce client risk, and if effectively structured and executed, provide opportunities for enhanced margins.'[94] A firm that sells hours earns less when its engineers work faster, unless it uses the saving to win more work. A firm that sells an outcome at a fixed price keeps what it saves. Before 2002 the leading Indian firm had decided that it wanted, where it could, to sell outcomes. When tools arrive that cut the hours needed for coding and testing, the work that Rajaraman says took 70 to 80 per cent of a project's time and was done in India, the way a contract is priced decides who keeps the gain.[95]

In 2002 that question was still far off. The more immediate change was in the size of the machine. In 1991 Infosys had advertised for sixty-four engineers.[96] In 2002 it employed about 10,740 people and had received nearly 200,000 applications in a year.[97] The firms' demand for new engineers had fallen for a season. The supply had not. The next chapter follows the graduates who kept coming, and the companies that learned to hire them by the thousand.

Notes to Chapter 11. Y2K and Nasdaq (1995-2002)

[1] *Business Today*, 'Infosys: Wealth & Values' (2001).

[2] *Business Today*, 'Infosys: Wealth & Values' (2001).

[3] *Business Today*, 'Infosys: Wealth & Values' (2001).

[4] *Business Today*, 'Infosys: Wealth & Values' (2001).

[5] *Business Today*, 'Infosys: Wealth & Values' (2001); Reserve Bank of India, *Handbook of Statistics on Indian Economy*, calendar-year average rate for 2000 (Rs 44.94 per US dollar); conversions by the author.

[6] *Business Today*, 'Infosys: Wealth & Values' (2001).

[7] *Business Today*, 'Infosys: Wealth & Values' (2001).

[8] *Business Today*, 'Infosys: Wealth & Values' (2001).

[9] Computer History Museum, 'Oral History of N.R. Narayana Murthy, Part 2' (2020), pp. 19-20. As Murthy recalled it in 2020, a cut for GE would have meant cuts for every other client, and less money for pay, tools and training.

[10] Infosys, Form F-1 (1999), p. 5.

[11] Infosys, Form F-1 (1999), p. 10.

[12] Infosys, Form 20-F FY2002, p. 5; Infosys, Form F-1 (1999), p. 6.

[13] Computer History Museum, 'Oral History of N.R. Narayana Murthy, Part 2' (2020), p. 16.

[14] Infosys, Form F-1 (1999), p. 8.

[15] Infosys, Form F-1 (1999), p. 27.

[16] Infosys, Form F-1 (1999), p. 5.

[17] Infosys, Form F-1 (1999), p. 12.

[18] Rajaraman, *History of Computing in India* (2012), p. 45.

[19] Rajaraman, *History of Computing in India* (2012), pp. 45-46.

[20] Motorola, Global Software Group media kit, Motorola GSG media kit (company self-description); Liu, 'Quick Win Strategy in CMMI Transition', Motorola (2005), slide 4 (a timeline slide the group showed at a conference of the American National Defense Industrial Association in 2005, which also places 'CMM 5: India' in 1993; slide image checked, the India Level 5 marker drops from the 1993 column); Rajaraman, *History of Computing in India* (2012), p. 45 (Rajaraman repeats the claim without a separate source).

[21] Babu, 'The Quality Bug', *Business Today* (2003), para. 4: 'Motorola India Electronics was the first company in India to get a SEI CMM certification in 1994'; the level is not given. Motorola's own material and the magazine differ on the year as well as the claim, and no record from the Software Engineering Institute itself has been located.

[22] Saxenian, 'Bangalore: The Silicon Valley of Asia?' (2001), p. 11. The two counts may cover different months or count organisations differently.

[23] Rajaraman, *History of Computing in India* (2012), p. 52.

[24] Saxenian, 'Bangalore: The Silicon Valley of Asia?' (2001), p. 11.

[25] Babu, 'The Quality Bug', *Business Today* (2003), para. 1.

[26] Babu, 'The Quality Bug', *Business Today* (2003), para. 7.

[27] Infosys, Form F-1 (1999), p. 6.

[28] Rajaraman, *History of Computing in India* (2012), p. 46.

[29] Infosys, Form F-1 (1999), p. 45.

[30] Infosys, Form F-1 (1999), p. 6.

[31] Infosys, Form F-1 (1999), p. 45.

[32] Infosys, Form 20-F FY2002, p. 4.

[33] Saxenian, 'Bangalore: The Silicon Valley of Asia?' (2001), p. 10 (Parthasarathy's series, 58 per cent at clients' sites in 1998); Khanna and Morales, 'The IT Boom and Other Unintended Consequences' (2016), p. 5 (56 per cent on site in 2000).

[34] Saxenian, 'Bangalore: The Silicon Valley of Asia?' (2001), p. 12.

[35] Saxenian, 'Bangalore: The Silicon Valley of Asia?' (2001), p. 12.

[36] *Business Today*, 'Infosys: Wealth & Values' (2001).

[37] *Business Today*, 'Infosys: Wealth & Values' (2001).

[38] Rajaraman, *History of Computing in India* (2012), p. 46.

[39] Infosys, Form F-1 (1999), pp. 32-33.

[40] Infosys, Form F-1 (1999), p. 15.

[41] Rajaraman, *History of Computing in India* (2012), p. 46. Rajaraman took the figure from an earlier history.

[42] Infosys, Form F-1 (1999), p. 15.

[43] Infosys, Form F-1 (1999), p. 15.

[44] *Computergram* (1999), 'Indian Software Exports Booming', 30 June 1999.

[45] *Computergram* (1999), 'Indian Software Firms Fear Post-Y2K Profit Slump', 14 June 1999.

[46] *Computergram* (1999), 'Indian Software Firms Fear Post-Y2K Profit Slump', 14 June 1999.

[47] *Computergram* (1999), 'Indian Software Firms Fear Post-Y2K Profit Slump', 14 June 1999.

[48] Rajaraman, *History of Computing in India* (2012), p. 47.

[49] Mukherjee, 'The Rs 1,800 Cr Hoax?' (2000).

[50] Mukherjee, 'The Rs 1,800 Cr Hoax?' (2000); Reserve Bank of India, *Handbook of Statistics on Indian Economy*, calendar-year average rate for 1999 (Rs 43.05 per US dollar); conversion by the author.

[51] Mukherjee, 'The Rs 1,800 Cr Hoax?' (2000).

[52] Mukherjee, 'The Rs 1,800 Cr Hoax?' (2000).

[53] Rajaraman, *History of Computing in India* (2012), p. 48.

[54] Rajaraman, *History of Computing in India* (2012), p. 46.

[55] Computer History Museum, 'Oral History of Natarajan Chandrasekaran' (2020), PDF pp. 1, 8 (interviewer's introduction for the dates; Chandrasekaran's recollection, 2020).

[56] Rajaraman, *History of Computing in India* (2012), p. 46.

[57] Mukherjee, 'The Rs 1,800 Cr Hoax?' (2000).

[58] Infosys, Form 20-F FY2002, p. 16 (year to March 2002: about 193,640 applications, about 6,730 interviewed, about 1,710 offers and about 1,550 accepted; recruiting focused 'on the top 20% of students from engineering departments of Indian schools'); Infosys, Form F-1 (1999), p. 6 (year to March 1998: about 70,870 applications and about 1,300 acceptances); *Business Today*, 'Infosys: Wealth & Values' (2001) (*Business Today*: 183,966 applications in 1999, 259,820 in 2000 up to 1 December, and 2,991 people hired from April to September 2000). Ratios calculated by the author.

[59] Infosys Consultants, recruitment advertisement, *Times of India*, 26 November 1991, p. 17 (the 1991 advertisement, for engineers of any discipline, with training promised); Infosys, Form F-1 (1999), p. 6.

[60] Infosys, Form F-1 (1999), p. 6.

[61] *Business Today*, 'Infosys: Wealth & Values' (2001).

[62] *Business Today*, 'Infosys: Wealth & Values' (2001).

[63] *Business Today*, 'Infosys: Wealth & Values' (2001).

[64] *Business Today*, 'Infosys: Wealth & Values' (2001).

[65] Computer History Museum, 'Oral History of N.R. Narayana Murthy, Part 2' (2020), p. 18.

[66] Infosys, prospectus (Form 424B4, 1999), cover page (dated 10 March 1999; filed 11 March 1999).

[67] Infosys, prospectus (Form 424B4, 1999), cover page: the equivalent price on the Mumbai exchange on 10 March 1999 was $37.18 per ADS, at Rs 42.50 to the dollar.

[68] Computer History Museum, 'Oral History of N.R. Narayana Murthy, Part 2' (2020), p. 18.

[69] *Computergram* (1999), 'Indian Software Exports Booming', 30 June 1999.

[70] Satyam Infoway, Form 20-F FY2002, p. 55. The filing appears to give the $4.50 on the basis it uses for its price history, which is stated after a four-for-one split of the ADSs in January 2000.

[71] Satyam Infoway, Form 20-F FY2002, p. 15.

[72] Satyam Infoway, Form 20-F FY2002, p. 10.

[73] Infosys, Form F-1 (1999), p. 28.

[74] Infosys, Form F-1 (1999), p. 28.

[75] Rajaraman, *History of Computing in India* (2012), p. 50.

[76] Reserve Bank of India, *Handbook of Statistics on Indian Economy*, calendar-year average rate for 2000 (Rs 44.94 per US dollar); conversion by the author.

[77] *Business Today*, 'Infosys: Wealth & Values' (2001).

[78] *Business Today*, 'Infosys: Wealth & Values' (2001).

[79] Computer History Museum, 'Oral History of N.R. Narayana Murthy, Part 2' (2020), p. 24.

[80] Satyam Infoway, Form 20-F FY2002, p. 18 (prices after the January 2000 split).

[81] Satyam Infoway, Form 20-F FY2002, p. 55.

[82] Satyam Infoway, Form 20-F FY2002, p. 13.

[83] Satyam Infoway, Form 20-F FY2002, p. 15.

[84] Infosys, Form 20-F FY2002, p. 5.

[85] Infosys, Form 20-F FY2002, p. 11.

[86] Reserve Bank of India, *Handbook of Statistics on Indian Economy*, balance-of-payments software services receipts ($6,341 million in FY2000-01, $7,556 million in FY2001-02), RBI Handbook edition id 10663.

[87] Raja Simhan, 'It's a Real Slowdown at Campuses' (2002).

[88] Raja Simhan, 'It's a Real Slowdown at Campuses' (2002).

[89] Raja Simhan, 'It's a Real Slowdown at Campuses' (2002).

[90] Raja Simhan, 'It's a Real Slowdown at Campuses' (2002); Reserve Bank of India, *Handbook of Statistics on Indian Economy*, financial-year average rate for FY2001-02 (Rs 47.69 per US dollar); conversion by the author.

[91] Infosys, Form F-1 (1999), pp. 32, 44.

[92] Infosys, Form F-1 (1999), p. 15.

[93] Infosys, Form F-1 (1999), p. 10.

[94] Infosys, Form 20-F FY2002, p. 6.

[95] Rajaraman, *History of Computing in India* (2012), p. 45.

[96] Infosys Consultants, recruitment advertisement, *Times of India*, 26 November 1991, p. 17.

[97] Infosys, Form 20-F FY2002, p. 16.

Chapter 12. The engineer factory (1998-2012)

After one year in his first software job, a young engineer whom the anthropologists Chris Fuller and Haripriya Narasimhan, in their published study, call Raghu earned 200,000 rupees a year (about $4,400). His father, after thirty years as an engineer at the nuclear power station at Kalpakkam, south of Chennai, earned 400,000.[1] Raghu had grown up in Kalpakkam, taken a degree in electronics and communication engineering at a college in Coimbatore, and joined one of India's six largest software companies, which the anthropologists call Indian Computer Services (ICS). When the researchers met him in Chennai he was 22, unmarried and a Tamil Brahman, a member of a community that made up about 2.5 per cent of Tamil Nadu's population and that, in the authors' 'strong impression', dominated many of the city's software firms.[2]

Raghu worked on production support for a British client's system. That is the work of keeping software that is already running in use: fixing what breaks, answering the client's requests, handling the next fault. He told the researchers that every day brought a new problem to solve, so he was never bored. But in their words he 'thinks he is not learning much', except how to turn down demanding foreign clients, and he wanted to leave the industry soon and study for an MBA. In the meantime he enjoyed the money.[3]

Raghu stands near the start of a machine that, between the late 1990s and the early 2010s, turned out software workers on a scale India had never seen. It had two halves. One was a mass of new engineering colleges, most of them private, which multiplied the number of young Indians with an engineering degree. The other was inside the firms: recruiting drives that tested hundreds of thousands of graduates a year, and training campuses that turned engineers of any branch into programmers in a few months. Both halves served the bargain on which the industry ran. The firms sold engineers' time, so their revenue grew with the number of engineers they could bill, and the factory existed to keep that number rising.

Colleges by the hundred

For forty years after independence India built engineering colleges slowly. The government's statistical compendium counts 31 engineering and technology colleges in 1950-51, 149 in 1980-81 and 289 in 1990-91.[4] Then the pace changed. There were 486 such colleges in 1997-98, 678 in 2000-01, 1,077 in 2001-02 and 1,562 in 2005-06.[5] The country added more engineering colleges in those eight years than in the previous five decades together.

V. Rajaraman, the historian of Indian computing, describes 'a rapid expansion of engineering colleges between 1998 and 2010', and 'a rapid growth of private engineering colleges and colleges offering MCA courses'.[6] The Master of Computer Applications (MCA) was the postgraduate course for graduates in science and commerce, combining computing, mathematics and management. The law made room for the private colleges. On 31 October 2002 eleven judges of the Supreme Court, in *T.M.A. Pai Foundation v. State of Karnataka*, overruled the fee and admission scheme the Court had laid down in the *Unni Krishnan* case of 1993 (Chapter 7).[7] For a college that took no government money, 'The decision on the fee to be charged must necessarily be left to the private educational institution'. Capitation fees and profiteering stayed forbidden, but 'a reasonable revenue surplus' for development and expansion was allowed.[8] A share of seats could again be filled by the management.[9] Private corporate bodies were allowed to set up universities in 2002, and under this policy, Rajaraman writes, private universities and private colleges 'proliferated'.[10] The supply of MCAs grew even faster than that of engineers, from about 7,000 graduates in 1998 to 120,000 by 2010.[11]

Quality did not keep pace, and the people who said so were the employers. Rajaraman, writing around 2012, notes that graduates of the Indian Institutes of Technology (IITs) and the National Institutes of Technology, about 10,000 a

year, were excellent, but that 'industry employers opine that the graduates of other institutions are not employable as soon as they graduate and need further training'.[12] The regulators responded with inspection. The All India Council for Technical Education (AICTE), the statutory body that approves technical colleges, set up a National Board of Accreditation, and the University Grants Commission an assessment council to grade institutions.[13]

The expansion was uneven across the map, and the south's colleges fed the south's software cities.[14] In Chennai, Fuller and Narasimhan found that of twenty software engineers under thirty on whom they had information, only three had studied outside Tamil Nadu. The rest could stay at home, because young men and women, Brahmans among them, could now get a good engineering education within the state.[15]

By the early 2010s the building had outrun demand. By 2012-13 AICTE had approved about 1.55 million undergraduate engineering seats, and about 37.6 per cent of them were empty, according to an analysis of the council's data by the education non-profit Avanti Fellows.[16] Approved undergraduate intake peaked at about 1.7 million in 2014-15. By 2016-17 almost half of the seats, 49.5 per cent, had no student in them.[17] The colleges had been built for a market that was already starting to change.

Sidebar: How many engineers does India really produce?

No figure about Indian education is quoted more often, or more loosely, than the number of engineers the country turns out each year. In the mid-2000s, a study at Duke University found, typical press articles said that in 2004 India had graduated 350,000 engineers and China 600,000, against about 70,000 in the United States.[18] Later, figures of well over a million a year appeared, including in Rajaraman's own history. Neither kind was a like-for-like count of four-year engineering graduates.

The Duke study was an early attempt to check. In December 2005 a team of graduate students led by Gary Gereffi, a sociologist, and Vivek Wadhwa, an executive teaching engineering management, found that the large Indian and Chinese figures 'include not only four-year degrees, but also three-year training programs and diploma holders', and that they had been set against American four-year degrees alone.[19] The figure they used for India in 2004, a projection by the National Association of Software and Service Companies (Nasscom), the industry's lobby group, was 215,000 graduates in engineering, computer science and information technology. Only 112,000 of them held four-year bachelor's degrees; the other 103,000 had three-year diplomas.[20] Per million people, they reckoned, the United States produced about 750 such graduates a year, China about 500 and India about 200.[21]

The team also found something more striking. Nasscom's projections rested on government data published with a 'three- to four-year lag', and 'the most recent official Indian graduation data are from 1993'.[22] Through much of the boom, on the Duke team's account, official counts of the engineers the industry was hiring were erratic and years out of date, and the industry's own lobby group was the 'de facto authority'.[23]

Later figures do not settle the question. Rajaraman gives the best-known growth series: 'In 1998 the number of engineering students was around 135,000 and in 2010 it grew to 1,300,000, a growth of almost 10 times.'[24] Elsewhere in the same history he describes what appears to be the same figure differently: 'Currently India graduates around 1.3 million engineers per year'.[25] Neither passage defines it. A total enrolment of 135,000 in 1998 is far too small for 486 colleges teaching four-year courses, so the 1998 figure is most likely one year's intake. By the same reasoning, 1.3 million in 2010 is most likely intake or approved seats rather than graduates. This book treats it as seats.

In 2014 Aspiring Minds, a company that sells employment tests to employers, wrote of 'the six hundred thousand engineers that graduate annually', without giving a source.[26] If the Duke and Aspiring Minds figures are roughly right, India's output of four-year engineering graduates grew about five times between 2004 and 2013, from about 112,000 to about 600,000. That is broadly in line with the more than threefold rise in college numbers between 1997-98 and 2005-06.[27]

The answer, then, depends on the question. By the early 2010s India approved about 1.5 million undergraduate engineering places a year, and a third or more of them stayed empty.[28] Its output of four-year engineering graduates was about 112,000 in 2004 and perhaps 600,000 by 2013, neither of them an official count. The often-repeated claim that India produces 1.5 million engineers a year matches the seats, not the graduates. The real number grew very fast. It was never as large as the figure that travelled.

Any branch will do

The second half of the factory explains why so many colleges were worth building. The large software firms did not hire computer scientists in any great number. They hired engineers, and then made them into programmers themselves. Rajaraman sets out the logic plainly: 'The IT companies' common complaint was that there were not enough employable graduates with computer science degrees. Instead they recruited all engineering graduates with aptitude in computing; however they could not be directly placed as members of project teams.'[29] So, he continues, 'All major IT companies had in-house training programmes ranging from 12 weeks to 6 months.'[30]

A mechanical, civil or electrical engineer would do, if he or she could pass the firm's aptitude test. That made a degree in any engineering branch a ticket to the best-paid jobs open to most graduates, and it gave families a reason to pay for one. A Nasscom projection, reported by the Duke team, was that 84.8 per cent of India's four-year engineering graduates in 2004 took branches it classed as IT-related, a broad category that included electrical, electronics and instrumentation engineering as well as computer science.[31]

The scale of recruitment is clearest in the filings of Infosys, the Bangalore (Bengaluru since 2014) software company, made to American securities regulators in much the same categories from year to year. In the year to March 1998 it made about 1,790 offers.[32] Ten years later, in the year to March 2008, it reported screening over 885,000 applications, testing over 227,000 applicants, interviewing more than 97,000 and making about 46,300 offers.[33] Offers had grown about twenty-six times, though the filings do not separate new graduates from experienced recruits.[34] Headcount tells the same story. Infosys had about 9,400 IT professionals in March 2002, about 36,800 employees in March 2005, about 91,200 in March 2008 (of whom about 18,700 worked outside India) and about 150,000 in March 2012.[35] A generation earlier the pattern had been very different. Natarajan Chandrasekaran, who joined Tata Consultancy Services (TCS) in 1986-87, recalled in a 2020 oral history for the Computer History Museum a firm of about 500 to 600 professionals that 'used to hire 20 people or 40 people a year', often by taking a whole class from IIT Bombay or IIT Delhi.[36]

Infosys was not unusual in kind, only in what it published. Rajaraman writes that 'every major software company was recruiting around 8000 graduates each year' in the years from 1998 to 2010, and that with attrition of 15 to 20 per cent and workforces 'of the order of 120,000', 'the replacement numbers themselves were over 15,000 each year'.[37]

How dependent the colleges had become on this hiring was shown by the one year it faltered: after the dot-com bust, campus placements at Anna University in Chennai by January 2002 were less than a third of the year before, as Chapter 11 describes.[38] In Bangalore, many of the engineers interviewed in 2004 and 2005 by the researchers Carol Upadhya and A.R. Vasavi of the National Institute of Advanced Studies (NIAS) described the downturn of 2000-02. Some, the researchers wrote, 'who had been campus recruited were later turned down by the companies, and remained unemployed for months'.[39] Infosys, at least, came back quickly, and on a much larger scale. In the year to March 2004 Infosys alone made about 13,600 offers of employment.[40]

Mysore

Before there was a training campus, there was a training building. In 2001 the magazine *Business Today* described the Infosys headquarters campus in Bangalore, where in a three-storey chrome and glass centre 'fifty of the country's best software instructors put every batch of engineers recruited by the company through a gruelling three-month training programme'.[41] By 2002, its filing said, Infosys employed 57 faculty members who ran three-month sessions for new recruits, and it had a Leadership Institute for its staff on a 200-acre campus in Mysore (Mysuru since 2014), elsewhere in Karnataka.[42]

The trainee campus came next, and Mysore became the factory floor. In its filing for the year to March 2005, Infosys reported that it had 'recently completed the construction of an employee training facility in Mysore'. The facility could 'house 4,500 trainees at any one time' and train about 12,000 employees a year.[43] The company said it did not fully use new employees on client work while they were in training, 'particularly during our 14-week training course for new employees'.[44] Three years later the capacity was 4,800 at a time and about 15,200 a year.[45] By the year to March 2012 the filing had a new name for it. 'We train the new engineering graduates that join us at our Global Education Center in Mysore,' the company wrote. 'With a total built-up area of 1.44 million square feet, the Infosys Global Education Center can train approximately 14,000 employees at a time.'[46] The course had lengthened

too. The same filing describes a '20-29 week training course for new employees'.[47] The filings give capacities and square feet, not the voices of the trainees.[48]

The firm also reached back into the colleges. Through a programme it called Campus Connect, Infosys worked with engineering colleges so that their own teachers could give industry-related training. In the year to March 2012 it reported more than 190 workshops for more than 4,500 college teachers.[49] The boundary between the college and the company, which in 1998 had been the campus interview, was becoming harder to see.

The pyramid and the bench

The shape of the workforce followed from the shape of a project: a small team at the client's site and a much larger one in India, the pyramid set out in Infosys's 1999 registration statement.[50] Its base had to be refilled every year with new graduates, and the growth of the base was what moved people upward.

The firms measured how much of the pyramid was earning money. Infosys called this utilisation. 'We define employee utilization as the proportion of total billed person months to total available person months excluding support personnel,' its filings said.[51] In the year to March 2012, utilisation of billable staff was 69.0 per cent including trainees and 75.9 per cent excluding them, down from 72.1 and 79.6 per cent the year before, a fall the company told investors had hurt its profitability.[52]

Put the other way round, on average in that year nearly a third of Infosys's billable staff, counting trainees, was not billed to a client. Some were in training. Others were between projects, or waiting for one to start. The industry called this the bench. In the firms' own terms it was not waste. Most work was still billed by the time spent on it: in the year to March 2008 only 31 per cent of Infosys's services revenue came from fixed-price projects, and the rest, 69 per cent, was billed on a 'time-and-materials basis'.[53] A firm paid that way must be able to staff a new contract at once, so it keeps trained people waiting. Here the book's bargain shows at the scale of one firm. Revenue followed the number of people billed, so growth meant hiring ahead of demand, and hiring ahead of demand meant a bench. But the bench is a cost that grows with every hiring wave, and it is the first thing cut when demand falls, as it was in 2001-02 and again in 2008-09.

Billing by time also made the engineer's time the thing to be managed. In Bangalore, Upadhya and Vasavi concluded that 'extracting the maximum amount of time (and "effort") out of software engineers is key to the profitability of services companies', and one respondent described the firms' systems for tracking hours as 'a system of "time slavery"'.[54] In their survey 61 per cent said they worked nine to ten hours a day on average, and staff of large Indian companies reported an average maximum of 13.8 hours, against 12.3 in multinationals.[55] The complaints they recorded were mundane: 'They'll give out work at 5:30 in the evening and expect you to have it done by the next morning, which isn't fair'.[56] One informant explained why flexitime did not shorten the day: 'What's the point of working till late if you aren't seen working?'[57]

Turnover was the other side of the pyramid. Infosys estimated the industry's attrition rate, the share of staff leaving each year, at 'approximately 20%'. Its own rate rose from 9.7 per cent in the year to March 2005 to 17.0 per cent in the year to March 2011.[58] People in Chennai talked about job-hopping constantly. Fuller and Narasimhan found less of it than the talk suggested: among the more senior ICS staff they knew, at least half had spent all or most of their working lives in the company.[59]

One in twelve

Who passed through the gate? The best evidence for these years comes from the Chennai study. Between August 2003 and February 2005, Fuller, an anthropologist at the London School of Economics, and Narasimhan, who did most of the fieldwork, studied IT professionals in the city.[60] Most of their interviews were at ICS.[61] They interviewed nineteen of its staff and described their sample as 'tiny' and 'the haphazard product of the problems of access' in an industry concerned with security. 'All names used are pseudonyms,' they wrote.[62] Everyone from the study named here is a real informant under a false name given by the authors, and most of their words come to us as the authors reported them.

At ICS the gate was a series of filters on school and college marks. Only students with a consistent record of high examination marks could sit the company's aptitude test and go on to an interview, where one in twelve was offered a job.[63] Starting salaries, in the middle of the range for large software firms, were 'around 15,000 rupees per month' for people in their early twenties (about $330), and could 'easily double within a few years'.[64] That put a new

recruit's own income near the level that only about 6 per cent of Indian households reached in 2001-02, according to estimates by the National Council of Applied Economic Research cited by the authors.[65]

The study also shows where these recruits came from. Most of the younger people the authors interviewed had fathers, and occasionally mothers, in professional or managerial posts in government or in public- or private-sector firms; the only exceptions were a few small businessmen and farmers.[66] In this sample, IT did not lift families into the middle class from below. It moved the children of the salaried middle class, many of them the children of government engineers and officials, into a richer, private-sector version of the class they were born into. That is one city and one firm. In Bangalore, Upadhya and Vasavi's survey of 132 software engineers found the same pattern: 84 per cent of their fathers worked in occupations 'that can be identified as solidly middle class'.[67] Together the two studies are the best evidence against the common claim that software created India's middle class.

Inside, the work was organised around appraisal and deadlines. Staff were appraised every six months; those who beat their targets could expect quick promotion, and those who fell short were told they would be 'let go'. Most worked a 40 to 50-hour, five-day week on flexitime, with extra hours when deadlines pressed. The offices were modern and open-plan, and young men and women, many of them unmarried, mixed there more freely than in most social settings in Chennai.[68]

The gate had a keeper. ICS's personnel manager, whom the authors call Ramakrishna, was in his late thirties, held an MCA, and described his job as 'tough and challenging'. In the 'war for talent' among the leading firms, it was his responsibility to make sure ICS recruited enough good people, above all by picking the best graduates from the top colleges. He earned about 70,000 rupees a month, plus stock options, and through much of 2003-04 he was negotiating for a rise while threatening to leave for a job in Bangalore.[69]

A young man the anthropologists call Shoaib shows what the move felt like. He was 29, ran the company's intranet and website, and had a degree in computer science from a Muslim college in Chennai and an MCA. His father had been an engineer in the state electricity board. After his father's early death Shoaib was offered a post at the board, and turned it down. He said he would never work in the public sector, because there you always 'have to bow your head to somebody'.[70] He was also, the authors noted, more worried than most of his colleagues about the future: 'there is no security in this job'.[71]

Older managers in Chennai's manufacturing firms watched the move with some resentment. Sampath, the personnel manager of a bus and lorry maker the study calls Chakra Vehicles, said the IT industry had 'disturbed the social fabric', with its 'quick promotions', 'fancy designations' and 'phenomenal compensation'.[72] One of his colleagues, the authors report, said that at campus interviews by the late 1990s only the weakest men still wanted to join Chakra, because everyone better was chasing IT salaries. Sampath himself, after more than twenty years, probably earned about 60,000 rupees a month, a level a competent IT professional could expect to reach in half the time.[73]

Where the money went is less glamorous than the stereotypes. As far as the authors could tell, most of it went on housing, bought, often with loans, much earlier than their parents had, and for those with children on private schooling.[74] Most of the young people they met had never even considered a government or public-sector job. One ICS assistant manager had turned down a post at the Indian Space Research Organisation because he thought it would be unchallenging and monotonous.[75] The city did not always welcome its new earners. Raghu shared a flat with two other young men in IT, and they were unpopular with their neighbours, who feared they would smoke, drink, cook meat and make a nuisance of themselves. In fact, the authors found, they never cooked at all. They ate every meal out, came home on weekdays only to sleep, and had friends round at weekends to drink beer.[76]

Marks, English and caste

The Chennai study's most discussed finding is about caste, and the authors were careful with it. ICS, like many other IT companies in the city, 'is said to be dominated by the high-status "Forward Castes"', they wrote, especially Tamil Brahmans, who made up about 2.5 per cent of the state's population, 'and it is our strong impression that this is true'. But 'no accurate figures exist', and the guesses of people in the industry varied widely.[77] In a later paper they repeated that Tamil Brahmans were 'disproportionately well represented' in south India's IT industry, and that 'no figures to confirm this assertion exist', although all the circumstantial evidence pointed that way.[78] The study's terms are the ones used here. In its usage, 'Forward Castes', Brahmans among them, are those that cannot benefit

from the reservation system of quotas in education and public employment, which is operated for the 'Backward' (and 'Most Backward') and 'Scheduled' Castes.[79]

The interest of the study lies less in the impression than in the explanation. Many people in Chennai said the reason was reservations. Tamil Nadu reserved 68 per cent of state-sector posts and college seats for the Other Backward Classes and the Scheduled Castes and Tribes, and many said this was why private companies employed so many Brahmans. 'Yet reservations alone cannot fully explain upper-caste predominance in IT companies.'[80]

The authors offered three further reasons. First, Brahmans had become an increasingly urban community, more likely than others to have the social skills and cultural knowledge, fluent English among them, that global IT companies wanted and that their selection tests demanded.[81] Second, the firms' own rules did the sorting. 'Because ICS allows only candidates with consistently high marks to apply for posts, its selection policy in practice favours Forward Caste candidates who, from a tender age, have been motivated to achieve repeated scholastic success.'[82] Third, Brahman families placed a high value on education, notably in mathematics and science, and they were the best-educated community in the state. The authors dismissed as 'fanciful' the popular idea, held by many Tamil Brahmans themselves, that the caste had a special aptitude for software because of its traditional religious learning.[83]

A footnote shows how small the measurable differences were. In 2004 the cut-off mark for admission to electronics and communication engineering at Anna University, the most competitive course in the state, was 293.28 out of 300 for Forward Caste candidates, 290.57 for Backward Castes, 283.67 for Most Backward Castes and 276.13 for Scheduled Castes.[84] The gap at the top was less than three marks in three hundred. The authors' point was that the belief in needing exceptionally high marks shaped how children were raised, even where the arithmetic was less harsh than families thought.[85]

The argument does not need the firms to have looked at caste at all. A selection system that screened on marks, then on an aptitude test, then on an interview in English, reproduced the advantages of urban, educated, upper-caste families on its own. The Chennai evidence is one city and one firm, and it is not a national count.[86] Bangalore points the same way. In Upadhya's account of the NIAS survey, 'Brahmins constituted 48 per cent of our sample', upper castes as a whole 71 per cent, and upper and dominant agricultural castes together 86 per cent. 'Only one respondent said that he belonged to a scheduled community.'[87] The authors warned that 132 engineers, drawn mainly from multinationals and large and medium Indian firms, were 'much too small to be representative'.[88] The mechanism is visible elsewhere. In 2014 Aspiring Minds argued that in screening applications 'the college name is a key signal, and resumes from unknown colleges aren't shortlisted', so that more than half of the graduates who met its benchmark for IT services jobs, studying outside the top 750 colleges, were 'invisible to most employers'.[89] It also estimated that a student from a third-tier college, with the same test score and marks as a student from a first-tier one, had 24 per cent lower odds of getting a job, and if hired was paid 66,000 rupees a year less.[90] The claims come from a firm with an interest in selling another way to find such students. They point in the same direction: a filter that worked by college name fell hardest on students from the new private colleges and the smaller towns.

Women in the factory

At ICS, 'approximately one-fifth' of new recruits were women, close to the share across the industry.[91] According to Nasscom, 79 per cent of India's software professionals were men in 2001-02, a share it expected to fall to 65 per cent by 2005.[92] In the colleges the gap was narrower. Among undergraduates enrolled in engineering and technology courses at AICTE-approved institutions in 2012, 33.7 per cent were women, according to the council's figures as compiled by the Society of Women Engineers.[93]

What the Chennai researchers heard about women differs from what studies of Western software firms had found. Almost everyone they met, men and women alike, 'insisted to us that both sexes have equal technical skills'.[94] The difference between men and women at ICS lay elsewhere: 'Married women professionals consistently give their families higher priority than their work.'[95] Many men wanted 'home-loving' wives who would not work. So women in IT preferred husbands in IT, and men in IT tended to prefer wives who were not.[96] One leading company, the authors reported, had started an in-house online matrimonial service to encourage its staff to marry each other, and so reduce the number of married women who resigned to follow husbands working elsewhere.[97] The managers of two small software firms, one of them a woman, admitted that they were reluctant to recruit women, because family duties took them away from work.[98]

The study's women show the range. A project manager in her early thirties, whom the authors call Anuradha, had made it a condition of her marriage that she and her husband return from America to Chennai. Her daughter was looked after by both sets of grandparents, and she 'firmly believes that the IT industry has helped to empower women'.[99] Jayashree, about 35, an assistant project manager with an MSc in physics from IIT Madras, had a different experience. Childcare for her two children was 'an endlessly negotiated series of arrangements' among hired help, her parents and her parents-in-law. Her husband, she complained, did not do his share at home; many chores, she joked, had been 'outsourced'. She later resigned from ICS and did contract work for it from home.[100]

Two younger engineers on Jayashree's team, called Sarojini and Parvati, were in their mid-twenties. Sarojini had an electrical and electronics degree from Coimbatore and joined ICS as a trainee in 2002. She told the researchers she found the work exciting and challenging, though her first ambition had been medicine, and she insisted that any husband would have to let her keep working. Her mother also worked, and she appeared, the authors thought, to put her determination down less to the IT industry than to her matrilineal Nayar background. Later in 2004 the company sent her to New Jersey.[101] Parvati, with a BSc in computer science and an MSc in information technology, found her work boring after six months. A year later she was a team leader, determined to get a posting in America. One of her weekly tasks was to search an internet matrimonial site for a husband for her elder sister, and the profiles asking for 'home-loving' wives increasingly irritated her.[102]

The authors' verdict held both halves together. 'Plainly, the IT industry has not revolutionised the position of Indian women.'[103] Yet when they compared their informants with the upper-middle-class Chennai women studied by Patricia Caplan in 1974-75, for most of whom 'employment outside the home is not considered at all desirable', they judged that 'real change has occurred'.[104] Young women in IT appeared to 'command greater respect and have more bargaining power within their families', particularly over whom they married.[105]

Trainable, not skilled

By the early 2010s a second number had joined the count of engineers. Only a quarter of Indian engineers were employable, it was said, or only a fifth. Both figures can be traced, and both come from parties with an interest.

The 'quarter' comes from the McKinsey Global Institute, the research arm of the consultancy. As the Duke study reported it, 'McKinsey concluded that only 10% of Chinese engineers and 25% of Indian engineers can compete in the global outsourcing arena', citing language, the quality of education, culture and other factors.[106] As quoted, its question was not whether Indian engineers could find work, but how many could compete for outsourced work from rich countries.

The 'fifth' comes from Aspiring Minds. Its 2014 report tested 'more than 1,20,000 engineering students from 520+ engineering colleges' who graduated in 2013, on the company's own test, a test it said was used by more than 650 companies.[107] It found that 'only 18.43% of them are employable for the Software Engineer-IT services role'.[108] These are the shares of a tested sample that met the vendor's benchmark for a role. They are not measures of whether graduates found jobs, and the company that produced them sold the test.

Read for what it measures, though, the report says something precise about the factory. The IT services figure, it explained, was calculated under the firms' 'current hiring philosophy', in which a recruit was not expected to have software skills already but was trained 'over a period of 3 to 6 months'. So 18.43 per cent was the share judged 'trainable into software engineers within a period of 3 to 6 months'.[109] The firms' bar was trainability, not skill. That was the bar Rajaraman described and the one the Mysore campus was built for.

The same report contained an early warning. 'The IT services industry is not growing at the same pace as before and the growth of entry-level jobs is diminishing,' it said. Because firms were hiring fewer people and wanted to cut training costs, there was 'a slow but definite trend towards hiring candidates who already have decent expertise in programming'.[110] It is a vendor's observation, from a firm that sold programming tests, but it is an early documented sign that the factory's demand curve was bending.

The price of a place in the factory points the same way. The figures are rough and measure different things. ICS's starting salary of about 15,000 rupees a month in 2003-05 came to about 180,000 rupees a year.[111] A study by the business website Moneycontrol, relayed by the trade site *HR Katha*, put the median annual pay of freshers at large IT firms in 2012 at about 245,000 rupees.[112] Adjusted for consumer prices, the first is worth about 660,000 rupees at 2024 prices and the second about 480,000.[113] One is a single firm and the other a median, so the comparison can only be suggestive. It suggests that as the gate widened, the price of entry fell.

Routine work

What were the factory's products made for? Fuller and Narasimhan answered in a sentence that matters for the rest of this book. 'Because software engineering in companies like ICS or TCS has progressively become more routinized and reliant on standard packages, IIT graduates are over-qualified and can secure more technically demanding, better-paid jobs.'[114] Most new recruits in the major firms, they went on, had only a bachelor's degree in engineering, usually from one of the leading colleges in Tamil Nadu.[115]

The people doing the work said similar things. 'Software engineering does not demand exceptional ability, as some experienced people acknowledge,' the authors wrote. Shoaib went further and claimed that any reasonably intelligent school-leaver could learn the job. He was exaggerating, they said, but they had come across several IT professionals bored by work that was too easy.[116] Raghu, keeping a British client's system running, thought he was learning little. Parvati was bored at six months, a team leader at eighteen and set on a posting in America. Sarojini found the work challenging, and was sent to New Jersey.[117] The firms did not need all their recruits to be stretched. They needed them to be trainable, reliable and available. Some managers said so, though not in the way Aspiring Minds' account of firms that looked only at top colleges would suggest. Several large Indian services companies, Upadhya reported, preferred graduates of lower-ranked colleges 'because the best students are not suitable for the kind of routinised work that is on offer'. As one HR manager put it, they required 'guys who can just sit and code and not ask questions'.[118]

Outside observers drew the same line. The Duke team divided engineering graduates into 'dynamic' engineers, capable of abstract thinking and high-level problem solving, and 'transactional' engineers, who might have the fundamentals but were 'typically responsible for rote and repetitive tasks in the workforce'.[119] It was 'the jobs of transactional engineers', the team wrote in 2005, that seemed 'easily outsourced' to relatively low-paid engineers in countries like India.[120] The phrase was American shorthand, and it was unkind to many Indian graduates who did far more than that. But it described the bulk of what the factory was designed to supply.

That was the factory's design, and it fitted the business the Indian firms had built. A client abroad specified a system; a small team onsite fixed the requirements; a large team in India built, tested and maintained it, to documented methods, on standard packages, billed by time. A trained graduate of almost any engineering branch could do most of that work after three months in Mysore, and every graduate the factory turned out was more time to sell. The design succeeded.

Notes to Chapter 12. The engineer factory (1998-2012)

[1] Fuller and Narasimhan, 'Information Technology Professionals and the New-Rich Middle Class' (2007), p. 132; dollar figure at the 2004 calendar-year average rate of Rs 45.32 per dollar, Reserve Bank of India, *Handbook of Statistics on Indian Economy*; the interview year is not stated, 2003-05.

[2] Fuller and Narasimhan, 'Information Technology Professionals and the New-Rich Middle Class' (2007), pp. 124-126, 132.

[3] Fuller and Narasimhan, 'Information Technology Professionals and the New-Rich Middle Class' (2007), p. 132.

[4] MoSPI, 'Number of Recognised Educational Institutions', Table 1.17, p. 19 (Table 1.17).

[5] MoSPI, 'Number of Recognised Educational Institutions', Table 1.17, p. 19 (Table 1.17).

[6] Rajaraman, *History of Computing in India* (2012), p. 56.

[7] *T.M.A. Pai Foundation v. State of Karnataka* (2002), para. 45 (majority opinion of B.N. Kirpal CJI).

[8] *T.M.A. Pai Foundation v. State of Karnataka* (2002), paras. 56-57; see also para. 69.

[9] *T.M.A. Pai Foundation v. State of Karnataka* (2002), para. 68.

[10] Rajaraman, *History of Computing in India* (2012), p. 51.

[11] Rajaraman, *History of Computing in India* (2012), p. 56.

[12] Rajaraman, *History of Computing in India* (2012), p. 12.

[13] Rajaraman, *History of Computing in India* (2012), p. 51.

[14] Saxenian, 'Bangalore: The Silicon Valley of Asia?' (2001), p. 14 n. 27. Saxenian reported an estimate that 60 per cent of India's computer science graduates came from Andhra Pradesh, Karnataka and Tamil Nadu; see Chapter 10.

[15] Fuller and Narasimhan, 'From Landlords to Software Engineers' (2008), pp. 189-190.

[16] Avanti Fellows, 'Technical Seats' (2026) (secondary analysis of AICTE and AISHE data; consulted 2026-10-12; post-cutoff material, analysis dated August 2026; live-checked 2026-10-13). The Avanti Fellows series of approved places begins in 2012-13.

[17] Avanti Fellows, 'Technical Seats' (2026) (secondary analysis of AICTE and AISHE data; consulted 2026-10-12; post-cutoff material, analysis dated August 2026; live-checked 2026-10-13).

[18] Gereffi and Wadhwa, *Framing the Engineering Outsourcing Debate* (2005), p. 3.

[19] Gereffi and Wadhwa, *Framing the Engineering Outsourcing Debate* (2005), pp. 1-2.

[20] Gereffi and Wadhwa, *Framing the Engineering Outsourcing Debate* (2005), p. 5.

[21] Gereffi and Wadhwa, *Framing the Engineering Outsourcing Debate* (2005), p. 6.

[22] Gereffi and Wadhwa, *Framing the Engineering Outsourcing Debate* (2005), p. 8.

[23] Gereffi and Wadhwa, *Framing the Engineering Outsourcing Debate* (2005), p. 8.

[24] Rajaraman, *History of Computing in India* (2012), p. 56.

[25] Rajaraman, *History of Computing in India* (2012), p. 12.

[26] Aspiring Minds, *National Employability Report: Engineers*, p. 7.

[27] Gereffi and Wadhwa, *Framing the Engineering Outsourcing Debate* (2005), p. 5; Aspiring Minds, *National Employability Report: Engineers*, p. 7; MoSPI, 'Number of Recognised Educational Institutions', Table 1.17, p. 19.

[28] Avanti Fellows, 'Technical Seats' (2026) (secondary analysis of AICTE and AISHE data; consulted 2026-10-12; post-cutoff material, analysis dated August 2026; live-checked 2026-10-13).

[29] Rajaraman, *History of Computing in India* (2012), p. 56.

[30] Rajaraman, *History of Computing in India* (2012), p. 56.

[31] Gereffi and Wadhwa, *Framing the Engineering Outsourcing Debate* (2005), p. 8.

[32] Infosys, Form F-1 (1999), p. 6.

[33] Infosys, Form 20-F FY2008, Item 4.

[34] Infosys, Form 20-F FY2005, Item 4; Infosys, Form 20-F FY2008, Item 4. The filings count all 'offers of employment' without separating new graduates from experienced recruits, so the growth in fresher hiring alone cannot be read from them. Offers are the steadier measure. The number of applications behaves oddly: the year to March 2005 shows about 1,348,800, more than the 885,000 of three years later, even though offers more than tripled in between. Both filings use 'screened' in one place and 'received' in another for the same figure, so the change of wording does not explain the fall.

[35] Infosys, Form 20-F FY2002; Infosys, Form 20-F FY2005, Item 6.D; Infosys, Form 20-F FY2008, Items 3 and 6.D; Infosys, Form 20-F FY2012, Items 3 and 6.D.

[36] Computer History Museum, 'Oral History of Natarajan Chandrasekaran' (2020), PDF p. 5 (recollection in 2020; the transcript spells IIT as 'ITT').

[37] Rajaraman, *History of Computing in India* (2012), p. 56.

[38] Raja Simhan, 'It's a Real Slowdown at Campuses' (2002).

[39] Upadhya and Vasavi, *Work, Culture, and Sociality in the Indian IT Industry* (2006), pp. 4, 57 (Bangalore fieldwork January 2004 to June 2005, p. 4; printed page numbers, PDF page minus 10).

[40] Infosys, Form 20-F FY2004, Items 4 and 6.D.

[41] *Business Today*, 'Infosys: Wealth & Values' (2001) (article by R. Sukumar).

[42] Infosys, Form 20-F FY2002, Items 4 and 6.D.

[43] Infosys, Form 20-F FY2005, Item 4.

[44] Infosys, Form 20-F FY2005, Item 5.

[45] Infosys, Form 20-F FY2008, Items 4 and 6.D.

[46] Infosys, Form 20-F FY2012, Item 4.

[47] Infosys, Form 20-F FY2012, Item 5.

[48] Infosys, Form 20-F FY2012, Item 5. No published account by a trainee at Mysore in this period has been located, and the company's press releases on the campus's opening have not been found.

[49] Infosys, Form 20-F FY2008, Item 4; Infosys, Form 20-F FY2012, Item 4.

[50] Infosys, Form F-1 (1999), p. 45. Chapter 11 describes the pyramid in more detail.

[51] Infosys, Form 20-F FY2005, Item 5; Infosys, Form 20-F FY2008, Item 5.

[52] Infosys, Form 20-F FY2012, Items 3 and 5.

[53] Infosys, Form 20-F FY2008, Items 3 and 5.

[54] Upadhya and Vasavi, *Work, Culture, and Sociality in the Indian IT Industry* (2006), p. 85.

[55] Upadhya and Vasavi, *Work, Culture, and Sociality in the Indian IT Industry* (2006), p. 85.

[56] Upadhya and Vasavi, *Work, Culture, and Sociality in the Indian IT Industry* (2006), p. 86.

[57] Upadhya and Vasavi, *Work, Culture, and Sociality in the Indian IT Industry* (2006), p. 87.

[58] Infosys, Form 20-F FY2005, Item 3; Infosys, Form 20-F FY2008, Item 3; Infosys, Form 20-F FY2012, Item 3.

[59] Fuller and Narasimhan, 'Information Technology Professionals and the New-Rich Middle Class' (2007), p. 137.

[60] Fuller and Narasimhan, 'From Landlords to Software Engineers' (2008), p. 170.

[61] Fuller and Narasimhan, 'Information Technology Professionals and the New-Rich Middle Class' (2007), p. 125.

[62] Fuller and Narasimhan, 'Information Technology Professionals and the New-Rich Middle Class' (2007), p. 127.

[63] Fuller and Narasimhan, 'Information Technology Professionals and the New-Rich Middle Class' (2007), p. 125.

[64] Fuller and Narasimhan, 'Information Technology Professionals and the New-Rich Middle Class' (2007), p. 125; dollar figure at the 2004 calendar-year average rate of Rs 45.32 per dollar, Reserve Bank of India, *Handbook of Statistics on Indian Economy*.

[65] Fuller and Narasimhan, 'Information Technology Professionals and the New-Rich Middle Class' (2007), pp. 125-126 and n. 9.

[66] Fuller and Narasimhan, 'Information Technology Professionals and the New-Rich Middle Class' (2007), p. 141.

[67] Upadhya and Vasavi, *Work, Culture, and Sociality in the Indian IT Industry* (2006), pp. 4, 37.

[68] Fuller and Narasimhan, 'Information Technology Professionals and the New-Rich Middle Class' (2007), p. 126.

[69] Fuller and Narasimhan, 'Information Technology Professionals and the New-Rich Middle Class' (2007), pp. 127-128.

[70] Fuller and Narasimhan, 'Information Technology Professionals and the New-Rich Middle Class' (2007), p. 130.

[71] Fuller and Narasimhan, 'Information Technology Professionals and the New-Rich Middle Class' (2007), p. 130.

[72] Fuller and Narasimhan, 'Information Technology Professionals and the New-Rich Middle Class' (2007), p. 133.

[73] Fuller and Narasimhan, 'Information Technology Professionals and the New-Rich Middle Class' (2007), p. 134.

[74] Fuller and Narasimhan, 'Information Technology Professionals and the New-Rich Middle Class' (2007), pp. 134, 142.

[75] Fuller and Narasimhan, 'Information Technology Professionals and the New-Rich Middle Class' (2007), p. 142.

[76] Fuller and Narasimhan, 'Information Technology Professionals and the New-Rich Middle Class' (2007), p. 132.

[77] Fuller and Narasimhan, 'Information Technology Professionals and the New-Rich Middle Class' (2007), p. 126.

[78] Fuller and Narasimhan, 'From Landlords to Software Engineers' (2008), p. 170.

[79] Fuller and Narasimhan, 'Information Technology Professionals and the New-Rich Middle Class' (2007), pp. 142-143; Fuller and Narasimhan, 'From Landlords to Software Engineers' (2008), p. 171.

[80] Fuller and Narasimhan, 'Information Technology Professionals and the New-Rich Middle Class' (2007), p. 142.

[81] Fuller and Narasimhan, 'Information Technology Professionals and the New-Rich Middle Class' (2007), p. 142.

[82] Fuller and Narasimhan, 'Information Technology Professionals and the New-Rich Middle Class' (2007), p. 143.

[83] Fuller and Narasimhan, 'Information Technology Professionals and the New-Rich Middle Class' (2007), pp. 142-143.

[84] Fuller and Narasimhan, 'Information Technology Professionals and the New-Rich Middle Class' (2007), p. 143 n. 13.

[85] Fuller and Narasimhan, 'Information Technology Professionals and the New-Rich Middle Class' (2007), p. 143.

[86] Baruah and Wankhar, 'The social gaps in India's IT workforce', NCAER (2025); Bairwa and Sofi, *International Journal of Social Economics* (2025). These national estimates of the caste composition of IT employment, drawn from labour-force survey data, could not be verified against their underlying figures and are not used.

[87] Upadhya, 'Employment, Exclusion and " in the Indian IT Industry' (2007)", p. 1864; the report gives the same survey as 71 per cent upper castes, 'about half' Brahmins, Upadhya and Vasavi, *Work, Culture, and Sociality in the Indian IT Industry* (2006), p. 38.

[88] Upadhya and Vasavi, *Work, Culture, and Sociality in the Indian IT Industry* (2006), p. 36.

[89] Aspiring Minds, *National Employability Report: Engineers*, pp. 7-8.

[90] Aspiring Minds, *National Employability Report: Engineers*, p. 8.

[91] Fuller and Narasimhan, 'Information Technology Professionals and the New-Rich Middle Class' (2007), pp. 125, 137.

[92] Fuller and Narasimhan, 'Information Technology Professionals and the New-Rich Middle Class' (2007), p. 125 n. 8.

[93] Society of Women Engineers, 'India Tertiary Education' (2025) (citing the AICTE enrolment dashboard).

[94] Fuller and Narasimhan, 'Information Technology Professionals and the New-Rich Middle Class' (2007), p. 138.

[95] Fuller and Narasimhan, 'Information Technology Professionals and the New-Rich Middle Class' (2007), p. 138.

[96] Fuller and Narasimhan, 'Information Technology Professionals and the New-Rich Middle Class' (2007), p. 138.

[97] Fuller and Narasimhan, 'Information Technology Professionals and the New-Rich Middle Class' (2007), p. 138 n. 12.

[98] Fuller and Narasimhan, 'Information Technology Professionals and the New-Rich Middle Class' (2007), p. 139.

[99] Fuller and Narasimhan, 'Information Technology Professionals and the New-Rich Middle Class' (2007), pp. 129-130.

[100] Fuller and Narasimhan, 'Information Technology Professionals and the New-Rich Middle Class' (2007), pp. 130-131.

[101] Fuller and Narasimhan, 'Information Technology Professionals and the New-Rich Middle Class' (2007), p. 131.

[102] Fuller and Narasimhan, 'Information Technology Professionals and the New-Rich Middle Class' (2007), pp. 131-132.

[103] Fuller and Narasimhan, 'Information Technology Professionals and the New-Rich Middle Class' (2007), p. 140.

[104] Fuller and Narasimhan, 'Information Technology Professionals and the New-Rich Middle Class' (2007), p. 140.

[105] Fuller and Narasimhan, 'Information Technology Professionals and the New-Rich Middle Class' (2007), pp. 140-141.

[106] Gereffi and Wadhwa, *Framing the Engineering Outsourcing Debate* (2005), p. 9. The McKinsey Global Institute study is cited as the Duke team quoted it; the study itself is not drawn on directly.

[107] Aspiring Minds, *National Employability Report: Engineers*, p. 9.

[108] Aspiring Minds, *National Employability Report: Engineers*, p. 7.

[109] Aspiring Minds, *National Employability Report: Engineers*, p. 12.

[110] Aspiring Minds, *National Employability Report: Engineers*, p. 13.

[111] Fuller and Narasimhan, 'Information Technology Professionals and the New-Rich Middle Class' (2007), p. 125.

[112] *HR Katha*, 'IT-Sector CEOs' Median Annual Salaries' (2022) (Moneycontrol study as relayed by *HR Katha*, 1 June 2022; the report gives no details of the study's method). Chapter 18 uses the same figure.

[113] OECD, Consumer Price Indices, India (India CPI, calendar-year average; 2004 and 2012 values multiplied by 3.659 and 1.948 to give 2024 rupees; the 2003-05 ICS figure is treated as 2004).

[114] Fuller and Narasimhan, 'From Landlords to Software Engineers' (2008), p. 189.

[115] Fuller and Narasimhan, 'From Landlords to Software Engineers' (2008), p. 189.

[116] Fuller and Narasimhan, 'Information Technology Professionals and the New-Rich Middle Class' (2007), p. 137.

[117] Fuller and Narasimhan, 'Information Technology Professionals and the New-Rich Middle Class' (2007), pp. 131-132.

[118] Upadhya, 'Employment, Exclusion and " in the Indian IT Industry' (2007)", p. 1865; Upadhya and Vasavi, *Work, Culture, and Sociality in the Indian IT Industry* (2006), pp. 35-36.

[119] Gereffi and Wadhwa, *Framing the Engineering Outsourcing Debate* (2005), pp. 3-4.

[120] Gereffi and Wadhwa, *Framing the Engineering Outsourcing Debate* (2005), p. 9.

Chapter 13. H-1B (1990-2015)

Reedhar Gunda, an electrical engineer from Bombay (Mumbai since 1995), came to the United States in 1998 on a temporary work visa sponsored by a body shopper, a middleman who hired Indian engineers and rented them out to American companies. When he arrived, the *Dallas Observer* reporter Miriam Rozen wrote that November, he found that the body shopper was taking $22 an hour out of his pay and leaving him $28. To get out he had to find another job and have the new employer petition for his visa all over again, which took several months.[1]

His friend Reddy Bandi, a 27-year-old electrical engineer from Hyderabad, the son of a rice farmer and the first in his family to go to college, had taken a different road. He had come on a student visa, taken a master's degree at Arizona State University within 18 months, and switched to the work visa.[2] He had kept away from body shops. 'I don't like to apply to companies who get people from the outside,' he said. 'I know what their tactics are.' He blamed both sides. Of those who came on the work visa he said, 'Their main intention is just to get in'; once they found they were underpaid, some switched employers 'two, three times a year'.[3] Working through a large American subcontractor, he thought he earned more cash than the permanent staff beside him at the Sabre Group, though they had better benefits and more authority. His wife, a physical therapist, could not work in her field on his visa.[4]

Rozen built her article around a body shop in suburban Plano. Its owner, an engineer from Madras (Chennai since 1996), recruited engineers and programmers in India, signed them to contracts with his firm and hired them out to American companies, often through a further middleman. By Rozen's account he took 'sometimes as much as 40 percent' of their pay.[5] The visa had been closed to new workers for almost five months after the annual quota ran out in May, she wrote, and he was 'making up for lost time'.[6] He said he had brought some 350 engineers and programmers from India since 1993.[7] By his own account, which he would not document, he paid salaries of $45,000 to $60,000; rival recruiters told Rozen he could bill a programmer at $90,000 to $200,000 a year.[8] He would not let her speak to any of his contract workers.[9] He had also begun sending American project specifications to his Madras office, where the work was done at Indian wages and emailed back. 'They are going to lose the whole business to India,' he said.[10] It was the forecast of an interested party, made in a year when the visas had run out by May.

Rozen doubted that the wage rules were much enforced. An enforcement coordinator in the Labor Department's Dallas office told her he received complaints about the visa about once a month; engineers, he believed, moved on rather than pursue a case. Nationally, she reported, prosecutors had filed only eight cases in six years alleging wilful breaches of the prevailing-wage rules for temporary workers.[11]

The body shop was the industry's bargain at its barest. What it sold was an engineer's hour. Its margin was the gap between the rate it billed and the wage it paid, and it could grow only by bringing in more engineers. The visa that made the trade possible was eight years old, and it was called the H-1B.

A visa with a cap

The Immigration Act of 1990 admitted foreign nationals 'to work temporarily in the United States if hired by a U.S. employer to perform work in a specialty occupation', in the words of the Congressional Research Service (CRS), the research arm of the US Congress. A specialty occupation required at least a bachelor's degree. The visa lasted up to three years and could be renewed for three more.[12] New visas were capped at 65,000 a year. That cap, the CRS noted in 2025, 'is the same now as when it was established in 1990', although Congress raised it for a few years around the turn of the century.[13]

The CRS has described the programme as one that 'epitomizes the tensions' between American employers competing for talent worldwide and the possible harm to American workers of their doing so.[14] The protection for

American workers was a promise on paper. Before petitioning for a worker, the employer filed a labour condition application with the Department of Labor, attesting that it would pay 'the greater of the actual wages paid to similar employees or the prevailing wages for that occupation'.[15] The CRS added a dry sentence that explains much of what followed: 'The LCA is a statement of intent rather than a documentation of actions taken'.[16] The department reviewed each application 'for completeness and obvious inaccuracies', and investigated only if someone later complained.[17]

For its first few years the cap did not bind. A table compiled from government data by the National Foundation for American Policy (NFAP), a pro-immigration policy group, shows 48,600 new petitions issued against the 65,000 cap in the US fiscal year 1992, and between 54,000 and 62,000 in each of the next four years.[18] Nor was it at first mainly a programmers' visa. The economists Gaurav Khanna and Nicolas Morales cite a 1992 government report finding that computing and related occupations accounted for only 11 per cent of H-1 visas in 1989. By 1999, according to the immigration service, close to two-thirds of H-1B visas went to computer-related occupations.[19] The visa was not free to use. A 2011 survey by the GAO, the US government's audit office, cited by Khanna and Morales, put the legal and administrative cost of each H-1B hire at between $2,300 and $7,500.[20] The new visa also changed the economics of sending Indian programmers to American client sites, the trade Chapter 6 follows. 'In 1993, the shift from B-1 to H-1 visas lowered the incentives to hire Indian engineers for on-site work,' Khanna and Morales write, 'as they were to be paid the prevailing market wage'.[21] The B-1 was a business visitor's visa, meant for meetings and negotiations, not for work. On this reading, one reason the Indian firms moved more of their work offshore in the 1990s was that the visa made on-site labour dearer.[22]

The cap began to bind as the American technology industry grew. Visa numbers ran out by September 1997, and the next year's in May 1998.[23] Both Congress and the Federal Reserve, the CRS wrote, worried 'that a scarcity of labor could curtail the pace of economic growth'.[24] Congress raised the ceiling to 115,000 for fiscal 1999 and 2000, and in 2000 to 195,000 for three years.[25] The government's own counting failed under the strain. The immigration service 'admitted in autumn 1999 that thousands of H-1B visas beyond the 115,000 ceiling were approved in FY1999'.[26] The 1998 law also charged employers $500 a worker, which, the *Observer* reported, would pay for scholarships for low-income students in mathematics, engineering and computer science, with about 5 per cent going to enforce the wage rules.[27] Firms that relied heavily on the visa faced extra duties. An 'H-1B dependent' employer, generally one with at least 15 per cent of its workforce on H-1Bs, had to attest that it had tried to recruit American workers and had not laid any off in the 90 days before or after hiring a visa holder.[28] Then the market turned. NFAP's table shows new petitions falling from 163,600 in fiscal 2001 to 79,100 in 2002 and 78,000 in 2003, against a cap of 195,000.[29] When the limit reverted to 65,000 in fiscal 2005, the CRS wrote, 'that limit was reached on the first day'.[30] In April 2007 the immigration service received about 119,193 petitions for the fiscal 2008 cap on the first two days it accepted them, and chose among them by a computer-generated random draw.[31] From the mid-2000s the visa was allocated by lottery.[32] A young engineer's chance of an American job now depended partly on luck.

How Indian the visa became

Indians did not dominate the visa at first. The sociologist AnnaLee Saxenian, writing in 2001 and citing a 2000 study of the visa by Lowell, reported that the number of Indian H-1B holders grew steadily from 1989, became the largest national group in 1994, doubled by 1996 and quintupled by 1999. In 1999 Indians received 47 per cent of all H-1B visas issued, 55,047 people; the next largest groups, from Britain and China, had 6 per cent or less each.[33]

The share kept rising, though the published figures measure different things. For fiscal 2005 the CRS reported that India was 'the leading country of origin for newly arriving H-1B workers, comprising almost half (49.0%) of all of the new arrivals', with China a distant second at 9.16 per cent.[34] The same report repeated an undated government estimate that 'nearly three-fourths of all of the systems analysts and programmers are from India'.[35] By 2014, according to immigration service data cited by Khanna and Morales, about 70 per cent of H-1B visas went to Indians; China, the next largest, had received only 8 per cent in 2012.[36] A visa that Congress had written for any skilled foreigner had become, in its largest use, a channel between Indian engineering colleges and American information technology.

Who sponsored those visas is a separate question, and the answer is less simple than the common belief that the visas mostly went to Indian outsourcing firms. In 2013, by Khanna and Morales's tabulation of immigration service reports, Infosys led with 6,298 approvals, just ahead of Tata Consultancy Services (TCS), with Cognizant third; the

year before, Cognizant had topped the list with 9,281. Eight more firms had over a thousand approvals each in 2013, among them Accenture, Wipro, IBM's Indian subsidiary and Deloitte.[37]

The authors drew the conclusion from the table: '10 out of the top 11 H-1B firms have Indians as their primary employment base'.[38] Three of those ten had their headquarters outside India: Cognizant and IBM in the United States, Accenture in the Bahamas. The table lists iGate under both the United States and India. Their main workforce, like that of Infosys or Wipro, was in India. Only Deloitte, tenth on the list, employed mainly Americans.[39] The largest sponsors, in other words, were firms built on the offshore services model, whoever owned them.[40]

The sponsors were also only part of the visa's use. Since fiscal 2000, the CRS noted, most H-1B workers had been exempt from the cap, because they were extending an existing status or worked for universities and research bodies.[41] The lottery decided who got in. Extensions, and for some a pending application for permanent residence, decided how long they stayed.[42]

Who went, and what they had learned

Most temporary visa workers had been educated before they arrived. A 2010 national survey of American college graduates, tabulated by Khanna and Morales, found that among foreign-born workers on temporary work visas whose highest degree was a bachelor's, 94 per cent had earned it abroad. Among those with a master's, 77 per cent had.[43] Since most H-1B holders were Indian, most of that education had been done in Indian colleges, without an American degree in between. Bandi, with his Arizona master's, was the exception.

That was a change from the earlier generation. The Indians already in Silicon Valley in 1990 had mostly come for graduate study. The 1990 census counted 28,520 Indians in the region's workforce, 98 per cent of them immigrants; 87 per cent of the Indians in its high-technology industries had arrived after 1970 and 60 per cent after 1980.[44] National Science Foundation figures that Saxenian cited show that 79 per cent of Indians who received American science and engineering doctorates in 1990-91 were still working in the United States in 1995.[45] Many were graduates of 'the prestigious Indian Institutes of Technology (IITs) or Indian Institutes of Science (IISs)', she wrote, and were united by professional as well as ethnic ties; a sizable group had grown up in Africa and never lived in India.[46]

The IITs had been feeding America for decades; one IIT Bombay study found over 40 per cent of a 1970s cohort of its electrical engineering graduates abroad.[47] The pattern outlasted the period of this chapter. A study published in 2023 by Prithwiraj Choudhury, Ina Ganguli and Patrick Gaulé, which followed students who sat the IIT entrance examination, found that 'among the top 1,000 scorers on the exam, 36% have migrated abroad, rising to 62% for the top 100 scorers'.[48] The H-1B also gave Indian students a reason to take an American degree first. The 2004 allowance of 20,000 extra visas for holders of American master's degrees, Khanna and Morales note, 'may incentivize students to obtain their degrees in US universities'.[49]

The two routes together supplied much of America's computing workforce. A national survey of college graduates showed that 55 per cent of foreigners working in computing fields in 2003 had arrived on a temporary work visa or a student visa, and a Department of Commerce estimate for the late 1990s was that 28 per cent of programmer jobs went to H-1B holders.[50] In 1990 immigrants (defined as people born abroad who arrived after the age of 18) made up 11.06 per cent of computer scientists in the United States with a bachelor's or master's degree. By 2000 they were 18.59 per cent, and by 2010 27.82 per cent.[51]

The pull on the colleges

The most important effect of the visa may have been on the people who never used it. Khanna and Morales built an economic model of students and firms in both countries to test a hypothesis: 'that Indian students enrolled in engineering schools to gain employment in the rapidly growing US IT industry via the H-1B visa program. Those who could not join the US workforce, due to the H-1B cap, remained in India, enabling the growth of an Indian IT sector'.[52] Those who went and came back when their visas expired brought skills and technology with them.[53] As the authors put it, workers who could not get a visa 'or return to India after their visa expires are a ready supply of labor for firms in India'.[54]

Their model's answer is that the visa 'induced Indians to switch to computer science (CS) occupations, increasing the CS workforce in India by 4.9% in 2010'.[55] The same model finds a cost on the other side. The visa 'induced US workers to switch to non-CS occupations, reducing the US native CS workforce by 7%'.[56] Indian IT output was 1.8 per cent higher, and the combined income of the two countries 1.5 per cent higher, than it would have been without the

programme.[57] These are estimates from a calibrated model, not counts. They describe a world without the H-1B that never existed. What they offer is a mechanism: the American prospect pulled Indian students towards engineering and computing, and the cap kept most of them at home, where the Indian firms hired them.

Over the same years, according to Nasscom (the National Association of Software and Service Companies), the industry's lobby group, information technology and business process outsourcing grew from 1.2 per cent of India's gross domestic product in 1998 to 7.5 per cent in 2012.[58] The workforce it produced was young. In 2002 Indian software engineers had 'a median age of 26.5 years, and 58% of them had less than three years of experience', according to Nasscom. A Nasscom-Hewitt survey 'found that 88% of firms visited engineering colleges to recruit, and 47% recruited only there'.[59] Drawing on a 2002 study by Arora and Athreye, Khanna and Morales report that the number of engineering doctorates in India fell from about 675 in 1987 to 375 in 1995.[60] The industry, they conclude, was attracting 'some of the brightest young graduates with little experience'.

Silicon Valley's Indians

By 2000, 53 per cent of Silicon Valley's scientists and engineers were foreign-born, Saxenian reported, and about 20,000 of them were Indian.[61] Many had found their careers stopped short. 'Many Indian engineers complained about a glass ceiling in the region's established companies, and responded by starting their own businesses', she wrote.[62] Prabhat Andleigh, quoted in the *Los Angeles Times* in 1996, put it more sharply: 'Why do you think there are so many Indian entrepreneurs in Silicon Valley? Because they know that sooner or later they will be held back'.[63]

Some of the older arrivals had already built companies. Some time in 1992 a group of Indian businessmen in the Valley gathered to meet a visitor from India, the Secretary of Electronics. His flight was delayed. While they waited, they 'began to informally share complaints about the difficulties of running a business'.[64] Among them were Suhas Patil, a former professor at the Massachusetts Institute of Technology who had founded the chip company Cirrus Logic; Prabhu Goel, founder of Gateway Design Automation; and Kanwal Rekhi, who had started and run the networking firm Excelan until it merged with Novell.[65] Patil and Goel had succeeded 'in spite of their lack of contacts or community support'.[66] Patil put the discovery in a sentence that the *San Jose Mercury News* printed two years later: 'as individuals we are brilliant, but collectively we amounted to nothing'.[67] Satish Gupta, another early member, said much the same in the same article: 'When some of us started our businesses we had nobody we could turn to for help'.[68]

Out of the delayed flight came a club with monthly meetings, called The Indus Entrepreneurs (TiE). Its founders chose 'Indus (rather than Indian)', Saxenian wrote, so that Pakistanis, Bangladeshis and Nepalese could belong too, though the Bay Area members were almost all Indian.[69] TiE's own history dates the founding to December 1992.[70] The first annual conference, in 1994, 'attracted over 500 people'. In 1997 more than 800 came to two days of talks on how to start a company, raise money and take it public.[71] TiE turned to mentoring and, through its members, investing in younger South Asian founders.[72] Its founders belonged to what Saxenian called 'an older generation of Indian immigrants'.[73] The engineers the H-1B brought were younger, trained in India, and hired for a few years rather than for a career.

By Saxenian's 1999 count, firms with Indian chief executives were 7 per cent of the Silicon Valley technology companies founded between 1980 and 1998: 774 firms, with $3.6 billion in sales and 16,598 jobs in 1998.[74] The Indian share of new start-ups rose from 3 per cent in 1980-84 to 9 per cent in 1995-98.[75] Her count, she was explicit, used chief executives as 'a proxy for immigrant-founded businesses'. A later study that counted Indian immigrants among the key founders of 1995-2005 start-ups produced the more often quoted figure of 15.5 per cent.[76] Any share of Indian start-ups is only as good as its definition.

Drain, then circulation

Saxenian's phrase 'brain circulation' is easily read back into the 1990s. Her own evidence for India in that decade points the other way.

The exception she described in detail was Radha Basu. Basu left 'her conservative South Indian family' for graduate study in computer science at the University of Southern California in the early 1970s and joined Hewlett-Packard.[77] In the mid-1980s, after the Indian government invited her to set up one of the country's first foreign subsidiaries, she spent four years building HP's centre in Bangalore (Bengaluru since 2014). When she went back to California in 1989 it employed 400 people.[78] The first office, she told Saxenian in 1997, 'consisted of a telex machine

on her dining room table'.[79] Customs officials could not accept that software could be exported over a satellite link without something physical to inspect, so for five years Basu copied HP's data onto tapes and shipped them to the United States so that the exports could be registered.[80] Doing business in India, she told Saxenian, needed patience: 'You can't just fly in and out and stay in a five-star hotel and expect to get things done like you can elsewhere'.[81] When officials talked of a 'single window of clearance' for foreign trade, she would joke: 'where is the window?'[82]

Basu was unusual. Saxenian wrote in 1999 that 'few Indian engineers choose to live and work permanently in India'; those who went home typically did so for a time.[83] Bangalore software companies reported turnover of 20-30 per cent a year, 'primarily because so many workers jump at the first opportunity to emigrate'. On balance, 'the "brain drain" of skilled workers to the United States continued unabated throughout the 1990s'.[84] The engineers who stayed in America played what she called 'an important, but largely arm's-length, role', persuading their employers to send work to India.[85] The reason was cost. Programmers' wages were 'ten times lower in India', and the fully loaded cost of an engineer there was 35-40 per cent of the American figure.[86]

Much of the work was still body shopping, which Saxenian defined narrowly as programming at the customer's site on 'time and material' contracts billed by the programmer-hour. In the series by Balaji Parthasarathy that she cites, on-site services were about 90 per cent of the value of Indian software exports in 1990 and still 61 per cent in 1995.[87] At the end of the decade much of the work in India was on the year 2000 problem, which earned exports but, in her judgement, created no new intellectual property.[88] The big Indian firms, she judged, 'have been so profitable playing the wage gap that they have had few incentives to address higher value-added segments of the market'.[89] Playing the wage gap was the bargain in another phrase: the firms sold hours, and the hours were cheap.

Writing in 2005, she repeated the point: 'few U.S.-educated emigrants returned to either India or China during the 1980s and 1990s'. It was the technology recession that 'triggered an upsurge in cross-regional entrepreneurship'.[90] In Taiwan and Israel, by contrast, returning immigrants had carried the Silicon Valley model of venture capital home from the early 1980s.[91] Indian entrepreneurs instead mobilised 'thousands of underemployed English-speaking Indian engineers to provide software development services for American corporations'.[92] On her account, India's diaspora had sent work home rather than companies. Inside India, the 1990s turnover did not spread ideas between firms as it did in Silicon Valley. It mainly pushed up wages, by 25-40 per cent a year. Learning happened 'primarily within the firm and through long-distance relationships', not with local universities or laboratories, even in Bangalore, home of the Indian Institute of Science.[93]

By the mid-2000s the trade itself was changing who travelled. In Chennai, the anthropologists Chris Fuller and Haripriya Narasimhan followed a group of engineers. Of their twenty younger software engineers, seven had been or were on overseas assignments in 2005, but only two had spent more than a year in the United States; 'the rest, as is now increasingly common, were away for only a few months'.[94] They argued that 'the very logic of "offshoring"' meant that flows of people might diminish 'as the work travels to the site of labor, instead of the reverse'.[95] For many engineers the American posting became a stint in a career spent in India.

At the Chennai firm they studied, which they called ICS, on-site assignments of several months or a year or two were a normal part of working life.[96] Their individual stories show the range. Ravi (a pseudonym, as are all names in the study), a project manager of 33 at a large Chennai firm, had spent a year on-site in England, 'where he earned plenty of money and got "good exposure"', but came home to be with his elderly father and had no wish to live abroad.[97] Balaram, 37, left the same company in 1995 for Australia, worked there in information technology for eight years, and came back in 2003 because 'the work available in Chennai is far more interesting and challenging, and the industry's growth prospects are much better'.[98] Anuradha, a project manager who had worked for ICS in America from 2000 to 2002, now had a young daughter and refused on-site assignments; she said she did not need any more 'exposure'.[99]

The parents

Migration also reached into Indian homes. Fuller and Narasimhan found that Chennai was 'said to be full of lonely old folk whose sons and daughters send them dollars, but only ever come on short visits'. Other parents visited their children so often that they became 'bored, exploited childminders marooned in American suburbia under "house arrest", to quote one elderly grandfather'.[100] One informant told them that overseas migration 'damages the proper reciprocal relationship between parents and children in a way that migration within India does not'.[101] Children

abroad could send money. They could not provide the daily care that sons and daughters were expected to give their parents.[102]

The B-1 channel

The H-1B was not the only door. The business visitor's visa that Khanna and Morales described as giving way to the H-1B in 1993 was still in use two decades later. The clearest record of how comes from a legal settlement.

On 30 October 2013 Infosys filed with the US Securities and Exchange Commission a settlement agreement with the US Attorney's Office for the Eastern District of Texas, the State Department and the Department of Homeland Security. Under it, 'Infosys agrees to pay a full settlement amount of $34,000,000'.[103] The agreement separates what both sides accepted from what the government alleged.

Both sides accepted that 'Infosys failed to maintain accurate I-9 records for many of its foreign nationals in the United States in 2010 and 2011', the forms that record whether an employee may legally work. This included 'a widespread failure to update and re-verify the employment authorization status of a substantial percentage of its foreign national employees'.[104]

The government alleged more, and Infosys denied it. The allegation was that Infosys 'knowingly and unlawfully used B-1 visa holders to perform skilled labor' in jobs that would otherwise have gone to American citizens or to legitimate H-1B holders, for purposes that included 'increasing profits, minimizing costs of securing visas, increasing flexibility of employee movement'.[105] According to the government, invitation letters described trips as 'customer discussions and related business development activities' when the work included 'coding and programming'.[106] It quoted a 'Do's and Don'ts' memorandum given to B-1 travellers: 'Do not mention activities like implementation, design & testing, consulting, etc., which sound like work.' Another line read: 'Please do not mention anything about contract rates'.[107] The government alleged that workers were told to give consular officers the destination named in the labour condition application when they were going elsewhere. In one case, it said, a worker was told to say Seattle when his real destination was Bentonville, Arkansas.[108] And it alleged that on 29 September 2009 Infosys staff were directed by email to convert a time-and-materials contract, which disclosed the names and billing rates of everyone on a project, into a fixed-price contract, which disclosed only the total, in a way that hid the B-1 holders' skilled work.[109]

Infosys's answer, in the same document, was that its use of B-1 visas 'was for legitimate business purposes and not in any way to circumvent the requirements of the H-1B program'. It said the memorandum 'was not intended to deceive U.S. officials' and had been withdrawn in 2011, and that contract changes were 'made to comply with internal Company guidelines and rules regarding billing practices'.[110] Infosys 'denies any allegations of civil or criminal wrongdoing and/or liability'.[111] The settlement resolved the claims without resolving that dispute.

Two points can be made safely. The first is that the older on-site channel had not closed when the H-1B opened, and the agreed facts show that one of the largest sponsors had not kept accurate track of whether its foreign employees in America were authorised to work. The second concerns the contract forms. Whatever happened in that 2009 email exchange, the government's description of the two forms is accurate. A time-and-materials contract names the people and their rates. A fixed-price contract names only the price. The Indian industry grew up on the first kind, the hour sold and billed. Who captures the savings under each kind of contract is a question Chapter 19 takes up.

Ninety days in Orlando

The other side of the trade has a name and a voice. Leo Perrero had a bachelor's degree in information technology and, by his account, 'twenty years of hard work' in the field.[112] He had worked for Disney in Orlando 'in some capacity for well over ten years', on the point-of-sale system through which visitors paid for food and merchandise.[113] In the holiday season of 2014 he received a meeting invitation from a senior executive and, he later told the Senate Judiciary Committee, his mind 'buzzed with thoughts of a promotion or a bonus'. He walked into a small conference room with about two dozen colleagues.[114] He recalled the executive's words: 'All of you in this room will be losing your jobs in the next 90 days. Your jobs have been given over to a foreign workforce.' Severance pay depended on training their replacements.[115] Later that day, he recalled, he had to tell his children at a church pumpkin sale that the family could not buy one.[116] Only '4 people would be directly rehired by Disney of the hundreds that were lost'.[117]

He described the handover in three stages. In the first, the newcomers sat beside the Disney staff and watched them work. 'Everything was recorded with audio and video. Then the very next day the recordings were put down on paper and read right back to us'.[118] In the second the incumbents worked 'side by side with our replacements'. In the third they watched 'the foreign workers completely take over our jobs'.[119] On 31 January 2015 they handed in their badges and laptops. His desk, chair and computer, he said, went to a worker 'who was just flown in to America weeks before'.[120] Of the severance he had accepted, he said: 'Sadly, I choose the money over America'.[121]

What he described in the first stage was a transfer of knowledge in the most literal sense. The tacit work of a team, what it knew but had never written down, was recorded, transcribed and read back, so that people paid less could do it, first beside the old staff and then without them. A quarter of a century of the visa had refined a method for moving work from one set of people to another by writing it down.

His complaint was aimed at two things together: visa holders 'physically being flown in from foreign countries' and 'the growing use of foreign remote offshore workers'.[122] The firms at the top of the sponsor tables earlier in this chapter ran both. To the worker in the conference room, the H-1B and the offshore centre were one system.

Notes to Chapter 13. H-1B (1990-2015)

[1] Rozen, 'Invasion of the Bodyshoppers', *Dallas Observer* (1998), para. 29 (paragraph count of the online text, newsletter boxes excluded).

[2] Rozen, 'Invasion of the Bodyshoppers', *Dallas Observer* (1998), paras 27, 30 (paragraph count of the online text, newsletter boxes excluded).

[3] Rozen, 'Invasion of the Bodyshoppers', *Dallas Observer* (1998), paras 27, 30 (paragraph count of the online text, newsletter boxes excluded).

[4] Rozen, 'Invasion of the Bodyshoppers', *Dallas Observer* (1998), paras 32, 34-35 (paragraph count of the online text, newsletter boxes excluded).

[5] Rozen, 'Invasion of the Bodyshoppers', *Dallas Observer* (1998), paras 1, 3-5 (paragraph count of the online text, newsletter boxes excluded). The article names the owner and his firm. He was a private individual, and he is not named here. The account of his business is Rozen's reporting and his own statements, as the text labels them; the article cites no documents for it.

[6] Rozen, 'Invasion of the Bodyshoppers', *Dallas Observer* (1998), para. 7 (paragraph count of the online text, newsletter boxes excluded).

[7] Rozen, 'Invasion of the Bodyshoppers', *Dallas Observer* (1998), para. 13 (paragraph count of the online text, newsletter boxes excluded).

[8] Rozen, 'Invasion of the Bodyshoppers', *Dallas Observer* (1998), para. 57 (paragraph count of the online text, newsletter boxes excluded).

[9] Rozen, 'Invasion of the Bodyshoppers', *Dallas Observer* (1998), para. 26 (paragraph count of the online text, newsletter boxes excluded). He refused, Rozen wrote, when told he would not control what appeared in print.

[10] Rozen, 'Invasion of the Bodyshoppers', *Dallas Observer* (1998), paras 75-76 (paragraph count of the online text, newsletter boxes excluded).

[11] Rozen, 'Invasion of the Bodyshoppers', *Dallas Observer* (1998), para. 44 (paragraph count of the online text, newsletter boxes excluded).

[12] Congressional Research Service, 'The H-1B Visa for Specialty Occupation Workers' (CRS IF12912, 2025), p. 1.

[13] Congressional Research Service, 'The H-1B Visa for Specialty Occupation Workers' (CRS IF12912, 2025), p. 2.

[14] Congressional Research Service, 'The H-1B Visa for Specialty Occupation Workers' (CRS IF12912, 2025), p. 1.

[15] Congressional Research Service, 'The H-1B Visa for Specialty Occupation Workers' (CRS IF12912, 2025), p. 1.

[16] Congressional Research Service, 'The H-1B Visa for Specialty Occupation Workers' (CRS IF12912, 2025), p. 1.

[17] Congressional Research Service, 'The H-1B Visa for Specialty Occupation Workers' (CRS RL30498, 2007 version), section 'Temporary Foreign Professional (H-1B) Workers'.

[18] NFAP, 'H-1B Petitions and Denial Rates in FY 2025' (2025), p. 23 (Appendix Table 1; NFAP, a pro-immigration policy group, compiling DHS data). Consulted 2026-10-12; post-cutoff material (published November 2025); live-checked 2026-10-13.

[19] Khanna and Morales, 'The IT Boom and Other Unintended Consequences' (2016), p. 8.

[20] Khanna and Morales, 'The IT Boom and Other Unintended Consequences' (2016), p. 8.

[21] Khanna and Morales, 'The IT Boom and Other Unintended Consequences' (2016), p. 5.

[22] Khanna and Morales, 'The IT Boom and Other Unintended Consequences' (2016), p. 5.

[23] Congressional Research Service, 'The H-1B Visa for Specialty Occupation Workers' (CRS RL30498, 2007 version), section 'Trends in H-1B Entries'.

[24] Congressional Research Service, 'The H-1B Visa for Specialty Occupation Workers' (CRS RL30498, 2007 version), section 'Summary'.

[25] Congressional Research Service, 'The H-1B Visa for Specialty Occupation Workers' (CRS RL30498, 2007 version), section 'Introduction'.

[26] Congressional Research Service, 'The H-1B Visa for Specialty Occupation Workers' (CRS RL30498, 2007 version), section 'Trends in H-1B Entries'.

[27] Rozen, 'Invasion of the Bodyshoppers', *Dallas Observer* (1998), paras 69, 71 (paragraph count of the online text, newsletter boxes excluded).

[28] Congressional Research Service, 'The H-1B Visa for Specialty Occupation Workers' (CRS RL30498, 2007 version), section 'Temporary Foreign Professional (H-1B) Workers'.

[29] NFAP, 'H-1B Petitions and Denial Rates in FY 2025' (2025), p. 23 (Appendix Table 1; NFAP, a pro-immigration policy group, compiling DHS data). Consulted 2026-10-12; post-cutoff material (published November 2025); live-checked 2026-10-13.

[30] Congressional Research Service, 'The H-1B Visa for Specialty Occupation Workers' (CRS RL30498, 2007 version), section 'Summary'.

[31] Congressional Research Service, 'The H-1B Visa for Specialty Occupation Workers' (CRS RL30498, 2007 version), section 'Recent H-1B Capped Petitions'.

[32] Congressional Research Service, 'The H-1B Visa for Specialty Occupation Workers' (CRS IF12912, 2025), p. 2.

[33] Saxenian, 'Bangalore: The Silicon Valley of Asia?' (2001), p. 12.

[34] Congressional Research Service, 'The H-1B Visa for Specialty Occupation Workers' (CRS RL30498, 2007 version), section 'Characteristics of Recent H-1B Nonimmigrants'.

[35] Congressional Research Service, 'The H-1B Visa for Specialty Occupation Workers' (CRS RL30498, 2007 version), section 'Characteristics of Recent H-1B Nonimmigrants'.

[36] Khanna and Morales, 'The IT Boom and Other Unintended Consequences' (2016), pp. 1, 11 and p. 1 n. 2.

[37] Khanna and Morales, 'The IT Boom and Other Unintended Consequences' (2016), Table 1, p. 54. Approvals in 2013: Infosys 6,298; Tata Consultancy Services 6,258; Cognizant 5,186; Accenture 3,346; Wipro 2,644; HCL Technologies 1,766; IBM's Indian subsidiary 1,624; Mahindra Satyam 1,589; Larsen & Toubro Infotech 1,580; Deloitte 1,491; iGate, which had absorbed Patni, 1,157.

[38] Khanna and Morales, 'The IT Boom and Other Unintended Consequences' (2016), p. 11.

[39] Khanna and Morales, 'The IT Boom and Other Unintended Consequences' (2016), Table 1, p. 54.

[40] Khanna and Morales, 'The IT Boom and Other Unintended Consequences' (2016), Table 1, p. 54. Khanna and Morales give no matching totals of initial and continuing petitions for 2012 and 2013, so whether these firms took most of all H-1B approvals in those years cannot be tested from their table.

[41] Congressional Research Service, 'The H-1B Visa for Specialty Occupation Workers' (CRS IF12912, 2025), p. 2.

[42] Congressional Research Service, 'The H-1B Visa for Specialty Occupation Workers' (CRS IF12912, 2025), p. 1.

[43] Khanna and Morales, 'The IT Boom and Other Unintended Consequences' (2016), Table 3, p. 54.

[44] Saxenian, *Silicon Valley's New Immigrant Entrepreneurs* (1999), p. 13.

[45] Saxenian, *Silicon Valley's New Immigrant Entrepreneurs* (1999), p. 3 n. 3.

[46] Saxenian, *Silicon Valley's New Immigrant Entrepreneurs* (1999), pp. 43-44.

[47] Bassett, 'Aligning India in the Cold War Era' (2009), p. 803.

[48] Choudhury, Ganguli and Gaulé, 'Top Talent, Elite Colleges, and Migration' (2023), abstract.

[49] Khanna and Morales, 'The IT Boom and Other Unintended Consequences' (2016), p. 8 n. 9 (the 2004 allowance) and p. 12.

[50] Khanna and Morales, 'The IT Boom and Other Unintended Consequences' (2016), p. 8.

[51] Khanna and Morales, 'The IT Boom and Other Unintended Consequences' (2016), Table 2, p. 54.

[52] Khanna and Morales, 'The IT Boom and Other Unintended Consequences' (2016), abstract.

[53] Khanna and Morales, 'The IT Boom and Other Unintended Consequences' (2016), abstract.

[54] Khanna and Morales, 'The IT Boom and Other Unintended Consequences' (2016), p. 11.

[55] Khanna and Morales, 'The IT Boom and Other Unintended Consequences' (2016), abstract.

[56] Khanna and Morales, 'The IT Boom and Other Unintended Consequences' (2016), abstract.

[57] Khanna and Morales, 'The IT Boom and Other Unintended Consequences' (2016), abstract.

[58] Khanna and Morales, 'The IT Boom and Other Unintended Consequences' (2016), p. 1.

[59] Khanna and Morales, 'The IT Boom and Other Unintended Consequences' (2016), p. 11.

[60] Khanna and Morales, 'The IT Boom and Other Unintended Consequences' (2016), p. 11.

[61] Saxenian, 'From Brain Drain to Brain Circulation' (2005), p. 2.

[62] Saxenian, *Silicon Valley's New Immigrant Entrepreneurs* (1999), p. 43.

[63] Saxenian, *Silicon Valley's New Immigrant Entrepreneurs* (1999), p. 43 and n. 17, citing Patrick J. McDonnel and Julie Pitta, ''Brain Gain' or Threat to US Jobs?', *Los Angeles Times*, 15 July 1996. Saxenian's note spells the first name 'Prahbat'; the text uses 'Prabhat'. His occupation is not given.

[64] Saxenian, *Silicon Valley's New Immigrant Entrepreneurs* (1999), p. 45.

[65] Saxenian, *Silicon Valley's New Immigrant Entrepreneurs* (1999), pp. 44-45.

[66] Saxenian, *Silicon Valley's New Immigrant Entrepreneurs* (1999), p. 45.

[67] Saxenian, *Silicon Valley's New Immigrant Entrepreneurs* (1999), p. 45 and n. 19, citing Julie Winokur, 'A Network for Sharing Success', *San Jose Mercury News*, 21 March 1994. Saxenian's main text gives the sentence to 'another local entrepreneur who subsequently organized the first meeting of TiE'; her note attributes it to Suhas Patil.

[68] Saxenian, *Silicon Valley's New Immigrant Entrepreneurs* (1999), p. 45.

[69] Saxenian, *Silicon Valley's New Immigrant Entrepreneurs* (1999), p. 46.

[70] TiE, 'The Founding of TiE' (TiE's own history; organisation self-description).

[71] Saxenian, *Silicon Valley's New Immigrant Entrepreneurs* (1999), p. 46.

[72] Saxenian, *Silicon Valley's New Immigrant Entrepreneurs* (1999), pp. 44-46.

[73] Saxenian, *Silicon Valley's New Immigrant Entrepreneurs* (1999), p. 44.

[74] Saxenian, *Silicon Valley's New Immigrant Entrepreneurs* (1999), p. 23.

[75] Saxenian, *Silicon Valley's New Immigrant Entrepreneurs* (1999), p. 24.

[76] Saxenian, *Silicon Valley's New Immigrant Entrepreneurs* (1999), pp. 7-8, 23; Wadhwa et al., *America's New Immigrant Entrepreneurs* (2007), pp. 5, 31. Saxenian's count used a Dun & Bradstreet database of 11,443 technology firms founded in Silicon Valley between 1980 and 1998, and treated as Indian-run those whose chief executives had Indian surnames (p. 7). It missed founders who had handed the chief executive's job to someone else, which her interviews suggested was common because venture capital 'has often been tied to the requirement that non-Asian senior executives be hired' (pp. 8, 23). The 15.5 per cent figure is from Wadhwa, Saxenian and colleagues (2007): the share of Silicon Valley start-ups of 1995-2005 with an Indian immigrant among the key founders, a different measure.

[77] Saxenian, *Silicon Valley's New Immigrant Entrepreneurs* (1999), p. 62.

[78] Saxenian, *Silicon Valley's New Immigrant Entrepreneurs* (1999), pp. 62-63.

[79] Saxenian, *Silicon Valley's New Immigrant Entrepreneurs* (1999), p. 64.

[80] Saxenian, *Silicon Valley's New Immigrant Entrepreneurs* (1999), p. 64 n. 19.

[81] Saxenian, *Silicon Valley's New Immigrant Entrepreneurs* (1999), p. 63.

[82] Saxenian, *Silicon Valley's New Immigrant Entrepreneurs* (1999), p. 64.

[83] Saxenian, *Silicon Valley's New Immigrant Entrepreneurs* (1999), p. 63.

[84] Saxenian, *Silicon Valley's New Immigrant Entrepreneurs* (1999), p. 65.

[85] Saxenian, *Silicon Valley's New Immigrant Entrepreneurs* (1999), pp. 65-66.

[86] Saxenian, *Silicon Valley's New Immigrant Entrepreneurs* (1999), p. 66.

[87] Saxenian, *Silicon Valley's New Immigrant Entrepreneurs* (1999), p. 66 n. 24. Other estimates of the on-site share, which measure it differently, are compared in Chapter 9.

[88] Saxenian, *Silicon Valley's New Immigrant Entrepreneurs* (1999), p. 67.

[89] Saxenian, *Silicon Valley's New Immigrant Entrepreneurs* (1999), p. 67.

[90] Saxenian, 'From Brain Drain to Brain Circulation' (2005), p. 6.

[91] Saxenian, 'From Brain Drain to Brain Circulation' (2005), pp. 3, 6.

[92] Saxenian, 'From Brain Drain to Brain Circulation' (2005), p. 6.

[93] Saxenian, 'From Brain Drain to Brain Circulation' (2005), p. 17.

[94] Fuller and Narasimhan, 'From Landlords to Software Engineers' (2008), p. 190.

[95] Fuller and Narasimhan, 'From Landlords to Software Engineers' (2008), p. 190.

[96] Fuller and Narasimhan, 'Information Technology Professionals and the New-Rich Middle Class' (2007), p. 126.

[97] Fuller and Narasimhan, 'Information Technology Professionals and the New-Rich Middle Class' (2007), pp. 128-129.

[98] Fuller and Narasimhan, 'Information Technology Professionals and the New-Rich Middle Class' (2007), p. 129.

[99] Fuller and Narasimhan, 'Information Technology Professionals and the New-Rich Middle Class' (2007), pp. 129-130.

[100] Fuller and Narasimhan, 'Information Technology Professionals and the New-Rich Middle Class' (2007), p. 146.

[101] Fuller and Narasimhan, 'Information Technology Professionals and the New-Rich Middle Class' (2007), p. 146.

[102] Fuller and Narasimhan, 'Information Technology Professionals and the New-Rich Middle Class' (2007), p. 146.

[103] Settlement Agreement between the United States and Infosys (2013), para. IV.A.

[104] Settlement Agreement between the United States and Infosys (2013), para. II.D.

[105] Settlement Agreement between the United States and Infosys (2013), para. II.E.2.

[106] Settlement Agreement between the United States and Infosys (2013), para. II.E.3.a.

[107] Settlement Agreement between the United States and Infosys (2013), para. II.E.3.b.

[108] Settlement Agreement between the United States and Infosys (2013), para. II.E.3.c.

[109] Settlement Agreement between the United States and Infosys (2013), para. II.E.3.d.

[110] Settlement Agreement between the United States and Infosys (2013), para. II.G.

[111] Settlement Agreement between the United States and Infosys (2013), para. II.I.

[112] Perrero, written testimony to the Senate Judiciary Committee (2016), p. 1.

[113] Perrero, written testimony to the Senate Judiciary Committee (2016), p. 1.

[114] Perrero, written testimony to the Senate Judiciary Committee (2016), p. 1.

[115] Perrero, written testimony to the Senate Judiciary Committee (2016), p. 1.

[116] Perrero, written testimony to the Senate Judiciary Committee (2016), p. 2.

[117] Perrero, written testimony to the Senate Judiciary Committee (2016), p. 1.

[118] Perrero, written testimony to the Senate Judiciary Committee (2016), p. 2.

[119] Perrero, written testimony to the Senate Judiciary Committee (2016), p. 2.

[120] Perrero, written testimony to the Senate Judiciary Committee (2016), p. 2.

[121] Perrero, written testimony to the Senate Judiciary Committee (2016), p. 2. Perrero's account was written testimony to a Senate Judiciary Committee hearing on high-skilled immigration on 25 February 2016. He named neither the contractor that supplied the replacements nor their nationality, and neither is supplied here.

[122] Perrero, written testimony to the Senate Judiciary Committee (2016), p. 2.

Chapter 14. Backlash and scale (2002-2011)

In February 2004 N. Gregory Mankiw, a Harvard economist then serving as chairman of the White House Council of Economic Advisers, made a remark about trade that the Princeton economist Alan Blinder would later call a 'textbook' statement. Offshoring, the movement of jobs but not of the people who do them from rich countries to poor ones, was in his words 'the latest manifestation of the gains from trade that economists have talked about at least since Adam Smith'. He added: 'More things are tradable than were tradable in the past, and that's a good thing.'[1] The remark caused what Blinder called a national uproar. 'Democratic and Republican politicians alike excoriated Mankiw', Blinder wrote, while economists lined up to defend him.[2]

The politicians were already acting. On 20 February 2004 the trade paper *Government Computer News* (GCN) reported that a provision in the omnibus budget bill President George W. Bush had signed the month before barred companies that won federal work through competitions under the government's Circular A-76 from shifting that work to other countries.[3] The measure was named the Thomas-Voinovich Amendment after its two Republican sponsors, Senator Craig Thomas of Wyoming and Senator George Voinovich of Ohio. GCN described it as 'part of a swell of legislation' against businesses and governments buying software and IT services from countries such as India and China.[4] For one year, work won by a vendor could not be 'performed by the contractor at a location outside the United States'.[5]

The states went further on paper. The National Conference of State Legislatures had counted 29 bills in 19 state legislatures that would require contractors to hire only American citizens or legal residents, regulate agencies' use of offshore call centres, or otherwise restrict foreign purchases.[6] Justin Marks, the conference's research analyst, said that state legislatures had had to fill a $200 billion budget gap over the previous three years and faced a choice between saving money by outsourcing and keeping jobs at home. Over the years, he said, legislators 'have seen manufacturing jobs go overseas and now they see white-collar jobs going abroad'.[7] The television presenter Lou Dobbs published a book that year with the title *Exporting America*.[8]

Not everyone thought the noise would matter. John Kost, a managing vice-president at the research firm Gartner, told GCN that government agencies had been buying offshore software for years. 'People are naive if they think it is not already happening in large quantities,' he said. 'Once the economy improves and politicians move to a different topic, I doubt it will be a potent political issue.'[9]

In India, the programmers whose work the bills were meant to keep at home were not persuaded. That same month *Wired* published a long report by the journalist Daniel Pink from Hexaware Technologies, a software exporter on the edge of Mumbai that he ranked eighteenth among India's software and services exporters. One of its project managers, Aparna Jairam, an engineering graduate of the University of Pune, wrote embedded software for handheld devices, and her main client was a utility company in the western United States. Her salary, Pink reported, was about $11,000 a year, and he set it beside 'your $70,000-a-year job', the job of the American reader he had in mind.[10] A colleague, Kavita Samudra, rejected the premise of the backlash. 'Back in the US, it's all about cheap, cheap, cheap. It's not only about India being cheap. It's quality services.'[11] The answer did not dispose of the charge. Whatever the quality, the client was buying an engineer's time, and in Mumbai that time cost a fraction of the American price. That is the bargain the rest of this book follows, and the backlash was an argument about its price.

Kost was right that the rules would not stop the work, and wrong about the argument. The federal restriction applied to a narrow class of contracts, and for one year.[12] The industry it was aimed at kept growing, and the argument about which work could be sent abroad did not fade. It became sharper, and one version of it, written in 2006, still sets the terms for the last part of this book.

The decade between the dot-com bust and the end of India's software park tax holiday was the one in which the factory reached full size, a second industry of back offices and call centres grew beside it, and the financial crisis of 2008 tested the model.

The wire test

Alan Blinder was no protectionist. He had served on the Council of Economic Advisers in 1993-94 and as vice-chairman of the Federal Reserve from 1994 to 1996.[13] In an essay in the March/April 2006 issue of *Foreign Affairs* he agreed with Mankiw's economics and disagreed with his sense of scale.[14]

He began modestly. There were no reliable national data, he wrote, but fragmentary studies suggested that 'well under a million service-sector jobs in the United States have been lost to offshoring to date', less than two weeks of normal gross job losses in the churn of the American labour market.[15] The future was another matter. Improvements in technology and communications, he argued, would bring much more offshoring of what he called 'impersonal services', which he defined as 'services that can be delivered electronically over long distances with little or no degradation in quality'.[16] He called the change 'a third Industrial Revolution'.[17]

His central claim was that the line between safe and exposed work would not follow the line between skilled and unskilled work. 'Janitors and crane operators are probably immune to foreign competition; accountants and computer programmers are not.'[18] Typing, a low-skill job, and security analysis, a high-skill one, were both already being delivered electronically from India, though on a small scale so far.[19] The telephone operator who arranged a conference call, or the clerk who took an airline booking by phone, might already be there. 'He or she may be in India already.'[20] His rough estimate of the American service jobs that could be offshored in the electronic future was two to three times the 14 million then in manufacturing, something like 28 to 42 million jobs. He thought the published estimates too low; one of them, he wrote, counted only 'the currently visible tip of the offshoring iceberg'.[21]

The usual answer to such fears in rich countries was more education. Blinder thought that answer incomplete, because the dividing line he described was not a line of education. 'The geeks may not inherit the earth after all', he wrote, 'at least not the highly paid geeks in the rich countries.'[22] He told Americans to worry less about China, whose challenge came mainly in manufacturing, and more about India, with its already strong advantage in English.[23]

One passage in the essay matters more to this book than the rest. 'The dividing line between personal and impersonal services will move over time. As information technology improves, more and more personal services will become impersonal services.'[24] Blinder's test, put shortly, was this: if a task can be sent down a wire without losing quality, it is exposed, however skilled the person who does it. The Indian industry had grown by answering yes for a lengthening list of tasks: first code, then accounts, then the voice on the end of a telephone line. Blinder was writing about the movement of work between countries. His test does not care where the work goes, and the last chapters of this book return to it.

Reconciliations and call centres

Software was half of the industry's story in these years. The other half was business process outsourcing (BPO): the transfer of a company's back-office work, such as accounts, claims, collections and customer calls, to another site or another firm. Its best-known Indian pioneer was Raman Roy, who helped build the Indian operations of American Express and General Electric before founding his own company. He told his story to *Knowledge at Wharton*, the University of Pennsylvania's business journal, in June 2003.[25]

At American Express's Indian division, by Roy's account, the cost per transaction turned out to be 'lower almost by 40% to 50%' than in the company's more advanced markets. Others in the company did not believe it. 'These guys in India obviously don't know how to make their reports; how could these numbers be true?'[26] Roy put the advantage down to the people. An associate doing bank reconciliations in the United States might have just completed high school and earn '$15,000 to $18,000 or perhaps $20,000'. At those salaries, he said, 'I was able to field professionals who were at least college graduates and who sometimes were CPAs', meaning certified public accountants.[27] This was the arbitrage of the software story applied to the back office: Indian graduates doing work that in rich countries was done by people with less schooling, for less money.

The savings were not only in wages. Paying suppliers faster earned American Express discounts for prompt settlement, and Roy once set those against his unit's salary bill. 'You've got us for free', he told it, 'because the

incremental discounts that we got alone were more than the salary bill in India.'[28]

The obstacles were physical. When the team asked the Indian telecommunications company VSNL for 100 per cent up-time, 'they looked at us as though we were from some other world because 90% up-time was considered outstanding'.[29] Roy also described a small piece of invention. His Delhi unit reconciled about 12,000 accounts in the spreadsheet program Lotus 1-2-3, where one slip of the hand could throw an account out of balance. Rather than wait fifteen months for head office's standard platform, Roy had an interim tool built by a local firm run by his former head of technology, Sunil Gujral. When the American package proved slower than expected on test data, the Indian tool reconciled the same files: 'It took only 25% of the time to reconcile the file.' Within three months, he said, it had become American Express's global reconciliation platform.[30] Once a process had been moved and written down, the people running it began to improve it.

Roy joined GE in 1996. Its first Indian project was work GE had already sent to Mexico, and Roy had to quote a lower price than Mexico's; his finance chief told him, 'we will probably lose money on every transaction.'[31] His unit became what he called 'the Five Sigma beta site for GE Capital', a testing ground for the company's quality programme.[32] When he proposed handling live customer calls from India, a very senior executive 'pulled out his pen and crossed out the investment', saying that 'call centers would never happen out of India'. Roy borrowed equipment from vendors for three months and found a client to fund part of a trial. 'It was nationalistic pride that drove me to see that a call center came into being,' he said. By 2003 he believed that more than half of GE Capital's Indian revenue came from its call centres.[33] He went on to found Spectramind, which the Indian software company Wipro bought in 2002.[34]

Whatever the dates, the kind of worker the model sought is clear. In the same interview Roy described Indian BPO staff as 'well qualified, highly educated people' who were 'not a transient workforce'. For them, he said, 'BPO is a full-time career', and they expected employers to think seriously about their careers.[35] V. Rajaraman, the historian of Indian computing, put the requirement more plainly. The back office meant accounts receivable, payroll, account reconciliation and similar work, and starting a BPO centre needed little more than overseas clients and 'trainable human resources with good knowledge of English (which was available in reasonable numbers)'.[36] That was the employers' view of the workforce. The workers' own view is harder to find.

The night shift

The fullest first-hand account of a call-centre floor in these years comes from an anonymous activist, recruited through a German job centre, who spent three months in 2005-06 as a foreign call-centre agent in Gurgaon (now Gurugram), a fast-growing satellite town south of Delhi, and published a report on the libcom.org website in January 2007.[37] It is a political tract, and its figures are the author's. The scholarly studies that follow set it beside other evidence.

The author worked for a research outsourcing firm whose Gurgaon office employed about 800 people, making about 100 outbound calls to companies in Europe in a ten-hour shift.[38] Gurgaon alone, the report estimated, probably had about 150,000 call-centre workers, and one building occupied by Genpact, the former GE Capital operation, held about 12,000.[39] The rules were set by the clients' world: officially, 'the only language allowed for internal communication in the call centre is English'.[40] The agents, mostly in their twenties and about half of them men, had nearly all worked in other call centres nearby.[41]

The report includes short interviews, recorded during breaks and given without names. A 22-year-old woman described her first job interview with a Gurgaon call centre in April 2004, while she was still living in Bhopal; she went to it with her mother. In the next year and a half, by her account, she worked in fourteen call centres and raised her monthly pay from Rs 8,000 to Rs 20,000 (about $180 to $440 at the 2006 average exchange rate) by changing jobs. She had worked six days a week from 2.30 in the morning to 12.30 in the afternoon, calling the United States, Canada and Britain, sometimes on automatic diallers that placed 400 to 500 calls a shift.[42] 'In call centres you mainly learn about working time and discipline,' she said, 'you are physically un-free, but mentally free.'[43]

A 21-year-old woman who had worked on GE Capital's Australian shift, with a cab collecting her at 2.30 in the morning, said she had answered about 100 calls a day and had five minutes to go to the toilet. She had to use an English name. 'I thought it would be fun, but it was more like a prison.'[44] A 22-year-old man from Calcutta (Kolkata since 2001), who described his background as Adivasi (a member of India's indigenous communities), took a technical-support job for Hewlett-Packard in a centre run by Daksh, later taken over by IBM. His training lasted two

months and was mainly about using the tools. 'I brought all my software books from the course, because I thought that I would work at IBM now, but I do not need them.'[45]

The same man's basic wage was Rs 10,500 a month. He was expected to sell software and computers to the callers whose problems he had solved; by his account he sold $1,000 to $2,000 worth in a month and received Rs 1,000 to Rs 1,500 in incentives for it.[46] A 24-year-old man who had worked on Citibank's American credit-card process for an outsourcing contractor said the contractor 'made sure that they got the people with the best accent' for that work. Agents on it wore ties, started at Rs 8,500 a month and, he said, could be dismissed for leaving a desk for a minute without logging out.[47]

The report put the basic wage for new agents at about Rs 8,000 a month and the average, with incentives, at Rs 12,000 to Rs 14,000 for a week of about 50 hours on night shifts, which is roughly $265 to $310 at the 2006 average rate.[48] The cleaners and canteen staff in the same building, many of them former small farmers and carpenters from Calcutta and Bihar, were paid about Rs 2,000 a month for 60 to 70 hours a week.[49] The author's impression was that upper-caste surnames were more frequent among the agents, though caste itself was never mentioned.[50] 'Not a single call centre employee I talked to had experience with manual labour,' the report says.[51] The workers were also clear about why the jobs had come: most of those the activists spoke to said that the main reason was wages of about a fifth of those in the United States or Britain, and some said that after five years in call centres you were finished.[52]

Schooling was part of the bargain. The agents were students or former students, several of whom had learned a second foreign language because it paid better, at courses only the upper-middle class could afford.[53] The woman from Bhopal had learned French when she decided that 'only speaking English is not enough of a qualification, because so many people speak English'. She estimated that about 40 per cent of call-centre workers were studying by correspondence, and said that many gave up after a while.[54]

Sociologists recorded the employer's side of that bargain. Carol Upadhya and A.R. Vasavi of the National Institute of Advanced Studies (NIAS) in Bangalore (Bengaluru since 2014), whose fieldwork ran from 2003 to 2006 and included structured interviews with 34 BPO workers, were told by one HR officer that a candidate with a 'very good family background with high percentage of marks in their degree' would not be taken, being likely to be 'looking at call centres as a stopgap arrangement'. One of their respondents had left a postgraduate degree off his c.v. to get a call-centre job.[55] Workers on British projects at one company had to understand 'fourteen different British dialects'.[56] One respondent told them: 'Initially I felt so ashamed that I worked in a call centre that I used to tell people that I'm working for an internet service provider without mentioning any details.'[57]

The pay looked different from inside. Ernesto Noronha and Premilla D'Cruz of the Indian Institute of Management Ahmedabad interviewed 18 telemarketing agents in Bangalore and Mumbai for a study published in 2007, and quote them without names. One set the pay against the alternatives: 'So repeating 10 lines a day, I will get paid 10000-12000 rupees - amazing, believe me, it is amazing.' Someone working from morning to evening 'in a garment or textile shop', the agent went on, 'wouldn't have been paid even more than 3000 rupees'.[58] Another described the targets that came with the pay: 'everyone knows that for one non-performer, there are five guys outside waiting to replace him.'[59] A third, with an engineering background, had 'got into telemarketing because typical engineering jobs did not pay'.[60]

Not every agent served foreign callers. A domestic industry, much of it answering calls for mobile and other telephone companies, grew beside the export one. Phil Taylor, D'Cruz, Noronha and Dora Scholarios, who surveyed 618 of its employees in 2008, cite a 2008 estimate by the investment bank Avendus Capital of average starting salaries of $2,000 a year there, against $4,500 for agents serving international customers.[61] A team leader at a Chennai outsourcer told them: 'I cannot show that I am under pressure but I have to give this pressure to my agents'.[62]

Some agents organised. UNITES, a union for BPO employees, was formed in September 2005 with backing from the global union federation UNI, and was, in the words of Taylor and his colleagues, 'the first explicit attempt to give BPO employees an independent voice'.[63] Despite attempts in several states, it managed to register only in Karnataka.[64] In 2007 the same researchers surveyed 879 of its members in six cities, a sample of union members rather than of agents at large.[65] A voice agent at a Chennai captive centre, one owned by the client company, told them that 'a technician is supposed to take at least 50-60 calls', that breaks were stopped or postponed, and that 'Once they find that the person has shouted at the customer, he is thrown out of the job.'[66] D'Cruz and Noronha, summarising eight years of fieldwork, found employers blocking the 'bank salary accounts of those absenting

themselves' and refusing relieving letters to those who quit, and members who feared 'having one's name blacklisted on NASSCOM's database'.[67]

The night shifts depended on cars. The report describes cab drivers 'tired after seventeen hours shifts' and a colleague killed in a car crash on her way home from work.[68] Transport became a public issue after December 2005, when a woman who worked the night shift for Hewlett-Packard's Bangalore operation was killed, in the words of Taylor and his colleagues, 'by a man purporting to be her driver'.[69] He was a cab driver, and he told her that the car on her usual route was not available and that she would have to travel in his.[70] A fast-track court in Bangalore convicted him in 2010, and the Karnataka High Court upheld the conviction and his life sentence in 2016. In 2023 the Supreme Court, hearing his appeal against sentence, ordered that he serve at least thirty years before he could be released.[71] Its judgment treated the case as one about the industry: 'The issue is of safety and security of women working with such companies.'[72] UNITES won its first tranche of recruits after campaigning for better regulation of transport.[69] One of Upadhya and Vasavi's informants described another risk: 'When a girl gets a drop at some odd hour in the night, neighbours switch on the lights and gaze at her with shock and make comments'.[73] The reviewer Bob Russell writes that Reena Patel's 2010 study of women in the industry found that company transport could 'extend the length of the working day by several hours', and that the mobile phone was used 'to keep track of women who work the night shift'.[74]

Two million

The industry's own figures for 2008 come from a Nasscom presentation, and they are the claims of an interested party. The number it put first was two million, the 'professionals employed in the sector' directly in FY2007-08, roughly ten times the 200,000 software professionals it had counted, on a narrower definition, for the year to March 1998.[75] The official record of exports, the Reserve Bank of India's balance-of-payments series for software services receipts, rose from $9.6bn in FY2002-03 to $53.1bn in FY2010-11.[76] Rajaraman gives 2.4 million directly employed in IT in 2010,[77] and by the industry's count, which he relays, 166 Indian IT companies had reached level 5, the top level, of the Software Engineering Institute's Capability Maturity Model, and more than 400 of the Fortune 500 companies were clients.[78]

The work was heavily concentrated. Nasscom's map of delivery locations in 2008 gave Bangalore 36 per cent, the New Delhi, Noida and Gurgaon region 17 per cent, Chennai 15 per cent, Mumbai and Pune 15 per cent, Hyderabad 14 per cent and all other places 5 per cent, though its caption said that the industry was 'well spread across multiple locations'.[79] The United States took 61 per cent of exports in FY2006-07, Britain 18 per cent and continental Europe 12 per cent.[80]

The presentation is more revealing about strain than about scale. Its list of the questions clients asked began with attrition and wage inflation: 'is India's talent pool running dry?'[81] Under 'Pressure Points' it listed 'High lateral hiring leading to attrition', meaning the recruiting of experienced staff from rival firms, and 'Tier 3 colleges yields low', and its first point was that talent was trainable but the industry needed 'to make them employable'.[82] Its proposed remedies were an education programme in miniature: an entry-level assessment for BPO and IT recruits, 'finishing schools' to prepare graduates for work, a faculty development programme to make teachers more suitable, and government plans for 20 new Indian Institutes of Information Technology.[83] Nasscom's cost comparison for 2007 put India at about $28,000 against $100,000 for the United States, and projected that India would keep most of that advantage to 2015.[84] That gap was the bargain in a single line, and every pressure point on the list, from wage inflation to lateral hiring, was a threat to it.

The last part of the presentation dealt with a different worry. 'With increasing levels of outsourcing,' it said, 'employees were viewed as a significant weak link'. Nasscom estimated that about 30 per cent of CVs contained false claims, and it noted that India had 'No law addressing privacy or employee rights'.[85] Its answer was the National Skills Registry, a database of employees' verified education and work histories, with photographs and fingerprints, meant, in its words, to 'instill confidence in client about employees'.[86] The industry's workforce had become something its foreign customers wanted to inspect.

The factory at full size

The largest firms show what this growth meant in practice. Infosys, whose filings with the American Securities and Exchange Commission give the most consistent figures, employed about 25,700 people at the end of March 2004.[87]

Its revenue rose from $1,062.6m in FY2003-04, 98.6 per cent of it from exports, to $4,176m in FY2007-08 and $4,663m in FY2008-09.[88] By the FY2008-09 filing its training campus at Mysore (Mysuru since 2014), which Chapter 12 follows, could house 13,500 trainees and train about 40,000 a year, with 476 full-time faculty.[89] One company was running a residential college for tens of thousands of engineers a year, because, as Nasscom put it, the industry needed 'to make them employable'.[82]

The quarter to March 2009

The financial crisis of 2008 reached the Indian firms through their largest customers, and the clearest record of it is the quarter that closed Infosys's financial year on 31 March 2009. About a third of the company's revenue, 33.9 per cent in FY2008-09, came from the financial services industry.[90] Its filing for that year described 'leading investment banks going bankrupt or being forced to sell themselves in distressed circumstances', and warned that clients might respond with 'contract terminations, deferrals of projects or delays in purchases'.[91]

In that quarter about 66.9 per cent of Infosys's technology professionals, trainees included, were billed to clients, against 70.2 per cent a year earlier.[92] Put the other way, about a third of the technical workforce was on the bench, the pool of trained staff not assigned to paying work, and was being paid without being billed. Pay stood still. 'Until 2009, when we instituted a freeze in salary increases,' the company said, it had reviewed compensation every year.[93] The gate narrowed: Infosys made about 16,900 offers in the year, against about 46,300 the year before.[94] Fewer people left, and attrition fell to 11.1 per cent.[95] Yet the filings show no mass layoffs. Infosys added about 12,400 employees, net of departures, and its headcount rose from about 91,200 in March 2008 to about 104,900 in March 2009 and 113,800 in March 2010.[96]

This was the bargain under strain. A firm that sold engineers' time had hired ahead of demand, and when demand paused it carried the unsold time on its own books rather than shed the people the next upturn would need. Part of the cost fell on the graduates who in a normal year would have been hired, and part on employees whose pay stood still. The FY2009-10 filing reported average rises that year of 8 per cent for offshore staff and 2 per cent for those working at client sites abroad, against 11-13 per cent and 4-5 per cent in FY2008-09, and the annual rise, normally made in April, was postponed to October 2009.[97]

Clients also pushed costs onto their suppliers. In response to smaller client budgets, Infosys said, it was offering more of its work at fixed prices and on fixed timetables, and such projects rose to 35.4 per cent of services revenue in FY2008-09 from 31.0 per cent the year before.[98] The same filing reported 'increased demand from our clients that all risks associated with foreign exchange fluctuations be borne by us'.[99] The filings also described clients consolidating their suppliers to cut costs, and long contracts with benchmarking provisions, under which a client could commission a study comparing the supplier's prices with those of its rivals and seek lower fees.[100]

The pattern had appeared before. In 2004, after the dot-com bust, Infosys had said that 'declines in demand, pricing pressures for our services, appreciation of the rupee against the dollar and increased wage pressures in India' had kept it from 'historical levels of profitability'.[101] On the evidence of its filings, Infosys met the second downturn by slowing hiring, carrying a larger bench, holding down pay and absorbing pressure on prices, while its headcount kept rising. By FY2010-11 utilisation including trainees was back to 72.1 per cent, and in FY2011-12 the company made about 41,460 offers.[102]

The crisis also showed how much of the model still depended on people abroad. At the end of March 2009 most of Infosys's technology professionals in the United States held either H-1B visas, about 8,200 people, or intra-company L-1 visas, about 1,400.[103] About 20,200 of its 104,900 employees worked outside India.[104] Those visa holders were the narrow top of the onsite-offshore pyramid, and the part of it most exposed to American politics (Chapter 13).

From parks to zones

The state's support for the industry changed form in these years. The tax holiday of the 1990s had been tied to software technology parks: export profits of units in the parks were exempt from income tax. The Finance Act of 2005 introduced a separate holiday for units in designated special economic zones (SEZs) that began providing services on or after 1 April 2005. Such units could deduct all their export profits from taxable income for five years and half of them for a further five, with more relief available on conditions.[105] The Special Economic Zones Act was

passed the same year.[106] Rajaraman counts 257 software and services companies set up in SEZs by 2007, and writes that many BPO centres were placed in the zones because profits earned there were tax exempt.[107]

The park holiday was extended and then allowed to end. Infosys reported that the Finance Act of 2009 extended the holiday for park units until 31 March 2011, and that for its remaining park units it would expire by the end of that fiscal year.[108] Four of its centres operated in SEZs by March 2009 and eight by March 2012, and 'many of our proposed development centers are likely to operate in SEZs'.[109] The FY2011-12 filing said that the expiry of tax holidays had already hurt its profit margin.[110]

The change favoured firms large enough to build or lease whole zones, and it moved the industry from office towers into walled campuses. It did not remove the old frustrations. Roy, who had set up centres for three companies, said in 2003 that the government promised a 'single-window clearance', but 'there are 25 single windows you need to go to'. For one new centre in Pune he counted 119 places where he had signed.[111]

By the end of the period the offshore model had largely won. The Reserve Bank's survey for 2010-11 found that about four-fifths of software and IT services export revenue, 79.3 per cent, was earned from work done in India, against 74.6 per cent in 2007-08; the United States took about 63.6 per cent.[112] The same survey put IT-enabled services (ITES) and BPO at about 26.3 per cent of these exports in 2010-11, down from 31.0 per cent the year before.[113] In the 1980s most Indian software export revenue had been earned by engineers working at clients' sites (Chapter 9). Now most of it was earned by engineers and agents in Indian cities, at desks a cab ride from home.

The other employers

Many of the people in this chapter worked for foreign companies. Roy built his first centres inside American Express and GE, and the Gurgaon call centres in the activist's report worked for Hewlett-Packard, Citibank, American Express and Deutsche Bank, sometimes in centres those companies owned and sometimes through contractors.[114] Nasscom's own chart divided the industry between Indian providers, the captive centres of global companies and global service providers.[115] Foreign companies had also been building engineering centres of their own in India since the mid-1980s, and the next chapter follows that second profession.[116]

Blinder's test applied to all of them, whoever owned the desk. It asked only whether the work could be sent down a wire, and in these years the list of work that could be sent kept growing. Blinder did not pretend to know where it would stop. 'No one knows how far this process will go.'[117]

Notes to Chapter 14. Backlash and scale (2002-2011)

[1] Blinder, 'Offshoring: The Next Industrial Revolution?' (2006), 'A Controversy Reconsidered'.

[2] Blinder, 'Offshoring: The Next Industrial Revolution?' (2006), 'A Controversy Reconsidered'.

[3] *Government Executive*, reports on federal offshoring restrictions (2004).

[4] *Government Executive*, reports on federal offshoring restrictions (2004).

[5] *Government Executive*, reports on federal offshoring restrictions (2004).

[6] *Government Executive*, reports on federal offshoring restrictions (2004).

[7] *Government Executive*, reports on federal offshoring restrictions (2004).

[8] Dobbs, *Exporting America* (2004) (bibliographic entry).

[9] *Government Executive*, reports on federal offshoring restrictions (2004).

[10] Pink, 'The New Face of the Silicon Age', *Wired* (2004), online text, paragraphs 8, 9, 12 and 41 (Hexaware's location, para. 12; its rank among exporters, para. 41). Journalist's account in contemporary press; *Wired*'s framing is favourable to outsourcing.

[11] Pink, 'The New Face of the Silicon Age', *Wired* (2004), online text, paragraph 16.

[12] *Government Executive*, reports on federal offshoring restrictions (2004).

[13] Blinder, 'Offshoring: The Next Industrial Revolution?' (2006), introductory note.

[14] Blinder, 'Offshoring: The Next Industrial Revolution?' (2006), 'A Controversy Reconsidered'.

[15] Blinder, 'Offshoring: The Next Industrial Revolution?' (2006), 'A Controversy Reconsidered'.

[16] Blinder, 'Offshoring: The Next Industrial Revolution?' (2006), 'A Controversy Reconsidered'.

[17] Blinder, 'Offshoring: The Next Industrial Revolution?' (2006), 'The Three Industrial Revolutions'.

[18] Blinder, 'Offshoring: The Next Industrial Revolution?' (2006), 'This Time It's Personal'.

[19] Blinder, 'Offshoring: The Next Industrial Revolution?' (2006), 'This Time It's Personal'.

[20] Blinder, 'Offshoring: The Next Industrial Revolution?' (2006), 'This Time It's Personal'.

[21] Blinder, 'Offshoring: The Next Industrial Revolution?' (2006), 'This Time It's Personal'. The 'currently visible tip' remark is made of a 2003 Berkeley study; Blinder set his own estimate against it, against a McKinsey study (both put 11 per cent of American jobs at risk) and against Forrester Research's prediction that 3.3 million American service jobs would move offshore by 2015.

[22] Blinder, 'Offshoring: The Next Industrial Revolution?' (2006), 'This Time It's Personal'; Blinder, 'Offshoring: The Next Industrial Revolution?' (2006), 'Is Forewarned Forearmed?'.

[23] Blinder, 'Offshoring: The Next Industrial Revolution?' (2006), 'Imperfect Vision' (India 'is learning to exploit its already strong comparative advantage in English').

[24] Blinder, 'Offshoring: The Next Industrial Revolution?' (2006), 'This Time It's Personal'.

[25] Knowledge at Wharton, 'The Little Start-Up That Could' (Raman Roy interview). Self-account (Knowledge at Wharton interview, 4 June 2003, part 1). All of Roy's story in this section is his own account.

[26] Knowledge at Wharton, 'The Little Start-Up That Could' (Raman Roy interview). Self-account (Knowledge at Wharton interview, 4 June 2003, part 1).

[27] Knowledge at Wharton, 'The Little Start-Up That Could' (Raman Roy interview). Self-account (Knowledge at Wharton interview, 4 June 2003, part 1).

[28] Knowledge at Wharton, 'The Little Start-Up That Could' (Raman Roy interview). Self-account (Knowledge at Wharton interview, 4 June 2003, part 1).

[29] Knowledge at Wharton, 'The Little Start-Up That Could' (Raman Roy interview). Self-account (Knowledge at Wharton interview, 4 June 2003, part 1).

[30] Knowledge at Wharton, 'The Little Start-Up That Could' (Raman Roy interview). Self-account (Knowledge at Wharton interview, 4 June 2003, part 1).

[31] Knowledge at Wharton, 'The Little Start-Up That Could' (Raman Roy interview). Self-account (Knowledge at Wharton interview, 4 June 2003, part 1). The dates of this history are disputed. The *Knowledge at Wharton* introduction places Roy's work at American Express in the mid-1980s (Knowledge at Wharton, 'The Little Start-Up That Could' (Raman Roy interview)). Rajaraman says that American Express was the first organisation to start BPO work in India, in 1994, followed by GE Capital International Services (GECIS) in 1997 (Rajaraman, *History of Computing in India* (2012), p. 53). Other accounts date GECIS to October 1996 and credit its founding both to Roy and to Pramod Bhasin, who headed it (William Davidson Institute, 'IT-Enabled Services (C): GE Capital International Services' (2003); the case study is cited from its catalogue summary, since no full copy has been located). The accounts can be partly reconciled, since Roy's mid-1980s findings concern the cost of the existing Indian card operation, not a dedicated export centre, but who founded GECIS, and when, remains open.

[32] Knowledge at Wharton, 'The Little Start-Up That Could' (Raman Roy interview). Self-account (Knowledge at Wharton interview, 4 June 2003, part 1).

[33] Knowledge at Wharton, 'The Little Start-Up That Could' (Raman Roy interview). Self-account (Knowledge at Wharton interview, 4 June 2003, part 1).

[34] Knowledge at Wharton, 'The Little Start-Up That Could' (Raman Roy interview). Self-account (Knowledge at Wharton interview, 4 June 2003, part 1).

[35] Knowledge at Wharton, 'The Little Start-Up That Could' (Raman Roy interview). Self-account (Knowledge at Wharton interview, 4 June 2003, part 1).

[36] Rajaraman, *History of Computing in India* (2012), p. 53.

[37] Anonymous, 'Working Life, Interviews and Leaflets in Delhi's Call Centre Cluster' (2007). Anonymous activist account; low reliability. The names it gives cannot be checked, and the first names it uses are not repeated here.

[38] Anonymous, 'Working Life, Interviews and Leaflets in Delhi's Call Centre Cluster' (2007). Anonymous activist account; low reliability.

[39] Anonymous, 'Working Life, Interviews and Leaflets in Delhi's Call Centre Cluster' (2007). Anonymous activist account; low reliability.

[40] Anonymous, 'Working Life, Interviews and Leaflets in Delhi's Call Centre Cluster' (2007). Anonymous activist account; low reliability.

[41] Anonymous, 'Working Life, Interviews and Leaflets in Delhi's Call Centre Cluster' (2007). Anonymous activist account; low reliability.

[42] Anonymous, 'Working Life, Interviews and Leaflets in Delhi's Call Centre Cluster' (2007); Reserve Bank of India, *Handbook of Statistics on Indian Economy*, calendar-year average rate for 2006 (Rs 45.31 per US dollar), conversions by the author. Anonymous activist account; low reliability.

[43] Anonymous, 'Working Life, Interviews and Leaflets in Delhi's Call Centre Cluster' (2007). Anonymous activist account; low reliability.

[44] Anonymous, 'Working Life, Interviews and Leaflets in Delhi's Call Centre Cluster' (2007). Anonymous activist account; low reliability.

[45] Anonymous, 'Working Life, Interviews and Leaflets in Delhi's Call Centre Cluster' (2007). Anonymous activist account; low reliability.

[46] Anonymous, 'Working Life, Interviews and Leaflets in Delhi's Call Centre Cluster' (2007). Anonymous activist account; low reliability.

[47] Anonymous, 'Working Life, Interviews and Leaflets in Delhi's Call Centre Cluster' (2007). Anonymous activist account; low reliability.

[48] Anonymous, 'Working Life, Interviews and Leaflets in Delhi's Call Centre Cluster' (2007); Reserve Bank of India, *Handbook of Statistics on Indian Economy*, calendar-year average rate for 2006 (Rs 45.31 per US dollar), conversions by the author. Anonymous activist account; low reliability.

[49] Anonymous, 'Working Life, Interviews and Leaflets in Delhi's Call Centre Cluster' (2007). Anonymous activist account; low reliability.

[50] Anonymous, 'Working Life, Interviews and Leaflets in Delhi's Call Centre Cluster' (2007). Anonymous activist account; low reliability.

[51] Anonymous, 'Working Life, Interviews and Leaflets in Delhi's Call Centre Cluster' (2007). Anonymous activist account; low reliability.

[52] Anonymous, 'Working Life, Interviews and Leaflets in Delhi's Call Centre Cluster' (2007). Anonymous activist account; low reliability.

[53] Anonymous, 'Working Life, Interviews and Leaflets in Delhi's Call Centre Cluster' (2007). Anonymous activist account; low reliability.

[54] Anonymous, 'Working Life, Interviews and Leaflets in Delhi's Call Centre Cluster' (2007). Anonymous activist account; low reliability.

[55] Upadhya and Vasavi, *Work, Culture, and Sociality in the Indian IT Industry* (2006), p. 129. Printed page numbers (PDF page minus 10). Anonymous HR officer and respondent reported by the researchers; fieldwork November 2003 to March 2006, 34 BPO workers interviewed (p. 4).

[56] Upadhya and Vasavi, *Work, Culture, and Sociality in the Indian IT Industry* (2006), p. 130.

[57] Upadhya and Vasavi, *Work, Culture, and Sociality in the Indian IT Industry* (2006), p. 153. Anonymous respondent quoted by the researchers.

[58] Noronha and D'Cruz, 'Reconciling Dichotomous Demands' (2007), p. 263. Anonymous agent quoted by the researchers; 18 agents in Bangalore and Mumbai (p. 255).

[59] Noronha and D'Cruz, 'Reconciling Dichotomous Demands' (2007), p. 264. Anonymous agent quoted by the researchers.

[60] Noronha and D'Cruz, 'Reconciling Dichotomous Demands' (2007), p. 270. Anonymous agent quoted by the researchers.

[61] Taylor et al., 'Domestic Labour' (2010), pre-publication manuscript, sections 'Indian domestic BPO and call centres' (Avendus 2008 starting salaries; Nasscom-Everest 2008, 81 per cent of domestic employment in telecoms) and 'Methods and sources' (618 questionnaires, April to June 2008, members of UNITES). Pre-publication manuscript without page numbers.

[62] Taylor et al., 'Domestic Labour' (2010), pre-publication manuscript, section 'Targets and the labour process'. Anonymous team leader quoted by the researchers. Pre-publication manuscript without page numbers.

[63] Taylor et al., 'Indian Call Centres and Business Process Outsourcing' (2009), accepted manuscript, accepted manuscript, p. 10; D'Cruz and Noronha, 'Hope to Despair' (2013), p. 480.

[64] D'Cruz and Noronha, 'Hope to Despair' (2013), p. 481.

[65] Taylor et al., 'Indian Call Centres and Business Process Outsourcing' (2009), accepted manuscript, accepted manuscript, p. 12 (survey April to July 2007 in Bangalore, Chennai, Hyderabad, Kochi, Mumbai and New Delhi; 45 interviews, p. 13).

[66] Taylor et al., 'Indian Call Centres and Business Process Outsourcing' (2009), accepted manuscript, accepted manuscript, p. 17. Anonymous agent quoted by the researchers.

[67] D'Cruz and Noronha, 'Hope to Despair' (2013), pp. 474 and 481.

[68] Anonymous, 'Working Life, Interviews and Leaflets in Delhi's Call Centre Cluster' (2007). Anonymous activist account; low reliability.

[69] Taylor et al., 'Indian Call Centres and Business Process Outsourcing' (2009), accepted manuscript, accepted manuscript, p. 16.

[70] Supreme Court of India, Criminal Appeal No. 942 of 2023 (28 March 2023), para. 14, p. 12. The High Court records that she worked for 'Hewlett Packard (HP) BPO' (Karnataka High Court, Crl. A. No. 245/2011 (31 May 2016), para. 7, p. 5).

[71] Karnataka High Court, Crl. A. No. 245/2011 (31 May 2016), paras. 1-3 and 126, pp. 3-4 and 74-75; Supreme Court of India, Criminal Appeal No. 942 of 2023 (28 March 2023), paras. 2, 15 and 17, pp. 1-2, 13 and 15. Fast Track (Sessions) Court-XI, Bengaluru, S.C. No. 523/2006, judgment of 6 October 2010: convicted under sections 366, 376 and 302 of the Indian Penal Code; for the murder, rigorous imprisonment for the rest of his life without release. Karnataka High Court, 31 May 2016 (H. Billappa and K.N. Phaneendra JJ.): conviction and sentence confirmed; his appeal and the State's appeal for the death penalty both dismissed. Supreme Court, 28 March 2023 (Abhay S. Oka and Rajesh Bindal JJ.): notice had been issued on sentence only; the trial court could not have directed that he never be released; sentence modified to imprisonment for life, with release only after thirty years of actual sentence, the appeal being partly allowed. By an order of 6 December 2023 the sentences for the other offences run concurrently (Supreme Court of India, Criminal Appeal No. 942 of 2023 (28 March 2023), M.A. No. 1752 of 2023).

[72] Supreme Court of India, Criminal Appeal No. 942 of 2023 (28 March 2023), para. 15, p. 13.

[73] Upadhya and Vasavi, *Work, Culture, and Sociality in the Indian IT Industry* (2006), p. 153. Anonymous informant quoted by the researchers.

[74] Russell, review of Patel, *Working the Night Shift* (2010), p. 524. Reviewer's account of Patel, *Working the Night Shift* (Stanford University Press, 2010); Patel's book is cited here from the review.

[75] Nasscom, *Strategic Review* and fact sheets, slides 2 and 9; Infosys, prospectus (Form 424B4, 1999), p. 42. Nasscom self-description (industry lobby group). Nasscom added seven to eight million indirect jobs by a multiplier of its own. On its estimates industry revenue rose from about $21.6bn in FY2003-04 to about $64bn in FY2007-08, exports being 62-66 per cent (slides 8, 10), and net additions to employment ran at 234,000 in FY2004-05, 328,000 in FY2005-06 and an estimated 375,000 in FY2007-08 (slide 18). Slide 2 describes Nasscom as having more than 1,200 members, including 250 global companies, which together accounted for over 90 per cent of exports. Other counts of employment for FY2007-08 are lower; Chapter 9 and the data appendix compare them.

[76] Reserve Bank of India, *Handbook of Statistics on Indian Economy*, balance-of-payments software services receipts, RBI Handbook edition ids 10663 (FY2002-03; $40.3bn in FY2007-08) and 15930 (FY2010-11). The bank's separate annual survey of software and IT services exporters, which covered 760 companies in 2010-11, put exports that year at about $47.6bn, up from $38.7bn the year before (Reserve Bank of India, *Survey on Computer Software and Information Technology Services Exports*, RBI Bulletin, October 2012, 'Highlights' and Table 1). The two series measure different things and should not be added or compared directly, and neither is wholly independent of the lobby: the bank noted in its survey for 2007-08 that it compiled the balance-of-payments software figures using Nasscom's data as a controlling total (Reserve Bank of India, *Survey on Computer Software and Information Technology Services Exports*, box 'Comparison of survey results with NASSCOM and Softex data').

[77] Rajaraman, *History of Computing in India* (2012), p. 80.

[78] Rajaraman, *History of Computing in India* (2012), pp. 52, 49. Industry claims relayed by Rajaraman. He adds (p. 49) that ten Indian IT companies were listed on foreign stock exchanges, with more than 400 delivery centres outside India. The Capability Maturity Model is described in Chapter 11.

[79] Nasscom, *Strategic Review* and fact sheets, slide 14. Nasscom self-description (industry lobby group). The caption also says that smaller 'Tier 2/3 cities' were emerging. The printed percentages add up to slightly more than 100.

[80] Nasscom, *Strategic Review* and fact sheets, slide 12. Nasscom self-description (industry lobby group).

[81] Nasscom, *Strategic Review* and fact sheets, slide 17. Nasscom self-description (industry lobby group).

[82] Nasscom, *Strategic Review* and fact sheets, slide 18. Nasscom self-description (industry lobby group).

[83] Nasscom, *Strategic Review* and fact sheets, slide 19. Nasscom self-description (industry lobby group).

[84] Nasscom, *Strategic Review* and fact sheets, slide 21. Nasscom self-description (industry lobby group).

[85] Nasscom, *Strategic Review* and fact sheets, National Skills Registry slides. Nasscom self-description (industry lobby group).

[86] Nasscom, *Strategic Review* and fact sheets, National Skills Registry slides. Nasscom self-description (industry lobby group).

[87] Infosys, Form 20-F FY2004, Item 4.

[88] Infosys, Form 20-F FY2004, Item 8; Infosys, Form 20-F FY2009, Item 4.

[89] Infosys, Form 20-F FY2009, Items 4 and 6.D.

[90] Infosys, Form 20-F FY2009, Item 3.

[91] Infosys, Form 20-F FY2009, Item 3.

[92] Infosys, Form 20-F FY2009, Item 3.

[93] Infosys, Form 20-F FY2009, Item 3.

[94] Infosys, Form 20-F FY2008, Item 4; Infosys, Form 20-F FY2009, Item 4. Over FY2008-09 Infosys received about 488,700 applications, tested about 77,600 applicants and interviewed about 46,600 before making its 16,900 offers; the 46,300 offers of FY2007-08 are from Infosys, Form 20-F FY2008. Chapter 12 follows the recruiting funnel.

[95] Infosys, Form 20-F FY2009, Item 3 (attrition 11.1% for fiscal 2009, 13.4% for fiscal 2008, excluding Infosys BPO and other subsidiaries).

[96] Infosys, Form 20-F FY2009, Items 4 and 6.D (about 12,400 net additions in fiscal 2009; headcount); Infosys, Form 20-F FY2010, Item 3.

[97] Infosys, Form 20-F FY2010, Item 5; Infosys, Form 6-K for the quarter ended 31 December 2009 (management's discussion of the nine months ended 31 December 2009). The annual filing dates the increase 'with effect from October 2010', which cannot be right for a rise paid within the year to March 2010; the quarterly report for the three months to December 2009 shows that the annual rise, normally made in April, had been postponed to October 2009. Utilisation including trainees averaged 67.5 per cent for FY2009-10 (Infosys, Form 20-F FY2010, Item 3). No press account of graduates whose joining dates were deferred in 2009 has been located.

[98] Infosys, Form 20-F FY2009, Item 3.

[99] Infosys, Form 20-F FY2009, Item 3.

[100] Infosys, Form 20-F FY2009, Item 3; Infosys, Form 20-F FY2010, Item 3.

[101] Infosys, Form 20-F FY2004, Item 3.

[102] Infosys, Form 20-F FY2012, Items 4 and 5.

[103] Infosys, Form 20-F FY2009, Item 3.

[104] Infosys, Form 20-F FY2009, Item 3.

[105] Infosys, Form 20-F FY2009, Item 3.

[106] Rajaraman, *History of Computing in India* (2012), p. 79.

[107] Rajaraman, *History of Computing in India* (2012), pp. 49, 53.

[108] Infosys, Form 20-F FY2010, Item 5.

[109] Infosys, Form 20-F FY2009, Item 3; Infosys, Form 20-F FY2012, Item 3.

[110] Infosys, Form 20-F FY2012, Item 3.

[111] Knowledge at Wharton, 'The Little Start-Up That Could' (Raman Roy interview). Self-account (Knowledge at Wharton interview, 4 June 2003, part 1).

[112] Reserve Bank of India, *Survey on Computer Software and Information Technology Services Exports*, RBI Bulletin, October 2012, survey for 2010-11, 'Highlights' and Table on type of service; Reserve Bank of India, *Survey on Computer Software and Information Technology Services Exports*, survey for 2007-08 (74.6% offsite).

[113] Reserve Bank of India, *Survey on Computer Software and Information Technology Services Exports*, RBI Bulletin, October 2012, survey for 2010-11, 'Highlights'.

[114] Knowledge at Wharton, 'The Little Start-Up That Could' (Raman Roy interview); Anonymous, 'Working Life, Interviews and Leaflets in Delhi's Call Centre Cluster' (2007). Anonymous activist account; low reliability. Self-account (Knowledge at Wharton interview, 4 June 2003, part 1).

[115] Nasscom, *Strategic Review* and fact sheets, slide 10. Nasscom self-description (industry lobby group).

[116] Rajaraman, *History of Computing in India* (2012), pp. 53-54 (Texas Instruments, 1985).

[117] Blinder, 'Offshoring: The Next Industrial Revolution?' (2006), 'This Time It's Personal'.

Part IV: Beyond services? (1985-2024)

Chapter 15. The other employers (1985-2024)

On 24 March 1998 Microsoft announced from its headquarters in Redmond, Washington, that it planned to open a software development centre in Hyderabad. The press release was short and hedged. The centre would open in the second half of the year, 'Subject to clearance by regulatory agencies', and an exact date was 'unavailable'.[1] It would work on one narrow problem: making Microsoft's Windows NT operating system and its BackOffice server programs work with computers that did not run Microsoft software. 'The development center initially will employ about 25 people,' the release said, in a company that then had more than 25,000 employees worldwide.[2]

Paul Maritz, the group vice-president for platforms and applications, chose his words with care. 'The majority of our development work will always be based in the United States, but we have a great deal of faith in the IT skills of India,' he said, adding that although the centre was 'initially relatively small', its work would be 'highly valued and strategically important'.[3] Moshe Dunie, a vice-president in the Windows division, called it 'an extension to our product team in Redmond'.[4] Its general manager would be Srini Koppolu, who had spent more than nine years in product development at Microsoft and would move from Redmond to Hyderabad. He would report to S. Somasegar, a director of Windows NT, who stayed at headquarters.[5] The company said the decision came at the end of 'a long evaluation process that began over a year ago', and only once it had found 'a compelling task for the center, the right location and the right management'.[6]

Twenty-five people, one task, a manager moved from the parent and a promise that most development work would stay at home: this was how a foreign product company entered India at the end of the 1990s. Earlier chapters followed the Indian services firms, which sold the hours of Indian engineers to clients abroad. Beside them, and less noticed, grew a second kind of employer: the Indian centres of foreign companies, which hired the same graduates to work on the parent's own products and systems. By 2024, on the count of the National Association of Software and Service Companies (Nasscom), the industry's lobby group, and Zinnov, a consultancy that sells advice to such centres, they employed about 1.9 million people.[7] They began with a satellite dish in Bangalore (Bengaluru since 2014) and ended as the 'global capability centres' of the 2020s. The question is what kind of profession they offered.

A dish on the roof

The first of them had arrived thirteen years before Microsoft. Texas Instruments (TI), the American chip maker, dates its start in India to 1985, and in a list of milestones published for its twentieth anniversary there it calls itself the first multinational to set up a software design centre in the country. Chapter 8 tells how TI won its satellite link, by working around a long list of government rules, and weighs that claim.[8]

What mattered to the engineers TI hired was the kind of work. None of it was billed by the hour. Within four years of arriving, TI India had a design centre for mixed-signal integrated circuits, chips that handle both analogue and digital signals, and over the next fifteen years it added work on disk-drive and wireless chips before moving, in 2004, into a seven-acre design facility in Bangalore.[9] It was engineering for TI's own products, done by TI's own employees, in India. At the twentieth-anniversary event in August 2005 TI's chairman, Tom Engibous, showed two mobile handsets that the company said were the first designed entirely in India, one developed with the Indian manufacturer BPL.[10]

Hewlett-Packard (HP) came through a returning engineer, Radha Basu, who in the account of the Berkeley scholar of regional development AnnaLee Saxenian spent four years building HP's software centre in Bangalore and left it with 400 people in 1989.[11] Saxenian adds that Indians educated in the United States were 'pivotal' in setting up the Indian facilities of Oracle, Novell, Bay Networks and other Silicon Valley companies.[12] Motorola later claimed

that its Indian centre had become, in 1993, the first commercial organisation in the world assessed at Level 5 of the Capability Maturity Model, a scale of software process discipline; Chapter 11 examines the claim.[13] Oracle opened a software development centre in Bangalore in 1994, on the dating of V. Rajaraman, the historian of Indian computing, and SAP (1998) and Intel (1999) are also reported to have opened development centres in India in that decade.[14]

Back offices came at the same time. Raman Roy, later called the father of Indian business process outsourcing, helped American Express and then, from 1996, GE to set up what an interviewer called 'captive shared-service centers' in India: units owned by the parent and working only for it.[15] Rajaraman counts American Express as the first to start back-office work for export in India, in 1994, followed by GE Capital International Services in 1997.[16] The word 'captive' stuck, and it describes the first wave well. These centres were owned by their clients. They did not compete for contracts.

Whether the work in these centres matched the work at headquarters is harder to show. Saxenian, writing in 2001, quoted the chief executive of one American company's offshore development centre in Bangalore, interviewed by another researcher in July 1996. He explained why his company had waited until the 1990s. Earlier, he said, he would have advised 'do anything, bodyshopping, subcontracting, modular work, but don't give full dependability here [in India] because nobody's ever done it.' By the mid-1990s new products and new versions of old products came from Bangalore. 'Technologically, there's zero difference,' he said.[17] It is a manager's claim, reported second-hand, and Saxenian called the evidence for such upgrading 'anecdotal'.[18] But it marks the ambition. Microsoft's Maritz, two years later, was more cautious about where the important work would be done.

The captive that became a vendor

The most revealing case of the first wave is a company usually counted among the services firms. Cognizant Technology Solutions went public on the Nasdaq stock market in June 1998. Its prospectus opened its corporate history with a plain sentence: 'The Company began its software development and maintenance services business in early 1994 as an in-house technology development center for The Dun & Bradstreet Corporation and its operating units.'[19] Those units included the market researcher A.C. Nielsen and Dun & Bradstreet's own information and software businesses. In November 1996 Dun & Bradstreet spun off a group of businesses, this one among them, to form a new parent company, Cognizant Corporation. During 1996 the development centre also set out to win clients outside the group.[20] The captive had become a vendor.

The prospectus shows who ran it. Its chairman and chief executive, Wijeyaraj (Kumar) Mahadeva, had been Dun & Bradstreet's vice-president for corporate strategy and, from 1993 to 1996, chairman of Dun & Bradstreet in India and China; he had led the team that set up the Indian business in 1994. He held a Harvard MBA and a master's degree in electrical engineering from Cambridge.[21] Its president, Lakshmi Narayanan, had joined the Indian subsidiary as chief technology officer in 1994 after nineteen years at Tata Consultancy Services (TCS), where he had been a regional head.[22] TCS had trained the man who ran the American captive's Indian arm.

The prospectus also shows where the work was done and by whom. At 31 March 1998 Cognizant had about 270 people in North America, at its offices and on clients' sites, of whom only fifteen were 'United States citizens or permanent residents'. It had about 30 in Europe, and 'approximately 870 persons on a full-time basis in its offshore software development centers in India'. Some 245 of its employees in the United States, 93 per cent of them, held H-1B temporary work visas.[23] About 74 per cent of its technical professionals were in India.[24] Its development centres, all in leased buildings, were in Chennai and Calcutta (Kolkata since 2001); its executive office in New York occupied 3,800 square feet.[25] Cognizant explained its model in words any Indian firm might have used: basing technical work in India gave it 'access to a large pool of skilled, English-speaking IT professionals with which to service customers on a cost basis significantly lower than in developed countries'.[26] It also warned investors that 'wages in India are increasing at a faster rate than in the United States'.[27]

The prospectus describes recruitment and training in detail, and it is one of the clearer records of how a foreign-owned employer turned Indian graduates into programmers in the 1990s. Each year the company visited 'approximately 45 premier colleges and technical schools in India'. It judged candidates on academic results, 'a written aptitude test measuring problem-solving skills' and a technical interview, and it also hired experienced staff from other firms.[28] New recruits went through 'an intensive orientation and training program' that taught them the company's own software engineering method, which it called QVIEW.[29] By 2002 Cognizant said it gave newly hired

programmers 'five months of combined classroom and on-the-job training'.[30] The degree got a graduate through the door. The employer taught the trade.

Cognizant then grew like the firms it was compared with. At the end of 2002 it had about 1,425 people in North America, 240 in Europe and 'approximately 4,500 persons' in its Indian development centres.[31] At the end of 2007 it had more than 55,400 staff worldwide, 'a majority' of its technical professionals in India.[32] At the end of 2012 it had about 156,700, of whom about 124,100 were in the Asia Pacific region.[33] In 2012 more American H-1B work visas were approved for Cognizant, 9,281, than for any other firm in a ranking two economists compiled from US immigration data.[34]

The lesson is that the line between a foreign company's captive and an 'Indian' services firm was never fixed. A captive could be spun out to sell to others, as Cognizant was. A services firm could be bought by a multinational: IBM acquired the Indian outsourcing firm Daksh.[35] What stayed constant was the work, done by engineers in India for clients or parents abroad.

A mirror in Bangalore

Product companies came for engineers, not for services. On 13 October 2004 Google's founders, Larry Page and Sergey Brin, spoke to reporters in Bangalore, the third stop of a visit that had taken them to New Delhi and Hyderabad. Google had opened offices in Bangalore and Hyderabad earlier that year.[36] 'One approach we are taking is that Bangalore is where we run a mirror exactly of what we have in the United States in terms of development,' Brin said.[37] Researchers in Bangalore would work with their American colleagues to devise new services and write the software, including search by speech, search in more languages, personalised search and features for the Gmail service.[38] Brin said Google had been 'too late' in tapping Indian talent: 'We would have preferred to do it sooner.'[39]

The man doing the hiring was Krishna Bharat, the engineer who had created Google's news service. 'We don't have a cap (on how many to hire),' he said, in the Associated Press report. 'We will take in as many people as we can, if they meet our global standards.'[40] The phrase 'global standards' marks the difference from Microsoft's careful start six years earlier. Google did not describe its Bangalore office as a satellite with a defined task. It described it, in development at least, as a mirror of the parent.

January 2007

Two reports three days apart, at the turn of January and February 2007, showed how far the multinationals had come. On 29 January Accenture's chairman and chief executive, William D. Green, told reporters in Bangalore that the consultancy would have 35,000 staff in India by the end of August, up 52 per cent on a year earlier. That would give it more employees in India than in the United States, where it would have just over 30,000.[41] The reporter, John Ribeiro, noted that multinational services companies including IBM, EDS and Capgemini were expanding fast in India, 'helping push up wages and attrition rates' among software and back-office staff.[42]

Three days later *InformationWeek* reported that IBM's 'Indian workforce now stands at 53,000, up from 39,000 a year ago and 23,000 two years ago'. If that rate continued, IBM would have more than 70,000 people in India during 2007, 'almost 25% of its current worldwide workforce of 300,000'.[43] 'IBM workers in India are engaged in everything from routine coding to high-end system design,' the article said; a year earlier the company had decided to consolidate all its development of software for service-oriented architectures, a way of building large systems out of networked services, in Bangalore.[44] India was also IBM's fastest-growing market, with sales up 37 per cent in 2006.[45] For a company that had shut its Indian operations in 1978, the return was complete.[46]

Accenture and IBM were not captives. They were services firms that happened to be American, and their Indian centres did the same kind of work as the Indian firms, for the same kind of clients. They belong in this chapter for another reason. They competed for the same engineers, and by 2007 each employed tens of thousands of people in India. For an engineer leaving college in 2007, 'working for a foreign company' could mean a chip-design team at TI, a product team at Google or Microsoft, a back-office captive at a bank, or a delivery centre at IBM or Accenture. The employers were foreign. The work ranged from research to routine maintenance.

Counting the centres

The first aggregate count of foreign companies' research and development centres came from Zinnov. In early 2008, according to a summary in the trade paper *EE Times*, it reported about 594 multinational R&D centres in India employing about 146,760 people, more than half of them (312) in Bangalore. It counted 93 new centres opened in 2004 and only 24 in 2007, amid rising costs and attrition of 8 to 20 per cent a year.[47]

Two scholars, Sarosh Kuruvilla and Aruna Ranganathan, who studied the Indian outsourcing industry from fieldwork in 2004-07, give a clearer view of the labour market. On Nasscom's figures, the number of software professionals employed in India rose from 297,000 in 2005 to about 562,000 in 2007.[48] Turnover in 2007 was about 20 per cent a year in software and 50-60 per cent in back-office work.[49] The authors named the multinationals as one cause: 'the entry and expansion of large MNCs such as IBM, Oracle, and Accenture into India have resulted in considerable "poaching" of developers, project managers, and high-level executives'.[50] Their own judgement, which they said rested on no systematic study, was that the main reason software engineers left was not pay but 'the lack of opportunities to work on cutting-edge technology'.[51] That is exactly what the product companies' centres claimed to offer.

In 2004, on a Nasscom survey they cite, the average software professional was about 28 years old, three in four were men, and 69 per cent were engineers.[52] Median pay that year, from a Nasscom-Hewitt survey, was about $6,560 for a software engineer and $30,816 for a project manager, with no separate figure for the multinational centres.[53] At that price the engineer's hour, which the services firms sold and the multinationals' centres used, was cheap. If more than two-thirds of software professionals were engineers, the industry, foreign centres included, was drawing most of its staff from the engineering colleges. The multinationals had come, in Rajaraman's account, 'to take advantage of the availability of high quality computer science graduates at reasonable cost'.[54] Kuruvilla and Ranganathan doubted that the colleges could keep up. Citing an estimate of 2006, they wrote that third-tier colleges produced 88 per cent of India's engineering undergraduates.[55] A manager at a well-known Wall Street firm with its own software support unit in India told them: 'we are really scraping the bottom of the barrel now and are hiring engineers from third and fourth tier schools who have very little engineering knowledge at all'.[56] The employers answered with training, like Cognizant's five months for new programmers, and by hiring experienced staff away from one another. Infosys told American investors in its annual report for the year to March 2009 that it had long faced strong competition for employees from the multinationals setting up in India.[57]

Captives and careers

Did the other employers offer a different career? The executives' statements describe product work: Brin's 'mirror', TI's chip design, IBM's 'high-end system design'.[58] TI reports that the Indian business magazine *Businessworld* named TI India the 'Best Company to Work for in India' in 2003 and second best in 2004.[59] These are employers describing themselves.

The one comparative study concerns back offices, not engineering. Kuruvilla and Ranganathan distinguished 'captives', 'BPO firms owned by U.S. client firms such as Dell, EDS, and HP that service only their owners', from 'third-party firms' that served many clients, and noted that consolidation was 'eroding the distinction'.[60] They found turnover of 15 to 20 per cent a year in five captive firms, 'although even some captives reported rates as high as 30%'. A survey of seven large third-party call centres found an average of 30 per cent, ranging from 12 to 62 per cent.[61] Their conclusion: 'captive call centers are able to provide more promotions to higher levels than "third-party" call centers, because the market stability they enjoy fosters longer career paths'.[62] A captive's work did not depend on winning the next contract. That stability, on this evidence, made room for a ladder. A pay premium at the multinational development centres is often asserted, but no published comparison of their pay with that of the Indian services firms has been located.[63]

Sidebar: Why did Eastern Europe become a nearshore rival, and why couldn't it scale?

In January 2007, on the same day he announced that Accenture would soon have more staff in India than in America, William Green looked at the alternatives. 'There will not be another phenomenon like India,' he said, and he expected no other country to overtake India in Accenture's headcount for at least ten years.[64] His colleague Basilio Rueda, a senior managing director of the firm's Global Delivery Network, explained why. Accenture had

opened service centres in Eastern European cities such as Prague and Bucharest, but, he said, it was not possible to increase the staff there easily, and the cost of staff was far higher than in India. The Eastern European centres were 'near-shore' locations, chosen to serve European clients from close by with cultural affinity.[65]

The firm that tested the nearshore model most fully was EPAM Systems, a services company listed in New York whose engineers worked mainly in the former Soviet bloc. In its annual report for 2021 EPAM wrote: 'Our largest delivery centers are located in Ukraine, Belarus and Russia'. At the end of that year it had 12,389 delivery professionals in Ukraine, 9,416 in Belarus and 8,933 in Russia, out of 52,617. India came next, with 4,349, followed by Poland, the United States, Hungary and Mexico.[66] EPAM said these countries were well placed for the work, in terms an Indian firm would have recognised: a supply of qualified software engineers, 'established educational infrastructure', an industrial base and government tax incentives.[67]

Then, on 24 February 2022, Russian forces attacked Ukraine.[68] EPAM's annual report for 2024 records the cost. Business continuity plans, relocation, the company's 'humanitarian commitment to our people in Ukraine' and 'the cost of our phased exit from Russia' had 'resulted in materially increased expenses'.[69] The company sold its remaining holdings in Russia on 26 July 2023.[70] And it reported a new fact about itself: 'During 2024, India became our largest delivery location, measured by the number of delivery professionals'. It had 10,072 there at the end of 2024, having added 3,006 during the year. Ukraine, with 8,764, down from 9,113 a year earlier, remained 'a significant delivery location'. EPAM's delivery staff as a whole numbered about 55,100.[71]

What do these figures show? They do not show that Eastern Europe failed a market test. A war is not a market. EPAM's Ukrainian engineers kept working through it, and the company said its operations there had not been significantly affected.[72] They show something narrower. Before 2022, a firm built on Eastern European engineers had about 30,000 of them in three neighbouring countries. When those countries became a risk, the company turned to India to grow, and more than doubled its Indian delivery staff in three years, to the point where India overtook Ukraine. Rueda's remark of 2007 points to one reason: staff numbers in Prague or Bucharest could not be raised easily, and staff there cost far more.[73] The comparison rests on one firm's executives and another firm's filings.[74]

A new name

In August 2019 Amazon opened a campus in Hyderabad that it described as 'our first owned campus outside USA and the single largest building in terms of area and head count for Amazon globally'. The company's vice-president for global real estate and facilities, John Schoettler, wrote on its Indian blog that the building had '15,000 work points across 1.8 million square feet' of office space, three amenity floors and twelve office floors, and lifts able to move 972 people at once.[75] The largest building of one of the world's largest companies was not in Seattle. It was in the city where Microsoft had started with twenty-five people.

By then the industry had a new name for such places. A 'global capability centre', or GCC, is a foreign company's own unit in India, whether it writes software, designs chips, runs finance and human resources, or analyses data. The term covers the old development centres and the old captives alike. In September 2024 Nasscom and Zinnov published a report on them. As Reuters summarised it, about 1,700 companies had GCCs in India, employing 1.9 million people, and the centres' market size was $64.6 billion in the year to March 2024. The 'revenue' of a GCC needs care. Reuters described the centres as supporting their parent organisations; they do not sell on the open market, so the figure is best read as a measure of the work they do for their parents, not of sales won.[76] Zinnov's own summary counted more than 2,975 separate units, over 400 new centres in five years, and claimed that leading multinationals now did nearly a third of their global engineering in India.[77]

The report also made projections. By 2030, it said, between 2,100 and 2,200 companies would have GCCs in India, employing between 2.5 and 2.8 million people, and the market would be worth $99-105 billion.[78] Those are forecasts by interested parties. A later edition counted 2,117 GCCs employing 2.36 million people in the year to March 2026, with revenue of about $98.4 billion (industry figures, as reported by *Business Today*), close to the figure the 2024 report had projected for 2030.[79]

The 1.9 million needs a denominator. Nasscom, in its review of February 2025, restated the size of the whole Indian technology industry after counting GCCs more fully: about 5.67 million employees in the year to March 2024. It said that the industry's export revenue was now split equally between 'Global MNCs (including GCCs)' and Indian service providers.[80] On these figures, roughly a third of India's technology workforce in 2024 worked for the other employers' own centres, an order of magnitude on the industry's own definition.

The work itself was said to be changing. Zinnov's summary claimed more than 500 centres with artificial intelligence and machine-learning capabilities, 185 of them with dedicated centres for the work, a pool of more than 120,000 people working on AI, and about 6,500 senior 'global roles' held in India, including more than 1,100 held by women.[81] It also pointed to smaller cities, where costs could be 25-30 per cent lower.[82] State governments competed for the centres. Karnataka's GCC policy for 2024-29, as Zinnov summarised it, aimed to double the number of GCCs in the state to 1,000 and to create 350,000 jobs by 2029.[83] The Union Budget of February 2025 announced that a national framework would be drawn up to guide states in promoting GCCs in smaller, 'tier 2' cities.[84]

None of this made the GCC new. The institution goes back to TI's dish in 1985, to HP's 400 people in 1989, to the captives of American Express and GE, and to the 594 R&D centres Zinnov counted in 2008.[85] Cognizant itself began as one. What changed by 2024 was the name, the scale, the spread from software and internet companies to banks and insurers, with retailers, health-care and drug companies expected next, and the claim that senior roles, not only tasks, were now based in India.[86]

The circle closes

The last document in this chapter is a services firm's annual report. At the end of 2024 Cognizant had 'approximately 336,800 employees', of whom 241,500 were in India and 42,800 in North America; women made up about 38 per cent of the workforce.[87] Seventy-two per cent of the staff of a company that had gone public in 1998 with an executive office in New York of 3,800 square feet now worked in India.[88] Accenture, IBM and EPAM had gone the same way, each in its own measure.[89]

Among its competitors Cognizant now listed, beside Accenture, Infosys, TCS and Wipro, 'clients' in-house technology resources, such as GCCs'.[90] Its risk factors were sharper. 'We face competition from clients' in-house technology resources, such as GCCs, which may provide a lower cost alternative to our services,' it told investors. And for staff: 'We compete for employees not only with other companies in our industry but also with companies in other industries, such as software services, engineering services and financial services companies, as well as our clients' GCCs.'[91]

A company that began in 1994 as one American firm's captive centre in India now named other firms' captive centres as a threat to its business and to its hiring.[92] Clients that had once sent work to Indian vendors were building their own Indian teams, staffed by the same graduates from the same colleges.

This matters for the argument of the rest of the book. A services firm sells its engineers' hours, and the price of an hour is set in a contract with a client. A GCC employs engineers to build and run its parent's own products and systems, and any gain in their output belongs directly to the parent. When tools arrive that let one engineer do the work of several, the two kinds of employer face different questions. The services firm must decide who keeps the saving, and the client will ask for it. The GCC's parent simply keeps it. Chapter 19 returns to that difference.

The multinational centres showed that Indian engineers could do a parent's core work, not only its maintenance; who would answer for the work when it went wrong, a question of trust and governance, is where the next chapter begins, inside the services industry itself and inside the state.

Notes to Chapter 15. The other employers (1985-2024)

[1] Microsoft, press releases on its India development centre (1998-1999).

[2] Microsoft, press releases on its India development centre (1998-1999).

[3] Microsoft, press releases on its India development centre (1998-1999).

[4] Microsoft, press releases on its India development centre (1998-1999).

[5] Microsoft, press releases on its India development centre (1998-1999).

[6] Microsoft, press releases on its India development centre (1998-1999).

[7] Suresh, 'India's Global Centre Market to Grow to $105 Billion by 2030' (2024).

[8] Texas Instruments, 'TI Celebrates 20 Years in India' (2005) (company milestone list, 2005); Saxenian, 'Bangalore: The Silicon Valley of Asia?' (2001), p. 6 (government rules; Chapter 8 gives the full account).

[9] Texas Instruments, 'TI Celebrates 20 Years in India' (2005). Milestones in the company's list: mixed-signal design centre 1989; application-specific products from 1990; hard-disk-drive signal-processing cores 1995; third-generation mobile and wireless network chipsets 2000; seven-acre Bangalore design facility 2004.

[10] Texas Instruments, 'TI Celebrates 20 Years in India' (2005). The same source says several chips then being designed in India used 65- and 90-nanometre processes, which TI called the most advanced in the industry (company claim).

[11] Saxenian, *Silicon Valley's New Immigrant Entrepreneurs* (1999), pp. 62-63. Basu's story is told more fully in Chapter 13.

[12] Saxenian, *Silicon Valley's New Immigrant Entrepreneurs* (1999), p. 63.

[13] Motorola, Global Software Group media kit.

[14] Rajaraman, *History of Computing in India* (2012), p. 48 (Oracle, Bangalore, 1994); *Computerworld*, 'Oracle Chooses India for Key Overseas Site'; *Managing Innovation from the Land of Ideas and Talent* (2009); Intel, press release (2000), and AP, 'Intel to Invest $1 Billion in India' (2005). The SAP and Intel primary texts were not consulted, so the text says 'are reported'.

[15] Knowledge at Wharton, 'The Little Start-Up That Could' (Raman Roy interview).

[16] Rajaraman, *History of Computing in India* (2012), p. 53. Chapter 14 follows the back-office captives and sets out the other datings of GE Capital International Services (October 1996).

[17] Saxenian, 'Bangalore: The Silicon Valley of Asia?' (2001), p. 11.

[18] Saxenian, 'Bangalore: The Silicon Valley of Asia?' (2001), p. 11.

[19] Cognizant, IPO prospectus (Form 424B4, 1998), p. 5.

[20] Cognizant, IPO prospectus (Form 424B4, 1998), p. 5.

[21] Cognizant, IPO prospectus (Form 424B4, 1998), p. 43.

[22] Cognizant, IPO prospectus (Form 424B4, 1998), p. 43.

[23] Cognizant, IPO prospectus (Form 424B4, 1998), p. 41.

[24] Cognizant, IPO prospectus (Form 424B4, 1998), p. 8.

[25] Cognizant, IPO prospectus (Form 424B4, 1998), p. 42.

[26] Cognizant, IPO prospectus (Form 424B4, 1998), p. 4.

[27] Cognizant, IPO prospectus (Form 424B4, 1998), p. 8.

[28] Cognizant, IPO prospectus (Form 424B4, 1998), p. 41.

[29] Cognizant, IPO prospectus (Form 424B4, 1998), p. 42.

[30] Cognizant, Form 10-K FY2002, p. 8.

[31] Cognizant, Form 10-K FY2002, p. 16.

[32] Cognizant, Form 10-K FY2007, pp. 4, 13.

[33] Cognizant, Form 10-K FY2012, p. 15.

[34] Khanna and Morales, 'The IT Boom and Other Unintended Consequences' (2016), Table 1 (p. 54).

[35] Kuruvilla and Ranganathan, 'Economic Development Strategies' (2008), p. 14.

[36] AP, 'Google Founders on Hiring Trip to India' (2004), and IIT Kanpur seminar listing (2004).

[37] AP, 'Google Founders on Hiring Trip to India' (2004), and IIT Kanpur seminar listing (2004).

[38] AP, 'Google Founders on Hiring Trip to India' (2004), and IIT Kanpur seminar listing (2004).

[39] AP, 'Google Founders on Hiring Trip to India' (2004), and IIT Kanpur seminar listing (2004).

[40] AP, 'Google Founders on Hiring Trip to India' (2004), and IIT Kanpur seminar listing (2004).

[41] *InfoWorld*, 'Accenture: More Staff in India than US by August' (2007) (IDG News, 29 January 2007).

[42] *InfoWorld*, 'Accenture: More Staff in India than US by August' (2007).

[43] *InformationWeek*, 'IBM Head Count in India Tops 50,000' (2006).

[44] *InformationWeek*, 'IBM Head Count in India Tops 50,000' (2006).

[45] *InformationWeek*, 'IBM Head Count in India Tops 50,000' (2006).

[46] *InformationWeek*, 'IBM Head Count in India Tops 50,000' (2006). IBM's 1978 exit is told in Chapter 5.

[47] *EE Times*, report of Zinnov data on R&D centres (2008) (EE Times, 17 March 2008). Zinnov's clients include the multinational centres it counts. The other centres were in Pune, the Delhi area, Hyderabad and Chennai, far behind Bangalore; the largest groups worked on telecommunications and networking, semiconductors and chip-design tools, and consumer electronics. The figures are given from the paper's summary.

[48] Kuruvilla and Ranganathan, 'Economic Development Strategies' (2008), p. 15.

[49] Kuruvilla and Ranganathan, 'Economic Development Strategies' (2008), p. 15.

[50] Kuruvilla and Ranganathan, 'Economic Development Strategies' (2008), p. 19.

[51] Kuruvilla and Ranganathan, 'Economic Development Strategies' (2008), p. 19.

[52] Kuruvilla and Ranganathan, 'Economic Development Strategies' (2008), p. 19.

[53] Kuruvilla and Ranganathan, 'Economic Development Strategies' (2008), p. 55 (table assembled from a Nasscom-Hewitt pay survey; industry-wide, in US dollars). The same table gives about $14,820 for a team leader and $63,125 for the head of a large development unit.

[54] Rajaraman, *History of Computing in India* (2012), p. 53.

[55] Kuruvilla and Ranganathan, 'Economic Development Strategies' (2008), p. 28, citing Cairncross (2006) for the third-tier shares. The authors add that the All India Council for Technical Education (AICTE), the regulator of technical colleges, had removed about 22,722 seats at 300 engineering colleges that lacked faculty, courses and infrastructure.

[56] Kuruvilla and Ranganathan, 'Economic Development Strategies' (2008), p. 29 (manager interviewed by the authors during their 2004-07 fieldwork; firm not named).

[57] Infosys, Form 20-F FY2009, Item 3 (risk factors). The same section warns that hiring by technology companies, especially in India, and worldwide competition for skilled professionals might cause a shortage of qualified staff where it hired.

[58] AP, 'Google Founders on Hiring Trip to India' (2004), and IIT Kanpur seminar listing (2004); Texas Instruments, 'TI Celebrates 20 Years in India' (2005); *InformationWeek*, 'IBM Head Count in India Tops 50,000' (2006).

[59] Texas Instruments, 'TI Celebrates 20 Years in India' (2005).

[60] Kuruvilla and Ranganathan, 'Economic Development Strategies' (2008), p. 14.

[61] Kuruvilla and Ranganathan, 'Economic Development Strategies' (2008), p. 19.

[62] Kuruvilla and Ranganathan, 'Economic Development Strategies' (2008), p. 21.

[63] The only pay figures in Kuruvilla and Ranganathan, 'Economic Development Strategies' (2008) (p. 55) are industry-wide medians and do not separate employer types.

[64] *InfoWorld*, 'Accenture: More Staff in India than US by August' (2007).

[65] *InfoWorld*, 'Accenture: More Staff in India than US by August' (2007).

[66] EPAM Systems, 10-K FY2021, p. 5.

[67] EPAM Systems, 10-K FY2021, p. 5.

[68] EPAM Systems, 10-K FY2024, p. 5.

[69] EPAM Systems, 10-K FY2024, p. 30.

[70] EPAM Systems, 10-K FY2024, p. 33.

[71] EPAM Systems, 10-K FY2024, p. 5.

[72] EPAM Systems, 10-K FY2024, p. 5.

[73] *InfoWorld*, 'Accenture: More Staff in India than US by August' (2007).

[74] *InfoWorld*, 'Accenture: More Staff in India than US by August' (2007) (Green and Rueda, Accenture); EPAM Systems (EPAM 10-Ks). Neither source gives the graduate output or wages of Poland, Ukraine or Romania, so Rueda's point about scale and cost cannot be tested from them.

[75] Amazon India, 'Owned and Loving It' (2019).

[76] Suresh, 'India's Global Centre Market to Grow to $105 Billion by 2030' (2024).

[77] Nasscom and Zinnov, *India GCC Landscape Report* (2024) (Zinnov report page).

[78] Suresh, 'India's Global Centre Market to Grow to $105 Billion by 2030' (2024).

[79] Agarwal, 'The Big Boom' (2026). Post-cutoff material, checked in October 2026.

[80] Nasscom, Strategic Review 2025 press release (2025).

[81] Nasscom and Zinnov, *India GCC Landscape Report* (2024).

[82] Nasscom and Zinnov, *India GCC Landscape Report* (2024).

[83] Nasscom and Zinnov, *India GCC Landscape Report* (2024); Government of Karnataka, *Karnataka GCC Policy 2024-2029* (ICAI summary confirms the 2024-29 policy; the targets are Zinnov's summary).

[84] Union Budget 2025-26 speech (2025), para. 91.

[85] Texas Instruments, 'TI Celebrates 20 Years in India' (2005); Saxenian, *Silicon Valley's New Immigrant Entrepreneurs* (1999), p. 63; Knowledge at Wharton, 'The Little Start-Up That Could' (Raman Roy interview); *EE Times*, report of Zinnov data on R&D centres (2008).

[86] Suresh, 'India's Global Centre Market to Grow to $105 Billion by 2030' (2024); Nasscom and Zinnov, *India GCC Landscape Report* (2024).

[87] Cognizant, Form 10-K FY2024, p. 10.

[88] Cognizant, Form 10-K FY2024, p. 10; Cognizant, IPO prospectus (Form 424B4, 1998), p. 42.

[89] *InfoWorld*, 'Accenture: More Staff in India than US by August' (2007); *InformationWeek*, 'IBM Head Count in India Tops 50,000' (2006); EPAM Systems, 10-K FY2024, p. 5.

[90] Cognizant, Form 10-K FY2024, p. 9.

[91] Cognizant, Form 10-K FY2024, pp. 15-16.

[92] Cognizant, IPO prospectus (Form 424B4, 1998), p. 5; Cognizant, Form 10-K FY2024, pp. 13, 15-16.

Chapter 16. Fraud and foundation (2008-2018)

On 7 January 2009 the board of Satyam Computer Services, a software company with its headquarters in Hyderabad, received a letter from its chairman and founder, B. Ramalinga Raju. It began: 'It is with deep regret, and tremendous burden that I am carrying on my conscience, that I would like to bring the following facts to your notice.' The first fact was that the company's balance sheet at 30 September 2008 carried 'Inflated (non-existent) cash and bank balances of Rs.5,040 crore (as against Rs. 5361 crore reflected in the books).'[1] Rs 5,040 crore is Rs 50.4 billion, or about $1.16 billion at the 2008 average exchange rate. On Raju's own account, nearly all the cash the company said it held did not exist.[2]

Copies went to the chairman of the Securities and Exchange Board of India (SEBI), the market regulator, and to the stock exchanges. Satyam filed the letter the same day with the United States Securities and Exchange Commission (SEC), because its shares also traded in New York.[3] By the end of that day, according to the order of the Company Law Board two days later, the share price had fallen from Rs 188 to Rs 38.40.[4]

Satyam was not a marginal firm. Its own filings called it the fourth-largest Indian IT services company by export revenue.[5] Its engineers were recruited, trained and deployed in much the same way as those of Infosys, Wipro and Tata Consultancy Services (TCS). The fraud therefore put a question to the whole industry this book describes. If the books of one of its four largest firms could be false for years, what exactly had clients, investors and graduates been buying? The answer that emerged over the next three months was that the work was real and the accounts were not.

The same month, in an unrelated decision, the government of India created an agency to give every resident a unique identity number. Within six months it had hired an Infosys co-founder to run it.[6] Satyam showed what happened when a software company's governance failed. Aadhaar, the identity system, showed what happened when the state borrowed the industry's people and methods for a project of its own, and when the Supreme Court then had to decide what that system could be made to do to Indian citizens.

The company on paper

The last annual report Satyam filed before the letter was its Form 20-F for the year to March 2008, lodged with the SEC that summer. It described a business growing very fast. 'As of March 31, 2008, we had 50,570 employees, whom we refer to as associates, worldwide as compared to 39,018 associates as of March 31, 2007.'[7] Revenue for the year was reported as $2,138.1 million, and the company said its revenue had grown at a compound rate of 39.1 per cent a year over three years.[8] It had 654 active customers, 'including 185 Fortune Global 500 or Fortune U.S. 500 companies'.[9]

The 20-F is also a record of how such a company made engineers. 'We recruit graduates from the engineering departments of India's leading universities, engineering and technical colleges and management institutes,' it said. It counted 'over 1,500 such institutions' in India, and described a selection process 'involving a series of tests and interviews'.[10] Each new technical recruit, the filing said, went through 'an intensive 12 week initial training program' and then at least 40 hours of training a year.[11] The company reported that 47,405 of its associates were technical staff in IT services, and that 9,941 of them were working onsite, at clients' premises abroad.[12] Attrition in IT services, the share of staff who left in a year, had been 13.1 per cent in the year to March 2008. 'None of our associates are represented by a union,' the filing said.[13]

That was the profession Satyam offered: a campus test, three months of company schooling, a project, perhaps a posting abroad. Nothing in it was unusual for the industry in 2008.

The financial pages were where the fiction lay, though no outside reader could see it. The 20-F reported 'cash and cash equivalents of $290.5 million' at 31 March 2008. It explained that cash used in investing in the year to March

2007 had risen 'primarily due to investment in bank deposits amounting to $745.6 million'.[14] Raju's letter, written five months after the filing, said that cash and bank balances of Rs 5,361 crore in the books included Rs 5,040 crore that did not exist.[15] A company growing at nearly 40 per cent a year, adding more than 11,000 people to its payroll in a year did not look strange from outside. That was the difficulty.

Riding a tiger

The letter, as filed with the SEC, runs to about two pages.[16] After the cash, Raju listed 'An accrued interest of Rs. 376 crore which is non-existent', 'An understated liability of Rs. 1,230 crore on account of funds arranged by me', and debtors (money owed by customers) overstated by Rs 490 crore.[17] For the quarter to September 2008, he wrote, the company had reported revenue of Rs 2,700 crore and an operating margin of Rs 649 crore, or 24 per cent of revenue. The actual figures were revenue of Rs 2,112 crore and an operating margin of Rs 61 crore, 3 per cent of revenue. In that quarter alone, he said, the false cash and bank balances had grown by Rs 588 crore.[18]

He explained how it had happened. 'What started as a marginal gap between actual operating profit and the one reflected in the books of accounts continued to grow over the years. It has attained unmanageable proportions as the size of company operations grew significantly.' The company, he wrote, had to carry extra staff and assets to justify the higher level of operations it was reporting, which raised its costs further. Every attempt to close the gap had failed. Because the promoters (the founding family) held only a small share of the equity, he feared that poor results would bring a takeover that would expose the gap.[19] Then he wrote: 'It was like riding a tiger, not knowing how to get off without being eaten.'[20]

The letter named the event that had brought the crisis into the open. According to the government's petition, as the Company Law Board summarised it, Satyam's board had recently approved a proposal to buy stakes, at huge cost, in two companies controlled by Raju's near relations; the Board itself recorded that four independent directors had resigned once the decision became public.[21] Raju called the 'aborted Maytas acquisition deal' 'the last attempt to fill the fictitious assets with real ones'.[22]

Much of the letter is Raju's defence of himself. He claimed that neither he nor the managing director, his brother B. Rama Raju, had sold shares in eight years except a small number for charity. He wrote that 'Every attempt was made to keep the wheel moving and to ensure prompt payment of salaries to the associates.' He claimed that 'neither me, nor the Managing Director took even one rupee/dollar from the company', and that none of the board members, past or present, had known. He listed seventeen senior executives who, he said, were also unaware.[23] These are the claims of the man who had confessed. SEBI later held him, his brother and three officials liable for fraud and insider trading, and an appeals tribunal upheld those findings in 2017.[24]

He closed with what he had built. 'I have seen it grow from few people to 53,000 people, with 185 Fortune 500 companies as customers and operations in 66 countries.' And: 'I sincerely apologize to all Satyamites and stakeholders'.[25] He recommended that the bank Merrill Lynch 'can be entrusted with the task of quickly exploring some Merger opportunities'.[26]

Two days in January

Merrill Lynch had already gone. The same SEC filing carried a letter from Kevan Watts, president of DSP Merrill Lynch, to the chairman of SEBI, dated 7 January. It said the bank had ended its advisory engagement with Satyam the day before. 'In the course of such engagement, we came to understand that there were material accounting irregularities, which prompted our aforesaid decision.'[27] An outside adviser had seen enough to walk away a day before the chairman confessed.

Satyam's own press release that day said its priorities included protecting 'the careers and security of its approximately 53,000 associates'.[28] SEBI appointed an investigating authority the same day to examine dealings in Satyam's shares, including possible fraud and insider trading.[29]

The state then moved quickly. On 8 January the Registrar of Companies for Andhra Pradesh petitioned the Special Judge for Economic Offences in Hyderabad for permission to search Satyam's offices and seize its books, papers and computer records. The judge's order records that the petition was filed 'at 04.30 PM at my residence'. The Registrar, the order says, apprehended that the company appeared to have shown 'artificial and inflated profits for the last several years'. The judge found a prima facie case and allowed the search.[30]

On 9 January the Union of India petitioned the Company Law Board in New Delhi to remove the directors. The petition, as the Board summarised it, described a listed company with about 300,000 shareholders and more than 53,000 employees, the fourth-largest IT company in India, which had 'received a number of awards for best corporate governance'.[31] The Board noted that the promoter group now held less than 4 per cent of the shares and financial institutions about 60 per cent. Its chairman, S. Balasubramanian, wrote that Raju's written admission established 'beyond any shadow of doubt' that there had been 'financial impropriety and jugglery of financial statements'. 'It appears that a serious fraud has been perpetrated on the society as a whole.'[32] He suspended the board with immediate effect and authorised the central government to appoint up to ten 'persons of eminence' as directors.[33]

Two years later the SEC summarised the order bluntly: 'the India government seized control of the company by dissolving its board of directors and appointing new government-nominated directors'.[34]

The people who did the work

For the engineers, the first question was whether they would be paid. *Business Today*, a business magazine, sent reporters to Satyam's staff in the weeks after the letter and published their accounts in its issue of 8 February 2009.[35] A week after Raju's arrest, the magazine reported, a European client had decided to move a project, with the team working on it, to another Indian vendor. Satyam staff in Europe told the magazine that '95 per cent of the team had agreed to join the competing vendor'.[36]

The rest of the industry did not rush to hire them. T. Sreedhar, managing director of TMI Network, a recruitment firm in Hyderabad, said his office was overrun by Satyam employees. They were, he said, 'stuck between the devil and the deep sea. They cannot quit the job because there is no option being made available by the large companies (who have chosen not to hire them)'. If they stayed, he added, they could not be sure of being paid that month.[37] The magazine reported that Nasscom (the National Association of Software and Service Companies), the industry's lobby group, and Infosys had issued statements against recruiting Satyam's people, 'not because they are tainted but to ensure that the company doesn't suffer because of an exodus'.[38] Rishi Das, chief executive of another recruiter, CareerNet Consulting, estimated that about 35,000 Satyam employees, 70 per cent of the workforce, were already looking for jobs with rivals. That was a recruiter's estimate, not a count.[39]

Some staff were tied in. Insiders told the magazine that about 7,000 employees had been required to give 'Rs 2 lakh bonds' (Rs 200,000) as a deposit under two-year contracts, which they would forfeit if they left.[40] The newest recruits were the most exposed. In 2007, the magazine said, Satyam had issued about 6,000 offer letters on campuses across the country. 'Those offers are not worth the paper they are written on.'[41]

The magazine printed five short first-person accounts, each 'as told to' one of its reporters. N. Kavitha, 22, had studied engineering at the Muthukumaran Institute of Technology near Mangadu, a suburb of Chennai, and had been selected by Satyam in a campus placement in 2007-08. 'I was told that I would get a joining letter by September 2008. This did not happen. The company HR said that I could definitely expect to join in January 2009.' Now, she said, she had lost hope of getting it. But she would still join if it came. 'In this recession, a fresher needs to grab any opportunity. I am not worried too much about Satyam's reputation in the market; I want the work experience first.'[42] Her position was close to the one Chapter 14 traces in the collapse of campus offers in 2009: a campus offer was a promise, not a job, and the graduate carried the risk when the company's plans changed.

S. Mastanamma, 24, had worked at Satyam's Chennai centre for two years after graduating from Newton's College of Engineering in Guntur, Andhra Pradesh. 'I am very positive about the company's future,' she said. She praised the project-level leadership, the employee portal and the training, and said human resources had opened a round-the-clock helpline for staff. 'I am not worried about lay-offs. The existing clients have to be serviced and the new board at Satyam is saying that people are needed to continue to service the clients.'[43]

A consultant in Europe, two years with the firm, who was not named, said the client had arranged to move the whole project team to another vendor. 'We, as employees, are very clear we want to save our livelihoods.' As an employer, the consultant said, Satyam had paid well. 'Raju was always elusive, I have never seen or heard from him in the two years I have been with the firm', except for one mail defending the Maytas deal.[44] A retail consultant five months into the job, also unnamed, said: 'There is a bit of uncertainty in every job.'[45]

In a bracketed note of its own in its issue of 22 February 2009, the magazine reported that employees had started to have their pay credited to their accounts.[46] When the board held a press conference in April, the company said

that on 20 March 2009, by its internal data, it had about 42,000 employees on a stand-alone basis (the parent company without its subsidiaries).[47]

Rs 58 a share

The government's nominees took over a company whose clients were still being served and whose cash was gone. The new directors included Kiran Karnik, who became chairman, and Deepak Parekh.[48] With Goldman Sachs and Avendus Capital as financial advisers, the board ran an auction under the supervision of Justice S.P. Bharucha. Bidders were judged first on technical criteria, including their record in corporate governance and in managing distressed companies, and then on price.[49] On 13 April 2009 the financial bids were opened. They were 'Rs. 58 (Tech Mahindra), Rs. 45.90 (Larsen & Toubro Limited) and Rs. 20 (consortium of WL Ross funds) per share'.[50]

Venturbay Consultants, a company controlled by Tech Mahindra, agreed to buy 31 per cent of the enlarged share capital at Rs 58 a share, putting Rs 1,756 crore into Satyam, which the company converted at about $351 million at a rate of Rs 50 to the dollar. It would then have to offer to buy at least a further 20 per cent from other shareholders.[51] Karnik said the result 'signals a new stage for the Company in its progress towards stabilization and growth'. He hoped it would reassure customers, 'who continue to be happy with Satyam's excellent service delivery'.[52] That was a claim by the chairman of a company being sold, but the bids themselves were evidence of a kind. The process, the company noted, 'has reached this significant stage within three months of the new Board's first meeting'.[53]

The speed of the sale says something about what an Indian services firm was. Its accounts had been falsified for years and its founder had been arrested. Yet within three months two large Indian companies offered real money for control of it. The board told the press that Satyam held about 450 acres of land and that two Hyderabad campuses had been valued at an average of Rs 1,700 crore.[54] But it was the engineers who kept the clients, and the clients who made the company worth buying.

The auditors

Satyam's auditors were Indian affiliates of PricewaterhouseCoopers (PwC), and the 20-F carried the consent of Price Waterhouse as its independent registered public accounting firm.[55] In April 2011 the SEC set out what it found they had done. Five Indian affiliates, which it called PW India, had 'repeatedly' conducted deficient audits of Satyam and enabled 'a massive accounting fraud to go undetected for several years'.[56] They agreed to settle and pay 'a $6 million penalty, the largest ever by a foreign-based accounting firm in an SEC enforcement action'. They also agreed not to take new American clients for six months, to train their staff, to revise their audit procedures and to appoint an independent monitor.[57]

The core finding was simple. When an auditor wants to know whether a company really holds the cash it reports, it writes to the banks and asks them to confirm the balances directly to the auditor. According to the SEC's order, PW India staff 'failed to conduct procedures to confirm Satyam's cash and cash equivalent balances or its accounts receivables'. They 'routinely relinquished control of the delivery and receipt of cash confirmations entirely to their audit clients and rarely, if ever, questioned the integrity of the confirmation responses they received from the client by following up with the banks'.[58] The SEC found that this was not limited to Satyam but was 'indicative of a much larger quality control failure throughout PW India'.[59] Robert Khuzami, the SEC's director of enforcement, said: 'PW India violated its most fundamental duty as a public watchdog'.[60]

These were findings in a settled case, not the verdict of a trial. The same day, Satyam itself, now under new management, agreed to settle the SEC's fraud charges, pay a $10 million penalty and carry out internal reforms.[61]

SEBI's own case against the auditors took nine years. In January 2018 it barred the firms practising under the Price Waterhouse name from certifying the accounts of listed companies for two years, and the two engagement partners on Satyam, S. Gopalakrishnan and Srinivas Talluri, for three. It ordered the partners and the Bangalore firm to disgorge Rs 13.09 crore in fees, with interest.[62] On 9 September 2019 the Securities Appellate Tribunal quashed the bans. 'In the instant case, mens rea was not proved,' it held. 'There is no finding that the auditors or the audit firms had knowledge or had intention to connive or collude with the management in the fabrication of the books of account of the company.' It did find a professional lapse, above all the failure to seek direct confirmation of bank balances, which 'amounted to negligence', and it upheld the disgorgement.[63] SEBI appealed. In November 2019 the Supreme Court issued notice and stayed the part of the ruling that limited SEBI's power to bar auditors; no decision

on that appeal has been found.[64] The tribunal was ruling on SEBI's powers under the securities laws; the criminal charges against the two partners were a matter for the criminal courts.

Findings and punishment

The Indian regulators moved more slowly. SEBI passed a final order on 15 July 2014 against Ramalinga Raju, Rama Raju and three officials: V. Srinivas, the former chief financial officer; G. Ramakrishna, a vice-president of finance; and V.S. Prabhakara Gupta, the head of internal audit. It held them liable for fraud and insider trading and barred them from the securities market for fourteen years.[65] On 12 May 2017 the Securities Appellate Tribunal upheld SEBI's findings on the merits but sent back the amounts and the length of the bars.[66] Its summary of SEBI's investigation is the clearest statement in the record of how the fraud worked: 'Investigation carried out by SEBI revealed that Ramalinga Raju as Chairman and Rama Raju as, MD of Satyam were instrumental in creating fictitious invoices, fictitious receipts etc. on the basis of which fictitious monthly bank statements were prepared'.[67]

SEBI's fresh order on the officials, in October 2018, recorded that the tribunal had traced their knowledge of the fraud to meetings with the Raju brothers from about 2001, at which, among other things, 'the need to show inflated results to attract customers/employees' was discussed.[68] The penalties in that order, and in a companion order of November 2018 on the Raju family, did not stand for those who appealed: on 2 February 2023 the tribunal set both aside and sent the amounts and bars back to SEBI.[69] SEBI's new order of 30 November 2023 barred the two brothers from the market until 14 July 2028 and ordered six parties, among them two of the officials, a third brother and the family company, to disgorge about Rs 624 crore in all (Rs 6.24 billion, about $76 million at the 2023 average rate). Because the brothers' and the two officials' appeals against the 2017 ruling were still pending in the Supreme Court, it was to take effect only from a date that court would direct.[70]

The criminal case ran on a separate track. The three charge-sheets of the Central Bureau of Investigation (CBI), India's federal police agency, were tried together, and on 9 April 2015 the XXI Additional Chief Metropolitan Magistrate-cum-Special Sessions Judge at Nampally, in Hyderabad, convicted ten men and sentenced each to seven years' rigorous imprisonment.[71] They were the Raju brothers; B. Suryanarayana Raju, a director of SRSR Advisory Services; V. Srinivas; the two Price Waterhouse partners, Gopalakrishnan and Talluri; and four former employees of the finance department, among them G. Ramakrishna.[72] Reports of the fines differ.[73]

After a dispute over which court should hear their appeals, settled by the Hyderabad High Court on 30 April 2015 in favour of the Metropolitan Sessions Judge at Nampally, that court on 11 May suspended all ten sentences and granted bail pending the appeals.[74] The CBI, by its own later account, appealed for heavier sentences, and in June 2022 it told the Telangana High Court that the brothers' appeals and its own were still pending.[75] No decision on any of the appeals had been found by October 2026, and in February 2025 the tax department told an income-tax tribunal that no order acquitting Raju existed.[76] Each of the ten, the two auditors among them, was therefore convicted by the trial court in April 2015 and had his sentence suspended in May 2015 pending appeal; for the brothers, the appeals were still undecided in June 2022, the latest record of their status.

A number for every resident

On 28 January 2009, three weeks after Raju's letter, the Planning Commission issued a notification creating the Unique Identification Authority of India (UIDAI).[77] The project had a longer history, which the Supreme Court later set out from government records. In March 2006 the Department of Information Technology had approved a project called 'Unique Identification for BPL Families', for households below the poverty line, to be carried out by the National Informatics Centre, the government's own computing agency.[78] The notification of January 2009 made the new authority an attached office of the Commission, and the Finance Minister approved Rs 147.31 crore for the first phase of enrolment.[79]

Then the state hired from the industry. By a notification of 2 July 2009, 'Mr. Nandan Nilekani was appointed as the Chairman of UIDAI for an initial tenure of five years in the rank and status of a Cabinet Minister. He assumed charge on July 24, 2009.'[80] Nilekani was a co-founder of Infosys and had been a director since 1981; from 1981 to 1987 he had managed its marketing and development in the United States.[81] When the Supreme Court decided the case about the system in 2018, Justice A.K. Sikri began the majority judgment with a line from the project's own message: 'It is better to be unique than the best.' Aadhaar, he wrote, 'means, in English, "foundation" or "base"'.[82]

Building it

The first part of the agency to start work was not in Delhi. 'Appropriately, the Technology Centre was the first unit of UIDAI to start functioning in 2009 in Bangalore,' its director general, Vijay Madan, later wrote in the agency's account of its own architecture.[83] *Forbes India*, which profiled the team in November 2010, described a hybrid: civil servants headed the departments and took the policy decisions, aided by experts from the private sector. Nilekani invited Srikanth Nadhamuni, an engineer who had spent fifteen years in Silicon Valley and with whom he had founded a non-profit in 2003 to help municipalities use IT, to head the technology centre. Raj Mashruwala, who had been ahead of Nilekani at IIT Bombay and had founded companies in manufacturing software in the United States, expected to be needed for three months at most. He ended up renting an apartment in Bangalore that doubled as the technology centre in its early days. Pramod Varma, 'an old Infosys hand', was the project's chief software architect.[84] The agency's own document later named Nadhamuni, Mashruwala, Varma, Sanjay Jain, Vivek Raghavan and Jagdish Babu among 'a band of dedicated domain experts' who had worked with the centre.[85]

They worked on unusual terms. By the magazine's count, of about 1,000 people who had applied in the project's first sixteen months, 23 had come as unpaid volunteers, 14 on sabbatical from their jobs, and '23 took massive pay cuts to join the Project Management Unit'; another 140 had come from the government. Because the volunteers drew no salaries, the magazine noted, the project had started before its budgets were sanctioned.[86] An industry that had sold its engineers' time by the hour to foreign clients was here lending some of them to the state for less, or for nothing. The magazine reported that MindTree, a Bangalore software firm, had built the enrolment software.[87] The difference between the two worlds was put plainly by K. Ganga, the deputy director general for finance, who came from the government's audit service: 'If an Infosys engineer wrote a program for Burger King, he isn't answerable to the public. I, on the other hand, have to answer parliamentary questions.'[88]

The architects described their aims in the industry's terms. 'The primary technical requirements of the project were of scale and vendor neutrality at all levels,' Regunath Balasubramanian, its principal architect, told a trade magazine in 2011.[89] The agency's document said the system was 'entirely built using open source components'.[90] Nadhamuni later called it 'a massive technology challenge, doing a 1000 trillion biometric matches per day'.[91] What it collected is recorded in Justice D.Y. Chandrachud's opinion in the 2018 case. Besides name, date of birth, gender, address and the names of parents or spouse, every resident would give 'All ten fingerprints, photograph and both iris scans'.[92]

The scheme was launched on 29 September 2010 at Tembhli, a village in Nandurbar district of Maharashtra, 'from where first Aadhaar was issued', according to the Comptroller and Auditor General (CAG), the government's auditor.[93] 'On 29 September 2011, the Aadhaar project completed one year. An announcement was made of the generation of ten crore enrolments and of more than 3.75 crore Aadhaar numbers', that is, 100 million enrolments and 37.5 million numbers.[94] For seven years the project rested on the Planning Commission's notification of 2009, which, Chandrachud pointed out, made no reference to biometrics at all. The Aadhaar Act came only in 2016, passed as a Money Bill, a form of legislation in which the upper house's part is limited.[95] By the time it was notified, UIDAI had generated about one billion numbers.[96] Twelve years after it was created, the agency itself was small. At the end of March 2021 it had 130 staff at its Delhi headquarters and 219 at its regional offices, and the work 'was being carried out by officers and staffs mostly either on deputation or from outsourced agencies'.[97]

The right to privacy

The challenge came early. In 2012, while Aadhaar still rested only on an executive notification, Justice K.S. Puttaswamy, a retired judge, and Pravesh Khanna filed a petition in the Supreme Court arguing that the scheme violated the right to privacy of 'innumerable citizens' under Article 21 of the Constitution, which protects life and personal liberty.[98] Two older decisions of the Court had been read as holding that the Constitution did not protect a right to privacy, and the question was referred to nine judges.[99] On 24 August 2017 they answered it unanimously: 'The right to privacy is protected as an intrinsic part of the right to life and personal liberty under Article 21 and as a part of the freedoms guaranteed by Part III of the Constitution.'[100] The judgment did not decide whether Aadhaar was lawful. It set the test that Aadhaar would have to pass.

Twenty-sixth of September, 2018

The Aadhaar case itself was decided on 26 September 2018 by five judges. Sikri wrote for Chief Justice Dipak Misra, Justice A.M. Khanwilkar and himself. Justice Ashok Bhushan wrote a separate concurring judgment. Chandrachud dissented.[101]

During the hearings the agency's chief executive, Dr Ajay Bhushan Pandey, had explained its engineering to the judges in a PowerPoint presentation.[102] The Authority claimed that authentication was 'purpose blind', returning only a yes or a no, and that the biometric system 'provides high accuracy of over 99.86%'.[103] Its own figures showed the other side. Of about 616 million Aadhaar numbers used for fingerprint authentication, 6.00 per cent were recorded as failed; for iris authentication the figure was 8.54 per cent. Failures, it said, 'do not mean exclusion or denial from subsidies, benefits or services', because agencies were obliged by law to provide other ways of establishing identity.[104] The CAG later found that 'nearly 4.75 Lakh duplicate Aadhaar numbers were cancelled as of November 2019', about 475,000, and 'instances of issue of Aadhaars with the same biometric data to different residents indicating flaws in the de-duplication process'.[105]

The majority upheld the Act and its central purpose, the use of Aadhaar to deliver subsidies and benefits paid for from the Consolidated Fund of India, and held that it had been validly passed as a Money Bill.[106] It then drew limits. School admission could not require Aadhaar, and 'CBSE, NEET, JEE, UGC etc. cannot make the requirement of Aadhaar mandatory as they are outside the purview of Section 7 and are not backed by any law.'[107] JEE is the Joint Entrance Examination for the engineering colleges. The Court struck down the provision that let 'any body corporate or person' use Aadhaar under a contract, which would have enabled, it said, 'commercial exploitation of an individual biometric and demographic information by the private entities'.[108] The compulsory linking of every bank account and every mobile number to Aadhaar went too.[109]

Chandrachud would have struck the Act down. He held that it was not a Money Bill, and that the programme violated 'essential norms pertaining to informational privacy, self-determination and data protection'.[110] Much of his opinion is about people the system failed. 'For persons, who are engaged in manual labour, and persons who are disabled or aged, fingerprints actually cannot be captured by biometric devices.'[111] He concluded: 'Dignity and rights of individuals cannot be based on algorithms or probabilities. Constitutional guarantees cannot be subject to the vicissitudes of technology.'[112] The majority recorded the government's assurance that no eligible person would lose a benefit because a fingerprint failed to match, and treated such failures as matters of the system's working rather than its validity.[113] Both sides accepted that the system made mistakes. They differed over whether a constitutional system could rest on a machine that was right most of the time.

Two lessons

The two halves of this chapter look unconnected: a private fraud in Hyderabad, a public database in New Delhi. They are connected by what the Indian software industry had become by 2009. It was large enough that its largest failure needed the state to rescue it, and capable enough that the state turned to it for a project of national scale.

Aadhaar lent the industry's people to a new client. For thirty years Indian programmers had built systems for foreign companies, under foreign contracts and foreign law. Here engineers from the same industry, some of them on reduced pay or none, built a system for the Indian state, meant for every resident. The Supreme Court's judgments of 2017 and 2018 were the first time Indian institutions had to decide, in detail, what Indian software could be made to do to Indian citizens.

Satyam showed where the value of an Indian services firm lay. By SEBI's investigation the books had been inflated from 2001 to 2008, and in the last quarter before the letter, by Raju's own figures, the real operating margin was 3 per cent of revenue.[114] Yet the work went on. Engineers like Mastanamma expected to go on servicing the clients, the new chairman said the customers were still happy with the service, and within three months Tech Mahindra bid Rs 58 a share for control.[115] What the bidder paid for was what the 20-F had described: some 50,000 associates recruited from the colleges, schooled for twelve weeks and placed on clients' work, nearly 10,000 of them at clients' sites abroad.[116] That is the bargain this book follows. The firm sold its engineers' time, so its worth lay in the people it could put in front of a client and the clients who would take them, and because revenue grew with headcount, reported growth was itself something to sell. SEBI's 2018 summary of the tribunal's findings listed 'the need to show inflated results to attract customers/employees' among the matters the officials had discussed with

the Raju brothers.[117] That suggests at least one thing the fiction was for. The books failed and the delivery machine did not, because the clients had never been buying the books. They had been buying the people.

The question the next chapter takes up follows from both. An industry that could run a client's systems so reliably that a fraud at the top did not stop them, and that could lend its people to build a national identity platform, had still produced very few software products of its own. Chapter 17 asks why.

Notes to Chapter 16. Fraud and foundation (2008-2018)

[1] Raju, letter to the Satyam board (2009).

[2] Raju, letter to the Satyam board (2009); Reserve Bank of India, *Handbook of Statistics on Indian Economy* (calendar-year 2008 average, Rs 43.50 per US dollar).

[3] Raju, letter to the Satyam board (2009); Satyam Computer Services, 6-K of 7 January 2009; 6-K of 9 January 2009, Ex. 99.3 (NYSE: SAY).

[4] Satyam Computer Services, 6-K of 9 January 2009, Ex. 99.4.

[5] Satyam Computer Services, Form 20-F FY2008, p. 19.

[6] *Justice K.S. Puttaswamy (Retd.) v. Union of India* (2018), Sikri J., pp. 11, 13; Infosys, prospectus (Form 424B4, 1999), p. 55.

[7] Satyam Computer Services, Form 20-F FY2008, p. 6.

[8] Satyam Computer Services, Form 20-F FY2008, p. 19.

[9] Satyam Computer Services, Form 20-F FY2008, p. 19.

[10] Satyam Computer Services, Form 20-F FY2008, p. 32.

[11] Satyam Computer Services, Form 20-F FY2008, p. 24.

[12] Satyam Computer Services, Form 20-F FY2008, p. 44.

[13] Satyam Computer Services, Form 20-F FY2008, p. 32.

[14] Satyam Computer Services, Form 20-F FY2008, p. 48.

[15] Raju, letter to the Satyam board (2009).

[16] Raju, letter to the Satyam board (2009). Press reproductions of the letter differ from the SEC copy in small ways; the chapter quotes the SEC copy throughout.

[17] Raju, letter to the Satyam board (2009).

[18] Raju, letter to the Satyam board (2009).

[19] Raju, letter to the Satyam board (2009).

[20] Raju, letter to the Satyam board (2009).

[21] Satyam Computer Services, 6-K of 9 January 2009, Ex. 99.4.

[22] Raju, letter to the Satyam board (2009).

[23] Raju, letter to the Satyam board (2009).

[24] SEBI, final order in the matter of Satyam Computer Services (2014-2018), pp. 1-2, 14.

[25] Raju, letter to the Satyam board (2009).

[26] Raju, letter to the Satyam board (2009).

[27] Satyam Computer Services, 6-K of 7 January 2009, Ex. 99.3.

[28] Satyam Computer Services, 6-K of 7 January 2009, Ex. 99.1.

[29] Satyam Computer Services, 6-K of 9 January 2009, Ex. 99.1.

[30] Satyam Computer Services, 6-K of 9 January 2009, Ex. 99.2.

[31] Satyam Computer Services, 6-K of 9 January 2009, Ex. 99.4.

[32] Satyam Computer Services, 6-K of 9 January 2009, Ex. 99.4.

[33] Satyam Computer Services, 6-K of 9 January 2009, Ex. 99.4.

[34] SEC, press release 2011-82 (2011).

[35] Sharma, 'When Your Boss Does You In' (2009).

[36] Sharma, 'When Your Boss Does You In' (2009).

[37] Sharma, 'When Your Boss Does You In' (2009).

[38] Sharma, 'When Your Boss Does You In' (2009).

[39] Sharma, 'When Your Boss Does You In' (2009).

[40] Sharma, 'When Your Boss Does You In' (2009).

[41] Sharma, 'When Your Boss Does You In' (2009).

[42] Sharma, 'When Your Boss Does You In' (2009).

[43] Sharma, 'When Your Boss Does You In' (2009).

[44] Sharma, 'When Your Boss Does You In' (2009).

[45] Sharma, 'When Your Boss Does You In' (2009).

[46] Sharma, 'Operation Satyam' (2009) (factual news note only; the article is a declared composite and nothing in it is quoted or attributed to a person).

[47] Satyam Computer Services, 6-K of 13 April 2009, Ex. 99.2. The figure is not directly comparable with the 53,000 in Raju's letter, which was a group figure from before the departures began, and the filings do not say how many had left.

[48] Sharma, 'When Your Boss Does You In' (2009); Satyam Computer Services, 6-K of 13 April 2009, Ex. 99.1.

[49] Satyam Computer Services, 6-K of 13 April 2009, Ex. 99.1.

[50] Satyam Computer Services, 6-K of 13 April 2009, Ex. 99.2.

[51] Satyam Computer Services, 6-K of 13 April 2009, Ex. 99.1.

[52] Satyam Computer Services, 6-K of 13 April 2009, Ex. 99.1.

[53] Satyam Computer Services, 6-K of 13 April 2009, Ex. 99.1.

[54] Satyam Computer Services, 6-K of 13 April 2009, Ex. 99.2.

[55] Satyam Computer Services, Form 20-F FY2008, p. 2 (exhibit list, Ex. 15.1).

[56] SEC, press release 2011-82 (2011).

[57] SEC, press release 2011-82 (2011).

[58] SEC, press release 2011-82 (2011).

[59] SEC, press release 2011-82 (2011).

[60] SEC, press release 2011-82 (2011).

[61] SEC, press release 2011-82 (2011).

[62] Securities Appellate Tribunal, *Price Waterhouse & Co. v. SEBI* (9 September 2019), para. 2 (summary of the SEBI order of 10 January 2018); para. 5 (Gopalakrishnan the engagement partner for 2000-07, Talluri for 2007-08); para. 133 ('nine long years' from the show-cause notices of 2009).

[63] Securities Appellate Tribunal, *Price Waterhouse & Co. v. SEBI* (9 September 2019), paras. 51, 136, 139-140.

[64] *LiveLaw*, 'SC Issues Notice On SEBI's Challenge' (2019) (*LiveLaw*, 22 November 2019; bench of Arun Mishra and Indira Banerjee JJ.). No later order of the Supreme Court has been found.

[65] SEBI, final order in the matter of Satyam Computer Services (2014-2018), pp. 1-2.

[66] SEBI, final order in the matter of Satyam Computer Services (2014-2018), p. 1.

[67] SEBI, final order in the matter of Satyam Computer Services (2014-2018), p. 2.

[68] SEBI, final order in the matter of Satyam Computer Services (2014-2018), p. 15.

[69] Securities Appellate Tribunal, *SRSR Holdings v. SEBI* (2 February 2023), paras. 1-2, 120 (Appeals 1-5 and 53 of 2019; the appellants were SRSR Holdings, B. Suryanarayana Raju, B. Rama Raju, B. Ramalinga Raju, V. Srinivas and G. Ramakrishna). The findings on the merits, upheld by the tribunal in 2017, were not reopened. V.S. Prabhakara Gupta was not a party to these appeals, and no later order on him has been located.

[70] SEBI, order in the matter of Satyam Computer Services (30 November 2023), para. 8, p. 5 (the appeals of G. Ramakrishna, V. Srinivas and the two brothers 'pending as on date'); para. 127, Table 19, p. 87 (Rs 624,09,82,520 in all, of which Rs 20,43,46,875 from each brother; the others were B. Suryanarayana Raju, SRSR Holdings, Srinivas and Ramakrishna); para. 128, Table 20, pp. 87-88 (restraint till 14 July 2028 for the brothers; no new restraint on the other four, though the brothers, Srinivas and Ramakrishna were to remain under the restraint the Supreme Court had directed until their appeals were decided); para. 129 (simple interest at 12 per cent a year from 7 January 2009); para. 131, p. 89 (effect from a date the Supreme Court directs). Reserve Bank of India, *Handbook of Statistics on Indian Economy* (2023 average, Rs 82.60 per US dollar). No decision of the Supreme Court in these appeals (C.A. Nos. 9493/2017 and connected) has been found. Checked against the SEBI order on 15 October 2026.

[71] Telangana High Court, Satyam-related orders (2022) (Telangana High Court, Crl. P. No. 13332 of 2011, order of 7 January 2022, para. 2, the petitioners' own account: three charge-sheets, joint trial as C.C. Nos. 1, 2 and 3 of 2010; W.P. No. 12595 of 2019, order of 6 June 2022, para. 3: 'The XXI ACMM-cum-Special Sessions Judge Nampally convicted the accused in the above case and sentenced them to imprisonment on 09.04.2015'); Associated Press (*Al Jazeera*) and *The Quint* on the Satyam verdict (2015) (Associated Press via *Al Jazeera*, 9 April 2015: ten convicted, seven years each); PTI, 'Satyam's Raju, 9 others get bail' (11 May 2015) (PTI, 11 May 2015: 'seven-year rigorous imprisonment sentence'). Checked against the Indian Kanoon texts on 15 October 2026.

[72] Securities Appellate Tribunal, *B. Appalanarasamma v. SEBI* (2017), minority view (Jog Singh), para. 76, listing the accused named in the CBI's charge-sheets (Talluri appears there as 'T. Srinivas').

[73] *Business Today*, 'B Ramalinga Raju Found Guilty' (2015) (Rs 5 crore each for Ramalinga Raju and Rama Raju; Rs 25 lakh each for the other eight); Associated Press (*Al Jazeera*) and *The Quint* on the Satyam verdict (2015) (Associated Press via *Al Jazeera*, 9 April 2015: 'The judge also fined Raju $806,000'; *The Quint*, 14 April 2015: 'a Rs 5.5-crore fine on Raju and his brother Rama Raju'); PTI, 'Satyam's Raju, 9 others get bail' (11 May 2015) (PTI, 11 May 2015: 'Rs 5.35 crore fine on Raju and Rama Raju, while others were fined Rs 25 lakh each'); *Gulf News*, 'Netflix stopped by court' (2020) (*Gulf News*, 2020: 'a fine of Rs 55 million on both the brothers'). No copy of the judgment in C.C. Nos. 1-3 of 2010 has been located online, so no figure is given as settled.

[74] *One Law Street*, 'High Court returns Appeals to Sessions Court' (2015) (a legal-news report that quotes the High Court's order: the appeals were to be presented 'before the Court of Metropolitan Sessions Judge, Nampally, Hyderabad, which alone has jurisdiction to entertain the appeals on all aspects'; the Sessions Court had returned them as not maintainable on 20 April 2015). The report misdates the conviction to 2000. No copy of the High Court's order has been located. PTI, 'Satyam's Raju, 9 others get bail' (11 May 2015) (PTI, carried by *Business Standard*, 11 May 2015; the same wire ran in *Business Today* and *The Quint*, so the three are one report): M. Laxman, special judge for economic offences, suspended the 'substantive and default' sentences on personal bonds and sureties, on condition that each paid one tenth of his fine within four weeks and undertook 'to cooperate for early disposal of the appeals without seeking undue adjournments'.

[75] Telangana High Court, Satyam-related orders (2022) (W.P. No. 12595 of 2019, 6 June 2022, para. 6, CBI's submission: 'The CBI filed appeal for enhancement of the sentence. The accused Nos.1 and 2 also filed appeal against the conviction', pending before the Metropolitan Sessions Judge, Nampally); *Gulf News*, 'Netflix stopped by court' (2020) (in a civil suit in 2020 Raju's counsel said the matter 'was pending in appeal'). No order deciding the appeals has been found in the Indian Kanoon database or in press reports to October 2026.

[76] Income Tax Appellate Tribunal, Hyderabad, *ACIT v. B. Ramalinga Raju* (11 February 2025), para. 27, p. 36: the Revenue 'categorically stated that no such order is in existence', and this 'remains unrebutted'. The tribunal does not say whether the appeals were then pending. Checked on 15 October 2026.

[77] *Justice K.S. Puttaswamy (Retd.) v. Union of India* (2018), Sikri J., pp. 11, 25; Comptroller and Auditor General, *Performance Audit on Functioning of UIDAI*, Report No. 24 (2021), p. 1 (creation on 28 January 2009 as an attached office of the Planning Commission).

[78] *Justice K.S. Puttaswamy (Retd.) v. Union of India* (2018), Sikri J., p. 8.

[79] *Justice K.S. Puttaswamy (Retd.) v. Union of India* (2018), Sikri J., pp. 11-12.

[80] *Justice K.S. Puttaswamy (Retd.) v. Union of India* (2018), Sikri J., p. 13.

[81] Infosys, prospectus (Form 424B4, 1999), p. 55.

[82] *Justice K.S. Puttaswamy (Retd.) v. Union of India* (2018), Sikri J., p. 3.

[83] UIDAI, *Aadhaar Technology & Architecture* (2014), p. 7 (message from the Director General).

[84] Jayashankar and Ramnath, 'UIDAI: Inside the World's Largest Data Management Project', *Forbes India* (2010) (*Forbes India*, 29 November 2010; the magazine's own account of the team, checked against the online text on 16 October 2026).

[85] UIDAI, *Aadhaar Technology & Architecture* (2014), pp. 9-10 (from the editor's desk, Ashok Dalwai, Deputy Director General, Technology Centre).

[86] Jayashankar and Ramnath, 'UIDAI: Inside the World's Largest Data Management Project', *Forbes India* (2010).

[87] Jayashankar and Ramnath, 'UIDAI: Inside the World's Largest Data Management Project', *Forbes India* (2010).

[88] Jayashankar and Ramnath, 'UIDAI: Inside the World's Largest Data Management Project', *Forbes India* (2010).

[89] Sharma, 'Aadhaar: A Testimony to Success of FOSS in India!', *Open Source For You* (2011).

[90] UIDAI, *Aadhaar Technology & Architecture* (2014), p. 80.

[91] Ranganna, 'Tech Guru, Srikanth Nadhamuni', *Man's World India* (2016).

[92] *Justice K.S. Puttaswamy (Retd.) v. Union of India* (2018), Chandrachud J., p. 198.

[93] Comptroller and Auditor General, *Performance Audit on Functioning of UIDAI*, Report No. 24 (2021), p. 2; UIDAI, *Aadhaar Technology & Architecture* (2014), p. 7 (first Aadhaar letter handed over at Nandurbar on 29 September 2010).

[94] *Justice K.S. Puttaswamy (Retd.) v. Union of India* (2018), Chandrachud J., p. 199.

[95] *Justice K.S. Puttaswamy (Retd.) v. Union of India* (2018), Chandrachud J., p. 415; Sikri J., p. 94.

[96] *Justice K.S. Puttaswamy (Retd.) v. Union of India* (2018), Chandrachud J., p. 416 (citing the Press Information Bureau).

[97] Comptroller and Auditor General, *Performance Audit on Functioning of UIDAI*, Report No. 24 (2021), Executive Summary, p. v.

[98] *Justice K.S. Puttaswamy (Retd.) v. Union of India* (2018), Sikri J., p. 7 (Writ Petition (Civil) No. 494 of 2012).

[99] *Justice K.S. Puttaswamy (Retd.) v. Union of India* (2017), order of the Court, para. 2.

[100] *Justice K.S. Puttaswamy (Retd.) v. Union of India* (2017), order of the Court, paras. 1-2 (lead judgment by Chandrachud J. for Khehar CJ, Agrawal J., Nazeer J. and himself; separate judgments by Chelameswar, Bobde, Sapre, Nariman and Kaul JJ.).

[101] *Justice K.S. Puttaswamy (Retd.) v. Union of India* (2018), Sikri J., pp. 3, 567; Bhushan J., paras 392-393.

[102] *Justice K.S. Puttaswamy (Retd.) v. Union of India* (2018), Sikri J., p. 56.

[103] *Justice K.S. Puttaswamy (Retd.) v. Union of India* (2018), Sikri J., pp. 55, 57, 60.

[104] *Justice K.S. Puttaswamy (Retd.) v. Union of India* (2018), Sikri J., p. 63.

[105] Comptroller and Auditor General, *Performance Audit on Functioning of UIDAI*, Report No. 24 (2021), p. 19, para. 3.2.2, and Executive Summary, p. vi (audit period 2014-15 to 2018-19, p. 7, para. 2.1). The duplicate figure is UIDAI's own, 'As per information provided by UIDAI Tech Centre'. Against the 129.04 crore (1.29 billion) numbers generated by March 2021 (p. 14, para. 3.1.2), the share was small.

[106] *Justice K.S. Puttaswamy (Retd.) v. Union of India* (2018), Sikri J., p. 565.

[107] *Justice K.S. Puttaswamy (Retd.) v. Union of India* (2018), Sikri J., pp. 555-556.

[108] *Justice K.S. Puttaswamy (Retd.) v. Union of India* (2018), Sikri J., pp. 559-560.

[109] *Justice K.S. Puttaswamy (Retd.) v. Union of India* (2018), Sikri J., pp. 512, 567.

[110] *Justice K.S. Puttaswamy (Retd.) v. Union of India* (2018), Chandrachud J., pp. 154, 469.

[111] *Justice K.S. Puttaswamy (Retd.) v. Union of India* (2018), Chandrachud J., pp. 375-376, para. 269.

[112] *Justice K.S. Puttaswamy (Retd.) v. Union of India* (2018), Chandrachud J., p. 376, para. 269.

[113] *Justice K.S. Puttaswamy (Retd.) v. Union of India* (2018), Sikri J., pp. 105-106.

[114] Securities Appellate Tribunal, *SRSR Holdings v. SEBI* (2 February 2023), para. 3; Raju, letter to the Satyam board (2009).

[115] Sharma, 'When Your Boss Does You In' (2009); Satyam Computer Services, 6-K of 13 April 2009, Ex. 99.1 (Karnik's statement, a claim by the chairman of a company being sold) and Ex. 99.2.

[116] Satyam Computer Services, Form 20-F FY2008, pp. 6, 24, 44.

[117] SEBI, final order in the matter of Satyam Computer Services (2014-2018), p. 15. The penalties in this order were set aside in 2023 (Securities Appellate Tribunal, *SRSR Holdings v. SEBI* (2 February 2023), para. 120); the tribunal's 2017 findings that it summarises were not reopened.

Chapter 17. The product question (1986-2024)

In 1992 Rajesh Hukku, then head of Citicorp Overseas Software, the bank's in-house programming unit in India, spun the unit off from the bank as a company of its own. Fifteen years later, in an interview with Ravi Aron of the Wharton School published in March 2007, he remembered one of the Citibank chiefs asking him, at the start, a blunt question: 'Why should I start the one-thousand-and-first software company in India?'[1] His answer, as he recalled it, was that the new firm would be the first to be wholly focused on understanding its customers' business, so that no banker could say of its staff, 'These guys are great techies, but they don't understand my business.'[2]

In Hukku's account, Citi had decided that writing software for sale was not its business. It invested $400,000 in the venture anyway, and the unit became Citicorp Information Technology Industries Ltd. (CITIL), later renamed i-flex solutions.[3] The bank, as he remembered it, liked the team but would give the new firm no 'management mind-share', and asked one thing more: 'Please don't come back asking for business from Citibank.'[4] In hindsight, he said, it was the best combination, 'because you get the money and you don't get controlled'.[5]

The question the Citibank chief put to Hukku is the question of this chapter, turned round. By 1992 India already had hundreds of software companies, and almost all of them sold the same thing: the time of Indian engineers, billed to foreign clients by the hour, the month or the project. A software product is something different. It is a program that a firm owns and sells many times over, so that the cost of writing it is paid once and the revenue keeps coming. The country that became the world's largest supplier of software labour built remarkably few products of its own. Why that happened, and what changed when it did, is the subject here.

The question matters beyond national pride. A firm that owns a product keeps the gains when its engineers become more productive. A firm that sells hours must negotiate with its clients over who keeps them. That difference decides how the tools of artificial intelligence, which make a programmer's hour more productive, land on the two kinds of firm.

An accounting package for small business

The longest-lived Indian software product grew from the home market. According to the history page on its own website, Tally Solutions began in 1986 as 'Peutronics Financial Accountant', founded by Bharat Goenka, 'inspired by his father, Shyam Sundar Goenka's vision to simplify business operations'.[6] The company says the firm grew 'from an accounting package for SMEs', small and medium enterprises, into business software for firms of all sizes.[7] Its timeline dates 'code-less accounting' to Tally 3.0 in 1988 and the formal establishment of Tally Solutions Private Limited to 1991.[8]

What the timeline shows clearly is how closely the product followed the Indian state's tax law rather than any foreign client's needs. Tally 6.3, in 1999, added support for several currencies and more advanced inventory management. Tally 7.2, in 2005, 'introduced VAT compliance, aiding businesses with tax regulations'. In 2017 the company launched software compliant with the new Goods and Services Tax (GST), and in 2020 a new generation of the product, TallyPrime.[9] Its buyers were small and medium Indian businesses, and it reached them not through onsite consultants but through a network of partners. It says it is now trusted by more than 2.5 million businesses.[10]

Tally's model was the opposite of the export firms' in almost every respect. It sold a packaged product, not people. Its customers were Indian, not American. It earned in rupees from small businesses rather than in dollars from large ones. Its survival over four decades is evidence that a domestic market for software existed through the years when the export industry was growing much faster. Its modest public profile beside Tata Consultancy Services (TCS) or

Infosys is a sign of how small that market was. Goenka himself would become a public critic of how the industry's institutions favoured services.

Selling to Africa first

i-flex took a different road, out of India and into the world, but not to the countries everyone else was going to. Hukku had studied electrical and electronics engineering at the Birla Institute of Technology and Science (BITS) at Pilani in Rajasthan, a desert campus where, he recalled, students with no television built their own radio station, and had worked for TCS, the export firm F.C. Kohli built, before joining Citicorp's in-house unit.[11]

The new company began poor, selling professional services and consulting and spending as little as it could.[12] It could not afford to chase clients in Britain or America. 'We decided early on that we were not going to go to the U.K. or the U.S. because we couldn't even afford the airfare. We said, "Let's go to Africa. That's all we can handle right now."'[13] The firm took the rights to an older Citicorp banking program called MicroBanker, improved it and sold it to African banks. By 1995, by Hukku's account, MicroBanker was number one in the league tables, sold to 117 banks, most of them in Africa.[14] He remembered a billboard in Nairobi in those years: 'Wanted. IT people, MicroBanker experience essential.'[15]

The services work and the old product paid for a new one. In Hukku's words, 'The money we were making on the services, consulting, MicroBanker, etc., was used to fund the creation of our dream product', which was FLEXCUBE, launched in 1997, an integrated banking package covering consumer, corporate, investment and internet banking and asset management.[16] This is the most important detail in his account for the argument of this chapter. India's best-known early global product was not built in spite of services. It was paid for by them.

Hukku's way of teaching his engineers the business they were automating was itself unusual. 'Right from the beginning, we said 10% to 15% of the staff have to be bankers, because how else will I learn the domain?'[17] The bankers came not from Wall Street but from Indian banks, among them Bank of India and the Reserve Bank of India, and many of them were not computer science graduates. The company asked them to teach.[18] In an industry whose training turned graduates into programmers for foreign clients' systems, i-flex was turning programmers into people who understood banking.

Two of the company's choices ran against the industry as a whole. One concerned markets. Hukku saw that the services business depended on America. 'We said that's a very big limitation for the whole industry to be dependent on one country.'[19] By 2007 i-flex claimed customers in more than 125 countries.[20] The other choice concerned the Y2K boom, the rush to fix two-digit dates in old programs before the year 2000, which brought Infosys about a fifth of its revenue at its height (Chapter 11). 'We're perhaps the only IT company from India who made $0 from Y2K,' Hukku claimed, looking back.[21]

FLEXCUBE climbed quickly. On Hukku's reading of an annual league table compiled by International Banking Systems, a British research firm, it was 16th of 60 or 70 core banking products in 1997, second in 1999 and first 'since 2002'.[22] By early 2007 it had been sold to more than 300 banks in 120 countries.[23] i-flex went public in 2002 on the National Stock Exchange and the Bombay Stock Exchange.[24] By the interviewer's account its shares doubled while almost every other technology stock fell in the dot-com bust, and Hukku put that down to the business model: 'We were a balanced product and services company with a globally hedged customer base.'[25]

The ending was a sale. In 2005 Oracle bought Citicorp's stake: 42 per cent for $592 million, in Wharton's introduction to the interview, or 41 per cent, held by Citigroup Venture Capital International, in Oracle's own announcement of 2 August 2005, which stated no price.[26] Either way, Hukku said, 'No one at the time, including me, ever dreamed that the $400,000 they put in would become almost $600 million in 13 years.'[27]

Hukku was clear that i-flex had never been a pure product company. 'We are a product company. We're also a services company,' he said.[28] As with Cognizant (Chapter 15), a foreign company's captive centre in India had become an independent firm, but this one sold a product, and its largest exit was to an American buyer.

A product inside the factory

Infosys, the archetype of the services firm, built a banking product of its own. Its annual report to American investors for the year to March 2002 said that, besides its IT services, it marketed and licensed 'our own centralized bank automation software application, known as FINACLE(TM)'.[29] It described Finacle as 'a fully web-enabled, integrated core banking solution that addresses the retail, corporate and trade finance activities of a bank'.[30]

The numbers show where the product sat in the firm. In the year to March 2009 Infosys told investors that it had earned 3.9 per cent of its total revenue from licensing software products, against 3.6 per cent the year before. 'The markets for our primary suite of software products which we call Finacle TM are competitive,' it added, and the costs of developing new products 'may not be recouped'.[31] Three years later the share was a little higher but still small: 4.6 per cent in the year to March 2012.[32]

The product was also sold with services wrapped round it. 'We also provide a core banking software solution, Finacle TM, for the banking industry and provide customization and implementation services around this solution,' the company said in 2009.[33] By March 2012, Infosys said, Finacle was used by 160 banks in 78 countries.[34]

One detail of the 2009 report shows how a services firm thought about its product. The director responsible for Finacle, T.V. Mohandas Pai, who had been chief financial officer from 1994 to 2006, was listed as head of 'Administration, Education and Research, Finacle, Human Resources Development' and the Infosys Leadership Institute.[35] The banking product shared a portfolio with the training machine. Finacle, like FLEXCUBE, was never a pure product business. It was a product nested inside a services company, contributing less than a twentieth of revenue.

A round table of its own

By 2013 some of the product companies had decided that the industry's lobby did not speak for them. The National Association of Software and Service Companies (Nasscom), the industry's lobby group, had represented the industry for about 25 years, as it grew from $100 million in 1992 to $108 billion in 2012.[36] In March 2013 Knowledge at Wharton reported that a group of 30 software product companies had come together to form the Indian Software Product Industry Round Table, known as iSPIRT, a think-tank for the products segment.[37]

It was a spin-off rather than a walk-out. The round table grew out of Nasscom's own annual product conclave, and its founders were Nasscom members, among them Goenka of Tally and Sharad Sharma.[38] Goenka told the business daily *Business Standard*: 'At Nasscom, the product companies do not have the same weightage as the services companies. The policy interventions Nasscom is trying to make are more in favor of the services industry.'[39] Sharma said: 'We believe that with this think-tank, we will get a louder voice in policy making and other initiatives.'[40]

Nasscom played the split down. Its chairman, N. Chandrasekaran, who was also chief executive of TCS, said: 'When you are on a growth paradigm, you cannot do everything that you want to do. I don't see any conflict.'[41] The association had set up an expert committee the previous July, chaired by N.R. Narayana Murthy, co-founder of Infosys, to chart a roadmap for the industry. Murthy said the industry should aim to reach $300 billion by 2020.[42] Of that, the committee suggested, the internet and mobile could account for $100 billion, the domestic market $75 billion, and software products 'around US$10 billion'.[43] Products were on the list, at about one-thirtieth of the target.

The round table did lobby for its members. In January 2019 it wrote to the prime minister on behalf of start-ups about the 'angel tax', asking that investments of up to Rs 10 crore a year from resident investors in recognised start-ups be exempt, and claiming that the number of angel investors had fallen 48 per cent since 2015.[44] But by then it was best known for something else. It describes itself as 'a non-profit think tank which champions digital public goods', and maintains the website of India Stack, a set of public digital services built around Aadhaar.[45] *Inc42*, a start-up news site, described it in 2023 as 'building the India Stack since 2013'.[46] That closeness to the state drew criticism. In 2017 Sharma, a member of its governing council, admitted to anonymously trolling critics of Aadhaar on Twitter and apologised.[47] The round table formed to speak for product companies had become, above all, an advocate of public platforms.

The village office

One Indian firm set out to make products from the start. Sridhar Vembu graduated from the Indian Institute of Technology Madras in 1989, took a doctorate at Princeton and worked as a wireless systems engineer at Qualcomm in San Diego. In 1996 he co-founded AdventNet with members of his family and friends, among them Tony Thomas, then at Bell Labs.[48] The company's own timeline begins: 'AdventNet Inc. is born in New Jersey, USA. It all begins with Network Management.'[49] Its first customers, Vembu recalled in 2010, were makers of network equipment such as Cisco.[50] He told *Forbes India* in 2020 what he had seen at the start: 'in India we have perfect talent but we produce almost zero software products.'[51]

On the company's account, the products followed one another: a sister brand, ManageEngine, in 2003, then online applications under the Zoho name, starting with a word processor and a customer relationship management (CRM) program in 2005. In 2009 AdventNet renamed itself Zoho Corporation.[52] It has never taken outside capital, by its own and its founder's statements. 'We have never taken any outside money,' Vembu said in 2010, and the company's website still says it has 'never taken money from investors'.[53] *Forbes India* reported that in 2019 Zoho made profits of Rs 516 crore on revenue of Rs 3,410 crore (Rs 34.1 billion, about $480 million at that year's average rate).[54]

In 2011, the company says, it opened an office in Tenkasi, a district in southern Tamil Nadu, and in 2018 another in Renigunta in Andhra Pradesh.[55] By 2020, *Forbes India* reported, 500 of its 9,300 employees, 8,800 of them in India, worked from those two rural offices, and Vembu himself had moved to Tenkasi in October 2019. 'I always thought that people migrating from villages to cities is not a good idea,' he said.[56]

Zoho also built its own school. Zoho University, which the company dates to 2005 and describes as 'an experiment in providing an alternative to the conventional college education', paid its students a stipend rather than charging fees, and by 2020, *Forbes India* reported, 875 of the firm's 9,300 staff had come through it.[57] Vembu said that 15 to 20 per cent of Zoho's engineers had no engineering degree.[58] One graduate, Saran Babu Paramasivam, told *Forbes India* that in 2005 he could not operate a computer; by 2020 he was a senior product manager, and he said 'a degree is not important anymore'.[59] Where the services firms recruited engineering graduates by the hundred thousand and billed their time, Zoho trained many of its own engineers and put their work into products it owned. When those engineers became more productive, the gain stayed with Zoho.

The warehouse in Chennai

The next generation of Indian product companies did not sell boxes of software to shopkeepers or core banking systems to banks. They sold subscriptions over the internet, software as a service (SaaS), mostly to small businesses abroad. The best documented is Freshworks, because it listed its shares in the United States and had to explain itself to American investors.

Its founder and chief executive, Girish Mathrubootham, began his letter in the 2021 registration statement with its origins: 'We started in a 700 square foot warehouse in Chennai in 2010. Our simple idea was to create a "fresh" helpdesk in response to a poor customer service experience I had.'[60] He listed the doubts he had heard along the way. Among them were 'You can't find talent in Chennai; you can't win unless you move to Bangalore' and 'You really can't build a global SaaS company from India.'[61]

The company's model needed none of the apparatus of the services industry. 'Freshworks was global from "day one" with our first few customers coming from four different continents,' Mathrubootham wrote. 'Our initial sales came through search-driven online acquisition, powered by robust product-led growth.'[62] Customers found the software by searching the web, tried it and paid for it, without a salesman on a plane. There were no onsite engineers, no visas, no time sheets and no client billing by the hour. By mid-2021, the company said, businesses from more than 120 countries used its products, and it had about 52,500 paying customers.[63]

It also lost money as it grew. Its revenue was $172.4 million in 2019 and $249.7 million in 2020, a rise of 45 per cent, and its net losses in those years were $31.1 million and $57.3 million.[64] The work was done in India. As of July 2021, Freshworks said, it had about 4,300 employees, and about 3,800 of them were in India, 'where most of engineering, product design, sales and marketing, customer support, and general and administrative personnel are located'. Its headquarters were in San Mateo, California.[65]

The rivalry between two Chennai product firms reached an American court. On 17 March 2020 Zoho sued Freshworks in the Northern District of California, alleging the misappropriation of trade secrets and violations of the Computer Fraud and Abuse Act.[66] In October 2020 the judge refused to dismiss the amended complaint, ruling that Zoho had 'clearly met the standard for pleading trade secret misappropriation', a ruling on the pleading rather than a finding of fact.[67] In December 2021 the companies settled and Zoho dismissed the suit. Freshworks's announcement and its annual report gave no terms. Freshworks said that a now-former sales employee had 'wrongfully accessed and used Zoho confidential information relating to sales leads', and that he had not acted at the company's direction.[68]

By then Freshworks was a public company. On 22 September 2021 the final prospectus fixed the price: 'The initial public offering price is $36.00 per share.' The company offered 28,500,000 shares of Class A common stock for listing

on the Nasdaq Global Select Market, about $1.03 billion before any extra shares the underwriters might buy.[69]

The listing settled something that the Chennai warehouse had left open. The most celebrated Indian software-as-a-service company was, legally and financially, an American one: incorporated in Delaware, headquartered in San Mateo and listed in New York, with its engineers in India.[70] The value it created was priced in dollars on an American exchange. Like i-flex's sale to Oracle, the reward for building a product in India was realised abroad.

Money arrives, and leaves

In Hukku's account, the shortage that had mattered most in 1992 was money. 'Even in 1992, there was not enough capital,' he said in 2007. By then, he believed, the position had reversed. 'So capital is not a problem today.'[71]

Fourteen years later capital arrived on a scale far beyond anything he described, and much of it went to consumer internet businesses rather than to software products in his sense. According to figures from Tracxn, a data firm, reported by *The Week* in January 2025, Indian start-ups raised $25.7 billion in 2022, a fall of 31 per cent on the year before, which implies about $37 billion in 2021.[72] On Tracxn's count, funding then fell by close to 58 per cent in 2023, to $10.8 billion, and rose by about 2 per cent in 2024, to $11.1 billion.[73] The emblem of the bust was Byju's, an education technology company, which by mid-2024 faced insolvency proceedings brought by the Board of Control for Cricket in India over unpaid dues.[74]

Sidebar: Why was India's most-used software never sold?

By the measure of daily use, the most successful Indian software of the 2010s was not sold by a private firm at all. Aadhaar, the biometric identity number, is one case (Chapter 16). The other is the Unified Payments Interface (UPI), a system run by the National Payments Corporation of India (NPCI) that lets a person pay another directly from a bank account using a phone.

The Reserve Bank of India's statistics show how fast it grew. In the year to March 2017, the year in which UPI was introduced, it carried about 18 million transactions, worth about Rs 6,961 crore (Rs 69.6 billion).[75] Seven years later, in the year to March 2024, it carried 131.13 billion transactions, worth about Rs 199.95 lakh crore (Rs 199.95 trillion, roughly $2.4 trillion at that year's average rate).[76] All India Radio, reporting NPCI's figures, said that in December 2024 alone UPI processed about 16.73 billion transactions worth around Rs 23.25 lakh crore (Rs 23.25 trillion), with the number of transactions about 39 per cent higher than a year earlier.[77]

No private Indian software product in this chapter came near that scale of use. Tally counts its users in millions of businesses, by its own claim; Freshworks counted its paying customers in tens of thousands.[78] UPI counts its transactions in billions each month.

Part of the answer lies in who built it. India Stack's own timeline, on the website iSPIRT maintains, credits the launch of UPI in 2016 to NPCI, and the identity and electronic-signature services beside it to other public bodies.[79] The press gives the volunteers a larger part: *Inc42* counted UPI among the projects iSPIRT 'has been behind'.[80] Either way the builders came from two worlds. The institutions were public, but some of the people around them came from the software industry: Nandan Nilekani, the Infosys co-founder who chaired the identity authority (Chapter 16), was a mentor to iSPIRT.[81]

The rest of the answer, this book argues, lies in what a payments network is. Such a network is worth most when every bank can join it and every phone can use it. A private firm that built one would have had reason to charge for it or close it to rivals. A body that the banks answered to could bring them all in at once.

That is a different answer to the product question. Where private Indian firms built products, they either sold them to small Indian businesses, as Tally did, or sold them abroad and realised their value abroad, as i-flex and Freshworks did. UPI was built for Indians, used at home, and run as infrastructure rather than sold as a product. The country's most-used software was the one piece that no one could sell.

Products and people

The 'services mindset' is the most common explanation in India for why the country built so few products. It is the participants' own. Hukku said Indian software companies had followed the model F.C. Kohli pioneered at TCS in the 1970s, 'of exporting the Indian brain, or doing software services projects offshore'. Kohli had worked very hard to make it succeed, Hukku said. 'The not-so-great news was that hundreds of companies that came after copied him.

Basically it is a human tendency to take the route which is already a proven success.'[82] When Aron asked whether i-flex was 'Made in Spite of India', he replied: 'It's not in spite of India, but it's in spite of whatever was successful when we came into existence.'[83] Goenka's complaint about Nasscom, and Nasscom's own roadmap, said the same thing about the industry's institutions.[84]

These are explanations from people with a stake in them. Several structural explanations fit the same facts. One is capital, which Hukku himself named: there was not enough of it in 1992, and venture funding at scale came only in the following decades.[85] Zoho's founders, by their own account, never raised any. A second is the market. Tally's history suggests that small Indian businesses would pay for software when the state's tax rules required it, but no series of Indian spending on software over the period has been located.[86] Building a product for India was no guarantee of success either. The Simputer, a low-cost multilingual hand-held computer designed in 2000 by a group from the Indian Institute of Science and Encore Software in Bangalore (Bengaluru since 2014), received worldwide attention but was, in the words of V. Rajaraman, the historian of Indian computing, 'not a commercial success'.[87] A third is talent. The services firms offered graduates secure, well-paid jobs with training attached, and a young product firm had to compete with that. Talent also meant knowledge of the customer's trade. i-flex's answer was to hire bankers to teach its programmers banking, and Zoho's was a school of its own.[88] A fourth is the exit. The largest rewards for building products in India were realised through a foreign buyer, as with Oracle and i-flex, or a foreign listing, as with Freshworks.[89] A 'mindset' may be the participants' name for these conditions rather than a separate cause. The question stays open.

What can be measured is the difference in scale, and it cuts both ways. At the end of 2024 Freshworks had about 4,400 employees, and approximately 3,700 of them, 83 per cent, lived in India.[90] Its revenue that year was $720.4 million, up from $596.4 million in 2023 and $498.0 million in 2022.[91] In the same year Cognizant, the American services firm born from a captive centre in India, had about 336,800 employees, 241,500 of them in India.[92] The figures are given only to show orders of magnitude. Freshworks earned its revenue with a workforce less than a fiftieth the size of Cognizant's in India.

The debate in India treated products as the higher path, and on revenue per engineer they were. But the engineer factory was turning out graduates in the hundreds of thousands each year. No set of product companies of the size documented here could have employed them. The services model was not only a habit of mind. It was the model in which the industry's growth turned into mass employment, because a firm that sold hours could grow only by hiring more people to sell.

Hukku had made a prediction in 2007: 'I can guarantee you that five years from now, there will be many famous product companies growing out of India.'[93] By the early 2020s there were some, and a flood of venture money that peaked in 2021. Yet in those same pandemic years the old model did not shrink. The firms that sold engineers' hours went on hiring, in numbers that no product company could match.

Freshworks's annual report for 2024 shows what happened when the product side met the new technology. On 5 November 2024, the company said, it had committed to a plan to realign its workforce 'that resulted in a 13% reduction in force', at a cost of about $9.7 million.[94] The same report described its generative AI offerings. 'Freddy AI Agent offers always-on, autonomous, personalized resolutions to customer and employee queries,' the company told investors, while Freddy AI Copilot provided 'always-on contextual assistance' for support, marketing and sales staff, 'designed to boost productivity'.[95] Among its risks it listed the chance that competitors 'may be able to develop new AI offerings that disrupt our customers' needs and negatively impact demand for our products'.[96]

Read beside Chapter 14, the description is pointed. An Indian-built product company was selling software designed to answer customers' and employees' questions without a person, which is much of what India's call centres and IT help desks had been paid by the hour to do. Products had not escaped the AI question. They met it from the other side, as sellers of the automation that the services firms would have to absorb. It was the services boom, and what it did to the people hired in it, that shaped the lives of most young Indian engineers, and the next chapter follows them.

Notes to Chapter 17. The product question (1986-2024)

[1] Knowledge at Wharton, interview with Rajesh Hukku (Knowledge at Wharton interview, 23 March 2007; founder's own account; web page, unpaginated).

[2] Knowledge at Wharton, interview with Rajesh Hukku.

[3] Knowledge at Wharton, interview with Rajesh Hukku.

[4] Knowledge at Wharton, interview with Rajesh Hukku.

[5] Knowledge at Wharton, interview with Rajesh Hukku.

[6] Tally Solutions, 'About Us' (Tally Solutions 'About us' page; company self-description; consulted 2026-10-12).

[7] Tally Solutions, 'About Us' (Tally Solutions 'About us' page; company self-description; consulted 2026-10-12).

[8] Tally Solutions, 'About Us' (Tally Solutions 'About us' page; company self-description; consulted 2026-10-12). No independent record of the incorporation date, such as the Registrar of Companies filing, has been located.

[9] Tally Solutions, 'About Us' (Tally Solutions 'About us' page; company self-description; consulted 2026-10-12).

[10] Tally Solutions, 'About Us' (company self-description, for the partner network and the 2.5 million businesses; consulted 2026-10-12).

[11] Knowledge at Wharton, interview with Rajesh Hukku (BITS Pilani, the campus radio station and closed-circuit television system, and TCS).

[12] Knowledge at Wharton, interview with Rajesh Hukku (Hukku: 'on day one we were selling professional services, we were doing consulting, and we were also trying to survive by spending very little money').

[13] Knowledge at Wharton, interview with Rajesh Hukku.

[14] Knowledge at Wharton, interview with Rajesh Hukku (founder's account; the league table for 1995 is not named).

[15] Knowledge at Wharton, interview with Rajesh Hukku (Hukku: so many banks there ran the program that there were not enough people who knew it).

[16] Knowledge at Wharton, interview with Rajesh Hukku.

[17] Knowledge at Wharton, interview with Rajesh Hukku.

[18] Knowledge at Wharton, interview with Rajesh Hukku (bankers 'from Bank of India, from Reserve Bank of India, from smaller financial institutions'; many were 'computer savvy' rather than computer science graduates; 'You come teach us what you know about banking').

[19] Knowledge at Wharton, interview with Rajesh Hukku.

[20] Knowledge at Wharton, interview with Rajesh Hukku (company claim as stated by its founder, who also put i-flex's share of the banks in several African countries at between a quarter and nearly half).

[21] Knowledge at Wharton, interview with Rajesh Hukku (founder's retrospective claim; he added: 'We made a conscious decision to stick to the value-addition paradigm and the original principles.' No figure in i-flex's accounts for the period has been located to confirm it).

[22] Knowledge at Wharton, interview with Rajesh Hukku (International Banking Systems league table as reported by Hukku, who also put FLEXCUBE ninth in its second year).

[23] Knowledge at Wharton, interview with Rajesh Hukku (Wharton's introduction to the interview).

[24] Knowledge at Wharton, interview with Rajesh Hukku (Wharton's introduction, which also gives revenue of Rs 5.5 billion, $126 million at the rate it used, in the December quarter of the year to March 2007).

[25] Knowledge at Wharton, interview with Rajesh Hukku (the share price is the interviewer's statement, as is the remark that in 2003 *Time* magazine picked i-flex as one of 15 survivors of the crash; the explanation is the founder's).

[26] Knowledge at Wharton, interview with Rajesh Hukku (Wharton's introduction: 42 per cent, $592 million); Oracle, press release on i-flex solutions (2005) (Oracle press release, 2 August 2005: 41 per cent, held by Citigroup Venture Capital International; no price stated; consulted 2026-10-14).

[27] Knowledge at Wharton, interview with Rajesh Hukku.

[28] Knowledge at Wharton, interview with Rajesh Hukku (he listed technology deployment, consulting and specialist business process outsourcing).

[29] Infosys, Form 20-F FY2002, Item 4 (Infosys 20-F for the year to March 2002).

[30] Infosys, Form 20-F FY2002, Item 4 (Finacle was sold alongside an internet banking program called BankAway).

[31] Infosys, Form 20-F FY2009, Item 3.D (Infosys 20-F for the year to March 2009, risk factors).

[32] Infosys, Form 20-F FY2012, Item 3.D (Infosys 20-F for the year to March 2012, risk factors; 4.9 per cent in the year to March 2011 and 4.2 per cent in the year to March 2010).

[33] Infosys, Form 20-F FY2009, Item 4.

[34] Infosys, Form 20-F FY2012, Item 4 (company claim; the same passage gives 48,500 branches). In a reorganisation of 2011 that merged seven business units into four by client industry, 'India and Finacle TM business units continue to be organized as before' (Infosys, Form 20-F FY2012, Item 4).

[35] Infosys, Form 20-F FY2009, Item 6 (directors and senior management).

[36] Press reports on the formation of iSPIRT (2013-2014) (Knowledge at Wharton, 14 March 2013; web page, unpaginated; the growth figures are given without a source).

[37] Press reports on the formation of iSPIRT (2013-2014) (Knowledge at Wharton, 14 March 2013; web page, unpaginated).

[38] *Inc42*, '10 years of iSPIRT' (2023) (Suprita Anupam, '10 Years Of iSPIRT: Inside The Making Of A Product Nation', *Inc42*, 27 February 2023; web page, unpaginated; consulted 2026-10-16): 'Set up as a spinoff from the NASSCOM Product Conclave (NPC) by 30 then (NASSCOM) members'. It names Sharma and Goenka among those at the first meeting, in Bengaluru on 4 February 2013. Venkataramakrishnan, *Scroll.in* (2017) (*Scroll.in*, 23 May 2017) calls iSPIRT 'an offshoot of industry body NASSCOM's annual product conclave'. Press reports on the formation of iSPIRT (2013-2014) names Goenka and Sharma as founder members.

[39] Press reports on the formation of iSPIRT (2013-2014) (quoting Business Standard).

[40] Press reports on the formation of iSPIRT (2013-2014) (quoting Business Standard).

[41] Press reports on the formation of iSPIRT (2013-2014) (Knowledge at Wharton, 14 March 2013; web page, unpaginated).

[42] Press reports on the formation of iSPIRT (2013-2014) (Knowledge at Wharton, 14 March 2013; web page, unpaginated). Murthy also said it was imperative to rebrand India as 'a proactive solutions provider, instead of a reactive solutions developer'.

[43] Press reports on the formation of iSPIRT (2013-2014) (Knowledge at Wharton, 14 March 2013; web page, unpaginated).

[44] *The News Minute* on the angel-tax letter (2019), and summary of the National Policy on Software Products (Shilpa S Ranipeta, *The News Minute*, 16 January 2019; consulted 2026-10-16). The letter was signed by nearly 70 entrepreneurs; the 48 per cent figure is iSPIRT's own claim. Rs 10 crore is Rs 100 million.

[45] India Stack (iSPIRT), home page and FAQ (India Stack website, FAQ; iSPIRT's self-description; consulted 2026-10-16). *Scroll.in* (Venkataramakrishnan, *Scroll.in* (2017)) describes India Stack as 'a set of software products that are explicitly aimed at building apps on top of the Aadhaar database'.

[46] *Inc42*, '10 years of iSPIRT' (2023).

[47] Venkataramakrishnan, *Scroll.in* (2017) (Rohan Venkataramakrishnan, *Scroll.in*, 23 May 2017; consulted 2026-10-16). Sharma wrote that 'Anonymity seemed easier than propriety' and 'I condoned tweets with uncivil comments.' The same report quotes the legal researcher Usha Ramanathan on 'a clear case of conflict of interest'; *Inc42* (*Inc42*, '10 years of iSPIRT' (2023)) also records accusations of conflicts of interest.

[48] Tripathi, 'Sridhar Vembu's vision from the village', *Forbes India* (2020) (Naandika Tripathi, 'Cover story: Sridhar Vembu's vision from the village', *Forbes India*, 3 June 2020; web page, unpaginated; consulted 2026-10-16), which says he co-founded AdventNet 'along with two of his brothers (he has three of them) and three friends'; *Mixergy*, interview with Sridhar Vembu (2010) (Andrew Warner, interview with Sridhar Vembu, *Mixergy*, 9 March 2010, transcript; founder's account), in which Vembu names a brother and Tony Thomas, a friend at Bell Labs: 'the three of us came together'.

[49] Zoho Corporation, 'About us' (timeline) (Zoho Corporation, 'About us', timeline; company self-description; consulted 2026-10-16).

[50] *Mixergy*, interview with Sridhar Vembu (2010) (founder's account). *Forbes India* (Tripathi, 'Sridhar Vembu's vision from the village', *Forbes India* (2020)) describes AdventNet as 'a software company serving network equipment vendors'.

[51] Tripathi, 'Sridhar Vembu's vision from the village', *Forbes India* (2020) (Vembu to *Forbes India*, 2020). The sentence begins: 'When we decided to launch this company, one thing that I noted was'.

[52] Zoho Corporation, 'About us' (timeline) (company timeline: ManageEngine, 2003; Zoho Writer and Zoho CRM, 2005; 'AdventNet is renamed to Zoho Corporation Pvt Ltd', 2009). *Forbes India* (Tripathi, 'Sridhar Vembu's vision from the village', *Forbes India* (2020)) says the renaming reflected 'the transition from a software company serving network equipment vendors to an innovative online applications provider'.

[53] *Mixergy*, interview with Sridhar Vembu (2010) (founder's account); Zoho Corporation, 'About us' (timeline) (company self-description: 'Since we have never taken money from investors, we have always been able to focus on what is best for the customer').

[54] Tripathi, 'Sridhar Vembu's vision from the village', *Forbes India* (2020) (figures as reported by *Forbes India*, which does not say whether 2019 is the calendar or the financial year); conversion at the 2019 calendar-year average rate, Rs 70.42 (Reserve Bank of India, *Handbook of Statistics on Indian Economy*).

[55] Zoho Corporation, 'About us' (timeline) (company timeline: 'We open an office in Tenkasi, India', 2011; 'Office opens in Renigunta, India', 2018). *Forbes India* (Tripathi, 'Sridhar Vembu's vision from the village', *Forbes India* (2020)) says the firm bought four acres in the village of Mathalamparai about nine years before 2020. The date of 2019 sometimes given for Tenkasi is that of Vembu's own move there (Tripathi, 'Sridhar Vembu's vision from the village', *Forbes India* (2020)).

[56] Tripathi, 'Sridhar Vembu's vision from the village', *Forbes India* (2020) (*Forbes India*'s figures; Vembu's words).

[57] Zoho Corporation, 'About us' (timeline) (company timeline: Zoho University 'launches as an experiment in providing an alternative to the conventional college education', 2005; 'Zoho University starts in Tenkasi, India', 2013); Tripathi, 'Sridhar Vembu's vision from the village', *Forbes India* (2020) (stipend of Rs 10,000 a month; 875 of 9,300 employees 'are students from Zoho Schools'). *Forbes India* dates the school's start to 2004 and gives its later name, Zoho Schools.

[58] Tripathi, 'Sridhar Vembu's vision from the village', *Forbes India* (2020) (Vembu: '15 to 20 percent of our engineers do not have an engineering degree. They are all trained in-house.').

[59] Tripathi, 'Sridhar Vembu's vision from the village', *Forbes India* (2020) (Paramasivam to *Forbes India*, 2020; he had then been with Zoho for 15 years).

[60] Freshworks, Form S-1 (2021), letter from the founder (Girish Mathrubootham).

[61] Freshworks, Form S-1 (2021), letter from the founder.

[62] Freshworks, Form S-1 (2021), letter from the founder.

[63] Freshworks, Form S-1 (2021), prospectus summary and business sections (company figures).

[64] Freshworks, Form S-1 (2021), prospectus summary.

[65] Freshworks, Form S-1 (2021), Business, 'Our Culture and Employees'.

[66] Freshworks, Form S-1 (2021), Risk Factors (allegations under federal and California trade-secret law and the Computer Fraud and Abuse Act; Freshworks said it believed it had 'meritorious defenses'). Zoho's headquarters in Chennai: Tripathi, 'Sridhar Vembu's vision from the village', *Forbes India* (2020).

[67] *Zoho Corporation v. Freshworks*, N.D. Cal., No. 3:20-cv-01869 (orders, 2020-2021) (US District Court, N.D. Cal., *Zoho Corporation Pvt. Ltd. v. Freshworks, Inc.*, No. 3:20-cv-01869-VC, Order Denying Motion to Dismiss, Dkt. 61, filed 27 October 2020, p. 1; consulted 2026-10-16). The order concerns the second amended complaint and Zoho's CRM, which it says 'is properly identified as a trade secret'.

[68] Freshworks, 'Freshworks and Zoho Reach Settlement' (2021), and Form 10-K for FY2021 (Freshworks press release, 'Freshworks and Zoho Reach Settlement to End Litigation', San Mateo, 23 December 2021, via Nasdaq; company statement; consulted 2026-10-16): 'These actions were not taken at the direction of Freshworks, and are contrary to Freshworks' policies.' Freshworks's annual report for 2021 (Freshworks, 'Freshworks and Zoho Reach Settlement' (2021), and Form 10-K for FY2021, Item 3) says: 'In December 2021, we reached an agreement with

Zoho to settle the litigation and Zoho dismissed the lawsuit.' Neither document states the terms. The same report's '$3.9 million related to a legal settlement' in 2021 does not name the other party and is not taken here to be the Zoho settlement.

[69] Freshworks, prospectus (Form 424B4, 2021), cover page (28,500,000 shares at $36.00; the gross figure, $1,026 million, is the author's calculation).

[70] Freshworks, Form 10-K FY2024, cover page and Item 1.

[71] Knowledge at Wharton, interview with Rajesh Hukku.

[72] Tracxn and Inc42 startup funding data, as reported (2021-2025) (Tracxn data as reported by *The Week*, Nachiket Kelkar, 14 January 2025; consulted 2026-10-14). Other trackers define a start-up differently and give different totals for 2021; their series are not joined here.

[73] Tracxn and Inc42 startup funding data, as reported (2021-2025) (Tracxn data as reported by *The Week*, 14 January 2025; consulted 2026-10-14). Six new 'unicorns', private firms valued at $1 billion or more, were created in 2024, against two in 2023.

[74] Byju's insolvency proceedings, as reported (2024-2026) (*LiveLaw*, 22 October 2024, and *The Week*, 23 October 2024; consulted 2026-10-14). Both date the National Company Law Tribunal's admission of the cricket board's insolvency petition, over dues of about Rs 158.9 crore (Rs 1.59 billion, about $19 million at the 2024 average rate of Rs 83.67, Reserve Bank of India, *Handbook of Statistics on Indian Economy*), to 16 June 2024. An appellate tribunal later closed the proceedings after a settlement, and on 23 October 2024 the Supreme Court set that order aside. The tribunals' orders are cited here from these reports.

[75] Reserve Bank of India, *Handbook of Statistics on Indian Economy* (RBI, Handbook of Statistics on Indian Economy, 2021 ed., Table 63). Accounts differ on whether UPI was launched in April or August 2016 (NPCI, UPI product statistics).

[76] Reserve Bank of India, *Handbook of Statistics on Indian Economy* (RBI, Handbook of Statistics on Indian Economy, 2026 ed., Table 58); dollar conversion at the 2023-24 average rate, Rs 82.79, Reserve Bank of India, *Handbook of Statistics on Indian Economy*.

[77] All India Radio News, 'UPI Transactions Surge to Record' (2025) (All India Radio, 2 January 2025, reporting NPCI data; consulted 2026-10-14).

[78] Tally Solutions, 'About Us' (company claim); Freshworks, Form S-1 (2021), prospectus summary.

[79] India Stack (iSPIRT), home page and FAQ (India Stack home page timeline; iSPIRT's self-description; consulted 2026-10-16): '2016 NPCI launches Unified Payments Interface'. It credits eKYC to UIDAI (2012), eSign to the Controller of Certifying Authorities (2015) and DigiLocker to MeitY (2015).

[80] *Inc42*, '10 years of iSPIRT' (2023) (*Inc42*, 2023).

[81] Venkataramakrishnan, *Scroll.in* (2017) (*Scroll.in*, 23 May 2017), which names Nilekani as a mentor and two members of the UIDAI team, Pramod Varma and Sanjay Jain, among iSPIRT's members.

[82] Knowledge at Wharton, interview with Rajesh Hukku.

[83] Knowledge at Wharton, interview with Rajesh Hukku.

[84] Press reports on the formation of iSPIRT (2013-2014) (Knowledge at Wharton, 14 March 2013; web page, unpaginated).

[85] Knowledge at Wharton, interview with Rajesh Hukku; Tracxn and Inc42 startup funding data, as reported (2021-2025) (Tracxn via *The Week*; consulted 2026-10-14).

[86] Tally Solutions, 'About Us' (company self-description).

[87] Rajaraman, *History of Computing in India* (2012), p. 55 (Rajaraman; Simputer designed 2000 by IISc and Encore Software, Bangalore; 'not a commercial success').

[88] Knowledge at Wharton, interview with Rajesh Hukku (founder's account; see note 18); Tripathi, 'Sridhar Vembu's vision from the village', *Forbes India* (2020) and Zoho Corporation, 'About us' (timeline) (Zoho University; see note 57).

[89] Knowledge at Wharton, interview with Rajesh Hukku; Freshworks, prospectus (Form 424B4, 2021), cover page.

[90] Freshworks, Form 10-K FY2024, Item 1A (and Item 1 for the total of about 4,400).

[91] Freshworks, Form 10-K FY2024, Item 7.

[92] Cognizant, Form 10-K FY2024, Item 1 (human capital).

[93] Knowledge at Wharton, interview with Rajesh Hukku.

[94] Freshworks, Form 10-K FY2024, Item 1A.

[95] Freshworks, Form 10-K FY2024, Item 1.

[96] Freshworks, Form 10-K FY2024, Item 1A.

Chapter 18. The pandemic boom (2020-2023)

On 21 September 2022 Rishad Premji, the chairman of Wipro, spoke at the convention of the All India Management Association. 'The reality is that there are people today working for Wipro and working directly for one of our competitors and we have actually discovered 300 people in the last few months who are doing exactly that,' he said.[1] Wipro had dismissed them.[2] The practice he described, working for two employers at once, depended on something that had changed two and a half years earlier, when the industry's engineers left their offices.

For thirty years the Indian services firm had sold its clients a place as well as people: secure floors, dedicated links and guarded campuses. In a few weeks of 2020 the place went away and the work carried on. Demand for the industry's services surged, and the firms hired on a scale they had never attempted. Engineers left their jobs at rates the companies called unprecedented, and some took second jobs on the side, while the real pay of a new graduate fell. Then demand turned, and campus recruits found their joining dates pushed back by months and years. The workers who complained began, for the first time in large numbers, to do so through unions. By the end of the period, one sentence in a company filing showed that something new had entered the terms on which India sold its engineers' time.

The factory goes home

On 25 March 2020 the Indian government put the country under lockdown.[3] Infosys, the second-largest of the Indian software services firms, had moved first.[4] In the annual report it filed with the United States Securities and Exchange Commission two months later, the company wrote: 'By March 20, 2020, we had enabled 93% of our employees to be able to work remotely.'[5] By the time it filed, the campuses in India were, on the company's figures, nearly empty. 'Only essential and critical services for our clients and our own operations are being supported from our campuses in India,' the filing said.[6]

The same report listed what else had changed, from secure laptops and wider bandwidth to 'information security controls for remote working and activity monitoring'.[7] It would 'Continue to monitor the productivity of employees as they work remotely'.[8] Even the training school moved. Infosys's next annual report recorded that 'The early days of the COVID-19 pandemic saw us relocate trainees from Mysuru and continue their training without missing a beat', using an in-house learning platform called Lex.[9] Mysuru (Mysore until 2014) was where the company trained its new graduates before they were placed on projects. Now the trainees were sent away and taught on screens.

The move was not as smooth as the headline figure suggests. Many of the industry's contracts specified where the work would be done. Infosys's annual report for the year to March 2020 warned that its ability to deliver while employees worked from home was 'sometimes constrained by contractual terms with our clients'.[10] Some clients had refused permission, or had been slow to give it: 'Since few of our clients did not give us the required permissions, or there were delays in obtaining them, our service delivery was marginally disrupted'.[11]

Within a year those doubts had mostly gone. 'As of March 31, 2021, we had enabled 99.3% of our employees with a secure remote working environment and 96.5% of our employees were working remotely,' Infosys reported.[12] Staff on critical client projects and on work involving sensitive data still went into the offices.[13]

The filings also recorded what home working cost the people doing it, in the language of risk. 'The productivity of our employees may be negatively impacted due to isolated remote working from home, quarantine requirements, negative social sentiment and personal anxiety,' Infosys wrote in 2020 and repeated in 2021.[14]

What the filings do show is the factory's front door changing. Infosys's recruiters had tested and interviewed candidates in person. In the year to March 2021, the company said, 'All interviews across the Globe in fiscal 2021

were conducted virtually', and the process from application to offer was digitised.[15] The training that followed was online too, and Infosys said it had 'repurposed Lex for over a million college students in India via our InfyTQ app', a platform that, in the company's words a year later, 'brings the best of our Mysuru training to the hands of the learners across the country'.[16] Students who passed its assessments could earn an Infosys certification.[17] The firm's training school, which had once been a campus that graduates travelled to after they were hired, was now partly a phone app that students used before.

These are the company's own descriptions, and they say nothing about how well remote training worked. They do establish the point that matters for the rest of the book. The work of the Indian services firm, and the teaching that prepared people for it, could be done almost anywhere. That opened a door the industry had kept shut: an engineer working at home could work for two employers at once.

Five million applications

Demand came back faster than the firms had planned for, and the clearest record of it is in the recruiting figures.

Infosys publishes its recruitment funnel every year. In the year to March 2020 (FY2019-20) it made 83,319 offers of employment, and in FY2020-21, the first pandemic year, 48,029.[18] Then came the surge. In FY2021-22 Infosys received 5,866,636 applications and made 222,500 offers.[19] The offers figure counts experienced recruits as well as new graduates, so it is not a count of freshers. Offers rose about four and a half times in a single year.

Headcount followed. Infosys added about 100,000 people in three years, from 242,371 employees in March 2020 to 343,234 in March 2023.[20] The other large firms grew in step. Wipro's rounded totals rose by about a quarter between March 2021 and March 2023.[21] Cognizant Technology Solutions, registered in the United States but with most of its staff in India, added about 66,000 people in the two years to the end of 2022, when 258,500 of its 355,300 employees were in India.[22]

Most of the new hires came from the same colleges that had fed the factory for two decades. A smaller stream came from elsewhere. Infosys said it had 'recruited over 7,280 employees locally in our markets' in FY2020-21, of whom 1,941 were fresh graduates, and that this hiring 'significantly de-risks our operations from regulatory changes related to immigration'.[23] The company ran versions of its foundation training in six countries and a training centre in Indianapolis.[24] Beside the hundreds of thousands hired in India these numbers are small. They show the firms insuring themselves against the visa politics of Chapter 13 by training some of their recruits in the client's own country.

A third of the workforce, every year

The boom was a labour-market event as much as a demand event. Engineers could leave, and they did.

Cognizant gives the clearest numbers, because, as an American company, it reports attrition (the share of staff who leave in a year) in its annual 10-K filing. Counting both resignations and dismissals, its attrition was 20.6 per cent in 2020, 30.8 per cent in 2021 and 31.7 per cent in 2022. In the last quarter of 2021, on an annualised basis, it reached 34.6 per cent.[25] 'In 2021, we experienced unprecedented attrition, which was considered industry-wide,' the company wrote. 'As a result, we hired over a hundred thousand new employees'.[26] It hired more than a hundred thousand again in 2022.[27] The departures were concentrated at the bottom of the pyramid. 'Our attrition is weighted towards our more junior employees,' Cognizant said, and in 2021 'voluntary attrition constituted the vast majority' of the total. In 2020, by contrast, only about half had been voluntary, because the company's own restructuring programme had raised the number of people it let go while the pandemic held down resignations.[28]

At those rates a large firm had to refill its pyramid almost from scratch every three years. Every departure meant another recruit to find, test, train and place on a project, and every replacement had to be paid more to stay. Cognizant's risk factors put the problem in the plainest terms of the business model: 'While we strive to adjust pricing to reduce the impact of compensation increases on our operating margin, we may not be successful in recovering these increases'.[29] Wipro's filing for FY2021-22 listed the same pressures on its margins: 'strong demand for digital and other niche skills are expected to result in wage inflation', 'Increased hiring and training costs due to higher attrition rates', and a 'Limited ability of the market to accept increase in prices for our offerings'.[30] This was the time-and-materials model under stress. A firm that sells hours at an agreed rate makes money only while the hours cost it less than the rate. When engineers' pay rose faster than clients would pay, the margin was squeezed from both sides.

Infosys gave its rates to investors on its quarterly results calls, whose transcripts it filed with the SEC, and they tell the same story with one caution: the company changed its measure partway through. In the quarter to March 2020 its executives reported voluntary attrition of 15.1 per cent, or 18.2 per cent counting dismissals. A year later voluntary attrition, 'calculated on annualized basis', was 15.2 per cent, and had risen, they said, 'as demand for talent increased'.[31] From FY2021-22 the company reported voluntary attrition over the previous twelve months instead, a measure that moves more slowly. On that basis it was 27.7 per cent in the year to March 2022, 20.9 per cent in the year to March 2023 and 12.6 per cent in the year to March 2024.[32] The first two figures are not strictly comparable with the last three, but the peak is not in doubt: in the year to March 2022 more than one Infosys employee in four chose to leave. Tata Consultancy Services (TCS), the largest of the firms by headcount, with 592,195 employees in March 2022, went through the same cycle. As reported in the press, its attrition was 12.1 per cent in the year to March 2020 and 7.2 per cent a year later, then rose to 17.4 per cent in the twelve months to March 2022 and 20.1 per cent in the twelve months to March 2023, before falling back to 12.5 per cent in the year to March 2024.[33]

Women's share of the workforce held steady through the churn, at least at the firms that report it. Cognizant said women made up 36 per cent of its workforce at the end of 2020 and 38 per cent at the end of 2021, the share it reported again for 2023.[34] Infosys said in its annual report for FY2023-24 that women made up 'nearly 40% of our workforce'.[35]

The price of a new engineer

For engineers who could change jobs, the boom was a seller's market. The demand for talent after the pandemic, *HR Katha* observed, had firms luring people with attractive packages and perks.[36] For those entering the industry, the real price moved the other way. A study by the business website Moneycontrol, published in 2022 and reported by the trade site *HR Katha* and by *Analytics India Magazine*, put the median annual pay of freshers at large IT firms at about Rs 245,000 in 2012 and Rs 355,000 in 2022, a rise of 45 per cent in ten years.[37] At each year's average exchange rate that is about $4,600 and $4,500.[38] Adjusted for Indian consumer prices, the 2022 figure was worth nearly a fifth less than the 2012 one.[39] Over the same decade, the study found, the median pay of chief executives at the large firms rose by 835 per cent.[40]

In this book's reading, these figures are the bargain seen from the bottom of the pyramid. A firm that sold engineers' time at rates its clients were reluctant to raise made its margin on the cheapest hours, the ones worked by new graduates, and it needed more of those graduates every year in order to grow. Whatever the boom did for engineers who changed jobs, on these figures it did not raise the real price of a new one.

Two employers, one laptop

Home working made a second job easy to hide. An engineer whose office was a bedroom, and whose manager saw her only on a video call, could in principle work for two companies at once. In 2022 the practice, known as moonlighting, became the subject of a public argument among the industry's leaders.

In August 2022 Premji wrote on Twitter that there was a lot of 'chatter about people moonlighting in the tech industry', and that 'it is cheating- plain and simple.'[41] His speech to the management association the next month, quoted at the start of this chapter, turned the complaint into dismissals. *Business Today* reported that Premji regarded the practice as a 'complete violation of integrity in its deepest form'.[2]

The industry did not speak with one voice. Infosys, the magazine reported, had told its staff the week before, in 'a strong and firm message', that moonlighting was not allowed. N. Ganapathy Subramaniam, the chief operating officer of TCS, had warned that employees chasing short-term gains in this way 'might lose out in the long term'.[42] T.V. Mohandas Pai, a former Infosys director, took the other side. 'No, moonlighting is not cheating,' he told the magazine. 'Employment is a contract between an employer who pays me for working for them for 'n' number of hours a day.' During those hours, he said, he was bound by the employer's conditions, including client confidentiality. 'Now what I do after that time is my freedom'.[43]

Infosys then wrote the problem into its annual report. From FY2022-23 its risk factors included a new line: 'Continued remote working exposes the company to the risk of some employees working for multiple employers simultaneously (moonlighting) without obtaining required prior approvals and this may result in potential conflict of interests, confidentiality breach, and reduced productivity.'[44]

The quarrel looks like one about loyalty. It was also about the unit in which the industry sold its work. The services firms billed their clients by the hour or the person-month. Pai's defence turned the same logic around: the employer bought a fixed number of hours, and the rest belonged to the worker. Premji's answer was that an engineer working directly for a competitor was breaking faith whatever the hours. When the work was done on a client's secure floor, the question did not arise, because the office day bounded what an engineer could do. At home that boundary was gone, and both sides found how much of the employment relationship had depended on it. The firms' answer, in the end, was to bring people back. Infosys described 2022 as the year it 'transitioned into a hybrid model of work for our employees and service providers following an extended period of remote work', and from FY2022-23 listed 'Our transition to a hybrid working model' among its risks.[45]

Packed bags

The turn in demand came in the second half of 2022. Cognizant dated it precisely: 'The rate of attrition began to decrease in the second half of 2022'.[46] For the engineers already employed, that meant fewer outside offers. For the graduates who had been promised jobs during the boom, it meant waiting.

On 5 October 2022 *Business Today* published complaints from new graduates whose start dates at Accenture kept moving. The magazine had already reported similar delays at Wipro, Infosys, HCLTech, Tech Mahindra and Capgemini.[47] The graduates it quoted were not named. One said: 'I was placed in Accenture as Associate Software Engineer (ASE) via campus placement in November 2021. I received my offer letter in the last week of April this year.' Since then, the start date had been 'pushed time after time'; the latest, 30 September, had now become 'somewhere in 2023'.[48] Another described a postponement two days before the joining date: 'I had packed my bags and booked all tickets to move to the new city, just two days before joining, they postponed the date again. When will they finally make me join, who will pay for my expenses in the meanwhile?'[49] A third had turned down other employers: 'I am riddled with depressive feelings because I rejected TCS, Wipro, and HCL for a job in Accenture.'[50] Others worried about the record it would leave: 'It's a year since I have been waiting to get onboarded. It is going to reflect as a gap year in my CV.'[51]

The graduates also blamed their colleges. The placement cell, the office that had invited the companies to campus and organised the tests, was the link between the college and the factory. 'The college placement cell should have helped us in this tough time,' said a graduate of CV Raman Global University in Bhubaneswar. A graduate of Galgotias University in Greater Noida said the placement office had stopped answering once students had graduated: 'they don't care for me anymore.'[52] For the colleges, a campus offer was a result to be counted at the end of the year. What happened after graduation was no longer their business.

None of this was new. The bust of 2001-02 had left recruits at Anna University waiting for deferred joining dates, and the collapse of Satyam in 2009 had stranded a batch of its campus hires (Chapters 11 and 16). This was the third time in about twenty years that the factory's newest products took the first shock of a turn in demand.

At Infosys the numbers turned with unusual speed. Offers of employment fell from 222,500 in FY2021-22 to 114,374 in FY2022-23, and to 26,975 in FY2023-24, a fall of about 88 per cent in two years. Applications fell from 5.9 million to 2.4 million.[53] Headcount, which had peaked at 343,234 in March 2023, fell to 317,240 a year later.[54] Wipro's rounded total fell from more than 250,000 to more than 234,000 over the same year.[55] Cognizant ended 2023 with about 347,700 employees, 7,600 fewer than a year before, and with 254,000 in India.[56] Its own measure of voluntary attrition in its technology services business fell from 25.6 per cent in 2022 to 13.8 per cent in 2023.[57] The engineers who had changed jobs freely in 2021 now stayed put.

Outside the factory laws

The people who took the delayed recruits' complaints to the government belonged to organisations that had barely existed a few years earlier. The industry had grown without unions, and partly under exemptions from the labour laws that governed factories.

A note published in November 2017 by Arvind Gurumurthy of the law firm Kochhar & Co. set out the position. The Industrial Disputes Act of 1947 required employers with more than 100 workers to get the labour commissioner's approval before retrenching staff, 'but this does not cover IT industry'.[58] The Industrial Employment (Standing Orders) Act of 1946 required industrial employers to publish certified terms of employment covering hours, shifts, leave and dismissal. 'However, by a notification published on January 25, 2014 the Karnataka

government exempted IT companies from applicability of Standing Orders Act for a period of five years.'[59] The engineer factory, in other words, was not in law a factory. The note suggested that the exemptions might have been meant to help the industry grow, and recorded that many employees' leaders saw them as giving firms room to treat layoffs as they liked.[60]

On Sunday 20 August 2017 close to 200 IT workers gathered at the YWCA Hall in Koramangala, Bengaluru, to found the Karnataka State IT/ITeS Employees Union (ITeS stands for IT-enabled services), known as KITU. *The News Minute* reported that the meeting came amid 'reports of large-scale layoffs in the IT sector'.[61] The union's general secretary was Vineeth, a testing lead at the Bengaluru software firm MindTree. He was also part of the IT Employees Centre, an association that had worked for IT employees' rights in Bengaluru for eight years and had received complaints about layoffs and maternity leave, but had never been registered as a union. 'What is currently happening is that if an employee is facing a problem, they have to fight in their individual capacity and the companies may not listen to them,' he told the reporter. 'The trade union will help them negotiate and bargain with the management, and even fight.'[62]

Others at the meeting spoke in the same terms. Sethumadhavan, a Wipro employee, said: 'As per our Constitution, we have the right to join a union.' The industry, he said, was facing many challenges, among them layoffs and automation, and 'This is the right time to form an organisation'.[63] Akshaya, an IBM employee on the union's executive committee, described the hours: 'sometimes we are given a project that requires us to work 15 hours a day and we are paid just Rs 25 per extra hour.'[64] The union's president, Amanullah Khan, answered the charge that unions held the country back: 'Trade unions are generally seen as anti-social or anti-national.' He said that neither the National Association of Software and Service Companies (Nasscom), the industry's lobby group, nor the government would be pleased. 'But we are not individuals anymore.'[65] The report noted that Mohandas Pai, among others, had opposed unions in the IT sector, on the grounds that they would harm a service industry.[66]

On 7 November 2017 the Karnataka labour commissioner certified KITU's formation under the Trade Unions Act of 1926.[67] Kochhar & Co. called it 'the first trade union in India, officially registered and recognised by the labour department, which is dedicated to the information technology ("IT") sector', with about 250 members by its own claim.[68] The same note acknowledged that 'unrecognised groups' had already existed, and the IT Employees Centre had been working for eight years before KITU was founded.[69] What is documented is narrower: KITU was the first IT employees' union registered in Karnataka, home to the city the reporter called India's Silicon Valley.

The pandemic gave such organisations work. A second body, the Nascent Information Technology Employees Senate (NITES), based in Pune, took up the cause of more than 2,000 graduates whom Infosys had recruited on campus in 2022 and had still not brought in.[70] It complained to the Union Ministry of Labour and Employment on 20 August 2024, and by the union's account Infosys issued about 2,000 offer letters with October joining dates on 2 and 3 September.[71] In February 2025 several hundred of those recruits, by then trainees at Mysuru, were let go after the company's internal assessments, over 300 by Infosys's count and 700 by NITES's, as the prologue tells.[72]

How many members either union had in these years is not on record. Their influence is easier to see than their size: complaints that would once have been one graduate's grievance against one company were now reaching a ministry in Delhi.

Offices, and the people in them

The bust brought restructuring as well as slower hiring. In the second quarter of 2023 Cognizant began what it called its NextGen programme, 'aimed at simplifying our operating model, optimizing corporate functions and consolidating and realigning office space to reflect the post-pandemic hybrid work environment'.[73] In 2023 it cost $229 million, about half of it for employee separations and the rest for leaving offices and other items.[74] The company warned that 'Our NextGen program and the associated reductions in headcount and consolidation of office space could disrupt our business'.[75]

The pandemic's lesson about place had come round again. In 2020 the firms had learned that the work could leave the campus. In 2023 one of them was giving up offices because hybrid working meant it needed fewer of them, and was cutting staff in the same programme.

The sentence in the risk factors

The most important single sentence of these years for the rest of this book appeared in Infosys's annual report for FY2023-24, filed in June 2024. In its risk factors, among the familiar warnings about competition and pricing, the company repeated a line it had carried in each annual report since 2020: 'due to competitive market conditions and pricing pressures, we are committing to higher productivity improvements in our contracts with our clients.' Then it added a new one: 'Due to the attention created by the code generation capabilities of AI technologies, clients have demanded higher commitments around productivity improvements.'[76] If those promised gains did not materialise, the filing said, the company's profitability could suffer.[77]

The sentence is short, and it is easy to read too much into it. It does not say that artificial intelligence had cut Infosys's hiring, and the same filing pointed elsewhere when it described pressure on demand, warning that clients 'may become cautious and reduce their discretionary spending in the short term or delay their decision-making process'.[78] Whether the fall in offers from 222,500 to 26,975 owed anything to AI is weighed in Chapter 19, which also follows how the sentence changed in later reports.

What the sentence does show is narrower and, for the book's argument, sharper. Clients had read about software that writes code. They had asked their Indian suppliers to commit, in contract, to doing the same work with less effort, and the supplier bore the risk if the gains did not come. Under time-and-materials billing, a productivity gain that the client captures means fewer billable hours. Under a fixed-price contract, it means a lower price for the same result. Either way the first share of the gain goes to the client, and the firm keeps something only if it beats its own promise.

The pandemic years had shown three things about the engineer factory. Its intake was set by clients' discretionary spending, not by the number of graduates: offers at one firm had risen four and a half times in a year and then fallen by nearly nine-tenths in two. Its workers, for a brief period, had the upper hand, and used it by leaving, by moonlighting and, in small numbers, by organising. And its work no longer needed a particular place. By 2024 the clients were asking whether it needed the same number of hours. For forty years the Indian services firms had grown by selling their engineers' time. Now they were being asked, in writing, to promise that the time would shrink.

Notes to Chapter 18. The pandemic boom (2020-2023)

[1] *Business Today*, 'Wipro Fires 300 Employees' (2022).

[2] *Business Today*, 'Wipro Fires 300 Employees' (2022) (dismissals and 'complete violation' as reported by *Business Today*).

[3] Infosys, Form 20-F FY2020, Item 5.

[4] *Business Today*, 'Wipro Fires 300 Employees' (2022); Infosys, Form 20-F FY2020, Item 5.

[5] Infosys, Form 20-F FY2020, Item 5.

[6] Infosys, Form 20-F FY2020, Item 5.

[7] Infosys, Form 20-F FY2020, Item 5.

[8] Infosys, Form 20-F FY2020, Item 5.

[9] Infosys, Form 20-F FY2021, Item 6.

[10] Infosys, Form 20-F FY2020, Item 3.

[11] Infosys, Form 20-F FY2020, Item 3.

[12] Infosys, Form 20-F FY2021, Item 5.

[13] Infosys, Form 20-F FY2021, Item 5.

[14] Infosys, Form 20-F FY2020, Item 3; Infosys, Form 20-F FY2021, Item 3.

[15] Infosys, Form 20-F FY2021, Item 6.

[16] Infosys, Form 20-F FY2021, Item 6; Infosys, Form 20-F FY2022, Item 6.

[17] Infosys, Form 20-F FY2022, Item 6.

[18] Infosys, Form 20-F FY2020, Items 4 and 6; Infosys, Form 20-F FY2021, Items 4 and 6. FY2019-20: 2,141,373 applications and 150,130 interviewed.

[19] Infosys, Form 20-F FY2022, Items 4 and 6 (523,385 applicants interviewed).

[20] Infosys, Form 20-F FY2020, Item 6; Infosys, Form 20-F FY2021, Item 6; Infosys, Form 20-F FY2023, Item 6 (242,371 in March 2020; 259,619 in 2021; 314,015 in 2022; 343,234 in 2023).

[21] Wipro, 20-F FY2023, Item 6 (more than 200,000 employees in March 2021, 240,000 in 2022, 250,000 in 2023).

[22] Cognizant, 10-K FY2021, Item 7; 10-K FY2022, Item 1 (about 289,500 at the end of 2020; 355,300 at the end of 2022, 258,500 of them in India).

[23] Infosys, Form 20-F FY2021, Item 4; Infosys, Form 20-F FY2022, Item 4; Infosys, Form 20-F FY2023, Item 4 (14,805 local hires, 3,650 of them fresh graduates, in FY2021-22; 10,169 and 2,216 in FY2022-23).

[24] Infosys, Form 20-F FY2022, Item 6 (Mexico, Britain, Germany, Australia, Singapore and Japan).

[25] Cognizant, 10-K FY2021, Item 1; 10-K FY2022, Item 1.

[26] Cognizant, 10-K FY2021, Item 1A.

[27] Cognizant, 10-K FY2022, Item 1A.

[28] Cognizant, 10-K FY2021, Item 1.

[29] Cognizant, 10-K FY2021, Item 1A.

[30] Wipro, 20-F FY2022, Item 5.

[31] Infosys, Forms 6-K, Q4 results filings (2020-2024), Form 6-K, Ex. 99.5, transcript of the Q4 FY2020 earnings call, 20 April 2020 (accession 0001067491-20-000029): 'Attrition on a standalone basis was slightly higher at 18.2%; however, voluntary attrition reduced further to 15.1% from 15.6% last quarter'; Ex. 99.5, Q4 FY2021 call, 14 April 2021 (0001067491-21-000030): 'Voluntary attrition for IT services calculated on annualized basis increased to 15.2% as demand for talent increased.' Company statements to investors; transcripts unpaginated.

[32] Infosys, Forms 6-K, Q4 results filings (2020-2024), Ex. 99.5, Q4 FY2022 call, 13 April 2022 (0001067491-22-000020): 'Voluntary LTM attrition increased to 27.7%'; Ex. 99.5, Q4 FY2023 call, 13 April 2023 (0001067491-23-000027): 'LTM voluntary attrition declined to 20.9%'; Ex. 99.4, Fact Sheet Q4 FY2024, 18 April 2024 (0001067491-24-000016), row 'Voluntary Attrition % (LTM - IT Services)': 12.6% (March 2024), 20.9% (March 2023). LTM = last twelve months. The basis changes between FY2021 (quarterly, annualised) and FY2022 (LTM). Earlier fact sheets are filed as images only. The series is in the data appendix.

[33] *HR Katha* reports of TCS results (2020-2024) (*HR Katha*, 17 April 2020, 13 April 2021, 21 April 2023 and 12 April 2024); *The News Minute* and *Communications Today* on TCS results (2020, 2022) (*Communications Today*, 17 April 2020, credited to *Business Standard*, for 12.1 per cent; *The News Minute*, 11 April 2022, for 17.4 per cent 'on a trailing 12-month basis' and 592,195 employees). The figures are as reported in the press accounts of TCS's quarterly results, not taken from the company's own fact sheets (Tata Consultancy Services, fact sheets and annual reports); those for 2021, 2023 and 2024 rest on *HR Katha* alone. The 2022 and 2023 reports give the basis as the last twelve months; the 2020 and 2021 reports do not state it. TCS's measure is not Infosys's voluntary attrition, and the two series should not be compared year by year. The series is in the data appendix.

[34] Cognizant, 10-K FY2021, Item 1; 10-K FY2023, Item 1.

[35] Infosys, Form 20-F FY2024, Item 6.

[36] *HR Katha*, 'IT-Sector CEOs' Median Annual Salaries' (2022) (*HR Katha*, 1 June 2022, on the demand for talent after the pandemic).

[37] *HR Katha*, 'IT-Sector CEOs' Median Annual Salaries' (2022) (*HR Katha*, 1 June 2022); Choudhary, 'Behind the Growing (and Huge) IT CEO-Fresher Pay Gap', *Analytics India Magazine* (2022) (*Analytics India Magazine*, 16 August 2022). Both report the Moneycontrol study 'IT CEO pay shot up 835% in a decade while freshers' salary grew 45%' (moneycontrol.com, 2022), which is cited here through them. Neither report gives the study's method, such as which firms it covered or whether pay means cost to company. Figures as in fresher_pay.csv.

[38] *HR Katha*, 'IT-Sector CEOs' Median Annual Salaries' (2022); Reserve Bank of India, *Handbook of Statistics on Indian Economy* (conversion at calendar-year average rates, Rs 53.44 per dollar in 2012 and Rs 78.60 in 2022).

[39] *HR Katha*, 'IT-Sector CEOs' Median Annual Salaries' (2022); OECD, Consumer Price Indices, India (deflated by the India CPI series: Rs 4.77 lakh and Rs 3.89 lakh in 2024 rupees; calculation by the author).

[40] *HR Katha*, 'IT-Sector CEOs' Median Annual Salaries' (2022); Choudhary, 'Behind the Growing (and Huge) IT CEO-Fresher Pay Gap', *Analytics India Magazine* (2022) (the 835 per cent rise as given by both reports; *HR Katha* gives the 2022 median as Rs 31.5 crore).

[41] *Business Today*, 'Wipro Fires 300 Employees' (2022).

[42] *Business Today*, 'Wipro Fires 300 Employees' (2022).

[43] *Business Today*, 'Wipro Fires 300 Employees' (2022).

[44] Infosys, Form 20-F FY2023, Item 3; Infosys, Form 20-F FY2024, Item 3.

[45] Infosys, Form 20-F FY2023, Item 3; Infosys, Form 20-F FY2024, Item 3.

[46] Cognizant, 10-K FY2022, Item 1A.

[47] Press reports on delayed onboarding of graduates (2022) (A. Chaturvedi, *Business Today*, 5 October 2022; speakers anonymous in the original).

[48] Press reports on delayed onboarding of graduates (2022) (A. Chaturvedi, *Business Today*, 5 October 2022; speakers anonymous in the original).

[49] Press reports on delayed onboarding of graduates (2022) (A. Chaturvedi, *Business Today*, 5 October 2022; speakers anonymous in the original).

[50] Press reports on delayed onboarding of graduates (2022) (A. Chaturvedi, *Business Today*, 5 October 2022; speakers anonymous in the original).

[51] Press reports on delayed onboarding of graduates (2022) (A. Chaturvedi, *Business Today*, 5 October 2022; speakers anonymous in the original).

[52] Press reports on delayed onboarding of graduates (2022) (A. Chaturvedi, *Business Today*, 5 October 2022; speakers anonymous in the original).

[53] Infosys, Form 20-F FY2022, Item 6; Infosys, Form 20-F FY2023, Item 6; Infosys, Form 20-F FY2024, Items 4 and 6.

[54] Infosys, Form 20-F FY2024, Item 6.

[55] Wipro, 20-F FY2024, Item 6.

[56] Cognizant, 10-K FY2023, Item 1.

[57] Cognizant, 10-K FY2023, Item 7.

[58] Mondaq and *The News Minute*, reports on the Karnataka IT union (2017-2024).

[59] Mondaq and *The News Minute*, reports on the Karnataka IT union (2017-2024). The note adds that some provisions of the shops and establishments law, on holidays and hours of work, had also been relaxed for IT companies in Karnataka and in some other states.

[60] Mondaq and *The News Minute*, reports on the Karnataka IT union (2017-2024).

[61] Das, 'Nearly 200 IT Employees Gather to Form a Trade Union' (2017).

[62] Das, 'Nearly 200 IT Employees Gather to Form a Trade Union' (2017).

[63] Das, 'Nearly 200 IT Employees Gather to Form a Trade Union' (2017).

[64] Das, 'Nearly 200 IT Employees Gather to Form a Trade Union' (2017).

[65] Das, 'Nearly 200 IT Employees Gather to Form a Trade Union' (2017); Hanna, *Exploiting Information Technology for Development* (1994), p. 105 (Nasscom's full name).

[66] Das, 'Nearly 200 IT Employees Gather to Form a Trade Union' (2017).

[67] Mondaq and *The News Minute*, reports on the Karnataka IT union (2017-2024).

[68] Mondaq and *The News Minute*, reports on the Karnataka IT union (2017-2024).

[69] Mondaq and *The News Minute*, reports on the Karnataka IT union (2017-2024); Das, 'Nearly 200 IT Employees Gather to Form a Trade Union' (2017) (IT Employees Centre; 'India's Silicon Valley').

[70] *The News Minute*, 'Infosys to Face Govt Probe over Delayed Recruitment' (2024) (*The News Minute*, 6 September 2024: more than 2,000 graduates selected in Infosys's 2022-23 campus drive; NITES 'a Pune-based IT union').

[71] *The News Minute*, 'Infosys to Face Govt Probe over Delayed Recruitment' (2024) (complaints filed on 20 August 2024); Ganeshan, '"Thrown on the Road"' (2025) (offer letters on 2 and 3 September 2024, according to NITES). *The News Minute* put the number still waiting for joining dates at about 700 in early September (*The News Minute*, 'Infosys to Face Govt Probe over Delayed Recruitment' (2024)), and *Deccan Herald* at about 500 by early October (Choudhary, 'IT Professionals' Union Accuses Karnataka Govt of Bias' (2024)).

[72] Press reports on the NITES complaint against Infosys (2024-2025) (*Deccan Herald*, February 2025: Infosys's 'over 300', NITES's 700; assessments as described by the company); Ganeshan, '"Thrown on the Road"' (2025) (recruited in 2022, onboarded at Mysuru in October 2024). *The News Minute*'s report (Ganeshan, '"Thrown on the Road"' (2025)) gives its own estimate of 400; the prologue sets out all three figures.

[73] Cognizant, 10-K FY2023, Item 1A.

[74] Cognizant, 10-K FY2023, Items 1A and 7 ($115 million in employee separation costs and $114 million in facility exit and other costs).

[75] Cognizant, 10-K FY2023, Item 1A.

[76] Infosys, Form 20-F FY2024, Item 3; Infosys, Form 20-F FY2020, Item 3; Infosys, Form 20-F FY2021, Item 3; Infosys, Form 20-F FY2022, Item 3; Infosys, Form 20-F FY2023, Item 3 (earlier appearances of the 'committing to higher productivity improvements' sentence).

[77] Infosys, Form 20-F FY2024, Item 3.

[78] Infosys, Form 20-F FY2024, Item 3; Infosys, Form 20-F FY2023, Item 3 (the same risk, there following a passage on clients' Russian operations and the conflict in Eastern Europe).

Part V: The break (2022-2026)

Chapter 19. The machines learn to code (2022-2026)

One sentence, three years

In June 2024 Infosys filed its annual report on Form 20-F with the US Securities and Exchange Commission, a document in which a misleading statement carries legal risk. Among its risk factors, in a passage about wages and productivity, was a new sentence: 'Due to the attention created by the code generation capabilities of AI technologies, clients have demanded higher commitments around productivity improvements.'[1] The sentence came back the next year and the year after, changed each time in small ways, and the changes trace the arrival of the problem. In the report filed on 1 July 2025 one phrase changed: clients 'have started demanding higher commitments around productivity improvements'.[2] In the report filed on 15 June 2026, for the year to March 2026, the cause was widened. The sentence now began 'Due to the attention created by evolving AI technologies', and the clients were still demanding.[3]

The sentences on either side explain what the demand meant. Before it, in both later reports, Infosys said that 'due to competitive market conditions and pricing pressures, we are committing to higher productivity improvements in our contracts with our clients'. After it, the company warned that failing to deliver those improvements 'may impact our profitability'.[4] Read together, the three sentences describe a mechanism. Clients hear that machines can write code. They ask for contracts that assume fewer hours of human work. The vendor promises the savings in advance and carries the loss if they do not arrive.

The 2026 report added a passage that had not appeared before. In its review of operations, Infosys described what its clients were doing: 'large technology platforms continue to expand their service layers, while clients themselves are selectively insourcing capabilities as AI improves productivity and reduces dependency on external labor-based models'.[5] An 'external labor-based model' is a fair description of what Infosys had sold since its founding in 1981. Here the company named, in a regulatory filing, the threat this chapter is about.

Its competitors' filings moved the same way. Wipro's report for the year to March 2026 added a risk factor: 'AI-driven automation, efficiency gains and increased client self-service may reduce demand for certain services we provide and may adversely affect pricing, margins, service levels or the mix of services delivered.'[6] It listed 'outcome-based pricing', meaning charging for a result delivered rather than for the hours spent on it, among the strategies firms were likely to adopt, and predicted that 'Hiring will shift from volume to skills, emphasizing AI fluency, domain expertise, and productivity-based workforce models'.[7] Cognizant, which is registered in the United States but employs most of its staff in India, wrote in its annual report for 2025 that 'demand for some services that we currently perform for our clients may be reduced' as AI evolved.[8] All three also told investors, at length, that AI would create new kinds of work. But risk factors are written for lawyers rather than customers, which is why they are useful here. By 2026 three of the largest employers of Indian software engineers had told an American regulator that the machines threatened the volume of work they sold.

That volume was the whole of the bargain this book has followed. For forty years the Indian software industry had sold the time of people who write code, priced by the hour and the head, and the colleges had grown to supply the heads. This chapter covers the years in which machines learned to write usable code, from 2022, when the pandemic hiring boom ended, to the autumn of 2026. When code became cheap to produce, both halves of the arrangement were exposed at once. The evidence for what happened is recent, thin in places and contested on the point that matters most, which is cause.

The tools on the desk

The same filings record which tools the engineers were given. Infosys's report for the year to March 2025 named one, GitHub Copilot, and claimed that Infosys was 'among the top GSIs in adoption', a GSI being a global systems integrator.[9] A year later the list was longer: 'We utilize leading AI solutions like Github Copilot, Cursor, Claude Code and Devin to name a few.'[10] All four are commercial products that use AI models to write or change code. Claude Code is made by Anthropic, which also makes the AI model that assisted with the research and drafting of this book, and the reader should weigh this chapter with that in mind; here it is treated like the other three, as a tool a company reported using. Infosys said it had 'infused AI throughout our software development lifecycle' and used its own platforms for what it called 'Agentic software engineering', meaning software agents that carry out programming tasks over several steps.[11]

Wipro gave a count. In the year to March 2025, 'over 20,000 associates' in its delivery teams were 'leveraging AI developer tools, boosting our overall productivity'.[12] A year later the same sentence said 'over 60,000'.[13] Wipro reported more than 240,000 employees at 31 March 2026, so on its own figures about a quarter of its workforce was using such tools; how the productivity was measured, the filing does not say.[14] Cognizant gave its staff 'leading vibe coding tools', meaning tools that let a person describe a program in ordinary language and have a model write it, and reported that they had produced 'more than 30,000 prototypes'.[15] The phrase thus entered an American annual report as a way of training a workforce of whom about 256,900, out of roughly 351,600 at the end of 2025, were in India.[16]

Every company also claimed that its workforce was being trained in AI, and none defined what training meant. The largest claim came from the National Association of Software and Service Companies (Nasscom), the industry's lobby group, which said that 2 million of the industry's 5.95 million direct employees had been 'upskilled in AI'.[17] The companies' own counts ran into the hundreds of thousands.[18] They show what the firms wanted investors and clients to believe, which was that their workforces were being remade rather than replaced.

The price of an hour

By 2026 the business press had a name for what Infosys had set out in 2024. Payal Ganguly's report for *Forbes India* on 11 May 2026 called it 'AI-led deflation, characterised by a productivity-based pricing structure demanded by customers and the passing on of productivity benefits to enterprises'.[19] She quoted HCL Technologies' (HCLTech's) chief executive, C. Vijayakumar, on his company's earnings call: 'A $100 million deal would be much less today, maybe $80 million, just on a rough ballpark.' Winning the same revenue, he added, took 'at least 25-30 percent more effort'.[20] K. Krithivasan, chief executive of Tata Consultancy Services (TCS), was more hopeful on his own call: 'Our expectation is AI will be net accretive.'[21] Salil Parekh of Infosys said that his company was signing multi-year deals priced on productivity.[22]

Who gained from a saved hour depended on the contract. A time-and-materials contract names the people and their rates; a fixed-price contract names only the price. Under the first, a saving in hours passes to the client, because the client pays for fewer hours. Under the second, the vendor keeps it, which is why Infosys had set out before 2002 to win more fixed-price work (Chapter 11). The evidence from 2026 suggests that both forms were being squeezed. Clients would not pay for hours that a machine could save, and under the productivity commitments Infosys described, the vendors had to promise the savings before they had made them.

There was a third form, and it was growing. Global capability centres (GCCs), the Indian offices that multinational companies own and staff themselves, employed about 2.36 million people by the year to March 2026, by the count of Nasscom and the consultancy Zinnov, and their revenue was already close to what the two had projected for 2030.[23] Those figures come from an industry body and from a consultancy that sells its services to GCCs, and they are not measured on the same basis as Nasscom's totals for the industry. What they suggest is a shift in who employed India's engineers. The client was moving its work in house, and the house was often in Bengaluru (Bangalore until 2014) or Hyderabad. That was the 'selectively insourcing' of which Infosys had warned, seen from the other side. A parent company that does the work in its own centre keeps whatever saving the tools produce, where the client of a services firm has to bargain for it.

This, in the book's reading, is where the bargain began to come apart. A firm that sells hours earns more only by selling more of them, which is why the industry had to keep hiring. A tool that shortens the hour does not merely make the firm more efficient. It hands the client a reason to buy fewer hours, and a captive centre a reason not to

buy them from outside at all. Writing and testing code had passed early the test that the economist Alan Blinder proposed in 2006 (Chapter 14): work that can be sent down a wire without losing quality is exposed, however skilled the person who does it, and that is how so much of it came to India. The filings of 2024 to 2026 show the same work being repriced in the light of what machines could do.

The class of 2026

The colleges asked a simpler question: would the entry-level job survive? The evidence is mixed, and it is easy to read too much into it.

At Infosys, offers of employment, which include experienced hires as well as graduates, had collapsed to 26,975 in the year to March 2024, when 2,436,929 people applied, about 90 applicants for each offer. They recovered to 83,207 and then 87,286 in the next two years, while applications rose to 4,457,748 and 5,884,480, about 54 and 67 for each offer on the book's calculation.[24] In April 2026 *Business Standard* reported that TCS had given 'a fresher hiring number of 25,000' for the year to March 2027, against the roughly 40,000 that the paper said TCS usually took from engineering colleges.[25] Krithivasan's own words, to the Press Trust of India earlier that month, put the figure in a different light: 'We have made 25,000 offers for freshers in FY27. Clarity on demand will lead to more hiring.'[26] On his account the 25,000 were offers already made, not a ceiling, and TCS had reportedly hired more than 44,000 freshers the year before.[27]

The other firms were uneven. Infosys said in April 2026 that it would add 20,000 freshers in the coming year, keeping its annual target, a plan for graduate hires that measures something different from its offers of all kinds.[28] Wipro's chief human resources officer, Saurabh Govil, declined to give a figure: 'Hiring will be as per demand.'[29] HCLTech's chief people officer, Ramachandran Sundararajan, said: 'Given the current environment, it is difficult to provide a number.'[30] Krithivasan added, in the newspaper's paraphrase, that freshers take 'up to nine months to come on the floor, while a lateral can deliver immediately'.[31] The complaint that graduates arrive unready is as old as the industry. What may have changed was that some of the junior work a fresher used to learn on was the work for which the companies now described AI tools. Infosys's 2024 report spoke of 'a code assistant' for 'tasks like coding, testing, and documentation'.[32] This is the kind of documented, standardised work Chapter 11 described, and the kind the tools were built for.

The dates, though, warn against a simple story. Infosys's offers had already collapsed in the year to March 2024, and the delayed joining dates began in 2022.[33] Yet the company's first annual report to mention the 'code generation capabilities of AI technologies' was the one filed in June 2024, and the first coding tool it named in a filing, GitHub Copilot, appeared in the report filed in July 2025; Wipro's first count of staff using such tools covered the year to March 2025.[34][35] On the companies' own records, the slowdown in graduate hiring began before AI coding tools were a reported part of their work. *Business Standard* itself blamed the conflict in West Asia for the gloomy outlook of 2026.[36] On the evidence available AI is at most one of several possible causes, alongside the end of the pandemic boom, weak demand from clients and geopolitics. On this book's reading, what the tools changed was the reason to expect the rung to come back.

What the slowdown felt like on a campus is documented by Yashraj Sharma of *Rest of World*, in a report published on 9 December 2025. Rishabh Mishra joined the Indian Institute of Information Technology, Design and Manufacturing in Jabalpur in 2022 with what the reporter called 'the most predictable dream in global tech': to 'study computer science, write code, and one day make it to Silicon Valley'.[37] His course was due to end in May 2026. Among his 400 classmates, according to the article, fewer than 25 per cent had secured job offers.[38] 'It is really bad out there,' Mishra said.[39] Some classmates were thinking of further study before looking for work. He was doubtful: 'But after one year, if you return to the job market, your degree is even more irrelevant.'[40] Another student named in the report, Nishant Kaushik, who had studied computer science at a well-ranked college in eastern India, had decided to look for jobs in sales or marketing instead.[41] *Rest of World* framed its report around AI and cited a finding by the consultancy EY that Indian IT services firms had cut entry-level roles by 20-25 per cent because of automation and AI.[42] One college in one year is a single data point. It is also a real student, in his own words, at the moment the argument about AI reached the campuses.

Those who did get jobs found that the starting salary had barely moved. Udit Hinduja of *ThePrint* reported in September 2025 on an unnamed TCS fresher in Ahmedabad who shared a room with three colleagues he had met in training, on a monthly salary of Rs 24,000 (about $270 at the year's average rate). 'I'm lucky that I had some savings

to buy a bike, otherwise I would have to walk to work,' he said.[43] Neeti Sharma, chief executive of the staffing firm TeamLease Digital, told the same reporter that starting salaries for science and engineering graduates had been Rs 2-2.5 lakh a year in 2014 and were Rs 3-3.5 lakh (about $3,400-4,000) 'today', 'not even enough to cover inflation over the last decade'.[44] The entry-level hour had always been the cheapest the industry sold. On the argument of this chapter, it was also the one a machine could most easily replace.

A stall, not a collapse

The fear in 2025 and 2026 was that the machines would empty the offices. The headcount figures do not show that. They show a stall after the boom and bust of the pandemic years.

Infosys had 317,240 employees at 31 March 2024, 323,578 a year later and 328,594 at 31 March 2026.[45] Wipro reported 'more than' 232,000, 230,000 and 240,000 at the same dates.[46] At Cognizant, voluntary attrition in technology services, the share of staff who chose to leave in a year, fell from 25.6 per cent in 2022 to 15.9 per cent in 2024 and 13.9 per cent in 2025, figures consistent with fewer outside offers.[47] Nasscom estimated 135,000 net additions across the industry in the year to March 2026, taking direct employees to 5.95 million, while revenue rose 6.1 per cent to $315 billion: the non-linearity with which the prologue began.[48] Nasscom's figures are an industry body's estimates and are revised.[49]

TCS was the exception. Avik Das of *Business Standard*, compiling the companies' results in April 2026, reported that the five largest firms' combined headcount fell by 6,981 in the year to March 2026, after rising by 12,718 the year before. TCS's net headcount fell by 23,460, while Infosys, Wipro and HCLTech added people.[50] Part of that fall was the result of a decision announced on a Sunday the summer before.

July 2025

On Sunday 27 July 2025 TCS said, in a statement printed by *Business Standard*, that it was 'on a journey to become a future-ready organisation'. Its initiatives included 'deploying AI at scale for our clients and ourselves' and 'realigning our workforce model'. 'We will be releasing associates from the organisation whose deployment may not be feasible,' it said. 'This will impact about 2 per cent of our global workforce, primarily in the middle and the senior grades, over the course of the year.'[51] The paper put the number at about 12,260 of TCS's 613,069 employees.[52] The company said it would provide, in *Deccan Herald*'s paraphrase, 'appropriate benefits, outplacement, counselling, and support' to those affected.[53]

Krithivasan explained the decision to the business website Moneycontrol. Reuters quoted him: 'This is not because of AI giving some 20% productivity gains. We are not doing that. This is driven by where there is a skill mismatch, or, where we think that we have not been able to deploy someone.'[54] *Outlook Business*, relaying the same interview, reported him saying that 'Some people, especially at senior levels, find it difficult to transition to tech-heavy roles', and that 'This is not a question of demand. It is about us being future-ready.'[55] He described a move away from 'waterfall' project management, in which a project passes through fixed stages under layers of managers, towards agile methods that needed fewer such layers.[56] Shritama Bose and Ujjaini Dutta of Reuters Breakingviews wrote that he was seeking to play down fears of an 'AI-induced jobs armageddon', and offered a reading of their own: TCS's headcount had risen 36 per cent since 2020, and while revenue per employee had risen 20 per cent, costs per employee had risen by a quarter.[57] On this reading the cut was about cost after over-hiring. The IT & ITES Democratic Employees Association (IIDEA), a union, offered a third: 'What we are witnessing is not transformation but cost-cutting masquerading as future-readiness.' It accused TCS of targeting experienced, better-paid staff.[58]

The three explanations are not exclusive, and AI could sit behind any of them. A skill mismatch can arise because clients want AI work; costs matter more when clients demand AI-driven discounts. No published record shows which roles were cut, or why, in enough detail to decide between them. Whether AI caused the TCS cuts remains an open question.

The bench

What the cut felt like is known from reporting in Bengaluru, almost entirely through employees who would not give their names. Anisha Reddy of *South First* reported on 2 August 2025 that more than 100 TCS employees had approached unions in the previous week.[59] Lenil Babu, secretary of the Karnataka State IT/ITeS Employees Union

(KITU), told her: 'So far, we have filed complaints with the Additional Labour Commissioner of Karnataka on behalf of 15 employees who were already let go. But another 100 have approached us after being put on the bench.'[60]

The bench is where an engineer waits, still employed and paid, between projects. In the industry's long boom, time on the bench was a normal part of a career and a cost the firms absorbed, because a firm that sells hours needs people ready to bill. *South First* reported that a new TCS policy, in force from 12 June 2025, allowed an employee at most 35 days a year without a project and required at least 225 billable days.[61] The reporter saw an email sent to one employee: 'Your prolonged un-allocation without any productive assignment is a matter of concern to the company and the company views this matter very seriously.'[62] The Nascent Information Technology Employees Senate (NITES), a union based in Pune, said the policy 'frames lack of allocation as a personal failure of the employee'.[63] TCS told *South First* that the restructuring was being planned with due care so that service to clients would not be affected.[64]

Korah Abraham of *The News Minute* reported on 7 August 2025 how the resignation meetings worked, again from anonymous accounts. A Bengaluru employee in middle management alleged: 'They offered three months' pay as an incentive to resign. But if someone refuses, they're told they'll be terminated without compensation, and won't receive a relieving letter.'[65] A relieving letter is the document an Indian employer gives a departing employee, and the next employer usually asks to see it. Two employees said they had been asked to switch off their phones and leave them on the table before meeting human resources staff.[66] A woman who had resigned said she would have fought the case if she could have afforded to: 'I come from a financially poor background.'[67] One employee estimated that more than 300 people had been made to resign from a single Bengaluru campus in a week.[68]

TCS denied that it had forced people out. When NITES wrote to the chief minister of Maharashtra in the autumn, claiming that about 2,500 employees in Pune had been made to resign or abruptly removed, the company replied: 'The misinformation shared here is inaccurate and purposefully mischievous. Only a limited number of employees have been affected by our recent initiative to realign skills in our organisation.' Those affected, it said, 'have been provided due care and severance'.[69]

KITU had taken its case to the state. It filed an industrial dispute on 29 July, two days after the announcement, accusing TCS of illegal mass retrenchment. Its argument was that the company was pushing people to resign so as to avoid the government approval that the Industrial Disputes Act requires before a firm with more than 100 employees retrenches workers.[70] Karnataka's labour minister, Santosh Lad, told reporters that his department was looking into the union's letter, but added: 'We cannot act in a hurry.'[71] On Wednesday 6 August two managers from TCS's human resources department sat down at the Labour Office in Bengaluru across from three KITU officials, among them its general secretary, Suhas Adiga, and its president, V.J.K. Nair. The Additional Labour Commissioner, G. Manjunath, presided as conciliation officer. The meeting ended without agreement, and a further round was set for 8 September.[72]

The company's own figures, two months later, sharpened the dispute. On the investor call after its results for the quarter to September, TCS's new chief human resources officer, Sudeep Kunnumal, said that 'the 20,000 headcount (reduction) is a factor of voluntary and involuntary attrition'; and, as *Deccan Herald* reported, he hinted that the involuntary part was 6,000 employees who had been 'released'. Headcount had fallen from 613,069 at the end of June to 593,314 at the end of September.[73] KITU and two other unions read that as a confession: 'By openly accepting that it has retrenched 6,000 employees, TCS has admitted to a violation of the Industrial Disputes Act'.[74] The company called the departures realignment; the unions called them retrenchment by another name. The word that decided which law applied was the one the two sides could not agree on.

The regulator's answer

The All India Council for Technical Education (AICTE), the regulator that approves engineering colleges, responded with announcements. On 23 December 2024 it declared 2025 the 'Year of Artificial Intelligence', with a campaign called 'AI for All', AI laboratories and workshops for teachers across what it said were more than 14,000 colleges with 40 million students.[75] On 6 June 2025 its chairman, T.G. Sitharam, told the Press Trust of India that AI would be brought into the model curricula of every branch of engineering, information technology and management. 'A committee of experts has been constituted to suggest necessary revisions in the model curriculums,' he said.[76] No circular making AI part of every branch, and no revised model curriculum, has been located. The regulator announced the change; whether the colleges carried it out is not on record.

The state builds a model

The Indian state answered the AI question as it had answered identity and payments, by building. The IndiaAI Mission was launched in March 2024 with an outlay of Rs 10,372 crore (Rs 103.7 billion, about $1.25 billion at the year's average rate), for computing capacity, data, models and research.[77] It selected twelve organisations and consortia to build foundation models, the large general-purpose models on which applications are built.[78] The first was Sarvam AI, founded in August 2023 by Vivek Raghavan and Pratyush Kumar, both graduates of the IITs.[79] On 26 April 2025, according to the trade journal *tele.net*, Sarvam was chosen to build 'an open source 120 billion parameter' model (parameters are the adjustable values in a model, a rough measure of its size).[80] The minister for electronics and information technology, Ashwini Vaishnaw, said: 'We are confident that Sarvam's models will be competitive with global models.'[81]

At the India AI Impact Summit in February 2026 Sarvam unveiled two models, Sarvam-30B and Sarvam-105B, the larger of them smaller than the 120 billion parameters announced in 2025.[82] Debarghya Sil of *Inc42* judged the models competitive in their size class and strong on Indian languages, but behind the frontier models of OpenAI, Anthropic, Google and Meta in 'raw reasoning depth'.[83] Raghavan called the launches a 'significant first step'.[84] The state was not trying to replace the services industry with a model of its own. It was trying to make sure that the country that had supplied so much of the world's programming labour owned some of the machines that were starting to do it.

More coders than ever

In October 2025 GitHub, the code-hosting service, projected in its annual *Octoverse* report, as *The Register* reported it, that India would have 57.5 million or more developers on the platform by 2030, overtaking the United States, and said that 80 per cent of new users began using its AI coding assistant in their first week.[85] A GitHub account is not a job, and the projection is not a count of engineers. But the two numbers belong together. More Indians were writing code than at any point in the history this book has traced, and most of the newest of them were writing it with a machine at their elbow. The industry built on selling their hours had told its regulator that clients would buy fewer of them. For half a century the problem had been how to make enough engineers. Now it was who would pay for their time.

As of October 2026

The account above rests on documents that will date quickly. What follows records where the numbers, the forecasts and the American visa stood when the book went to press.

In the year to March 2026 two of the largest firms shrank in dollar terms. According to *Forbes India*, TCS's revenue fell 0.5 per cent in US dollars and Wipro's 0.32 per cent.[86] At the same time the companies reported growing AI revenue: TCS an annualised figure above $2.3 billion by the end of the year, HCLTech $620 million of what it called 'Advanced AI' revenue, and Infosys $275 million in a single quarter.[87] Nasscom made its first estimate of the industry's AI revenue, at $10-12 billion.[88] Each company defines AI revenue its own way, so the figures cannot be added up or compared, and Gaurav Vasu of the research firm UnearthInsight put it at under 5-7 per cent of the top five firms' revenue.[89] Vijayakumar spoke of a deflationary effect of 3 to 5 per cent on HCLTech's guidance for the year to March 2027.[90] The brokers were blunter. A report by Motilal Oswal Financial Services, dated 4 May 2026, estimated that 12-15 per cent of the sector's revenue faced 'direct AI-driven displacement risk', and said: 'The cost of delivering an outcome has fallen sharply; these savings are currently being extracted almost entirely by enterprise buyers through productivity demands and vendor price pressure.'[91] Kotak Securities expected the industry's revenue growth to stay 'range-bound at 3 percent'.[92]

The route to America narrowed at the same time. On 19 September 2025 President Donald Trump signed Proclamation 10973, which said the H-1B programme, the visa that had carried Indian engineers to client sites since the 1990s, had been 'deliberately exploited to replace, rather than supplement, American workers', and that 'Information technology (IT) firms in particular have prominently manipulated the H-1B system'.[93] From 21 September, for twelve months, it barred the entry of H-1B workers whose petitions were not 'accompanied or supplemented by a payment of $100,000'.[94] The traffic was already falling: the National Foundation for American Policy, a pro-immigration research group, counted 4,573 initial-employment approvals in the US fiscal year 2025 for

the top seven Indian-based companies, 70 per cent fewer than in 2015.[95] On 18 September 2026 Proclamation 11069 extended the restriction for another year, and claimed that 'the largest IT staffing and outsourcing firms' had cut 'their combined H-1B registrations from 24,946 to 2,055, a 92 percent decrease'.[96] Those are a party's figures, offered to justify the policy.

The courts then stopped the fee without striking down either proclamation. On 8 June 2026 Judge Leo T. Sorokin of the federal district court in Massachusetts, in a suit brought by twenty states, held that the agencies' policy 'imposes a tax on H-1B petitions without the requisite delegation by Congress' and vacated it.[97] The First Circuit refused to suspend his order pending appeal.[98] A federal court in Washington, DC, had upheld the fee in 2025, and that ruling too was on appeal.[99] On 30 September 2026 Judge Haywood S. Gilliam Jr. in the Northern District of California barred the agencies from enforcing their policies under both proclamations until they had gone through the formal process of public notice and comment.[100] By mid-October the $100,000 could not be collected, three appeals were pending, and the Department of Homeland Security had proposed a fee of about $103,000 through the process the courts said had been skipped.[101]

The factory, meanwhile, had not stopped. By TCS's quarterly results as reported, about 14,000 freshers joined in the April-June quarter of 2026 and about 10,000 in July-September, and its headcount had risen for three quarters in a row, to 598,056.[102] Its chief human resources officer put the policy in five words: 'We are hiring for demand'.[103] In the long boom the firms had carried the bench as the cost of always being ready to bill. Now the largest of them was hiring only for the work in hand.

Notes to Chapter 19. The machines learn to code (2022-2026)

[1] Infosys, Form 20-F FY2024, Item 3.D.

[2] Infosys, Form 20-F FY2025, Item 3.D, HTML p. 15; post-cutoff material, checked in October 2026.

[3] Infosys, Form 20-F FY2026, Item 3.D, HTML p. 15; post-cutoff material, checked in October 2026.

[4] Infosys, Form 20-F FY2025, Item 3.D, HTML p. 15; Infosys, Form 20-F FY2026, Item 3.D, HTML p. 15; post-cutoff material, checked in October 2026.

[5] Infosys, Form 20-F FY2026, Item 5, HTML p. 61; post-cutoff material, checked in October 2026.

[6] Wipro, Form 20-F, FY2026, p. 11; post-cutoff material, checked in October 2026.

[7] Wipro, Form 20-F, FY2026, p. 33; post-cutoff material, checked in October 2026.

[8] Cognizant, Form 10-K 2025, p. 29; post-cutoff material, checked in October 2026.

[9] Infosys, Form 20-F FY2025, Item 4, HTML p. 47; post-cutoff material, checked in October 2026.

[10] Infosys, Form 20-F FY2026, Item 4, HTML p. 48; post-cutoff material, checked in October 2026.

[11] Infosys, Form 20-F FY2026, Item 4, HTML p. 48; post-cutoff material, checked in October 2026.

[12] Wipro, Form 20-F, FY2025, p. 33; post-cutoff material, checked in October 2026.

[13] Wipro, Form 20-F, FY2026, p. 36; post-cutoff material, checked in October 2026.

[14] Wipro, Form 20-F, FY2026, p. 73; post-cutoff material, checked in October 2026.

[15] Cognizant, Form 10-K 2025, p. 10; post-cutoff material, checked in October 2026.

[16] Cognizant, Form 10-K 2025, p. 10; post-cutoff material, checked in October 2026.

[17] PTI, 'India FY26 Tech Revenues Seen 6.1 Pc Up' (2026); post-cutoff material, checked in October 2026.

[18] Infosys, Form 20-F FY2025, HTML p. 52 (Infosys: 'Over 2,75,000 of our people are trained on AI at different levels of proficiency', in Indian digit grouping); Cognizant, Form 10-K 2025, p. 10 (Cognizant: 'upskilled more than 330,000 associates on GenAI' between July 2023 and the end of 2025); Infosys, Form 20-F FY2026, HTML p. 118 (Infosys, 2026: 84 per cent of its workforce 'now AI-aware'); Press reports on TCS job cuts (2025) (Outlook Business, 30 July 2025, relaying TCS: more than 550,000 employees trained in basic AI and more than 100,000 in advanced skills); post-cutoff material, checked in October 2026.

[19] Ganguly, 'Indian IT Braces for AI-Deflation' (2026); post-cutoff material, checked in October 2026.

[20] Ganguly, 'Indian IT Braces for AI-Deflation' (2026); post-cutoff material, checked in October 2026.

[21] Ganguly, 'Indian IT Braces for AI-Deflation' (2026); post-cutoff material, checked in October 2026.

[22] Ganguly, 'Indian IT Braces for AI-Deflation' (2026) (paraphrase); post-cutoff material, checked in October 2026.

[23] Suresh, 'India's Global Centre Market to Grow to $105 Billion by 2030' (2024) (the 2024 projection for 2030, set out in Chapter 15); Agarwal, 'The Big Boom' (2026); post-cutoff material, checked in October 2026.

[24] Infosys, Form 20-F FY2024, Item 6.D; Infosys, Form 20-F FY2025, Item 6.D; Infosys, Form 20-F FY2026, Item 6.D (offers and applications for the years to March 2024, 2025 and 2026); post-cutoff material, checked in October 2026. Applications per offer are the book's calculation.

[25] Das, 'Top 5 IT Cos' Net Hiring Down for FY26' (2026); post-cutoff material, checked in October 2026.

[26] *Business Today*, 'TCS Fresher Hiring Hits 25,000 in FY27' (2026); post-cutoff material, checked in October 2026.

[27] *Business Today*, 'TCS Fresher Hiring Hits 25,000 in FY27' (2026); post-cutoff material, checked in October 2026.

[28] Das, 'Top 5 IT Cos' Net Hiring Down for FY26' (2026); post-cutoff material, checked in October 2026.

[29] Das, 'Top 5 IT Cos' Net Hiring Down for FY26' (2026); post-cutoff material, checked in October 2026.

[30] Das, 'Top 5 IT Cos' Net Hiring Down for FY26' (2026); post-cutoff material, checked in October 2026.

[31] Das, 'Top 5 IT Cos' Net Hiring Down for FY26' (2026); post-cutoff material, checked in October 2026.

[32] Infosys, Form 20-F FY2024, Item 4.

[33] Press reports on delayed onboarding of graduates (2022); Infosys, Form 20-F FY2024, Item 6.D.

[34] Infosys, Form 20-F FY2024, Item 3.D and Item 4; Infosys, Form 20-F FY2023, Item 3.D; Infosys, Form 20-F FY2022, Item 3.D.

[35] Infosys, Form 20-F FY2025, Item 4, HTML p. 47; Wipro, Form 20-F, FY2025, p. 33; post-cutoff material, checked in October 2026.

[36] Das, 'Top 5 IT Cos' Net Hiring Down for FY26' (2026); post-cutoff material, checked in October 2026.

[37] Sharma, '"Everyone Is So Panicked"' (2025); post-cutoff material, checked in October 2026.

[38] Sharma, '"Everyone Is So Panicked"' (2025) (the article does not give the source of the figure); post-cutoff material, checked in October 2026.

[39] Sharma, '"Everyone Is So Panicked"' (2025); post-cutoff material, checked in October 2026.

[40] Sharma, '"Everyone Is So Panicked"' (2025); post-cutoff material, checked in October 2026.

[41] Sharma, '"Everyone Is So Panicked"' (2025); post-cutoff material, checked in October 2026.

[42] Sharma, '"Everyone Is So Panicked"' (2025); post-cutoff material, checked in October 2026. EY's finding is cited as *Rest of World* reported it.

[43] Hinduja, 'India's IT Freshers Grapple with Stagnant Pay' (2025); Reserve Bank of India, *Handbook of Statistics on Indian Economy* (exchange rate at the financial year's average, RBI series); post-cutoff material, checked in October 2026.

[44] Hinduja, 'India's IT Freshers Grapple with Stagnant Pay' (2025); Reserve Bank of India, *Handbook of Statistics on Indian Economy* (exchange rate at the financial year's average, RBI series); post-cutoff material, checked in October 2026.

[45] Infosys, Form 20-F FY2024, Item 6.D; Infosys, Form 20-F FY2025, Item 6.D; Infosys, Form 20-F FY2026, Item 6.D; post-cutoff material, checked in October 2026.

[46] Wipro, Form 20-F, FY2026, p. 73; post-cutoff material, checked in October 2026. The FY2026 report restates the March 2024 figure; Wipro's FY2024 report (Wipro) gave 'more than 234,000' for that date, the figure used in Chapter 18.

[47] Cognizant, Form 10-K 2025, p. 11 (2024 and 2025); Cognizant (10-K for 2023), p. 10 (2022); post-cutoff material, checked in October 2026.

[48] PTI, 'India FY26 Tech Revenues Seen 6.1 Pc Up' (2026) (net additions, direct employees, revenue and growth for the year to March 2026; Nambiar's 'non-linearity' between job growth of 2.3 per cent and revenue growth of 6.1 per cent, as paraphrased by the Press Trust of India); post-cutoff material, checked in October 2026.

[49] Nasscom, Strategic Review 2025 press release (2025); PTI, 'India FY26 Tech Revenues Seen 6.1 Pc Up' (2026); post-cutoff material, checked in October 2026. Nasscom had estimated in February 2025 that the industry added 126,000 people in the year to March 2025, taking its workforce to 5.8 million (Nasscom, Strategic Review 2025 press release (2025)). A year later its estimate of revenue for that year was raised from $282.6 billion to $297 billion; the report of the revision does not give the reason (PTI, 'India FY26 Tech Revenues Seen 6.1 Pc Up' (2026)).

[50] Das, 'Top 5 IT Cos' Net Hiring Down for FY26' (2026) (the companies' reported figures as compiled by *Business Standard*; Tech Mahindra's headcount also fell, by 1,108); post-cutoff material, checked in October 2026. TCS's own quarterly fact sheets were not available from the company's website when checked.

[51] Shinde and Das, 'TCS to lay off 2% of its workforce', *Business Standard* (2025) (*Business Standard*, 27 July 2025), quoting TCS's statement; live-checked 2026-10-15; post-cutoff material. *Deccan Herald* (29 July 2025, also Shinde and Das, 'TCS to lay off 2% of its workforce', *Business Standard* (2025)) printed the same statement with small differences: it adds "Towards this, a number of reskilling and redeployment initiatives have been underway', reads 'we will also be releasing associates', and has 'infrastructure, and realigning' where *Business Standard* has 'next-gen infrastructure and realigning'. The text here follows *Business Standard*. Reuters paraphrased the same statement as a plan to 'deploy artificial intelligence at scale, and realign its workforce model' (Bose and Dutta, 'India's New IT Playbook Begins with Layoffs' (2025)).

[52] Shinde and Das, 'TCS to lay off 2% of its workforce', *Business Standard* (2025) (*Business Standard*'s figures; *Deccan Herald* gives 12,261); Bose and Dutta, 'India's New IT Playbook Begins with Layoffs' (2025) (Reuters: roughly 12,200 out of more than 613,000); live-checked 2026-10-15; post-cutoff material.

[53] Shinde and Das, 'TCS to lay off 2% of its workforce', *Business Standard* (2025) (*Deccan Herald*, 29 July 2025, paraphrasing the company); live-checked 2026-10-15; post-cutoff material.

[54] Bose and Dutta, 'India's New IT Playbook Begins with Layoffs' (2025); post-cutoff material, checked in October 2026.

[55] Press reports on TCS job cuts (2025); post-cutoff material, checked in October 2026.

[56] Press reports on TCS job cuts (2025); post-cutoff material, checked in October 2026.

[57] Bose and Dutta, 'India's New IT Playbook Begins with Layoffs' (2025); post-cutoff material, checked in October 2026.

[58] Abraham, 'TCS Employees Allege Forced Resignations' (2025) (IIDEA's statement, and its claim that experienced, better-paid staff were targeted); post-cutoff material, checked in October 2026.

[59] Reddy, 'Forced Resignations, Unfair Bench Policy' (2025); post-cutoff material, checked in October 2026.

[60] Reddy, 'Forced Resignations, Unfair Bench Policy' (2025); post-cutoff material, checked in October 2026.

[61] Reddy, 'Forced Resignations, Unfair Bench Policy' (2025); post-cutoff material, checked in October 2026.

[62] Reddy, 'Forced Resignations, Unfair Bench Policy' (2025); post-cutoff material, checked in October 2026.

[63] Reddy, 'Forced Resignations, Unfair Bench Policy' (2025); post-cutoff material, checked in October 2026.

[64] Reddy, 'Forced Resignations, Unfair Bench Policy' (2025); post-cutoff material, checked in October 2026.

[65] Abraham, 'TCS Employees Allege Forced Resignations' (2025); post-cutoff material, checked in October 2026.

[66] Abraham, 'TCS Employees Allege Forced Resignations' (2025); post-cutoff material, checked in October 2026.

[67] Abraham, 'TCS Employees Allege Forced Resignations' (2025); post-cutoff material, checked in October 2026.

[68] Abraham, 'TCS Employees Allege Forced Resignations' (2025) (a single employee's estimate); post-cutoff material, checked in October 2026.

[69] *Business Standard* and *Outlook Business* on the TCS layoffs (October 2025) (*Business Standard*, 1 October 2025: NITES's email to Chief Minister Devendra Fadnavis and TCS's statement in reply); live-checked 2026-10-15; post-cutoff material. *Outlook Business* (2 October 2025, citing NDTV Profit, also *Business Standard* and *Outlook Business* on the TCS layoffs (October 2025)) gives the same first two sentences but ends the third 'as applicable to each individual circumstance', where *Business Standard* has 'as is due to them in each of the individual circumstances'. The statement answered NITES's claims about Pune; no TCS statement addressed specifically to KITU or to the Bengaluru conciliation has been located.

[70] Abraham, 'TCS Employees Allege Forced Resignations' (2025) (KITU's claims: the dispute filed on 29 July, the charge of illegal mass retrenchment, and the argument that resignations were sought to avoid the approval the Industrial Disputes Act requires of a firm with more than 100 employees); post-cutoff material, checked in October 2026.

[71] *Deccan Herald* on the unions' TCS complaint (2025) (*Deccan Herald*, 31 July 2025: 'Three days ago, a union submitted a letter. Our department is looking into it'; 'There are conditions. We cannot act in a hurry.'); live-checked 2026-10-15; post-cutoff material.

[72] Abraham, 'TCS Employees Allege Forced Resignations' (2025) (*The News Minute*, 7 August 2025: the meeting of 6 August, the participants and the adjournment to 8 September; TCS was represented by a general manager and an assistant manager in human resources, and KITU also by its secretary, Sooraj Nidiyanga); post-cutoff material, checked in October 2026. No report of the outcome of the conciliation, no failure-of-conciliation report, no reference to a labour court and no finding by the Karnataka Labour Department has been located in the press or in published court records as of 15 October 2026. Summaries of reports in *Deshabhimani* mention later rounds, but the reports themselves could not be retrieved and are not relied on here.

[73] *Deccan Herald* on the unions' TCS complaint (2025) (*Deccan Herald* with PTI, 12 October 2025; the parenthesis is as printed, after an opening ellipsis, and the 6,000 figure its paraphrase of what Kunnumal 'hinted'; the headcount figures are from TCS's fact sheet for the quarter as reported, the paper noting that TCS's stock-exchange filing on the results omitted headcount and attrition); live-checked 2026-10-15; post-cutoff material.

[74] *Deccan Herald* on the unions' TCS complaint (2025) (joint statement of KITU, the Association of IT Employees and the Union of IT and ITES Employees, as reported by *Deccan Herald*, 12 October 2025); live-checked 2026-10-15; post-cutoff material.

[75] AICTE, AI model curricula and 2025 announcements (Careers360, 23 December 2024); the figures for colleges and students are AICTE's own.

[76] AICTE, AI model curricula and 2025 announcements (PTI via Careers360, 6 June 2025). Sitharam also noted that AICTE had introduced a B.Tech degree in AI and data science in 2017. No AICTE circular or revised model curriculum giving effect to the announcement had been located when checked in October 2026.

[77] Sil, 'Sarvam and the Sovereign AI Dream' (2026); Reserve Bank of India, *Handbook of Statistics on Indian Economy* (exchange rate at the financial year's average, RBI series); consulted 2026-10-12; post-cutoff material; live-checked 2026-10-14.

[78] Sil, 'Sarvam and the Sovereign AI Dream' (2026); post-cutoff material, checked in October 2026.

[79] Sil, 'Sarvam and the Sovereign AI Dream' (2026); tele.net, 'IndiaAI Mission Selects Three New Start-Ups' (2025); post-cutoff material, checked in October 2026.

[80] tele.net, 'IndiaAI Mission Selects Three New Start-Ups' (2025).

[81] *Business Today*, 'India Selects Sarvam to Build Sovereign AI Model' (2025).

[82] Sil, 'Sarvam and the Sovereign AI Dream' (2026); post-cutoff material, checked in October 2026. The government's own release on what Sarvam was contracted to build has not been located, so the difference between the 120 billion parameters announced in 2025 and the 105 billion released in 2026 is left unexplained.

[83] Sil, 'Sarvam and the Sovereign AI Dream' (2026) (the magazine's assessment, based partly on the company's own evaluations); post-cutoff material, checked in October 2026.

[84] Sil, 'Sarvam and the Sovereign AI Dream' (2026) (as reported); post-cutoff material, checked in October 2026.

[85] GitHub, *Octoverse 2025* (*The Register*'s report of GitHub's *Octoverse* 2025 figures: India to add more than 35.6 million developers by 2030, reaching 57.5 million or more, against 54.7 million projected for the United States; 80 per cent of new users began using Copilot in their first week, which GitHub called 'part of the default developer experience'). These are a company's figures about its own platform; post-cutoff material, checked in October 2026.

[86] Ganguly, 'Indian IT Braces for AI-Deflation' (2026); post-cutoff material, checked in October 2026.

[87] Ganguly, 'Indian IT Braces for AI-Deflation' (2026) (company figures); post-cutoff material, checked in October 2026.

[88] PTI, 'India FY26 Tech Revenues Seen 6.1 Pc Up' (2026); post-cutoff material, checked in October 2026.

[89] Das, 'Top 5 IT Cos' Net Hiring Down for FY26' (2026); post-cutoff material, checked in October 2026.

[90] Ganguly, 'Indian IT Braces for AI-Deflation' (2026) (paraphrase); post-cutoff material, checked in October 2026.

[91] Ganguly, 'Indian IT Braces for AI-Deflation' (2026); post-cutoff material, checked in October 2026. The Motilal Oswal and Kotak Securities reports are cited as *Forbes India* reported them.

[92] Ganguly, 'Indian IT Braces for AI-Deflation' (2026); post-cutoff material, checked in October 2026.

[93] Proclamation 10973 (2025), p. 46027 (Proclamation 10973, 'Restriction on Entry of Certain Nonimmigrant Workers', signed 19 September 2025). The proclamation also cited a rise in the share of H-1B workers in IT from 32 per cent in the US fiscal year 2003 to more than 65 per cent on average in the previous five years, a '36 percent discount' for H-1B entry-level positions, unemployment rates of 6.1 and 7.5 per cent among recent graduates in computer science and computer engineering (p. 46027), and an unnamed 2017 study of the effect of foreign workers on American computer scientists' wages and employment (p. 46028); post-cutoff material, checked in October 2026.

[94] Proclamation 10973 (2025), p. 46028 (from 12:01 a.m. Eastern Daylight Time on 21 September 2025, for twelve months, and for entries after that date only); p. 46029 (new rules on wages ordered); post-cutoff material, checked in October 2026.

[95] NFAP, 'H-1B Petitions and Denial Rates in FY 2025' (2025), p. 6 (4,573 approvals; 70 per cent fewer than in 2015 and 37 per cent fewer than in 2024); p. 5 (TCS: 846 approvals in 2025, against 1,452 in 2024); post-cutoff material, checked in October 2026.

[96] Proclamation 11069 (2026), pp. 60497-60498 (Proclamation 11069, 18 September 2026: restriction extended to 21 September 2027; the $100,000 payment made for 'over 700 petitions'; a new rule weighting the H-1B lottery towards higher-paid workers for the 2027 season; the 92 per cent figure on p. 60498, where the firms are not named and the years compared are not stated in that sentence; the next sentence compares the 2025 and 2027 application seasons on a different measure); post-cutoff material, checked in October 2026.

[97] *California v. Mullin*, D. Mass., Memorandum and Order (8 June 2026), p. 1 (*State of California v. Mullin*, Civ. No. 25-13829-LTS, D. Mass., Memorandum and Order, Doc. 106, 8 June 2026: twenty states sued federal departments and officials 'for their role in implementing the Proclamation'); p. 26 (tax; separation of powers, Count IV); pp. 32, 33, 36 (Administrative Procedure Act: no notice and comment, excess of statutory authority, arbitrary and capricious); p. 42 ('The Policy implementing the Proclamation is declared unlawful and is VACATED in its entirety'); *California v. Mullin*, D. Mass., Judgment (8 June 2026), pp. 1-2 (Judgment, Doc. 107: ten agency documents, among them memoranda of 20 September 2025, question-and-answer pages, a fee schedule and the payment website, vacated 'insofar as they impose a $100,000 payment obligation on H-1B petitions'); checked against the court record on 15 October 2026; post-cutoff material. Proclamation 10973 itself was not vacated; an early summary by the law firm Clark Hill (Law-firm summaries of the H-1B fee litigation (2026), 9 June 2026) which said that it was is mistaken on that point.

[98] *Global Nurse Force v. Trump*, N.D. Cal., Order (30 September 2026), p. 2 (*Global Nurse Force v. Trump*, No. 25-cv-08454-HSG, N.D. Cal., Order, Dkt. 130, 30 September 2026, citing *California v. Mullin*, 183 F.4th 42, 46-52 (1st Cir. 2026), stay pending appeal denied); checked against the court record on 15 October 2026; post-cutoff material.

[99] *Global Nurse Force v. Trump*, N.D. Cal., Order (30 September 2026), p. 2 (citing *Chamber of Commerce v. DHS*, 815 F. Supp. 3d 73, 79 (D.D.C. 2025), 'currently on appeal' in the D.C. Circuit); checked against the court record on 15 October 2026; post-cutoff material.

[100] *Global Nurse Force v. Trump*, N.D. Cal., Order (30 September 2026), pp. 2, 33-34 (the judge's opening words, that the case was 'not about whether the challenged Proclamation is good or bad'; the Agency Policies vacated and remanded; enforcement of policies under Proclamations 10973 and 11069 barred 'until notice-and-comment rulemaking under the APA and regulatory flexibility analyses' are carried out; after the Massachusetts ruling 'the $100,000 fee is no longer in effect'); merits at pp. 26-30; Ninth Circuit appeal no. 26-6667 (Dkt. 138, 9 October 2026); checked against the court record on 15 October 2026; post-cutoff material.

[101] Law-firm summaries of the H-1B fee litigation (2026) (DiRaimondo & Schroeder, 5 October 2026: the proposed rule); post-cutoff material, checked in October 2026.

[102] Thapliyal, 'TCS Adds 4,258 Employees in Q2' (2026); Shinde, 'TCS Onboards 10,000 Freshers in Q2 FY27' (2026); post-cutoff material, checked in October 2026.

[103] Shinde, 'TCS Onboards 10,000 Freshers in Q2 FY27' (2026); post-cutoff material, checked in October 2026.

Epilogue. India in 2035

Rishabh Mishra's course at the Indian Institute of Information Technology, Design and Manufacturing in Jabalpur was due to end in May 2026. He had joined in 2022 to study computer science, write code and, in the words of the *Rest of World* reporter who told his story, 'one day make it to Silicon Valley'. By December 2025, according to the article, fewer than a quarter of his 400 classmates had job offers. 'It is really bad out there,' he said.[1] Earlier that year, at the other end of the same pipeline, the trainee whom the prologue called Gayatri had been let go from Infosys's training campus in Mysuru, four months after a joining date she had waited nearly two years for. 'They wasted two and a half years of my career,' she said.[2] Between them they stand at the two ends of the book's present: a graduate who cannot get in, and a recruit who got in and was sent home.

What their degrees will be worth by 2035 is the question this epilogue takes up. The furthest official projection is a NITI Aayog report of October 2025, *Roadmap for Job Creation in the AI Economy*, prepared by the Indian government's policy think-tank with the National Association of Software and Service Companies (Nasscom), the industry's lobby group, and the consultancy BCG.[3] It offers two endpoints for 2031. In what it calls a business-as-usual case, 'the headcount in the tech services sector could go down from 7.5 to 8 million in 2023 to 6 million by 2031'. If India acted on the report's advice, the total 'could swell to 10 million'.[4] The numbers need caution. On one page the report calls its base the employment of the tech services industry, and on another a 'total tech workforce' that includes 'tech jobs in other sectors', which is not the industry headcount that Nasscom publishes.[5] And the report was written to argue for a new national programme of AI teaching and retraining, so its two paths are framed to show the cost of not adopting it.[6] That makes the numbers advocacy, published under a government's name and co-written by the industry that would benefit from the spending. The 2035 of this epilogue's title is therefore the book's own horizon, about ten years from the time of writing.[7]

What the history shows

The book began with a working argument. India's software profession, it proposed, was shaped less by technology than by three constraints: scarce foreign exchange, a surplus of English-educated engineers, and an American labour market hungry for them. The result priced engineering by the hour and grew by adding people, and AI attacks exactly that model. On the evidence of the chapters, the verdict is mixed.

The first constraint holds best. In the 1970s the Department of Electronics allowed computers to be imported 'exclusively meant for software export'.[8] The policy of 1986 still required an importer to 'earn 250% of the cost of the imported computer within 4 years', and the budget of July 1991 extended a tax concession to software so that 'the exports of this industry should register rapid growth'.[9][10] Selling programmers' time abroad was the export the rules made easiest. But scarcity built skills as well as exporting labour: because software could not be imported, in the account of V. Rajaraman, the historian of Indian computing, Indian hardware firms wrote their own, and 'a large number of systems programmers in India became experts in UNIX development and usage'.[11] And a trade justified by dollars kept fewer of them than its headline figures suggested: the development scholar Richard Heeks put net earnings at around 40 per cent of the gross.[12]

The second constraint is where the strongest counter-evidence lies. In the early 1990s the World Bank found not a surplus but a shortage: 'the demand for trained software personnel far outstrips the supply'.[13] The surplus came later, and it was built: on one reading, by decades of public investment in science and engineering in Bangalore (Bengaluru since 2014), and then by private colleges that multiplied the supply until, by 2016-17, on an analysis of the regulator's data by the education charity Avanti Fellows, 49.5 per cent of approved undergraduate engineering

seats had no student in them.[14][15] On this book's reading, the surplus was less a starting condition than something the industry's demand called into being.

The third constraint is real, but it did not stay a demand for people. The economists Gaurav Khanna and Nicolas Morales built a model in which Indian students enrolled in engineering 'to gain employment in the rapidly growing US IT industry via the H-1B visa program', with those the visa cap kept out 'enabling the growth of an Indian IT sector'.[16] The share of work done at the client's site then fell from 90 per cent in 1988 to 56 per cent in 2000.[17] The constraints set the price of the work; the wire decided how much of it could be done in India.

The middle of the argument is the best supported. Heeks wrote in 1998 that in custom software services 'earnings are directly proportional to the number of people working in the industry', and the pyramid, the bench and the training campuses were built on that arithmetic.[18] Even in the year to March 2008, fixed-price projects brought Infosys only 31 per cent of its services revenue.[19] Whether AI ends that model is the last clause, and the one the rest of this epilogue weighs.

The book's judgement

What follows is the book's own argument, built from the evidence of its chapters. It is that the hourly bargain is ending as a way to grow; that what replaces it depends on who keeps the hour a machine saves; and that the cost falls first on the bottom rung.

The first part rests on the companies' own words. In three successive annual reports to the American regulator, from 2024 to 2026, Infosys said that clients were demanding higher commitments around productivity because of what AI could do, and that it was promising those improvements in its contracts in advance.[20] In 2026 it added that clients were 'selectively insourcing capabilities as AI improves productivity and reduces dependency on external labor-based models'.[21] Wipro predicted that 'Hiring will shift from volume to skills', and TCS's head of human resources said in October 2026 that the company was 'hiring for demand'.[22] Nasscom's own figures show the result: in the year to March 2026 the industry's revenue rose 6.1 per cent and its jobs 2.3 per cent.[23] For fifty years revenue and headcount had risen together because revenue was, roughly, heads multiplied by a billing rate. A gap between them is small, but it is the bargain coming apart. It does not mean the offices empty. By 2026 headcount had stalled, not collapsed. It means that growth no longer requires the factory to keep hiring at the old pace.

The second part is the pricing mechanism the book has traced since the Y2K years. Under time-and-materials billing a saved hour belongs to the client, who simply pays for fewer of them. Under a fixed price the vendor keeps what it saves, which is why Infosys told investors in 1999 that it wanted more fixed-price work and reported in 2002 that the share of such work had risen.[24] In a captive centre owned by the client there is nothing to bargain over: the parent keeps the saving outright (Chapter 15). And a product firm can sell the saving itself. Freshworks, an Indian-built software company with most of its staff in India, told investors that its Freddy AI Agent 'offers always-on, autonomous, personalized resolutions to customer and employee queries', which is much of what India's help desks had been paid by the hour to do.[25] Each of these owners would produce a different industry. In 2026 the evidence pointed one way. The vendors had promised the savings before making them, deals for the same work were worth, by HCLTech's chief executive's rough estimate, about a fifth less, and a broker's report said the savings were 'currently being extracted almost entirely by enterprise buyers through productivity demands and vendor price pressure'.[26] For now, the client owns the saving.

The companies' answer deserves its strongest statement. They argue that AI creates more work than it removes, and they had numbers to show for it. By the end of the year to March 2026 TCS reported AI revenue at an annualised rate above $2.3 billion, HCLTech reported $620 million for the year and Infosys $275 million in a single quarter, and Nasscom made a first estimate of the industry's AI revenue at $10-12 billion. 'Our expectation is AI will be net accretive,' TCS's chief executive said.[27] Each firm defines AI revenue its own way, and one analyst put it at under 5-7 per cent of the five largest firms' revenue.[28] The new work is real. But on the evidence so far it is small beside the old work being repriced, and Wipro expected the industry to move towards 'outcome-based pricing', charging for a result rather than for the hours spent on it.[22] New work sold that way is the point of the judgement, not an exception to it.

The third part follows from the second. The hours a client is most willing to stop paying for are the junior ones: the coding, testing and documentation that the book's chapters followed from the Y2K years, broken into documented steps that any trained graduate could follow. Those are the tasks for which Infosys described its code

assistant in 2024, and the tasks the NITI roadmap expects to go first: in an IT services company, it says, AI 'typically translates to a shrinking of entry-level roles involving QA and support'.[29][30] The report illustrates this with invented characters, a junior test engineer in Pune and a support agent in Chennai.[31] The starting salary had barely moved in a decade, and TCS's chief executive said, in a newspaper's paraphrase, that freshers take 'up to nine months to come on the floor, while a lateral can deliver immediately'.[32] Infosys's offers had recovered from their trough, but in the year to March 2026 they were still well under half the 222,500 of the boom year to March 2022.[33] The rung is still there. It is narrower, and the reason to expect it to widen again is weaker than it was.

The machines have been forecast to take this work before, and the forecast was wrong for thirty years. In 1994 the World Bank's study of the Indian industry warned that 'Demand for these skills is likely to be eroded by tools that automate many of these tasks', and added that 'a long-term decline in demand for routine or low-level programmers has been forecasted'.[34] In 1998 Heeks named 'automation from new technology' among the risks to the trade and saw a trap in it: clients sent work offshore only when it was tightly specified, and such projects were 'more amenable to automated software tools'.[35] Instead exports kept growing fast, and by 2026 Nasscom counted 5.95 million direct employees.[36][23] The warnings were wrong because demand absorbed the productivity. Each tool that made an engineer faster made software cheaper, cheaper software found new buyers, and a firm that sold hours could use the saving to win more work. The book's one documented case of the pattern in India is small but exact. When the railway booking office in New Delhi was computerised in 1986, Rajaraman records, there was 'no reduction of staff; in fact there was an increase as more counters were operated with more clerks working simultaneously'.[37] On the book's reading, the machine raised each clerk's output and demand that the queues had held down absorbed it.

For that to happen again, three things would have to be true. Demand for software at the new, lower price would have to grow faster than the hours needed for each piece of it fall. The saving would have to stay with someone who reinvests it in hiring, rather than being taken by the client in advance. And the new work would have to reach the firms and the graduates the factory was built to supply. On the evidence of 2026 the first is uncertain: the industry still grew, but HCLTech said winning the same revenue took 'at least 25-30 percent more effort', TCS's and Wipro's revenue fell in dollar terms, and one broker expected growth to stay 'range-bound at 3 percent'.[38] The second points the other way, for as long as the clients keep the saving. The third is partly true: Indian engineers were still being hired in large numbers, but increasingly by the captive centres, which by the year to March 2026 employed about 2.36 million people by an industry count, and the graduates still had to get past a narrower bottom rung.[39] The book's reading is that demand may again absorb much of the productivity, and that Indian engineers may again be employed in large numbers, but not through the bargain that built the factory. Heeks's trap has closed on the hourly vendor, and most tightly at the bottom rung.

Four things to watch

The four futures that follow are signposts, not forecasts. They are not exclusive, and they are not equally likely. The evidence in this book currently favours the third, in which the client moves in, and the first, in which the pyramid thins, because both follow from the client keeping the saving. It gives least support to the second, which needs the price of each engineer to rise while the price of each deal falls. The fourth is real but small.

The thinning pyramid

An Indian services firm has long been described as a pyramid, with a wide base of junior engineers billed at rates well above their pay. In this future the base shrinks. The firms survive, smaller at the bottom and older on average, and the colleges keep producing graduates trained for a base that is narrowing, so that the prologue's waiting fresher becomes the normal case. The evidence for it is the pricing evidence above. The supply side had begun to adjust before the tools arrived: by the Avanti Fellows analysis, approved undergraduate intake peaked at about 1.7 million seats in 2014-15 and fell to about 1.25 million by 2021-22, with roughly a third of the seats left empty.[40] The counter-evidence is that by 2026 the pyramid had not visibly thinned: Infosys's offers had tripled from their trough, TCS was still taking freshers by the tens of thousands, and Nasscom still reported net hiring, while the delayed joining dates began in 2022, two years before Infosys's filings first mentioned AI code generation.[41][42] The signposts are fresher offers as a share of headcount at the five largest firms falling year after year, revenue per employee rising while

headcount stays flat, and the share of engineering graduates in engineering jobs falling, which is why the missing placement data matter.

The diamond

In this future the shape becomes a diamond: fewer people at the bottom and a wide, better-paid middle that directs and checks what the tools produce. The image is the staffing executive Kamal Karanth's, in a Reuters Breakingviews column's paraphrase.[43] The NITI roadmap expects 'the price-per-FTE for the industry to grow 5-8% annually over the next few years', FTE meaning one full-time person's working time.[44] The weight of the evidence is against it for now. Price per worker and value per deal measure different things, but the deflation reported seven months after NITI's estimate points the other way, and nothing published by 2026 settles which is right.[45] For education this future is kinder but harder: it needs fewer graduates who can follow a specification and more who can judge whether a system is right. The signposts would be revenue per employee and average pay rising together, lateral hiring outpacing campus hiring, and a revised curriculum actually taught: the All India Council for Technical Education (AICTE), the regulator, declared 2025 its 'Year of Artificial Intelligence', but no revised model curriculum had appeared by the time of writing.[46]

The client moves in

In this future multinationals stop buying Indian engineers' time from Indian vendors and employ the engineers themselves, in global capability centres (GCCs), the Indian offices that a foreign company owns and staffs. It has a long history, from the software design centre that Texas Instruments, by its own account, began operating in Bangalore in 1985 to the centres of Chapter 15.[47] It is the future the evidence most clearly shows arriving. By the year to March 2026, on Nasscom and Zinnov's count, GCC revenue was already close to what the two had projected for 2030; Infosys listed clients 'setting up or expanding their own GCC in India' as a risk; and Karnataka set a target of 500 new GCCs by 2029.[39][48][49] Indian engineers keep their jobs but change employers, and the national champions matter less. The counter-evidence is about the numbers themselves: every GCC figure comes from Nasscom and from Zinnov, a consultancy that sells advice to GCCs, and Nasscom has restated its counts before.[50] The signposts would be GCC headcount growing while the services firms' stays flat, services firms earning more from building and running centres for clients, and states beyond Karnataka adopting their own policies under the national framework that the 2025-26 Union Budget promised.[51]

The builder state

In this future barriers abroad and ambitions at home turn the industry inward. The H-1B route was contested by 2026: a $100,000 fee proclaimed in September 2025, the administration's claim of a 92 per cent fall in the largest outsourcing firms' registrations, and courts that had stopped the fee being collected without striking down the proclamations.[52] The Indian firms' approved petitions had begun to fall before the fee, by the count of the National Foundation for American Policy, a pro-immigration research group.[53] At home the state that had built Aadhaar, the biometric identity system, and UPI, the payments network, had launched the IndiaAI Mission and backed Sarvam's first foundation models, and Nasscom expected domestic revenue to grow faster than the industry as a whole.[54][55] The counter-evidence is serious. *Inc42* judged that Sarvam's models trailed the leading foreign ones in 'raw reasoning depth', and the NITI report records that India's share of granted AI patents fell from 8-10 per cent in 2010 to under 5 per cent in 2023.[56][57] Building identity and payments systems was a feat of engineering and organisation; building frontier AI needs far more capital. The signposts would be spending on AI sustained beyond one budget cycle, Indian-language models in use in state services, and a rising share of graduates hired by domestic product firms and public bodies. GitHub's projection of Indian developers on its platform counts accounts on one company's service, not jobs, but a rising number of Indians writing code for Indian users would show up there first.[58]

Reading the signposts

Several of these signposts are numbers that India does not collect well. AICTE's data on graduates placed in jobs are unusable, by the Avanti Fellows analysis; Nasscom's headcount is an industry body's estimate; GCC counts come

from parties that sell to GCCs; AI revenue has no common definition.[59] Projections in this industry have also been overtaken within a few years: Nasscom and Zinnov's GCC projection for 2030 was nearly reached by 2026, and NITI's rising price per engineer was contradicted by reported deflation seven months later.[60][61] A reader in 2030 who wants to know which future is arriving will need, first, an independent count of software employment, a placement series for engineering graduates, and revenue per employee for the large firms on comparable terms.

The joining date

The book began with a joining date that kept moving. For Gayatri the date came after nearly two years, and the job lasted four months. She told the reporter it had been her dream job.[62] For Rishabh Mishra and his classmates in Jabalpur, the reporting ends before any date. Some of them were thinking of further study instead. Mishra was doubtful: 'after one year, if you return to the job market, your degree is even more irrelevant'.[63]

His sentence is about education as much as about jobs. For half a century the engineering degree in India was a ticket to a particular kind of work: billable hours, learned first on a company training campus and then sold to a client by the month. The colleges, the entrance examinations, the coaching towns and the company campuses were built around that ticket. If the book's judgement is right, the ticket will still buy work in 2035, but less often at the bottom, more often from a client's own centre, and on terms set by whoever keeps the hour the machine saves. The engineer factory was built on purpose, by the state and the companies and millions of families, in answer to the questions of its time. The question for the next ten years is what a joining date means when the factory's product is no longer hours.

Notes to Epilogue. India in 2035

[1] Sharma, '"Everyone Is So Panicked"' (2025); post-cutoff material, checked in October 2026.

[2] Ganeshan, '"Thrown on the Road"' (2025).

[3] NITI Aayog, *Roadmap for Job Creation in the AI Economy* (2025), PDF pp. 1, 5; post-cutoff material, checked in October 2026.

[4] NITI Aayog, *Roadmap for Job Creation in the AI Economy* (2025), PDF p. 6; post-cutoff material, checked in October 2026.

[5] NITI Aayog, *Roadmap for Job Creation in the AI Economy* (2025), PDF p. 11, printed p. 12; post-cutoff material, checked in October 2026.

[6] NITI Aayog, *Roadmap for Job Creation in the AI Economy* (2025), PDF p. 7; post-cutoff material, checked in October 2026.

[7] NITI Aayog, *Roadmap for Job Creation in the AI Economy* (2025), printed p. 14. The report's only mention of 2035 is a proposed vision statement for its talent mission: 'To position India as the global epicentre of AI talent by 2035'. That is a goal, not an estimate; no forecast of Indian software employment for 2035 has been located. The report is also inconsistent about its upside: 'by 2030, we can add 2 million more new jobs' on one page (PDF p. 11), up to 4 million by 2031 in the foreword of Debjani Ghosh, a NITI Distinguished Fellow (PDF p. 5); post-cutoff material, checked in October 2026.

[8] Rajaraman, *History of Computing in India* (2012), pp. 18, 28.

[9] Rajaraman, *History of Computing in India* (2012), p. 32.

[10] Singh, *Budget 1991-92* speech (1991), p. 20.

[11] Rajaraman, *History of Computing in India* (2012), p. 40 (Rajaraman's account of the years 1978-84, in which he was a participant).

[12] Heeks, 'The Uneven Profile of Indian Software Exports' (1998), p. 2.

[13] Hanna, *Exploiting Information Technology for Development* (1994), p. 50.

[14] RoyChowdhury, 'Livelihood and Income' (c. 2011), p. 1.

[15] Avanti Fellows, 'Technical Seats' (2026); post-cutoff material, checked in October 2026.

[16] Khanna and Morales, 'The IT Boom and Other Unintended Consequences' (2016), abstract.

[17] Khanna and Morales, 'The IT Boom and Other Unintended Consequences' (2016), p. 5.

[18] Heeks, 'The Uneven Profile of Indian Software Exports' (1998), pp. 14-15.

[19] Infosys, Form 20-F FY2008, Item 3.D.

[20] Infosys, Form 20-F FY2024, Item 3.D (filed June 2024); Infosys, Form 20-F FY2025, Item 3.D, HTML p. 15 (filed 1 July 2025); Infosys, Form 20-F FY2026, Item 3.D, HTML p. 15 (filed 15 June 2026); post-cutoff material, checked in October 2026.

[21] Infosys, Form 20-F FY2026, Item 5, HTML p. 61; Ganguly, 'Indian IT Braces for AI-Deflation' (2026); post-cutoff material, checked in October 2026.

[22] Wipro, Form 20-F, FY2026, p. 33 (Wipro); Shinde, 'TCS Onboards 10,000 Freshers in Q2 FY27' (2026) (Sudeep Kunnumal, *Business Standard*, 8 October 2026); post-cutoff material, checked in October 2026.

[23] PTI, 'India FY26 Tech Revenues Seen 6.1 Pc Up' (2026) (Nasscom's estimates for the year to March 2026: revenue $315 billion, up 6.1 per cent; 5.95 million direct employees, jobs up 2.3 per cent); post-cutoff material, checked in October 2026.

[24] Infosys, Form F-1 (1999), pp. 32, 44 (1999); Infosys, Form 20-F FY2002, p. 6 (2002).

[25] Freshworks, Form 10-K FY2024, Item 1 (Freddy AI Agent); Item 1 also gives about 83 per cent of employees in India at the end of 2024.

[26] Infosys, Form 20-F FY2025, Item 3.D; Infosys, Form 20-F FY2026, Item 3.D (productivity commitments); Ganguly, 'Indian IT Braces for AI-Deflation' (2026) (C. Vijayakumar's rough estimate that a $100 million deal 'would be much less today'; the Motilal Oswal Financial Services report of 4 May 2026, as reported by *Forbes India*); post-cutoff material, checked in October 2026.

[27] Ganguly, 'Indian IT Braces for AI-Deflation' (2026) (company figures for AI revenue, as reported by *Forbes India*; Krithivasan on TCS's earnings call); PTI, 'India FY26 Tech Revenues Seen 6.1 Pc Up' (2026) (Nasscom's AI revenue estimate); post-cutoff material, checked in October 2026.

[28] Das, 'Top 5 IT Cos' Net Hiring Down for FY26' (2026) (Gaurav Vasu of UnearthInsight, in *Business Standard*); post-cutoff material, checked in October 2026.

[29] Infosys, Form 20-F FY2024, Item 4 ('a code assistant' for 'tasks like coding, testing, and documentation').

[30] NITI Aayog, *Roadmap for Job Creation in the AI Economy* (2025), printed p. 8; post-cutoff material, checked in October 2026.

[31] NITI Aayog, *Roadmap for Job Creation in the AI Economy* (2025), printed p. 7 (a junior QA engineer in Pune named Rekha and a first-line support agent in Chennai named Aman; the report gives no surnames and no sign that they are real people); post-cutoff material, checked in October 2026.

[32] Hinduja, 'India's IT Freshers Grapple with Stagnant Pay' (2025) (Neeti Sharma of TeamLease Digital: Rs 2-2.5 lakh a year in 2014, Rs 3-3.5 lakh in 2025); Das, 'Top 5 IT Cos' Net Hiring Down for FY26' (2026) (Krithivasan, in *Business Standard*'s paraphrase); post-cutoff material, checked in October 2026.

[33] Infosys, Form 20-F FY2022, Item 6 (222,500 offers, year to March 2022); Infosys, Form 20-F FY2026, Item 6.D (87,286, year to March 2026); post-cutoff material, checked in October 2026.

[34] Hanna, *Exploiting Information Technology for Development* (1994), p. 33.

[35] Heeks, 'The Uneven Profile of Indian Software Exports' (1998), pp. 7, 14.

[36] Heeks, 'The Uneven Profile of Indian Software Exports' (1998), p. 2.

[37] Rajaraman, *History of Computing in India* (2012), p. 38.

[38] Ganguly, 'Indian IT Braces for AI-Deflation' (2026) (HCLTech's Vijayakumar; TCS revenue down 0.5 per cent and Wipro's 0.32 per cent in US dollars in the year to March 2026; Kotak Securities, as reported by *Forbes India*); post-cutoff material, checked in October 2026.

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[57] NITI Aayog, *Roadmap for Job Creation in the AI Economy* (2025), printed pp. 9-10; post-cutoff material, checked in October 2026.

[58] GitHub, *Octoverse 2025*; post-cutoff material, checked in October 2026.

[59] Avanti Fellows, 'Technical Seats' (2026); post-cutoff material, checked in October 2026.

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[61] NITI Aayog, *Roadmap for Job Creation in the AI Economy* (2025), printed p. 13; Ganguly, 'Indian IT Braces for AI-Deflation' (2026); post-cutoff material, checked in October 2026.

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Timeline

This timeline lists 162 dated events that the chapters describe, from the founding of the Indian Statistical Institute in 1931 to the autumn of 2026. Each entry gives the date as the chapters state it, a short description, the chapters in which the event appears, and the source IDs from the book's source register, where full references are given. Where the sources disagree on a date, the entry follows the chapter and notes the other date. Indian fiscal years run from April to March and are written FY1997-98. Figures from Nasscom, the industry's lobby group, from Zinnov, a consultancy that sells to global capability centres, and from companies about themselves are labelled as theirs.

Date	Event (Chapter)	Sources
1931	Indian Statistical Institute (ISI) founded in Calcutta under P.C. Mahalanobis (Ch. 1)	S0010
1944	Government of British India begins paying for Indian students at American universities (Ch. 2)	S0003
1946	Sarker Committee's interim report proposes four higher technical institutes to the standard of MIT (Ch. 2)	S0049, S0003
1950	National Sample Survey set up; ISI opens a computing laboratory under S.K. Mitra (Ch. 1)	S0007, S0010
1950-1954	Mahalanobis's talks in Washington to buy a UNIVAC fail, partly after a 1953 US Embassy dispatch calls him a Soviet sympathiser (Ch. 1)	S0007
1951	First Indian Institute of Technology opens at Kharagpur (Ch. 2)	S0003
1954-02	At a lunch hosted by Nehru, Mahalanobis presses the Soviet ambassador for help with a computer (Ch. 1)	S0007
1955-07	HEC-2M reaches Calcutta and is working by August 1955 (Rajaraman; a historian dates arrival to early 1956) (Ch. 1)	S0001, S0010, S0007
1956-1961	Second Five Year Plan, drafted by Mahalanobis (Ch. 1)	S0001
1958-03	Government adopts the Scientific Policy Resolution (Chs. 1, 3)	S0001
1958	IIT Bombay opens with Soviet support (Ch. 2)	S0003
1958-12	Soviet URAL computer unveiled at ISI, Calcutta (Ch. 1)	S0007, S0010, S0001
1959-02	TIFR completes assembly of its full-scale computer; Morton Nadler arrives at ISI on a two-year contract (Ch. 1)	S0001, S0025, S0007
1960-02	TIFR computer (later TIFRAC) commissioned for routine work (Ch. 1)	S0001, S0052
1960-07	IIT Kanpur admits its first 100 undergraduates (Ch. 2)	S0031
1961	India, the United States and nine American universities agree to set up the Kanpur Indo-American Program (KIAP) (Ch. 2)	S0003

Date	Event (Chapter)	Sources
1962	Border war with China (Chs. 1, 3)	S0001
1963	Government appoints an electronics committee under Homi Bhabha (Chs. 1, 3)	S0052
1963-07	IBM 1620 flown to Kanpur on a chartered DC-7 (installed August 1963); V. Rajaraman joins the Kanpur faculty (Ch. 2)	S0003, S0001
1964 (mid)	CDC 3600 installed at TIFR, bought for $1.5 million with USAID funds (Chs. 1, 3)	S0001
1965	IIT Kanpur begins master's-level teaching with a computer-science option, the first in India (1965 per IIT Kanpur; 1966 per Rajaraman) (Ch. 2)	S0001, S0043
1965-05	IIT Kanpur's first class graduates: 66 of the original 100 (Ch. 2)	S0031
1966	Bhabha Committee report ready; Bhabha dies in an air crash before formally submitting it (Ch. 3)	S0001, S0052
1967	ECIL set up at Hyderabad under the Department of Atomic Energy (Ch. 3)	S0001
1967-03-23	George Fernandes asks in the Lok Sabha about LIC's computers; Morarji Desai says there will be no retrenchment (Ch. 3)	S0019
1968	Tata Consultancy Services (TCS) constituted, building on a Tata group computer centre running since 1967 (Ch. 4)	S0039, S0235, S0231
1969	F.C. Kohli moves from Tata Electric to TCS as general manager (Ch. 4)	S0228, S0038, S0036
1969-07-19	Labour Ministry sets up the Committee on Automation under R. Venkataraman (Ch. 3)	S0001, S0020
1970	Department of Electronics (DoE) established; computer import licensing moves to it (Chs. 2, 3, 4, 8)	S0001
1970	IIT Kanpur undergraduates write what Bassett calls the first compiler written in India (Ch. 2)	S0003
1971	India-Pakistan war and creation of Bangladesh; US embargoes electronics exports to India (Chs. 3, 4)	S0001, S0003
1972	Software Export Scheme eases hardware imports for exporters (Aspray; Rajaraman dates the first policy paper to 1974) (Ch. 4)	S0230
1972	NCSDCT set up at TIFR with DoE and UNDP funds (formally launched 1975) (Chs. 3, 4)	S0001, S0013
1972	Narendra Patni's Data Conversion Inc. begins sending American documents to Poona to be keyed onto tape (Ch. 4)	S0229, S0044
1972	KIAP ends (Ch. 2)	S0031, S0030
1972-05-18	Committee on Automation signs its recommendations, including prior agreement with workers (Ch. 3)	S0020, S0021
1973	FERA amendment requires most foreign firms to dilute equity to 40 per cent (Chs. 4, 5)	S0056, S0026
1973-1974	TCS-Burroughs partnership formed; first offshore work for Burroughs clients (Chs. 4, 6)	S0039, S0038
1974	First policy paper on importing computers for software export (Chs. 3, 4; Epilogue)	S0001
1975-01	Kohli's Computer Society of India address at Ahmedabad (Ch. 4)	S0231
1976	Computer Maintenance Corporation (CMC) created by the DoE (Chs. 3, 5)	S0066

Date	Event (Chapter)	Sources
1976-08	Hindustan Computers Ltd (HCL) founded by Shiv Nadar and colleagues; the founders' calculator firm Microcomp held 49 per cent (Ch. 5)	S0081, S0358
1977-03	Janata government takes office (Chs. 3, 5)	S0056
1978	IIT Kanpur starts India's first undergraduate degree in computer science, with 20 seats; over 100,000 sit the IIT entrance exam for about 1,500 places (Chs. 2, 5, 7)	S0001
1978	Minicomputer policy opens computer manufacture to private Indian firms (Chs. 3, 4, 5, 8)	S0001
1978-06-01	IBM closes its Indian operations; CMC takes over maintenance (Chs. 3, 4, 5, 6, 8, 9, 15; Epilogue)	S0055, S0066, S0001, S0237
1980	Panel on Computer Manpower Development under Rajaraman proposes the MCA degree (Chs. 5, 6, 7)	S0001
1980	Twenty-one Indian firms export about $4 million of software, mostly onsite (Chs. 4, 5, 6)	S0230
1980-1985	Sixth Plan: Karnataka, Maharashtra, Tamil Nadu and Andhra Pradesh permit self-financing private technical institutions (Ch. 7)	S0247
1980-81	India has 149 engineering and technology colleges (Chs. 7, 8, 12)	S0248
1981	NIIT founded by Rajendra Pawar and Vijay Thadani (Chs. 5, 6, 7)	S0065, S0244
1981-07	Infosys incorporated by Murthy and six colleagues (Chs. 2, 4, 5, 6, 9, 10)	S0089, S0034
1982	Asian Games in Delhi computerised by NIC on DCM terminals under Rajiv Gandhi (Chs. 5, 8)	S0001
1984	Bank unions observe an 'anti-computerisation year' (Chs. 5, 8)	S0001
1984	Indira Gandhi assassinated; Rajiv Gandhi becomes prime minister (Chs. 5, 8)	S0001
1984-11-19	New computer policy; government expects software exports of $258 million by 1990 (Chs. 5, 6, 8)	S0237, S0241
1985	Texas Instruments begins operating a software design centre in Bangalore (Chs. 4, 6, 7, 8, 10, 14, 15; Epilogue)	S0061
1985	TI's satellite equipment delivered by bullock cart to Sona Towers, Bangalore (Chs. 8, 10)	S0249, S0250
1985-10-28	Citicorp Overseas Software Ltd (COSL) inaugurated at SEEPZ, Bombay (Chs. 6, 8)	S0240
1985-11-15	Pilot computerised passenger reservation system (IMPRESS) launched at New Delhi (Ch. 8)	S0396
1986	VSNL set up for overseas communications (Ch. 8)	S0001
1986	Computerised railway reservation opens at the New Delhi booking office with 50 counters; staff numbers rise (Ch. 8; Epilogue)	S0001
1986-12	Policy on Computer Software Export: duty-free imports against a 250 per cent export obligation; satellite export allowed (Chs. 6, 8; Epilogue)	S0241, S0001
1987	AICTE Act gives the All India Council for Technical Education statutory powers (Ch. 7)	S0076
1988	C-DAC set up at Pune with Rs 300 million (Ch. 8)	S0001
1988	Nasscom founded with 38 members (Ch. 8)	S0001, S0060

Date	Event (Chapter)	Sources
1990	GE begins working with Infosys (Ch. 9)	S0034
1990	Immigration Act of 1990 creates the H-1B visa with an annual cap of 65,000 (Chs. 9, 13)	S0105, S0254
1990-08	First Software Technology Park opens at Pune; Bangalore follows in October and Bhubaneswar in December (Chs. 8, 9, 10)	S0241, S0091, S0001
1990-91	Software exports reach $131.2 million (Heeks; $128 million on Bhatnagar's series) (Chs. 5, 6, 8, 9)	S0085, S0001
1991-05	Rajiv Gandhi assassinated; Infosys postpones its share issue (Chs. 8, 9)	S0034
1991-07-01	Rupee devalued in two steps, on 1 and 3 July (Ch. 9)	S0088, S0001
1991-07-24	Manmohan Singh's budget extends the section 80HHC export tax concession to software (Ch. 9; Epilogue)	S0088
1991-11-26	Infosys advertises in the Times of India for 64 engineers: 'A computer software background is not essential' (Chs. 9, 11)	S0252
1992	Citicorp spins off its Indian software unit (later i-flex) with $400,000 under Rajesh Hukku (Ch. 17)	S0176
1992-07-30	Supreme Court rules in Mohini Jain that capitation fees are impermissible (Chs. 7, 12)	S0064
1993-02-04	Supreme Court's Unni Krishnan judgment sets up free and payment seats (Chs. 7, 12)	S0063
1993-02-24	Infosys initial public offering opens (shares listed June 1993) (Chs. 9, 11)	S0034, S0089
1994	World Bank study by Nagy Hanna warns that automation tools may erode demand for body shopping (Ch. 6; Epilogue)	S0241
1995	Onsite work falls to 66 per cent of software revenue (90 per cent in 1988, 56 per cent in 2000) (Ch. 9; Epilogue)	S0254
1995-05	General Electric seeks price cuts; Infosys declines and loses much of the business (Chs. 9, 11)	S0034, S0089
1996-10	GE Capital International Services set up (1997 in other accounts) (Chs. 14, 15)	S0110
1997-09	H-1B cap of 65,000 reached for the first time (Ch. 13)	S0105
FY1997-98	Y2K work is 23.3 per cent of Infosys revenue (Chs. 9, 11)	S0089, S0001
1998	Infosys receives about 70,870 applications and hires about 1,300 (year to March 1998) (Chs. 9, 11, 12)	S0089
1998	American Competitiveness and Workforce Improvement Act raises the H-1B cap to 115,000 (Ch. 13)	S0105
1998-03-24	Microsoft announces a software development centre in Hyderabad (Chs. 10, 15)	S0195
1998-06	Cognizant Technology Solutions goes public on Nasdaq; about 870 of its staff are in India (Ch. 15)	S0274
1999	Indians receive 47 per cent of H-1B visas (55,047) (Chs. 10, 13)	S0255
1999-03-11	Infosys lists on Nasdaq; 1.8 million ADSs priced at $34 (Ch. 11)	S0259, S0090

Date	Event (Chapter)	Sources
2000	Simputer hand-held designed by IISc and Encore Software (Chs. 10, 17)	S0001
2000-01	Y2K rollover passes without crisis; Nasscom puts India's Y2K cost at Rs 1,800 crore (Ch. 11)	S0108
2000-12-15	Infosys share price reaches Rs 7,160 (Rs 63.75 in April 1996) (Ch. 11)	S0262
2002-01	Anna University campus placements fall to about 320 from 1,200; some firms defer joining dates and cut promised pay (Chs. 11, 12, 18)	S0263
2004	Google opens offices in Bangalore and Hyderabad (Ch. 15)	S0200
2004-01	Thomas-Voinovich provision restricting offshoring of some federal work signed into law (Ch. 14)	S0111
2004-02	Gregory Mankiw calls offshoring 'the latest manifestation of the gains from trade' (Ch. 14)	S0113, S0111
2004-10	H-1B cap reverts to 65,000 for FY2005 and is reached on the first day; India 49 per cent of new arrivals (Ch. 13)	S0105
2004-10-13	Sergey Brin in Bangalore: Google will run 'a mirror exactly' of US development there (Ch. 15)	S0200
2005	Infosys completes its Mysore trainee campus for 4,500 trainees at a time (Ch. 12)	S0264, S0115
2005-12	Duke University study challenges press counts of Indian engineering graduates (Ch. 12)	S0213
2006-03	Alan Blinder's essay on offshoring in Foreign Affairs (Chs. 14, 19; Epilogue)	S0113
2007-01-29	Accenture says its India headcount will reach 35,000 and pass its US headcount (Ch. 15)	S0193
2007-04	FY2008 H-1B cap reached in two days (about 119,193 petitions) (Ch. 13)	S0105
FY2007-08	Infosys makes about 46,300 job offers (Chs. 12, 14)	S0265, S0089
2008	Nasscom: nearly 2 million direct IT-BPO employees and about $64 billion revenue (lobby figures) (Ch. 14)	S0107
2009	Infosys freezes salary increases; job offers fall to about 16,900 (Ch. 14)	S0271
2009-01-07	Ramalinga Raju's letter to the Satyam board admits falsified accounts (Chs. 16, 18)	S0122
2009-01-28	Planning Commission notification creates UIDAI (Ch. 16)	S0127
2009-04-13	Tech Mahindra wins the Satyam auction at Rs 58 a share (Ch. 16)	S0279
2009-07-02	Nandan Nilekani appointed UIDAI chairman with Cabinet rank (assumes charge 24 July) (Chs. 16, 17)	S0127
2010	Freshworks (as Freshdesk) founded in Chennai (Ch. 17)	S0173, S0174
2010-11	RBI survey: software and IT services exports $47.6 billion; 79.3 per cent delivered from India (Ch. 14)	S0106
2011-03-31	Income-tax holiday for Software Technology Park units expires (Ch. 14)	S0114
2012-13	AICTE approves about 1.55 million undergraduate engineering seats; 37.6 per cent empty (Ch. 12)	S0267

Date	Event (Chapter)	Sources
2013-03	About 30 product companies form iSPIRT, saying Nasscom favours services (Ch. 17)	S0188
2013-10-30	Infosys settles with the US government for $34 million over visa and I-9 issues, denying wrongdoing (Ch. 13)	S0268
2014	Aspiring Minds reports 18.43 per cent of 2013 engineering graduates 'employable' for IT services (vendor test) (Ch. 12)	S0153
2014-01-25	Karnataka exempts IT companies from the Standing Orders Act for five years (Ch. 18)	S0132
2014-07-15	SEBI bars Satyam's promoters and officials for 14 years (Ch. 16)	S0123
2014-15	Approved undergraduate engineering intake peaks at about 1.7 million (Ch. 12; Epilogue)	S0267
2015-01-31	Disney IT staff in Orlando leave after training their replacements (Ch. 13)	S0269
2015-04-09	CBI special court in Hyderabad convicts Raju and others in the Satyam case (Ch. 16)	S0123, S0124
2015-05-11	Sentences of the ten convicted in the Satyam case suspended and bail granted pending appeal (Ch. 16)	S0406
2016	Unified Payments Interface (UPI) launched (April or August disputed) (Ch. 17)	S0129, S0148
2017-08-20	KITU founded at a meeting of about 200 IT workers in Bengaluru (Ch. 18)	S0294
2017-08-24	Nine-judge Supreme Court bench holds privacy a fundamental right (Puttaswamy) (Ch. 16)	S0127
2018-09-26	Supreme Court upholds the Aadhaar Act but strikes down private use, bank and mobile linking and data retention beyond six months (Ch. 16)	S0127
2019-09-09	Securities Appellate Tribunal quashes SEBI's bans on Price Waterhouse and two partners, upholding disgorgement (Ch. 16)	S0404
2020-03-20	Infosys has enabled 93 per cent of employees to work remotely (Ch. 18)	S0287
2021	Indian start-up funding peaks (about $37 billion on Tracxn's implied count; other vendors higher) (Ch. 17)	S0147
FY2021-22	Infosys makes 222,500 job offers (48,029 the year before) (Ch. 18)	S0289, S0288
2021-09-22	Freshworks lists on Nasdaq at $36 a share (Ch. 17)	S0174
2021-12	Zoho's trade-secrets suit against Freshworks settled and dismissed; terms undisclosed (Ch. 17)	S0411, S0412
2021-12-31	EPAM's largest delivery centres are in Ukraine, Belarus and Russia; India has 4,349 (Ch. 15)	S0170
2022-08	Rishad Premji calls moonlighting 'cheating' (Ch. 18)	S0131
2022-10	Business Today reports delayed joining dates for campus recruits at Accenture and other firms (Prologue; Ch. 18)	S0134
2023	Cognizant's NextGen restructuring costs $229 million; voluntary attrition falls to 13.8 per cent (Ch. 18)	S0293
FY2023-24	Infosys offers fall to 26,975; headcount falls to 317,240 (Prologue; Chs. 18, 19; Epilogue)	S0291

Date	Event (Chapter)	Sources
2023-02-02	Securities Appellate Tribunal sets aside SEBI's 2018 Satyam orders for those who appealed (Ch. 16)	S0402
2024-03	IndiaAI Mission launched with an outlay of about Rs 10,372 crore (Ch. 19; Epilogue)	S0186
2024	Karnataka adopts a GCC policy for 2024-29 (Ch. 15; Epilogue)	S0204
2024-06	NITES says more than 10,000 graduates await joining dates at five IT firms (union estimate) (Prologue)	S0133
2024-06	Infosys's annual report says clients demand productivity commitments because of AI code generation (Prologue; Chs. 18, 19; Epilogue)	S0291
2024-08-20	NITES complains to the Union Labour Ministry about 2,000+ unonboarded Infosys recruits of 2022 (Prologue; Ch. 18)	S0295, S0311
2024-09	Nasscom-Zinnov count about 1,700 GCCs employing 1.9 million; project $99-105 billion revenue by 2030 (industry figures) (Chs. 15, 19; Epilogue)	S0141, S0317
2024-10-07	Delayed Infosys recruits join the Mysuru campus (21 October in another account) (Prologue)	S0311, S0313
2024-12-23	AICTE declares 2025 the 'Year of Artificial Intelligence' (Ch. 19; Epilogue)	S0145
2025-02	Union Budget announces a national framework to guide states on GCCs (Ch. 15; Epilogue)	S0205
2025-02-07	Infosys lets go Mysuru trainees from the delayed 2022 batch (about 300 by its count, 700 by NITES's) (Prologue; Ch. 18; Epilogue)	S0133, S0311, S0312
2025-02-13	Union Labour Ministry directs Karnataka to take 'urgent necessary actions' on the Infosys terminations (Prologue)	S0133
2025-04-26	IndiaAI selects Sarvam AI to build a sovereign language model (Ch. 19)	S0308, S0307, S0186
2025-06-06	AICTE chairman says a committee is revising model curricula to bring AI into all branches (Ch. 19; Epilogue)	S0145
2025-07-01	Infosys annual report: clients 'have started demanding' productivity commitments (Ch. 19)	S0296
2025-07-27	TCS announces a 2 per cent cut (about 12,200 jobs); its chief executive says AI is not the cause (Ch. 19)	S0135, S0301
2025-09-19	Proclamation 10973 imposes a $100,000 payment on new H-1B petitions from 21 September (Ch. 19; Epilogue)	S0136
2025-10	NITI Aayog roadmap: tech workforce could fall to 6 million or rise to 10 million by 2031 (Epilogue)	S0316
2025-12-09	Rest of World reports fewer than a quarter of one Jabalpur class have job offers (Ch. 19; Epilogue)	S0305
2026-02	Nasscom estimates FY2025-26 industry revenue at $315 billion and direct employees at 5.95 million (industry body) (Prologue; Ch. 19; Epilogue)	S0306
2026-04	TCS net headcount down 23,460 in FY2025-26; TCS says it has made 25,000 fresher offers for FY2026-27 (Prologue; Ch. 19)	S0139, S0309

Date	Event (Chapter)	Sources
2026-05	Zinnov-Nasscom count 2,117 GCCs employing 2.36 million in FY2025-26 (industry figures) (Chs. 15, 19; Epilogue)	S0318
2026-05-11	Forbes India reports 'AI-led deflation'; HCLTech's chief executive says a $100 million deal is 'maybe $80 million' today (Ch. 19; Epilogue)	S0142
2026-06-08	US District Court in Massachusetts rules the $100,000 H-1B fee unlawful (Ch. 19; Epilogue)	S0137
2026-06-15	Infosys annual report says clients are 'selectively insourcing' as AI raises productivity; lists Claude Code among its tools (Ch. 19; Epilogue)	S0297
2026-09-18	Proclamation 11069 extends the H-1B fee to 21 September 2027 (Ch. 19; Epilogue)	S0300
2026-10	TCS reports 10,000 freshers onboarded in the July-September quarter (14,000 in April-June); headcount 598,056 (Prologue; Ch. 19)	S0138, S0310

Data appendix

This appendix describes the 27 data files built for this book. Each section says what a file measures, how it is defined, what years it covers, where the figures come from and what limits them, and gives a short table of selected values. The full files, with a source ID and usually a confidence grade on every row, are in the book's repository under `book/research/data/`, together with notes on gaps (`notes/`) and the scripts that rebuild them (`scripts/`). Source IDs refer to the source register, `book/research/sources.md`. A few conventions apply throughout. Indian fiscal years run from April to March and are written FY2023-24 (the year to 31 March 2024); where a file gives a single year for a fiscal year, it is the calendar year in which the fiscal year ends. Lakh (100,000) and crore (10 million) are converted to international units. Rupee amounts are converted to US dollars at annual average rates from `inr_usd.csv` (calendar years) and `inr_usd_fy.csv` (fiscal years). Real values are deflated as each file states: rupee series by an Indian consumer price index (`cpi_india_cy.csv` or `cpi_iw.csv`), dollar series by the US CPI-U (`us_cpi_u.csv`), with the base year given in the column name. Series with different definitions are kept in separate columns and never joined. Figures from Nasscom, the industry's lobby group, from Zinnov, a consultancy that sells to the global capability centres it counts, from Aspiring Minds, which sells the test it reports on, and from the National Foundation for American Policy (NFAP), a pro-immigration group, are marked as such, as are projections.

Reference series

Rupees per US dollar, calendar year (`inr_usd.csv`)

What it measures. The annual average exchange rate of the Indian rupee against the US dollar, used for every calendar-year conversion in the book.

Definition and unit. Rupees per US dollar. Three independent series sit in separate columns: the Reserve Bank of India (RBI) Handbook (1987-2025), the Federal Reserve's G.5A noon buying rates (1973-2025) and the Penn World Table (1950-2023). A preferred column takes RBI where available, then the Federal Reserve, then the Penn World Table.

Coverage. 1950-2025 (76 years).

Sources. S0155, S0320, S0319.

Caveats. The series agree within 0.3 per cent in the overlap years checked. The rupee was pegged for much of the period, and the regime dates in the file's header (fixed parity from 1949, devaluation of June 1966, sterling link, basket peg, unified market rate from 1993) are from background knowledge and not yet sourced. Annual averages hide the within-year jumps of 1966 and July 1991.

Year	Rupees per dollar (preferred)	Series used
1950	4.76	Penn World Table
1960	4.76	Penn World Table
1966	6.36	Penn World Table (devaluation in June)
1970	7.50	Penn World Table
1975	8.41	Federal Reserve
1980	7.89	Federal Reserve
1985	12.33	Federal Reserve
1991	22.69	RBI (devaluations of 1 and 3 July)
2000	44.94	RBI
2010	45.73	RBI
2020	74.10	RBI
2025	87.17	RBI

Full file: `book/research/data/inr_usd.csv`.

Rupees per US dollar, fiscal year (`inr_usd_fy.csv`)

What it measures. The exchange rate averaged over the Indian fiscal year, and the rate at its end, used for converting fiscal-year flows such as export earnings and company revenue.

Definition and unit. Rupees per US dollar, annual average and end-year (31 March) rate, as published in RBI Handbook Table 133.

Coverage. FY1986-87 to FY2025-26.

Sources. S0155.

Caveats. The basis changes from official rates (to FY1991-92) to indicative market rates and then RBI and FBIL reference rates. Averages are used for flows, end-year rates for stocks. No RBI series was obtained for fiscal years before FY1986-87.

Fiscal year	Average	End-year (31 March)
FY1986-87	12.78	12.89
FY1991-92	24.47	31.23
FY1995-96	33.45	34.35
FY2000-01	45.68	46.64
FY2005-06	44.27	44.61
FY2008-09	45.99	50.95
FY2013-14	60.50	60.10
FY2015-16	65.47	66.33
FY2020-21	74.23	73.50
FY2022-23	80.36	82.22
FY2024-25	84.58	85.58
FY2025-26	88.31	94.65

Full file: `book/research/data/inr_usd_fy.csv`.

Consumer prices in India, calendar year (`cpi_india_cy.csv`)

What it measures. A long consumer price index for India, used to deflate calendar-year rupee figures and any figure before FY1993-94.

Definition and unit. OECD Main Economic Indicators series for India, index 2015 = 100. A multiplier column converts a nominal rupee value of a given year into 2024 rupees.

Coverage. 1957-2024.

Sources. S0321 (OECD via FRED).

Caveats. The OECD documentation was not read. The series moves with the Consumer Price Index for Industrial Workers (CPI-IW) rather than the newer CPI Combined, and is treated here as CPI-IW-based. Base changes in the underlying national index (1960, 1982, 2001, 2016) are linked by the OECD by a method not documented in the file.

Year	Index (2015 = 100)	Multiply by, for 2024 rupees
1960	1.70	91.71
1970	3.09	50.41
1975	5.40	28.89
1980	6.55	23.81
1985	10.23	15.25
1991	17.59	8.87
1995	25.40	6.14
2000	36.57	4.27
2005	44.44	3.51
2010	67.29	2.32
2015	100.00	1.56
2024	155.99	1.00

Full file: `book/research/data/cpi_india_cy.csv`.

Consumer Price Index for Industrial Workers, fiscal year (`cpi_iw.csv`)

What it measures. The Labour Bureau's CPI-IW, the index the book's house rules prefer for deflating fiscal-year rupee series.

Definition and unit. All-India general index, fiscal-year average, as reproduced in RBI Handbook Table 34. The file gives the value on the base in force each year and a series linked to 2016 = 100 with the published linking factors (4.63 and 2.88), plus a multiplier into FY2025-26 rupees.

Coverage. FY1993-94 to FY2025-26.

Sources. S0156 via S0155.

Caveats. Bases change in FY2005-06 and FY2020-21. Values on the older bases are published as whole numbers, so the linked series carries rounding error of up to about 0.2 per cent. Earlier years are not in this table; `cpi_india_cy.csv` covers them by calendar year.

Fiscal year	Base	Published value	Linked (2016 = 100)	Multiply by, for FY2025-26 rupees
FY1993-94	1982	258	19.35	7.60
FY1995-96	1982	313	23.47	6.26
FY2000-01	1982	444	33.30	4.41
FY2005-06	2001	117	40.62	3.62
FY2008-09	2001	145	50.35	2.92
FY2010-11	2001	180	62.50	2.35
FY2013-14	2001	236	81.94	1.79
FY2015-16	2001	265	92.01	1.60
FY2020-21	2016	118	118.00	1.25
FY2022-23	2016	131	131.00	1.12
FY2024-25	2016	143	143.00	1.03
FY2025-26	2016	147	147.00	1.00

Full file: `book/research/data/cpi_iw.csv`.

US consumer prices (`us_cpi_u.csv`)

What it measures. The US Consumer Price Index for All Urban Consumers, used to express dollar figures in 2025 dollars.

Definition and unit. BLS series CUUR0000SA0, US city average, all items, annual average, index 1982-84 = 100. A multiplier converts a nominal dollar value of a given year into 2025 dollars. Fiscal-year dollar figures elsewhere are deflated with a weighted average of the two calendar years they span, as each file states.

Coverage. 1950-2025. 2026 is omitted as a partial year.

Sources. S0157.

Caveats. The 2025 average rests on 11 months, because October 2025 is missing from the BLS data.

Year	CPI-U (1982-84 = 100)	Multiply by, for 2025 dollars
1950	24.1	13.36
1960	29.6	10.88
1970	38.8	8.30
1975	53.8	5.98
1980	82.4	3.91
1985	107.6	2.99
1991	136.2	2.36
2000	172.2	1.87
2005	195.3	1.65
2010	218.1	1.48
2020	258.8	1.24
2025	321.9	1.00

Full file: `book/research/data/us_cpi_u.csv`.

Computers and early computing education

Computers installed in India (`computers_installed.csv`)

What it measures. How many computers India had, from the first counts in the 1960s to 2010.

Definition and unit. Machines, as counted by each source. The scope differs from row to row: all digital computers, IBM installations, ECIL sales, personal computers in use.

Coverage. Points from 1964 to 2010. A table of separate counts, not a series.

Sources. S0007, S0001, S0020, S0003, S0237.

Caveats. Definitions change between rows. The sources disagree on IBM's installed base in 1977-78 (over 150 machines in Rajaraman's account; 800 installations maintained by CMC in Dataquest's), probably because they count different things. The 1990 figure does not define its scope.

Date	Measure	Value
End 1964	Digital computers in India	15
1965	Computers in India (Committee on Automation)	30
1968	Computers in India	88
1967-71	Computers introduced (5 referred for labour screening)	94
1970	IBM 1401 installations	about 80
Aug 1971	Computers in India (government 10, public sector 39, private 55, research and education 36)	140
1971-78	ECIL computers sold (4 to private buyers)	98
1974	IBM share of installations	65% of 217
1977-78	IBM installations (disputed)	over 150
1978	Computers in India	fewer than 1,000
1990	Computers in India	80,000
2010	Personal computers in use	over 60 million

Full file: `book/research/data/computers_installed.csv`.

Early computer-science degrees (`cs_degrees_early.csv`)

What it measures. When computer science began as a degree subject in India, and the few counts of graduates available before 2010.

Definition and unit. A table of dated events with counts where the source gives them; persons or programmes.

Coverage. 1965-2010.

Sources. S0001 (Rajaraman 2012), S0003 (Bassett 2009), S0043.

Caveats. Sources disagree on the start of Kanpur's master's teaching (1965 or 1966) and on the rank of the last student admitted to its first B.Tech class (44 or 40). Rajaraman's 1998 figure of 135,000 'engineering students' is probably an intake, not a count of all students.

Date	Event or count	Value
1965	IIT Kanpur M.Tech with a computer-science option, first CS teaching as a discipline (1966 in Rajaraman's chronology)	
1978	IIT Kanpur's first B.Tech in computer science, the first at any IIT	20 seats
1978-80	Other IITs start CS degrees	
1980	Rajaraman panel proposes the Master of Computer Applications (MCA)	
1980s	Institutions fully funded to start the MCA	10
1998	MCA graduates	about 7,000
1998	Engineering students (probably intake)	about 135,000
2010	MCA graduates	about 120,000

Full file: `book/research/data/cs_degrees_early.csv`.

Engineering education and graduates

Engineering colleges, intake, enrolment and vacancy (`engineering_intake.csv`)

What it measures. The capacity of India's engineering education: how many colleges, how many approved places, how many students took them up and how many places stayed empty.

Definition and unit. Institutions; persons; per cent. 'Approved intake' is the maximum admissions the All India Council for Technical Education (AICTE) approves, excluding supernumerary seats, at AICTE institutions only (the IITs are outside AICTE). First-year enrolment at all levels is the figure given to the Lok Sabha. Vacancy is the share of approved places left empty.

Coverage. College counts 1950-51 to 2005-06; intake and enrolment 2012-13 to 2021-22; IIT seats 1978.

Sources. S0248 (statistics ministry), S0323 (Lok Sabha reply), S0267 (Avanti Fellows analysis of AICTE and AISHE data), S0324, S0001.

Caveats. College counts are institutions, not students. There is a gap in all series between 2006 and 2012. AICTE's 2022-23 data are an incomplete load and are not used, and its data on graduates placed in jobs are unusable. A 2023 press table of 'BTech' intake does not match either official series and is kept apart at low confidence.

Year	Colleges	Approved intake, all levels	Approved intake, undergraduate	First-year enrolment, all levels	Undergraduate seats empty
1950-51	31				
1960-61	76				
1970-71	107				
1980-81	149				
1990-91	289				
1997-98	486				
2000-01	678				
2005-06	1,562				
2012-13		2,695,333	1,549,000	1,751,538	37.6%
2014-15			1,705,000 (peak)	1,739,357	
2016-17				1,559,229	49.5%
2021-22		2,366,728	1,253,000		34.1%

Full file: `book/research/data/engineering_intake.csv`.

Engineering graduates (`engineering_graduates.csv`)

What it measures. How many engineers India produces each year, a much-quoted and poorly measured figure.

Definition and unit. Persons per year, with the measure stated on each row; four-year degrees and three-year diplomas are kept apart.

Coverage. 2004 and about 2013 only. The official AISHE out-turn tables were not obtained.

Sources. S0213 (Duke University study, 2005), S0153 (Aspiring Minds, 2014), S0001.

Caveats. There is no consistent series. The 2004 figures are a Nasscom projection reported by the Duke team. The 600,000 figure is Aspiring Minds' and is unsourced. Rajaraman's 1.3 million is kept only to document a dispute: he calls it both students and annual graduates, and it is most likely intake or seats.

Period	Measure	Value
2004	Bachelor's degrees in engineering, CS and IT (Nasscom projection)	112,000
2004	Three-year diplomas in the same fields (Nasscom projection)	103,000
2004	Total, degrees and diplomas	215,000
c. 2013	Engineers graduating annually (Aspiring Minds, unsourced)	600,000
c. 2010	Rajaraman's 1.3 million (probably intake; not graduates)	1,300,000

Full file: `book/research/data/engineering_graduates.csv`.

Women in engineering education (`women_engineering_enrolment.csv`)

What it measures. Women's share of engineering students and graduates.

Definition and unit. Per cent, with the base stated on every row: AICTE first-year undergraduate enrolment, AISHE enrolment across all years of study, or AISHE degrees awarded. One workforce figure is kept for contrast.

Coverage. 2012-13, 2020-21 and 2021-22 points; no consistent annual series.

Sources. S0325 (Society of Women Engineers, citing AICTE and AISHE), S0152, S0117.

Caveats. The bases differ in scope and are never combined. A widely reported rise from 8 per cent (2016) to 19.72 per cent (2021) appears to concern the IITs and NITs, not all colleges; its scope could not be verified and it is excluded.

Year	Base	Women
2012-13	AICTE first-year undergraduate engineering enrolment	33.7% (303,229)
2020-21	AICTE first-year undergraduate engineering enrolment	30.2% (200,865)
2020-21	AISHE B.Tech enrolment, all years	28.7%
2021-22	AISHE undergraduate computer engineering enrolment	34.2%
2021-22	AISHE undergraduate computer engineering degrees awarded	38.5% (83,696)
2001-02	Software professionals in industry (Nasscom; contrast only)	21%

Full file: `book/research/data/women_engineering_enrolment.csv`.

'Employability' estimates (`employability_tests.csv`)

What it measures. The share of tested engineering students who met a test vendor's or consultant's benchmark for particular jobs.

Definition and unit. Per cent of the tested sample meeting the vendor's own benchmark for a role. This is not a labour-market outcome: it does not measure whether graduates found work.

Coverage. Aspiring Minds reports of 2014 and 2016; one McKinsey figure of about 2005 as reported by the Duke study.

Sources. S0153 (Aspiring Minds National Employability Report), S0213.

Caveats. Aspiring Minds sells the AMCAT test it uses, and the benchmarks are its own. Samples and benchmarks differ between editions, and press reports often mix them. The McKinsey study itself was not read. Every use names the vendor and the role.

Report	Role or measure	Meeting benchmark
2014 (2013 graduates)	Software engineer, IT services ('trainable within 3 to 6 months')	18.43%
2014	Software engineer, IT products	3.21%
2014	Non-functional role such as BPO	39.84%
2016	Software engineer, IT services	17.91%
2016	Software engineer, IT products	3.67%
2016	Business process outsourcing	40.57%
2016	Start-up software engineering	3.84%
2016	'Software-industry ready' (programming)	4.51%
c. 2005	Indian engineers able to 'compete in the global outsourcing arena' (McKinsey, via Duke)	25%

Samples: over 120,000 students at 520+ colleges (2014); over 150,000 at 650+ colleges (2016). Full file: `book/research/data/employability_tests.csv`.

Graduate unemployment (`graduate_unemployment.csv`)

What it measures. Unemployment among Indian graduates, with a few comparison points.

Definition and unit. Unemployment rate, per cent of the labour force. Indian rows use the Periodic Labour Force Survey (PLFS), usual status (principal and subsidiary), persons aged 15 and over with graduate education, unless stated. PLFS years run July to June.

Coverage. India 2019-20 to 2023-24; one all-persons and one under-25 point; US 2025 points.

Sources. S0326 (Press Information Bureau), S0327, S0328, S0329 (statistics ministry), S0136.

Caveats. PLFS rates for 2017-18 and 2018-19 were not obtained (a register note giving 17.2 per cent for 2017-18 was wrong; that is the 2019-20 figure). The 2023-24 graduate rate of 11.2 per cent comes from a press report and conflicts with 13.0 per cent in another note; it is low confidence. PLFS is never joined to the older NSSO surveys. The under-25 figure uses a different method and was not read in the original. The US figures are as cited in a US presidential proclamation.

Year	Country	Population	Rate
2019-20	India	Graduates aged 15+ (PLFS)	17.2%
2020-21	India	Graduates aged 15+ (PLFS)	15.5%
2021-22	India	Graduates aged 15+ (PLFS)	14.9%
2022-23	India	Graduates aged 15+ (PLFS)	13.4%
2023-24	India	Graduates aged 15+ (PLFS; press report, disputed)	11.2%
2023-24	India	All persons aged 15+ (PLFS)	3.2%
2023	India	Graduates under 25 (State of Working India 2026)	39.3%
2025	United States	Recent graduates aged 22-27, computer science (NY Fed)	6.1%
2025	United States	Recent graduates aged 22-27, computer engineering (NY Fed)	7.5%

Full file: `book/research/data/graduate_unemployment.csv`.

Emigration of IIT graduates (`iit_emigration.csv`)

What it measures. The share of IIT graduates and top entrance-exam scorers who went abroad.

Definition and unit. Share of a cohort living or working abroad, in per cent or as stated.

Coverage. 1970s IIT Bombay cohorts; the JEE 2010 cohort.

Sources. S0003 (Bassett), S0046 (Sukhatme survey), S0172 (Choudhury, Ganguli and Gaulé, NBER Working Paper 31308, 2023).

Caveats. Institutions, cohorts and methods differ, so the rows cannot be read as a trend. The second row is a secondary report of the Sukhatme survey.

Cohort	Measure	Value
1973-1977 graduates	IIT Bombay electrical engineering graduates living abroad	slightly over 40%
Since 1970	IIT Bombay graduates settled abroad (as reported in Nature, 1993)	about 3 in 10
JEE 2010, top 1,000	Migrated abroad by about 2020	36%
JEE 2010, top 100	Migrated abroad	62%
JEE 2010 cohort	Extra migration for graduate school at the original 'Top 5' IITs	+5 percentage points

Full file: `book/research/data/iit_emigration.csv`.

Indian students in the United States (`indian_students_us.csv`)

What it measures. How many students from India were enrolled in American higher education, including those on post-study work.

Definition and unit. Persons, academic year, from the Institute of International Education's Open Doors India fact sheet. A level split is given for the latest two years.

Coverage. 2017/18 to 2024/25. Earlier Open Doors years were not obtained.

Sources. S0158 (IIE is a non-profit with State Department support).

Caveats. Totals include Optional Practical Training (OPT), the post-study work route, which rose from 97,556 to 143,740 in the last year; graduate students fell 9.5 per cent in 2024/25 while the total rose.

Academic year	Total	Undergraduate	Graduate	OPT
2017/18	196,271			
2018/19	202,014			
2019/20	193,124			
2020/21	167,582			
2021/22	199,182			
2022/23	268,923			
2023/24	331,602	36,053	196,567	97,556
2024/25	363,019	40,135	177,892	143,740

Full file: `book/research/data/indian_students_us.csv`.

The industry

Software and services exports (`software_exports.csv`)

What it measures. India's export earnings from software and related services, in three families of series that measure different things.

Definition and unit. US$ million, nominal, with real columns in 2025 dollars (deflated by `us_cpi_u.csv`). Family one is early gross foreign-exchange earnings from software: Heeks (citing his own research and Dataquest) and Bhatnagar (via Rajaraman). Family two is Nasscom's industry exports, whose scope widens over time from 'software exports' to 'IT-BPO exports' to 'technology industry' exports including business process management and engineering services. Family three is the RBI: a survey of software and IT services exporters (2009-11) and the balance-of-payments 'software services' receipts (from FY2000-01).

Coverage. 1980 to FY2025-26, with gaps (no series for FY1998-99 to FY1999-2000 except one Nasscom point). Heeks rows for 1980-1987 are calendar years; all others are fiscal years.

Sources. S0085, S0001, S0109, S0107, S0140, S0306, S0106, S0155 (six editions of the RBI Handbook).

Caveats. The families are not reconciled and must not be joined. Heeks's figures are gross; net foreign-exchange earnings were estimated at about 40 per cent of gross. Nasscom's figures are a lobby group's and change definition; its FY2025-26 figure is an estimate. The RBI balance-of-payments series is compiled with Nasscom data as a controlling total, so it is not wholly independent. The FY2025-26 RBI value is preliminary. Revisions between RBI editions were small.

The 1990s: two series compared. Two series cover the decade of the reforms. One, compiled by Subhash Bhatnagar and reproduced by V. Rajaraman, gives $128 million in 1990-91, $164 million in 1991-92, $225 million in 1992-93, $330 million in 1993-94, $450 million in 1994-95, $734 million in 1995-96, $1,100 million in 1996-97 and $1,759 million in 1997-98 (S0001, p. 48, Table 2). The other, Richard Heeks's, drawn from his own research and the trade magazine *Dataquest*, gives $131.2 million, $173.9 million, $219.8 million, $314.0 million, $480.9 million, $668.0 million, $997.0 million and $1,650.0 million for the same years (S0085, p. 2, Table 1). The two differ by up to about a tenth in some years: Bhatnagar's figure is about 10 per cent higher in 1995-96 and 1996-97, and Heeks's about 7 per cent higher in 1994-95. Both show annual growth of between about a quarter and two-thirds (Bhatnagar 28 to 63 per cent, Heeks 24 to 65 per cent). On Heeks's longer series, growth ran at 29 to 51 per cent a year in 1988-90, slowed to 24 to 33 per cent in 1990-93, and from 1993-94 ran at 39 to 65 per cent, faster than in 1990-93 and at least as fast as in the late 1980s (S0085, p. 2). Heeks notes that these 'headline' figures are gross foreign-exchange earnings (S0085, p. 2); as the caveat above says, net earnings were a much smaller share. The narrative chapters give one figure from these series and cite it.

Period	Heeks	Bhatnagar	Nasscom (lobby)	RBI survey	RBI balance of payments	RBI BoP, 2025 dollars
1980	4.0					
1984	25.3					
FY1990-91	131.2	128				
FY1995-96	668.0	734				
FY1997-98	1,650.0	1,759				
FY1998-99			2,650 (software exports)			
FY2000-01					6,341	11,771
FY2005-06					23,600	38,592
FY2010-11				47,600	53,100	77,784
FY2015-16					74,153	100,406
FY2020-21					99,997	122,945
FY2025-26			246,000 (estimate)		203,675 (preliminary)	

Full file: `book/research/data/software_exports.csv`.

Industry employment (`industry_workforce.csv`)

What it measures. How many people the Indian software and IT industry employs.

Definition and unit. Persons, point estimates in separate columns by source. The industry's scope widens over time, from 'software professionals' to 'IT-BPO' to the 'technology industry' including business process management, engineering services, products and global capability centres.

Coverage. FY1989-90 to FY2025-26, sparse points only; no annual series was obtained.

Sources. S0259 (Infosys prospectus, quoting Nasscom), S0254, S0001, S0107, S0140, S0306, S0316 (NITI Aayog).

Caveats. Almost every figure is Nasscom's, a lobby group's estimate on the definition current at the time; FY2023-24 was restated upwards after a 2024 reassessment of global capability centres. NITI Aayog's range is an all-sector 'tech workforce' including technology jobs outside the industry and must not be compared with Nasscom's. PLFS-based counts were not obtained.

Period	Measure	Persons
FY1989-90	Software professionals (Nasscom)	56,000
1996	Software professionals (Nasscom)	160,000
FY1997-98	Software professionals (Nasscom)	200,000
FY2007-08	IT-BPO direct employees (Nasscom, 'nearly 2 million')	2,000,000
2010	Direct employment in IT (Rajaraman, from Nasscom)	2,400,000
FY2023-24	Technology industry direct employees (Nasscom, restated)	5,670,000
2023	All-sector tech workforce (NITI Aayog; not comparable)	7.5-8 million
FY2024-25	Technology industry (Nasscom estimate; revised to 5,815,000)	5,800,000
FY2025-26	Technology industry (Nasscom estimate; net additions 135,000)	5,950,000

Full file: `book/research/data/industry_workforce.csv`.

The largest firms (`top_firms.csv`)

What it measures. Revenue, headcount and revenue per employee at selected Indian-heritage IT firms.

Definition and unit. One row per firm-year. Revenue in US$ million, nominal and in 2025 dollars; Wipro also in rupees, converted at the RBI fiscal-year average rate (the book's conversion, not Wipro's). Employees at year end. Revenue per employee in US$ thousand, using the average of opening and closing headcount. Infosys, Wipro and TCS years end on 31 March (the year shown is the calendar year in which the fiscal year ends); Cognizant uses calendar years.

Coverage. Infosys FY1999-FY2026 with gaps; Wipro FY2016-FY2026; Cognizant 2001-2025 with headcount gaps; TCS headcount points.

Sources. Infosys 20-F and F-1 filings (S0322, S0264-S0266, S0270-S0272, S0287-S0291, S0296, S0297); Wipro 20-F (S0292, S0298); Cognizant 10-K (S0192, S0275-S0277, S0293, S0299); SEC XBRL company facts; TCS (S0160, S0139, S0310); S0155; S0157.

Caveats. Infosys reported under US GAAP to FY2008 and IFRS from FY2009. Wipro states headcount as 'more than' a round number and revises it between filings. Cognizant's 2001-02 counts cover technical professionals only. TCS and HCLTech revenue were not compiled (the TCS site was blocked to the research tools); the FY2026 TCS headcount is derived.

Firm	Year	Revenue, US$m \| *Revenue, 2025 US$m*	Employees	Revenue per employee, US$k	
Infosys	FY1999	121	238	3,800	
Infosys	FY2005	1,592	2,691	36,800	52.5
Infosys	FY2009	4,663	6,979	104,900	47.6
Infosys	FY2012	6,994	9,959	150,000	
Infosys	FY2020	12,780	16,044	242,371	54.3
Infosys	FY2023	18,212	19,831	343,234	55.4
Infosys	FY2024	18,562	19,469	317,240	56.2
Infosys	FY2026	20,158		328,594	61.8
Wipro	FY2026	10,489		more than 240,000	44.6
Cognizant	2012	7,347	10,302	156,700	49.9
Cognizant	2025	21,108	21,108	351,600 (256,900 in India)	61.3
TCS	FY2025			607,979	

Full file: `book/research/data/top_firms.csv`.

Recruitment and fresher hiring (`fresher_hiring.csv`)

What it measures. How many people applied to, were interviewed by and were offered jobs at the largest firms, and how many new graduates were hired.

Definition and unit. Persons. Infosys rows are the company's annual recruitment statistics as filed with the SEC. From FY2003-04 'offers' counts all offers of employment, experienced (lateral) and fresher alike. Offers per 1,000 applications is calculated for this book. Other rows are fresher-specific press figures, each labelled.

Coverage. Infosys FY1997-98 to FY2025-26 with gaps (FY1998-99 to FY1999-2000, FY2002-03, FY2005-06 to FY2006-07, FY2010-11, FY2012-13 to FY2018-19); TCS, HCLTech and Infosys press points for FY2022-23 to FY2026-27.

Sources. Infosys filings (S0259, S0322, S0260, S0270, S0264, S0265, S0271, S0272, S0266, S0287-S0291, S0296, S0297); press (S0309, S0310, S0138, S0139, S0133).

Caveats. Offers, hires and joiners are different measures and must not be compared across columns. 'Tested' is not reported from FY2011-12. Infosys used 'received' and 'screened' for applications in different filings. The press figures are low to medium confidence; the Infosys figures of 50,000 and 11,900 fresh graduates for FY2022-23 and FY2023-24 come from a trade site citing other reports. The TCS figure for the July-September 2026 quarter postdates the book's research cut-off.

Fiscal year (Infosys)	Applications	Interviewed	Offers	Offers per 1,000 applications
FY1997-98	70,870	5,800	1,790	25.3
FY2000-01	385,200	20,800	7,650	19.9
FY2001-02	193,640	6,730	1,710	8.8
FY2004-05	1,348,800	36,600	14,500	10.8
FY2007-08	885,000	97,600	46,300	52.3
FY2008-09	488,700	46,600	16,900	34.6
FY2011-12	622,970	60,860	41,460	66.6
FY2019-20	2,141,373	150,130	83,319	38.9
FY2020-21	1,427,618	135,216	48,029	33.6
FY2021-22	5,866,636	523,385	222,500	37.9
FY2023-24	2,436,929	194,367	26,975	11.1
FY2025-26	5,884,480	450,901	87,286	14.8

Press points on freshers: TCS 'reportedly' hired over 44,000 in FY2025-26 and had made 25,000 fresher offers for FY2026-27 by April 2026, with about 14,000 joining in April-June 2026 and 10,000 in July-September; HCLTech hired over 11,000 graduates in FY2025-26; Infosys planned to add 20,000 freshers in FY2026-27. Full file: `book/research/data/fresher_hiring.csv`.

Starting pay (`fresher_pay.csv`)

What it measures. The annual pay of new graduates joining Indian IT services firms.

Definition and unit. Rupees lakh a year, nominal, as reported; medians, single-firm packages and quoted ranges are different measures and labelled. A real column gives the value, or the midpoint of a range, in 2024 rupees, deflated by `cpi_india_cy.csv`.

Coverage. Points from 2007 to 2026; no consistent series. Years after 2024 are not deflated.

Sources. S0330, S0331, S0332, S0333, S0334, S0153.

Caveats. Offer, cost-to-company and take-home pay differ. Premium tracks (such as TCS 'Digital' or HCLTech 'Elite') are excluded. The 2007 TCS figure rests on one anecdotal source. The Moneycontrol study's method was not read. The Aspiring Minds figures are a vendor's estimates.

Year	Measure	Rs lakh a year	2024 Rs lakh
2007	TCS trainee package (single anecdotal source)	3.16	9.86
2012	Median fresher pay, large IT firms (Moneycontrol study)	2.45	4.77
2014	STEM graduate starting salaries (TeamLease Digital estimate)	2.0-2.5	3.72
2015	Median salary (Aspiring Minds, vendor estimate)	2.82	4.40
2016	Median salary (Aspiring Minds, vendor estimate)	3.13	4.65
2022	Median fresher pay, large IT firms (Moneycontrol study)	3.55	3.89
2022	Infosys fresher pay (annual report, as reported)	3.6	3.94
2022	Freshers at large firms (T.V. Mohandas Pai)	3.5-4.0	4.11
2025	STEM graduate starting salaries (TeamLease Digital estimate)	3.0-3.5	
2025	Wipro entry-level salaries (company)	3.0-4.0	
2026	TCS trainee package	3.36	

Full file: `book/research/data/fresher_pay.csv`.

Attrition and women's share, by firm (`firm_women_attrition.csv`)

What it measures. The share of staff leaving each year, and the share of women in the workforce, at firms that report them.

Definition and unit. Per cent. Each row states the firm's own definition: total or voluntary attrition, with or without subsidiaries, IT services only, trailing twelve months.

Coverage. Infosys attrition FY2005-06 to FY2011-12 and FY2019-20 to FY2023-24, women FY2023-24; Cognizant 2020-2025; TCS FY2019-20 to FY2024-25 and the quarter to September 2026.

Sources. Infosys 20-F (S0265, S0271, S0272, S0266, S0291) and 6-K results filings (S0381); Cognizant 10-K (S0293, S0192, S0299); TCS (S0160, S0310), and for FY2019-20 to FY2023-24 press reports of TCS's quarterly results (S0384, S0385).

Caveats. Definitions differ between firms and change over time; rows with different definitions must not be compared. Cognizant switched its headline measure in its 10-K for 2023 from total to voluntary attrition in technology services, so 2022 appears on both bases. The Infosys women's figure is 'nearly 40%'. The TCS figures for FY2019-20 to FY2023-24 are as reported in the press, not taken from the company's fact sheets; those for FY2020-21, FY2022-23 and FY2023-24 rest on one trade publication (*HR Katha*). The reports give the basis as the last twelve months only for FY2021-22 and FY2022-23. They are not voluntary-only rates and should not be set against Infosys's voluntary series.

Firm	Period	Attrition	Definition	Women
Infosys	FY2005-06	11.2%	Comparable, excluding BPO and subsidiaries	
Infosys	FY2007-08	13.4%	Comparable, excluding BPO and subsidiaries	
Infosys	FY2010-11	17.0%	Twelve months to 31 March	
Infosys	FY2011-12	14.7%	Twelve months to 31 March	
Infosys	FY2019-20	15.1%	Voluntary, standalone, fourth quarter	
Infosys	FY2020-21	15.2%	Voluntary, IT services, fourth quarter annualised	
Infosys	FY2021-22	27.7%	Voluntary, last twelve months	
Infosys	FY2022-23	20.9%	Voluntary, last twelve months, IT services	
Infosys	FY2023-24	12.6%	Voluntary, last twelve months, IT services	39.3%
Cognizant	2020	20.6%	Total	36%
Cognizant	2021	30.8%	Total	38%
Cognizant	2022	31.7%	Total	38%
Cognizant	2022	25.6%	Voluntary, technology services	
Cognizant	2023	13.8%	Voluntary, technology services	38%
Cognizant	2025	13.9%	Voluntary, technology services	39%
TCS	FY2019-20	12.1%	As reported; basis not stated	
TCS	FY2020-21	7.2%	As reported (one outlet); basis not stated	
TCS	FY2021-22	17.4%	As reported; trailing twelve months	
TCS	FY2022-23	20.1%	As reported (one outlet); last twelve months	
TCS	FY2023-24	12.5%	As reported (one outlet); basis not stated	

Firm	Period	Attrition	Definition	Women
TCS	FY2024-25	13.3%	Last twelve months, IT services	35.2%

Full file: `book/research/data/firm_women_attrition.csv`.

Foreign firms' Indian headcount (`foreign_firm_india_headcount.csv`)

What it measures. How many people foreign-headquartered IT firms employ in India.

Definition and unit. Employees in India, as reported by the firm or by the press from company statements; persons.

Coverage. Sparse points, 2005-2025.

Sources. S0193, S0194, S0170, S0293, S0192, S0299.

Caveats. Most firms do not disclose Indian headcount. The IBM and Accenture points are press reports of company statements, and the Accenture figure was a plan.

Date	Firm	Employees in India	Total employees
Early 2005	IBM	23,000	
Early 2006	IBM	39,000	
Feb 2007	IBM	53,000	about 300,000
Aug 2007 (planned)	Accenture	35,000	160,000
2021	EPAM	4,349	52,617 (delivery)
2024	EPAM	10,072	about 55,100 (delivery)
2022	Cognizant	258,500	355,300
2023	Cognizant	254,000	347,700
2024	Cognizant	241,500	336,800
2025	Cognizant	256,900	351,600

Full file: `book/research/data/foreign_firm_india_headcount.csv`.

Global capability centres (`gcc.csv`)

What it measures. The number, staff and revenue of multinational companies' own centres in India: research and development centres in the 2000s, 'global capability centres' (GCCs) in the 2020s.

Definition and unit. Centres; persons; US$ billion, as each source defines them. The 2008 'MNC R&D centre' is narrower than the later GCC, which includes back-office captives. GCC 'revenue' measures work done for the parent, not sales.

Coverage. 2004 and 2007 (new centres), 2008, FY2023-24 and FY2025-26, with projections and a state target kept separate.

Sources. S0207 (Zinnov via EE Times), S0141, S0317 (Nasscom-Zinnov via Reuters), S0318 (Zinnov-Nasscom via Business Today), S0140, S0204.

Caveats. Every figure comes from Nasscom, the industry's lobby group, or Zinnov, a consultancy that sells to GCCs; both have an interest in a growing count. Nasscom restated GCC revenue in September 2024. The 2004-08 figures were read through a summary. Projections and the Karnataka target are not outcomes.

Period	Measure	Centres	Employees	Revenue, US$bn
2004	New MNC R&D centres opened (Zinnov)	93		
2007	New MNC R&D centres opened (Zinnov)	24		
2008	MNC R&D centres (Zinnov)	594	146,760	
FY2023-24	GCCs (Nasscom-Zinnov)	about 1,700 companies (2,975 units)	1,900,000	64.6
FY2025-26	GCCs (Zinnov-Nasscom)	2,117	2,360,000	98.4
2030	PROJECTION made in 2024 (Nasscom-Zinnov)	2,100-2,200	2.5-2.8 million	99-105
2029	TARGET: new GCCs sought by Karnataka's policy	500		

Full file: `book/research/data/gcc.csv`.

Start-up funding (`startup_funding.csv`)

What it measures. Venture funding raised by Indian start-ups each year.

Definition and unit. US$ billion, calendar year, as reported by each data vendor.

Coverage. 2021-2024.

Sources. S0147 (Tracxn via The Week; Inc42).

Caveats. Vendors define 'start-up' and 'tech' differently, and their series must not be joined. Tracxn's 2021 figure is implied from its reported 31 per cent fall in 2022. Some rows were read through a summary.

Year	Vendor	US$ billion
2021	Tracxn (implied)	about 37
2021	Inc42	about 42
2021	Other vendor, as reported	43.5
2022	Tracxn	25.7
2023	Tracxn	10.8
2024	Tracxn	11.1

Full file: `book/research/data/startup_funding.csv`.

Unified Payments Interface (`upi.csv`)

What it measures. The volume and value of payments made through UPI, the state-backed mobile payments network.

Definition and unit. Annual (April-March) volume in billion transactions and value in Rs lakh crore (Rs trillion), from the RBI's payment system indicators.

Coverage. FY2016-17 to FY2023-24, plus one monthly point for August 2025.

Sources. S0155 (RBI Handbook, 2021, 2025 and 2026 editions), S0129.

Caveats. UPI began in FY2016-17, so that year is partial. FY2024-25 was not in the editions read; the NPCI site could not be reached. The August 2025 figure is monthly and from the press.

Period	Transactions, billion	Value, Rs lakh crore
FY2016-17	0.02	0.07
FY2017-18	0.92	1.10
FY2018-19	5.39	8.77
FY2019-20	12.52	21.32
FY2020-21	22.33	41.04
FY2021-22	45.96	84.16
FY2022-23	83.71	139.15
FY2023-24	131.13	199.95
August 2025 (one month)	20.01	

Full file: `book/research/data/upi.csv`.

The H-1B visa

Cap and policy events (`h1b_policy.csv`)

What it measures. The annual cap on new H-1B visas, how many new petitions were issued against it, and the main policy changes.

Definition and unit. Petitions, US fiscal years (October-September). Numeric rows are NFAP's table of new petitions issued against the cap; other rows are dated events.

Coverage. 1990-2026.

Sources. S0105 (Congressional Research Service), S0338 (NFAP), S0335, S0136, S0137, S0300.

Caveats. NFAP is a pro-immigration group; its table is compiled from government data. The start year of the extra 20,000 places for holders of US master's degrees is not in the sources read (Chapter 13 gives 2004). Rulings of June and September 2026 on the $100,000 fee are summarised differently by different law firms, and the court orders were not read; the 2026 events postdate the research cut-off.

US fiscal year	Cap	New petitions issued against the cap
1992	65,000	48,600
1995	65,000	54,200
1997	65,000	65,000 (cap first reached, September 1997)
1999	115,000	115,000
2000	115,000	115,000
2001	195,000	163,600
2002	195,000	79,100
2003	195,000	78,000
2004-2026	65,000	65,000 (cap reached every year)

Events in the file: the Immigration Act of 1990 creates the visa; the FY2005 cap is reached on the first day of filing; the FY2008 cap is reached in two days in April 2007 (about 119,193 petitions); denial rates for initial employment fall from 24 per cent (FY2018) to 13 per cent (FY2020); the first lottery by electronic registration is held for FY2021; Proclamation 10973 of 19 September 2025 requires a $100,000 payment for new petitions; a DHS rule of 29 December 2025 weights selection towards higher wage levels; a Massachusetts court rules the fee unlawful on 8 June 2026; Proclamation 11069 of 18 September 2026 extends the fee to September 2027; and a California court vacates the implementing policies on 30 September 2026. Full file: `book/research/data/h1b_policy.csv`.

H-1B approvals by country of birth (`h1b_country.csv`)

What it measures. The share of approved H-1B petitions that went to workers born in India.

Definition and unit. Approved petitions and per cent, by beneficiary's country of birth; initial (new employment), continuing (extensions) or all, as labelled.

Coverage. October 1999-February 2000; FY2005 (new arrivals); FY2023; FY2024. USCIS reports for FY2000-22 exist but could not be reached.

Sources. S0104, S0105, S0336 (USCIS FY2023 report), S0335 (USCIS FY2024 report, via a verified mirror).

Caveats. Petitions are not people: one worker can appear more than once. The FY2005 row counts new arrivals, a different measure. The USCIS FY2024 report gives two slightly different totals for India.

Period	Measure	India	Share
Oct 1999-Feb 2000	Approved petitions (INS)	124,697	48.5%
FY2005	New arrivals		49.0%
FY2023	All approved petitions	279,386	72.3%
FY2023	Approved, initial employment	68,825	57.9%
FY2023	Approved, continuing employment	210,561	78.8%
FY2024	All approved petitions	283,397	71.0%
FY2024	Approved, initial employment	80,449	57.0%
FY2024	Approved, continuing employment	203,306	78.7%
FY2023	All approved petitions, China (for comparison)	45,344	11.7%
FY2024	All approved petitions, China (for comparison)	46,680	11.7%

Full file: `book/research/data/h1b_country.csv`.

H-1B approvals by employer (`h1b_employer.csv`)

What it measures. Approved H-1B petitions for initial employment at selected employers.

Definition and unit. Petitions approved for new and concurrent employment, by employer, as compiled by NFAP from the USCIS H-1B Employer Data Hub, with related entities grouped by NFAP.

Coverage. FY2022-FY2025 for selected employers, and one administration claim on registrations.

Sources. S0337, S0338 (NFAP, a pro-immigration group); S0300; underlying data S0159 (USCIS hub, not reachable).

Caveats. NFAP revises earlier years between briefs (Cognizant FY2024 appears as 2,837 and 2,873). 'Indian-based' is NFAP's grouping and includes Cognizant, which is headquartered in the United States. The administration's claim that the largest outsourcing firms cut registrations from 24,946 to 2,055 concerns registrations, not approvals, names no firms and is a party's figure.

Employer	FY2023	FY2024	FY2025
Amazon	4,052	3,871	4,644
Meta Platforms	735	920	1,555
Microsoft	987	1,264	1,394
Google	1,267	1,058	1,050
TCS	1,174	1,452	846
Cognizant	2,597	2,873 (2,837 in an earlier brief)	743
IBM	979	1,348	501
LTIMindtree	914	798	401
HCL America	715	1,248	379
Infosys		2,504	263
Wipro			267
Top seven Indian-based companies (NFAP grouping)			4,573 (70% below FY2015)

Full file: `book/research/data/h1b_employer.csv`.

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Index term list

This list gives 487 terms for the index: people, organisations, places, laws and policies, events and concepts. Each term is followed by the parts of the book in which it appears: the Prologue, Chapters 1 to 19 (by number) and the Epilogue. The locations were found by searching the text of each chapter, excluding its endnotes, so they record mentions rather than page numbers; page references will be added when the book is typeset. Names are given as the text gives them, with people listed surname first. Where a place or body has more than one name, the variants are merged under one heading, with the current name first and the older one in brackets, as in Mumbai (Bombay); the entry covers mentions under either name. Pseudonyms given by the sources are marked as such. Abbreviations used in the text follow the full name in brackets.

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